

Introduction to Differential Geometry

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Notation

$\mathcal{M}_+(X, \mathcal{A})$	finite measures on $(X, \mathcal{A})$
v	velocity

Preface

cc-by-nc-sa

Part I

Prerequisites

Chapter 1

Review of Fundamentals

1.1 Set Theory

merge with set theory from analysis notes

Definition 1.1.0.1. Let $\{A_i\}_{i \in I}$ be a collection of sets. The **disjoint union** of $\{A_i\}_{i \in I}$, denoted $\coprod_{i \in I} A_i$, is defined by

$$\coprod_{i \in I} A_i = \bigcup_{i \in I} \{i\} \times A_i$$

We define the **natural projection map**, denoted $\pi : \coprod_{i \in I} A_i \rightarrow I$, by $\pi(i, a) = i$.

Definition 1.1.0.2. Let E and M be sets, $\pi : E \rightarrow M$ a surjection and $\sigma : M \rightarrow E$. Then σ is said to be a section of (E, M, π) if $\pi \circ \sigma = \text{id}_M$.

Note 1.1.0.3. Let $\{A_i\}_{i \in I}$ be a collection of sets and $\sigma : I \rightarrow \coprod_{i \in I} A_i$. We will typically be interested in sections σ of $\left(\coprod_{i \in I} A_i, I, \pi\right)$.

Exercise 1.1.0.4. Let $\{A_i\}_{i \in I}$ be a collection of sets and $\sigma : I \rightarrow \coprod_{i \in I} A_i$. Then σ is a section of $\coprod_{i \in I} A_i$ iff for each $i \in I$, $\sigma(i) \in A_i$

Proof. Clear. □

1.2 Linear Algebra

Note 1.2.0.1. We denote the standard basis on $\mathbb{R}^n$ by $(e_1, \dots, e_n)$.

Definition 1.2.0.2. Let $A \in \mathbb{R}^{n \times n}$. Then A is said to be **invertible** if $\det(A) \neq 0$. We denote the set of $n \times n$ invertible matrices by $GL(n, \mathbb{R})$.

$$O(n)$$

Exercise 1.2.0.3. Let $A, B \in \mathbb{R}^{n \times n}$. Then $AB = I$ iff $BA = I$.

Proof.

- ($\implies$):
Suppose that $AB = I$. Then

$$\begin{aligned} \ker B &\subset \ker AB \\ &= \ker I \\ &= \{0\} \end{aligned}$$

so that $\ker B = \{0\}$. Hence $\text{Im } B = \mathbb{R}^n$ and B is surjective. Then

$$\begin{aligned} IB &= BI \\ &= B(AB) \\ &= (BA)B \end{aligned}$$

Since B is surjective, $I = BA$.

- ($\impliedby$):
Immediate by the previous part.

□

Definition 1.2.0.4. Let $A \in \mathbb{R}^{n \times p}$. Then A is said to be an **orthogonal matrix** if $A^*A = I$. We denote the set of $n \times p$ orthogonal matrices by $O(n, p)$. We write $O(n)$ in place of $O(n, n)$.

$$O(n)$$

Exercise 1.2.0.5. Define $\phi : S_n \rightarrow GL(n, \mathbb{R})$ by

$$\phi(\sigma) = \begin{pmatrix} e_{\sigma(1)}^* \\ \vdots \\ e_{\sigma(n)}^* \end{pmatrix}$$

Then

1. for each $A \in \mathbb{R}^{n \times p}$,

$$(\phi(\sigma)A)_{i,j} = A_{\sigma(i),j}$$

i.e. left multiplying A by $\phi(\sigma)$ the the same as permuting the rows of A by σ

2. ϕ is a group homomorphism

Proof. 1. Let $A \in \mathbb{R}^{n \times p}$. Then

$$\begin{aligned} (\phi(\sigma)A)_{i,j} &= \langle e_{\sigma(i)}^*, Ae_j \rangle \\ &= A_{\sigma(i),j} \end{aligned}$$

2. Let $\sigma, \tau \in S_n$. Part (1) implies that

$$\begin{aligned}\phi(\sigma\tau) &= \begin{pmatrix} e_{\sigma\tau(1)}^* \\ \vdots \\ e_{\sigma\tau(n)}^* \end{pmatrix} \\ &= \begin{pmatrix} e_{\sigma(1)}^* \\ \vdots \\ e_{\sigma(n)}^* \end{pmatrix} \begin{pmatrix} e_{\tau(1)}^* \\ \vdots \\ e_{\tau(n)}^* \end{pmatrix} \\ &= \phi(\sigma)\phi(\tau)\end{aligned}$$

Since $\sigma, \tau \in S_n$ are arbitrary, ϕ is a group homomorphism. □

Definition 1.2.0.6. Define $\phi : S_n \rightarrow GL(n, \mathbb{R})$ as in the previous exercise. Let $P \in GL(n, \mathbb{R})$. Then P is said to be a **permutation matrix** if there exists $\sigma \in S_n$ such that $P = \phi(\sigma)$. We denote the set of $n \times n$ permutation matrices by $\text{Perm}(n)$.

Exercise 1.2.0.7. We have that

1. $\text{Perm}(n)$ is a subgroup of $GL(n, \mathbb{R})$
2. $\text{Perm}(n)$ is a subgroup of $O(n)$

Proof.

1. By definition, $\text{Perm}(n) = \text{Im } \phi$. Since $\phi : S_n \rightarrow GL(n, \mathbb{R})$ is a group homomorphism, $\text{Im } \phi$ is a subgroup of $GL(n, \mathbb{R})$. Hence $\text{Perm}(n)$ is a subgroup of $GL(n, \mathbb{R})$.
2. Let $P \in \text{Perm}(n)$. Then there exists $\sigma \in S_n$ such that $P = \phi(\sigma)$. Then

$$\begin{aligned}PP^* &= \begin{pmatrix} e_{\sigma(1)}^* \\ \vdots \\ e_{\sigma(n)}^* \end{pmatrix} \begin{pmatrix} e_{\sigma(1)}^* \\ \vdots \\ e_{\sigma(n)}^* \end{pmatrix}^* \\ &= \begin{pmatrix} e_{\sigma(1)}^* \\ \vdots \\ e_{\sigma(n)}^* \end{pmatrix} (e_{\sigma(1)} \quad \cdots \quad e_{\sigma(n)}) \\ &= (\langle e_{\sigma(i)}, e_{\sigma(j)} \rangle)_{i,j} \\ &= I\end{aligned}$$

A previous exercise implies that $P^*P = I$. Hence $P \in O(n)$. Since $P \in \text{Perm}(n)$ is arbitrary, $\text{Perm}(n) \subset O(n)$. Part (1) implies that $\text{Perm}(n)$ is a group. Hence $\text{Perm}(n)$ is a subgroup of $O(n)$. □

Note 1.2.0.8. We will write P_σ in place of $\phi(\sigma)$.

Exercise 1.2.0.9. Let $Z \in \mathbb{R}^{p \times n}$. If $\text{rank } Z = k$, then there exist $\sigma \in S_n$, $\tau \in S_p$ and $A \in GL(k, \mathbb{R})$, such that for each $i, j \in \{1, \dots, k\}$,

$$(P_\tau Z P_\sigma^*)_{i,j} = A_{i,j}$$

Proof. Suppose that $\text{rank } Z = k$. Then there exist $i_1, \dots, i_k \in \{1, \dots, p\}$ such that $i_1 < \dots < i_k$ and $\{e_{i_1}^* Z, \dots, e_{i_k}^* Z\}$ is linearly independent. Set

$$Z' = \begin{pmatrix} e_{i_1}^* Z \\ \vdots \\ e_{i_k}^* Z \end{pmatrix}$$

Then $\text{rank } Z' = k$. Hence there exist $j_1, \dots, j_k \in \{1, \dots, n\}$ such that $j_1 < \dots < j_k$, and $\{Z'e_{i_1}, \dots, Z'e_{i_k}\}$ is linearly independent. Set

$$A = (Z'e_{i_1} \quad \dots \quad Z'e_{i_k})$$

Then $A \in \mathbb{R}^{k \times k}$ and $\text{rank } A = k$. Thus $A \in GL(k, \mathbb{R})$. Choose $\sigma \in S_n$ and $\tau \in S_p$ such that $\sigma(1) = j_1, \dots, \sigma(k) = j_k$ and $\tau(1) = i_1, \dots, \tau(k) = i_k$. Let $a, b \in \{1, \dots, k\}$. By construction,

$$\begin{aligned} (P_\tau Z P_\sigma^*)_{a,b} &= Z_{\tau(a), \sigma(b)} \\ &= Z_{i_a, j_b} \\ &= A_{a,b} \end{aligned}$$

□

Definition 1.2.0.10. Let $A \in \mathbb{R}^{n \times p}$. Then A is said to be a **diagonal matrix** if for each $i \in [n]$ and $j \in [p]$, $i \neq j$ implies that $A_{i,j} = 0$. We denote the set of $n \times p$ diagonal matrices by $D(n, p, \mathbb{R})$. We write $D(n, \mathbb{R})$ in place of $D(n, n, \mathbb{R})$.

Definition 1.2.0.11. For $(n, k), (m, l)$ $\text{diag}_{p, (n \times p)} : \mathbb{R}^p \rightarrow \mathbb{R}^{n \times p}$ and $\text{diag}_{n, (n \times p)} : \mathbb{R}^p \rightarrow \mathbb{R}^{n \times p}$ by $\text{diag}(v)$ **FINISH!!!**

Definition 1.2.0.12. Let $A \in \mathbb{R}^{n \times n}$ and $\lambda \in \sigma(A)$. Suppose that A is symmetric. We define the **geometric multiplicity** of λ , denoted $\mu(\lambda)$, by

$$\mu(\lambda) = \dim \ker([\phi_\alpha] - \lambda I)$$

Definition 1.2.0.13. Let V be an n -dimensional vector space, $U \subset V$ a k -dimensional subspace and $(e_j)_{j=1}^n \subset V$ a be a basis. Then $(e_j)_{j=1}^n$ is said to be **adapted to** U if $(e_j)_{j=1}^k$ is a basis for U .

1.3 Calculus

1.3.1 Differentiation

maybe assume differentiation of Banach spaces so we can skip this stuff?

Definition 1.3.1.1. Let $n \geq 1$. For $i = 1, \dots, n$, define $x^i : \mathbb{R}^n \rightarrow \mathbb{R}$ by $x^i(a^1, \dots, a^n) = a^i$. The functions $(x^i)_{i=1}^n$ are called the **standard coordinate functions on $\mathbb{R}^n$** .

Definition 1.3.1.2. Let $U \subset \mathbb{R}^n$ be open, $f : U \rightarrow \mathbb{R}$ and $a \in U$. Then f is said to be

- **differentiable with respect to x^i at a** if

$$\lim_{h \rightarrow 0} \frac{f(a + he^i) - f(a)}{h}$$

exists. If f is differentiable with respect to x^i at a , we define the **partial derivative of f with respect to x^i at a** , denoted

$$\frac{\partial f}{\partial x^i}(a) \text{ or } \frac{\partial}{\partial x^i} f$$

to be the limit above.

- **differentiable at a** if there exists $L \in L(\mathbb{R}^n; \mathbb{R})$ such that $F(a + h) = F(a) + L(h) + o(h)$ as $h \rightarrow 0$.

Definition 1.3.1.3. Let $U \subset \mathbb{R}^n$ be open and $f : U \rightarrow \mathbb{R}$. Then f is said to be **differentiable with respect to x^i** if for each $a \in U$, f is differentiable with respect to x^i at a .

Exercise 1.3.1.4. Let $U \subset \mathbb{R}^n$ be open, $f : U \rightarrow \mathbb{R}$ and $a \in U$. Suppose that $\frac{\partial^2 f}{\partial x^i \partial x^j}$ and $\frac{\partial^2 f}{\partial x^j \partial x^i}$ exist and are continuous at a . Then

$$\frac{\partial^2 f}{\partial x^i \partial x^j}(a) = \frac{\partial^2 f}{\partial x^j \partial x^i}(a)$$

Proof.

□

Definition 1.3.1.5. Let $U \subset \mathbb{R}^n$ be open and $f : U \rightarrow \mathbb{R}$. Then f is said to be **smooth** if for each $i_1, \dots, i_k \in \{1, \dots, n\}$, $\frac{\partial^k f}{\partial i_1 \dots i_k}$ exists and is continuous on U .

Definition 1.3.1.6. Let $U \subset \mathbb{R}^n$, $f : U \rightarrow \mathbb{R}$. Then f is said to be **smooth** if there exists $U' \subset \mathbb{R}^n$ and $f' : U' \rightarrow \mathbb{R}$ such that $U \subset U'$, U' is open, $f'|_U = f$ and f' is smooth. The set of smooth functions on U is denoted $C^\infty(U)$.

Theorem 1.3.1.7. Taylor's Theorem:

Let $U \subset \mathbb{R}^n$ be open and convex, $p \in U$, $f \in C^\infty(U)$ and $T \in \mathbb{N}$. Then there exist $(g_\alpha)_{|\alpha|=T+1} \subset C^\infty(U)$ such that for each $x \in U$,

$$f(x) = \sum_{k=0}^T \frac{1}{k!} \left[\sum_{|\alpha|=k} \partial^\alpha f(p)(x-p)^\alpha \right] + \frac{1}{(T+1)!} \sum_{|\alpha|=T+1} g_\alpha(x)(x-p)^\alpha$$

and for each $|\alpha| = T+1$,

$$g_\alpha(p) = \partial^\alpha f(p)$$

FIX!!!

Proof. See analysis notes **FINISH!!!**

□

Definition 1.3.1.8. Let $U \subset \mathbb{R}^n$ and $F : U \rightarrow \mathbb{R}^m$. Let $x^1, \dots, x^n$ be the standard coordinate functions on $\mathbb{R}^n$ and $y_1, \dots, y_m$ be the standard coordinate functions on $\mathbb{R}^m$. For $i \in \{1, \dots, m\}$, we define the **i th component of F** , denoted $F^i : U \rightarrow \mathbb{R}$, by

$$F^i = y^i \circ F$$

Thus $F = (F_1, \dots, F_m)$

Definition 1.3.1.9. Let $U \subset \mathbb{R}^n$ be open and $F : U \rightarrow \mathbb{R}^m$. Then F is said to be **smooth** if for each $i \in \{1, \dots, m\}$, the i th component of F , $F^i : U \rightarrow \mathbb{R}$, is smooth.

Definition 1.3.1.10. Let $U \subset \mathbb{R}^n$ and $F : U \rightarrow \mathbb{R}^m$. Then F is said to be **smooth** if for each $x \in U$, there exists $U_x \in \mathcal{N}_x$ and $\tilde{F} : U_x \rightarrow \mathbb{R}^m$ such that U_x is open, $\tilde{F}$ is smooth and $\tilde{F}|_{U \cap U_x} = F|_{U \cap U_x}$.

Definition 1.3.1.11. Let $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ and $F : U \rightarrow V$. Then F is said to be a **diffeomorphism** if F is a bijection and F, F^{-1} are smooth.

Exercise 1.3.1.12. Let $U \subset \mathbb{R}^n$ and $V \subset \mathbb{R}^m$ and $F : U \rightarrow V$. If F is a diffeomorphism, then F is a homeomorphism.

Proof. Suppose that F is a diffeomorphism. By definition, F is a bijection and F and F^{-1} are smooth. Thus, F and F^{-1} are continuous and F is a homeomorphism. $\square$

Definition 1.3.1.13. Let $U \subset \mathbb{R}^n$ be open, $p \in U$ and $F : U \rightarrow \mathbb{R}^m$. We define the **Jacobian of F at p** , denoted $\frac{\partial F}{\partial x}(p) \in \mathbb{R}^{m \times n}$, by

$$\left(\frac{\partial F}{\partial x}(p) \right)_{i,j} = \frac{\partial F^i}{\partial x^j}(p)$$

Exercise 1.3.1.14. Inverse Function Theorem:

Let $U, V \subset \mathbb{R}^n$ be open and $F : U \rightarrow V$.

1.3.2 Differentiation on Subspaces

Exercise 1.3.2.1. make this an ex in analysis notes Let $U_F, U_G \in \mathcal{T}_X$, $F : U_F \rightarrow Y$, $G : U_G \rightarrow Y$ and $a \in U_1 \cap U_2$. If there exists $U \in \mathcal{T}_X$ such that $a \in U$, $U \subset U_1 \cap U_2$ and $F|_U = G|_U$, then $DF(a) = DG(a)$.

Proof. Suppose that there exists $U \in \mathcal{T}_X$ such that $a \in U$, $U \subset U_1 \cap U_2$ and $F|_U = G|_U$. Then (make a prev ex that for each $x \in X$, $D\iota_U(x) = \text{id}_X$)

$$\begin{aligned}
 DF(a) &= DF(a) \circ \text{id}_X & &= DF(a) \circ D\iota_U(a) \\
 &= D(F \circ \iota_U)(a) \\
 &= DF|_U(a) \\
 &= DG|_U(a) \\
 &= D(G \circ \iota_U)(a) \\
 &= DG(a) \circ D\iota_U(a) \\
 &= DF(a) \circ \text{id}_X \\
 &= DG(a).
 \end{aligned}$$

□

Definition 1.3.2.2. Let $A \subset \mathbb{R}^m$ and $F : A \rightarrow \mathbb{R}^n$. Then F is said to be **smooth** if for each $a \in A$, there exists $B \in \mathcal{T}_{\mathbb{R}^m}$ and $g : B \rightarrow \mathbb{R}^n$ such that $a \in B$, g is smooth and $g|_{A \cap B} = F|_{A \cap B}$.

Exercise 1.3.2.3. Let $A \subset \mathbb{R}^m$ and $F : A \rightarrow \mathbb{R}^n$. If F is smooth, then F is continuous.

Proof. Suppose that F is smooth. Let $a \in A$. Since F is smooth, there exists $B \subset \mathbb{R}^m$ such that $a \in B$, B is open in $\mathbb{R}^m$, G is smooth and $G|_{A \cap B} = F|_{A \cap B}$. Since G is smooth, G is continuous. Let $V \subset \mathbb{R}^n$. Suppose that V is open in $\mathbb{R}^n$ and $F(a) \in V$. Since $F(a) = G(a)$ and G is continuous, there exists $U_G \subset B$ such that U_G is open in B , $a \in U_G$ and $G(U_G) \subset V$. Since B is open in $\mathbb{R}^m$ and U_G is open in B , we have that U_G is open in $\mathbb{R}^m$. Set $U_F = U_G \cap A$. Then $a \in U_F$, U_F is open in A and

$$\begin{aligned}
 F(U_F) &= F(U_G \cap A) \\
 &= G(U_G \cap A) \\
 &\subset G(U_G) \\
 &\subset V
 \end{aligned}$$

Since $V \subset \mathbb{R}^n$ such that V is open in $\mathbb{R}^n$ and $F(a) \in V$ is arbitrary, we have that for each $V \subset \mathbb{R}^n$, if V is open in $\mathbb{R}^n$ and $F(a) \in V$, then there exists $U_F \subset A$ such that U_F is open in A , $a \in U_F$ and $F(U_F) \subset V$. Thus F is continuous at a . Since $a \in A$ is arbitrary, F is continuous. □

Exercise 1.3.2.4. Let $A \subset \mathbb{R}^m$, $B \subset A$ and $f : A \rightarrow \mathbb{R}^n$. If f is smooth, then $f|_B$ is smooth.

Proof. Suppose that f is smooth. Let $b \in B$. Since $B \subset A$, $b \in A$. Since $b \in A$ and f is smooth, there exists $U \subset \mathbb{R}^m$ and $F : U \rightarrow \mathbb{R}^n$ such that $b \in U$, U is open in $\mathbb{R}^m$, F is smooth and $F|_{U \cap A} = f|_{U \cap A}$. Define $g : B \rightarrow \mathbb{R}^n$ by $g := f|_B$. Since $B \subset A$,

$$\begin{aligned}
 F|_{U \cap B} &= f|_{U \cap B} \\
 &= g|_{U \cap B}
 \end{aligned}$$

Since $b \in B$ is arbitrary, we have that for each $b \in B$, there exists $U \subset \mathbb{R}^m$ and $F : U \rightarrow \mathbb{R}^n$ such that $b \in U$, U is open in $\mathbb{R}^m$, F is smooth and $F|_{U \cap B} = g|_{U \cap B}$. Thus g is smooth. □

Exercise 1.3.2.5. Let $A \subset \mathbb{R}^m$ and $f : A \rightarrow \mathbb{R}^n$. Then f is smooth iff for each $a \in A$, there exists $U \in \mathcal{T}_{\mathbb{R}^m} \cap A$ such that $a \in U$ and $f|_U$ is smooth.

Proof.

• ($\Rightarrow$) :

Suppose that f is smooth. Let $a \in A$. Set $U := A$. Then $a \in U$, U is open in A and $f|_U = f$ which is smooth.

• ($\Leftarrow$) :

Suppose that for each $a \in A$, there exists $U \subset A$ such that $a \in U$ and $f|_U$ is smooth. Let $a \in A$. By assumption, there exists $U \subset A$ such that $a \in U$, U is open in A and $f|_U$ is smooth. Define $h : U \rightarrow \mathbb{R}^n$ by $h := f|_U$. Since $a \in U$ and h is smooth, there exists $U_0 \subset \mathbb{R}^m$ and $g_0 : U_0 \rightarrow \mathbb{R}^n$ such that $a \in U_0$, U_0 is open in $\mathbb{R}^m$ and $g_0|_{U \cap U_0} = h|_{U \cap U_0}$. Since U is open in A , there exists $\tilde{U} \subset \mathbb{R}^m$ such that $\tilde{U}$ is open in $\mathbb{R}^m$ and $U = \tilde{U} \cap A$. Define $B \subset \mathbb{R}^m$ and $g : B \rightarrow \mathbb{R}^n$ by $B := U_0 \cap \tilde{U}$ and $g = g_0|_B$. Then $a \in B$ and B is open in $\mathbb{R}^m$. The previous exercise implies that g is smooth. Furthermore,

$$\begin{aligned} g|_{B \cap A} &= g|_{U_0 \cap \tilde{U} \cap A} \\ &= g|_{U_0 \cap U} \\ &= h|_{U_0 \cap U} \\ &= f|_{U_0 \cap U} \\ &= f|_{U_0 \cap \tilde{U} \cap A} \\ &= f|_{B \cap A} \end{aligned}$$

Since $a \in A$ is arbitrary, we have that for each $a \in A$, there exists $B \subset \mathbb{R}^m$ and $g : B \rightarrow \mathbb{R}^n$ such that $a \in B$, B is open in $\mathbb{R}^m$, g is smooth and $g|_{A \cap B} = f|_{A \cap B}$. Hence f is smooth. □

Exercise 1.3.2.6. Let $A \subset \mathbb{R}^m$, $B \subset \mathbb{R}^n$, $f : A \rightarrow B$ and $g : B \rightarrow \mathbb{R}^p$. If f and g are smooth, then $g \circ f$ is smooth.

Proof. Suppose that f and g are smooth. Let $a \in A$. Set $b = f(a)$. Then $b \in B$. Since f is smooth, there exists $U \subset \mathbb{R}^m$ and $F : U \rightarrow \mathbb{R}^n$ such that $a \in U$, U is open in $\mathbb{R}^m$, F is smooth and $F|_{U \cap A} = f|_{U \cap A}$. Since g is smooth, there exists $V \subset \mathbb{R}^n$ and $G : V \rightarrow \mathbb{R}^p$ such that $b \in V$, V is open in $\mathbb{R}^n$, G is smooth and $G|_{V \cap B} = g|_{V \cap B}$. We define $W \subset \mathbb{R}^m$ and $H : W \rightarrow \mathbb{R}^p$ by $W := U \cap F^{-1}(V)$ and $H := G \circ F|_W$.

- By construction, $a \in W$.
- Since F is smooth, F is continuous. Thus $F^{-1}(V)$ is open in $\mathbb{R}^m$ which implies that W is open in $\mathbb{R}^m$.
- Since F is smooth, [an exercise in the section on differentiation](#) implies that $F|_W$ is smooth. Since $F|_W$ and G are smooth, [a previous exercise in the section on differentiation](#) implies that H is smooth.
- Let $x \in W \cap A$. Since $W \cap A \subset A \cap U$, $f(x) = F(x)$. Since $f(x) \in B$ and $W \subset F^{-1}(V)$, we have that $F(x) \in V \cap B$. Thus

$$\begin{aligned} g \circ f(x) &= g(F(x)) \\ &= G(F(x)) \\ &= H(x) \end{aligned}$$

Since $x \in W \cap A$ is arbitrary, we have that $H|_{W \cap A} = (g \circ f)|_{W \cap A}$.

Thus $g \circ f$ is smooth. □

Exercise 1.3.2.7. Let $A \subset \mathbb{R}^m$ and $F : A \rightarrow \mathbb{R}^n$. If F is smooth, then $[DF] : A \rightarrow \mathbb{R}^{n \times m}$ is smooth.

Proof. Suppose that □

1.3.3 Calculus and Permutations

Exercise 1.3.3.1. Let $U, V \subset \mathbb{R}^n$ and $F : U \rightarrow V$. Then F is a diffeomorphism iff for each $p \in U$, there exists a relatively open neighborhood $N \subset U$ of p such that $F|_N : N \rightarrow F(N)$ is a diffeomorphism

Proof. content... **FIX or get rid** □

Definition 1.3.3.2.

- Let $\sigma \in S_n$ and $x = (x^1, \dots, x^n) \in \mathbb{R}^n$. We define $\sigma \cdot x \in \mathbb{R}^n$ by

$$\sigma \cdot x = (x^{\sigma(1)}, \dots, x^{\sigma(n)})$$

- We define the **permutation action** of S_n on $\mathbb{R}^n$ to be the map $S_n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ given by $(\sigma, x) \mapsto \sigma \cdot x$.
- Let $\sigma \in S_n$. We define $\Phi_\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $\Phi_\sigma(x) := \sigma \cdot x$.

Exercise 1.3.3.3. Let $\sigma \in S_n$. Then

1. $D\Phi_\sigma = P_\sigma$.
2. $\Phi_\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a diffeomorphism,

Proof.

- 1.

$$\begin{aligned}
 D(\Phi_\sigma)(p) &= \left(\frac{\partial \pi_i \circ \Phi_\sigma}{\partial x^j}(p) \right)_{i,j} \\
 &= \left(\frac{\partial \pi_{\sigma(i)}}{\partial x^j}(p) \right)_{i,j} \\
 &= P_\sigma \left(\frac{\partial \pi_i}{\partial x^j}(p) \right)_{i,j} \\
 &= P_\sigma \left(\frac{\partial \pi_i \circ \text{id}_{\mathbb{R}^n}}{\partial x^j}(p) \right)_{i,j} \\
 &= P_\sigma D \text{id}_{\mathbb{R}^n}(p) \\
 &= P_\sigma I \\
 &= P_\sigma
 \end{aligned}$$

2. Clear.

□

Definition 1.3.3.4.

- Let $\sigma \in S_n$, U a set, $V \subset \mathbb{R}^n$ and $\phi : U \rightarrow \mathbb{R}^n$ with $\phi = (x^1, \dots, x^m)$. We define $\sigma \cdot \phi : U \rightarrow \mathbb{R}^n$ by

$$(\sigma \cdot \phi)(x) := \phi(\sigma \cdot x)$$

- We define the **permutation action** of S_n on $(\mathbb{R}^n)^U$ to be the map $S_n \times (\mathbb{R}^n)^U \rightarrow \mathbb{R}^n$ given by $(\sigma, \phi) \mapsto \sigma \cdot \phi$.

Exercise 1.3.3.5. Let $\sigma \in S_m$. Then for each $p \in \mathbb{R}^n$, $D(\sigma \text{id}_{\mathbb{R}^n})(p) = P_\sigma$.

Proof. Note that since $\text{id}_{\mathbb{R}^n} = (\pi_1, \dots, \pi_n)$, we have that $\sigma \text{id}_{\mathbb{R}^n} = (\pi_{\sigma(1)}, \dots, \pi_{\sigma(n)})$. Let $p \in \mathbb{R}^n$. Then

□

1.3.4 Integration

1.4 Topology

Definition 1.4.0.1. Let $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ be topological spaces and $f : X \rightarrow Y$. Then f is said to be **continuous** if for each $U \in \mathcal{T}$, $f^{-1}(U) \in \mathcal{T}_X$.

Definition 1.4.0.2. Let $(X, \mathcal{T}_X), (Y, \mathcal{T}_Y)$ be topological spaces and $f : X \rightarrow Y$. Then f is said to be a **homeomorphism** if f is a bijection and f, f^{-1} are continuous.

Definition 1.4.0.3. Let X, Y be topological spaces. Then X and Y are said to be **homeomorphic** if there exists $f : X \rightarrow Y$ such that f is a homeomorphism. If X and Y are homeomorphic, we write $X \cong Y$.

Theorem 1.4.0.4. Let $m, n \in \mathbb{N}$. If $m \neq n$, then $\mathbb{R}^m \not\cong \mathbb{R}^n$

1.5 Group Actions

1.5.1 Subactions

Exercise 1.5.1.1. Let X be a set, G a group and $\triangleleft : G \times X \rightarrow X$ a group action. Then

1. for each $x \in X$, $\triangleright(\bar{x} \times G) = \bar{x}$,
2. for each $x \in X$, $\triangleright|_{\bar{x} \times G} : \bar{x} \times G \rightarrow \bar{x}$ is a group action.

Proof. content... □

Definition 1.5.1.2. Let X be a set, G a group and $\triangleleft : G \times X \rightarrow X$ a group action. For each $x \in X$, we define **action of G on $\bar{x}$ induced by $\triangleleft$** $\triangleright_x : G \times \bar{x} \rightarrow \bar{x}$ by $g \triangleright_x := g \triangleleft x$.

Exercise 1.5.1.3. Let X be a set, G a group and $\triangleleft : G \times X \rightarrow X$ a group action.

is free iff for each $x \in M$, $\triangleleft|_{P_x \times G}$ is free. given a left action $\triangleright : G \times X \rightarrow X$ and $x \in X$, such that $\triangleright(\times G) \subset Y$, show that $\triangleright(Y \times G) = Y$ and $\triangleright|_{Y \times G}$ is a group action and $\triangleright|_{Y \times G}$ is free iff

Proof. Suppose that $\triangleleft$ is free. Let $x \in M$, $p \in P_x$ and $g \in G$. Suppose that $p \triangleleft_x g = p$. Then $p \triangleleft g = p$. Thus $g = e$. Since $p \in P_x$ and $g \in G$ are arbitrary, $\triangleleft$ is free

Conversely, suppose that for each $x \in M$, $\triangleleft|_{P_x \times G}$ is free. Let $g \in G$ and $p \in P$. □

Chapter 2

Multilinear Algebra

2.1 Tensor Products

Let V and W be vector spaces.

2.2 (r, s) -Tensors

Definition 2.2.0.1. Let $V_1, \dots, V_k, W$ be vector spaces and $\alpha : \prod_{i=1}^n V_i \rightarrow W$. Then α is said to be **multilinear** if for each $i \in \{1, \dots, k\}$, $v \in V$, $c \in \mathbb{R}$ and $v_1, \dots, v_k \in V$,

$$\alpha(v_1, \dots, v_i + cv, \dots, v_k) = \alpha(v_1, \dots, v_i, \dots, v_k) + c\alpha(v_1, \dots, v, \dots, v_k)$$

We define

$$L(V_1, \dots, V_k; W) = \left\{ \alpha : \prod_{i=1}^n V_i \rightarrow W : \alpha \text{ is multilinear} \right\}$$

Note 2.2.0.2. For the remainder of this section we let V denote an n -dimensional vector space with basis $\{e^1, \dots, e^n\}$ with dual space V^* and dual basis $\{\epsilon_1, \dots, \epsilon_n\}$ defined by $\epsilon^i(e^j) = \delta_{i,j}$. We identify V with V^{**} by the isomorphism $V \rightarrow V^{**}$ defined by $v \mapsto \hat{v}$ where $\hat{v}(\alpha) = \alpha(v)$ for each $\alpha \in V^*$.

Definition 2.2.0.3. Let $\alpha : (V^*)^r \times V^s \rightarrow \mathbb{R}$. Then α is said to be an (r, s) -tensor on V if $\alpha \in L(\underbrace{V^*, \dots, V^*}_r, \underbrace{V, \dots, V}_s; \mathbb{R})$. The set

of all (r, s) -tensors on V is denoted $T_s^r(V)$.

When $r = s = 0$, we set $T_s^r = \mathbb{R}$.

Exercise 2.2.0.4. We have that $T_s^r(V)$ is a vector space.

Proof. Clear. □

Exercise 2.2.0.5. Under the identification of V with V^{**} as noted above, we have that $V = T_0^1(V)$.

Proof. By definition,

$$\begin{aligned} V &= V^{**} \\ &= L(V^*; \mathbb{R}) \\ &= T_0^1(V) \end{aligned}$$

□

Definition 2.2.0.6. Let $\alpha \in T_{s_1}^{r_1}(V)$ and $\beta \in T_{s_2}^{r_2}(V)$. We define the **tensor product of α with β** , denoted $\alpha \otimes \beta \in T_{s_1+s_2}^{r_1+r_2}(V)$, by

$$\alpha \otimes \beta(v^*, w^*, v, w) = \alpha(v^*, v)\beta(w^*, w)$$

for each $v^* \in (V^*)^{r_1}$, $w^* \in (V^*)^{r_2}$, $v \in V^{s_1}$ and $w \in V^{s_2}$.

When $r_1 = s_1 = r_2 = s_2 = 0$ (so that $\alpha, \beta \in \mathbb{R}$), we set $\alpha \otimes \beta = \alpha\beta$.

Definition 2.2.0.7. We define the **tensor product**, denoted $\otimes : T_{s_1}^{r_1}(V) \times T_{s_2}^{r_2}(V) \rightarrow T_{s_1+s_2}^{r_1+r_2}(V)$ by

$$(\alpha, \beta) \mapsto \alpha \otimes \beta$$

Exercise 2.2.0.8. The tensor product $\otimes : T_{s_1}^{r_1}(V) \times T_{s_2}^{r_2}(V) \rightarrow T_{s_1+s_2}^{r_1+r_2}(V)$ is well defined.

Proof. Tedious but straightforward. □

Exercise 2.2.0.9. The tensor product $\otimes : T_{s_1}^{r_1}(V) \times T_{s_2}^{r_2}(V) \rightarrow T_{s_1+s_2}^{r_1+r_2}(V)$ is associative.

Proof. Let $\alpha \in T_{s_1}^{r_1}(V)$, $\beta \in T_{s_2}^{r_2}(V)$ and $\gamma \in T_{s_3}^{r_3}(V)$. Then for each $u^* \in (V^*)^{r_1}$, $v^* \in (V^*)^{r_2}$, $w^* \in (V^*)^{r_3}$, $u \in V^{s_1}$, $v \in V^{s_2}$, $w \in V^{s_3}$,

$$\begin{aligned} (\alpha \otimes \beta) \otimes \gamma(u^*, v^*, w^*, u, v, w) &= (\alpha \otimes \beta)(u^*, v^*, u, v) \gamma(w^*, w) \\ &= [\alpha(u^*, u) \beta(v^*, v)] \gamma(w^*, w) \\ &= \alpha(u^*, u) [\beta(v^*, v) \gamma(w^*, w)] \\ &= \alpha(u^*, u) (\beta \otimes \gamma)(v^*, w^*, v, w) \\ &= \alpha \otimes (\beta \otimes \gamma)(u^*, v^*, w^*, u, v, w) \end{aligned}$$

So that

$$(\alpha \otimes \beta) \otimes \gamma = \alpha \otimes (\beta \otimes \gamma)$$

□

Exercise 2.2.0.10. The tensor product $\otimes : T_{s_1}^{r_1}(V) \times T_{s_2}^{r_2}(V) \rightarrow T_{s_1+s_2}^{r_1+r_2}(V)$ is bilinear.

Proof.

1. Linearity in the first argument:

Let $\alpha, \beta \in T_{s_1}^{r_1}(V)$, $\gamma \in T_{s_2}^{r_2}(V)$, $\lambda \in \mathbb{R}$, $v^* \in (V^*)^{r_1}$, $w^* \in (V^*)^{r_2}$, $v \in V^{s_1}$ and $w \in V^{s_2}$. To see that the tensor product is linear in the first argument, we note that

$$\begin{aligned} [(\alpha + \lambda\beta) \otimes \gamma](v^*, w^*, v, w) &= (\alpha + \lambda\beta)(v^*, v) \gamma(w^*, w) \\ &= [\alpha(v^*, v) + \lambda\beta(v^*, v)] \gamma(w^*, w) \\ &= \alpha(v^*, v) \gamma(w^*, w) + \lambda\beta(v^*, v) \gamma(w^*, w) \\ &= \alpha \otimes \gamma(v^*, w^*, v, w) + \lambda(\beta \otimes \gamma)(v^*, w^*, v, w) \\ &= [\alpha \otimes \gamma + \lambda(\beta \otimes \gamma)](v^*, w^*, v, w) \end{aligned}$$

So that

$$(\alpha + \lambda\beta) \otimes \gamma = \alpha \otimes \gamma + \lambda(\beta \otimes \gamma)$$

2. Linearity in the second argument:

Similar to (1). □

Definition 2.2.0.11.

1. Define $\mathcal{I}_n^{\otimes k} = \{(i_1, i_2, \dots, i_k) \in \mathbb{N}^k : i_1, \dots, i_k \leq n\}$. Each element $I \in \mathcal{I}_n^{\otimes k}$ is called an **unordered index of length k in $[n]$** . Recall that $\#\mathcal{I}_n^{\otimes k} = n^k$.
2. Define $\mathcal{I}_n^{\wedge k} = \{(i_1, i_2, \dots, i_k) \in \mathbb{N}^k : i_1 < i_2 < \dots < i_k \leq n\}$. Each element $I \in \mathcal{I}_k$ is called an **ordered index of length k in $[n]$** . Recall that $\#\mathcal{I}_n^{\wedge k} = \binom{n}{k}$.

need to discuss difference between multi indices $\alpha \in \mathbb{N}_0^m$ and tuple $I \in \mathcal{I}_n^{\otimes k}$

Definition 2.2.0.12. Let $I = \{(i_1, i_2, \dots, i_k) \in \mathcal{I}_n^{\otimes k}\}$.

1. Define $\epsilon^I \in (V^*)^k$ and $e_I \in V^k$ by

$$\epsilon^I = (\epsilon^{i_1}, \dots, \epsilon^{i_k})$$

and

$$e^I = (e^{i_1}, \dots, e^{i_k})$$

2. Define $e^{\otimes I} \in T_0^k(V)$ and $\epsilon^{\otimes I} \in T_k^0(V)$ by

$$e^{\otimes I} = e^{i_1} \otimes \dots \otimes e^{i_k}$$

and

$$\epsilon^{\otimes I} = \epsilon^{i_1} \otimes \dots \otimes \epsilon^{i_k}$$

Exercise 2.2.0.13. Let $\alpha, \beta \in T_s^r(V)$. If for each $I \in \mathcal{I}_r, J \in \mathcal{I}_s$, $\alpha(\epsilon^I, e^J) = \beta(\epsilon^I, e^J)$, then $\alpha = \beta$.

Proof. Suppose that for each $I \in \mathcal{I}_r, J \in \mathcal{I}_s$, $\alpha(\epsilon^I, e^J) = \beta(\epsilon^I, e^J)$. Let $v_1^*, \dots, v_r^* \in V^*$ and $v_1, \dots, v_s \in V$. For each $i \in \{1, \dots, r\}$ and $j \in \{1, \dots, s\}$, write

$$v_i^* = \sum_{k_i=1}^n a_{i,k_i} \epsilon^{k_i}$$

and

$$v_j = \sum_{l_j=1}^n b_{j,l_j} e^{l_j}$$

Then

$$\begin{aligned} \alpha(v_1^*, \dots, v_r^*, v_1, \dots, v_s) &= \sum_{k_1, \dots, k_r=1}^n \sum_{l_1, \dots, l_s=1}^n \prod_{i=1}^r \prod_{j=1}^s a_{i,k_i} b_{j,l_j} \alpha(\epsilon^{k_1}, \dots, \epsilon^{k_r}, e^{l_1}, \dots, e^{l_s}) \\ &= \sum_{k_1, \dots, k_r=1}^n \sum_{l_1, \dots, l_s=1}^n \prod_{i=1}^r \prod_{j=1}^s a_{i,k_i} b_{j,l_j} \beta(\epsilon^{k_1}, \dots, \epsilon^{k_r}, e^{l_1}, \dots, e^{l_s}) \\ &= \beta(v_1^*, \dots, v_r^*, v_1, \dots, v_s) \end{aligned}$$

So that $\alpha = \beta$. □

Exercise 2.2.0.14. Let $I, K \in \mathcal{I}_r$ and $J, L \in \mathcal{I}_s$. Then $e^{\otimes I} \otimes \epsilon^{\otimes J}(\epsilon^K, e^L) = \delta_{I,K} \delta_{J,L}$.

Proof. Write $I = (i_1, \dots, i_r), K = (k_1, \dots, k_r)$ and $J = (j_1, \dots, j_s), L = (l_1, \dots, l_s)$. Then

$$\begin{aligned} e^{\otimes I} \otimes \epsilon^{\otimes J}(\epsilon^K, e^L) &= e^{\otimes I}(\epsilon^K) \epsilon^{\otimes J}(e^L) \\ &= e^{i_1} \otimes \dots \otimes e^{i_r}(\epsilon^{k_1}, \dots, \epsilon^{k_r}) \epsilon^{j_1} \otimes \dots \otimes \epsilon^{j_s}(e^{l_1}, \dots, e^{l_s}) \\ &= \left[\prod_{m=1}^r e^{i_m}(\epsilon^{k_m}) \right] \left[\prod_{n=1}^s \epsilon^{j_n}(e^{l_n}) \right] \\ &= \left[\prod_{m=1}^r \delta_{i_m, k_m} \right] \left[\prod_{n=1}^s \delta_{j_n, l_n} \right] \\ &= \delta_{I,K} \delta_{J,L} \end{aligned}$$

□

Exercise 2.2.0.15. The set $\{e^{\otimes I} \otimes \epsilon^{\otimes J} : I \in \mathcal{I}_r, J \in \mathcal{I}_s\}$ is a basis for $T_s^r(V)$ and $\dim T_s^r(V) = n^{r+s}$.

Proof. Let $(a_J^I)_{I \in \mathcal{I}_r, J \in \mathcal{I}_s} \subset \mathbb{R}$. Let $\alpha = \sum_{(I,J) \in \mathcal{I}_r \times \mathcal{I}_s} a_J^I e^{\otimes I} \otimes \epsilon^{\otimes J}$. Suppose that $\alpha = 0$. Then for each $(I, J) \in \mathcal{I}_r \times \mathcal{I}_s$, $\alpha(\epsilon^I, e^J) = a_J^I = 0$.

Thus $\{e^{\otimes I} \otimes \epsilon^{\otimes J} : I \in \mathcal{I}_r, J \in \mathcal{I}_s\}$ is linearly independent. Let $\beta \in T_s^r(V)$. For $(I, J) \in \mathcal{I}_r \times \mathcal{I}_s$, put $b_J^I = \beta(\epsilon^J, e^I)$. Define $\mu = \sum_{(I,J) \in \mathcal{I}_r \times \mathcal{I}_s} b_J^I e^{\otimes I} \otimes \epsilon^{\otimes J} \in T_s^r(V)$. Then for each $(I, J) \in \mathcal{I}_r \times \mathcal{I}_s$, $\mu(\epsilon^I, e^J) = b_J^I = \beta(\epsilon^I, e^J)$. Hence $\mu = \beta$ and therefore

$\beta \in \text{span}\{e^{\otimes I} \otimes \epsilon^{\otimes J}\}$. □

2.3 Covariant k -Tensors

2.3.1 Symmetric and Alternating Covariant k -Tensors

Definition 2.3.1.1. Let $\alpha : V^k \rightarrow \mathbb{R}$. Then α is said to be a **covariant k -tensor on V** if $\alpha \in T_k^0(V)$. We denote the set of covariant k -tensors by $T_k(V)$.

Definition 2.3.1.2. For $\sigma \in S_k$ and $\alpha \in T_k(V)$, define the $\sigma\alpha : V^k \rightarrow \mathbb{R}$ by

$$\sigma\alpha(v_1, \dots, v_k) = \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)})$$

We define the **permutation action** of S_k on $T_k(V)$ to be the map $S_k \times T_k(V) \rightarrow T_k(V)$ given by $(\sigma, \alpha) \mapsto \sigma\alpha$

Exercise 2.3.1.3. The permutation action of S_k on $T_k(V)$ is a group action.

Proof.

1. Clearly for each $\sigma \in S_k$ and $\alpha \in T_k(V)$, $\sigma\alpha \in T_k(V)$.
2. Clearly for each $\alpha \in T_k(V)$, $e\alpha = \alpha$.
3. Let $\tau, \sigma \in S_k$ and $\alpha \in T_k(V)$. Then for each $v_1, \dots, v_k \in V$,

$$\begin{aligned} (\tau\sigma)\alpha(v_1, \dots, v_k) &= \alpha(v_{\tau\sigma(1)}, \dots, v_{\tau\sigma(k)}) \\ &= \tau\alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \\ &= \tau(\sigma\alpha)(v_1, \dots, v_k) \end{aligned}$$

□

Exercise 2.3.1.4. Let $\sigma \in S_k$. Then $L_\sigma : T_k(V) \rightarrow T_k(V)$ given by $L_\sigma(\alpha) = \sigma\alpha$ is a linear transformation.

Proof. Let $\alpha, \beta \in T_k(V)$, $c \in \mathbb{R}$ and $v_1, \dots, v_k \in V$. Then

$$\begin{aligned} \sigma(c\alpha + \beta)(v_1, \dots, v_k) &= (c\alpha + \beta)(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \\ &= c\alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)}) + \beta(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \\ &= c\sigma\alpha(v_1, \dots, v_k) + \sigma\beta(v_1, \dots, v_k) \end{aligned}$$

So $\sigma(c\alpha + \beta) = c\sigma\alpha + \sigma\beta$.

□

Definition 2.3.1.5. Let $\alpha \in T_k(V)$. Then α is said to be

- **symmetric** if for each $\sigma \in S_k$, $\sigma\alpha = \alpha$
- **antisymmetric** if for each $\sigma \in S_k$, $\sigma\alpha = \text{sgn}(\sigma)\alpha$
- **alternating** if for each $v_1, \dots, v_k \in V$, if there exists $i, j \in \{1, \dots, k\}$ such that $v_i = v_j$, then $\alpha(v_1, \dots, v_k) = 0$.

We denote the set of symmetric k -tensors on V by $\Sigma^k(V)$. We denote the set of alternating k -tensors on V by $\Lambda^k(V)$. [update language here](#)

Exercise 2.3.1.6. Let $\alpha \in T_k(V)$. Then α is antisymmetric iff α is alternating.

Proof. Suppose that α is antisymmetric. Let $v_1, \dots, v_k \in V$. Suppose that there exists $i, j \in \{1, \dots, k\}$ such that $v_i = v_j$. Define $\sigma \in S_k$ by $\sigma = (i, j)$. Then

$$\begin{aligned} \alpha(v_1, \dots, v_i, \dots, v_j, \dots, v_k) &= \alpha(v_1, \dots, v_j, \dots, v_i, \dots, v_k) \\ &= \sigma(\alpha)(v_1, \dots, v_i, \dots, v_j, \dots, v_k) \\ &= \text{sgn}(\sigma)\alpha(v_1, \dots, v_i, \dots, v_j, \dots, v_k) \\ &= -\alpha(v_1, \dots, v_i, \dots, v_j, \dots, v_k) \end{aligned}$$

Therefore $2\alpha(v_1, \dots, v_i, \dots, v_j, \dots, v_k) = 0$ which implies that $\alpha(v_1, \dots, v_i, \dots, v_j, \dots, v_k) = 0$. Hence α is alternating. Conversely, suppose that α is alternating. Let $i, j \in \{1, \dots, k\}$ and $v_1, \dots, v_k \in V$. Then

$$\begin{aligned} 0 &= \alpha(v_1, \dots, v_i + v_j, \dots, v_i + v_j, \dots, v_k) \\ &= \alpha(v_1, \dots, v_i, \dots, v_j, \dots, v_k) + \alpha(v_1, \dots, v_j, \dots, v_i, \dots, v_k) \end{aligned}$$

Since $i, j \in \{1, \dots, k\}$ and $v_1, \dots, v_k \in V$ are arbitrary, we have that for each $\tau \in S_k$, τ is a transposition implies that

$$\begin{aligned} \tau\alpha &= -\alpha \\ &= \text{sgn}(\tau)\alpha \end{aligned}$$

Let $n \in \mathbb{N}$. Suppose that for each $\tau_1, \dots, \tau_{n-1} \in S_k$ if for each $j \in \{1, \dots, n-1\}$, τ_j is a transposition, then $(\tau_1 \cdots \tau_{n-1})\alpha = \sigma(\tau_1 \cdots \tau_{n-1})\alpha$. Let $\tau_1, \dots, \tau_n \in S_k$. Suppose that for each $j \in \{1, \dots, n\}$, τ_j is a transposition. Then

$$\begin{aligned} (\tau_1 \cdots \tau_n)\alpha &= (\tau_1 \cdots \tau_{n-1})(\tau_n\alpha) \\ &= (\tau_1 \cdots \tau_{n-1})(\text{sgn}(\tau_n)\alpha) \\ &= (\text{sgn}(\tau_n)(\tau_1 \cdots \tau_{n-1})\alpha) \\ &= (\text{sgn}(\tau_n) \text{sgn}((\tau_1 \cdots \tau_{n-1})\alpha)) \\ &= \text{sgn}(\tau_1 \cdots \tau_n)\alpha \end{aligned}$$

By induction, for each $n \in \mathbb{N}$ and $\tau_1, \dots, \tau_n \in S_k$, if for each $j \in \{1, \dots, n\}$, τ_j is a transposition, then $(\tau_1 \cdots \tau_n)\alpha = \text{sgn}(\tau_1 \cdots \tau_n)\alpha$. Now let $\sigma \in S_k$. Then there exist $n \in \mathbb{N}$ and $\tau_1, \dots, \tau_n \in S_k$ such that $\sigma = \tau_1 \cdots \tau_n$ and for each $j \in \{1, \dots, n\}$, τ_j is a transposition. Hence

$$\begin{aligned} \sigma\alpha &= (\tau_1 \cdots \tau_n)\alpha \\ &= \text{sgn}(\tau_1 \cdots \tau_n)\alpha \\ &= \text{sgn}(\sigma)\alpha \end{aligned}$$

Therefore α is antisymmetric. □

Definition 2.3.1.7. Define the **symmetric operator** $S : T_k(V) \rightarrow \Sigma^k(V)$ by

$$\text{Sym}(\alpha) = \frac{1}{k!} \sum_{\sigma \in S_k} \sigma\alpha$$

Define the **alternating operator** $A : T_k(V) \rightarrow \Lambda^k(V)$ by

$$\text{Alt}(\alpha) = \frac{1}{k!} \sum_{\sigma \in S_k} \text{sgn}(\sigma)\sigma\alpha$$

Exercise 2.3.1.8.

1. For $\alpha \in T_k(V)$, $\text{Sym}(\alpha)$ is symmetric.
2. For $\alpha \in T_k(V)$, $\text{Alt}(\alpha)$ is alternating.

Proof.

1. Let $\alpha \in T_k(V)$ and $\sigma \in S_k$. Then

$$\begin{aligned} \sigma \text{Sym}(\alpha) &= \sigma \left[\frac{1}{k!} \sum_{\tau \in S_k} \tau\alpha \right] \\ &= \frac{1}{k!} \sum_{\tau \in S_k} \sigma\tau\alpha \\ &= \frac{1}{k!} \sum_{\tau \in S_k} \tau\alpha \\ &= \text{Sym}(\alpha) \end{aligned}$$

2. Let $\alpha \in T_k(V)$ and $\sigma \in S_k$. Then

$$\begin{aligned}
 \sigma \operatorname{Alt}(\alpha) &= \sigma \left[\frac{1}{k!} \sum_{\tau \in S_k} \operatorname{sgn}(\tau) \tau \alpha \right] \\
 &= \frac{1}{k!} \sum_{\tau \in S_k} \operatorname{sgn}(\tau) \sigma \tau \alpha \\
 &= \frac{1}{k!} \sum_{\tau \in S_k} \operatorname{sgn}(\sigma) \operatorname{sgn}(\sigma \tau) \sigma \tau \alpha \\
 &= \operatorname{sgn}(\sigma) \frac{1}{k!} \sum_{\tau \in S_k} \operatorname{sgn}(\sigma \tau) \sigma \tau \alpha \\
 &= \operatorname{sgn}(\sigma) \frac{1}{k!} \sum_{\tau \in S_k} \operatorname{sgn}(\tau) \tau \alpha \\
 &= \operatorname{sgn}(\sigma) \operatorname{Alt}(\alpha)
 \end{aligned}$$

□

Exercise 2.3.1.9.

1. For $\alpha \in \Sigma^k(V)$, $\operatorname{Sym}(\alpha) = \alpha$.
2. For $\alpha \in \Lambda^k(V)$, $\operatorname{Alt}(\alpha) = \alpha$.

Proof.

1. Let $\alpha \in \Sigma^k(V)$. Then

$$\begin{aligned}
 \operatorname{Sym}(\alpha) &= \frac{1}{k!} \sum_{\sigma \in S_k} \sigma \alpha \\
 &= \frac{1}{k!} \sum_{\sigma \in S_k} \alpha \\
 &= \alpha
 \end{aligned}$$

2. Let $\alpha \in \Lambda^k(V)$. Then

$$\begin{aligned}
 \operatorname{Alt}(\alpha) &= \frac{1}{k!} \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma) \sigma \alpha \\
 &= \frac{1}{k!} \sum_{\sigma \in S_k} \operatorname{sgn}(\sigma)^2 \alpha \\
 &= \alpha
 \end{aligned}$$

□

Exercise 2.3.1.10. The symmetric operator $S : T_k(V) \rightarrow \Sigma^k(V)$ and the alternating operator $A : T_k(V) \rightarrow \Lambda^k(V)$ are linear.

Proof. Clear.

□

Exercise 2.3.1.11. Let $\alpha \in T_k(V)$ and $\beta \in T_l(V)$. Then

1. $\operatorname{Alt}(\operatorname{Alt}(\alpha) \otimes \beta) = \operatorname{Alt}(\alpha \otimes \beta)$
2. $\operatorname{Alt}(\alpha \otimes \operatorname{Alt}(\beta)) = \operatorname{Alt}(\alpha \otimes \beta)$

Proof. First note that if we fix $\mu \in S_{k+l}$, then for each $\tau \in S_k$, choosing $\sigma = \mu\tau^{-1}$ yields $\sigma\tau = \mu$. For each $\mu \in S_{k+l}$, the map $\phi_\mu : S_k \rightarrow S_{k+l}$ given by $\phi_\mu(\tau) = \mu\tau^{-1}$ is injective. Thus for each $\mu \in S_{k+l}$, we have that $\#\{(\sigma, \tau) \in S_{k+l} \times S_k : \mu = \sigma\tau\} = k!$

1. Then

$$\begin{aligned}
 \text{Alt}(\text{Alt}(\alpha) \otimes \beta) &= \frac{1}{(k+l)!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma \left[\text{Alt}(\alpha) \otimes \beta \right] \\
 &= \frac{1}{(k+l)!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma \left[\left(\frac{1}{k!} \sum_{\tau \in S_k} \text{sgn}(\tau) \tau \alpha \right) \otimes \beta \right] \\
 &= \frac{1}{(k+l)!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma \left[\frac{1}{k!} \sum_{\tau \in S_k} \text{sgn}(\tau) (\tau \alpha) \otimes \beta \right] \\
 &= \frac{1}{(k+l)!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma \left[\frac{1}{k!} \sum_{\tau \in S_k} \text{sgn}(\tau) \tau (\alpha \otimes \beta) \right] \\
 &= \frac{1}{k!(k+l)!} \sum_{\sigma \in S_{k+l}} \sum_{\tau \in S_k} \text{sgn}(\sigma\tau) \sigma\tau (\alpha \otimes \beta) \\
 &= \frac{k!}{k!(k+l)!} \sum_{\mu \in S_{k+l}} \text{sgn}(\mu) \mu (\alpha \otimes \beta) \\
 &= \frac{1}{(k+l)!} \sum_{\mu \in S_{k+l}} \text{sgn}(\mu) \mu (\alpha \otimes \beta) \\
 &= \text{Alt}(\alpha \otimes \beta)
 \end{aligned}$$

2. Similar to (1).

□

2.3.2 Exterior Product

Definition 2.3.2.1. Let $\alpha \in \Lambda^k(V)$ and $\beta \in \Lambda^l(V)$. The **exterior product** of α and β is defined to be the map $\alpha \wedge \beta \in \Lambda^{k+l}(V)$ given by

$$\alpha \wedge \beta = \frac{(k+l)!}{k!l!} \text{Alt}(\alpha \otimes \beta)$$

Thus $\wedge : \Lambda^k(V) \times \Lambda^l(V) \rightarrow \Lambda^{k+l}(V)$.

Exercise 2.3.2.2. The exterior product $\wedge : \Lambda^k(V) \times \Lambda^l(V) \rightarrow \Lambda^{k+l}(V)$ is bilinear.

Proof. Clear.

□

Exercise 2.3.2.3. The exterior product $\wedge : \Lambda^k(V) \times \Lambda^l(V) \rightarrow \Lambda^{k+l}(V)$ is associative.

Proof. Let $\alpha \in \Lambda^k(V)$, $\beta \in \Lambda^l(V)$ and $\gamma \in \Lambda^m(V)$. Then

$$\begin{aligned}
 (\alpha \wedge \beta) \wedge \gamma &= \left[\frac{(k+l)!}{k!l!} \text{Alt}(\alpha \otimes \beta) \right] \wedge \gamma \\
 &= \frac{(k+l+m)!}{(k+l)!m!} \text{Alt} \left(\left[\frac{(k+l)!}{k!l!} \text{Alt}(\alpha \otimes \beta) \right] \otimes \gamma \right) \\
 &= \frac{(k+l+m)!}{(k+l)!m!} \frac{(k+l)!}{k!l!} \text{Alt}(\text{Alt}(\alpha \otimes \beta) \otimes \gamma) \\
 &= \frac{(k+l+m)!}{m!} \frac{1}{k!l!} \text{Alt}((\alpha \otimes \beta) \otimes \gamma) \\
 &= \frac{(k+l+m)!}{k!(l+m)!} \frac{(l+m)!}{l!m!} \text{Alt}(\alpha \otimes (\beta \otimes \gamma)) \\
 &= \frac{(k+l+m)!}{k!(l+m)!} \frac{(l+m)!}{l!m!} \text{Alt}(\alpha \otimes \text{Alt}(\beta \otimes \gamma)) \\
 &= \frac{(k+l+m)!}{k!(l+m)!} \text{Alt}(\alpha \otimes \frac{(l+m)!}{l!m!} \text{Alt}(\beta \otimes \gamma)) \\
 &= \frac{(k+l+m)!}{k!(l+m)!} \text{Alt}(\alpha \otimes (\beta \wedge \gamma)) \\
 &= \alpha \wedge (\beta \wedge \gamma)
 \end{aligned}$$

□

Exercise 2.3.2.4. Let $\alpha_i \in \Lambda^{k_i}(V)$ for $i = 1, \dots, m$. Then

$$\bigwedge_{i=1}^m \alpha_i = \frac{(\sum_{i=1}^m k_i)!}{\prod_{i=1}^m k_i!} \text{Alt} \left(\bigotimes_{i=1}^m \alpha_i \right)$$

Proof. To see that the statment is true in the case $m = 3$, the proof of the previous exercise tells us that indeed

$$\alpha_1 \wedge \alpha_2 \wedge \alpha_3 = \frac{(k_1 + k_2 + k_3)!}{k_1!k_2!k_3!} \text{Alt}(\alpha_1 \otimes \alpha_2 \otimes \alpha_3)$$

Now, suppose that the statement is true for each $3 \leq m \leq m_0$. Then the proof of the previous exercise tells us the

$$\begin{aligned}
 \bigwedge_{i=1}^{m_0+1} \alpha_i &= \left(\bigwedge_{i=1}^{m_0-1} \alpha_i \right) \wedge \alpha_{m_0} \wedge \alpha_{m_0+1} \\
 &= \frac{(\sum_{i=1}^{m_0-1} k_i + k_{m_0} + k_{m_0+1})!}{(\sum_{i=1}^{m_0-1} k_i)!k_{m_0}!k_{m_0+1}!} \text{Alt} \left(\left[\bigwedge_{i=1}^{m_0-1} \alpha_i \right] \otimes \alpha_{m_0} \otimes \alpha_{m_0+1} \right) \\
 &= \frac{(\sum_{i=1}^{m_0-1} k_i + k_{m_0} + k_{m_0+1})!}{(\sum_{i=1}^{m_0-1} k_i)!k_{m_0}!k_{m_0+1}!} \text{Alt} \left(\left[\frac{(\sum_{i=1}^{m_0-1} k_i)!}{\prod_{i=1}^{m_0-1} k_i!} \text{Alt} \left(\bigotimes_{i=1}^{m_0-1} \alpha_i \right) \right] \otimes \alpha_{m_0} \otimes \alpha_{m_0+1} \right) \\
 &= \frac{(\sum_{i=1}^{m_0+1} k_i)!}{\prod_{i=1}^{m_0+1} k_i!} \text{Alt} \left(\text{Alt} \left[\bigotimes_{i=1}^{m_0-1} \alpha_i \right] \otimes \alpha_{m_0} \otimes \alpha_{m_0+1} \right) \\
 &= \frac{(\sum_{i=1}^{m_0+1} k_i)!}{\prod_{i=1}^{m_0+1} k_i!} \text{Alt} \left(\left[\bigotimes_{i=1}^{m_0-1} \alpha_i \right] \otimes \alpha_{m_0} \otimes \alpha_{m_0+1} \right) \\
 &= \frac{(\sum_{i=1}^{m_0+1} k_i)!}{\prod_{i=1}^{m_0+1} k_i!} \text{Alt} \left(\bigotimes_{i=1}^{m_0+1} \alpha_i \right)
 \end{aligned}$$

□

Exercise 2.3.2.5. Define $\tau \in S_{k+l}$ by

$$\tau = \begin{pmatrix} 1 & 2 & \cdots & l & l+1 & l+2 & \cdots & l+k \\ 1+k & 2+k & \cdots & l+k & 1 & 2 & \cdots & k \end{pmatrix}$$

Then the inversion number of τ is kl . (Hint: inversion number)

Proof.

$$\begin{aligned} N(\tau) &= \sum_{i=1}^l k \\ &= kl \end{aligned}$$

Since $\text{sgn}(\tau) = (-1)^{N(\tau)}$ we know that $\text{sgn}(\tau) = (-1)^{kl}$. □

Exercise 2.3.2.6. Let $\alpha \in \Lambda^k(V)$, $\beta \in \Lambda^l(V)$. Then

$$\alpha \wedge \beta = (-1)^{kl} \beta \wedge \alpha$$

Proof. Define $\tau \in S_{k+l}$ as in the previous exercise. Note that For $\sigma \in S_{k+l}$ and $v_1, \dots, v_{k+l} \in V$, we have that

$$\begin{aligned} \sigma\tau(\beta \otimes \alpha)(v_1, \dots, v_l, v_{l+1}, \dots, v_{l+k}) &= \beta \otimes \alpha(v_{\sigma\tau(1)}, \dots, v_{\sigma\tau(l)}, v_{\sigma\tau(l+1)}, \dots, v_{\sigma\tau(l+k)}) \\ &= \beta(v_{\sigma\tau(1)}, \dots, v_{\sigma\tau(l)})\alpha(v_{\sigma\tau(l+1)}, \dots, v_{\sigma\tau(l+k)}) \\ &= \beta(v_{\sigma(1+k)}, \dots, v_{\sigma(l+k)})\alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)}) \\ &= \alpha(v_{\sigma(1)}, \dots, v_{\sigma(k)})\beta(v_{\sigma(1+k)}, \dots, v_{\sigma(l+k)}) \\ &= \alpha \otimes \beta(v_{\sigma(1)}, \dots, v_{\sigma(k)}, v_{\sigma(1+k)}, \dots, v_{\sigma(l+k)}) \\ &= \sigma(\alpha \otimes \beta)(v_1, \dots, v_k, v_{1+k}, \dots, v_{l+k}) \end{aligned}$$

Thus $\sigma\tau(\beta \otimes \alpha) = \sigma(\alpha \otimes \beta)$. Then

$$\begin{aligned} \beta \wedge \alpha &= \frac{(k+l)!}{k!l!} \text{Alt}(\beta \otimes \alpha) \\ &= \frac{(k+l)!}{k!l!} \frac{1}{(k+l)!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma(\beta \otimes \alpha) \\ &= \frac{(k+l)!}{k!l!} \frac{1}{(k+l)!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma\tau) \sigma\tau(\beta \otimes \alpha) \\ &= \text{sgn}(\tau) \frac{(k+l)!}{k!l!} \frac{1}{(k+l)!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma(\alpha \otimes \beta) \\ &= \text{sgn}(\tau) \frac{(k+l)!}{k!l!} \text{Alt}(\alpha \otimes \beta) \\ &= \text{sgn}(\tau) \alpha \wedge \beta \\ &= (-1)^{kl} \alpha \wedge \beta \end{aligned}$$

□

Exercise 2.3.2.7. Let $\alpha \in \Lambda^k(V)$. If k is odd, then $\alpha \wedge \alpha = 0$.

Proof. Suppose that k is odd. The previous exercise tells us that

$$\begin{aligned} \alpha \wedge \alpha &= (-1)^{k^2} \alpha \wedge \alpha \\ &= -\alpha \wedge \alpha \end{aligned}$$

Thus $\alpha \wedge \alpha = 0$. □

Exercise 2.3.2.8. Fundamental Example:

Let $\alpha_1, \dots, \alpha_m \in \Lambda^1(V)$ and $v_1, \dots, v_m \in V$. Then

$$\left(\bigwedge_{i=1}^m \alpha_i \right) (v_1, \dots, v_m) = \det(\alpha_i(v_j))$$

Proof. The previous exercises tell us that

$$\begin{aligned} \left(\bigwedge_{i=1}^m \alpha_i \right) (v_1, \dots, v_m) &= m! \operatorname{Alt} \left(\bigotimes_{i=1}^m \alpha_i \right) (v_1, \dots, v_m) \\ &= m! \left[\frac{1}{m!} \sum_{\sigma \in S_m} \operatorname{sgn}(\sigma) \sigma \left(\bigotimes_{i=1}^m \alpha_i \right) \right] (v_1, \dots, v_m) \\ &= \sum_{\sigma \in S_m} \operatorname{sgn}(\sigma) \left(\bigotimes_{i=1}^m \alpha_i \right) (v_{\sigma(1)}, \dots, v_{\sigma(m)}) \\ &= \sum_{\sigma \in S_m} \operatorname{sgn}(\sigma) \prod_{i=1}^m \alpha_i(v_{\sigma(i)}) \\ &= \det(\alpha_i(v_j)) \end{aligned}$$

□

Note 2.3.2.9. Recall that $\mathcal{I}_n^{\wedge k} = \{(i_1, i_2, \dots, i_k) \in \mathbb{N}^k : i_1 < i_2 < \dots < i_k \leq n\}$ and that $\#\mathcal{I}_n^{\wedge k} = \binom{n}{k}$.

Definition 2.3.2.10. Let $I = \{(i_1, i_2, \dots, i_k) \in \mathcal{I}_n^{\wedge k}\}$.

Define $\epsilon^{\wedge I} \in \Lambda^k(V)$ by

$$\epsilon^{\wedge I} = \epsilon^{i_1} \wedge \dots \wedge \epsilon^{i_k}$$

Exercise 2.3.2.11. Let $I = (i_1, \dots, i_k)$ and $J = (j_1, \dots, j_k) \in \mathcal{I}_n^{\wedge k}$. Then $\epsilon^{\wedge I}(e^J) = \delta_{I,J}$.

Proof. Put $A = \begin{pmatrix} \epsilon^{i_1}(e^{j_1}) & \dots & \epsilon^{i_1}(e^{j_k}) \\ \vdots & & \vdots \\ \epsilon^{i_k}(e^{j_1}) & \dots & \epsilon^{i_k}(e^{j_k}) \end{pmatrix}$. A previous exercise tells us that $\epsilon^{\wedge I}(e^J) = \det A$. If $I = J$, then $A = I_{k \times k}$ and therefore

$\epsilon^I(e^J) = 1$. Suppose that $I \neq J$. Put $l_0 = \min\{l : 1 \leq l \leq k, i_l \neq j_l\}$. If $i_{l_0} < j_{l_0}$, then all entries on the l_0 -th row of A are 0. If $i_{l_0} > j_{l_0}$, then all entries on the l_0 -th column of A are 0. □

Exercise 2.3.2.12. Let $\alpha, \beta \in \Lambda^k(V)$. If for each $I \in \mathcal{I}_n^{\wedge k}$, $\alpha(e^I) = \beta(e^I)$, then $\alpha = \beta$.

Proof. Suppose that for each $I \in \mathcal{I}_n^{\wedge k}$, $\alpha(e^I) = \beta(e^I)$. Let $v_1, \dots, v_k \in V$. For $i = 1, \dots, k$, write $v_i = \sum_{j_i=1}^n a_{i,j_i} e^{j_i}$. Then

$$\begin{aligned} \alpha(v_1, \dots, v_k) &= \sum_{j_1, \dots, j_k=1}^n \left(\prod_{i=1}^k a_{i,j_i} \right) \alpha(e^{j_1}, \dots, e^{j_k}) \\ &= \sum_{j_1 \neq \dots \neq j_k}^n \left(\prod_{i=1}^k a_{i,j_i} \right) \alpha(e^{j_1}, \dots, e^{j_k}) \\ &= \sum_{J \in \mathcal{I}_n^{\wedge k}} \left[\sum_{\sigma \in S_J} \operatorname{sgn}(\sigma) \left(\prod_{i=1}^k a_{i, \sigma(j_i)} \right) \right] \alpha(e^J) \\ &= \sum_{J \in \mathcal{I}_n^{\wedge k}} \left[\sum_{\sigma \in S_J} \operatorname{sgn}(\sigma) \left(\prod_{i=1}^k a_{i, \sigma(j_i)} \right) \right] \beta(e^J) \\ &= \sum_{j_1, \dots, j_k=1}^n \left(\prod_{i=1}^k a_{i,j_i} \right) \beta(e^{j_1}, \dots, e^{j_k}) \\ &= \beta(v_1, \dots, v_k) \end{aligned}$$

□

Exercise 2.3.2.13. The set $\{\epsilon^{\wedge I} : I \in \mathcal{I}_n^{\wedge k}\}$ is a basis for $\Lambda^k(V)$ and $\dim \Lambda^k(V) = \binom{n}{k}$.

Proof. Let $(a_I)_{I \in \mathcal{I}_n^{\wedge k}} \subset \mathbb{R}$. Let $\alpha = \sum_{I \in \mathcal{I}_n^{\wedge k}} a_I \epsilon^{\wedge I}$. Suppose that $\alpha = 0$. Then for each $J \in \mathcal{I}_n^{\wedge k}$, $\alpha(e^J) = a_J = 0$. Thus $\{\epsilon^{\wedge I} : I \in \mathcal{I}_n^{\wedge k}\}$ is linearly independent. Let $\beta \in \Lambda^k(V)$. For $I \in \mathcal{I}_n^{\wedge k}$, put $b_I = \beta(e^I)$. Define $\mu = \sum_{I \in \mathcal{I}_n^{\wedge k}} b_I \epsilon^{\wedge I} \in \Lambda^k(V)$. Then for each $J \in \mathcal{I}_n^{\wedge k}$, $\mu(e^J) = b_J = \beta(e^J)$. Hence $\mu = \beta$ and therefore $\beta \in \text{span}\{\epsilon^{\wedge I} : I \in \mathcal{I}_n^{\wedge k}\}$. □

2.3.3 Interior Product

Definition 2.3.3.1. Let V be a finite dimensional vector space and $v \in V$. We define **interior multiplication by v** , denoted $\iota_v : T_k \rightarrow T_{k-1}$, by

$$\iota_v \alpha(w_1, \dots, w_{k-1}) = \alpha(v, w_1, \dots, w_{k-1})$$

Exercise 2.3.3.2. Let V be a finite dimensional vector space and $v \in V$. Then $\iota_v|_{\Lambda^k(V)} : \Lambda^k(V) \rightarrow \Lambda^{k-1}(V)$.

Proof. Let $\alpha \in \Lambda^k(V)$. Define $\beta \in \Lambda^k(V)$ by $\beta(w_1, \dots, w_k) = \alpha(w_k, w_1, \dots, w_{k-1})$. Let $\sigma \in S_{k-1}$. Define $\tau \in S_k$ by $\tau(j) = \begin{cases} 1 & j = k \\ \sigma(j) & j \neq k \end{cases}$. Let $w_1, \dots, w_{k-1} \in V$. Set $w_k = v$. Then

$$\begin{aligned} \sigma(\iota_v \alpha)(w_1, \dots, w_{k-1}) &= \iota_v \alpha(w_{\sigma(1)}, \dots, w_{\sigma(k-1)}) \\ &= \alpha(v, w_{\sigma(1)}, \dots, w_{\sigma(k-1)}) \\ &= \beta(w_{\sigma(1)}, \dots, w_{\sigma(k-1)}, v) \\ &= \beta(w_{\sigma(1)}, \dots, w_{\sigma(k-1)}, w_k) \\ &= \beta(w_{\tau(1)}, \dots, w_{\tau(k-1)}, w_{\tau(k)}) \\ &= \text{sgn}(\tau) \beta(w_1, \dots, w_{k-1}, w_k) \\ &= \text{sgn}(\sigma) \beta(w_1, \dots, w_{k-1}, v) \\ &= \text{sgn}(\sigma) \alpha(v, w_1, \dots, w_{k-1}) \\ &= \text{sgn}(\sigma) (\iota_v \alpha)(w_1, \dots, w_{k-1}) \end{aligned}$$

Since $w_1, \dots, w_{k-1} \in V$ are arbitrary, $\sigma(\iota_v \alpha) = \text{sgn}(\sigma) \iota_v \alpha$. Hence $\iota_v \alpha \in \Lambda^{k-1}(V)$. □

2.4 (0, 2)-Tensors

Definition 2.4.0.1. Let V be a finite dimensional vector space, $v \in V$ and $\alpha \in T_2^0(V)$. Then α is said to be **degenerate** if there exists $v \in V$ such that $v \neq 0$ and for each $w \in V$, $\alpha(v, w) = 0$.

Definition 2.4.0.2. Let V be a finite dimensional vector space, $\alpha \in T_2^0(V)$. We define $\phi_\alpha : V \rightarrow V^*$ by

$$\phi_\alpha(v) = \iota_v \alpha$$

Exercise 2.4.0.3. Let V be a finite dimensional vector space, $\alpha \in T_2^0(V)$. Then $\phi_\alpha \in L(V; V^*)$.

Proof. Let $v_1, v_2 \in V$ and $\lambda \in \mathbb{R}$. Then for each $w \in V$,

$$\begin{aligned} \phi_\alpha(v_1 + \lambda v_2)(w) &= (\iota_{v_1 + \lambda v_2} \alpha)(w) \\ &= \alpha(v_1 + \lambda v_2, w) \\ &= \alpha(v_1, w) + \lambda \alpha(v_2, w) \\ &= (\iota_{v_1} \alpha)(w) + \lambda (\iota_{v_2} \alpha)(w) \\ &= \phi_\alpha(v_1)(w) + \lambda \phi_\alpha(v_2)(w) \\ &= [\phi_\alpha(v_1) + \lambda \phi_\alpha(v_2)](w) \end{aligned}$$

Therefore, $\phi_\alpha(v_1 + \lambda v_2) = \phi_\alpha(v_1) + \lambda \phi_\alpha(v_2)$. Thus $\phi_\alpha \in L(V; V^*)$. □

Exercise 2.4.0.4. Let V be a finite dimensional vector space and $\alpha \in T_2^0(V)$. Then α is nondegenerate iff ϕ_α is an isomorphism.

Proof.

• ($\implies$:)

Suppose that α is nondegenerate. Let $v \in \ker \phi_\alpha$. Then for each $w \in V$,

$$\begin{aligned} \alpha(v, w) &= (\iota_v \alpha)(w) \\ &= \phi_\alpha(v)(w) \\ &= 0 \end{aligned}$$

Since α is nondegenerate, $v = 0$. Since $v \in \ker \phi_\alpha$ is arbitrary, $\ker \phi_\alpha = \{0\}$. Hence ϕ_α is injective. Since $\dim V = \dim V^*$, ϕ_α is surjective. Hence ϕ_α is an isomorphism.

• ($\impliedby$:)

Suppose that ϕ_α is an isomorphism. Let $v \in V$. Suppose that for each $w \in V$, $\alpha(v, w) = 0$. Then for each $w \in V$,

$$\begin{aligned} \phi_\alpha(v)(w) &= (\iota_v \alpha)(w) \\ &= \alpha(v, w) \\ &= 0 \end{aligned}$$

Thus $\phi_\alpha(v) = 0$ which implies that $v \in \ker \phi_\alpha$. Since ϕ_α is an isomorphism, $v = 0$. Hence α is nondegenerate. □

Exercise 2.4.0.5. Let V be a finite dimensional vector space and $\alpha \in T_2^0(V)$. Then

$$1. [\phi_\alpha]_{i,j} = \alpha(e_j, e_i)$$

$$2. \text{ for each } v, w \in V,$$

$$\alpha(v, w) = [w]^* [\phi_\alpha] [v]$$

Proof.

1. Set $A = [\phi_\alpha]$. Let $i, j \in \{1, \dots, n\}$. By definition,

$$\phi_\alpha(e_j) = \sum_{k=1}^n A_{k,j} \epsilon^k$$

Then

$$\begin{aligned} \phi_\alpha(e_j)(e_i) &= \sum_{k=1}^n A_{k,j} \epsilon^k(e_i) \\ &= \sum_{k=1}^n A_{k,j} \delta_{k,i} \\ &= A_{i,j} \end{aligned}$$

2. Let $v, w \in V$. Then there exist $(v^i)_{i=1}^n, (w^j)_{j=1}^n \subset \mathbb{R}$ such that $v = \sum_{i=1}^n v^i e_i$ and $w = \sum_{j=1}^n w^j e_j$. Part (1) implies that

$$\begin{aligned} \alpha(v, w) &= \sum_{i=1}^n \sum_{j=1}^n v^i w^j \alpha(e_i, e_j) \\ &= \sum_{i=1}^n \sum_{j=1}^n v^i w^j [\phi_\alpha]_{j,i} \\ &= \sum_{i=1}^n \sum_{j=1}^n [v]_i [w]_j [\phi_\alpha]_{j,i} \\ &= [w]^* [\phi_\alpha] [v] \end{aligned}$$

□

2.4.1 Scalar Product Spaces

pg 40 of Lee's intro to riemannian manifolds

Definition 2.4.1.1. Let V be a finite dimensional vector space and $\alpha \in \Sigma^2(V)$ (define $\Sigma^2(V)$ i.e. symmetric (0,2)-tensors). Then α is said to be

- **positive semidefinite** if for each $v \in V$, $\alpha(v, v) \geq 0$
- **positive definite** if for each $v \in V$, $v \neq 0$ implies that $\alpha(v, v) > 0$
- **negative semidefinite** if $-\alpha$ is positive semidefinite
- **negative definite** if $-\alpha$ is positive definite

Exercise 2.4.1.2. Let V be a finite dimensional vector space and $\alpha \in \Sigma^2(V)$. Then

1. α is positive semidefinite iff for each $\lambda \in \sigma([\phi_\alpha])$, $\lambda \geq 0$
2. α is positive definite iff for each $\lambda \in \sigma([\phi_\alpha])$, $\lambda > 0$

Proof.

1. • ($\implies$) :
Suppose that there exists $\lambda \in \sigma([\phi_\alpha])$ such that $\lambda < 0$. Then there exists $v_\lambda \in \mathbb{R}^n$ $v_\lambda^* [\phi_\alpha] v_\lambda$
- ($\impliedby$) :

Suppose that α is positive semidefinite. Write $\sigma(\phi_\alpha) = \{\lambda_1, \dots, \lambda_n\}$. Define $\Lambda \in \mathbb{R}^{n \times n}$ by $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. Since α is symmetric, $[\phi_\alpha]$ is symmetric. There exists $U \in O(n)$ such that $[\phi_\alpha] = U \Lambda U^*$. **FINISH!!!**

□

Definition 2.4.1.3. Let V be a finite dimensional vector space and $\alpha \in \Sigma^2(V)$. Then α is said to be a **scalar product** if α is nondegenerate. In this case, (V, α) is said to be a **scalar product space**.

Definition 2.4.1.4. Let V be a finite dimensional vector space and $\alpha \in \Sigma^2(V)$ a scalar product on V . We define the **index** of α , denoted $\text{ind } \alpha$ by

$$\text{ind } \alpha = \max\{\dim W : W \text{ is a subspace of } V \text{ and } \alpha|_{W \times W} \text{ is negative definite}\}$$

Definition 2.4.1.5. Let (V, α) be a scalar product space.

- Let $v_1, v_2 \in V$. Then v_1 and v_2 are said to be **orthogonal** if $\alpha(v_1, v_2) = 0$.
- Let $U \subset V$ be a subspace. We define the **orthogonal subspace of U** , denoted by $U^\perp$, by

$$U^\perp = \{v \in V : \text{for each } u \in U, \alpha(u, v) = 0\}$$

Exercise 2.4.1.6. Let (V, α) be a scalar product space and $U \subset V$ a subspace. Then $U^\perp$ is a subspace of V .

Proof. We note that since $U^\perp = \bigcap_{u \in U} \ker \phi_\alpha(u)$, $U^\perp$ is a subspace of V . □

Exercise 2.4.1.7. Let (V, α) be an n -dimensional scalar product space, $U \subset V$ a k -dimensional subspace and $(e_j)_{j=1}^n \subset V$ a basis for V . Suppose that $(e_j)_{j=1}^k$ is a basis for U . Then for each $v \in V$, $v \in U^\perp$ iff for each $j \in [k]$, $\alpha(v, e_j) = 0$.

Proof. Let $v \in V$.

- ($\implies$): Suppose that $v \in U^\perp$. Since $(e_j)_{j=1}^k \subset U$, we have that for each $j \in [k]$, $\alpha(v, e_j) = 0$.
- ($\impliedby$): Suppose that for each $j \in [k]$, $\alpha(v, e_j) = 0$. Let $u \in U$. Then there exist $(a^j)_{j=1}^k \subset \mathbb{R}$ such that $u = \sum_{j=1}^k a^j e_j$. This implies that

$$\begin{aligned} \alpha(v, u) &= \sum_{j=1}^k a^j \alpha(v, e_j) \\ &= 0 \end{aligned}$$

Since $u \in U$ is arbitrary, we have that $v \in U^\perp$. □

Exercise 2.4.1.8. Let (V, α) be a scalar product space and $U \subset V$ a subspace. Then

1. $\dim V = \dim U + \dim U^\perp$
2. $(U^\perp)^\perp = U$

Proof.

1. Set $n = \dim V$ and $k = \dim U$. Choose a basis $(e_j)_{j=1}^n$ such that $(e_j)_{j=1}^k$ is a basis for U .
2. □

Exercise 2.4.1.9. Let V be a finite dimensional vector space and $\alpha \in \Sigma^2(V)$. Set $\sigma([\phi_\alpha])^- = \{\lambda \in \sigma([\phi_\alpha]) : \lambda < 0\}$. Then

$$\text{ind } \alpha = \sum_{\lambda \in \sigma([\phi_\alpha])^-} \mu(\lambda)$$

Proof. Since α is symmetric, there exist $U \in O(n)$ and $\Lambda \in D(n, \mathbb{R})$ such that $[\phi_\alpha] = U\Lambda U^*$. Define $(u_j)_{j=1}^n \subset V$ by $u_j = \sum_{i=1}^n U_{i,j} e_i$. Define $J^- = \{j \in [n] : \Lambda_{j,j} < 0\}$, $n^- = \#J^-$ and $V^- = \text{span}\{u_j : j \in J^-\}$. Let $v \in V^-$. Then there exist $(a^j)_{j \in J^-}$ such that $v = \sum_{j \in J^-} a^j u_j$. We note that

$$\begin{aligned} U^*[\phi_\alpha]U &= U^*(U\Lambda U^*)U \\ &= (U^*U)\Lambda(U^*U) \\ &= I\Lambda I \\ &= \Lambda \end{aligned}$$

A previous exercise implies that

$$\begin{aligned} \alpha(v, v) &= \sum_{j \in J^-} \sum_{k \in J^-} a^j a^k \alpha(u_j, u_k) \\ &= \sum_{j \in J^-} \sum_{k \in J^-} a^j a^k [u_j]^* [\phi_\alpha] [u_k] \\ &= \sum_{j \in J^-} \sum_{k \in J^-} a^j a^k ([e_j]^* U^*) [\phi_\alpha] (U[e_k]) \\ &= \sum_{j \in J^-} \sum_{k \in J^-} a^j a^k (U^*[\phi_\alpha]U)_{j,k} \\ &= \sum_{j \in J^-} \sum_{k \in J^-} a^j a^k (\Lambda)_{j,k} \\ &= \sum_{j \in J^-} |a^j|^2 \Lambda_{j,j} \\ &< 0 \end{aligned}$$

Since $v \in V^-$ is arbitrary, $\alpha|_{V^- \times V^-}$ is negative definite. Thus

$$\begin{aligned} \text{ind } \alpha &\geq \dim V^- \\ &= n^- \end{aligned}$$

Set $J^+ = (J^-)^c$. Let $W \subset V$ be a subspace. Suppose that $\alpha|_{W \times W}$ is negative definite. For the sake of contradiction, suppose that there exists $j_0 \in J^+$ such that $u_{j_0} \in W$. Then

$$\begin{aligned} \alpha(u_{j_0}, u_{j_0}) &= [u_{j_0}]^* [\phi_\alpha] [u_{j_0}] \\ &= [u_{j_0}]^* U\Lambda U^* [u_{j_0}] \\ &= \Lambda_{j_0, j_0} \\ &\geq 0 \end{aligned}$$

which is a contradiction since $\alpha|_{W \times W}$ is negative definite. Thus for each $j \in J^+$, $u_j \notin W$. □

Definition 2.4.1.10. Let (V, α) be an n -dimensional scalar product space. We define the **scalar norm associated to α** , denoted $\|\cdot\|_\alpha : V \rightarrow \mathbb{R}$ by $\|v\|_\alpha := |\alpha(v, v)|^{1/2}$.

Note 2.4.1.11.

- When the context is clear, we write $\|\cdot\|$ in place of $\|\cdot\|_\alpha$.
- α is not positive definite iff $\|\cdot\|_\alpha$ is not a norm.

alternatively, define GS algorithm in terms of orthogonal projections

Exercise 2.4.1.12. Gram-Schmidt Algorithm:

Let (V, α) be an n -dimensional scalar product space and $(v_j)_{j \in [n]} \subset V$ a basis for V . For $j \in [n]$, define $u_j, e_j \in V$. If α is nondegenerate, then there exists $(e_j)_{j=1}^n \subset V$ such that $(e_j)_{j=1}^n$ is an orthonormal basis for V .

Proof. Suppose that α is nondegenerate. Then for each $v \in V$, $\alpha(v, v) \neq 0$. Choose $(v_j)_{j=1}^n \subset V$ such that $(v_j)_{j=1}^n$ is a basis for V . For each $j \in [n]$, we define

$$u_j := \begin{cases} v_1, & j = 1 \\ v_j - \sum_{k=1}^{j-1} [\alpha(v_j, u_k) / \alpha(u_k, u_k)] u_k, & j \geq 2 \end{cases}$$

$$e_j := u_j / \|u_j\|_\alpha.$$

Let $j_1, j_2 \in [n]$. Suppose that $j_1 \leq j_2$. Then $\alpha(e_{j_1}, e_{j_2})$

- Clearly,

$$\begin{aligned} \alpha(u_1, u_2) &= \alpha(v_1, v_2 - \sum_{k=1}^{j_1} [\alpha(v_2, u_k) / \alpha(u_k, u_k)] u_k) \\ &= \alpha(v_1, v_2 - \frac{\alpha(v_2, u_1)}{\alpha(u_1, u_1)} u_1) \\ &= \alpha(v_1, v_2 - \frac{\alpha(v_2, v_1)}{\alpha(v_1, v_1)} v_1) \\ &= \alpha(v_1, v_2) - \alpha(v_1, \frac{\alpha(v_2, v_1)}{\alpha(v_1, v_1)} v_1) \\ &= \alpha(v_1, v_2) - \frac{\alpha(v_2, v_1)}{\alpha(v_1, v_1)} \alpha(v_1, v_1) \\ &= \alpha(v_1, v_2) - \alpha(v_2, v_1) \end{aligned}$$

•

$$\begin{aligned} \alpha(u_1, u_2) &= \alpha(v_1, v_2 - \sum_{k=1}^{j_1} [\alpha(v_2, u_k) / \alpha(u_k, u_k)] u_k) \\ &= \alpha(v_1, v_2 - \frac{\alpha(v_2, u_1)}{\alpha(u_1, u_1)} u_1) \\ &= \alpha(v_1, v_2 - \frac{\alpha(v_2, v_1)}{\alpha(v_1, v_1)} v_1) \\ &= \alpha(v_1, v_2) - \alpha(v_1, \frac{\alpha(v_2, v_1)}{\alpha(v_1, v_1)} v_1) \\ &= \alpha(v_1, v_2) - \frac{\alpha(v_2, v_1)}{\alpha(v_1, v_1)} \alpha(v_1, v_1) \\ &= \alpha(v_1, v_2) - \alpha(v_2, v_1) \end{aligned}$$

FINISH!!! proof by induction?

□

2.4.2 Symplectic Vector Spaces

Definition 2.4.2.1. Let V be a finite dimensional vector space and $\omega \in \Lambda^2(V)$. Then ω is said to be a **symplectic form** if ω is nondegenerate. In this case (V, ω) is said to be a **symplectic space**.

Exercise 2.4.2.2. Let V be a $2n$ -dimensional vector space with basis $(a_j, b_j)_{j=1}^n$ and corresponding dual basis $(\alpha^j, \beta^j)_{j=1}^n$. Define $\omega \in \Lambda^2(V)$ by

$$\omega = \sum_{j=1}^n \alpha^j \wedge \beta^j$$

Then

1. for each $j, k \in \{1, \dots, n\}$,
 - (a) $\omega(a_j, a_k) = 0$
 - (b) $\omega(b_j, b_k) = 0$
 - (c) $\omega(a_j, b_k) = \delta_{j,k}$
2. (V, ω) is a symplectic space

Proof.

1. Let $j, k \in \{1, \dots, n\}$.

(a)

$$\begin{aligned} \omega(a_j, a_k) &= \sum_{l=1}^n \alpha^l \wedge \beta^l(a_j, a_k) \\ &= \sum_{l=1}^n [\alpha^l(a_j)\beta^l(a_k) - \alpha^l(a_k)\beta^l(a_j)] \\ &= 0 \end{aligned}$$

(b) Similar to (a)

(c)

$$\begin{aligned} \omega(a_j, b_k) &= \sum_{l=1}^n \alpha^l \wedge \beta^l(a_j, b_k) \\ &= \sum_{l=1}^n [\alpha^l(a_j)\beta^l(b_k) - \alpha^l(b_k)\beta^l(a_j)] \\ &= \sum_{l=1}^n \alpha^l(a_j)\beta^l(b_k) \\ &= \sum_{l=1}^n \delta_{j,l}\delta_{l,k} \\ &= \delta_{j,k} \end{aligned}$$

2. Let $v \in V$. Then there exist $(q^j, p^j)_{j=1}^n \subset \mathbb{R}$ such that $v = \sum_{j=1}^n q^j a_j + p^j b_j$. Suppose that for each $w \in V$, $\omega(v, w) = 0$. Let $k \in \{1, \dots, n\}$. Then

$$\begin{aligned} 0 &= \omega(v, a_k) \\ &= \sum_{j=1}^n q^j \omega(a_j, a_k) + p^j \omega(b_j, a_k) \\ &= \sum_{j=1}^n p^j \delta_{j,k} \\ &= p^k \end{aligned}$$

Similarly,

$$\begin{aligned}
 0 &= \omega(v, b_k) \\
 &= \sum_{j=1}^n q^j \omega(a_j, b_k) + p^j \omega(b_j, b_k) \\
 &= \sum_{j=1}^n q^j \delta_{j,k} \\
 &= q^k
 \end{aligned}$$

Since $k \in \{1, \dots, n\}$ is arbitrary, $v = 0$. Hence ω is nondegenerate. Therefore (V, ω) is symplectic. □

Exercise 2.4.2.3. Let (V, ω) be a symplectic space. Then $\dim V$ is even.

Proof. Set $n = \dim V$. Let $(e_j)_{j=1}^n$ be a basis for V . Define $[\omega] \in \mathbb{R}^{n \times n}$ by $[\omega]_{i,j} = \omega(e_i, e_j)$. Since $\omega \in \Lambda^2(V)$, $[\omega]^* = -[\omega]$. Therefore

$$\begin{aligned}
 \det[\omega] &= \det[\omega]^* \\
 &= \det(-[\omega]) \\
 &= (-1)^n \det[\omega]
 \end{aligned}$$

For the sake of contradiction, suppose that n is odd. Then $\det[\omega] = -\det[\omega]$ which implies that $\det[\omega] = 0$. Since ω is nondegenerate, $[\omega] \in GL(n, \mathbb{R})$. This is a contradiction. Hence n is even. □

Definition 2.4.2.4. Let (V, ω) be a symplectic space and $S \subset V$ a subspace. We define the **symplectic complement of V** , denoted $S^\perp$, by

$$S^\perp = \{v \in V : \text{for each } w \in S, \omega(v, w) = 0\}$$

Exercise 2.4.2.5. Let (V, ω) be a symplectic space and $S \subset V$ a subspace. Then $S^\perp$ is a subspace.

Proof. We note that

$$S^\perp = \bigcap_{v \in S} \ker \iota_v \omega$$

Hence $S^\perp$ is a subspace. □

Exercise 2.4.2.6. Let (V, ω) be a symplectic space and $S \subset V$ a subspace. Then

$$\dim V = \dim S + \dim S^\perp$$

Proof. □

Exercise 2.4.2.7. Let (V, ω) be a symplectic space and $S \subset V$ a subspace. Then $(S^\perp)^\perp = S$.

Proof. Let $v \in (S^\perp)^\perp$. Then for each $w \in S^\perp$, $\omega(v, w) = 0$. □

Part II

Manifolds

Chapter 3

Topological Manifolds

3.1 Introduction

- redo in terms of all charts (U, ϕ) where for some j , $\phi(U) \in \mathcal{T}_{\mathbb{H}_j^n}$ or $\phi(U) \in \mathcal{T}_{\mathbb{R}^n}$ and then make an exercise about equivalently being $\phi(U) \in \mathcal{T}_{\mathbb{H}_j^n}$ and if $\phi(U) \in \mathcal{T}_{\mathbb{R}^n}$ iff interior chart.
- show $\emptyset$ is a top manifold of every dimension

Exercise 3.1.0.1. We have that $\mathbb{R}$ is homeomorphic to $(0, \infty)$

Proof. Define $f : \mathbb{R} \rightarrow (0, \infty)$ by $f(x) = e^x$. Then f is a homeomorphism. □

Definition 3.1.0.2. Let $n \in \mathbb{N}$ and $j \in [n]$. We define the j -th coordinate upper half space of $\mathbb{R}^n$, denoted $\mathbb{H}_j^n$, by

$$\mathbb{H}_j^n = \{(x^1, x^2, \dots, x^n) \in \mathbb{R}^n : x^j \geq 0\}$$

and we define

$$\partial \mathbb{H}_j^n = \{(x^1, x^2, \dots, x^n) \in \mathbb{R}^n : x^j = 0\}$$

$$\text{Int } \mathbb{H}_j^n = \{(x^1, x^2, \dots, x^n) \in \mathbb{R}^n : x^j > 0\}$$

We endow $\mathbb{H}_j^n$, $\partial \mathbb{H}_j^n$ and $\text{Int } \mathbb{H}_j^n$ with the subspace topology inherited from $\mathbb{R}^n$.

We define the projection map $\pi_{\partial \mathbb{H}_j^n} : \partial \mathbb{H}_j^n \rightarrow \mathbb{R}^{n-1}$ by

$$\pi_{\partial \mathbb{H}_j^n}(x^1, \dots, x^{j-1}, x^j, x^{j+1}, \dots, x^n) = (x^1, \dots, x^{j-1}, 0, x^{j+1}, \dots, x^{n-1})$$

Definition 3.1.0.3. We define $\mathbb{R}^0 := \{0\}$, $\mathbb{H}^0 := \{0\}$, $\partial \mathbb{H}^0 := \emptyset$, and $\mathbb{H}_1^{-1} = \emptyset$ endowed with the discrete topology.

Note 3.1.0.4. show in calculus section that $\lambda_{n,k} : \mathbb{H}_j^n \rightarrow \mathbb{H}_k^n$ is a diffeo

Exercise 3.1.0.5. Let $n \in \mathbb{N}$ and $j \in [n]$. Then

1. $\partial \mathbb{H}_j^n$ is homeomorphic to $\mathbb{R}^{n-1}$,
2. $\text{Int } \mathbb{H}_j^n$ is homeomorphic to $\mathbb{R}^n$.

Proof.

1. Clearly $\pi_{\partial \mathbb{H}_j^n}$ is a homeomorphism.
2. Define $f_j : \mathbb{R}^n \rightarrow \text{Int } \mathbb{H}_j^n$ by $f_j(x^1, \dots, x^{j-1}, x^j, x^{j+1}, \dots, x^n) = (x^1, \dots, x^{j-1}, e^{x^j}, x^{j+1}, \dots, x^n)$. Then f is a homeomorphism. □

Exercise 3.1.0.6. Let $A \subset \mathbb{H}_j^n$. Suppose that A is open in $\mathbb{H}_j^n$. Then A is open in $\mathbb{R}^n$ iff $A \cap \partial \mathbb{H}_j^n = \emptyset$.

Hint: simply connected? **FINISH!!!**

Proof.

- ($\implies$) :

Suppose that A is open in $\mathbb{R}^n$. For the sake of contradiction, suppose that $A \cap \partial\mathbb{H}_j^n \neq \emptyset$. Then there exists $x \in A$ such that $x \in \partial\mathbb{H}_j^n$. Since A is open in $\mathbb{R}^n$, there exists $B \subset A$ such that B is open in $\mathbb{R}^n$, $x \in B$ and B is simply connected. Set $B' := B \setminus \{x\}$. Then B' is not simply connected. **FINISH!!! Just show that you cant get a ball in $\mathbb{R}^n$ around x which is contained in $\mathbb{H}_j^n$.**

- ($\impliedby$) :

Suppose that $A \cap \partial\mathbb{H}_j^n = \emptyset$. Then $A \subset \text{Int } \mathbb{H}_j^n$. Since $\text{Int } \mathbb{H}_j^n$ is open in $\mathbb{R}^n$, we have that

$$\begin{aligned} \mathcal{T}_{\text{Int } \mathbb{H}_j^n} &= \mathcal{T}_{\mathbb{R}^n} \cap \text{Int } \mathbb{H}_j^n \\ &\subset \mathcal{T}_{\mathbb{R}^n} \end{aligned}$$

An exercise in the section on subspace topology in the analysis notes implies that

$$\begin{aligned} \mathcal{T}_{\text{Int } \mathbb{H}_j^n} &= \mathcal{T}_{\mathbb{R}^n} \cap \text{Int } \mathbb{H}_j^n \\ &= (\mathcal{T}_{\mathbb{R}^n} \cap \mathbb{H}_j^n) \cap \text{Int } \mathbb{H}_j^n \\ &= \mathcal{T}_{\mathbb{H}_j^n} \cap \text{Int } \mathbb{H}_j^n \end{aligned}$$

Since $A \in \mathcal{T}_{\mathbb{H}_j^n}$ and $A \subset \text{Int } \mathbb{H}_j^n$, we have that

$$\begin{aligned} A &\in \mathcal{T}_{\mathbb{H}_j^n} \cap \text{Int } \mathbb{H}_j^n \\ &= \mathcal{T}_{\text{Int } \mathbb{H}_j^n} \\ &\subset \mathcal{T}_{\mathbb{R}^n} \end{aligned}$$

Thus A is open in $\mathbb{R}^n$.

□

Definition 3.1.0.7. Let $(M, \mathcal{T})$ be a topological space, $n \in \mathbb{N}$, $j \in [n]$, $U \subset M$, $V \subset \mathbb{R}^n$ and $\phi : U \rightarrow V$. Then

- (U, ϕ) is said to be an $\mathbb{R}^n$ -**coordinate chart on $(M, \mathcal{T})$** if

- $U \in \mathcal{T}$
- $V \in \mathcal{T}_{\mathbb{R}^n}$
- ϕ is a $(\mathcal{T} \cap U, \mathcal{T}_{\mathbb{R}^n} \cap V)$ -homeomorphism

- (U, ϕ) is said to be an $\mathbb{H}_j^n$ -**coordinate chart on $(M, \mathcal{T})$** if

- $U \in \mathcal{T}$
- $V \in \mathcal{T}_{\mathbb{H}_j^n}$
- ϕ is a $(\mathcal{T} \cap U, \mathcal{T}_{\mathbb{H}_j^n} \cap V)$ -homeomorphism

- (U, ϕ) is said to be an n -**coordinate chart on $(M, \mathcal{T})$** if (U, ϕ) is an $\mathbb{R}^n$ -coordinate chart on $(M, \mathcal{T})$ or there exists $j \in [n]$ such that (U, ϕ) is an $\mathbb{H}_j^n$ -coordinate chart on $(M, \mathcal{T})$.

- We define

$$X^{n,j}(M, \mathcal{T}) := \{(U, \phi) : (U, \phi) \text{ is an } \mathbb{H}_j^n\text{-coordinate chart on } (M, \mathcal{T})\}$$

and

$$X^n(M, \mathcal{T}) := \{(U, \phi) : (U, \phi) \text{ is an } n\text{-coordinate chart on } (M, \mathcal{T})\}$$

Note 3.1.0.8. From Definition 1.3.3.2, Exercise 1.3.3.3 and Definition 1.3.3.4, we recall

- the definition of the action $S_n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ given by $(\sigma, x) \mapsto \sigma \cdot x$,

- for $\sigma \in S_n$, the definition of the map $\Phi_\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^n$,
- that Φ_σ is a diffeomorphism,
- for $U \subset \mathbb{R}^n$, the definition of the action $S_n \times (\mathbb{R}^n)^U \rightarrow (\mathbb{R}^n)^U$ given by $(\sigma, \phi) \mapsto \sigma \cdot \phi$.

Exercise 3.1.0.9. Let $(M, \mathcal{T})$ be a topological space, $n \in \mathbb{N}$, $j \in [n]$ and $(U, \phi) \in X^{n,j}(M, \mathcal{T})$. For each $\sigma \in S_n$, $\sigma \cdot \phi \in X^{n,\sigma(j)}(M, \mathcal{T})$.

Proof. Let $\sigma \in S_n$. We note the following:

1. By definition, $\sigma \cdot \phi = \Phi_\sigma \circ \phi$. Since $\Phi_\sigma(\mathbb{H}_j^n) = \mathbb{H}_{\sigma(j)}^n$, we have that $(\sigma \cdot \phi)(U) \subset \mathbb{H}_{\sigma(j)}^n$.
2. Since Φ_σ is a diffeomorphism, $\Phi_\sigma|_{\mathbb{H}_j^n}$ is a $(\mathcal{T}_{\mathbb{H}_j^n}, \mathcal{T}_{\mathbb{H}_{\sigma(j)}^n})$ -homeomorphism. Since $(U, \phi) \in X^{n,j}(M, \mathcal{T})$, ϕ is a $(\mathcal{T} \cap U, \mathcal{T}_{\mathbb{H}_j^n} \cap \phi(U))$ -homeomorphism. Thus $\sigma \cdot \phi$ is a $(\mathcal{T} \cap U, \mathcal{T}_{\mathbb{H}_{\sigma(j)}^n} \cap (\sigma \cdot \phi)(U))$ -homeomorphism.

Since $(U, \phi) \in X^{n,j}(M, \mathcal{T})$, $U \in \mathcal{T}$. Since $\sigma \cdot \phi$ is a homeomorphism, we have that $(\sigma \cdot \phi)(U) \in \mathcal{T}_{\mathbb{H}_{\sigma(j)}^n}$. Summarizing, we have that

- $U \in \mathcal{T}$,
- $(\sigma \cdot \phi)(U) \in \mathcal{T}_{\mathbb{H}_{\sigma(j)}^n}$,
- $\sigma \cdot \phi$ is a $(\mathcal{T} \cap U, \mathcal{T}_{\mathbb{H}_{\sigma(j)}^n} \cap \Phi_\sigma(U))$ -homeomorphism.

Hence $(U, \sigma \cdot \phi) \in X^{n,\sigma(j)}(M, \mathcal{T})$. □

Exercise 3.1.0.10. Let $(M, \mathcal{T})$ be a topological space, $n \in \mathbb{N}$ and $j, k \in [n]$. For each $p \in M$, there exists $(U, \phi) \in X^{n,j}(M, \mathcal{T})$ such that $p \in U$ iff there exists $(V, \psi) \in X^{n,k}(M, \mathcal{T})$ such that $p \in V$.

Proof. Let $p \in M$.

- $(\implies)$:
Suppose that there exists $(U, \phi) \in X^{n,j}(M, \mathcal{T})$ such that $p \in U$. Choose $\sigma \in S_n$ such that $\sigma(j) = k$. Define $V := U$ and $\psi := \sigma \cdot \phi$. Then $(V, \psi) \in X^{n,k}(M, \mathcal{T})$ and $p \in V$.
- $(\impliedby)$:
Suppose that there exists $(V, \psi) \in X^{n,k}(M, \mathcal{T})$ such that $p \in V$. Choose $\tau \in S_n : \tau(k) = j$. Define $U := V$ and $\phi = \tau \cdot \psi$. Then $(U, \phi) \in X^{n,j}(M, \mathcal{T})$ and $p \in U$. □

Note 3.1.0.11. So if there is at least one coordinate chart to the j -th upper half-space, then there are coordinate charts to all upper half spaces.

need to define $[n] = \{1, \dots, n\}$ if $n \geq 1$ and $[n] = \{1\}$ if $n \in \{-1, 0\}$.

Definition 3.1.0.12. Let $(M, \mathcal{T})$ be a topological space and $n \in \mathbb{N}$. We define

$$X^n(M, \mathcal{T}) := \bigcup_{j=1}^n X^{n,j}(M, \mathcal{T})$$

add case $n = 0$.

Note 3.1.0.13. We will write $X^n(M)$ in place of $X^n(M, \mathcal{T})$ when the topology is not ambiguous.

Definition 3.1.0.14. Let M be a topological space and $n \in \mathbb{N}$. Then M is said to be **locally Euclidean of dimension n** if for each $p \in M$, there exists $(U, \phi) \in X^n(M)$ such that $p \in U$.

Definition 3.1.0.15. Let M be a topological space and $n \in \mathbb{N}_{-1}$. Then M is said to be an **n -dimensional topological manifold** if

1. M is Hausdorff

2. M is second-countable
3. M is locally Euclidean of dimension n

Exercise 3.1.0.16. Let $n \in \mathbb{N}_{-1}$. Then

1. $(\mathbb{R}^n, \text{id}_{\mathbb{R}^n}) \in X^n(\mathbb{R}^n)$
2. $(\mathbb{H}_j^n, \text{id}_{\mathbb{H}_j^n}) \in X^n(\mathbb{H}_j^n)$. **fix**

Proof.

- 1.
- 2.

□

Exercise 3.1.0.17. Let $n \in \mathbb{N}_0$. Then

1. $\mathbb{R}^n$ is an n -dimensional topological manifold of dimension n ,
2. if $n \geq 1$, then $\mathbb{H}_j^n$ is an n -dimensional topological manifold of dimension n . **fix**

Proof.

- 1.
- 2.

□

Theorem 3.1.0.18. **Invariance of Domain**

Theorem 3.1.0.19. Topological Invariance of Dimension:

Let $n \in \mathbb{N}_0$, M an m -dimensional topological manifold and N a n -dimensional topological manifold. If M and N are homeomorphic, then $m = n$.

try to prove, first for subsets of $\mathbb{R}^m$ and $\mathbb{R}^n$, then the general case, see math stack exchange for short proof <https://math.stackexchange.com/questions/1111111/proving-topological-invariance-of-dimension-using-brouwer-fixed-point-theorem> the idea is that suppose $U \subset \mathbb{R}^m$, $V \subset \mathbb{R}^n$ are open and $f : U \rightarrow V$ is homeo. If $n < m$, then $\iota \circ f$ is a topological embedding onto its image where $\iota : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the inclusion, since $n < m$, no subset of $\iota(\mathbb{R}^n)$ (besides the empty set) is open in $\mathbb{R}^m$. Now use Invariance of domain theorem from algebraic topology.

Note 3.1.0.20. In light of the previous theorem, we write $X(M)$ in place of $X^n(M)$ and refer to n -coordinate charts as coordinate charts when the context is clear.

Exercise 3.1.0.21. Let $n \in \mathbb{N}$, $j, k \in [n]$, $U \in \mathcal{T}_{\mathbb{H}_j^n}$, $V \in \mathcal{T}_{\mathbb{H}_k^n}$ and $\phi : U \rightarrow V$. Suppose that ϕ is a $(\mathcal{T}_{\mathbb{H}_j^n} \cap U, \mathcal{T}_{\mathbb{H}_k^n} \cap V)$ -homeomorphism. Then for each $p \in U$,

1. $p \in \partial \mathbb{H}_j^n$ iff $\phi(p) \in \partial \mathbb{H}_k^n$,
2. $p \in \text{Int } \mathbb{H}_j^n$ iff $\phi(p) \in \text{Int } \mathbb{H}_k^n$.

Proof. Let $p \in U$.

1. • $(\implies)$:
For the sake of contradiction, suppose that $p \in \partial \mathbb{H}_j^n$ and $\phi(p) \notin \partial \mathbb{H}_k^n$. Then

$$\begin{aligned} \phi(p) &\in (\partial \mathbb{H}_k^n)^c \\ &= \text{Int } \mathbb{H}_k^n \end{aligned}$$

Since $\text{Int } \mathbb{H}_k^n \cap V \in \mathcal{T}_{\mathbb{H}_k^n} \cap V$ and $\phi(p) \in \text{Int } \mathbb{H}_k^n \cap V$, there exists $B_V \in \mathcal{T}_{\mathbb{H}_k^n} \cap V$ such that $B_V \subset \text{Int } \mathbb{H}_k^n \cap V$, $\phi(p) \in B_V$ and B_V is simply connected. Define $B_U := \phi^{-1}(B_V)$. Since ϕ is a $(\mathcal{T}_{\mathbb{H}_j^n} \cap U, \mathcal{T}_{\mathbb{H}_k^n} \cap V)$ -homeomorphism, $\phi|_{B_U} : B_U \rightarrow B_V$ is a $(\mathcal{T}_{\mathbb{H}_j^n} \cap B_U, \mathcal{T}_{\mathbb{H}_k^n} \cap B_V)$ -homeomorphism. Therefore $B_U \in \mathcal{T}_{\mathbb{H}_j^n} \cap U$, $p \in B_U$ and B_U is simply connected.

Define $B'_U \in \mathcal{T}_{\mathbb{H}_j^n} \cap U$ and $B'_V \in \mathcal{T}_{\mathbb{H}_k^n} \cap V$ by $B'_U := B_U \setminus \{p\}$ and $B'_V := B_V \setminus \{\phi(p)\}$. Since $p \in \partial \mathbb{H}_j^n$, B'_U is simply connected. Since ϕ is a $(\mathcal{T}_{\mathbb{H}_j^n} \cap U, \mathcal{T}_{\mathbb{H}_k^n} \cap V)$ -homeomorphism, $\phi|_{B'_U} : B'_U \rightarrow B'_V$ is a $(\mathcal{T}_{\mathbb{H}_j^n} \cap B'_U, \mathcal{T}_{\mathbb{H}_k^n} \cap B'_V)$ -homeomorphism. Therefore B'_V is simply connected.

Since $\phi(p) \in \text{Int } \mathbb{H}_k^n$, B'_V is not simply connected. This is a contradiction. Hence $p \in \partial \mathbb{H}_j^n$ implies that $\phi(p) \in \partial \mathbb{H}_k^n$.

• ($\Leftarrow$) :

Suppose that $\phi(p) \in \partial \mathbb{H}_k^n$. Set $q = \phi(p)$. Then $\phi^{-1} : V \rightarrow U$ is a $(\mathcal{T}_{\mathbb{H}_k^n} \cap V, \mathcal{T}_{\mathbb{H}_j^n} \cap U)$ -homeomorphism. The previous part implies that

$$\begin{aligned} p &= \phi^{-1}(q) \\ &\in \partial \mathbb{H}_j^n \end{aligned}$$

2. By part (1), we have that

$$\begin{aligned} p \in \text{Int } \mathbb{H}_j^n &\iff p \notin \partial \mathbb{H}_j^n \\ &\iff \phi(p) \notin \partial \mathbb{H}_k^n \\ &\iff \phi(p) \in \text{Int } \mathbb{H}_k^n \end{aligned}$$

□

Definition 3.1.0.22. Let $n \in \mathbb{N}$, $(M, \mathcal{T})$ be an n -dimensional topological manifold and $(U, \phi) \in X^n(M, \mathcal{T})$. Then (U, ϕ) is said to be

- an **interior chart** if there exists $j \in [n]$ such that $(U, \phi) \in X^{n,j}(M, \mathcal{T})$ and $\phi(U) \cap \partial \mathbb{H}_j^n = \emptyset$,
- a **boundary chart** if there exists $j \in [n]$ such that $(U, \phi) \in X^{n,j}(M, \mathcal{T})$ and $\phi(U) \cap \partial \mathbb{H}_j^n \neq \emptyset$.

We set

- $X_{\text{Int}}^n(M, \mathcal{T}) := \{(U, \phi) \in X^n(M, \mathcal{T}) : (U, \phi) \text{ is an interior chart}\}$
- $X_{\partial}^n(M, \mathcal{T}) := \{(U, \phi) \in X^n(M, \mathcal{T}) : (U, \phi) \text{ is a boundary chart}\}$

For $j \in [n]$, we define

- $X_{\text{Int}}^{n,j}(M, \mathcal{T}) := X_{\text{Int}}^n(M, \mathcal{T}) \cap X^{n,j}(M, \mathcal{T})$,
- $X_{\partial}^{n,j}(M, \mathcal{T}) := X_{\partial}^n(M, \mathcal{T}) \cap X^{n,j}(M, \mathcal{T})$.

Exercise 3.1.0.23. Let $n \in \mathbb{N}$, M be an n -dimensional topological manifold, $j \in [n]$ and $(U, \phi) \in X^{n,j}(M, \mathcal{T})$. Then

1. $(U, \phi) \in X_{\text{Int}}^{n,j}(M, \mathcal{T})$ iff for each $k \in [n]$

Proof.

- 1.
2. for each $p \in M$, there exists $(U, \phi) \in X_{\text{Int}}^{n,j}(M)$ such that $p \in U$ iff there exists $(V, \psi) \in X_{\text{Int}}^{n,k}(M, \mathcal{T})$ such that $p \in V$.
3. for each $p \in M$, there exists $(U, \phi) \in X_{\partial}^{n,j}(M)$ such that $p \in U$ iff there exists $(V, \psi) \in X_{\partial}^{n,k}(M, \mathcal{T})$ such that $p \in V$.

□

Exercise 3.1.0.24. Let $n \in \mathbb{N}$, $(M, \mathcal{T})$ be an n -dimensional topological manifold and $j \in [n]$. Then

1. $X^n(M, \mathcal{T}) = X_{\text{Int}}^n(M, \mathcal{T}) \cup X_{\partial}^n(M, \mathcal{T})$
2. $X_{\text{Int}}^n(M, \mathcal{T}) \cap X_{\partial}^n(M, \mathcal{T}) = \emptyset$

Proof. **FIX**

1. By definition, $X_{\text{Int}}^n(M, \mathcal{T}) \cup X_{\partial}^n(M, \mathcal{T}) \subset X^n(M, \mathcal{T})$. Let $(U, \phi) \in X^n(M, \mathcal{T})$. By definition, there exists $j \in [n]$ such that $(U, \phi) \in X^{n,j}(M, \mathcal{T})$. If $\phi(U) \cap \partial\mathbb{H}_j^n = \emptyset$, then

$$\begin{aligned} (U, \phi) &\in X_{\text{Int}}^{n,j}(M) \\ &\subset X_{\text{Int}}^{n,j}(M) \cup X_{\partial}^{n,j}(M) \end{aligned}$$

If $\phi(U) \cap \partial\mathbb{H}_j^n \neq \emptyset$, then

$$\begin{aligned} (U, \phi) &\in X_{\partial}^{n,j}(M) \\ &\subset X_{\text{Int}}^{n,j}(M) \cup X_{\partial}^{n,j}(M) \end{aligned}$$

Since $(U, \phi) \in X^n(M, \mathcal{T})$ is arbitrary, $X^n(M, \mathcal{T}) \subset X_{\text{Int}}^n(M) \cup X_{\partial}^n(M)$. Therefore $X^n(M) = X_{\text{Int}}^n(M) \cup X_{\partial}^n(M)$.

2. For the sake of contradiction, suppose that $X_{\text{Int}}^n(M) \cap X_{\partial}^n(M) \neq \emptyset$. Then there exists $(U, \phi) \in X^n(M, \mathcal{T})$ such that $(U, \phi) \in X_{\text{Int}}^n(M, \mathcal{T})$ and $(U, \phi) \in X_{\partial}^n(M, \mathcal{T})$. Therefore

- there exists $j \in [n]$ such that $(U, \phi) \in X^{n,j}(M, \mathcal{T})$ and $\phi(U) \cap \partial\mathbb{H}_j^n = \emptyset$,
- there exists $k \in [n]$ such that $(U, \phi) \in X^{n,k}(M, \mathcal{T})$ and $\phi(U) \cap \partial\mathbb{H}_k^n \neq \emptyset$.

Since $(U, \phi) \in X^{n,j}(M, \mathcal{T})$, we have that $\phi(U) \in \mathcal{T}_{\mathbb{H}_j^n}$ and ϕ is a $(\mathcal{T} \cap U, \mathcal{T}_{\mathbb{H}_j^n} \cap \phi(U))$ -homeomorphism. Similarly, since $(U, \phi) \in X^{n,k}(M, \mathcal{T})$, we have that $\phi(U) \in \mathcal{T}_{\mathbb{H}_k^n}$ and ϕ is a $(\mathcal{T} \cap U, \mathcal{T}_{\mathbb{H}_k^n} \cap \phi(U))$ -homeomorphism. Therefore $\text{id}_{\phi(U)} = \phi \circ \phi^{-1}$ is a $(\mathcal{T}_{\mathbb{H}_j^n} \cap \phi(U), \mathcal{T}_{\mathbb{H}_k^n} \cap \phi(U))$ -homeomorphism.

Since $\phi(U) \cap \partial\mathbb{H}_k^n \neq \emptyset$, there exists $p \in U$ such that $\phi(p) \in \partial\mathbb{H}_k^n$. Exercise 3.1.0.21 implies that

$$\begin{aligned} \phi(p) &= \text{id}_{\phi(U)}(\phi(p)) \\ &= \phi \circ \phi^{-1}(\phi(p)) \\ &\in \partial\mathbb{H}_j^n \end{aligned}$$

This is a contradiction since $\phi(U) \cap \partial\mathbb{H}_j^n = \emptyset$. Hence $X_{\text{Int}}^n(M, \mathcal{T}) \cap X_{\partial}^n(M, \mathcal{T}) = \emptyset$.

□

Definition 3.1.0.25. Let M be an n -dimensional topological manifold. We define the

- **interior** of M , denoted $\text{Int } M$, by

$$\text{Int } M = \{p \in M : \text{there exists } (U, \phi) \in X_{\text{Int}}(M) \text{ such that } p \in U\}$$

- **boundary** of M , denoted ∂M , by

$$\partial M = \{p \in M : \text{there exists } (V, \psi) \in X_{\partial}(M) \text{ such that } p \in V \text{ and } \psi(p) \in \partial\mathbb{H}_j^n\}$$

FINISH!!!

Exercise 3.1.0.26. Let M be an n -dimensional topological manifold. Let $(U, \phi) \in X_{\text{Int}}(M)$. Then $U \subset \text{Int } M$.

Proof. Let $p \in U$. Since $(U, \phi) \in X_{\text{Int}}(M)$ and $p \in U$, by definition, $p \in \text{Int } M$. Since $p \in U$ is arbitrary, $U \subset \text{Int } M$. □

Exercise 3.1.0.27. Let M be an n -dimensional topological manifold and $(U, \phi) \in X(M)$. Then $(U, \phi) \in X_{\text{Int}}(M)$ iff $\phi(U)$ is open in $\mathbb{R}^n$.

Proof. Suppose that $(U, \phi) \in X_{\text{Int}}(M)$. Then there exists $j \in [n]$ such that $(U, \phi) \in X^{n,j}(M)$ and $\phi(U) \cap \partial\mathbb{H}_j^n = \emptyset$. Since $\phi(U) \in \mathcal{T}_{\mathbb{H}_j^n}$, Exercise 3.1.0.6 implies that $\phi(U) \in \mathcal{T}_{\mathbb{R}^n}$.

Conversely, suppose that $\phi(U) \in \mathcal{T}_{\mathbb{R}^n}$. Since $(U, \phi) \in X^n(M)$, there exists $j \in [n]$ such that $(U, \phi) \in X^{n,j}(M)$. Therefore $\phi(U) \in \mathcal{T}_{\mathbb{H}_j^n}$. Since $\phi(U) \in \mathcal{T}_{\mathbb{R}^n}$, Exercise 3.1.0.6 implies that $\phi(U) \cap \partial\mathbb{H}_j^n = \emptyset$. Thus $(U, \phi) \in X_{\text{Int}}(M)$. □

Exercise 3.1.0.28. Let M be an n -dimensional topological manifold, $(U, \phi) \in X_\partial(M)$ and $p \in U$. If $\phi(p) \notin \partial\mathbb{H}_j^n$, then $p \in \text{Int } M$.

Proof. Suppose that $\phi(p) \notin \partial\mathbb{H}_j^n$. Then $\phi(p) \in \text{Int } \mathbb{H}_j^n$. Hence there exists $B' \subset \phi(U)$ such that B' is open in $\mathbb{R}^n$ and $\phi(p) \in B'$. Set $U' = \phi^{-1}(B')$ and $\phi' = \phi|_{U'}$. Then U' is open in M and $\phi' : U' \rightarrow B'$ is a homeomorphism. Hence $(U', \phi') \in X_{\text{Int}}(M)$. Since $\phi(p) \in B'$, we have that $p \in U'$. By definition, $p \in \text{Int } M$. $\square$

Exercise 3.1.0.29. Let M be an n -dimensional topological manifold. Then

1. $M = \text{Int } M \cup \partial M$
 2. $\text{Int } M \cap \partial M = \emptyset$
- Hint:** simply connected

Proof.

1. By definition, $\text{Int } M \cup \partial M \subset M$. Let $p \in M$. Since M is a manifold, there exists $(U, \phi) \in X(M)$ such that $p \in U$. A previous exercise implies that $(U, \phi) \in X_{\text{Int}}(M) \cup X_\partial(M)$. If $(U, \phi) \in X_{\text{Int}}(M)$, then by definition,

$$\begin{aligned} p &\in \text{Int } M \\ &\subset \text{Int } M \cup \partial M \end{aligned}$$

Suppose that $(U, \phi) \in X_\partial(M)$. If $\phi(p) \in \partial\mathbb{H}_j^n$, then by definition,

$$\begin{aligned} p &\in \partial M \\ &\subset \text{Int } M \cup \partial M \end{aligned}$$

Suppose that $\phi(p) \notin \partial\mathbb{H}_j^n$. The previous exercise implies that $p \in \text{Int } M$. Therefore,

$$\begin{aligned} p &\in \text{Int } M \\ &\subset \text{Int } M \cup \partial M \end{aligned}$$

Since $p \in M$ is arbitrary, $M \subset \text{Int } M \cup \partial M$. Therefore $M = \text{Int } M \cup \partial M$.

2. For the sake of contradiction, suppose that $\text{Int } M \cap \partial M \neq \emptyset$. Then there exists $p \in M$ such that $p \in \text{Int } M \cap \partial M$. By definition, there exists $(U, \phi) \in X_{\text{Int}}(M)$, $(V, \psi) \in X_\partial(M)$ such that $p \in U \cap V$ and $\psi(p) \in \partial\mathbb{H}_j^n$. Note that $\psi(U \cap V)$ is open in $\mathbb{H}_j^n$, $\phi(U \cap V)$ is open in $\mathbb{R}^n$ and $\phi|_{U \cap V} \circ (\psi|_{U \cap V})^{-1} : \psi^{-1}(U \cap V) \rightarrow \phi(U \cap V)$ is a homeomorphism. Since $\psi(U \cap V)$ is open in $\mathbb{H}_j^n$, there exists an $B_\psi \subset \psi(U \cap V)$ such that B_ψ is open in $\mathbb{H}_j^n$, B_ψ is simply connected and $\psi(p) \in B_\psi$. Set $B_\phi = \phi \circ \psi^{-1}(B_\psi)$. Since $\phi(U \cap V)$ is open in $\mathbb{R}^n$, B_ϕ is open in $\mathbb{R}^n$. Since B_ψ is simply connected and $\phi|_{U \cap V} \circ (\psi|_{U \cap V})^{-1} : \psi^{-1}(U \cap V) \rightarrow \phi(U \cap V)$ is a homeomorphism, B_ϕ is simply connected. Set $B'_\phi = B_\phi \setminus \{\phi(p)\}$ and $B'_\psi = B_\psi \setminus \{\psi(p)\}$. Then $\phi \circ \psi^{-1} : B'_\psi \rightarrow B'_\phi$ is a homeomorphism. Since $\psi(p) \in \partial\mathbb{H}_j^n$, B'_ψ is simply connected. Since B_ϕ is open in $\mathbb{R}^n$, B'_ϕ is not simply connected. This is a contradiction since B'_ϕ is homeomorphic to B'_ψ . So $\partial M \cap \text{Int } M = \emptyset$. $\square$

Exercise 3.1.0.30. Let M be an n -dimensional topological manifold. Then

1. $\text{Int } M$ is open
2. ∂M is closed

Proof.

1. Let $p \in \text{Int } M$. Then there exists $(U, \phi) \in X_{\text{Int}}(M)$ such that $p \in U$. By definition, U is open and a previous exercise implies that $U \subset \text{Int } M$. Since $p \in \text{Int } M$ is arbitrary, we have that for each $p \in \text{Int } M$, there exists $U \subset \text{Int } M$ such that U is open. Hence $\text{Int } M$ is open.
2. Since $\partial M = (\text{Int } M)^c$, and $\text{Int } M$ is open, we have that ∂M is closed. $\square$

Exercise 3.1.0.31. Let M be an n -dimensional topological manifold, $(U, \phi) \in X(M)$ and $p \in U$. If $p \in \partial M$, then $(U, \phi) \in X_{\partial}(M)$.
Hint: simply connected

Proof. Suppose that $p \in \partial M$. Then there exists a $(V, \psi) \in X_{\partial}(M)$ such that $p \in V$ and $\psi(p) \in \partial \mathbb{H}_j^n$. Note that $\psi(U \cap V)$ is open in $\mathbb{H}_j^n$, $\phi(U \cap V)$ is open in $\mathbb{R}^n$ and $\phi|_{U \cap V} \circ (\psi|_{U \cap V})^{-1} : \psi^{-1}(U \cap V) \rightarrow \phi(U \cap V)$ is a homeomorphism. Since $\psi(U \cap V)$ is open in $\mathbb{H}_j^n$, there exists $B_{\psi} \subset \psi(U \cap V)$ such B_{ψ} is open in $\mathbb{H}_j^n$, B_{ψ} is simply connected and $\psi(p) \in B_{\psi}$. Set $B_{\phi} = \phi \circ \psi^{-1}(B_{\psi})$.

For the sake of contradiction, suppose that $(U, \phi) \in X_{\text{Int}}(M)$. Then $\phi(U)$ is open in $\mathbb{R}^n$. Hence $\phi(U \cap V)$ is open in $\mathbb{R}^n$ and B_{ϕ} is open in $\mathbb{R}^n$. Since $\phi|_{U \cap V} \circ (\psi|_{U \cap V})^{-1} : \psi^{-1}(U \cap V) \rightarrow \phi(U \cap V)$ is a homeomorphism, B_{ϕ} is simply connected. Set $B'_{\phi} = B_{\phi} \setminus \{\phi(p)\}$ and $B'_{\psi} = B_{\psi} \setminus \{\psi(p)\}$. Since $\psi(p) \in \partial \mathbb{H}_j^n$, B'_{ψ} is simply connected. Since B_{ϕ} is open in $\mathbb{R}^n$, B'_{ϕ} is not simply connected. This is a contradiction since B'_{ϕ} is homeomorphic to B'_{ψ} . So $(U, \phi) \notin X_{\text{Int}}(M)$. Since $(X_{\text{Int}}(M))^c = X_{\partial}(M)$, we have that $(U, \phi) \in X_{\partial}(M)$. $\square$

Exercise 3.1.0.32. Let M be an n -dimensional topological manifold, $(U, \phi) \in X_{\partial}(M)$ and $p \in U$. Then

1. $p \in \partial M$ iff $\phi(p) \in \partial \mathbb{H}_j^n$ for some j .
2. $p \in \text{Int } M$ iff $\phi(p) \in \text{Int } \mathbb{H}_j^n$

Proof.

1. Suppose that $p \in \partial M$. For the sake of contradiction, suppose that $\phi(p) \notin \partial \mathbb{H}^n$. Then $\phi(p) \in \text{Int } \mathbb{H}^n$. Hence there exists $B' \subset \phi(U)$ such that B' is open in $\mathbb{R}^n$ and $\phi(p) \in B'$. Set $U' = \phi^{-1}(B')$ and $\phi' = \phi|_{U'}$. Then $p \in U'$ and $(U', \phi') \in X_{\text{Int}}(M)$. Since $p \in U$, the previous exercise implies that $(U', \phi') \in X_{\partial}(M)$. This is a contradiction since $X_{\text{Int}}(M) \cap X_{\partial}(M) = \emptyset$. So $\phi(p) \in \partial \mathbb{H}^n$.

Conversely, suppose that $\phi(p) \in \partial \mathbb{H}^n$. By definition, $p \in \partial M$.

2. A previous exercise implies that $\text{Int } M = (\partial M)^c$. Part (1) implies that

$$\begin{aligned} p &\in (\partial M)^c \\ &= \text{Int } M \end{aligned}$$

if and only if

$$\begin{aligned} \phi(p) &\in (\partial \mathbb{H}^n)^c \\ &= \text{Int } \mathbb{H}^n \end{aligned}$$

$\square$

Exercise 3.1.0.33. Let M be an n -dimensional topological manifold and $p \in M$. Then $p \in \partial M$ iff for each $(U, \phi) \in X(M)$, $p \in U$ implies that $(U, \phi) \in X_{\partial}(M)$ and $\phi(p) \in \partial \mathbb{H}^n$.

Proof. Suppose that $p \in \partial M$. Let $(U, \phi) \in X(M)$. Suppose that $p \in U$. The previous two exercises imply that $(U, \phi) \in X_{\partial}(M)$ and $\phi(p) \in \partial \mathbb{H}^n$.

Conversely, suppose that for each $(U, \phi) \in X(M)$, $p \in U$ implies that $(U, \phi) \in X_{\partial}(M)$ and $\phi(p) \in \partial \mathbb{H}^n$. Since M is a manifold, there exists $(U, \phi) \in X(M)$ such that $p \in U$. By assumption, $(U, \phi) \in X_{\partial}(M)$ and $\phi(p) \in \partial \mathbb{H}^n$. By definition, $p \in \partial M$. $\square$

Exercise 3.1.0.34. Let M be an n -dimensional topological manifold. Let $(U, \phi) \in X_{\partial}(M)$. Then

1. $\phi(U \cap \partial M) = \phi(U) \cap \partial \mathbb{H}^n$
2. $\phi(U \cap \text{Int } M) = \phi(U) \cap \text{Int } \mathbb{H}^n$

Proof.

1. Since $(U, \phi) \in X_{\partial}(M)$, a previous exercise implies that for each $p \in U$, $p \in \partial M$ iff $\phi(p) \in \partial \mathbb{H}^n$. Let $q \in \phi(U \cap \partial M)$. Then there exists $p \in U \cap \partial M$ such that $\phi(p) = q$. Since $p \in \partial M$, $\phi(p) \in \partial \mathbb{H}^n$. Hence

$$\begin{aligned} q &= \phi(p) \\ &\in \phi(U) \cap \partial \mathbb{H}^n \end{aligned}$$

Since $q \in \phi(U \cap \partial M)$ is arbitrary, $\phi(U \cap \partial M) \subset \phi(U) \cap \partial \mathbb{H}^n$.

Let $q \in \phi(U) \cap \partial \mathbb{H}^n$. Then there exists $p \in U$ such that $q = \phi(p)$. Since $\phi(p) \in \partial \mathbb{H}^n$, we have that $p \in \partial M$. Hence $p \in U \cap \partial M$ and

$$\begin{aligned} q &= \phi(p) \\ &\in \phi(U \cap \partial M) \end{aligned}$$

Since $q \in \phi(U) \cap \partial \mathbb{H}^n$ is arbitrary, $\phi(U) \cap \partial \mathbb{H}^n \subset \phi(U \cap \partial M)$. Thus $\phi(U \cap \partial M) = \phi(U) \cap \partial \mathbb{H}^n$.

2. Since $(U, \phi) \in X_{\partial}(M)$, a previous exercise implies that for each $p \in U$, $p \in \text{Int } M$ iff $\phi(p) \in \text{Int } \mathbb{H}^n$. Let $q \in \phi(U \cap \text{Int } M)$. Then there exists $p \in U \cap \text{Int } M$ such that $\phi(p) = q$. Since $p \in \text{Int } M$, $\phi(p) \in \text{Int } \mathbb{H}^n$. Hence

$$\begin{aligned} q &= \phi(p) \\ &\in \phi(U) \cap \text{Int } \mathbb{H}^n \end{aligned}$$

Since $q \in \phi(U \cap \text{Int } M)$ is arbitrary, $\phi(U \cap \text{Int } M) \subset \phi(U) \cap \text{Int } \mathbb{H}^n$.

Let $q \in \phi(U) \cap \text{Int } \mathbb{H}^n$. Then there exists $p \in U$ such that $q = \phi(p)$. Since $\phi(p) \in \text{Int } \mathbb{H}^n$, we have that $p \in \text{Int } M$. Hence $p \in U \cap \text{Int } M$ and

$$\begin{aligned} q &= \phi(p) \\ &\in \phi(U \cap \text{Int } M) \end{aligned}$$

Since $q \in \phi(U) \cap \text{Int } \mathbb{H}^n$ is arbitrary, $\phi(U) \cap \text{Int } \mathbb{H}^n \subset \phi(U \cap \text{Int } M)$. Thus $\phi(U \cap \text{Int } M) = \phi(U) \cap \text{Int } \mathbb{H}^n$.

□

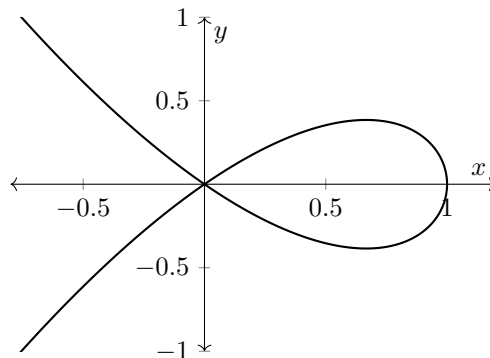
Exercise 3.1.0.35. Graph of Continuous Function:

Let $f \in C(\mathbb{R})$. Set $M = \{(x, y) \in \mathbb{R}^2 : f(x) = y\}$ (i.e. the graph of f). Then M is a 1-dimensional manifold.

Proof. Set $U = \mathbb{R}$ and define $\phi : U \rightarrow M$ by $\phi(x) = (x, f(x))$. Then $\phi^{-1} = \pi_1$. Since f is continuous, ϕ is continuous. Since π_1 is continuous, ϕ is a homeomorphism. □

Exercise 3.1.0.36. Nodal Cubic:

Let $M = \{(x, y) \in \mathbb{R}^2 : y^2 = x^2 - x^3\}$. We equip M with the subspace topology.



Then M is not a 1-dimensional topological manifold.

Hint: connected components

Proof. Suppose that M is a 1-dimensional manifold. Set $p = (0, 0)$. Then there exists $(U, \phi) \in X(M)$ such that $p \in U$. Since $\phi(U)$ is open (in $\mathbb{R}$ or $\mathbb{H}$), there exists a $B \subset \phi(U)$ such that B is open (in $\mathbb{R}$ or $\mathbb{H}$), B is connected and $\phi(p) \in B$. Set $V = \phi^{-1}(B)$, $V' = V \setminus \{p\}$ and $B' = B \setminus \{\phi(p)\}$. Then $\phi : V \rightarrow B$ and $\phi' : V' \rightarrow B'$ are homeomorphisms. Since B is open (in $\mathbb{R}$ or $\mathbb{H}$) and connected, B' has at most two connected components. Then V' This is a contradiction since V' has four connected components and B' and V' are homeomorphic. $\square$

Exercise 3.1.0.37. Topological Manifold Chart Lemma:

Let M be a set, Γ an index set and for each $\alpha \in \Gamma$, $U_\alpha \subset M$ and $\phi_\alpha : U_\alpha \rightarrow \mathbb{H}^n$. Suppose that

- for each $\alpha \in \Gamma$, $\phi_\alpha(U_\alpha) \in \mathcal{T}_{\mathbb{H}^n}$
- for each $\alpha, \beta \in \Gamma$, $\phi_\alpha(U_\alpha \cap U_\beta) \in \mathcal{T}_{\mathbb{H}^n}$
- for each $\alpha \in \Gamma$, $\phi_\alpha : U_\alpha \rightarrow \phi_\alpha(U_\alpha)$ is a bijection
- for each $\alpha, \beta \in \Gamma$, $\phi_\beta|_{U_\alpha \cap U_\beta} \circ (\phi_\alpha|_{U_\alpha \cap U_\beta})^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \rightarrow \phi_\beta(U_\alpha \cap U_\beta)$ is continuous
- there exists $\Gamma' \subset \Gamma$ such that Γ' is countable and $M \subset \bigcup_{\alpha \in \Gamma'} U_\alpha$
- for each $p, q \in M$, there exists $\alpha \in \Gamma$ such that $p, q \in U_\alpha$ or there exist $\alpha, \beta \in \Gamma$ such that $p \in U_\alpha$, $q \in U_\beta$ and $U_\alpha \cap U_\beta = \emptyset$

Define

- $\mathcal{B} = \{\phi_\alpha^{-1}(V) : V \in \mathcal{T}_{\mathbb{H}^n} \text{ and } \alpha \in \Gamma\}$
- $\mathcal{T}_M = \tau(\mathcal{B})$

Then

1. $\mathcal{B}$ is a basis for $\mathcal{T}_M$
Hint: For $B_1, B_2 \subset \mathbb{H}^n$, $\phi_{\alpha_1}^{-1}(B_1) \cap \phi_{\alpha_2}^{-1}(B_2) = \phi_{\alpha_1}^{-1}(B_1 \cap [\phi_{\alpha_1}|_{U_{\alpha_1} \cap U_{\alpha_2}} \circ (\phi_{\alpha_2}|_{U_{\alpha_1} \cap U_{\alpha_2}})^{-1}(B_2)])$
2. $(M, \mathcal{T}_M)$ is an n -dimensional topological manifold
3. $\mathcal{T}_M$ is the unique topology $\mathcal{T}$ on M such that $(U_\alpha, \phi_\alpha)_{\alpha \in \Gamma} \subset X^n(M, \mathcal{T})$

Proof.

1. • By assumption, $M \subset \bigcup_{\alpha \in \Gamma} U_\alpha$
- Let $A_1, A_2 \in \mathcal{B}$ and $p \in A_1 \cap A_2$. By definition, there exist $\alpha_1, \alpha_2 \in \Gamma$ and $B_1, B_2 \subset \mathbb{H}^n$ such that B_1, B_2 are open in $\mathbb{H}^n$ and

$$\begin{aligned} A_1 &= \phi_{\alpha_1}^{-1}(B_1) & A_2 &= \phi_{\alpha_2}^{-1}(B_2) \\ &\subset U_{\alpha_1} & &\subset U_{\alpha_2} \end{aligned}$$

Set $\psi_1 = \phi_{\alpha_1}|_{U_{\alpha_1} \cap U_{\alpha_2}}$ and $\psi_2 = \phi_{\alpha_2}|_{U_{\alpha_1} \cap U_{\alpha_2}}$. We note that

$$\begin{aligned} \psi_1^{-1}(B_1) &= U_{\alpha_2} \cap \phi_{\alpha_1}^{-1}(B_1) & \psi_2^{-1}(B_2) &= U_{\alpha_1} \cap \phi_{\alpha_2}^{-1}(B_2) \\ &= U_{\alpha_2} \cap A_1 & &= U_{\alpha_1} \cap A_2 \\ &\subset U_{\alpha_1} \cap U_{\alpha_2} & &\subset U_{\alpha_1} \cap U_{\alpha_2} \end{aligned}$$

Let $q \in \phi_{\alpha_1}^{-1}(B_1 \cap [\psi_1 \circ \psi_2^{-1}(B_2)])$. Then $\phi_{\alpha_1}(q) \in B_1 \cap [\psi_1 \circ \psi_2^{-1}(B_2)]$. Hence $\phi_{\alpha_1}(q) \in B_1$ and $\phi_{\alpha_1}(q) \in \psi_1 \circ \psi_2^{-1}(B_2)$. This implies that

$$\begin{aligned} q &\in \phi_{\alpha_1}^{-1}(B_1) \\ &= A_1 \end{aligned}$$

and since $\psi_2^{-1}(B_2) \subset U_{\alpha_1} \cap U_{\alpha_2}$ and $\phi_{\alpha_1} : U_{\alpha_1} \rightarrow \phi_{\alpha_1}(U_{\alpha_1})$ is a bijection, we have that

$$\begin{aligned} q &\in \phi_{\alpha_1}^{-1}(\psi_1 \circ \psi_2^{-1}(B_2)) \\ &= \psi_2^{-1}(B_2) \\ &= U_{\alpha_1} \cap A_2 \end{aligned}$$

Thus

$$\begin{aligned} q &\in A_1 \cap (U_{\alpha_1} \cap A_2) \\ &= A_1 \cap A_2 \end{aligned}$$

Since $q \in \phi_{\alpha_1}^{-1}(B_1 \cap [\psi_1 \circ \psi_2^{-1}(B_2)])$ is arbitrary, we have that $\phi_{\alpha_1}^{-1}(B_1 \cap [\psi_1 \circ \psi_2^{-1}(B_2)]) \subset A_1 \cap A_2$. Conversely, let

$$\begin{aligned} q &\in A_1 \cap A_2 \\ &= \phi_{\alpha_1}^{-1}(B_1) \cap \phi_{\alpha_2}^{-1}(B_2) \end{aligned}$$

Then $\phi_{\alpha_1}(q) \in B_1$ and $\phi_{\alpha_2}(q) \in B_2$. Since $A_1 \cap A_2 \subset U_{\alpha_1} \cap U_{\alpha_2}$, we have that

$$\begin{aligned} \psi_2(q) &= \phi_{\alpha_2}(q) \\ &\in B_2 \end{aligned}$$

which implies that $q \in \psi_2^{-1}(B_2)$. Therefore

$$\begin{aligned} \phi_{\alpha_1}(q) &= \psi_1(q) \\ &\in \psi_1(\psi_2^{-1}(B_2)) \\ &= \psi_1 \circ \psi_2^{-1}(B_2) \end{aligned}$$

Hence $\phi_{\alpha_1}(q) \in B_1 \cap [\psi_1 \circ \psi_2^{-1}(B_2)]$. This implies that $q \in \phi_{\alpha_1}^{-1}(B_1 \cap [\psi_1 \circ \psi_2^{-1}(B_2)])$. Since $q \in A_1 \cap A_2$ is arbitrary, we have that $A_1 \cap A_2 \subset \phi_{\alpha_1}^{-1}(B_1 \cap [\psi_1 \circ \psi_2^{-1}(B_2)])$. Thus

$$\begin{aligned} A_1 \cap A_2 &= \phi_{\alpha_1}^{-1}(B_1 \cap [\psi_1 \circ \psi_2^{-1}(B_2)]) \\ &\in \mathcal{B} \end{aligned}$$

Thus $\mathcal{B}$ is a basis for $\mathcal{T}_M$.

2. (a) **(locally Euclidean of dimension n):**

Let $\alpha \in \Gamma$. By definition, for each $B \subset \mathcal{T}_{\mathbb{H}^n}$,

$$\begin{aligned} \phi_{\alpha}^{-1}(B) &\in \mathcal{B} \\ &\subset \mathcal{T}_M \end{aligned}$$

Hence ϕ_{α} is continuous.

Let $A \in \mathcal{T}_{U_{\alpha}}$. Then there exists $U \subset \mathcal{T}_M$ such that $A = U \cap U_{\alpha}$. Since $\mathcal{B}$ is a basis for $\mathcal{T}_M$, there exists $\Gamma' \subset \Gamma$, $(V_{\beta})_{\beta \in \Gamma'} \subset \mathcal{T}_{\mathbb{H}^n}$ such that $U = \bigcup_{\beta \in \Gamma'} \phi_{\beta}^{-1}(V_{\beta})$. Thus

$$\begin{aligned} A &= U \cap U_{\alpha} \\ &= \left[\bigcup_{\beta \in \Gamma'} \phi_{\beta}^{-1}(V_{\beta}) \right] \cap U_{\alpha} \\ &= \bigcup_{\beta \in \Gamma'} [\phi_{\beta}^{-1}(V_{\beta}) \cap U_{\alpha}] \end{aligned}$$

Let $\beta \in \Gamma'$. Since $\phi_\alpha(U_\alpha \cap U_\beta) \subset \phi_\alpha(U_\alpha)$ and $\phi_\alpha(U_\alpha \cap U_\beta) \in \mathcal{T}_{\mathbb{H}^n}$, we have that

$$\begin{aligned}\phi_\alpha(U_\alpha \cap U_\beta) &= \phi_\alpha(U_\alpha) \cap \phi_\alpha(U_\alpha \cap U_\beta) \\ &\in \mathcal{T}_{\phi_\alpha(U_\alpha)}\end{aligned}$$

Therefore $\mathcal{T}_{\phi_\alpha(U_\alpha \cap U_\beta)} \subset \mathcal{T}_{\phi_\alpha(U_\alpha)}$. Since $(\phi_\beta|_{U_\alpha \cap U_\beta}) \circ (\phi_\alpha|_{U_\alpha \cap U_\beta})^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \rightarrow \phi_\beta(U_\alpha \cap U_\beta)$ is continuous, we have that $(\phi_\beta|_{U_\alpha \cap U_\beta}) \circ (\phi_\alpha|_{U_\alpha \cap U_\beta})^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \rightarrow \mathbb{H}^n$ is continuous and therefore

$$\begin{aligned}[(\phi_\beta|_{U_\alpha \cap U_\beta}) \circ (\phi_\alpha|_{U_\alpha \cap U_\beta})^{-1}]^{-1}(V_\beta) &\in \mathcal{T}_{\phi_\alpha(U_\alpha \cap U_\beta)} \\ &\subset \mathcal{T}_{\phi_\alpha(U_\alpha)}\end{aligned}$$

Since $\beta \in \Gamma'$ is arbitrary, we have that

$$\begin{aligned}\phi_\alpha(A) &= \phi_\alpha\left(\bigcup_{\beta \in \Gamma'} [\phi_\beta^{-1}(V_\beta) \cap U_\alpha]\right) \\ &= \bigcup_{\beta \in \Gamma'} \phi_\alpha(\phi_\beta^{-1}(V_\beta) \cap U_\alpha) \\ &= \bigcup_{\beta \in \Gamma'} (\phi_\alpha|_{U_\alpha \cap U_\beta}) \circ (\phi_\beta|_{U_\alpha \cap U_\beta})^{-1}(V_\beta) \\ &= \bigcup_{\beta \in \Gamma'} [(\phi_\beta|_{U_\alpha \cap U_\beta}) \circ (\phi_\alpha|_{U_\alpha \cap U_\beta})^{-1}]^{-1}(V_\beta) \\ &\in \mathcal{T}_{\phi_\alpha(U_\alpha)}\end{aligned}$$

Since $A \in \mathcal{T}_{U_\alpha}$ is arbitrary, $\phi_\alpha^{-1} : \phi_\alpha(U_\alpha) \rightarrow U_\alpha$ is continuous. Hence $\phi_\alpha : U_\alpha \rightarrow \phi_\alpha(U_\alpha)$ is a homeomorphism and $(U_\alpha, \phi_\alpha) \in X^n(M)$. Since $M = \bigcup_{\alpha \in \Gamma} U_\alpha$, we have that M is locally Euclidean of dimension n .

(b) **(Hausdorff):**

Let $p, q \in M$. Suppose that $p \neq q$. Then there exists $\alpha \in \Gamma$ such that $p, q \in U_\alpha$ or there exist $\alpha, \beta \in \Gamma$ such that $p \in U_\alpha$, $q \in U_\beta$ and $U_\alpha \cap U_\beta = \emptyset$.

- Suppose that there exists $\alpha \in \Gamma$ such that $p, q \in U_\alpha$. Since $p \neq q$, $\phi_\alpha(p) \neq \phi_\alpha(q)$. Since $\mathbb{H}^n$ is Hausdorff, there exist $V_p, V_q \subset \phi(U_\alpha)$ such that V_p and V_q are open in $\mathbb{H}^n$, $p \in V_p$, $q \in V_q$ and $V_p \cap V_q = \emptyset$. Set $U_p = \phi_\alpha^{-1}(V_p)$ and $U_q = \phi_\alpha^{-1}(V_q)$. Then U_p, U_q are open, $p \in U_p$, $q \in U_q$ and $U_p \cap U_q = \emptyset$.
- Suppose that there exist $\alpha, \beta \in \Gamma$ such that $p \in U_\alpha$, $q \in U_\beta$ and $U_\alpha \cap U_\beta = \emptyset$. Set $U_p = U_\alpha$ and $U_q = U_\beta$. Then U_p, U_q are open, $p \in U_p$, $q \in U_q$ and $U_p \cap U_q = \emptyset$.

Thus for each $p, q \in M$ there exist $U_p, U_q \subset M$ such that U_p, U_q are open, $p \in U_p$, $q \in U_q$ and $U_p \cap U_q = \emptyset$. Hence

(c) **(second-countable):**

By assumption, there exists $\Gamma' \subset \Gamma$ such that Γ' is countable and $M \subset \bigcup_{\alpha \in \Gamma'} U_\alpha$. Let $\alpha \in \Gamma'$. Since $\phi_\alpha(U_\alpha) \in \mathcal{T}_{\mathbb{H}^n}$ and $\mathbb{H}^n$ is second-countable, we have that $\phi_\alpha(U_\alpha)$ is second-countable. Since $\phi_\alpha : U_\alpha \rightarrow \phi_\alpha(U_\alpha)$ is a homeomorphism, we have that U_α is second-countable. Since $M = \bigcup_{\alpha \in \Gamma'} U_\alpha$, an exercise in topology [cite](#) implies that M is second-countable.

3. Let $\mathcal{T}$ be a topology on M . Suppose that $(U_\alpha, \phi_\alpha)_{\alpha \in \Gamma} \subset X^n(M, \mathcal{T})$. Then for each $\alpha \in \Gamma$, $U_\alpha \in \mathcal{T}$ and $\phi_\alpha : U_\alpha \rightarrow \phi_\alpha(U_\alpha)$ is a $(\mathcal{T} \cap U_\alpha, \mathcal{T}_{\mathbb{H}^n} \cap \phi_\alpha(U_\alpha))$ -homeomorphism.

Let $U \in \mathcal{B}$. By definition, there exists $\alpha \in \Gamma$ and $V \in \mathcal{T}_{\mathbb{H}^n}$ such that $U = \phi_\alpha^{-1}(V)$. Since $U_\alpha \in \mathcal{T}$, we have that $\mathcal{T} \cap U_\alpha \subset \mathcal{T}$. Since $V \cap \phi_\alpha(U_\alpha) \in \mathcal{T}_{\mathbb{H}^n} \cap \phi_\alpha(U_\alpha)$, and ϕ_α is a $(\mathcal{T} \cap U_\alpha, \mathcal{T}_{\mathbb{H}^n} \cap \phi_\alpha(U_\alpha))$ -homeomorphism, we have that

$$\begin{aligned}U &= \phi_\alpha^{-1}(V) \\ &= \phi_\alpha^{-1}(V \cap \phi_\alpha(U_\alpha)) \\ &\in \mathcal{T} \cap U_\alpha \\ &\subset \mathcal{T}\end{aligned}$$

Since $U \in \mathcal{B}$ is arbitrary, $\mathcal{B} \subset \mathcal{T}$. Therefore

$$\begin{aligned}\mathcal{T}_M &= \tau(\mathcal{B}) \\ &\subset \tau(\mathcal{T}) \\ &= \mathcal{T}\end{aligned}$$

Conversely, Let $U \in \mathcal{T}$ and $\alpha \in \Gamma$. Then $U \cap U_\alpha \in \mathcal{T} \cap U_\alpha$. Since $\phi_\alpha : U_\alpha \rightarrow \phi_\alpha(U_\alpha)$ is a $(\mathcal{T} \cap U_\alpha, \mathcal{T}_{\mathbb{H}^n} \cap \phi_\alpha(U_\alpha))$ -homeomorphism, we have that $\phi_\alpha(U \cap U_\alpha) \in \mathcal{T}_{\mathbb{H}^n} \cap \phi_\alpha(U_\alpha)$. Since $U_\alpha \in \mathcal{T}_M$, $\mathcal{T}_M \cap U_\alpha \subset \mathcal{T}_M$. Since $\phi_\alpha : U_\alpha \rightarrow \phi_\alpha(U_\alpha)$ is a $(\mathcal{T}_M \cap U_\alpha, \mathcal{T}_{\mathbb{H}^n} \cap \phi_\alpha(U_\alpha))$ -homeomorphism, we have that

$$\begin{aligned}U \cap U_\alpha &= \phi_\alpha^{-1}(\phi_\alpha(U \cap U_\alpha)) \\ &\in \mathcal{T}_M \cap U_\alpha \\ &\subset \mathcal{T}_M\end{aligned}$$

Then

$$\begin{aligned}U &= U \cap M \\ &= U \cap \left(\bigcup_{\alpha \in \Gamma} U_\alpha \right) \\ &= \bigcup_{\alpha \in \Gamma} (U \cap U_\alpha) \\ &\in \mathcal{T}_M\end{aligned}$$

Since $U \in \mathcal{T}$ is arbitrary, $\mathcal{T} \subset \mathcal{T}_M$. Thus $\mathcal{T} = \mathcal{T}_M$. □

Exercise 3.1.0.38. Let M be a set, Γ an index set and for each $\alpha \in \Gamma$, $U_\alpha \subset M$ and $\phi_\alpha : U_\alpha \rightarrow \mathbb{H}^n$. Suppose that

- for each $\alpha \in \Gamma$, $\phi_\alpha(U_\alpha) \in \mathcal{T}_{\mathbb{H}^n}$
- for each $\alpha, \beta \in \Gamma$, $\phi_\alpha(U_\alpha \cap U_\beta) \in \mathcal{T}_{\mathbb{H}^n}$
- for each $\alpha \in \Gamma$, $\phi_\alpha : U_\alpha \rightarrow \phi_\alpha(U_\alpha)$ is a bijection
- for each $\alpha, \beta \in \Gamma$, $\phi_\beta|_{U_\alpha \cap U_\beta} \circ (\phi_\alpha|_{U_\alpha \cap U_\beta})^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \rightarrow \phi_\beta(U_\alpha \cap U_\beta)$ is continuous
- there exists $\Gamma' \subset \Gamma$ such that Γ' is countable and $M \subset \bigcup_{\alpha \in \Gamma'} U_\alpha$
- for each $p, q \in M$, there exists $\alpha \in \Gamma$ such that $p, q \in U_\alpha$ or there exist $\alpha, \beta \in \Gamma$ such that $p \in U_\alpha$, $q \in U_\beta$ and $U_\alpha \cap U_\beta = \emptyset$

Then there exists a unique topology $\mathcal{T}_M$ on M such that $(M, \mathcal{T}_M)$ is an n -dimensional topological manifold and $(U_\alpha, \phi_\alpha)_{\alpha \in \Gamma} \subset X^n(M, \mathcal{T}_M)$.

Proof. Immediate by previous exercise. □

3.2 Submanifolds

3.2.1 Open Submanifolds

Note 3.2.1.1. Let $(M, \mathcal{T})$ be an n -dimensional topological manifold and $U \subset M$. Suppose that U is open in M . Unless otherwise specified, we equip U with $\mathcal{T} \cap U$.

Exercise 3.2.1.2. Let M be an n -dimensional topological manifold, $(U, \phi) \in X(M)$ and $U' \subset U$. If U' is open in M , then $(U', \phi|_{U'}) \in X^n(M)$.

Proof. Suppose that U' is open in M . Set $\phi' = \phi|_{U'}$.

- By assumption U' is open in M .
- Since U' is open in M , we have that $U' = U' \cap U$ is open in U . Since ϕ is a homeomorphism and U' is open in U , we have that $\phi(U')$ is open in $\phi(U)$. By assumption $\phi(U)$ is open in $\mathbb{R}^n$ or $\phi(U)$ is open in $\mathbb{H}^n$. Therefore $\phi'(U')$ is open in $\mathbb{R}^n$ or $\phi'(U')$ is open in $\mathbb{H}^n$.
- Since $\phi : U \rightarrow V$ is a homeomorphism, $\phi' : U' \rightarrow \phi'(U')$ is a homeomorphism.

So $(U', \phi') \in X^n(M)$. □

Note 3.2.1.3. Since U is open in M , U' being open in U is equivalent to U' being open in M , so we could have also assumed that U' is open in U .

Exercise 3.2.1.4. Let M be an n -dimensional topological manifold and $U \subset M$. If U is open, then

$$X^n(U) = \{(V, \psi) \in X^n(M) : V \subset U\}$$

Proof. Suppose that U is open and set $A = \{(V, \psi) \in X^n(M) : V \subset U\}$. Let $(V, \psi) \in X^n(U)$. By definition of $X^n(U)$, V is open in U . Thus, there exists $W \subset M$ such that W is open in M and $V = U \cap W$. Since U is open in M , we have that $V = U \cap W$ is open in M . Hence $(V, \psi) \in X^n(M)$ which implies that $(V, \psi) \in A$. Since $(V, \psi) \in X^n(U)$ is arbitrary, $X^n(U) \subset A$. Conversely, suppose that $(V, \psi) \in A$. Then $(V, \psi) \in X^n(M)$ and $V \subset U$. By definition of $X^n(M)$, V is open in M . Since $V \subset U$, we have that $V = V \cap U$ is open in U . Hence $(V, \psi) \in X^n(U)$. Since $(V, \psi) \in X^n(U)$ is arbitrary, $A \subset X^n(U)$. Hence $X^n(A) = A$. □

Exercise 3.2.1.5. Let M be an n -dimensional topological manifold, $(U, \phi) \in X(M)$ and $U' \subset U$. If U' is open in M , then $(U', \phi|_{U'}) \in X^n(U)$.

Proof. Suppose that U' is open in M . A previous exercise implies that $(U', \phi') \in X^n(M)$. The previous exercise implies that $(U', \phi') \in X^n(U)$. □

Exercise 3.2.1.6. Topological Open Submanifolds:

Let M be an n -dimensional topological manifold and $U \subset M$ open. Then U is an n -dimensional topological manifold.

Proof.

1. Since M is Hausdorff, U is Hausdorff.
2. Since M is second-countable, U is second countable.
3. Let $p \in U$. Since then there exists $(V, \psi) \in X^n(M)$ such that $p \in V$. Set $V' = U \cap V$ and $\psi' = \psi|_{U \cap V}$. The previous exercise implies that $(V', \psi') \in X^n(U)$. Therefore U is locally Euclidean of dimension n .

Hence U is an n -dimensional topological manifold. □

Exercise 3.2.1.7. Let M be an n -dimensional topological manifold and $U \subset M$. If U is open, then

1. $X_{\text{Int}}(U) = \{(V, \psi) \in X_{\text{Int}}(M) : V \subset U\}$
2. $X_{\partial}(U) = \{(V, \psi) \in X_{\partial}(M) : V \subset U\}$

Proof. Suppose that U is open in M .

1. Set $A = \{(V, \psi) \in X_{\text{Int}}(M) : V \subset U\}$. Let $(V, \psi) \in X_{\text{Int}}(U)$. By definition of $X_{\text{Int}}(U)$, V is open in U and $\phi(V)$ is open in $\mathbb{R}^n$. Since U is open in M , V is open in M . Hence $(V, \psi) \in X_{\text{Int}}(M)$. Since U is open in M , V is open in M . Hence $(V, \psi) \in X_{\text{Int}}(M)$ which implies that $(V, \psi) \in A$. Since $(V, \psi) \in X_{\text{Int}}(U)$ is arbitrary, $X_{\text{Int}}(U) \subset A$. Conversely, let $(V, \psi) \in A$. Then $(V, \psi) \in X_{\text{Int}}(M)$ and $V \subset U$. By definition of $X_{\text{Int}}(M)$, V is open in M and $\phi(V)$ is open in $\mathbb{R}^n$. Thus $V = V \cap U$ is open in U . So $(V, \psi) \in X_{\text{Int}}(U)$. Since $(V, \psi) \in A$ is arbitrary, $A \subset X_{\text{Int}}(U)$. Thus $X_{\text{Int}}(U) = A$.
2. Set $B = \{(V, \psi) \in X_{\partial}(M) : V \subset U\}$. Let $(V, \psi) \in X_{\partial}(U)$. By definition of $X_{\partial}(U)$, V is open in U , $\phi(V)$ is open in $\mathbb{H}^n$ and $\partial\mathbb{H}_j^n \cap \phi(V) \neq \emptyset$. Since U is open in M , V is open in M . Hence $(V, \psi) \in X_{\partial}(M)$, which implies that $(V, \psi) \in B$. Since $(V, \psi) \in X_{\partial}(U)$ is arbitrary, $X_{\partial}(U) \subset B$. Conversely, let $(V, \psi) \in B$. Then $(V, \psi) \in X_{\partial}(M)$ and $V \subset U$. By definition of $X_{\partial}(M)$, V is open in M , $\phi(V)$ is open in $\mathbb{H}^n$ and $\partial\mathbb{H}_j^n \cap \phi(V) \neq \emptyset$. Thus $V = V \cap U$ is open in U . So $(V, \psi) \in X_{\partial}(U)$. Since $(V, \psi) \in B$ is arbitrary, $B \subset X_{\partial}(U)$. Thus $X_{\partial}(U) = B$.

□

Exercise 3.2.1.8. Let M be an n -dimensional topological manifold and $U \subset M$. If U is open, then $\partial U = \partial M \cap U$.

Proof. Suppose that U is open. Let $p \in \partial U$. Then there exists $(V, \psi) \in X_{\partial}(U)$ such that $p \in V$ and $\psi(p) \in \partial\mathbb{H}^n$. Since U is open, the previous exercise implies that $(V, \psi) \in X_{\partial}(M)$. Thus $p \in \partial M$. Since $p \in \partial U$ is arbitrary, $\partial U \subset \partial M$. Since $\partial U \subset U$, we have that $\partial U \subset \partial M \cap U$.

Conversely, let $p \in \partial M \cap U$. Since $p \in \partial M$, there exists $(V, \psi) \in X_{\partial}(M)$ such that $p \in V$ and $\psi(p) \in \partial\mathbb{H}^n$. Set $V' = V \cap U$ and $\psi' = \psi|_{V'}$. Then $p \in V'$ since V and U are open in M , V' is open in M . A previous exercise implies that $(V', \psi') \in X(M)$. Since $p \in \partial M$, a previous exercise implies that $(V', \psi') \in X_{\partial}(M)$. The previous exercise implies that $(V', \psi') \in X_{\partial}(U)$. Since $\psi'(p) \in \partial\mathbb{H}^n$, $p \in \partial U$. Since $p \in \partial M \cap U$ is arbitrary, $\partial M \cap U \subset \partial U$. Hence $\partial U = \partial M \cap U$.

label exercises and reference them!!!

□

3.2.2 Boundary Submanifolds

Note 3.2.2.1. Let $(M, \mathcal{T})$ be an n -dimensional topological manifold. Unless otherwise specified, we equip ∂M with $\mathcal{T} \cap \partial M$.

Definition 3.2.2.2. Let M be an n -dimensional topological manifold and $\pi : \partial\mathbb{H}_j^n \rightarrow \mathbb{R}^{n-1}$ the projection map. For $(U, \phi) \in X_{\partial}(M)$, we define $\bar{U} \subset \partial M$ and $\bar{\phi} : \bar{U} \rightarrow \pi(\phi(\bar{U}))$ by $\bar{U} = U \cap \partial M$ and $\bar{\phi} = \pi \circ \phi|_{\bar{U}}$ respectively.

Exercise 3.2.2.3. Let M be an n -dimensional topological manifold, and $\lambda : \partial\mathbb{H}_j^n \rightarrow \mathbb{R}^{n-1}$ a homeomorphism. Then $\{(\bar{U}, \bar{\phi}) : (U, \phi) \in X_{\partial}(M)\} \subset X_{\text{Int}}^{n-1}(\partial M)$.

Proof. Let $(U, \phi) \in X_{\partial}(M)$.

1. Since U is open in M , $\bar{U} = U \cap \partial M$ is open in ∂M .
2. Since $(U, \phi) \in X_{\partial}(M)$, $\phi(U)$ is open in $\mathbb{H}^n$. A previous exercise implies that $\phi(\bar{U}) = \phi(U) \cap \partial\mathbb{H}^n$ which is open in $\partial\mathbb{H}^n$. Since $\pi : \partial\mathbb{H}_j^n \rightarrow \mathbb{R}^{n-1}$ is a homeomorphism, we have that $\pi(\phi(\bar{U}))$ is open in $\mathbb{R}^{n-1}$.
3. Since $\phi|_{\bar{U}} : \bar{U} \rightarrow \phi(U) \cap \partial\mathbb{H}^n$ and $\pi|_{\phi(\bar{U})} : \phi(\bar{U}) \rightarrow \pi(\phi(\bar{U}))$ are homeomorphisms, we have that $\bar{\phi} = \pi|_{\phi(\bar{U})} \circ \phi|_{\bar{U}}$ is a homeomorphism.

Hence $(\bar{U}, \bar{\phi}) \in X_{\text{Int}}^{n-1}(\partial M)$.

□

Exercise 3.2.2.4. Topological Boundary Submanifold:

Let M be an n -dimensional topological manifold. Then

1. ∂M is an $(n-1)$ -dimensional topological manifold
2. $\partial(\partial M) = \emptyset$

Proof.

1. (a) Since M is Hausdorff, ∂M is Hausdorff.
- (b) Since M is second-countable, ∂M is second countable.

- (c) Let $p \in \partial M$. Then there exists $(U, \phi) \in X_{\partial}(M)$ such that $\phi(p) \in \partial \mathbb{H}^n$. Then $p \in \bar{U}$ and the previous exercise implies that $(\bar{U}, \bar{\phi}) \in X_{\text{Int}}^{n-1}(\partial M)$. Thus ∂M is locally Euclidean of dimension $n - 1$.

Hence ∂M is an $(n - 1)$ -dimensional topological manifold.

2. Let $p \in \partial M$. Part (1) implies that there exists $(U, \phi) \in X_{\text{Int}}^{n-1}(\partial M)$ such that $p \in U$. Thus $p \in \text{Int } \partial M$. Since $p \in \partial M$ is arbitrary, $\text{Int } \partial M = \partial M$. Hence

$$\begin{aligned} \partial(\partial M) &= (\text{Int}(\partial M))^c \\ &= (\partial M)^c \\ &= \emptyset \end{aligned}$$

□

3.2.3 Embedded Submanifolds

Definition 3.2.3.1. Let $M, N \in \text{Obj}(\mathbf{Man}^0)$ and $F \in \text{Hom}_{\mathbf{Top}}(N, M)$. Then F is said to be a **Top**-embedding if $F|^{F(M)} \in \text{Iso}_{\mathbf{Top}}(M, F(M))$. **here $F|^{F(M)} : M \rightarrow F(M)$ is the restriction on the codomain, make sure to use standardized notation if it changes.**

Exercise 3.2.3.2. Let $M, N \in \text{Obj}(\mathbf{Man}^0)$ and $F \in \text{Hom}_{\mathbf{Top}}(N, M)$. Define

$$F_*X^n(N, \mathcal{T}_N) := \{(F(V), \psi \circ F|_V^{-1}) : (V, \psi) \in X^n(N, \mathcal{T}_N)\}.$$

If F is a **Top**-embedding, then

1. $F_*X^n(N, \mathcal{T}_N) \subset X^n(F(N), \mathcal{T}_M \cap F(N))$.
2. $(F(N), \mathcal{T}_M \cap F(N)) \in \text{Obj}(\mathbf{Man}^0)$.

Proof. Suppose that F is a **Top**-embedding. Set $n := \dim N$. Since F is a **Top**-embedding, $F \in \text{Iso}_{\mathbf{Top}}((N, \mathcal{T}_N), (F(N), \mathcal{T}_M \cap F(N)))$.

1. Let $(U, \phi) \in F_*X^n(N, \mathcal{T}_N)$. Then there exists $(V, \psi) \in \mathcal{A}_N$ such that $U = F(V)$ and $\phi = \psi \circ F|_V^{-1}$. Since $(V, \psi) \in \mathcal{A}_N$ and $\mathcal{A}_N \subset X^n(N, \mathcal{T}_N)$, we have that (V, ψ) is an $\mathbb{R}^n$ -coordinate chart on $(N, \mathcal{T}_N)$ or there exists $j \in [n]$ such that (V, ψ) is an $\mathbb{H}_j^n$ -coordinate chart on $(N, \mathcal{T}_N)$.

- Suppose that (V, ψ) is an $\mathbb{R}^n$ -coordinate chart on $(N, \mathcal{T}_N)$.
 - Since $V \in \mathcal{T}_N$, we have that

$$\begin{aligned} U &= F(V) \\ &\in \mathcal{T}_M \cap F(N). \end{aligned}$$

- Since $F \in \text{Iso}_{\mathbf{Top}}((N, \mathcal{T}_N), (F(N), \mathcal{T}_M \cap F(N)))$, we have that

$$\begin{aligned} F|_V &\in \text{Iso}_{\mathbf{Top}}((V, \mathcal{T}_N \cap V), (F(V), [\mathcal{T}_M \cap F(N)] \cap F(V))) \\ &= \text{Iso}_{\mathbf{Top}}((V, \mathcal{T}_N \cap V), (F(V), \mathcal{T}_M \cap F(V))). \end{aligned}$$

Since (V, ψ) is an $\mathbb{R}^n$ -coordinate chart on $(N, \mathcal{T}_N)$, we have that $\psi(V) \in \mathcal{T}_{\mathbb{R}^n}$. Thus

$$\begin{aligned} \phi(U) &= \psi \circ F|_V^{-1}(F(V)) \\ &= \psi(V) \\ &\in \mathcal{T}_{\mathbb{R}^n}. \end{aligned}$$

- Since $\psi \in \text{Iso}_{\mathbf{Top}}((V, \mathcal{T}_N \cap V), (\psi(V), \mathcal{T}_{\mathbb{R}^n} \cap \psi(V)))$, and $F|_V^{-1} \in \text{Iso}_{\mathbf{Top}}((F(V), \mathcal{T}_M \cap F(V)), (V, \mathcal{T}_N \cap V))$, we have that $\psi \circ F|_V^{-1} \in \text{Iso}_{\mathbf{Top}}((F(V), \mathcal{T}_M \cap F(V)), (\psi(V), \mathcal{T}_{\mathbb{R}^n} \cap \psi(V)))$.

Hence $(U, \phi) \in X^n(F(N), \mathcal{T}_M \cap F(N))$.

- Similarly, if there exists $j \in [n]$ such that (V, ψ) is an $\mathbb{H}_j^n$ -coordinate chart on $(N, \mathcal{T}_N)$, then $(U, \phi) \in X^n(F(N), \mathcal{T}_M \cap F(N))$.

Since $(U, \phi) \in F_*X^n(N, \mathcal{T}_N)$ is arbitrary, we have that $F_*X^n(N, \mathcal{T}_N) \subset X^n(F(N), \mathcal{T}_M \cap F(N))$.

2. (a) Since $F \in \text{Iso}_{\mathbf{Top}}((N, \mathcal{T}_N), (F(N), \mathcal{T}_M \cap F(N)))$ and $(N, \mathcal{T}_N)$ is Hausdorff, $(F(N), \mathcal{T}_M \cap F(N))$ is Hausdorff.
- (b) Since $F \in \text{Iso}_{\mathbf{Top}}((N, \mathcal{T}_N), (F(N), \mathcal{T}_M \cap F(N)))$ and $(N, \mathcal{T}_N)$ is second-countable, $(F(N), \mathcal{T}_M \cap F(N))$ is second-countable.
- (c) Let $p \in F(N)$. Then there exists $q \in N$ such that $F(q) = p$. Since $N \in \text{Obj}(\mathbf{Man}^0)$, there exists $(V, \psi) \in X^n(N, \mathcal{T}_N)$ such that $q \in V$. Define $(U, \phi) \in F_*X^n(N, \mathcal{T}_N)$ by $U := F(V)$ and $\phi := \psi \circ F|_V^{-1}$. By definition, $(U, \phi) \in F_*X^n(N, \mathcal{T}_N)$. Furthermore,

$$\begin{aligned} p &= F(q) \\ &\in F(V) \\ &= U. \end{aligned}$$

Since $p \in F(N)$ is arbitrary, we have that for each $p \in F(N)$, there exists $(U, \phi) \in F_*X^n(N, \mathcal{T}_N)$ such that $p \in U$. Hence $(F(N), \mathcal{T}_M \cap F(N))$ is locally Euclidean of dimension n .

Thus $(F(N), \mathcal{T}_M \cap F(N)) \in \text{Obj}(\mathbf{Man}^0)$.

□

3.3 Product Manifolds

Note 3.3.0.1. Let $(M, \mathcal{T}_M)$ and $(N, \mathcal{T}_N)$ be m -dimensional and n -dimensional topological manifold respectively. Unless otherwise specified, we equip $M \times N$ with $\mathcal{T}_M \otimes \mathcal{T}_N$.

Definition 3.3.0.2. Let $m \in \mathbb{N}$ and $n \in \mathbb{N}_0$. Define $\lambda_0 : \mathbb{H}_j^m \times \text{Int } \mathbb{H}_j^n \rightarrow \mathbb{H}^{m+n}$ by $\lambda((x^1, \dots, x^{m-1}, x^m), (y^1, \dots, y^n)) := (x^1, \dots, x^{m-1}, y^1, \dots, y^n, \log y^n, x^m)$.

Exercise 3.3.0.3. Let $m \in \mathbb{N}$ and $n \in \mathbb{N}_0$. Then

1. λ_0 is a $(\mathcal{T}_{\mathbb{H}^m} \otimes \mathcal{T}_{\text{Int } \mathbb{H}^n}, \mathcal{T}_{\mathbb{H}^{m+n}})$ -homeomorphism,
2. $\lambda_0(\partial \mathbb{H}^m \times \text{Int } \mathbb{H}^n) = \partial \mathbb{H}^{m+n}$,
3. $(\mathbb{H}^m \times \text{Int } \mathbb{H}^n, \lambda_0) \in X^{m+n}(\mathbb{H}^m \times \text{Int } \mathbb{H}^n, \mathcal{T}_{\mathbb{H}^n} \otimes \mathcal{T}_{\text{Int } \mathbb{H}^n})$.

Proof.

1. Clearly λ_0 is a homeomorphism.
2. Clearly $\lambda_0(\partial \mathbb{H}^m \times \text{Int } \mathbb{H}^n) = \partial \mathbb{H}^{m+n}$
3. We note that

- $\mathbb{H}^m \times \text{Int } \mathbb{H}^n \in \mathcal{T}_{\mathbb{H}^n} \otimes \mathcal{T}_{\text{Int } \mathbb{H}^n}$,
- $\mathbb{H}^{m+n} \in \mathcal{T}_{\mathbb{H}^{m+n}}$,
- part (1) implies that λ_0 is a $(\mathcal{T}_{\mathbb{H}^m} \otimes \mathcal{T}_{\text{Int } \mathbb{H}^n}, \mathcal{T}_{\mathbb{H}^{m+n}})$ -homeomorphism.

Thus $(\mathbb{H}^m \times \text{Int } \mathbb{H}^n, \lambda_0) \in X^{m+n}(\mathbb{H}^m \times \mathbb{H}^n, \mathcal{T}_{\mathbb{H}^n} \otimes \mathcal{T}_{\text{Int } \mathbb{H}^n})$.

□

Exercise 3.3.0.4. Let $m, n \in \mathbb{N}_0$. Then $\mathbb{H}^m \times \text{Int } \mathbb{H}^n$ is an $m + n$ -dimensional topological manifold.

Proof.

1. Clearly $\mathbb{H}^m \times \text{Int } \mathbb{H}^n$ is Hausdorff.
2. Clearly $\mathbb{H}^m \times \text{Int } \mathbb{H}^n$ is second-countable.
3. Since $\lambda_0 \in X^{m+n}(\mathbb{H}^m \times \mathbb{H}^n, \mathcal{T}_{\mathbb{H}^n} \otimes \mathcal{T}_{\text{Int } \mathbb{H}^n})$, we have that for each $p \in \mathbb{H}^m \times \text{Int } \mathbb{H}^n$, there exists $(U, \phi) \in X^{m+n}(\mathbb{H}^m \times \text{Int } \mathbb{H}^n, \mathcal{T}_{\mathbb{H}^n} \otimes \mathcal{T}_{\text{Int } \mathbb{H}^n})$ such that $p \in U$. Thus $(\mathbb{H}^m \times \mathbb{H}^n, \mathcal{T}_{\mathbb{H}^n} \otimes \mathcal{T}_{\text{Int } \mathbb{H}^n})$ is locally Euclidean of dimension $m + n$.

Thus $(\mathbb{H}^m \times \mathbb{H}^n, \mathcal{T}_{\mathbb{H}^n} \otimes \mathcal{T}_{\text{Int } \mathbb{H}^n})$ is an $m + n$ -dimensional topological manifold.

□

Exercise 3.3.0.5. Let $(M, \mathcal{T}_M)$, $(N, \mathcal{T}_N)$ be topological manifolds. Set $m = \dim M$ and $n = \dim N$. Suppose that $\partial N = \emptyset$. Then for each $(U, \phi) \in X^m(M, \mathcal{T}_M)$, $(V, \psi) \in X^n(N, \mathcal{T}_N)$,

$$(U \times V, \lambda_0|_{\phi(U) \times \psi(V)} \circ [\phi \times \psi]) \in X^{m+n}(M \times N, \mathcal{T}_M \otimes \mathcal{T}_N)$$

Proof. Let $(U, \phi) \in X^m(M, \mathcal{T}_M)$ and $(V, \psi) \in X^n(N, \mathcal{T}_N)$.

- Since $U \in \mathcal{T}_M$ and $V \in \mathcal{T}_N$, $U \times V \in \mathcal{T}_M \otimes \mathcal{T}_N$.
- Since $\phi(U) \in \mathcal{T}_{\mathbb{H}^m}$ and $\psi(V) \in \mathcal{T}_{\mathbb{H}^n}$, $\phi(U) \times \psi(V) \in \mathcal{T}_{\mathbb{H}^m} \otimes \mathcal{T}_{\mathbb{H}^n}$. Since $\partial N = \emptyset$, $(V, \psi) \in X_{\text{Int}}^n(N, \mathcal{T}_N)$ and therefore $\psi(V) \subset \text{Int } \mathbb{H}^n$. Since $\lambda_0 : \mathbb{H}^m \times \text{Int } \mathbb{H}^n \rightarrow \mathbb{H}^{m+n}$ is a homeomorphism,

$$\begin{aligned} \lambda_0|_{\phi(U) \times \psi(V)} \circ [\phi \times \psi](U \times V) &= \lambda_0(\phi(U) \times \psi(V)) \\ &\in \mathcal{T}_{\mathbb{H}^{m+n}} \end{aligned}$$

- Since $\phi : U \rightarrow \phi(U)$ is a $(\mathcal{T}_M \cap U, \mathcal{T}_{\mathbb{H}^m} \cap \phi(U))$ -homeomorphism and $\psi : V \rightarrow \psi(V)$ is a $(\mathcal{T}_N \cap V, \mathcal{T}_{\mathbb{H}^n} \cap \psi(V))$ -homeomorphism, [an exercise in the section on product topologies in the analysis notes](#) implies that $\phi \times \psi : U \times V \rightarrow \phi(U) \times \psi(V)$ is a $([\mathcal{T}_M \otimes \mathcal{T}_N] \cap [U \times V], [\mathcal{T}_{\mathbb{H}^m} \otimes \mathcal{T}_{\mathbb{H}^n}] \cap [\phi(U) \times \psi(V)])$ -homeomorphism. Since $\lambda_0|_{\phi(U) \times \psi(V)} : \phi(U) \times \psi(V) \rightarrow \lambda_0(\phi(U) \times \psi(V))$ is a $([\mathcal{T}_{\mathbb{H}^m} \otimes \mathcal{T}_{\text{Int } \mathbb{H}^n}] \cap [\phi(U) \times \psi(V)], \mathcal{T}_{\mathbb{H}^{m+n}} \cap \lambda_0(\phi(U) \times \psi(V)))$ -homeomorphism, $\lambda_0|_{\phi(U) \times \psi(V)} \circ (\phi \times \psi)$ is a $([\mathcal{T}_M \otimes \mathcal{T}_N] \cap [U \times V], \mathcal{T}_{\mathbb{H}^{m+n}} \cap \lambda_0(U \times V))$ -homeomorphism.

Hence $(U \times V, \lambda_0|_{\phi(U) \times \psi(V)} \circ [\phi \times \psi]) \in X^{m+n}(M \times N, \mathcal{T}_M \otimes \mathcal{T}_N)$. Since $(U, \phi) \in X^m(M, \mathcal{T}_M)$ and $(V, \psi) \in X^n(N, \mathcal{T}_N)$ are arbitrary, we have that for each $(U, \phi) \in X^m(M, \mathcal{T}_M)$ and $(V, \psi) \in X^n(N, \mathcal{T}_N)$,

$$(U \times V, \lambda_0|_{\phi(U) \times \psi(V)} \circ [\phi \times \psi]) \in X^{m+n}(M \times N, \mathcal{T}_M \otimes \mathcal{T}_N)$$

□

Exercise 3.3.0.6. Let M, N be topological manifolds. Set $m = \dim M$ and $n = \dim N$. Suppose that $\partial N = \emptyset$. Then for each $(U, \phi) \in X_{\partial}^m(M, \mathcal{T}_M)$, $(V, \psi) \in X^n(N, \mathcal{T}_N)$,

$$(U \times V, \lambda_0|_{\phi(U) \times \psi(V)} \circ [\phi \times \psi]) \in X_{\partial}^{m+n}(M \times N, \mathcal{T}_M \otimes \mathcal{T}_N)$$

Proof. Let $(U, \phi) \in X_{\partial}^m(M)$ and $(V, \psi) \in X^n(N)$. Define $\eta : U \times V \rightarrow \lambda_0(\phi(U) \times \psi(V))$ by

$$\eta := \lambda_0|_{\phi(U) \times \psi(V)} \circ [\phi \times \psi]$$

Since $(U, \phi) \in X_{\partial}^m(M)$, $\phi(U) \cap \partial \mathbb{H}^m \neq \emptyset$. Then there exists $p \in U$ such that $\phi(p) \in \partial \mathbb{H}^m$. So $\eta(p, q) \in \partial \mathbb{H}^{m+n}$. Thus $\eta(U \times V) \cap \partial \mathbb{H}^{m+n} \neq \emptyset$ and $(U \times V, \eta) \in X_{\partial}^{m+n}(M \times N)$. Since $(U, \phi) \in X_{\partial}^m(M)$ and $(V, \psi) \in X^n(N, \mathcal{T}_N)$ are arbitrary, we have that for each $(U, \phi) \in X_{\partial}^m(M, \mathcal{T}_M)$ and $(V, \psi) \in X^n(N, \mathcal{T}_N)$,

$$(U \times V, \lambda_0|_{\phi(U) \times \psi(V)} \circ [\phi \times \psi]) \in X_{\partial}^{m+n}(M \times N, \mathcal{T}_M \otimes \mathcal{T}_N)$$

□

Note 3.3.0.7. The above is still true if $\partial N \neq \emptyset$

Exercise 3.3.0.8. Let M, N be topological manifolds. Suppose that $\partial N = \emptyset$. Then

1. $M \times N$ is a topological manifold
2. $\partial(M \times N) = \partial M \times N$

Proof. Set $m = \dim M$ and $n = \dim N$.

1.
 - Since M and N are Hausdorff, $M \times N$ is Hausdorff.
 - Since M and N are second-countable, $M \times N$ is second-countable.
 - Let $a \in M \times N$. Then there exist $p \in M$ and $q \in N$ such that $a = (p, q)$. Since M and N are locally Euclidean, there exist $(U, \phi) \in X^m(M)$ and $(V, \psi) \in X^n(N)$ such that $p \in U$ and $q \in V$. Then $(p, q) \in U \times V$. Exercise 3.3.0.5 implies that $(U \times V, \lambda_0 \circ [\phi \times \psi]) \in X^{m+n}(M \times N)$. Since $a \in M \times N$ is arbitrary, $M \times N$ is locally Euclidean of dimension $m + n$.

Thus $M \times N$ is an $(m + n)$ -dimensional topological manifold.

2.
 - Let $a \in \partial(M \times N)$. Then there exists $p \in M$ and $q \in N$ such that $a = (p, q)$. Since $(M, \mathcal{T}_M)$ and (N) are locally Euclidean, there exist $(U, \phi) \in X^m(M)$ and $(V, \psi) \in X^n(N)$ such that $p \in U$ and $q \in V$. Define $\eta : U \times V \rightarrow \lambda_0(\phi(U) \times \psi(V))$ by

$$\eta := \lambda_0|_{\phi(U) \times \psi(V)} \circ [\phi \times \psi]$$

Exercise 3.3.0.5 implies that $\eta \in X^{m+n}(M \times N)$. Since $(p, q) \in \partial(M \times N)$, Exercise 3.3.0.6 implies that $\eta \in X_{\partial}^{m+n}(M \times N)$ and $\eta(p, q) \in \partial \mathbb{H}^{m+n}$. Therefore

$$\begin{aligned} \phi \times \psi(p, q) &= \lambda_0|_{\phi(U) \times \psi(V)}^{-1} \circ \eta \\ &\in \partial \mathbb{H}^m \times \text{Int } \mathbb{H}^n \end{aligned}$$

Hence $\phi(p) \in \partial \mathbb{H}^m$ and $\psi(q) \in \text{Int } \mathbb{H}^n$. Thus $(U, \phi) \in X_{\partial}^m(M)$ and $p \in \partial M$. Therefore

$$\begin{aligned} a &= (p, q) \\ &\in \partial M \times N \end{aligned}$$

Since $a \in \partial(M \times N)$ is arbitrary, we have that $\partial(M \times N) \subset \partial M \times N$.

- Let $a \in \partial M \times N$. Then there exists $p \in \partial M$ and $q \in N$ such that $a = (p, q)$. By definition, there exists $(U, \phi) \in X_{\partial}^m(M)$ and $(V, \psi) \in X^n(N)$ such that $p \in U$, $q \in V$ and $\phi(p) \in \partial \mathbb{H}^m$. Since $\partial N = \emptyset$, $\psi(q) \in \text{Int } \mathbb{H}^n$. Define $\eta : U \times V \rightarrow \lambda_0(\phi(U) \times \psi(V))$ by

$$\eta := \lambda_0|_{\phi(U) \times \psi(V)} \circ [\phi \times \psi]$$

Exercise 3.3.0.5 implies that $(U \times V, \eta) \in X^{m+n}(M \times N, \mathcal{T}_M \otimes \mathcal{T}_N)$. Then

$$\begin{aligned} \eta(a) &= \eta(p, q) \\ &= \lambda_0(\phi(p), \psi(q)) \\ &\in \partial \mathbb{H}^{m+n} \end{aligned}$$

Thus $\eta \in X_{\partial}^{m+n}(M \times N)$ and $a \in \partial(M \times N)$. Since $a \in \partial M \times N$ is arbitrary, $\partial M \times N \subset \partial(M \times N)$.

Thus $\partial(M \times N) = \partial M \times N$.

□

3.4 Submanifolds

Definition 3.4.0.1. *topological embedding*

Definition 3.4.0.2. Let M, N be topological manifolds of dimensions m, n respectively and $F : N \rightarrow M$ a topological embedding. Then $\{(F(V), \psi \circ F^{-1}) : (V, \psi) \in X^n(N)\} \subset X^n(M)$.

Proof. Since

□

Chapter 4

Smooth Manifolds

use smooth manifold chart lemma to show that $\mathbb{H}^n$, $\text{Int } \mathbb{H}^n$ and $\mathbb{H}^m \times \text{Int } \mathbb{H}^n$ are smooth manifolds.

4.1 Introduction

Definition 4.1.0.1. Let M be an n -dimensional topological manifold and $(U, \phi), (V, \psi) \in X(M)$. Then (U, ϕ) and (V, ψ) are said to be **smoothly compatible** if

$$\psi|_{U \cap V} \circ (\phi|_{U \cap V})^{-1} : \phi(U \cap V) \rightarrow \psi(U \cap V) \text{ is a diffeomorphism}$$

Definition 4.1.0.2. Let $(M, \mathcal{T})$ be an n -dimensional topological manifold.

- Let $\mathcal{A} \subset X^n(M, \mathcal{T})$. Then $\mathcal{A}$ is said to be an **atlas on M** if $M \subset \bigcup_{(U, \phi) \in \mathcal{A}} U$.
- Let $\mathcal{A}$ be an atlas on M . Then $\mathcal{A}$ is said to be **smooth** if for each $(U, \phi), (V, \psi) \in \mathcal{A}$, (U, ϕ) and (V, ψ) are smoothly compatible.
- Let $\mathcal{A}$ be a smooth atlas on M . Then $\mathcal{A}$ is said to be **maximal** if for each smooth atlas $\mathcal{B}$ on M , $\mathcal{A} \subset \mathcal{B}$ implies that $\mathcal{A} = \mathcal{B}$. A maximal smooth atlas on M is called a **smooth structure on M** .
- Let $\mathcal{A}$ be an atlas on M . Then $(M, \mathcal{T}, \mathcal{A})$ is said to be an **n -dimensional smooth manifold** if $\mathcal{A}$ is a smooth structure on M .

Note 4.1.0.3. When the context is clear, we write M or $(M, \mathcal{A})$ in place of $(M, \mathcal{T}, \mathcal{A})$.

Definition 4.1.0.4. Let M be a topological manifold and $\mathcal{B}$ a smooth atlas on M . We define the **smooth structure on M generated by $\mathcal{B}$** , denoted $\alpha_M(\mathcal{B})$, by

$$\alpha_M(\mathcal{B}) = \{(U, \phi) \in X(M) : \text{for each } (V, \psi) \in \mathcal{B}, (U, \phi) \text{ and } (V, \psi) \text{ are smoothly compatible}\}$$

Note 4.1.0.5. When the context is clear, we write $\alpha(\mathcal{B})$ in place of $\alpha_M(\mathcal{B})$.

Exercise 4.1.0.6. Let M be an n -dimensional topological manifold and $\mathcal{B}$ a smooth atlas on M . Then $\alpha(\mathcal{B})$ is the unique smooth structure $\mathcal{A}$ on M such that $\mathcal{B} \subset \mathcal{A}$.

Proof. Clearly $\mathcal{B} \subset \alpha(\mathcal{B})$. Let (U, ϕ) and $(V, \psi) \in \alpha(\mathcal{B})$. Define $F : \phi(U \cap V) \rightarrow \psi(U \cap V)$ by

$$F = \psi|_{U \cap V} \circ (\phi|_{U \cap V})^{-1}$$

Let $q \in \phi(U \cap V)$. Set $p = \phi^{-1}(q)$. Since $\mathcal{B}$ is an atlas and $p \in U \cap V \subset M$, there exists $(W, \chi) \in \mathcal{B}$ such that $p \in W$. By definition of $\alpha(\mathcal{B})$, $\psi|_{W \cap V} \circ (\chi|_{W \cap V})^{-1} : \chi(W \cap V) \rightarrow \psi(W \cap V)$ and $\chi|_{U \cap W} \circ (\phi|_{U \cap W})^{-1} : \phi(U \cap W) \rightarrow \chi(U \cap W)$ are diffeomorphisms. Set $N = U \cap W \cap V$. Then $q \in \phi(N) \subset \phi(U \cap V)$ and

$$\begin{aligned} F|_{\phi(N)} &= \psi|_N \circ (\phi|_N)^{-1} \\ &= [\psi|_N \circ (\chi|_N)^{-1}] \circ [\chi|_N \circ (\phi|_N)^{-1}] \end{aligned}$$

is a diffeomorphism. Thus, for each $q \in \phi(U \cap V)$, there exists $N' \subset \phi(U \cap V)$ such that $F|_{N'}$ is a diffeomorphism. Hence F is a diffeomorphism and $(U, \phi), (V, \psi)$ are smoothly compatible. Therefore $\alpha(\mathcal{B})$ is a smooth atlas.

To see that $\alpha(\mathcal{B})$ is maximal, let $\mathcal{B}'$ be a smooth atlas on M . Suppose that $\alpha(\mathcal{B}) \subset \mathcal{B}'$ and let $(U, \phi) \in \mathcal{B}'$. By definition, for each chart $(V, \psi) \in \mathcal{B}'$, (U, ϕ) and (V, ψ) are smoothly compatible. Since $\mathcal{B} \subset \alpha(\mathcal{B}) \subset \mathcal{B}'$, we have that $(U, \phi) \in \alpha(\mathcal{B})$. So $\alpha(\mathcal{B}) = \mathcal{B}'$ and $\alpha(\mathcal{B})$ is a maximal smooth atlas on M . $\square$

Exercise 4.1.0.7. Let $(M, \mathcal{A})$ be an n -dimensional smooth manifold. Then for each $\sigma \in S_n$, and $(U, \phi) \in \mathcal{A}$, $(U, \sigma \cdot \phi) \in \mathcal{A}$.

Proof. content... $\square$

Definition 4.1.0.8. Let $n \in \mathbb{N}_0$. We define the **standard smooth structure** on $\mathbb{H}^n$, denoted $\mathcal{A}_{\mathbb{H}^n}$, by $\mathcal{A}_{\mathbb{H}^n} = \alpha_{\mathbb{H}^n}(\mathbb{H}^n, \text{id}_{\mathbb{H}^n})$.

Note 4.1.0.9. Unless otherwise specified we equip $\mathbb{H}^n$ with $\mathcal{A}_{\mathbb{H}^n}$.

Note 4.1.0.10. Let $n \in \mathbb{N}$. We recall the definition of $\eta_0 : \mathbb{R}^n \rightarrow \text{Int } \mathbb{H}^n$ in Definition 9.4.2.3 given by $\eta_0(a^1, \dots, a^{n-1}, a^n) := (a^1, \dots, a^{n-1}, e^{a^n})$. We know from Exercise ?? that η_0 is a homeomorphism.

Definition 4.1.0.11. Let $n \in \mathbb{N}_0$. Define $\mathcal{A}_{\mathbb{R}^n}$: We define the **standard smooth structure** on $\mathbb{R}^n$, denoted $\mathcal{A}_{\mathbb{R}^n}$, by $\mathcal{A}_{\mathbb{R}^n} = \alpha_{\mathbb{R}^n}(\mathbb{R}^n, \text{id}_{\mathbb{R}^n})$. **finish**

Exercise 4.1.0.12. Define $U \subset \mathbb{R}$ and $\phi : U \rightarrow \mathbb{R}$ by $U := \mathbb{R}$ and $\phi(x) := x^3$. Then

1. $(U, \phi) \in X^1(\mathbb{R})$
2. $(U, \phi) \notin \mathcal{A}_{\mathbb{R}}$

Proof.

1.
 - Trivially, U is open in $\mathbb{R}$.
 - Trivially, $\mathbb{R}$ is open in $\mathbb{R}$
 - Clearly ϕ is continuous. Also, ϕ is a bijection. and since for each $x \in \mathbb{R}$, $\phi^{-1}(x) = x^{1/3}$, ϕ^{-1} is continuous. Hence ϕ is a homeomorphism.
 So $(U, \phi) \in X^1(\mathbb{R})$.
2. Define $V \subset M$ and $\psi : V \rightarrow \mathbb{R}$ by $V := \mathbb{R}$ and $\psi := \text{id}_{\mathbb{R}}$. By definition, $(V, \psi) \in \mathcal{A}_{\mathbb{R}}$. Since ϕ^{-1} is not differentiable at $x = 0$ and $\psi \circ \phi^{-1} = \phi^{-1}$, we have that $\psi \circ \phi^{-1}$ is not smooth and therefore $\psi \circ \phi^{-1}$ is not a diffeomorphism. Hence (U, ϕ) and (V, ψ) are not smoothly compatible. Thus $(U, \phi) \notin \mathcal{A}_{\mathbb{R}}$. $\square$

Exercise 4.1.0.13. Let $(M, \mathcal{A})$ be a smooth manifold and $\mathcal{A}_0 \subset \mathcal{A}$. Suppose that $\mathcal{A}_0$ is an atlas on M . Let $(U, \phi) \in X(M)$. Then $(U, \phi) \in \mathcal{A}$ iff for each $(V, \psi) \in \mathcal{A}_0$, (U, ϕ) and (V, ψ) are smoothly compatible.

Proof. Set $n := \dim M$.

- ($\implies$):
Suppose that $(U, \phi) \in \mathcal{A}$. Since $\mathcal{A}$ is smooth, for each $(V, \psi) \in \mathcal{A}$, (U, ϕ) and (V, ψ) are smoothly compatible. Since $\mathcal{A}_0 \subset \mathcal{A}$, we have that for each $(V, \psi) \in \mathcal{A}_0$, (U, ϕ) and (V, ψ) are smoothly compatible.
- ($\impliedby$):
Suppose that for each $(V, \psi) \in \mathcal{A}_0$, (U, ϕ) and (V, ψ) are smoothly compatible. Let $(V, \psi) \in \mathcal{A}$ and $a \in \phi(U \cap V)$. Set $p := \phi^{-1}(a)$. Since $\mathcal{A}_0$ is an atlas on M , there exists $(W_0, \alpha_0) \in \mathcal{A}_0$ such that $p \in W_0$. Define $f : \phi(U \cap W_0) \rightarrow \alpha_0(U \cap W_0)$, $g : \alpha_0(W_0 \cap V) \rightarrow \psi(W_0 \cap V)$ and $h : \phi(U \cap V) \rightarrow \psi(U \cap V)$ by $f := \alpha_0|_{U \cap W_0} \circ \phi|_{U \cap W_0}^{-1}$, $g := \psi|_{W_0 \cap V} \circ \alpha_0|_{W_0 \cap V}^{-1}$ and $h := \psi|_{U \cap V} \circ \phi|_{U \cap V}^{-1}$. By assumption, (U, ϕ) and (W_0, α_0) are smoothly compatible. Thus f is a diffeomorphism and therefore f is smooth. Since $(W_0, \alpha_0), (V, \psi) \in \mathcal{A}$, we have that (W_0, α_0) and (V, ψ) are smoothly compatible. Thus g is a diffeomorphism and therefore g is smooth. Define $A \subset M$ and $A' \subset \mathbb{R}^n$ by $A := U \cap V \cap W_0$ and $A' = \phi(A)$. Since $p \in A$, $a \in A'$. Since A is open in $U \cap V$ and ϕ is a homeomorphism, A' is open in $\phi(U \cap V)$. Exercise 1.3.2.4 implies that $f|_{A'}$ is smooth. Since $h|_{A'} = g \circ f|_{A'}$, $h|_{A'}$ is smooth. Since $a \in \phi(U \cap V)$ is arbitrary, we have that for each $a \in \phi(U \cap V)$, there exists $A' \subset \phi(U \cap V)$ such that $a \in A'$, A' is open in $\phi(U \cap V)$ and $h|_{A'}$ is smooth. Exercise 1.3.2.5 implies that h is smooth. Thus (U, ϕ) and (V, ψ) are smoothly compatible. Since $(V, \psi) \in \mathcal{A}$ is arbitrary, we have that $\mathcal{A} \cup \{(U, \phi)\}$ is a smooth atlas on M . Since $\mathcal{A}$ is maximal, $\mathcal{A} \cup \{(U, \phi)\} = \mathcal{A}$. Thus $(U, \phi) \in \mathcal{A}$.

□

Exercise 4.1.0.14. Smooth Manifold Chart Lemma:

Let M be a set, Γ an index set and for each $\alpha \in \Gamma$, $U_\alpha \subset M$ and $\phi_\alpha : U_\alpha \rightarrow \mathbb{H}^n$. Suppose that

- (a) for each $\alpha \in \Gamma$, $\phi_\alpha(U_\alpha) \in \mathcal{T}_{\mathbb{H}^n}$
- (b) for each $\alpha, \beta \in \Gamma$, $\phi_\alpha(U_\alpha \cap U_\beta) \in \mathcal{T}_{\mathbb{H}^n}$
- (c) for each $\alpha \in \Gamma$, $\phi_\alpha : U_\alpha \rightarrow \phi_\alpha(U_\alpha)$ is a bijection
- (d) for each $\alpha, \beta \in \Gamma$, $\phi_\beta|_{U_\alpha \cap U_\beta} \circ (\phi_\alpha|_{U_\alpha \cap U_\beta})^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \rightarrow \phi_\beta(U_\alpha \cap U_\beta)$ is smooth
- (e) there exists $\Gamma' \subset \Gamma$ such that Γ' is countable and $M \subset \bigcup_{\alpha \in \Gamma'} U_\alpha$
- (f) for each $p, q \in M$, there exists $\alpha \in \Gamma$ such that $p, q \in U_\alpha$ or there exist $\alpha, \beta \in \Gamma$ such that $p \in U_\alpha$, $q \in U_\beta$ and $U_\alpha \cap U_\beta = \emptyset$

Then there exists a unique topology $\mathcal{T}_M$ on M and smooth structure $\mathcal{A}_M$ on $(M, \mathcal{T}_M)$ such that $(M, \mathcal{T}_M, \mathcal{A}_M)$ is an n -dimensional smooth manifold and $(U_\alpha, \phi_\alpha)_{\alpha \in \Gamma} \subset \mathcal{A}_M$.

Proof. Define

- $\mathcal{B} = \{\phi_\alpha^{-1}(V) : \alpha \in \Gamma \text{ and } V \in \mathcal{T}_{\mathbb{H}^n}\}$
- $\mathcal{T}_M = \tau(\mathcal{B})$
- $\mathcal{A}' = \{(U_\alpha, \phi_\alpha) : \alpha \in \Gamma\}$.

Exercise 3.1.0.37 (the topological manifold chart lemma) implies that $\mathcal{T}_M$ is the unique topology on M such that $(M, \mathcal{T}_M)$ is an n -dimensional topological manifold and $\mathcal{A}' \subset X^n(M, \mathcal{T}_M)$. Since $M = \bigcup_{\alpha \in \Gamma} U_\alpha$, $\mathcal{A}'$ is an atlas on M . Since for each $\alpha, \beta \in \Gamma$, $\phi_\beta|_{U_\alpha \cap U_\beta} \circ (\phi_\alpha|_{U_\alpha \cap U_\beta})^{-1} : \phi_\alpha(U_\alpha \cap U_\beta) \rightarrow \phi_\beta(U_\alpha \cap U_\beta)$ is smooth, we have that $\mathcal{A}'$ is smooth. Set $\mathcal{A}_M = \alpha(\mathcal{A}')$. A previous exercise implies that $\mathcal{A}_M$ is the unique smooth structure $\mathcal{A}$ on M such that $\mathcal{A}' \subset \mathcal{A}$. Hence $(M, \mathcal{A}_M)$ is an n -dimensional smooth manifold and $\mathcal{A}' \subset \mathcal{A}_M$. [link exercises](#) □

4.2 Open and Boundary Submanifolds

4.2.1 Open Submanifolds

Exercise 4.2.1.1. Let $(M, \mathcal{A})$ be an n -dimensional smooth manifold, $(U, \phi) \in \mathcal{A}$ and $U' \subset U$. If U' is open, then $(U', \phi|_{U'}) \in \mathcal{A}$.

Proof. Set $\phi' = \phi|_{U'}$. A previous exercise implies that $(U', \phi') \in X(U)$. Define $\mathcal{B} = \mathcal{A} \cup \{(U', \phi')\}$. Let $(V, \psi) \in \mathcal{B}$. If $(V, \psi) = (U', \phi')$, then

$$\phi' \circ \psi^{-1} = \text{id}_{U'}$$

which is a diffeomorphism. Thus (U', ϕ') , (V, ψ) are smoothly compatible. Suppose that $(V, \psi) \in \mathcal{A}$. Since $\mathcal{A}$ is smooth, $\psi|_{U \cap V} \circ (\phi|_{U \cap V})^{-1} : \phi(U \cap V) \rightarrow \psi(U \cap V)$ is a diffeomorphism. Therefore $\psi|_{U' \cap V} \circ (\phi'|_{U' \cap V})^{-1} : \phi'(U' \cap V) \rightarrow \psi(U' \cap V)$ is a diffeomorphism and (U', ϕ') , (V, ψ) are smoothly compatible. Since $(V, \psi) \in \mathcal{B}$ is arbitrary, $\mathcal{B}$ is smooth. Since $\mathcal{A}$ is maximal and $\mathcal{A} \subset \mathcal{B}$, we have that $\mathcal{A} = \mathcal{B}$ and $(U', \phi') \in \mathcal{A}$. $\square$

Exercise 4.2.1.2. Let $(M, \mathcal{A})$ be a n -dimensional smooth manifold and $U \subset M$ open. Set $\mathcal{B} = \{(V, \psi) \in \mathcal{A} : V \subset U\}$. Then $\mathcal{B}$ is a smooth atlas on U .

Proof.

- Some previous exercises imply that U is an n -dimensional topological manifold and $X(U) = \{(V, \psi) \in X(M) : V \subset U\}$. Since

$$\begin{aligned} \mathcal{B} &\subset \mathcal{A} \\ &\subset X(M) \end{aligned}$$

we have that $\mathcal{B} \subset X(U)$. Let $p \in U$. Then there exists $(V, \psi) \in \mathcal{A}$ such that $p \in V$. Set $V' = U \cap V$ and $\psi' = \psi|_{V'}$. The previous exercise implies that $(V', \psi') \in \mathcal{A}$. By definition, $(V', \psi') \in \mathcal{B}$. Since $p \in U$ is arbitrary, we have that for each $p \in U$, there exists $(V', \psi') \in \mathcal{B}$ such that $p \in V'$. Hence $\mathcal{B}$ is an atlas on U .

- Let $(V_1, \psi_1), (V_2, \psi_2) \in \mathcal{B}$. Then $(V_1, \psi_1), (V_2, \psi_2) \in \mathcal{A}$. Since $\mathcal{A}$ is smooth, (V_1, ψ_1) and (V_2, ψ_2) are smoothly compatible. Since $(V_1, \psi_1), (V_2, \psi_2) \in \mathcal{B}$ are arbitrary, $\mathcal{B}$ is smooth. $\square$

Definition 4.2.1.3. Smooth Open Submanifold:

Let $(M, \mathcal{A})$ be an n -dimensional smooth manifold and $U \subset M$ open. A previous exercise implies that U is an n -dimensional topological manifold. We define the **induced smooth structure on U** , denoted $\mathcal{A}|_U \subset X(U)$, by

$$\mathcal{A}|_U = \alpha_U(\{(V, \psi) \in \mathcal{A} : V \subset U\})$$

Then $(U, \mathcal{A}|_U)$ is said to be a **smooth open submanifold of $(M, \mathcal{A})$** .

Exercise 4.2.1.4. Let $(M, \mathcal{A})$ be an n -dimensional smooth manifold and $U \subset M$ open. Then

1. $\mathcal{A}|_U \subset \mathcal{A}$,
2. $\mathcal{A}|_U = \{(V, \psi) \in \mathcal{A} : V \subset U\}$.

Proof.

1. Set $\mathcal{B} = \{(V, \psi) \in \mathcal{A} : V \subset U\}$. Let $(U', \phi) \in \mathcal{A}|_U$, $(V, \psi) \in \mathcal{A}$ and $a \in \phi(U' \cap V)$. Set $p = \phi^{-1}(a)$. Exercise 4.2.1.2 implies that $\mathcal{B}$ is a smooth atlas on U . Thus there exists $(W, \alpha) \in \mathcal{B}$ such that $p \in W$. Set $A := W \cap U' \cap V$ and $A_0 := \phi(A)$. Then $p \in A$, $a \in A_0$, A is open in M , A_0 is open in $\phi(U' \cap V)$ and A_0 is open in $\phi(W \cap U')$. Define $f : \phi(W \cap U') \rightarrow \alpha(W \cap U')$, $g : \alpha(W \cap V) \rightarrow \psi(W \cap V)$ and $h : \phi(U' \cap V) \rightarrow \psi(U' \cap V)$ by $f := \alpha|_{W \cap U'} \circ \phi|_{W \cap U'}^{-1}$, $g := \psi|_{W \cap V} \circ \alpha|_{W \cap V}^{-1}$ and $h := \psi|_{U' \cap V} \circ \phi|_{U' \cap V}^{-1}$. Since $\mathcal{B} \subset \mathcal{A}$, g is smooth. Since $\mathcal{B} \subset \mathcal{A}|_U$, f is smooth. Exercise 1.3.2.4 implies that $f|_{A_0}$ is smooth. Since $h|_{A_0} = g \circ f|_{A_0}$, Exercise 1.3.2.6 implies that $h|_{A_0}$ is smooth. Since $a \in \phi(U' \cap V)$ is arbitrary, we have that for each $a \in \phi(U' \cap V)$, there exists $A_0 \subset \phi(U' \cap V)$ such that $a \in A_0$, A_0 is open in $\phi(U' \cap V)$ and $h|_{A_0}$ is smooth. Exercise 1.3.2.5 implies that h is smooth. Similarly h^{-1} is smooth. Thus h is a diffeomorphism. Therefore (V, ψ) and (U', ϕ) are smoothly compatible. Since $(V, \psi) \in \mathcal{A}$ is arbitrary, we have that $\{(U', \phi)\} \cup \mathcal{A}$ is a smooth atlas. Since $\mathcal{A}$ is maximal, $\{(U', \phi)\} \cup \mathcal{A} = \mathcal{A}$. Thus $(U', \phi) \in \mathcal{A}$. Since $(U', \phi) \in \mathcal{A}|_U$ is arbitrary, we have that $\mathcal{A}|_U \subset \mathcal{A}$.

2. By definition,

$$\begin{aligned}\mathcal{B} &\subset \alpha_U(\mathcal{B}) \\ &= \mathcal{A}|_U\end{aligned}$$

Since $\mathcal{A}|_U \subset \mathcal{A}$, the definition of $\mathcal{B}$ implies that $\mathcal{A}|_U \subset \mathcal{B}$. Hence $\mathcal{A}|_U = \mathcal{B}$. □

Note 4.2.1.5. Let $(M, \mathcal{A})$ be an n -dimensional smooth manifold and $U \subset M$. Suppose that U is open in M . Unless otherwise specified, we equip U with $\mathcal{A}|_U$.

4.2.2 Boundary Submanifolds

Exercise 4.2.2.1. Let $\pi : \partial\mathbb{H}^n \rightarrow \mathbb{R}^{n-1}$ be the projection map given by $\pi(x^1, \dots, x^{n-1}, 0) = (x^1, \dots, x^{n-1})$. Then π is a diffeomorphism.

Proof. Define projection map $\pi' : \mathbb{R}^n \rightarrow \mathbb{R}^{n-1}$ by $\pi'(x^1, \dots, x^{n-1}, x^n) = (x^1, \dots, x^{n-1})$. Then $\mathbb{R}^n$ is an open neighborhood of $\partial\mathbb{H}^n$, $\pi'|_{\partial\mathbb{H}^n} = \pi$ and π' is smooth. Then by definition, π is smooth. Clearly, π^{-1} is smooth. So π is a diffeomorphism. □

Definition 4.2.2.2. Let $(M, \mathcal{A})$ be a n -dimensional smooth manifold and $\pi : \partial\mathbb{H}^n \rightarrow \mathbb{R}^{n-1}$ the projection map. Recall that for $(U, \phi) \in X_\partial^n(M)$, the $(n-1)$ -coordinate chart $(\bar{U}, \bar{\phi}) \in X_{\text{Int}}^{n-1}(\partial M)$ is defined by $\bar{U} = U \cap \partial M$ and $\bar{\phi} = \pi|_{\phi(\bar{U})} \circ \phi|_{\bar{U}}$. We define

$$\bar{\mathcal{A}} = \{(\bar{U}, \bar{\phi}) \in X_{\partial}^{n-1}(M) : (U, \phi) \in \mathcal{A}\}$$

Exercise 4.2.2.3. Let $(M, \mathcal{A})$ be a n -dimensional smooth manifold. Then $\bar{\mathcal{A}}$ is a smooth atlas on ∂M .

Proof.

- A previous exercise implies that ∂M is an $(n-1)$ -dimensional topological manifold. Let $p \in \partial M$. Then there exists $(U, \phi) \in \mathcal{A}$ such that $p \in U$. Since $\mathcal{A} \subset X^n(M)$ and $p \in \partial M$, we have that $p \in \bar{U}$ and a previous exercise implies that $(U, \phi) \in X_\partial^n(M)$. By definition of $\bar{\mathcal{A}}$, $(\bar{U}, \bar{\phi}) \in \bar{\mathcal{A}}$. Since $p \in \partial M$ is arbitrary, $\bar{\mathcal{A}}$ is an atlas on ∂M .
- Let $(\bar{U}, \bar{\phi}), (\bar{V}, \bar{\psi}) \in \bar{\mathcal{A}}$. Since (U, ϕ) and (V, ψ) are smoothly compatible, $\psi|_{U \cap V} \circ (\phi|_{U \cap V})^{-1}$ is a diffeomorphism. Thus $\psi|_{\bar{U} \cap \bar{V}} \circ (\phi|_{\bar{U} \cap \bar{V}})^{-1}$ is a diffeomorphism. Since $\pi|_{\phi(U \cap V)}$ and $\pi|_{\psi(U \cap V)}$ are diffeomorphisms, $\pi|_{\phi(\bar{U} \cap \bar{V})}$ and $\pi|_{\psi(\bar{U} \cap \bar{V})}$ are diffeomorphisms. Then

$$\begin{aligned}\bar{\psi}|_{\bar{U} \cap \bar{V}} \circ (\bar{\phi}|_{\bar{U} \cap \bar{V}})^{-1} &= \left[\pi|_{\psi(\bar{U} \cap \bar{V})} \circ \psi|_{\bar{U} \cap \bar{V}} \right] \circ \left[(\phi|_{\bar{U} \cap \bar{V}})^{-1} \circ (\pi|_{\phi(\bar{U} \cap \bar{V})})^{-1} \right] \\ &= \pi|_{\psi(\bar{U} \cap \bar{V})} \circ \left[\psi|_{\bar{U} \cap \bar{V}} \circ (\phi|_{\bar{U} \cap \bar{V}})^{-1} \right] \circ (\pi|_{\phi(\bar{U} \cap \bar{V})})^{-1}\end{aligned}$$

is a diffeomorphism. Therefore $(\bar{U}, \bar{\phi})$ and $(\bar{V}, \bar{\psi})$ are smoothly compatible. Since $(\bar{U}, \bar{\phi}), (\bar{V}, \bar{\psi}) \in \bar{\mathcal{A}}$ are arbitrary, $\bar{\mathcal{A}}$ is smooth. □

Definition 4.2.2.4. Let $(M, \mathcal{A})$ be a n -dimensional smooth manifold. We define the **induced smooth structure on the boundary**, denoted $\mathcal{A}|_{\partial M}$, by

$$\mathcal{A}|_{\partial M} = \alpha(\bar{\mathcal{A}})$$

We define the **smooth boundary submanifold of M** to be $(\partial M, \mathcal{A}|_{\partial M})$.

Note 4.2.2.5. Let $(M, \mathcal{A})$ be an n -dimensional smooth manifold. Unless otherwise specified, we equip ∂M with $\mathcal{A}|_{\partial M}$.

4.3 Product Manifolds

Note 4.3.0.1. Let $m \in \mathbb{N}$ and $n \in \mathbb{N}_0$. We recall the definition of $\lambda_0 : \mathbb{H}^m \times \text{Int } \mathbb{H}^n \rightarrow \mathbb{H}^{m+n}$ in Definition 3.3.0.2 by $\lambda((x^1, \dots, x^{m-1}, x^m), (y^1, \dots, y^{n-1}, \log y^n, x^m))$ and from Exercise 3.3.0.3, we know that

- $\lambda_0(\partial \mathbb{H}^m \times \text{Int } \mathbb{H}^n) = \partial \mathbb{H}^{m+n}$,
- $(\mathbb{H}^m \times \text{Int } \mathbb{H}^n, \lambda_0) \in X^{m+n}(\mathbb{H}^m \times \text{Int } \mathbb{H}^n)$.

Definition 4.3.0.2. Let M, N be topological manifolds of dimension m and n respectively, $\mathcal{A} \subset X^m(M)$ and $\mathcal{B} \subset X^n(N)$. Suppose that $\mathcal{A}$ and $\mathcal{B}$ are smooth atlases on M and N respectively and $\partial N = \emptyset$. We define the **product atlas of $\mathcal{A}$ and $\mathcal{B}$ on $M \times N$** , denoted $\mathcal{A} \otimes_0 \mathcal{B}$, by

$$\mathcal{A} \otimes_0 \mathcal{B} = \{(U \times V, \lambda_0|_{\phi(U) \times \psi(V)} \circ [\phi \times \psi]) : (U, \phi) \in \mathcal{A} \text{ and } (V, \psi) \in \mathcal{B}\}$$

Exercise 4.3.0.3. Let M, N be topological manifolds of dimension m and n respectively, $\mathcal{A} \subset X^m(M)$ and $\mathcal{B} \subset X^n(N)$. Suppose that $\mathcal{A}$ and $\mathcal{B}$ are smooth atlases on M and N respectively and $\partial N = \emptyset$. Then $\mathcal{A} \otimes_0 \mathcal{B}$ is a smooth atlas on $M \times N$.

Proof.

- Exercise 3.3.0.5 and the proof of Exercise 3.3.0.6 implies that $\mathcal{A} \otimes_0 \mathcal{B}$ is an atlas on $M \times N$.
- Let $(W_1, \eta_1), (W_2, \eta_2) \in \mathcal{A} \otimes_0 \mathcal{B}$. Then there exist $(U_1, \phi_1), (U_2, \phi_2) \in \mathcal{A}$, $(V_1, \psi_1), (V_2, \psi_2) \in \mathcal{B}$ such that $W_1 = U_1 \times V_1$, $W_2 = U_2 \times V_2$, $\eta_1 = \lambda_0|_{\phi_1(U_1) \times \psi_1(V_1)} \circ [\phi_1 \times \psi_1]$ and $\eta_2 = \lambda_0|_{\phi_2(U_2) \times \psi_2(V_2)} \circ [\phi_2 \times \psi_2]$. For notational convenience, set $U := U_1 \cap U_2$ and $V := V_1 \cap V_2$. Then $W_1 \cap W_2 = U \cap V$ and

$$\begin{aligned} \eta_2|_{W_1 \cap W_2} \circ \eta_1|_{W_1 \cap W_2}^{-1} &= \eta_2|_{U \cap V} \circ \eta_1|_{U \cap V}^{-1} \\ &= \lambda_0|_{\phi_2(U) \times \psi_2(V)} \circ [\phi_2 \times \psi_2]|_{U \times V} \circ [\phi_1 \times \psi_1]|_{U \times V}^{-1} \circ \lambda_0|_{\phi_1(U) \times \psi_1(V)}^{-1} \\ &= \lambda_0|_{\phi_2(U) \times \psi_2(V)} \circ [\phi_2|_U \times \psi_2|_V] \circ [\phi_1|_U^{-1} \times \psi_1|_V^{-1}] \circ \lambda_0|_{\phi_1(U) \times \psi_1(V)}^{-1} \\ &= \lambda_0|_{\phi_2(U) \times \psi_2(V)} \circ [(\phi_2|_U \circ \phi_1|_U^{-1}) \times (\psi_2|_V \circ \psi_1|_V^{-1})] \circ \lambda_0|_{\phi_1(U) \times \psi_1(V)}^{-1} \end{aligned}$$

Write $\phi_2 = (x_2^1, \dots, x_2^m)$ and $\psi_2 = (y_2^1, \dots, y_2^n)$. Since $\phi_2|_U \circ \phi_1|_U^{-1}$ and $\psi_2|_V \circ \psi_1|_V^{-1}$ are smooth, **reference components of smooth tuples are smooth** implies that for each $j \in [m]$ and $k \in [n]$, $x_2^j \circ \phi_1|_U^{-1}$ and $y_2^k \circ \psi_1|_V^{-1}$ are smooth. Let $(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) \in \eta_1(W_1 \cap W_2)$. Then

$$\begin{aligned} \eta_2|_{W_1 \cap W_2} \circ \eta_1|_{W_1 \cap W_2}^{-1}(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) &= (x_2^1 \circ \phi_1^{-1}(a^1, \dots, a^m), \dots, x_2^{m-1} \circ \phi_1^{-1}(a^1, \dots, a^m), \\ &\quad y_2^1 \circ \psi_1^{-1}(b^1, \dots, b^{n-1}, e^{b^n}), \dots, y_2^{n-1} \circ \psi_1^{-1}(b^1, \dots, b^{n-1}, e^{b^n}), \\ &\quad \log y_2^n \circ \psi_1^{-1}(b^1, \dots, b^{n-1}, e^{b^n}), x_2^m \circ \phi_1^{-1}(a^1, \dots, a^m)) \end{aligned}$$

Hence **reference tuples of smooth maps are smooth** $\eta_2|_{W_1 \cap W_2} \circ \eta_1|_{W_1 \cap W_2}^{-1}$ is smooth. Since $(W_1, \eta_1), (W_2, \eta_2) \in \mathcal{A} \otimes_0 \mathcal{B}$ are arbitrary, we have that $\mathcal{A} \otimes_0 \mathcal{B}$ is smooth. □

Definition 4.3.0.4. Let $(M, \mathcal{A}), (N, \mathcal{B})$ be smooth manifolds. Suppose that $\partial N = \emptyset$. We define the **product smooth structure**, denoted $\mathcal{A} \otimes \mathcal{B}$, by

$$\mathcal{A} \otimes \mathcal{B} = \alpha_{M \times N}(\mathcal{A} \otimes_0 \mathcal{B})$$

We define the **smooth product manifold of $(M, \mathcal{A})$ and $(N, \mathcal{B})$** to be $(M \times N, \mathcal{A} \otimes \mathcal{B})$.

Note 4.3.0.5. Let $(M, \mathcal{A})$ and $(M, \mathcal{B})$ be an n -dimensional smooth manifolds. Unless otherwise specified, we equip $M \times N$ with $\mathcal{A} \otimes \mathcal{B}$.

Exercise 4.3.0.6. Show that if $U \subset M$ is open, $V \subset N$ open, then $(\mathcal{A} \otimes \mathcal{B})|_{U \times V} = \mathcal{A}|_U \otimes \mathcal{B}|_V$.

Proof. **FINISH!!!** □

Chapter 5

Smooth Maps

5.1 Smooth Maps between Manifolds

Definition 5.1.0.1. Let $(M, \mathcal{A}), (N, \mathcal{B}) \in \text{Man}^\infty$.

- Let $F : M \rightarrow N$. Then F is said to be $(\mathcal{A}, \mathcal{B})$ -**smooth** if for each $p \in M$, there exist $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U$, $F(U) \subset V$ and $\psi \circ F|_U \circ \phi^{-1} \in C^\infty(\phi(U), \psi(V))$.
- We define $C^\infty((M, \mathcal{A}), (N, \mathcal{B})) := \{F \in N^M : F \text{ is } (\mathcal{A}, \mathcal{B})\text{-smooth}\}$.

Note 5.1.0.2. When the context is clear,

- we may say “smooth” instead of “ $(\mathcal{A}, \mathcal{B})$ -smooth”,
- we may write $C^\infty(M, N)$ in place of $C^\infty((M, \mathcal{A}), (N, \mathcal{B}))$,
- we may write $C^\infty(M)$ in place of $C^\infty(M, \mathbb{R})$.

Intuitively, if the **ref prev def** were an appropriate generalization of our previous notion of smoothness for maps between Euclidean (half) spaces discussed in **ref def in prereq material**, we would need the two definitions to be equivalent when M and N are Euclidean (half) spaces. We would also need the two definitions to be compatible in the **natural?** intuitive way when only one of M or N is a Euclidean (half) space. Explicitly, we would need the generalized definition to satisfy that for each $U \in \mathcal{T}_{\mathbb{H}^m}$, $V \in \mathcal{T}_{\mathbb{H}^n}$ and $F : U \rightarrow V$, F is $(\mathcal{A}_U, \mathcal{A}_V)$ -smooth (as in **the prev def**) iff F is smooth (as in **ref prereq def**). The following exercise states that the **prev def** indeed satisfies this requirement.

Exercise 5.1.0.3. Let $m, n \in \mathbb{N}_0$, $M \in \mathcal{T}_{\mathbb{H}^m}$, $N \in \mathcal{T}_{\mathbb{H}^n}$. Then $C^\infty((M, \mathcal{A}_M), (N, \mathcal{A}_N)) = C^\infty(M, N)$.

Proof.

- Let $F \in C^\infty((M, \mathcal{A}_M), (N, \mathcal{A}_N))$ and $p \in M$. By **prev def**, there exists $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$ such that $p \in U$, $F(U) \subset V$ and $\psi \circ F|_U \circ \phi^{-1} \in C^\infty(\phi(U), \psi(V))$. Define $F_0 \in C^\infty(\phi(U), \psi(V))$ by $F_0 := \psi \circ F|_U \circ \phi^{-1}$. By **ref definition of standard smooth structure on open subsets of $\mathbb{H}^m$** , $(M, \text{id}_M) \in \mathcal{A}_M$. Since $(U, \phi), (M, \text{id}_M) \in \mathcal{A}_M$ and $\mathcal{A}_M \in \text{Atlasinf}(M)$, we have that (U, ϕ) and (M, id_M) are smoothly compatible. Hence

$$\begin{aligned} \phi \circ \text{id}_U &= \phi \circ (\text{id}_M|_U) - 1 \\ &\in \text{Diffeo}(U, \phi(U)). \end{aligned}$$

Similarly, $\text{id}_V \circ \psi^{-1} \in \text{Diffeo}(\psi(V), V)$. **ref ex in prereq material about composition of smooth functions being smooth** then implies that

$$\begin{aligned} F|_U &= (\psi^{-1} \circ \psi) \circ F|_U \circ (\phi^{-1} \circ \phi) \\ &= \psi^{-1} \circ (\psi \circ F|_U \circ \phi^{-1}) \circ \phi \\ &= \psi^{-1} \circ F_0 \circ \phi \\ &= (\text{id}_V \circ \psi^{-1}) \circ F_0 \circ (\phi \circ \text{id}_U) \\ &\in C^\infty(U, V). \end{aligned}$$

Since $p \in M$ is arbitrary, we have that for each $p \in M$, there exists $U \in \mathcal{N}^{\text{opn}}(p, \mathcal{T}_M)$ and $V \in \mathcal{N}^{\text{opn}}(p, \mathcal{T}_N)$ such that $F(U) \subset V$ and $F|_U \in C^\infty(U, V)$. **make/ref ex in prereq material showing that if for each $p \in M$, there exists $U \in \mathcal{N}^{\text{opn}}(p, \mathcal{T}_M)$ and $V \in \mathcal{N}^{\text{opn}}(p, \mathcal{T}_N)$ such that $F(U) \subset V$ and $F|_U \in C^\infty(U, V)$, then $F \in C^\infty(M, N)$ implies that $F \in C^\infty(M, N)$.** Since $F \in C^\infty((M, \mathcal{A}_M), (N, \mathcal{A}_N))$ is arbitrary, we have that for each $F \in C^\infty((M, \mathcal{A}_M), (N, \mathcal{A}_N))$, $F \in C^\infty(M, N)$. Thus $C^\infty((M, \mathcal{A}_M), (N, \mathcal{A}_N)) \subset C^\infty(M, N)$.

• ($\Leftarrow$):

Let $F \in C^\infty(M, N)$, and $p \in M$. Define $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$ by $(U, \phi) := (M, \text{id}_M)$ and $(V, \psi) := (N, \text{id}_N)$. Then $p \in U$,

$$\begin{aligned} F(U) &= F(M) \\ &\subset N \\ &= V, \end{aligned}$$

and

$$\begin{aligned} \psi \circ F|_U \circ \phi^{-1} &= \text{id}_N \circ F \circ \text{id}_M \\ &= F \\ &\in C^\infty(M, N) \\ &= C^\infty(\phi(U), \psi(V)). \end{aligned}$$

Since $p \in M$ is arbitrary, we have that for each $p \in M$, there exists $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$ such that $p \in U$, $F(U) \subset V$ and $\psi \circ F|_U \circ \phi^{-1} \in C^\infty(\phi(U), \psi(V))$. Thus $F \in C^\infty((M, \mathcal{A}_M), (N, \mathcal{A}_N))$. Since $F \in C^\infty(M, N)$ is arbitrary, we have that for each $F \in C^\infty(M, N)$, $F \in C^\infty((M, \mathcal{A}_M), (N, \mathcal{A}_N))$. Therefore $C^\infty(M, N) \subset C^\infty((M, \mathcal{A}_M), (N, \mathcal{A}_N))$.

Since $C^\infty((M, \mathcal{A}_M), (N, \mathcal{A}_N)) \subset C^\infty(M, N)$ and $C^\infty(M, N) \subset C^\infty((M, \mathcal{A}_M), (N, \mathcal{A}_N))$, we have that $C^\infty((M, \mathcal{A}_M), (N, \mathcal{A}_N)) = C^\infty(M, N)$. $\square$

As a consequence of the compatibility between **ref prev def** and **ref old def in prereq material**, we are provide an equivalent formulation of smoothness for mappings of a smooth manifold into a Euclidean space which is a sort of hybrid of the new and old definition. We formalize this in the following exercise.

Exercise 5.1.0.4. Let $M \in \text{Man}^\infty$, $n \in \mathbb{N}_0$. Then $C^\infty((M, \mathcal{A}_M), (\mathbb{R}^n, \mathcal{A}_{\mathbb{R}^n}))$ iff for each $(U, \phi) \in \mathcal{A}_M$, $F|_U \circ \phi \in C^\infty(\phi(U), \mathbb{R}^n)$.

Proof. **FINISH!!!, similar to prev ex** $\square$

Exercise 5.1.0.5. Let $M, N \in \text{Man}^\infty$. Then $C^\infty(M, N) \subset C(M, N)$.

Proof. Let $F \in C^\infty(M, N)$ and $p \in M$. Since $\mathcal{A}_M \in \text{Atlasinf}(M)$ and $\mathcal{A}_N \in \text{Atlasinf}(N)$, there exist $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$ such that $p \in U$, $F(U) \subset V$ and $\psi \circ F \circ \phi^{-1}$ is smooth. Define $F_0 : \phi(U) \rightarrow \psi(V)$ by

$$F_0 = \psi \circ F \circ \phi^{-1}$$

By definition, F_0 is smooth. Exercise 1.3.2.3 implies that $F_0 \in C(\phi(U), \psi(V))$. Since $\phi \in \text{Homeo}(U, \phi(U))$ and $\psi \in \text{Homeo}(V, \psi(V))$ and $F|_U = \psi^{-1} \circ F_0 \circ \phi$, we have that $F|_U \in C(U, N)$. Thus, F is continuous at p . Since $p \in M$ is arbitrary, we have that for each $p \in M$, F is continuous at p . Hence $F \in C(M, N)$. Since $F \in C^\infty(M, N)$ is arbitrary, we have that for each $F \in C^\infty(M, N)$, $F \in C(M, N)$. $\square$

Definition 5.1.0.6. Let $(M, \mathcal{A}), (N, \mathcal{B}) \in \text{Man}^\infty$.

- Let $F : M \rightarrow N$. Then F is said to be a **$(\mathcal{A}, \mathcal{B})$ -diffeomorphism** if F is a bijection, $F \in C^\infty((M, \mathcal{A}), (N, \mathcal{B}))$ and $F^{-1} \in C^\infty((N, \mathcal{B}), (M, \mathcal{A}))$.
- We define $\text{Diffeo}((M, \mathcal{A}), (N, \mathcal{B})) := \{F \in C^\infty((M, \mathcal{A}), (N, \mathcal{B})) : F \text{ is a } (\mathcal{A}, \mathcal{B})\text{-diffeomorphism}\}$.

Note 5.1.0.7. When the context is clear,

- we may say “diffeomorphism” instead of $(\mathcal{A}, \mathcal{B})$ -diffeomorphism,

- we may write $\text{Diffeo}(M, N)$ in place of $\text{Diffeo}((M, \mathcal{A}), (N, \mathcal{B}))$.

Exercise 5.1.0.8. Let $U \in \mathcal{T}_{\mathbb{H}^m}$, $V \in \mathcal{T}_{\mathbb{H}^n}$ and $F : U \rightarrow V$. Then $F \in \text{Diffeo}(U, V)$ iff F is a diffeomorphism.

Proof. **FINISH!!!**, ref prev ex on equivalence of old and new definitions of smoothness □

Exercise 5.1.0.9. Let $M \in \text{Man}^\infty$ and $(U, \phi) \in \mathcal{A}_M$. Then $\phi \in \text{Diffeo}(U, \phi(U))$.

Proof. By definition, $\phi \in \text{Homeo}(U, \phi(U))$. By definition, for each $(V, \psi) \in \mathcal{A}_M$, $\text{id}_{\mathbb{R}^n} \circ \psi$ **FINISH!!!** □

Exercise 5.1.0.10. Equivalence of Smoothness:

Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds and $F : M \rightarrow N$. Then the following are equivalent:

1. $F : M \rightarrow N$ is smooth
2. for each $\mathcal{A}_0 \subset \mathcal{A}$ and $\mathcal{B}_0 \subset \mathcal{B}$, if $\mathcal{A}_0$ is an atlas on M and $\mathcal{B}_0$ is an atlas on N , then for each $(U, \phi) \in \mathcal{A}_0$ and $(V, \psi) \in \mathcal{B}_0$, $U \cap F^{-1}(V)$ is open in M and $\psi \circ F \circ \phi|_{U \cap F^{-1}(V)}^{-1}$ is smooth.
3. for each $p \in M$, there exist $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U$, $F(p) \in V$, $U \cap F^{-1}(V)$ is open in M and $\psi \circ F \circ \phi|_{U \cap F^{-1}(V)}^{-1}$ is smooth.
4. F is continuous and there exist $\mathcal{A}_0 \subset \mathcal{A}$ and $\mathcal{B}_0 \subset \mathcal{B}$ such that $\mathcal{A}_0$ is an atlas on M , $\mathcal{B}_0$ is an atlas on N and for each $(U, \phi) \in \mathcal{A}_0$ and $(V, \psi) \in \mathcal{B}_0$, $\psi \circ F \circ \phi|_{U \cap F^{-1}(V)}^{-1}$ is smooth

Proof. Set $m := \dim M$ and $n := \dim N$.

1. (1) $\implies$ (2):

Suppose that F is smooth. Let $\mathcal{A}_0 \subset \mathcal{A}$ and $\mathcal{B}_0 \subset \mathcal{B}$. Suppose that $\mathcal{A}_0$ is an atlas on M and $\mathcal{B}_0$ is an atlas on N . Let $(U_0, \phi_0) \in \mathcal{A}_0$ and $(V_0, \psi_0) \in \mathcal{B}_0$. Since $\mathcal{A}_0 \subset \mathcal{A}$ and $\mathcal{B}_0 \subset \mathcal{B}$, we have that $(U_0, \phi_0) \in \mathcal{A}$ and $(V_0, \psi_0) \in \mathcal{B}$. Since F is smooth, Exercise 5.1.0.5 implies that F is continuous and therefore $U_0 \cap F^{-1}(V_0) \in \mathcal{T}_M$. Define $F_0 : \phi_0(U_0 \cap F^{-1}(V_0)) \rightarrow \psi_0(V_0)$ by $F_0 := \psi_0 \circ F \circ \phi_0|_{U_0 \cap F^{-1}(V_0)}^{-1}$. Let $a \in \phi_0(U_0 \cap F^{-1}(V_0))$. Define $p \in M$ by $p := \phi_0^{-1}(a)$. Since F is smooth, there exists $(U_1, \phi_1) \in \mathcal{A}$ and $(V_1, \psi_1) \in \mathcal{B}$ such that $p \in U_1$, $F(U_1) \subset V_1$ and $\psi_1 \circ F \circ \phi_1^{-1}$ is smooth. Define $U := U_0 \cap U_1 \cap F^{-1}(V_0 \cap V_1)$, $\alpha : \phi_1(U \cap U_1) \rightarrow \phi_0(U \cap U_1)$, $\beta : \psi_1(V_0 \cap V_1) \rightarrow \psi_0(V_0 \cap V_1)$ and $F_1 : \phi_1(U_1) \rightarrow \psi_1(V_1)$ by $U := U_0 \cap U_1 \cap F^{-1}(V_0 \cap V_1)$, $\alpha := \phi_0|_{U \cap U_1} \circ \phi_1|_{U \cap U_1}^{-1}$, $\beta := \psi_0|_{V_0 \cap V_1} \circ \psi_1|_{V_0 \cap V_1}^{-1}$ and $F_1 := \psi_1 \circ F \circ \phi_1^{-1}$. We note the following:

- since F is continuous, $V_0, V_1 \in \mathcal{T}_N$ and $U_0, U_1 \in \mathcal{T}_M$, we have that $U \in \mathcal{T}_M$,
- since $p \in U$ and $a = \phi_0(p)$, we have that $a \in \phi_0(U)$,
- since $\phi_0(U) \subset \phi_0(U_0 \cap F^{-1}(V_0))$, $\phi_0(U) \in \mathcal{T}_{\mathbb{H}^m}$ and $\phi_0(U_0 \cap F^{-1}(V_0)) \in \mathcal{T}_{\mathbb{H}^m}$, we have that $\phi_0(U) \in \mathcal{T}_{\mathbb{H}^m} \cap \phi_0(U_0 \cap F^{-1}(V_0))$,
- since $(U_0, \phi_0), (U_1, \phi_1) \in \mathcal{A}$, (U_0, ϕ_0) and (U_1, ϕ_1) are smoothly compatible and α is a diffeomorphism,
- since $(V_0, \psi_0), (V_1, \psi_1) \in \mathcal{B}$, (V_0, ψ_0) and (V_1, ψ_1) are smoothly compatible and β is a diffeomorphism,
- since $F_1 = \psi_1 \circ F \circ \phi_1^{-1}$, F_1 is smooth,
- since α^{-1} is smooth, Exercise 1.3.2.4 implies that $\alpha|_{\phi_1(U)}^{-1}$ is smooth,
- since $F_0|_{\phi_0(U)} = \beta \circ F_1 \circ \alpha|_{\phi_1(U)}^{-1}$, Exercise 1.3.2.6 implies that $F_0|_{\phi_0(U)}$ is smooth.

Therefore we have the following:

- Since $a \in \phi_0(U_0 \cap F^{-1}(V_0))$ is arbitrary, we have that for each $a \in \phi_0(U_0 \cap F^{-1}(V_0))$, there exists $A \in \mathcal{T}_{\mathbb{H}^m} \cap \phi_0(U_0 \cap F^{-1}(V_0))$ such that $a \in A$, and $F_0|_A$ is smooth. Exercise 1.3.2.5 implies that F_0 is smooth.
- Since $(U_0, \phi_0) \in \mathcal{A}_0$ and $(V_0, \psi_0) \in \mathcal{B}_0$ are arbitrary, we have that for each $(U, \phi) \in \mathcal{A}_0$ and $(V, \psi) \in \mathcal{B}_0$, $\psi \circ F \circ \phi|_{U \cap F^{-1}(V)}^{-1}$ is smooth.
- Since $\mathcal{A}_0 \subset \mathcal{A}$ and $\mathcal{B}_0 \subset \mathcal{B}$ such that $\mathcal{A}_0$ is an atlas on M and $\mathcal{B}_0$ is an atlas on N are arbitrary, we have that for each $\mathcal{A}_0 \subset \mathcal{A}$ and $\mathcal{B}_0 \subset \mathcal{B}$, if $\mathcal{A}_0$ is an atlas on M and $\mathcal{B}_0$ is an atlas on N , then for each $(U, \phi) \in \mathcal{A}_0$ and $(V, \psi) \in \mathcal{B}_0$, $U \cap F^{-1}(V)$ is open in M and $\psi \circ F \circ \phi|_{U \cap F^{-1}(V)}^{-1}$ is smooth.

2. (2) $\implies$ (3):

Suppose that for each $\mathcal{A}_0 \subset \mathcal{A}$ and $\mathcal{B}_0 \subset \mathcal{B}$, if $\mathcal{A}_0$ is an atlas on M and $\mathcal{B}_0$ is an atlas on N , then for each $(U, \phi) \in \mathcal{A}_0$ and $(V, \psi) \in \mathcal{B}_0$, $U \cap F^{-1}(V)$ is open in M and $\psi \circ F \circ \phi|_{U \cap F^{-1}(V)}^{-1}$ is smooth. Let $p \in M$. Since $\mathcal{A}$ is an atlas on M and $\mathcal{B}$ is an atlas on N , there exists $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U$ and $F(p) \in V$. By assumption, $U \cap F^{-1}(V)$ is open in M and $\psi \circ F \circ \phi|_{U \cap F^{-1}(V)}^{-1}$ is smooth. Since $p \in M$ is arbitrary, we have that for each $p \in M$, there exist $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U$, $F(p) \in V$, $U \cap F^{-1}(V)$ is open in M and $\psi \circ F \circ \phi|_{U \cap F^{-1}(V)}^{-1}$ is smooth.

3. (3) $\implies$ (4):

Suppose that for each $p \in M$, there exist $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U$, $F(p) \in V$, $U \cap F^{-1}(V)$ is open in M and $\psi \circ F \circ \phi|_{U \cap F^{-1}(V)}^{-1}$ is smooth.

- Let $p \in M$. By assumption, there exist $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U$, $F(p) \in V$, $U \cap F^{-1}(V)$ is open in M and $\psi \circ F \circ \phi|_{U \cap F^{-1}(V)}^{-1}$ is smooth. Define $A \subset M$, $A_1 \subset \mathbb{H}^m$ and $F_1 : A_1 \rightarrow \mathbb{R}^n$ by $A := U \cap F^{-1}(V)$, $A_1 := \phi(A)$ and $F_1 := \psi \circ F \circ \phi|_A^{-1}$. Since F_1 is smooth, Exercise 1.3.2.3 implies that $F_1 : A_1 \rightarrow \mathbb{R}^n$ is continuous. Since $\phi|_A$ and ψ are homeomorphisms,

$$\begin{aligned} F|_A &= \psi^{-1} \circ (\psi \circ F \circ \phi|_A) \circ \phi|_A^{-1} \\ &= \psi^{-1} \circ F_1 \circ \phi|_A^{-1} \end{aligned}$$

which is continuous. We note that $p \in A$ and A is open in M . Since $p \in M$ is arbitrary, we have that for each $p \in M$, there exists $A \subset M$ such that $p \in A$, A is open in M and $F|_A$ is continuous. Thus F is continuous.

- – By assumption, for each $p \in M$, there exists $(U_p, \phi_p) \in \mathcal{A}$ and $(V_p, \psi_p) \in \mathcal{B}$ such that $p \in U_p$, $F(p) \in V_p$, $U_p \cap F^{-1}(V_p)$ is open in M and $\psi_p \circ F \circ \phi_p|_{U_p \cap F^{-1}(V_p)}^{-1}$ is smooth. The axiom of choice implies that there exist $(U_p, \phi_p)_{p \in M} \subset \mathcal{A}$ and $(V_p, \psi_p)_{p \in M} \subset \mathcal{B}$ such that for each $p \in M$, $p \in U_p$, $F(p) \in V_p$, $U_p \cap F^{-1}(V_p)$ is open in M and $\psi_p \circ F \circ \phi_p|_{U_p \cap F^{-1}(V_p)}^{-1}$ is smooth. Define $\mathcal{A}_0 \subset \mathcal{A}$ and $\mathcal{B}_0 \subset \mathcal{B}$ by $\mathcal{A}_0 := (U_p, \phi_p)_{p \in M}$ and $\mathcal{B}_0 := (V_p, \psi_p)_{p \in M}$ respectively. By construction, $\mathcal{A}_0$ is an atlas on M and $\mathcal{B}_0$ is an atlas on N .
- Let $(U, \phi) \in \mathcal{A}_0$ and $(V, \psi) \in \mathcal{B}_0$. Define $\tilde{A} \subset \mathbb{H}^m$ and $\tilde{F} : \tilde{A} \rightarrow \mathbb{R}^n$ by $\tilde{A} = \phi(U \cap F^{-1}(V))$ and $\tilde{F} = \psi \circ F \circ \phi|_{U \cap F^{-1}(V)}^{-1}$. Since F is continuous, $U \cap F^{-1}(V)$ is open in M . Since ϕ is a homeomorphism, $\tilde{A}$ is open in $\mathbb{H}^m$. Let $a \in \tilde{A}$. Set $p := \phi^{-1}(a)$. Define $A \subset M$ by $A := U \cap U_p \cap F^{-1}(V \cap V_p)$. We note that $p \in A$ and since F is continuous, A is open in M . Define $A_0 \subset \mathbb{H}^m$ and $F_0 : A_0 \rightarrow \mathbb{R}^n$ by $A_0 = \phi_p(A)$ and $F_0 = \psi_p \circ F \circ \phi_p|_A^{-1}$. By construction, $\psi_p \circ F \circ \phi_p|_{U_p \cap F^{-1}(V_p)}^{-1}$ is smooth. [An exercise about restriction in the section on differentiation on subspaces](#) implies that F_0 is smooth. We define $\alpha : \phi_p(U \cap U_p) \rightarrow \phi(U \cap U_p)$ and $\beta : \psi_p(V \cap V_p) \rightarrow \psi(V \cap V_p)$ by

$$\alpha := \phi|_{U \cap U_p} \circ \phi_p|_{U \cap U_p}^{-1}, \quad \beta := \psi|_{V \cap V_p} \circ \psi_p|_{V \cap V_p}^{-1}$$

Since $\phi, \phi_p \in \mathcal{A}$, we know that ϕ and ϕ_p are smoothly compatible. Therefore α is a diffeomorphism. Similarly, β is a diffeomorphism. [the restriction exercise again implies that](#) $\alpha|_{A_0}$ is a diffeomorphism. Since $\tilde{F}|_{\phi(A)} = \beta \circ F_0 \circ \alpha|_{A_0}^{-1}$, we have that $\tilde{F}|_{\phi(A)}$ is smooth. We note that $a \in \phi(A)$, $\phi(A)$ is open in $\tilde{A}$. Since $a \in \tilde{A}$ is arbitrary, we have that for each $a \in \tilde{A}$, there exists $E \subset \tilde{A}$ such that $a \in E$, E is open in $\tilde{A}$ and $\tilde{F}|_E$ is smooth. [An exercise in the section on differentiation on subspaces](#) implies that $\tilde{F}$ is smooth. Since $(U, \phi) \in \mathcal{A}_0$ and $(V, \psi) \in \mathcal{B}_0$ are arbitrary, we have that for each $(U, \phi) \in \mathcal{A}_0$ and $(V, \psi) \in \mathcal{B}_0$, $\psi \circ F \circ \phi|_{U \cap F^{-1}(V)}^{-1}$ is smooth.

4. (4) $\implies$ (1):

Suppose that F is continuous and there exist $\mathcal{A}_0 \subset \mathcal{A}$ and $\mathcal{B}_0 \subset \mathcal{B}$ such that $\mathcal{A}_0$ is an atlas on M , $\mathcal{B}_0$ is an atlas on N and for each $(U, \phi) \in \mathcal{A}_0$ and $(V, \psi) \in \mathcal{B}_0$, $\psi \circ F \circ \phi|_{U \cap F^{-1}(V)}^{-1}$ is smooth. Let $p \in M$. Since $\mathcal{A}_0$ is an atlas on M and $\mathcal{B}_0$ is an atlas on N , there exists $(U', \phi') \in \mathcal{A}_0$ and $(V, \psi) \in \mathcal{B}_0$ such that $p \in U'$ and $F(p) \in V$. Define $A_0 \subset \mathbb{H}^m$ and $F_0 : A_0 \rightarrow \mathbb{R}^n$ by $A_0 = \phi'(U' \cap F^{-1}(V))$ and $F_0 = \psi \circ F \circ \phi'|_{U' \cap F^{-1}(V)}^{-1}$. By assumption F_0 is smooth. Since F is continuous, $F(p) \in V$ and V is open in N , we have that there exists $U_0 \subset M$ such that $p \in U_0$, U_0 is open in M and $F(U_0) \subset V$. Define $U \subset M$ and $\phi : U \rightarrow \phi'(U)$ by $U := U' \cap U_0$ and $\phi = \phi'|_U$. Then $p \in U$, U is open in M and

$$\begin{aligned} F(U) &= F(U' \cap U_0) \\ &\subset F(U_0) \\ &\subset V \end{aligned}$$

An exercise in the section on smooth manifolds implies that $(U, \phi) \in \mathcal{A}$. Since F_0 is smooth, an exercise in the section on subspace differentiation implies that $F_0|_{\phi(U)}$ is smooth. Since $\psi \circ F \circ \phi^{-1} = F_0|_{\phi(U)}$, we have that $\psi \circ F \circ \phi^{-1}$ is smooth. Since $p \in M$ is arbitrary, we have that for each $p \in M$, there exist $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U$, $F(p) \in V$, $F(U) \subset V$ and $\psi \circ F \circ \phi^{-1}$ is smooth. Hence F is smooth. $\square$

Exercise 5.1.0.11. Let $(M, \mathcal{A})$, $(N, \mathcal{B})$, $(E, \mathcal{C})$ be smooth manifolds and $F : M \rightarrow N$, $G : N \rightarrow E$. If F and G are smooth, then $G \circ F : M \rightarrow E$ is smooth.

Proof. Set $m = \dim M$, $n = \dim N$ and $e = \dim E$. Suppose that F and G are smooth. Let $p_0 \in M$. Since F is smooth, there exists $(U_0, \phi_0) \in \mathcal{A}$ and $(V_0, \psi_0) \in \mathcal{B}$ such that $p_0 \in U_0$, $F(p_0) \in V_0$, $F(U_0) \subset V_0$ and $\psi_0 \circ F \circ \phi_0^{-1}$ is smooth. Set $p_1 = F(p_0)$. Since G is smooth, there exists $(U_1, \phi_1) \in \mathcal{B}$ and $(V_1, \psi_1) \in \mathcal{C}$ such that $p_1 \in U_1$, $G(p_1) \in V_1$, $G(U_1) \subset V_1$ and $\psi_1 \circ G \circ \phi_1^{-1}$ is smooth. Define $f : \phi_0(U_0) \rightarrow \mathbb{H}^n$ and $g : \phi_1(U_1) \rightarrow \mathbb{H}^e$ by $f = \psi_0 \circ F \circ \phi_0^{-1}$ and $g = \psi_1 \circ G \circ \phi_1^{-1}$ respectively. Set $W_1 = U_1 \cap V_0$ and $W_0 = F^{-1}(W_1)$. Since W_1 is open in N and F is continuous, W_0 is open in M . An exercise in the section on open submanifolds implies that

$$\begin{aligned} (W_0, \phi_0|_{W_0}) &\in \mathcal{A}|_{W_0} \\ &\subset \mathcal{A} \end{aligned}$$

Since $p_1 \in W_1$, $p_0 \in W_0$. Furthermore,

$$\begin{aligned} G \circ F(p_0) &= G(p_1) \\ &\in V_1 \end{aligned}$$

and

$$\begin{aligned} G \circ F(W_0) &= G(F(W_0)) \\ &\subset G(W_1) \\ &\subset G(U_1) \\ &\subset V_1 \end{aligned}$$

Since $(U_1, \phi_1), (V_0, \psi_0) \in \mathcal{B}$, (U_1, ϕ_1) and (V_0, ψ_0) are smoothly compatible. Thus $\phi_1|_{W_1} \circ \psi_0|_{W_1}^{-1} : \psi_0(W_1) \rightarrow \phi_1(W_1)$ is smooth. Since f and g are smooth, we have that $f|_{\phi_0(W_0)}$ is smooth and therefore

$$\begin{aligned} \psi_1 \circ (G \circ F) \circ \phi_0|_{W_0}^{-1} &= (\psi_1 \circ G \circ \phi_1|_{W_1}^{-1}) \circ (\phi_1|_{W_1} \circ \psi_0|_{W_1}^{-1}) \circ (\psi_0 \circ F \circ \phi_0|_{W_0}^{-1}) \\ &= g \circ (\phi_1|_{W_1} \circ \psi_0|_{W_1}^{-1}) \circ f|_{\phi_0(W_0)} \end{aligned}$$

is smooth. Since $p_0 \in M$ is arbitrary, we have that for each $p_0 \in M$, there exists $(W_0, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{C}$ such that $p_0 \in W_0$, $G \circ F(p_0) \in V$, $G \circ F(W_0) \subset V$ and $\psi \circ (G \circ F) \circ \phi^{-1}$ is smooth. Thus $G \circ F$ is smooth. $\square$

5.2 Smooth Maps on Open and Boundary Submanifolds

Exercise 5.2.0.1. Locality of Smoothness:

Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds and $F : M \rightarrow N$. Then the following are equivalent:

1. F is smooth
2. for each $U \subset M$, if U is open in M , then $F|_U : U \rightarrow N$ is smooth.
3. for each $p \in M$, there exists $U \subset M$ such that $p \in U$, U is open in M and $F|_U : U \rightarrow N$ is smooth.

Proof.

- (1) $\implies$ (2):

Suppose that F is smooth. Let $U \subset M$. Suppose that U is open in M . Let $p \in U$. Since $\mathcal{A}|_U$ is an atlas on U and $\mathcal{B}$ is an atlas on N , there exist $(U_0, \phi_0) \in \mathcal{A}|_U$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U_0$ and $F(p) \in V$. Since $p \in U$, we have that

$$\begin{aligned} F|_U(p) &= F(p) \\ &\in V \end{aligned}$$

An exercise in the section on open submanifolds implies that $\mathcal{A}|_U \subset \mathcal{A}$. Thus $(U_0, \phi_0) \in \mathcal{A}$. Since F is smooth a previous exercise implies that $U_0 \cap F^{-1}(V)$ is open in M and $\psi \circ F \circ \phi_0|_{U_0 \cap F^{-1}(V)}$ is smooth. Since $U_0 \subset U$, we have that

$$\begin{aligned} U_0 \cap F|_U^{-1}(V) &= U_0 \cap (U \cap F^{-1}(V)) \\ &= U_0 \cap F^{-1}(V) \end{aligned}$$

and $\psi \circ F|_U \circ \phi_0|_{U_0 \cap F|_U^{-1}(V)}^{-1} = \psi \circ F \circ \phi_0|_{U_0 \cap F^{-1}(V)}^{-1}$. Thus $U_0 \cap F|_U^{-1}(V)$ is open in U and $\psi \circ F|_U \circ \phi_0|_{U_0 \cap F|_U^{-1}(V)}^{-1}$ is smooth. Since $p \in U$ is arbitrary, we have that for each $p \in U$, there exists $(U_0, \phi_0) \in \mathcal{A}|_U$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U_0$, $F|_U(p) \in V$, $U_0 \cap F|_U^{-1}(V)$ is open in U and $\psi \circ F|_U \circ \phi_0|_{U_0 \cap F|_U^{-1}(V)}^{-1}$ is smooth. (3) in smooth equivalence implies that $F|_U$ is smooth. Since $U \subset M$ with U open in M is arbitrary, we have that for each $U \subset M$, if U is open in M , then $F|_U : U \rightarrow N$ is smooth.

- (2) $\implies$ (3):

Suppose that for each $U \subset M$, if U is open in M , then $F|_U : U \rightarrow N$ is smooth. Let $p \in M$. Since $\mathcal{A}$ is an atlas on M , there exists $(U, \phi) \in \mathcal{A}$ such that $p \in U$. Since $(U, \phi) \in \mathcal{A}$, U is open in M . By assumption, $F|_U : U \rightarrow N$ is smooth. Since $p \in M$ is arbitrary, we have that for each $p \in M$, there exists $U \subset M$ such that $p \in U$, U is open in M and $F|_U : U \rightarrow N$ is smooth.

- (3) $\implies$ (1):

Suppose that for each $p \in M$, there exists $U \subset M$ such that $p \in U$, U is open in M and $F|_U : U \rightarrow N$ is smooth. Let $p \in M$. Let $p \in M$. By assumption, there exists $U \subset M$ such that $p \in U$, U is open in M and $F|_U : U \rightarrow N$ is smooth. Since $F|_U$ is smooth, there exist $(U', \phi) \in \mathcal{A}|_U$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U'$, $F(p) \in V$, $F|_U(U') \subset V$ and $\psi \circ F|_U \circ \phi^{-1}$ is smooth. An exercise in the section on open submanifolds implies that $\mathcal{A}|_U \subset \mathcal{A}$. Thus $(U', \phi) \in \mathcal{A}$. Since $p \in M$ is arbitrary, we have that for each $p \in M$, there exists $(U', \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U'$, $F(p) \in V$, $F(U') \subset V$ and $\psi \circ F \circ \phi^{-1}$ is smooth. Thus F is smooth. □

Exercise 5.2.0.2. Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds, $U \subset M$ and $F : M \rightarrow N$. Suppose that U is open in M . If F is a diffeomorphism, then $F|_U : U \rightarrow F(U)$ is a diffeomorphism.

Proof. Suppose that F is a diffeomorphism. Then F and F^{-1} are smooth. Hence F is a homeomorphism and $F(U)$ is open in N . By definition, F and F^{-1} are smooth. A previous exercise about locality of smoothness implies that $F|_U$ and $F^{-1}|_{F(U)}$ are smooth. Since $F|_U^{-1} = F^{-1}|_{F(U)}$, $F|_U$ is a diffeomorphism. □

Exercise 5.2.0.3. Let $(M, \mathcal{A})$ be a smooth manifold and $(U, \phi) \in \mathcal{A}$. Then $\phi : U \rightarrow \phi(U)$ is a diffeomorphism.

Proof. Set $n := \dim M$. Let $(V, \psi) \in \mathcal{A}$. By definition, ϕ is continuous. Since $(U, \phi), (V, \psi) \in \mathcal{A}$, we have that (U, ϕ) and (V, ψ) are smoothly compatible. Hence $\phi|_{U \cap V} \circ \psi|_{U \cap V}^{-1}$ is a diffeomorphism. Define $\alpha : \psi(U \cap V) \rightarrow \phi(U \cap V)$ by $\alpha = \phi|_{U \cap V} \circ \psi|_{U \cap V}^{-1}$. Since $V \cap \phi^{-1}(\phi(U)) = U \cap V$ and $\phi(U) \cap (\phi^{-1})^{-1}(V) = \phi(U \cap V)$, we have that $V \cap \phi^{-1}(\phi(U))$ and $\phi(U) \cap (\phi^{-1})^{-1}(V)$ are open. Furthermore,

$$\begin{aligned} \text{id}_{\phi(U)} \circ \phi \circ \psi|_{V \cap \phi^{-1}(\phi(U))}^{-1} &= \text{id}_{\phi(U)} \circ \phi \circ \psi|_{V \cap U}^{-1} \\ &= \text{id}_{\phi(U)} \circ \alpha \\ &= \alpha \end{aligned}$$

and

$$\begin{aligned} \psi \circ \phi^{-1} \circ \text{id}_{\phi(U)}|_{\phi(U) \cap (\phi^{-1})^{-1}(V)} &= \psi \circ \phi^{-1} \circ \text{id}_{\phi(U)}|_{\phi(U \cap V)} \\ &= \alpha^{-1} \circ \text{id}_{\phi(U \cap V)} \\ &= \alpha^{-1} \end{aligned}$$

Since α is a diffeomorphism, we have that $\text{id}_{\phi(U)} \circ \phi \circ \psi|_{V \cap \phi^{-1}(\phi(U))}^{-1}$ and $\psi \circ \phi^{-1} \circ \text{id}_{\phi(U)}|_{\phi(U) \cap (\phi^{-1})^{-1}(V)}$ are smooth. Since $(\mathcal{A}|_{\mathbb{H}^n})_{\phi(U)} = \alpha(\text{id}_{\phi(U)})$, $\mathcal{A} = \alpha(\mathcal{A})$ and $(V, \psi) \in \mathcal{A}$ is arbitrary, [a previous exercise about smoothness depending on a smooth atlas](#) implies that ϕ and ϕ^{-1} are smooth. Hence ϕ is a diffeomorphism. $\square$

Exercise 5.2.0.4. Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds and $F : M \rightarrow N$ a diffeomorphism. Then

1. for each $(V, \psi) \in \mathcal{B}$, $(F^{-1}(V), \psi \circ F|_{F^{-1}(V)}) \in \mathcal{A}$
2. for each $(U, \phi) \in \mathcal{A}$, $(F(U), \phi \circ F|_{F^{-1}(U)}) \in \mathcal{B}$

Proof. Set $n := \dim M$.

1. Let $(V, \psi) \in \mathcal{B}$. Since $F^{-1}(V)$ is open in M , [a previous exercise](#) implies that $F|_{F^{-1}(V)}^{-1}$ is a diffeomorphism. [A previous exercise implies that \$\psi\$ is a diffeomorphism.](#) Therefore $\psi \circ F|_{F^{-1}(V)}^{-1}$ is a diffeomorphism.

(a) Since $(V, \psi) \in \mathcal{B}$ and $F|_{F^{-1}(V)}^{-1}$ is a homeomorphism, we have that

- $F^{-1}(V)$ is open in M .
- $\psi(V)$ is open in $\mathbb{H}^n$
- $\psi \circ F|_{F^{-1}(V)} : F^{-1}(V) \rightarrow \psi(V)$ is a homeomorphism

So $(F^{-1}(V), \psi \circ F|_{F^{-1}(V)}) \in \mathcal{A}$.

- (b) Let $(U, \phi) \in \mathcal{A}$. [A previous exercise implies that \$\psi\$ is a diffeomorphism.](#) [A previous exercise](#) implies that $\phi|_{U \cap F^{-1}(V)}$ and $\psi \circ F|_{U \cap F^{-1}(V)}$ are diffeomorphisms. Hence $(\psi \circ F|_{F^{-1}(V)}^{-1})|_{U \cap F^{-1}(V)} \circ \phi|_{U \cap F^{-1}(V)}^{-1}$ is a diffeomorphism. Therefore $(F(U), \psi \circ F|_{F^{-1}(V)}^{-1})$ and (V, ψ) are smoothly compatible. Since $(U, \phi) \in \mathcal{A}$ is arbitrary, we have that for each $(U, \phi) \in \mathcal{A}$, (U, ϕ) and $(F^{-1}(V), \psi \circ F|_{F^{-1}(V)})$ are smoothly compatible. Since $\mathcal{A}$ is maximal, $(F^{-1}(V), \psi \circ F|_{F^{-1}(V)}) \in \mathcal{A}$.

2. Similar to (1).

$\square$

Exercise 5.2.0.5. Let $M \in \text{Obj}(\mathbf{Man}^0)$ and $\mathcal{A}_1, \mathcal{A}_2$ smooth structures on M . Define $\iota : M \rightarrow M$ by $\iota(p) = p$. If $\iota \in \text{Iso}_{\mathbf{ManBnd}^\infty}[(M, \mathcal{A}_1), (M, \mathcal{A}_2)]$, then $\mathcal{A}_1 = \mathcal{A}_2$.

Proof. Set $n := \dim M$. Suppose that ι is a $(\mathcal{A}_1, \mathcal{A}_2)$ -diffeomorphism. Exercise 5.2.0.4 implies that $\mathcal{A}_1 = \mathcal{A}_2$. [maybe give more details.](#) $\square$

Exercise 5.2.0.6. Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds and $F : M \rightarrow N$. Then F is smooth iff for each $(V, \psi) \in \mathcal{B}$ with $\psi = (y^1, \dots, y^n)$, $y^i \circ F$ is smooth.

Proof. Suppose that F is smooth. Let $(V, \psi) \in \mathcal{B}$ with $\psi = (y^1, \dots, y^n)$. Then for each $i \in \{1, \dots, n\}$, F^i is smooth. Conversely, suppose that for each $(V, \psi) \in \mathcal{B}$ with $\psi = (y^1, \dots, y^n)$ and $i \in \{1, \dots, n\}$, $y^i \circ F$ is smooth. $\square$

Definition 5.2.0.7. Let $(N, \mathcal{B})$ be a smooth n -dimensional manifold, $F : M \rightarrow N$ smooth and $(V, \psi) \in \mathcal{B}$ with $\psi = (y^1, \dots, y^n)$. For $i \in \{1, \dots, n\}$, We define the **i -th component of F with respect to (V, ψ)** , denoted $F^i : V \rightarrow \mathbb{R}$, by

$$F^i = y^i \circ F$$

Exercise 5.2.0.8. Let $(M, \mathcal{A})$ be a smooth manifold, $(U, \phi) \in \mathcal{A}$ with $\phi = (x^1, \dots, x^n)$, $p \in U$ and $f \in C^\infty(M, \mathcal{A})$. Then $f|_U \in C^\infty(U, \mathcal{A}|_U)$.

Proof. Let

□

5.3 Smooth Maps and Product Manifolds

Note 5.3.0.1. Let $m \in \mathbb{N}$ and $n \in \mathbb{N}_0$. We recall the definition of $\lambda_0 : \mathbb{H}^m \times \text{Int } \mathbb{H}^n \rightarrow \mathbb{H}^{m+n}$ in Definition 3.3.0.2 by $\lambda((x^1, \dots, x^{m-1}, x^m), (y^1, \dots, y^{n-1}, \log y^n, x^m))$.

Exercise 5.3.0.2. Let $(M, \mathcal{A}), (N, \mathcal{B}), (E, \mathcal{C})$ be smooth manifolds and $F : M \times N \rightarrow E$. Suppose that $\partial N = \emptyset$. Then the following are equivalent:

1. F is smooth
2. there exist $\mathcal{A}_0 \subset \mathcal{A}, \mathcal{B}_0 \subset \mathcal{B}, \mathcal{C}_0 \subset \mathcal{C}$, such that $\mathcal{A}_0$ is an atlas on M , $\mathcal{B}_0$ is an atlas on N , $\mathcal{C}_0$ is an atlas on E and for each $(U, \phi) \in \mathcal{A}_0, (V, \psi) \in \mathcal{B}_0, (W, \chi) \in \mathcal{C}_0, (U \times V) \cap F^{-1}(W)$ is open and $\chi \circ F \circ [\lambda_0 \circ (\phi \times \psi)]|_{(U \times V) \cap F^{-1}(W)}^{-1}$ is smooth.
3. for each $(p, q) \in M \times N$, there exist $(U, \phi) \in \mathcal{A}, (V, \psi) \in \mathcal{B}$ and $(W, \chi) \in \mathcal{C}$ such that $(p, q) \in U \times V, F(p, q) \in W, (U \times V) \cap F^{-1}(W)$ is open in $M \times N$ and $\chi \circ F \circ [\lambda_0 \circ (\phi \times \psi)]|_{(U \times V) \cap F^{-1}(W)}^{-1}$ is smooth.

Proof. Set $m := \dim M, n = \dim N$ and $e = \dim E$.

1. • ($\implies$):
 Suppose that F is smooth. Let $(U, \phi) \in \mathcal{A}_0, (V, \psi) \in \mathcal{B}_0$ and $(W, \chi) \in \mathcal{C}_0$. Set $\eta := \lambda_0|_{\phi(U) \times \psi(V)} \circ (\phi \times \psi)$. By Definition 4.3.0.2 and Definition 4.3.0.4, $\eta \in \mathcal{A} \otimes \mathcal{B}$. Since F is smooth the second characterization in Exercise 5.1.0.10 implies that $(U \times V) \cap F^{-1}(W)$ is open in $M \times N$ and $\chi \circ F \circ \eta|_{(U \times V) \cap F^{-1}(W)}^{-1}$ is smooth.
 Since $(U, \phi) \in \mathcal{A}_0, (V, \psi) \in \mathcal{B}_0$ and $(W, \chi) \in \mathcal{C}_0$ are arbitrary, we have that for each $(U, \phi) \in \mathcal{A}_0, (V, \psi) \in \mathcal{B}_0, (W, \chi) \in \mathcal{C}_0, (U \times V) \cap F^{-1}(W)$ is open in $M \times N$ and $\chi \circ F \circ [\lambda_0 \circ (\phi \times \psi)]|_{(U \times V) \cap F^{-1}(W)}^{-1}$ is smooth.
- ($\impliedby$):
 Suppose that for each $(U, \phi) \in \mathcal{A}_0, (V, \psi) \in \mathcal{B}_0, (W, \chi) \in \mathcal{C}_0, (U \times V) \cap F^{-1}(W)$ is open and $\chi \circ F \circ [\lambda_0 \circ (\phi \times \psi)]|_{(U \times V) \cap F^{-1}(W)}^{-1}$ is smooth. Let $(p, q) \in M \times N$. Since $\mathcal{A}_0$ is an atlas on $M, \mathcal{B}_0$ is an atlas on N and $\mathcal{C}_0$ is an atlas on E , there exist $(U, \phi) \in \mathcal{A}_0, (V, \psi) \in \mathcal{B}_0, (W, \chi) \in \mathcal{C}_0$ such that $p \in U, q \in V$ and $F(p, q) \in W$. Define $\eta := \lambda_0 \circ (\phi \times \psi)|_{(U \times V) \cap F^{-1}(W)}$. Definition 4.3.0.2 and Definition 4.3.0.4 imply that $\eta \in \mathcal{A} \otimes \mathcal{B}$. Set $F_0 := \chi \circ F \circ \eta|_{(U \times V) \cap F^{-1}(W)}^{-1}$. By assumption, $(U \times V) \cap F^{-1}(W)$ is open and F_0 is smooth.
 Since $(p, q) \in M \times N$ is arbitrary, the third characterization in Exercise 5.1.0.10 implies that F is smooth. **FINISH!!!**
2. Similar to (1).

□

Exercise 5.3.0.3. Let $(M, \mathcal{A}), (N, \mathcal{B}), (E, \mathcal{C})$ be smooth manifolds, $G : E \rightarrow M \times N$. Suppose that $\partial N = \emptyset$. Then the following are equivalent:

1. G is smooth iff
2. there exist $\mathcal{A}_0 \subset \mathcal{A}, \mathcal{B}_0 \subset \mathcal{B}, \mathcal{C}_0 \subset \mathcal{C}$ such that $\mathcal{A}_0$ is an atlas on $M, \mathcal{B}_0$ is an atlas on $N, \mathcal{C}_0$ is an atlas on E and for each $(U, \phi) \in \mathcal{A}_0, (V, \psi) \in \mathcal{B}_0, (W, \chi) \in \mathcal{C}_0, [\lambda_0 \circ (\phi \times \psi)] \circ G \circ \chi|_{W \cap G^{-1}(U \times V)}^{-1}$ is smooth.
3. for each $p \in E$, there exist $(W, \chi) \in \mathcal{C}, (U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$ such that $p \in W, G(p) \in U \times V, W \cap F^{-1}(U \times V)$ is open in E and $[\lambda_0 \circ (\phi \times \psi)] \circ G \circ \chi|_{W \cap G^{-1}(U \times V)}^{-1}$ is smooth.

Proof.

1. **FINISH!!!**, need to add detail about set to which we restrict is open or that G is continuous like in the above result
- 2.

□

Exercise 5.3.0.4. We have that $\lambda_0 : \mathbb{H}^m \times \text{Int } \mathbb{H}^n \rightarrow \mathbb{H}^{m+n}$ is a diffeomorphism.

Proof. Define $(U, \phi) \in \mathcal{A}$, $(V, \psi) \in \mathcal{A}_{\mathbb{H}^n} |_{\text{Int } \mathbb{H}^n}$ and $(W, \chi) \in \mathcal{A}_{\mathbb{H}^{m+n}}$ by $(U, \phi) := (\mathbb{H}^m, \text{id}_{\mathbb{H}^m})$, $(V, \psi) := (\text{Int } \mathbb{H}^n, \text{id}_{\text{Int } \mathbb{H}^n})$ and $(W, \chi) := (\mathbb{H}^{m+n}, \text{id}_{\mathbb{H}^{m+n}})$. Set $\mathcal{A}_0 = \{(U, \phi)\}$, $\mathcal{B}_0 = \{(V, \psi)\}$ and $\mathcal{C}_0 := \{(W, \chi)\}$. Then $\mathcal{A}_0$ is a smooth atlas on $\mathbb{H}^m$, $\mathcal{B}_0$ is a smooth atlas on $\text{Int } \mathbb{H}^n$ and $\mathcal{C}_0$ is a smooth atlas on $\mathbb{H}^m$.

Define $F := \lambda_0$, $\eta := \lambda_0 \circ (\phi \times \psi)$ and $F_0 := \chi \circ F \circ \eta|_{(U \times V) \cap F^{-1}(W)}$. We note that for each $(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) \in \lambda_0[\phi \times \psi(U \times V \cap F^{-1}(W))]$,

$$\begin{aligned} F_0(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) &= \chi \circ F \circ \eta|_{(U \times V) \cap \text{proj}_1^{-1}(W)}^{-1}(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) \\ &= \text{id}_{\mathbb{H}^m} \circ \lambda_0 \circ \lambda_0^{-1}(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) \\ &= (a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) \\ &= \text{id}_{\mathbb{H}^{m+n}}(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) \end{aligned}$$

Hence F_0 is smooth. Exercise 5.2.0.1 implies that λ_0 is smooth. Similarly, λ_0^{-1} is smooth. Thus λ_0 is a diffeomorphism. $\square$

Exercise 5.3.0.5. Let $m, n \in \mathbb{N}$. Then

1. $\text{proj}_1 : \mathbb{H}^m \times \text{Int } \mathbb{H}^n \rightarrow \mathbb{H}^m$ is smooth
2. $\text{proj}_2 : \mathbb{H}^m \times \text{Int } \mathbb{H}^n \rightarrow \mathbb{H}^n$ is smooth

Proof.

1. Define $(U, \phi) \in \mathcal{A}$, $(V, \psi) \in \mathcal{A}_{\mathbb{H}^n} |_{\text{Int } \mathbb{H}^n}$ and $(W, \chi) \in \mathcal{A}_{\mathbb{H}^m}$ by $(U, \phi) := (\mathbb{H}^m, \text{id}_{\mathbb{H}^m})$, $(V, \psi) := (\text{Int } \mathbb{H}^n, \text{id}_{\text{Int } \mathbb{H}^n})$ and $(W, \chi) := (\mathbb{H}^m, \text{id}_{\mathbb{H}^m})$. Set $\mathcal{A}_0 = \{(U, \phi)\}$, $\mathcal{B}_0 = \{(V, \psi)\}$ and $\mathcal{C}_0 := \{(W, \chi)\}$. Then $\mathcal{A}_0$ is a smooth atlas on $\mathbb{H}^m$, $\mathcal{B}_0$ is a smooth atlas on $\text{Int } \mathbb{H}^n$ and $\mathcal{C}_0$ is a smooth atlas on $\mathbb{H}^m$.

Define $F := \text{proj}_1$, $\eta := \lambda_0 \circ (\phi \times \psi)$ and $F_0 := \chi \circ F \circ \eta|_{(U \times V) \cap F^{-1}(W)}$. We note that for each $(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) \in \lambda_0[\phi \times \psi(U \times V \cap F^{-1}(W))]$,

$$\begin{aligned} F_0(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) &= \chi \circ F \circ \eta|_{(U \times V) \cap \text{proj}_1^{-1}(W)}^{-1}(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) \\ &= \text{id}_{\mathbb{H}^m} \circ \text{proj}_1 \circ \lambda_0^{-1}(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) \\ &= \text{proj}_1(a^1, \dots, a^m, e^{b^1}, \dots, e^{b^n}) \\ &= (a^1, \dots, a^m) \end{aligned}$$

Hence F_0 is smooth. Exercise 5.2.0.1 implies that proj_1 is smooth.

2. Similar to (1). $\square$

Definition 5.3.0.6. Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds. We define the **projection maps onto M and N** , denoted by $\pi_M : M \times N \rightarrow M$ and $\pi_N : M \times N \rightarrow N$ respectively, by

- $\pi_M(p, q) = p$
- $\pi_N(p, q) = q$

Exercise 5.3.0.7. Let M and N be smooth manifolds. Suppose that $\partial N = \emptyset$. Then

1. $\pi_M : M \times N \rightarrow M$ is smooth,
2. $\pi_N : M \times N \rightarrow N$ is smooth.

Proof.

1. Set $m = \dim M$ and $n = \dim N$.

Let $(p, q) \in M \times N$. Then there exists $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U$ and $q \in V$.

Define $F := \pi_M$, $\eta := \lambda_0 \circ (\phi \times \psi)$ and $F_0 := \phi \circ F \circ \eta|_{(U \times V) \cap F^{-1}(W)}$. We note that for each $(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) \in \lambda_0[\phi \times \psi(U \times V \cap F^{-1}(W))]$,

$$\begin{aligned} F_0(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) &= \chi \circ F \circ \eta|_{(U \times V) \cap \text{proj}_1^{-1}(W)}^{-1}(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) \\ &= \text{id}_{\mathbb{H}^m} \circ \pi_M \circ \lambda_0^{-1} \\ &= (a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) \\ &= \text{id}_{\mathbb{H}^{m+n}}(a^1, \dots, a^{m-1}, b^1, \dots, b^n, a^m) \end{aligned}$$

Hence F_0 is smooth. Exercise 5.2.0.1 implies that λ_0 is smooth. Similarly, λ_0^{-1} is smooth. Thus λ_0 is a diffeomorphism.

Let $(U, \phi), (U', \phi') \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$. Then for each $(a, b) \in \phi(U) \times \psi(V)$

$$\begin{aligned} \phi'|_{U' \cap U} \circ \pi_M \circ [\phi \times \psi]^{-1}|_{\phi(U) \times \psi(V)}(a, b) &= \phi'|_{U' \cap U} \circ \pi_M \circ [\phi|_{\phi(U)}^{-1} \times \psi|_{\psi(V)}^{-1}](a, b) \\ &= \phi' \circ \phi^{-1}(a) \\ &= (\phi' \circ \phi^{-1}) \circ \text{proj}_1(a, b) \end{aligned}$$

Since $(a, b) \in \phi(U) \times \psi(V)$ is arbitrary,

$$\phi'|_{U' \cap U} \circ \pi_M \circ [\phi \times \psi]^{-1}|_{\phi(U \cap U') \times \psi(V)} = \phi'|_{U' \cap U} \circ \phi|_{U \cap U'}^{-1} \circ \text{proj}_1|_{\phi(U \cap U') \times \psi(V)}$$

where $\text{proj}_1 : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the usual projection map. Since $(U, \phi), (U', \phi') \in \mathcal{A}_M$, (U, ϕ) and (U', ϕ') are smoothly compatible. Hence $\phi'|_{U' \cap U} \circ \phi|_{U \cap U'}^{-1}$ is smooth. Since proj_1 is smooth **need to show smooth functions in the calculus sense are smooth in the manifold sense, what does it mean for a projection to be smooth?, BIG ISSSUE, may need to define differentiation on product spaces in calculus section and redo product manifold stuff**, therefore $\phi'|_{U' \cap U} \circ \pi_M \circ [\phi \times \psi]^{-1}|_{\phi(U) \times \psi(V)}$ is smooth. Since **fix here** and $(V, \psi) \in \mathcal{A}_N$ are arbitrary, we have that $\pi_M : M \times N \rightarrow M$ is smooth. we have that (U, ϕ) and (U', ϕ') are smoothly compatible. Thus $\phi'|_{U \cap U'} \circ \phi^{-1}|_{U \cap U'}^{-1}$ is smooth. **FINISH!!!**

2. Similar to (1).

□

Exercise 5.3.0.8. Let M, N, E be smooth manifolds. Suppose that $\partial N = \emptyset$. Let $F : E \rightarrow M \times N$. Then F is smooth iff $\pi_M \circ F$ is smooth and $\pi_N \circ F$ is smooth.

Proof.

- ($\implies$):

Suppose that F is smooth. Exercise ??**previous exercise** implies that π_M and π_N are smooth. Hence $\pi_M \circ F$ and $\pi_N \circ F$ are smooth.

- ($\impliedby$):

Suppose that $\pi_M \circ F$ and $\pi_N \circ F$ are smooth. Let $(U, \phi) \in \mathcal{A}_M$, $(V, \psi) \in \mathcal{A}_N$ and $(W, \eta) \in \mathcal{A}_E$. We note that $(F, G)^{-1}(U \times V) = F^{-1}(U) \cap G^{-1}(V)$. Since F, G are smooth, F, G are continuous. Thus $W \cap F^{-1}(U) \cap G^{-1}(V)$ is open. Set $W' := W \cap F^{-1}(U) \cap G^{-1}(V)$. Then

$$\begin{aligned} (\phi \times \psi) \circ (F, G) \circ \eta|_{W \cap (F, G)^{-1}(U \times V)}^{-1} &= (\phi \circ F, \psi \circ G) \circ \eta|_{W'}^{-1} \\ &= (\phi \circ F \circ \eta|_{W'}^{-1}, \psi \circ G \circ \eta|_{W'}^{-1}). \end{aligned}$$

Since F and G are smooth, $\phi \circ F \circ \eta|_{W'}^{-1}$ and $\psi \circ G \circ \eta|_{W'}^{-1}$ are smooth. Exercise ?? **(make exercise in review of fundys section showing that (F, G) is smooth iff F and G are smooth, where F, G are maps $\mathbb{R}^n \rightarrow \mathbb{R}_1^n$ and $\mathbb{R}^n \rightarrow \mathbb{R}_1^n$ respectively.)** then implies that $(\phi \circ F \circ \eta|_{W'}^{-1}, \psi \circ G \circ \eta|_{W'}^{-1})$ is smooth. Since $(U, \phi) \in \mathcal{A}_M$, $(V, \psi) \in \mathcal{A}_N$ and $(W, \eta) \in \mathcal{A}_E$ are arbitrary, we have that for each $(U, \phi) \in \mathcal{A}_M$, $(V, \psi) \in \mathcal{A}_N$ and $(W, \eta) \in \mathcal{A}_E$, $(\phi \times \psi) \circ (F, G) \circ \eta|_{W \cap (F, G)^{-1}(U \times V)}^{-1}$ is smooth. Exercise 5.3.0.3 then implies that (F, G) is smooth.

FINISH!!! □

Exercise 5.3.0.9. Let M, N, E be smooth manifolds. Suppose that $\partial N = \emptyset$. Let $F : E \rightarrow M$ and $G : E \rightarrow N$. Then (F, G) is smooth iff F and G are smooth.

Proof. Since $\pi_M \circ (F, G) = F$ and $\pi_N \circ (F, G) = G$, Exercise 5.3.0.9 implies that (F, G) is smooth iff F and G are smooth. □

Definition 5.3.0.10. Let M and N be smooth manifolds and $(p, q) \in M \times N$. We define the **slice maps at q and p** , denoted by $\iota_q^M : M \rightarrow M \times N$ and $\iota_p^N : N \rightarrow M \times N$ respectively, by

- $\iota_q^M(a) = (a, q)$
- $\iota_p^N(b) = (p, b)$

Exercise 5.3.0.11. Let M and N be smooth manifolds and $(p, q) \in M \times N$. Then

1. $\iota_q^M : M \rightarrow M \times N$ is smooth,
2. $\iota_p^N : N \rightarrow M \times N$ is smooth.

Proof. Let $()$ □

5.4 Smooth Functions on Manifolds

Definition 5.4.0.1. Let $(M, \mathcal{A})$ be a smooth manifold and $f : M \rightarrow \mathbb{R}$. Then f is said to be **smooth** if for each $(U, \phi) \in \mathcal{A}$, $f \circ \phi^{-1}$ is smooth. The set of all smooth functions on M is denoted $C^\infty(M, \mathcal{A})$.

Note 5.4.0.2. When the context is clear, we write $C^\infty(M)$ in place of $C^\infty(M, \mathcal{A})$.

Exercise 5.4.0.3. Let $(M, \mathcal{A})$ be a smooth manifold and $f : M \rightarrow \mathbb{R}$. Then f is $(\mathcal{A}, \mathcal{A}_{\mathbb{R}})$ -smooth iff for each $(U, \phi) \in \mathcal{A}$, $f \circ \phi^{-1}$ is smooth.

Proof.

- $(\implies)$:
Suppose that for each $(U, \phi) \in \mathcal{A}$, $f \circ \phi^{-1}$ is smooth. Let $(U, \phi) \in \mathcal{A}$. Since $\text{id}_{\mathbb{R}} \circ f \circ \phi^{-1} = f \circ \phi^{-1}$ and $f \circ \phi^{-1}$ is smooth, we have that $\text{id}_{\mathbb{R}} \circ f \circ \phi^{-1}$ is smooth. Since $\mathcal{A} = \alpha(\mathcal{A})$ and $\mathcal{A}_{\mathbb{R}} = \alpha((\mathbb{R}, \text{id}_{\mathbb{R}}))$, [an exercise in the section on smooth maps](#) implies that f is $(\mathcal{A}, \mathcal{A}_{\mathbb{R}})$ -smooth.
- $(\impliedby)$:
Suppose that f is $(\mathcal{A}, \mathcal{A}_{\mathbb{R}})$ -smooth. Let $(U, \phi) \in \mathcal{A}$. Since $(\mathbb{R}, \text{id}_{\mathbb{R}}) \in \mathcal{A}_{\mathbb{R}}$ and $f \circ \phi^{-1} = \text{id}_{\mathbb{R}} \circ f \circ \phi^{-1}$, we have that $f \circ \phi^{-1}$ is smooth. Since $(U, \phi) \in \mathcal{A}$ is arbitrary, we have that for each $(U, \phi) \in \mathcal{A}$, $f \circ \phi^{-1}$ is smooth.

□

Note 5.4.0.4. When the context is clear, we write $C^\infty(M, \mathcal{A})$ in place of $C^\infty(M)$.

Exercise 5.4.0.5. Let $(M, \mathcal{A})$ be a smooth manifold, $\mathcal{A}_0 \subset \mathcal{A}$. Suppose that $\mathcal{A}_0$ is an atlas on M and $f : M \rightarrow \mathbb{R}$. Then f is smooth iff for each $(U, \phi) \in \mathcal{A}_0$, $f \circ \phi^{-1}$ is smooth.

Proof.

- $(\implies)$:
Suppose that f is smooth. Let $(U, \phi) \in \mathcal{A}_0$. Since $\mathcal{A}_0 \subset \mathcal{A}$, $(U, \phi) \in \mathcal{A}$. Since f is smooth, $f \circ \phi^{-1}$ is smooth. Since $(U, \phi) \in \mathcal{A}_0$ is arbitrary, we have that for each $(U, \phi) \in \mathcal{A}_0$, $f \circ \phi^{-1}$ is smooth.
- $(\impliedby)$:
Suppose that for each $(U, \phi) \in \mathcal{A}_0$, $f \circ \phi^{-1}$ is smooth. Then for each $(U, \phi) \in \mathcal{A}_0$, $\text{id}_{\mathbb{R}} \circ f \circ \phi^{-1}$ is smooth. Since $\mathcal{A} = \alpha(\mathcal{A}_0)$ and $\mathcal{A}_{\mathbb{R}} = \alpha((\mathbb{R}, \text{id}_{\mathbb{R}}))$, [an exercise in the section on smooth maps](#) implies that f is $(\mathcal{A}, \mathcal{A}_{\mathbb{R}})$ -smooth. [A previous exercise](#) implies that f is smooth.

□

Exercise 5.4.0.6. Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds and $F : M \rightarrow N$. Then F is smooth iff F is continuous and for each $g \in C^\infty(N)$, $g \circ F$ is smooth.

Proof.

- $(\implies)$:
Suppose that F is smooth. Then F is continuous. Let $g \in C^\infty(N)$. Then $g \circ F$ is smooth. Since $g \in C^\infty(N)$ is arbitrary, we have that for each $g \in C^\infty(N)$, $g \circ F$ is smooth.
- $(\impliedby)$:
Suppose that F is continuous and for each $g \in C^\infty(N)$, $g \circ F$ is smooth. Let $p \in U$.
Let $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$. Set $W = U \cap F^{-1}(V)$. Since F is continuous, W is open in M . Define $G : W \rightarrow V$ by $G := F|_W$.
[FINISH!!!, maybe use bump functions to go from a smooth \$g\$ on \$V\$ to \$N\$](#)

□

Exercise 5.4.0.7. Let M be a smooth manifold. Then $C^\infty(M)$ is a vector space.

Proof. Let $f, g \in C^\infty(M)$, $\lambda \in \mathbb{R}$ and $(U, \phi) \in \mathcal{A}$. By assumption, $f \circ \phi^{-1}$ and $g \circ \phi^{-1}$ are smooth. Hence

$$(f + \lambda g) \circ \phi^{-1} = f \circ \phi^{-1} + \lambda g \circ \phi^{-1}$$

is smooth. Since $(U, \phi) \in \mathcal{A}$ is arbitrary, $f + \lambda g \in C^\infty(M)$. Since $f, g \in C^\infty(M)$ and $\lambda \in \mathbb{R}$ are arbitrary, $C^\infty(M)$ is a vector space. $\square$

Definition 5.4.0.8. Let $(M, \mathcal{A})$ be a smooth manifold, $(U, \phi) \in \mathcal{A}$ with $\phi = (x^1, \dots, x^n)$, $f \in C^\infty(U)$ and $i \in \{1, \dots, n\}$. We define the **partial derivative of f with respect to x^i** , denoted

$$\partial f / \partial x^i \in C^\infty(M),$$

by

$$\frac{\partial f}{\partial x^i}(p) = \frac{\partial}{\partial u^i}[f \circ \phi^{-1}](\phi(p))$$

or equivalently,

$$\frac{\partial f}{\partial x^i} = \left(\frac{\partial}{\partial u^i}[f \circ \phi^{-1}] \right) \circ \phi$$

Note 5.4.0.9. Another common notation for $\partial f / \partial x^i$ is $\partial_i f$.

Exercise 5.4.0.10. Let $(M, \mathcal{A})$ be a smooth manifold, $(U, \phi) \in \mathcal{A}$ with $\phi = (x^1, \dots, x^n)$, $f \in C^\infty(U)$ and $i \in \{1, \dots, n\}$. Then $\partial / \partial x^i : C^\infty(U) \rightarrow C^\infty(U)$ is linear.

Proof. **FINISH!!!** $\square$

Exercise 5.4.0.11. Let $(M, \mathcal{A})$ be a smooth manifold, $(U, \phi) \in \mathcal{A}$ with $\phi = (x^1, \dots, x^n)$, $f \in C^\infty(U)$ and $i, j \in \{1, \dots, n\}$. Then

$$\frac{\partial}{\partial x^i} \frac{\partial}{\partial x^j} f = \left(\frac{\partial}{\partial u^i} \frac{\partial}{\partial u^j} [f \circ \phi^{-1}] \right) \circ \phi$$

Proof.

$$\begin{aligned} \frac{\partial}{\partial x^i} \frac{\partial}{\partial x^j} f &= \frac{\partial}{\partial x^i} \left(\frac{\partial}{\partial x^j} f \right) \\ &= \frac{\partial}{\partial x^i} \left(\left[\frac{\partial}{\partial u^j} [f \circ \phi^{-1}] \right] \circ \phi \right) \\ &= \left(\frac{\partial}{\partial u^i} \left[\left(\left[\frac{\partial}{\partial u^j} [f \circ \phi^{-1}] \right] \circ \phi \right) \circ \phi^{-1} \right] \right) \circ \phi \\ &= \left(\frac{\partial}{\partial u^i} \left[\frac{\partial}{\partial u^j} [f \circ \phi^{-1}] \right] \right) \circ \phi \\ &= \left(\frac{\partial}{\partial u^i} \frac{\partial}{\partial u^j} [f \circ \phi^{-1}] \right) \circ \phi \end{aligned}$$

$\square$

Exercise 5.4.0.12. Let $(M, \mathcal{A})$ be a smooth manifold, $(U, \phi) \in \mathcal{A}$ with $\phi = (x^1, \dots, x^n)$ and $i, j \in \{1, \dots, n\}$. Then

$$\frac{\partial}{\partial x^i} \frac{\partial}{\partial x^j} = \frac{\partial}{\partial x^j} \frac{\partial}{\partial x^i}$$

Proof. Let $f \in C^\infty(U)$. Since $f \circ \phi^{-1}$ is smooth,

$$\frac{\partial}{\partial u^i} \frac{\partial}{\partial u^j} [f \circ \phi^{-1}] = \frac{\partial}{\partial u^j} \frac{\partial}{\partial u^i} [f \circ \phi^{-1}]$$

The previous exercise implies that

$$\begin{aligned}
\frac{\partial}{\partial x^i} \frac{\partial}{\partial x^j} f &= \left(\frac{\partial}{\partial u^i} \frac{\partial}{\partial u^j} [f \circ \phi^{-1}] \right) \circ \phi \\
&= \left(\frac{\partial}{\partial u^j} \frac{\partial}{\partial u^i} [f \circ \phi^{-1}] \right) \circ \phi \\
&= \frac{\partial}{\partial x^j} \frac{\partial}{\partial x^i} f
\end{aligned}$$

□

Exercise 5.4.0.13. Let $(M, \mathcal{A})$ be a smooth manifold, $(U, \phi) \in \mathcal{A}$ with $\phi = (x^1, \dots, x^n)$ and $f \in C^\infty(U)$. Then for each $\alpha \in \mathbb{N}_0^n$,

$$\partial^\alpha f = (\partial^\alpha [f \circ \phi^{-1}]) \circ \phi$$

Proof. The claim is clearly true when $|\alpha| = 0$ or by definition if $|\alpha| = 1$. Let $n \in \mathbb{N}$ and suppose the claim is true for each $|\alpha| \in \{1, \dots, n-1\}$. Then there exists $i \in \{1, \dots, n\}$ such that $\alpha_i \geq 1$. Hence

$$\begin{aligned}
\partial^\alpha f &= \partial^{e^i} (\partial^{\alpha-e^i} f) \\
&= \partial^{e^i} (\partial^{\alpha-e^i} [f \circ \phi^{-1}] \circ \phi) \\
&= (\partial^{e^i} [(\partial^{\alpha-e^i} [f \circ \phi^{-1}] \circ \phi) \circ \phi^{-1}]) \circ \phi \\
&= (\partial^{e^i} [\partial^{\alpha-e^i} [f \circ \phi^{-1}]] \circ \phi) \\
&= (\partial^\alpha [f \circ \phi^{-1}]) \circ \phi
\end{aligned}$$

□

Exercise 5.4.0.14. Taylor's Theorem:

Let $(M, \mathcal{A})$ be a smooth manifold, $(U, \phi) \in \mathcal{A}$ with $\phi = (x^1, \dots, x^n)$ and $\phi(U)$ convex, $p \in U$, $f \in C^\infty(U)$ and $T \in \mathbb{N}$. Then there exist $(g_\alpha)_{|\alpha|=T+1} \subset C^\infty(U)$ such that for each $q \in U$,

$$f(q) = \sum_{k=0}^T \frac{1}{k!} \left[\sum_{|\alpha|=k} \partial^\alpha f(p) (\phi(q) - \phi(p))^\alpha \right] + \frac{1}{(T+1)!} \sum_{|\alpha|=T+1} g_\alpha(q) (\phi(q) - \phi(p))^\alpha$$

and for each $|\alpha| = T+1$,

$$g_\alpha(p) = \partial^\alpha f(p)$$

FIX!!! and put reference.

Proof. Since $\phi(U)$ is open and convex and $f \circ \phi^{-1} \in C^\infty(\phi(U))$, Taylors thorem in section 2.1 implies that there exist $(\tilde{g}_\alpha)_{|\alpha|=T+1} \subset C^\infty(\phi(U))$ such that for each $q \in U$,

$$f \circ \phi^{-1}(\phi(q)) = \sum_{k=0}^T \frac{1}{k!} \left[\sum_{|\alpha|=k} \partial^\alpha [f \circ \phi^{-1}](\phi(p)) (\phi(q) - \phi(p))^\alpha \right] + \frac{1}{(T+1)!} \sum_{|\alpha|=T+1} \tilde{g}_\alpha(\phi(q)) (\phi(q) - \phi(p))^\alpha$$

and for each $|\alpha| = T+1$,

$$\begin{aligned}
\tilde{g}_\alpha(\phi(p)) &= \partial^\alpha [f \circ \phi^{-1}](\phi(p)) \\
&= \partial^\alpha f(p)
\end{aligned}$$

For each $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = T+1$, define $g_\alpha \in C^\infty(U)$ by $g_\alpha := \tilde{g}_\alpha \circ \phi$. Then

$$\begin{aligned}
f(q) &= f \circ \phi^{-1}(\phi(q)) \\
&= \sum_{k=0}^T \frac{1}{k!} \left[\sum_{|\alpha|=k} \partial^\alpha [f \circ \phi^{-1}](\phi(p)) (\phi(q) - \phi(p))^\alpha \right] + \frac{1}{(T+1)!} \sum_{|\alpha|=T+1} \tilde{g}_\alpha(\phi(q)) (\phi(q) - \phi(p))^\alpha \\
&= \sum_{k=0}^T \frac{1}{k!} \left[\sum_{|\alpha|=k} \partial^\alpha f(p) (\phi(q) - \phi(p))^\alpha \right] + \frac{1}{(T+1)!} \sum_{|\alpha|=T+1} g_\alpha(q) (\phi(q) - \phi(p))^\alpha.
\end{aligned}$$

and for each $\alpha \in \mathbb{N}_0^n$ with $|\alpha| = T + 1$,

$$\begin{aligned} g_\alpha(p) &= \tilde{g}_\alpha \circ \phi(p) \\ &= \partial^\alpha f(p). \end{aligned}$$

□

Exercise 5.4.0.15. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $p \in M$. Set $n := \dim N$. Then for each $b \in (U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$. Then for each $b \in \text{FINISH}$ (diff eq question for jet bundles)

Proof. content...

□

Exercise 5.4.0.16. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $(U, \phi) \in \mathcal{A}_M$. Write $\phi = (x^1, \dots, x^n)$. Then for each $i, j \in \{1, \dots, n\}$,

$$\frac{\partial x^k}{\partial x^j} = \delta_{j,k}$$

Proof. Let $i, j \in \{1, \dots, n\}$. Then for each $p \in U$,

$$\begin{aligned} \frac{\partial x^k}{\partial x^j}(p) &= \frac{\partial}{\partial u^j} \Big|_{\phi(p)} x^k \circ \phi^{-1} \\ &= \frac{\partial}{\partial u^j} \Big|_{\phi(p)} u^k \circ \phi \circ \phi^{-1} \\ &= \frac{\partial}{\partial u^j} \Big|_{\phi(p)} u^k \\ &= \delta_{j,k} \end{aligned}$$

□

Exercise 5.4.0.17. Change of Coordinates:

Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $(U, \phi), (V, \psi) \in \mathcal{A}_M$. Write $\phi = (x^1, \dots, x^n)$ and $\psi = (y^1, \dots, y^n)$. Then for each $j \in \{1, \dots, n\}$, $p \in U \cap V$ and $f \in C^\infty(M)$

$$\frac{\partial f}{\partial y^j}(p) = \sum_{k=1}^n \frac{\partial x^k}{\partial y^j}(p) \frac{\partial f}{\partial x^k}(p).$$

Proof. Let $f \in C^\infty(M)$. Set $h := \phi \circ \psi^{-1}$ and write $h = (h^1, \dots, h^n)$. Then $\phi = h \circ \psi$ and $\psi^{-1} = \phi^{-1} \circ h$. By definition and the chain rule, we have that

$$\begin{aligned} \frac{\partial f}{\partial y^j}(p) &= \frac{\partial}{\partial u^j} \Big|_{\psi(p)} f \circ \psi^{-1} \\ &= \frac{\partial}{\partial u^j} \Big|_{\psi(p)} f \circ \phi^{-1} \circ h \\ &= \sum_{k=1}^n \left(\frac{\partial}{\partial u^k} \Big|_{h \circ \psi(p)} f \circ \phi^{-1} \right) \left(\frac{\partial}{\partial u^j} \Big|_{\psi(p)} h^k \right) \\ &= \sum_{k=1}^n \left(\frac{\partial}{\partial u^k} \Big|_{\phi(p)} f \circ \phi^{-1} \right) \left(\frac{\partial}{\partial u^j} \Big|_{\psi(p)} x^k \circ \psi^{-1} \right) \\ &= \sum_{k=1}^n \left(\frac{\partial}{\partial x^k} \Big|_p f \right) \left(\frac{\partial}{\partial y^j} \Big|_p x^k \right) \\ &= \sum_{k=1}^n \frac{\partial x^k}{\partial y^j}(p) \frac{\partial f}{\partial x^k}(p). \end{aligned}$$

□

Exercise 5.4.0.18. Chain Rule:

Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $(U, \phi), (V, \psi) \in \mathcal{A}_M$. Write $\phi = (x^1, \dots, x^n)$ and $\psi = (y^1, \dots, y^n)$. Then for each $j \in \{1, \dots, n\}$, $p \in U \cap V$ and $f \in C^\infty(M)$

$$\frac{\partial f}{\partial y^j}(p) = \sum_{k=1}^n \frac{\partial x^k}{\partial y^j}(p) \frac{\partial f}{\partial x^k}(p).$$

DO CHAIN RULE

Definition 5.4.0.19. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $(U, \phi), (V, \psi) \in \mathcal{A}_M$. Write $\phi = (x^1, \dots, x^n)$ and $\psi = (y^1, \dots, y^n)$.

Definition 5.4.0.20. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$, $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$. Set $m := \dim M$, $n := \dim N$ and write $\phi = (x^1, \dots, x^m)$ and $\psi = (y^1, \dots, y^n)$. Let $I, J \in \mathcal{I}_n^{\otimes k}$. Write $I = (i_1, \dots, i_k)$ and $J = (j_1, \dots, j_k)$. We define $\partial(y^J \circ F)/\partial x^I \in C^\infty(U)$ by

$$\frac{\partial(y^J \circ F)}{\partial x^I} := \prod_{r=1}^k \frac{\partial(y^{i_r} \circ F)}{\partial x^{j_r}}$$

Note 5.4.0.21. If $F = \text{id}_M$, we write $\partial y^J / \partial x^I$ in place of $\partial(y^J \circ \text{id}_M) / \partial x^I$.

Exercise 5.4.0.22. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$, (U, ϕ) and $(V, \psi) \in \mathcal{A}_M$. Set $n := \dim M$ and write $\phi = (x^1, \dots, x^n)$ and $\psi = (y^1, \dots, y^n)$. Let $I, J \in \mathcal{I}_n^{\otimes k}$. Write $I = (i_1, \dots, i_k)$ and $J = (j_1, \dots, j_k)$. Then

$$\frac{\partial}{\partial x^I} = \sum_{J \in \mathcal{I}_{\otimes k}} \frac{\partial y^J}{\partial x^I} \frac{\partial}{\partial y^J}$$

need to redefine/carefully handle notation for $I \in \mathcal{I}_{\otimes k}^n$ and $\alpha \in \mathbb{N}_0^n$ and partial derivatives, we can send $I \mapsto \alpha$ by $\alpha_j := \#\{l \in [k] : i_l = j\}$

Proof. A previous exercise implies that for each $p \in U \cap V$,

$$\begin{aligned} \frac{\partial}{\partial x^I} &= \prod_{r=1}^k \frac{\partial}{\partial x^{i_r}} \\ &= \prod_{r=1}^k \left[\sum_{s_r=1}^n \frac{\partial y^{s_r}}{\partial x^{i_r}} \frac{\partial}{\partial y^{s_r}} \right] \end{aligned}$$

FINISH!!!!

□

5.5 Local Information

5.5.1 Partitions of Unity

Definition 5.5.1.1. Let $p \in M$, $U \in \mathcal{N}^{\text{opn}}(p)$ and $\rho \in C_c^\infty(M)$. Then ρ is said to be a **bump function at p supported in U** if

1. $\rho \geq 0$
2. there exists $V \in \mathcal{N}^{\text{opn}}(p)$ such that $\rho|_V = 1$
3. $\text{supp } \rho \subset U$

maye need to change, typically bump functions ρ are 1 on a compact K and $\text{supp } \rho \subset U$ where U is open and $K \subset U$

Exercise 5.5.1.2. Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(t) = \begin{cases} e^{-\frac{1}{1-t^2}} & t \in (-1, 1) \\ 0 & t \notin (-1, 1) \end{cases}$$

Then $f \in C_c^\infty(\mathbb{R})$.

Proof. □

5.5.2 Germs of Functions

Definition 5.5.2.1. Let $M, N \in \text{Man}^\infty$ and $p \in M$.

- Let $F, G \in C^\infty(M, N)$. Then F is said to **share the same germ at p** , denoted $F \sim_p G$, if there exists $U \in \mathcal{N}^{\text{opn}}(p)$ such that $F|_U = G|_U$.
- We define the **germ of $C^\infty(M, N)$ at p** , denoted $\text{germ}_p^\infty(M, N) \subset C^\infty(M, N) \times C^\infty(M, N)$ by

$$\text{germ}_p^\infty(M, N) := \{(F, G) \in C^\infty(M, N) \times C^\infty(M, N) : F \sim_p G\}.$$

Note 5.5.2.2. When the context is clear, we may write germ_p^∞ in place of $\text{germ}_p^\infty(M, N)$.

Exercise 5.5.2.3. Let $M, N \in \text{Man}^\infty$ and $p \in M$. Then $\text{germ}_p^\infty \in \text{Equiv}(C^\infty(M))$.

Proof. Clear. **FINISH!!!** □

Definition 5.5.2.4. Let $M, N \in \text{Man}^\infty$ and $p \in M$. We define $C_p^\infty(M, N) := C^\infty(M, N) / \text{germ}_p^\infty$.

Exercise 5.5.2.5. We have that

1. $\text{add} : C^\infty(M) \times C^\infty(M) \rightarrow C^\infty(M)$ is $((\text{germ}_p^\infty)^{\times 2}, \text{germ}_p^\infty)$ -invariant
- 2.
3. $\text{scalar_mult} : C^\infty(M) \times C^\infty(M) \rightarrow C^\infty(M)$ is $((\text{germ}_p^\infty)^{\times 2}, \text{germ}_p^\infty)$ -invariant
4. $\text{mult} : C^\infty(M) \times C^\infty(M) \rightarrow C^\infty(M)$ is $((\text{germ}_p^\infty)^{\times 2}, \text{germ}_p^\infty)$ -invariant

Proof. Clear. **FINISH!!!** □

Definition 5.5.2.6. Let $M \in \text{Man}^\infty$ and $p \in M$. Since $\text{add}, \text{scalar_mult}, \text{mult}$ are $((\text{germ}_p^\infty)^{\times 2}, \text{germ}_p^\infty)$ -invariant, **ref universal algebra notes** implies that

- there exists a unique $\text{add}_p : C_p^\infty(M) \times C_p^\infty(M) \rightarrow C_p^\infty(M)$ such that $\pi_{\text{germ}_p^\infty} \circ \text{add} = \text{add}_p \circ \pi_{\text{germ}_p^\infty}^{\times 2}$, i.e. the following diagram commutes:

$$\begin{array}{ccc} C^\infty(M) \times C^\infty(M) & \xrightarrow{\text{add}} & C^\infty(M) \\ \pi_{\text{germ}_p^\infty}^{\times 2} \downarrow & & \downarrow \pi_{\text{germ}_p^\infty} \\ C_p^\infty(M) \times C_p^\infty(M) & \xrightarrow{\text{add}_p} & C_p^\infty(M) \end{array}$$

- **FINISH!!!** for `scalar_multp`
- there exists a unique $\text{mult}_p : C_p^\infty(M) \times C_p^\infty(M) \rightarrow C_p^\infty(M)$ such that $\pi_{\text{germ}_p^\infty} \circ \text{mult} = \text{mult}_p \circ \pi_{\text{germ}_p^\infty}^{\times 2}$, i.e. the following diagram commutes:

$$\begin{array}{ccc}
 C^\infty(M) \times C^\infty(M) & \xrightarrow{\text{mult}} & C^\infty(M) \\
 \pi_{\text{germ}_p^\infty}^{\times 2} \downarrow & & \downarrow \pi_{\text{germ}_p^\infty} \\
 C_p^\infty(M) \times C_p^\infty(M) & \xrightarrow{\text{mult}_p} & C_p^\infty(M)
 \end{array}$$

Note 5.5.2.7. When the context is clear, we may write $f + g$, λf and fg in place of $\text{add}_p(f, g)$, $\text{scalar_mult}_p(\lambda, f)$ and $\text{mult}_p(f, g)$ respectively.

Exercise 5.5.2.8. Let $M \in \text{Man}^\infty$ and $p \in M$. Then

1. $(C_p^\infty(M), \text{add}_p, \text{scalar_mult}_p, \text{mult}_p) \in \text{Alg}_{\mathbb{R}}$.
2. $\pi_{\text{germ}_p^\infty} \in \text{Hom}_{\text{Alg}_{\mathbb{R}}}(C^\infty(M), C_p^\infty(M))$
3. $\ker \pi_{\text{germ}_p^\infty} = \pi_{\text{germ}_p^\infty}(0)$.

Proof. **should be clear from universal algebra**

- 1.
- 2.
- 3.

FINISH!!!

□

Chapter 6

The Tangent and Cotangent Spaces

6.1 The Tangent Space

6.1.1 Derivations

Definition 6.1.1.1. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $(U, \phi) \in \mathcal{A}$ and $p \in U$. Set $n := \dim M$ and write $\phi = (x^1, \dots, x^n)$. For $j \in [n]$, we define the **partial derivative operator with respect to x^i at p** , denoted

$$\left. \frac{\partial}{\partial x^i} \right|_p : C^\infty(M) \rightarrow \mathbb{R},$$

by

$$\left. \frac{\partial}{\partial x^i} \right|_p f := \frac{\partial f}{\partial x^i}(p).$$

Note 6.1.1.2. Another common notation for $\partial/\partial x^i|_p$ is $\partial_i|_p$.

Exercise 6.1.1.3. Change of Coordinates:

Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $(U, \phi), (V, \psi) \in \mathcal{A}$, $p \in U \cap V$ and $f \in C^\infty(M)$. Set $n := \dim M$ and write $\phi = (x^1, \dots, x^n)$, $\psi = (y^1, \dots, y^n)$. Then for each $j \in \{1, \dots, n\}$,

$$\left. \frac{\partial}{\partial y^j} \right|_p = \sum_{k=1}^n \frac{\partial x^k}{\partial y^j}(p) \left. \frac{\partial}{\partial x^k} \right|_p.$$

Proof. Clear by [exercise in previous section on smooth functions on manifolds](#) □

Definition 6.1.1.4. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $p \in M$.

- Let $v : C^\infty(M) \rightarrow \mathbb{R}$. Then v is said to be

- **Leibnizian at p** if for each $f, g \in C^\infty(M)$,

$$v(fg) = v(f)g(p) + f(p)v(g)$$

- a **derivation on $C^\infty(M)$ at p** if for each $f, g \in C^\infty(M)$ and $a \in \mathbb{R}$,

1. v is linear
2. v is Leibnizian

- We define $\text{Deriv}^\infty_p(M) \subset \mathbb{R}^{C^\infty(M)}$ by $\text{Deriv}^\infty_p(M) := \{v \in \mathbb{R}^{C^\infty(M)} : v \text{ is a derivation on } C^\infty(M) \text{ at } p\}$.

Exercise 6.1.1.5. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $p \in M$ and $v \in \text{Deriv}^\infty_p(M)$. Then v is germ_p^∞ -invariant.

Hint: bump function and Leibnizianity

Proof. Let $f, g \in C^\infty(M)$. Suppose that $f \sim_p g$. Then there exists $U \in \mathcal{N}^{\text{opn}}(p)$ such that $f|_U = g|_U$. Define $h \in C^\infty(M)$ by $h := f - g$. Then $h|_U = 0$. Since $\{p\} \in \mathcal{K}(M)$ and $U \in \mathcal{N}^{\text{opn}}(p)$, **make/ref ex on bump functions and partition of unity** implies that there exists $\rho \in C^\infty(M)$ such that $\rho(p) = 0$ and $\rho|_{U^c} \equiv 1$. Then $\rho h = h$ and

$$\begin{aligned} v(h) &= v(\rho h) \\ &= v(\rho)h(p) + \rho(p)v(h) \\ &= v(\rho)0 + 0v(h) \\ &= 0. \end{aligned}$$

Thus

$$\begin{aligned} v(f) &= v(g + h) \\ &= v(g) + v(h) \\ &= v(g). \end{aligned}$$

Since $f, g \in C^\infty(M)$ with $f \sim_p g$ are arbitrary, we have that for each $f, g \in C^\infty(M)$, $f \sim_p g$ implies that $v(f) = v(g)$. Thus v is germ_p^∞ -invariant. $\square$

Note 6.1.1.6. Let $v \in \text{Deriv}_p^\infty(M)$. Since v is germ_p^∞ -invariant, **(ref ex)** implies that there exists a unique $\bar{v} : C_p^\infty \rightarrow \mathbb{R}$ such that $v = \bar{v} \circ \pi_{\text{germ}_p^\infty}$, i.e the following diagram commutes:

$$\begin{array}{ccc} C^\infty(M) & & \\ \pi_{\text{germ}_p^\infty} \downarrow & \searrow v & \\ C_p^\infty(M) & \xrightarrow{\bar{v}} & \mathbb{R} \end{array}$$

Definition 6.1.1.7. Let $M \in \text{Man}^\infty$ and $p \in M$.

Let $v \in \text{Deriv}_p^\infty(M)$. We define the **tangent vector to M at p induced by v** , denoted $v_p : C_p^\infty(M) \rightarrow \mathbb{R}$, to be the unique $v_p : C_p^\infty \rightarrow \mathbb{R}$ such that $v = v_p \circ \pi_{\text{germ}_p^\infty}$, i.e the following diagram commutes:

$$\begin{array}{ccc} C^\infty(M) & & \\ \pi_{\text{germ}_p^\infty} \downarrow & \searrow v & \\ C_p^\infty(M) & \xrightarrow{v_p} & \mathbb{R} \end{array}$$

We define the **tangent space of M at p** , denoted $T_p M$, by

$$T_p M = \{v_p \in \mathbb{R}^{C_p^\infty(M)} : v \in \text{Deriv}_p^\infty(M)\}$$

Definition 6.1.1.8. Let $M \in \text{Obj}(\text{ManBnd}^\infty)$ and $p \in M$.

Exercise 6.1.1.9. $T_p M$ is a vector space

Proof. **FINISH!!!**

$\square$

Note 6.1.1.10. We often identify $\text{Deriv}_p^\infty(M)$ with $T_p M$. **and abuse notation**

Exercise 6.1.1.11. Exercise 5.4.0.10

Exercise 6.1.1.12. Let $f \in C^\infty(M)$ and $v \in T_p M$. If f is constant, then $vf = 0$.

Proof. Suppose that $f = 1$. Then $f^2 = f$ and $v(f^2) = 2v(f)$. So $v(f) = 2v(f)$ which implies that $v(f) = 0$. If $f \neq 1$, then there exists $c \in \mathbb{R}$ such that $f = c$. Since v is linear, $v(f) = cv(1) = 0$. $\square$

Exercise 6.1.1.13. Let $(U, \phi) \in \mathcal{A}$ with $\phi = (x^1, \dots, x^n)$ and $p \in U$. Then

$$\left\{ \frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p \right\}$$

is a basis for $T_p M$ and $\dim T_p M = n$.

Proof. Clearly $\frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p \in T_p M$. Let $a_1, \dots, a_n \in \mathbb{R}$. Suppose that

$$v = \sum_{i=1}^n a_i \frac{\partial}{\partial x^i} \Big|_p = 0$$

Then

$$\begin{aligned} 0 &= vx^j \\ &= \sum_{i=1}^n a_i \frac{\partial}{\partial x^i} \Big|_p x^j \\ &= a_j \end{aligned}$$

Hence $\left\{ \frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p \right\}$ is independent.

Now, let $v \in T_p M$ and $f \in C^\infty(M)$. By Taylor's theorem, there exist $g_1, \dots, g_n \in C_p^\infty(M)$ such that

$$f = f(p) + \sum_{i=1}^n (x^i - x^i(p))g_i$$

and for each $i \in \{1, \dots, n\}$,

$$g_i(p) = \frac{\partial}{\partial x^i} \Big|_p f.$$

Then

$$\begin{aligned} v(f) &= \sum_{i=1}^n v(x^i - x^i(p))g_i(p) + \sum_{i=1}^n (x^i(p) - x^i(p))v(g_i) \\ &= \sum_{i=1}^n v(x^i)g_i(p) \\ &= \sum_{i=1}^n v(x^i) \frac{\partial}{\partial x^i} \Big|_p f \\ &= \left[\sum_{i=1}^n v(x^i) \frac{\partial}{\partial x^i} \Big|_p \right] f \end{aligned}$$

Thus

$$v = \sum_{i=1}^n v(x^i) \frac{\partial}{\partial x^i} \Big|_p$$

and

$$v \in \text{span} \left\{ \frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p \right\}$$

□

Definition 6.1.1.14. Let $(N, \mathcal{B})$ be a smooth manifold, $F : M \rightarrow N$ smooth and $p \in M$. We define the **derivative of F at p** , denoted $DF_p : T_p M \rightarrow T_{F(p)} N$, by

$$\left[DF_p(v) \right] (f) = v(f \circ F)$$

for $v \in T_p M$ and $f \in C^\infty(N)$.

Exercise 6.1.1.15. Let $(N, \mathcal{B})$ be a smooth manifold, $F : M \rightarrow N$ smooth and $p \in M$. Then for each $v \in T_p M$, $DF_p(v)$ is a derivation.

Proof. Let $v \in T_p M$, $f, g \in C_{F(p)}^\infty(N)$ and $c \in \mathbb{R}$. Then

1.

$$\begin{aligned} DF_p(v)(f + cg) &= v((f + cg) \circ F) \\ &= v(f \circ F + cg \circ F) \\ &= v(f \circ F) + cv(g \circ F) \\ &= DF_p(v)(f) + cDF_p(v)(g) \end{aligned}$$

So $DF_p(v)$ is linear.

2.

$$\begin{aligned} DF_p(v)(fg) &= v(fg \circ F) \\ &= v((f \circ F) * (g \circ F)) \\ &= v(f \circ F) * (g \circ F)(p) + (f \circ F)(p) * v(g \circ F) \\ &= DF_p(v)(f) * g(F(p)) + f(F(p)) * DF_p(v)(g) \end{aligned}$$

So $DF_p(v)$ is Leibnizian and hence $DF_p(v) \in T_{F(p)} N$ □

Exercise 6.1.1.16. Let $(N, \mathcal{B})$ be a smooth manifold, $F : M \rightarrow N$ smooth and $p \in M$. If F is a diffeomorphism, then DF_p is an isomorphism.

Proof. Suppose that F is a diffeomorphism. Since F is a homeomorphism, $\dim N = n$. Choose $(U, \phi) \in \mathcal{A}$ such that $p \in U$. A previous exercise tells us that $(F(U), \phi \circ F^{-1}) \in \mathcal{B}$. Write $\phi = (x^1, \dots, x^n)$ and $\phi \circ F^{-1} = (y^1, \dots, y^n)$. Let $f \in C^\infty(N)$. Then

$$\begin{aligned} \left. \frac{\partial}{\partial y^i} \right|_{F(p)} f &= \left. \frac{\partial}{\partial u^i} \right|_{\phi \circ F^{-1}(F(p))} f \circ (\phi \circ F^{-1})^{-1} \\ &= \left. \frac{\partial}{\partial u^i} \right|_{\phi(p)} f \circ F \circ \phi^{-1} \\ &= \left. \frac{\partial}{\partial x^i} \right|_p f \circ F \end{aligned}$$

Therefore

$$\begin{aligned} \left[DF(p) \left(\left. \frac{\partial}{\partial x^i} \right|_p \right) \right] (f) &= \left. \frac{\partial}{\partial x^i} \right|_p f \circ F \\ &= \left. \frac{\partial}{\partial y^i} \right|_{F(p)} f \end{aligned}$$

Hence

$$DF(p) \left(\left. \frac{\partial}{\partial x^i} \right|_p \right) = \left. \frac{\partial}{\partial y^i} \right|_{F(p)}$$

Since $\left\{ \left. \frac{\partial}{\partial x^1} \right|_p, \dots, \left. \frac{\partial}{\partial x^n} \right|_p \right\}$ is a basis for $T_p M$ and $\left\{ \left. \frac{\partial}{\partial y^1} \right|_{F(p)}, \dots, \left. \frac{\partial}{\partial y^n} \right|_{F(p)} \right\}$ is a basis for $T_{F(p)} N$, $DF(p)$ is an isomorphism. □

Exercise 6.1.1.17. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $(U, \phi) \in \mathcal{A}_M$ and $p \in U$. Write $\phi = (x^1, \dots, x^n)$. Then for each $j \in [n]$,

$$D\phi(p) \left(\frac{\partial}{\partial x^j} \Big|_p \right) = \frac{\partial}{\partial u^j} \Big|_{\phi(p)}$$

Proof. Let $j \in [n]$, $f \in C_{\phi(p)}^\infty(\phi(U))$. Then

$$\begin{aligned} D\phi(p) \left(\frac{\partial}{\partial x^j} \Big|_p \right) (f) &= \frac{\partial}{\partial x^j} \Big|_p [f \circ \phi] \\ &= \frac{\partial}{\partial u^j} \Big|_{\phi(p)} [f \circ \phi \circ \phi^{-1}] \\ &= \frac{\partial}{\partial u^j} \Big|_{\phi(p)} (f). \end{aligned}$$

Since $f \in C_{\phi(p)}^\infty(\phi(U))$ is arbitrary, we have that for each $f \in C_{\phi(p)}^\infty(\phi(U))$,

$$D\phi(p) \left(\frac{\partial}{\partial x^j} \Big|_p \right) (f) = \frac{\partial}{\partial u^j} \Big|_{\phi(p)} (f).$$

Thus

$$D\phi(p) \left(\frac{\partial}{\partial x^j} \Big|_p \right) = \frac{\partial}{\partial u^j} \Big|_{\phi(p)}.$$

□

Definition 6.1.1.18. Let $(M, \mathcal{A})$ be a smooth m -dimensional manifold, $(N, \mathcal{B})$ a n -dimensional smooth manifold, $F : M \rightarrow N$ smooth, $(U, \phi) \in \mathcal{A}$, $(V, \psi) \in \mathcal{B}$ and $p \in U \cap F^{-1}(V)$. Write $\phi = (x^1, \dots, x^m)$ and $\psi = (y^1, \dots, y^n)$. Define the ordered bases $B_\phi := \left\{ \frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^m} \Big|_p \right\}$ of $T_p M$ and $B_\psi := \left\{ \frac{\partial}{\partial y^1} \Big|_{F(p)}, \dots, \frac{\partial}{\partial y^n} \Big|_{F(p)} \right\}$ of $T_{F(p)} N$. We define the **matrix representation of $DF(p)$ with respect to ϕ and ψ** , denoted $[DF(p)]_{\phi, \psi} \in \mathbb{R}^{n \times m}$, by

$$[DF(p)]_{\phi, \psi} := [DF(p)]_{B_\phi, B_\psi}$$

Exercise 6.1.1.19. Let $(M, \mathcal{A})$ be a smooth m -dimensional manifold, $(N, \mathcal{B})$ a n -dimensional smooth manifold, $F : M \rightarrow N$ smooth, $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$. Write $\phi = (x^1, \dots, x^m)$ and $\psi = (y^1, \dots, y^n)$. Suppose that $p \in U \cap F^{-1}(V)$. Then the matrix representation of DF_p with respect to the bases B_ϕ and B_ψ is

$$([DF(p)]_{\phi, \psi})_{j, k} = \frac{\partial(y^j \circ F)}{\partial x^k}(p)$$

Proof. Let $[DF(p)]_{\phi, \psi} = (a_{j, k})_{j, k} \in \mathbb{R}^{n \times m}$. Then for each $k \in [m]$,

$$DF(p) \left(\frac{\partial}{\partial x^k} \Big|_p \right) = \sum_{j=1}^n a_{j, k} \frac{\partial}{\partial y^j} \Big|_{F(p)}$$

This implies that for each $k, l \in [m]$,

$$\begin{aligned} DF(p) \left(\frac{\partial}{\partial x^k} \Big|_p \right) (y^l) &= \sum_{j=1}^n a_{j, k} \frac{\partial}{\partial y^j} \Big|_{F(p)} (y^l) \\ &= \sum_{j=1}^n a_{j, k} \delta_{j, l} \\ &= a_{l, k} \end{aligned}$$

By definition,

$$\begin{aligned} a_{j,k} &= DF_p \left(\frac{\partial}{\partial x^k} \Big|_p \right) (y^j) \\ &= \frac{\partial}{\partial x^k} \Big|_p (y^j \circ F) \\ &= \frac{\partial (y^j \circ F)}{\partial x^k} (p). \end{aligned}$$

□

Note 6.1.1.20. Since $\text{rank } DF_p$ is independent of basis, it is independent of coordinate charts $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$.

Exercise 6.1.1.21. need exercise giving $\sigma\phi$ has derivative $P_\sigma D\phi$.

6.1.2 Velocities along Paths

Definition 6.1.2.1. define calc I derivative and identiyfy with new manifold derifivative definition, show $f'(0)$ can be identified with

Definition 6.1.2.2. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$. We define the **paths on** M , denoted $\text{Paths}(M)$, by $\text{Paths}(M) := \coprod_{I \in \mathcal{T}_\mathbb{R}} \text{Hom}_{\mathbf{ManBnd}^\infty}(I, M)$

Definition 6.1.2.3. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $(I, \gamma) \in \text{Paths}(M)$ and $t_0 \in I$. We define the **velocity of** γ **at** t_0 , denoted $\gamma'(t_0) \in T_p M$, by

$$\gamma'(t_0) := D\gamma(t_0) \left(\frac{d}{dt} \Big|_{t_0} \right).$$

Note 6.1.2.4. Another common notation for $\gamma'(t_0)$ is $\dot{\gamma}(t_0)$.

Definition 6.1.2.5. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $(U, \phi) \in \mathcal{A}_M$. Set $n := \dim M$ and write $\phi = (x^1, \dots, x^n)$. Let $(I, \gamma) \in \text{Paths}(M)$. Suppose that $I \subset U$. For each $k \in [n]$, we define the **k -th component functions of γ with respect to (U, ϕ)** , denoted $\gamma^k \in \text{Hom}_{\mathbf{ManBnd}^\infty}(I, \mathbb{R})$, by $\gamma^k(t) := \text{proj}^{\lfloor n \rfloor}_k \circ \phi \circ \gamma$.

Let $\gamma \in \text{Hom}_{\mathbf{ManBnd}^\infty}(I, M)$, $t_0 \in I$ and $(U, \phi) \in \mathcal{A}_M$. Write $\phi = (x^1, \dots, x^n)$. Define $p \in M$ by $p := \gamma(t_0)$. If $p \in U$, then $\gamma'(t_0) =$

Exercise 6.1.2.6. Let $I \in \mathcal{T}_\mathbb{R}$, $t_0 \in \mathbb{R}$. Then

Exercise 6.1.2.7. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $I \in \mathcal{T}_\mathbb{R}$, $\gamma \in \text{Hom}_{\mathbf{ManBnd}^\infty}(I, M)$ and $t_0 \in I$. Define $p \in M$ by $p := \gamma(t_0)$. Then $(f \circ \gamma)'(t_0) = Df(p)(\gamma'(t_0))$.

Proof. By definition,

$$\begin{aligned} (f \circ \gamma)'(t_0) &= D(f \circ \gamma)(t_0) \left(\frac{d}{dt} \Big|_{t_0} \right) \\ &= Df(\gamma(t_0)) \circ D\gamma(t_0) \left(\frac{d}{dt} \Big|_{t_0} \right) \\ &= Df(p) \circ D\gamma(t_0) \left(\frac{d}{dt} \Big|_{t_0} \right) \\ &= Df(p) \left[D\gamma(t_0) \left(\frac{d}{dt} \Big|_{t_0} \right) \right] \\ &= Df(p)(\gamma'(t_0)). \end{aligned}$$

□

Definition 6.1.2.8. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$.

- For each $\epsilon > 0$, we define the **open interval of radius ϵ centered at 0**, denoted $I_\epsilon \subset \mathbb{R}$, by $I_\epsilon := (-\epsilon, \epsilon)$.
- We define the **paths on M** , denoted $\text{Paths}(M)$, by $\text{Paths}(M) := \bigcup_{\epsilon > 0} \text{Hom}_{\mathbf{ManBnd}^\infty}(I_\epsilon, M)$
- For each $p \in M$, we define
 - the **paths on M centered at p** , denoted $\text{Paths}_p(M) \subset \text{Paths}(M)$, by $\text{Paths}_p(M) := \{\gamma \in \text{Paths}(M) : \gamma(0) = p\}$
 - $\sim_p^v \subset \text{Paths}_p(M) \times \text{Paths}_p(M)$ by $\gamma_1 \sim_p^v \gamma_2$ iff for each $f \in C^\infty(M)$, $[f \circ \gamma_1]'(0) = [f \circ \gamma_2]'(0)$.

Exercise 6.1.2.9. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $p \in M$. Then $\sim_p^v$ is an equivalence relation on $\text{Paths}_p(M)$.

Proof. Clear. **FINISH!!!** □

Exercise 6.1.2.10. Define $\tan_p : \text{Paths}_p(M) \rightarrow T_p M$ by $\tan_p(\gamma) := \gamma'(0)$. Then $\tan_p$ is $\sim_p^v$ -invariant.

Proof. Let $\gamma_1, \gamma_2 \in \text{Paths}_p(M)$. Suppose that $\gamma_1 \sim_p^v \gamma_2$. Then for each $f \in C^\infty(M)$,

$$\begin{aligned}
 Df(p) \circ D\gamma_1(0) \left(\left(\frac{d}{dt} \right) \Big|_{t_0} \right) &= Df(\gamma_1(t_0)) \circ D\gamma_1(t_0)(0) \\
 &= D(f \circ \gamma_1)'(0) \\
 &= (f \circ \gamma_1)'(0) \\
 &= D(f \circ \gamma_2)'(0) \\
 &= Df(\gamma_2(t_0)) \circ D\gamma_2(t_0) \left(\left(\frac{d}{dt} \right) \Big|_{t_0} \right) \\
 &= Df(p) \circ D\gamma_2(0) \left(\left(\frac{d}{dt} \right) \Big|_{t_0} \right).
 \end{aligned}$$
□

Definition 6.1.2.11. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $p \in M$. We define $V_p M \subset \mathcal{P}(\text{Paths}_p(M))$ by $V_p M := \text{Paths}_p(M) / \sim_p^v$.

Exercise 6.1.2.12. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $p \in M$. Then $\text{Iso}_{\mathbf{Vect}_\mathbb{R}}(V_p M, T_p M) \neq \emptyset$.

Proof. Define $A : V_p M \rightarrow T_p M$ by A/π_p . □

6.2 Tangent Space of Product Manifolds

Exercise 6.2.0.1. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$. Set $m := \dim M$ and $n := \dim N$. Let $(U_M, \phi_M) \in \mathcal{A}_M$ and $(U_N, \phi_N) \in \mathcal{A}_N$. Write $\phi_M = (x^1, \dots, x^m)$ and $\phi_N = (y^1, \dots, y^n)$. Define $(U, \phi) \in \mathcal{A}_M \otimes \mathcal{A}_N$ by $U := U_M \times U_N$ and $\phi := \phi_M \times \phi_N$. Write $\phi = (\tilde{x}^1, \dots, \tilde{x}^m, \tilde{y}^1, \dots, \tilde{y}^n)$. Then

1. for each $j \in [m]$, $k \in [n]$ and $(p, q) \in M \times N$,

$$\begin{aligned}
 \frac{\partial}{\partial \tilde{x}^k} \Big|_{(p,q)} (x^j \circ \pi_M) &= \frac{\partial}{\partial x^k} \Big|_p (x^j), & \frac{\partial}{\partial \tilde{y}^k} \Big|_{(p,q)} (x^j \circ \pi_M) &= 0, \\
 \frac{\partial}{\partial \tilde{x}^k} \Big|_{(p,q)} (y^j \circ \pi_N) &= 0, & \frac{\partial}{\partial \tilde{y}^k} \Big|_{(p,q)} (y^j \circ \pi_N) &= \frac{\partial}{\partial y^k} \Big|_q (y^j).
 \end{aligned}$$

2. $[D\pi_M(p, q)]_{\phi_M, \phi} = (I_m \ 0)$ and $[D\pi_N(p, q)]_{\phi_N, \phi} = (0 \ I_n)$

Proof.

1. Let $j \in [m]$, $k \in [n]$ and $(p, q) \in M \times N$. Let $(u^i, v^j) \in \mathbb{R}^{m+n}$ denote the usual coordinates (use wording used elsewhere). Then Exercise ?? implies that

$$\begin{aligned}
 \frac{\partial}{\partial \tilde{x}^k} \Big|_{(p,q)} (x^j \circ \pi_M) &= \frac{\partial}{\partial u^k} \Big|_{\phi(p,q)} (x^j \circ \pi_M \circ \phi^{-1}) \\
 &= \frac{\partial}{\partial u^k} \Big|_{\phi(p,q)} (x^j \circ \phi_M^{-1} \circ \text{proj}_{[m]}) \\
 &= \sum_{l=1}^m \frac{\partial(x^j \circ \phi_M^{-1})}{\partial u^l}(\phi_M(p)) \frac{\partial(u^l \circ \text{proj}_{[m]})}{\partial u^k}(\phi(p, q)) \\
 &= \sum_{l=1}^m \frac{\partial(x^j \circ \phi_M^{-1})}{\partial u^l}(\phi_M(p)) \delta_{l,k} \\
 &= \frac{\partial(x^j \circ \phi_M^{-1})}{\partial u^k}(\phi_M(p)) \\
 &= \frac{\partial}{\partial u^k} \Big|_{\phi_M(p)} x^j \circ \phi_M^{-1} \\
 &= \frac{\partial}{\partial x^k} \Big|_p x^j
 \end{aligned}$$

and

$$\begin{aligned}
 \frac{\partial}{\partial \tilde{y}^k} \Big|_{(p,q)} (x^j \circ \pi_M) &= \frac{\partial}{\partial v^k} \Big|_{\phi(p,q)} (x^j \circ \pi_M \circ \phi^{-1}) \\
 &= \frac{\partial}{\partial v^k} \Big|_{\phi(p,q)} (x^j \circ \phi_M^{-1} \circ \text{proj}_{[m]}) \\
 &= \sum_{l=1}^m \frac{\partial(x^j \circ \phi_M^{-1})}{\partial u^l}(\phi_M(p)) \frac{\partial(u^l \circ \text{proj}_{[m]})}{\partial v^k}(\phi(p, q)) \\
 &= \sum_{l=1}^m \frac{\partial(x^j \circ \phi_M^{-1})}{\partial u^l}(\phi_M(p)) 0 \\
 &= 0
 \end{aligned}$$

Similarly,

$$\frac{\partial}{\partial \tilde{x}^k} \Big|_{(p,q)} (y^j \circ \pi_N) = 0, \quad \text{and} \quad \frac{\partial}{\partial \tilde{y}^k} \Big|_{(p,q)} (y^j \circ \pi_N) = \frac{\partial}{\partial y^k} \Big|_q (y^j)$$

2. The previous part implies that

$$\begin{aligned}
 ([D\pi_M(p, q)]_{\phi_M, \phi})_{j,k} &= \left(\left(\frac{\partial}{\partial \tilde{x}^j} \Big|_{(p,q)} (x^i \circ \pi_M) \right)_{i,j} \left(\frac{\partial}{\partial \tilde{y}^j} \Big|_{(p,q)} (x^i \circ \pi_M) \right)_{i,j} \right) \\
 &= \begin{pmatrix} \frac{\partial}{\partial x^1} \Big|_p (x^1) & \cdots & \frac{\partial}{\partial x^m} \Big|_p (x^1) & 0 & \cdots & 0 \\ & & \vdots & & & \\ \frac{\partial}{\partial x^1} \Big|_p (x^m) & \cdots & \frac{\partial}{\partial x^m} \Big|_p (x^m) & 0 & \cdots & 0 \end{pmatrix} \\
 &= (I_m \quad 0).
 \end{aligned}$$

Similarly, $([D\pi_N(p, q)]_{\phi_N, \phi})_{j,k} = (0 \quad I_n)$.

□

Exercise 6.2.0.2. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $p \in M$ and $q \in N$. Set $m := \dim M$ and $n := \dim N$. Define $\alpha \in \text{Hom}_{\mathbf{Vect}_\mathbb{R}}(T_{(p,q)}(M \times N), T_p M \times T_q N)$ by $\alpha := (D\pi_M(p, q), D\pi_N(p, q))$. Then

1. Let $(U_M, \phi_M) \in \mathcal{A}_M$ and $(U_N, \phi_N) \in \mathcal{A}_N$. Write $\phi_M = (x^1, \dots, x^m)$ and $\phi_N = (y^1, \dots, y^n)$. Define $(U, \phi) \in \mathcal{A}_M \otimes \mathcal{A}_N$ by $U := U_M \times U_N$ and $\phi := \phi_M \times \phi_N$. Write $\phi = (\tilde{x}^1, \dots, \tilde{x}^m, \tilde{y}^1, \dots, \tilde{y}^n)$. Then for each $j \in [m]$ and $k \in [n]$,

$$\alpha\left(\frac{\partial}{\partial \tilde{x}^j}\bigg|_{(p,q)}\right) = \left(\frac{\partial}{\partial x^j}\bigg|_p, 0\right), \quad \alpha\left(\frac{\partial}{\partial \tilde{y}^k}\bigg|_{(p,q)}\right) = \left(0, \frac{\partial}{\partial y^k}\bigg|_q\right)$$

2. $\alpha \in \text{Iso}_{\mathbf{Vect}_\mathbb{R}}(T_{p,q}(M \times N), T_p M \times T_q N)$.

Proof.

1. Clear by previous exercise

2. The previous part implies that $\text{Im } \alpha = T_p M \times T_q N$ and α is surjective. Since

$$\begin{aligned} \dim T_{(p,q)}(M \times N) &= m + n \\ &= \dim(T_p M \times T_q N), \end{aligned}$$

we have that α is surjective and therefore α is an isomorphism and $\alpha \in \text{Iso}_{\mathbf{Vect}_\mathbb{R}}(T_{p,q}(M \times N), T_p M \times T_q N)$.

□

Exercise 6.2.0.3. Let $M_1, M_2, N_1, N_2 \in \text{Obj}(\mathbf{Man}^\infty)$, $F_1 \in \text{Hom}_{\mathbf{Man}^\infty}(M_1, N_1)$, $F_2 \in \text{Hom}_{\mathbf{Man}^\infty}(M_2, N_2)$, $(U_1, \phi_1) \in \mathcal{A}_{M_1}$, $(U_2, \phi_2) \in \mathcal{A}_{M_2}$, $(V_1, \psi_1) \in \mathcal{A}_{N_1}$, $(V_2, \psi_2) \in \mathcal{A}_{N_2}$ and $(p_1, p_2) \in M_1 \times M_2$. Set $m_1 := \dim M_1$, $m_2 := \dim M_2$, $n_1 := \dim N_1$ and $n_2 := \dim N_2$. Write $\phi_1 = (x_1^1, \dots, x_1^{m_1})$ and $\phi_2 = (x_2^1, \dots, x_2^{m_2})$, $\psi_1 = (y_1^1, \dots, y_1^{n_1})$ and $\psi_2 = (y_2^1, \dots, y_2^{n_2})$. Define $(U, \phi) \in \mathcal{A}_{M_1} \otimes \mathcal{A}_{M_2}$ and $(V, \psi) \in \mathcal{A}_{N_1} \otimes \mathcal{A}_{N_2}$ by $U := U_1 \times U_2$, $\phi := \phi_1 \times \phi_2$, $V := V_1 \times V_2$ and $\psi := \psi_1 \times \psi_2$. Write $\phi = (\tilde{x}_1^1, \dots, \tilde{x}_1^{m_1}, \tilde{x}_2^1, \dots, \tilde{x}_2^{m_2})$ and $\psi = (\tilde{y}_1^1, \dots, \tilde{y}_1^{n_1}, \tilde{y}_2^1, \dots, \tilde{y}_2^{n_2})$. Then for each $i \in [m_1]$, $j \in [m_2]$, $k \in [n_1]$ and $l \in [n_2]$,

$$\begin{aligned} \frac{\partial[\tilde{y}_1^k \circ (F_1 \times F_2)]}{\partial \tilde{x}_1^i}(p_1, p_2) &= \frac{\partial[y_1^k \circ F_1]}{\partial x_1^i}(p_1), & \frac{\partial[\tilde{y}_1^k \circ (F_1 \times F_2)]}{\partial \tilde{x}_2^j}(p_1, p_2) &= 0, \\ \frac{\partial[\tilde{y}_2^l \circ (F_1 \times F_2)]}{\partial \tilde{x}_1^i}(p_1, p_2) &= 0, & \frac{\partial[\tilde{y}_2^l \circ (F_1 \times F_2)]}{\partial \tilde{x}_2^j}(p_1, p_2) &= \frac{\partial[y_2^l \circ F_2]}{\partial x_2^j}(p_2). \end{aligned}$$

Proof. Denote the usual coordinates on $\mathbb{R}^{m_1+m_2}$ by $(u_1^1, \dots, u_1^{m_1}, u_2^1, \dots, u_2^{m_2})$. Denote the standard basis of $\mathbb{R}^{m_1}$, $\mathbb{R}^{m_2}$ and $\mathbb{R}^{m_1+m_2}$

by $(e_1^1, \dots, e_1^{m_1})$, $(e_2^1, \dots, e_2^{m_2})$ and $(\tilde{e}_1^1, \dots, \tilde{e}_1^{m_1}, \tilde{e}_2^1, \dots, \tilde{e}_2^{m_2})$ respectively. Let $i \in [m_1]$, $j \in [m_2]$, $k \in [n_1]$ and $l \in [n_2]$. Then

$$\begin{aligned}
\frac{\partial[\tilde{y}_1^k \circ (F_1 \times F_2)]}{\partial \tilde{x}_1^i}(p_1, p_2) &= \frac{\partial[\tilde{y}_1^k \circ (F_1 \times F_2) \circ \phi^{-1}]}{\partial u_1^i}(\phi(p_1, p_2)) \\
&= \frac{\partial[\tilde{y}_1^k \circ (F_1 \times F_2) \circ (\phi_1^{-1} \times \phi_2^{-1})]}{\partial u_1^i}(\phi_1(p_1), \phi_2(p_2)) \\
&= \frac{\partial[\tilde{y}_1^k \circ ([F_1 \circ \phi_1^{-1}] \times [F_2 \circ \phi_2^{-1})]}{\partial u_1^i}(\phi_1(p_1), \phi_2(p_2)) \\
&= \frac{d}{dt} \Big|_{t=0} [\tilde{y}_1^k \circ ([F_1 \circ \phi_1^{-1}] \times [F_2 \circ \phi_2^{-1}]) (\phi_1(p_1), \phi_2(p_2)) + t\tilde{e}_1^i] \\
&= \frac{d}{dt} \Big|_{t=0} [\tilde{y}_1^k \circ ([F_1 \circ \phi_1^{-1}] \times [F_2 \circ \phi_2^{-1}]) (\phi_1(p_1) + te_1^i, \phi_2(p_2))] \\
&= \frac{d}{dt} \Big|_{t=0} [\tilde{y}_1^k([F_1 \circ \phi_1^{-1}](\phi_1(p_1) + te_1^i), [F_2 \circ \phi_2^{-1}](\phi_2(p_2)))] \\
&= \frac{d}{dt} \Big|_{t=0} [\text{proj}_k^{n_1+n_2} \circ (\psi_1 \times \psi_2)([F_1 \circ \phi_1^{-1}](\phi_1(p_1) + te_1^i), [F_2 \circ \phi_2^{-1}](\phi_2(p_2)))] \\
&= \frac{d}{dt} \Big|_{t=0} [\text{proj}_k^{n_1+n_2}(\psi_1([F_1 \circ \phi_1^{-1}](\phi_1(p_1) + te_1^i)), (\psi_2[F_2 \circ \phi_2^{-1}](\phi_2(p_2)))] \\
&= \frac{d}{dt} \Big|_{t=0} [\text{proj}_k^{n_1}(\psi_1([F_1 \circ \phi_1^{-1}](\phi_1(p_1) + te_1^i)))] \\
&= \frac{d}{dt} \Big|_{t=0} [\text{proj}_k^{n_1} \circ \psi_1 \circ [F_1 \circ \phi_1^{-1}](\phi_1(p_1) + te_1^i)] \\
&= \frac{\partial[y_1^k \circ (F_1 \circ \phi_1^{-1})]}{\partial u_1^i}(\phi_1(p_1)) \\
&= \frac{\partial[y_1^k \circ F_1]}{\partial x_1^i}(p_1)
\end{aligned}$$

The other claims follow similarly. □

Exercise 6.2.0.4. Let $M_1, M_2, N_1, N_2 \in \text{Obj}(\mathbf{Man}^\infty)$, $F_1 \in \text{Hom}_{\mathbf{Man}^\infty}(M_1, N_1)$, $F_2 \in \text{Hom}_{\mathbf{Man}^\infty}(M_2, N_2)$ and $(p_1, p_2) \in M_1 \times M_2$. Define $\alpha_M \in \text{Iso}_{\mathbf{Vect}_{\mathbb{R}}}(T_{(p_1, p_2)}(M_1 \times M_2), T_{p_1}M_1 \times T_{p_2}M_2)$ and $\alpha_N \in \text{Iso}_{\mathbf{Vect}_{\mathbb{R}}}(T_{(F_1(p_1), F_2(p_2))}(N_1 \times N_2), T_{F_1(p_1)}N_1 \times T_{F_2(p_2)}N_2)$ as in Exercise 6.2.0.2. Then

$$\alpha_N \circ D(F_1 \times F_2)(p_1, p_2) = DF_1(p_1) \times DF_2(p_2) \circ \alpha_M,$$

i.e. following diagram commutes:

$$\begin{array}{ccc}
T_{(p_1, p_2)}(M_1 \times M_2) & \xrightarrow{D(F_1 \times F_2)(p_1, p_2)} & T_{(F_1(p_1), F_2(p_2))}(N_1 \times N_2) \\
\downarrow \alpha_M & & \downarrow \alpha_N \\
T_{p_1}M_1 \times T_{p_2}M_2 & \xrightarrow{DF_1(p_1) \times DF_2(p_2)} & T_{F_1(p_1)}N_1 \times T_{F_2(p_2)}N_2
\end{array}$$

Proof. Set $m_1 := \dim M_1$, $m_2 := \dim M_2$, $n_1 := \dim N_1$ and $n_2 := \dim N_2$. Choose $(U_1, \phi_1) \in \mathcal{A}_{M_1}$, $(U_2, \phi_2) \in \mathcal{A}_{M_2}$, $(V_1, \psi_1) \in \mathcal{A}_{N_1}$ and $(V_2, \psi_2) \in \mathcal{A}_{N_2}$ such that $p_1 \in U_1, p_2 \in U_2$, $F_1(p_1) \in V_1$ and $F_2(p_2) \in V_2$. Write $\phi_1 = (x_1^1, \dots, x_1^{m_1})$, $\phi_2 = (x_2^1, \dots, x_2^{m_2})$, $\psi_1 = (y_1^1, \dots, y_1^{n_1})$ and $\psi_2 = (y_2^1, \dots, y_2^{n_2})$. Define $(U, \phi) \in \mathcal{A}_{M_1} \otimes \mathcal{A}_{M_2}$ and $(V, \psi) \in \mathcal{A}_{N_1} \otimes \mathcal{A}_{N_2}$ by $U := U_1 \times U_2$, $\phi := \phi_1 \times \phi_2$, $V := V_1 \times V_2$ and $\psi = \psi_1 \times \psi_2$. Write $\phi = (\tilde{x}_1^1, \dots, \tilde{x}_1^{m_1}, \tilde{x}_2^1, \dots, \tilde{x}_2^{m_2})$ and $\psi = (\tilde{y}_1^1, \dots, \tilde{y}_1^{n_1}, \tilde{y}_2^1, \dots, \tilde{y}_2^{n_2})$. Let $i \in [m_1]$. The chain rule

implies that for each $f \in C^\infty(N_1 \times N_2)$,

$$\begin{aligned}
 D(F_1 \times F_2)(p_1, p_2) \left(\frac{\partial}{\partial \tilde{x}_1^i} \Big|_{(p_1, p_2)} \right) (f) &= \frac{\partial}{\partial \tilde{x}_1^i} \Big|_{(p_1, p_2)} [f \circ (F_1 \times F_2)] \\
 &= \sum_{k=1}^{n_1} \frac{\partial f}{\partial \tilde{y}_1^k} ([F_1 \times F_1](p_1, p_2)) \frac{\partial [\tilde{y}_1^k \circ (F_1 \times F_2)]}{\partial \tilde{x}_1^i} (p_1, p_2) \\
 &\quad + \sum_{l=1}^{n_2} \frac{\partial f}{\partial \tilde{y}_2^l} ([F_1 \times F_1](p_1, p_2)) \frac{\partial [\tilde{y}_2^l \circ (F_1 \times F_2)]}{\partial \tilde{x}_1^i} (p_1, p_2) \\
 &= \left[\sum_{k=1}^{n_1} \frac{\partial [\tilde{y}_1^k \circ (F_1 \times F_2)]}{\partial \tilde{x}_1^i} (p_1, p_2) \frac{\partial}{\partial \tilde{y}_1^k} \Big|_{F_1 \times F_2(p_1, p_2)} \right. \\
 &\quad \left. + \sum_{l=1}^{n_2} \frac{\partial [\tilde{y}_2^l \circ (F_1 \times F_2)]}{\partial \tilde{x}_1^i} (p_1, p_2) \frac{\partial}{\partial \tilde{y}_2^l} \Big|_{F_1 \times F_2(p_1, p_2)} \right] (f).
 \end{aligned}$$

The previous exercise then implies that

$$\begin{aligned}
 D(F_1 \times F_2)(p_1, p_2) \left(\frac{\partial}{\partial \tilde{x}_1^i} \Big|_{(p_1, p_2)} \right) &= \sum_{k=1}^{n_1} \frac{\partial [\tilde{y}_1^k \circ (F_1 \times F_2)]}{\partial \tilde{x}_1^i} (p_1, p_2) \frac{\partial}{\partial \tilde{y}_1^k} \Big|_{F_1 \times F_2(p_1, p_2)} \\
 &\quad + \sum_{l=1}^{n_2} \frac{\partial [\tilde{y}_2^l \circ (F_1 \times F_2)]}{\partial \tilde{x}_1^i} (p_1, p_2) \frac{\partial}{\partial \tilde{y}_2^l} \Big|_{F_1 \times F_2(p_1, p_2)} \\
 &= \sum_{k=1}^{n_1} \frac{\partial [y_1^k \circ F_1]}{\partial x_1^i} (p_1) \frac{\partial}{\partial \tilde{y}_1^k} \Big|_{F_1 \times F_2(p_1, p_2)}.
 \end{aligned}$$

Therefore

$$\begin{aligned}
 \alpha_N \circ D(F_1 \times F_2)(p_1, p_2) \left(\frac{\partial}{\partial \tilde{x}_1^i} \Big|_{(p_1, p_2)} \right) &= \sum_{k=1}^{n_1} \frac{\partial [y_1^k \circ F_1]}{\partial x_1^i} (p_1) \alpha_N \left(\frac{\partial}{\partial \tilde{y}_1^k} \Big|_{F_1 \times F_2(p_1, p_2)} \right) \\
 &= \sum_{k=1}^{n_1} \frac{\partial [y_1^k \circ F_1]}{\partial x_1^i} (p_1) \left(\frac{\partial}{\partial y_1^k} \Big|_{F_1(p_1)}, 0 \right) \\
 &= \left(\sum_{k=1}^{n_1} \frac{\partial [y_1^k \circ F_1]}{\partial x_1^i} (p_1) \frac{\partial}{\partial y_1^k} \Big|_{F_1(p_1)}, 0 \right) \\
 &= \left(DF_1(p_1) \left(\frac{\partial}{\partial x_1^i} \Big|_{p_1} \right), DF_2(p_2)(0) \right) \\
 &= DF_1(p_1) \times DF_2(p_2) \left(\frac{\partial}{\partial x_1^i} \Big|_{p_1}, 0 \right) \\
 &= DF_1(p_1) \times DF_2(p_2) \circ \alpha_M \left(\frac{\partial}{\partial \tilde{x}_1^i} \Big|_{(p_1, p_2)} \right).
 \end{aligned}$$

Since $i \in [m_1]$ is arbitrary, we have that for each $i \in [m_1]$,

$$\alpha_N \circ D(F_1 \times F_2)(p_1, p_2) \left(\frac{\partial}{\partial \tilde{x}_1^i} \Big|_{(p_1, p_2)} \right) = DF_1(p_1) \times DF_2(p_2) \circ \alpha_M \left(\frac{\partial}{\partial \tilde{x}_1^i} \Big|_{(p_1, p_2)} \right)$$

Similarly, for each $j \in [m_2]$,

$$\alpha_N \circ D(F_1 \times F_2)(p_1, p_2) \left(\frac{\partial}{\partial \tilde{x}_2^j} \Big|_{(p_1, p_2)} \right) = DF_1(p_1) \times DF_2(p_2) \circ \alpha_M \left(\frac{\partial}{\partial \tilde{x}_2^j} \Big|_{(p_1, p_2)} \right)$$

Since

$$\left(\frac{\partial}{\partial \tilde{x}_1^i} \Big|_{(p_1, p_2)}, \frac{\partial}{\partial \tilde{x}_2^j} \Big|_{(p_1, p_2)} : i \in [m_1], j \in [m_2] \right)$$

is a basis for $T_{(p_1, p_2)M_1 \times M_2}$, we have that

$$\alpha_N \circ D(F_1 \times F_2)(p_1, p_2) = DF_1(p_1) \times DF_2(p_2) \circ \alpha_M.$$

□

Exercise 6.2.0.5. Let $M, N_1, N_2 \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $F_1 \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N_1)$ and $F_2 \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N_2)$. Set $m := \dim M$, $n_1 := \dim N_1$ and $n_2 := \dim N_2$. Define $G \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N_1 \times N_2)$ by $G := (F_1, F_2)$. Let $(U, \phi) \in \mathcal{A}_M$, $(V_1, \psi_1) \in \mathcal{A}_{N_1}$ and $(V_2, \psi_2) \in \mathcal{A}_{N_2}$. Write $\phi = (x^1, \dots, x^m)$, $\psi_1 = (y^1, \dots, y^{n_1})$ and $\psi_2 = (z^1, \dots, z^{n_2})$. Define $(V, \psi) \in \mathcal{A}_{N_1} \otimes \mathcal{A}_{N_2}$ by $V := V_1 \times V_2$ and $\psi := \psi_1 \times \psi_2$. Write $\psi = (\tilde{y}^1, \dots, \tilde{y}^{n_1}, \tilde{z}^1, \dots, \tilde{z}^{n_2})$. Then for each $p \in U \cap G^{-1}(V)$,

$$[DG(p)]_{\phi, \psi} = \begin{pmatrix} [DF_1(p)]_{\phi, \psi_1} \\ [DF_2(p)]_{\phi, \psi_2} \end{pmatrix}.$$

Proof. Let $p \in G^{-1}(V)$. Then for each $j \in [n_1]$ and $k \in [m]$,

$$\frac{\partial}{\partial x^k} \Big|_p [\tilde{y}^j \circ G] = \frac{\partial}{\partial x^k} \Big|_p [y^j \circ F_1].$$

Similarly, for each $j \in [n_2]$ and $k \in [m]$.

$$\frac{\partial}{\partial x^k} \Big|_p [\tilde{z}^j \circ G] = \frac{\partial}{\partial x^k} \Big|_p [z^j \circ F_2].$$

Hence

$$\begin{aligned} [DG(p)]_{\phi, \psi} &= \begin{pmatrix} \left(\frac{\partial [\tilde{y}^j \circ G]}{\partial x^k} (p) \right)_{j,k} \\ \left(\frac{\partial [\tilde{z}^j \circ G]}{\partial x^k} (p) \right)_{j,k} \end{pmatrix} \\ &= \begin{pmatrix} [DF_1(p)]_{\phi, \psi_1} \\ [DF_2(p)]_{\phi, \psi_2} \end{pmatrix} \end{aligned}$$

□

Exercise 6.2.0.6. Let $M, N_1, N_2 \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $F_1 \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N_1)$, $F_2 \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N_2)$ and $p \in M$. Define $G \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N_1 \times N_2)$ by $G := (F_1, F_2)$ and define $\alpha_N \in \text{Iso}_{\mathbf{Vect}_{\mathbb{R}}}(T_{(F_1(p), F_2(p))}N_1 \times N_2, T_{F_1(p)}N_1 \times T_{F_2(p)}N_2)$ as in Exercise 6.2.0.2. Then

$$\alpha_N \circ DG(p) = (DF_1(p), DF_2(p)).$$

Proof. Set $m := \dim M$, $n_1 := \dim N_1$ and $n_2 := \dim N_2$. Then there exists $(U, \phi) \in \mathcal{A}_M$, $(V_1, \psi_1) \in \mathcal{A}_{N_1}$ and $(V_2, \psi_2) \in \mathcal{A}_{N_2}$ such that $p \in U$, $F_1(p) \in V_1$ and $F_2(p) \in V_2$. Write $\phi = (x^1, \dots, x^m)$, $\psi_1 = (y^1, \dots, y^{n_1})$ and $\psi_2 = (z^1, \dots, z^{n_2})$. Define $(V, \psi) \in \mathcal{A}_{N_1} \otimes \mathcal{A}_{N_2}$ by $V := V_1 \times V_2$ and $\psi := \psi_1 \times \psi_2$. Write $\psi = (\tilde{y}^1, \dots, \tilde{y}^{n_1}, \tilde{z}^1, \dots, \tilde{z}^{n_2})$. Define $q \in N_1 \times N_2$ by $q := G(p)$. Let $k \in [m]$. Then Exercise 6.2.0.5 implies that

$$\begin{aligned} DG(p) \left(\frac{\partial}{\partial x^k} \right) &= [DG(p)]_{\phi, \psi} e_k^\phi \\ &= \sum_{j=1}^{n_1+n_2} ([DG(p)]_{\phi, \psi})_{j,k} e_j^\psi \\ &= \sum_{j=1}^{n_1} ([DG(p)]_{\phi, \psi})_{j,k} e_j^\psi + \sum_{j=1}^{n_2} ([DG(p)]_{\phi, \psi})_{n_2+j,k} e_{n_2+j}^\psi \\ &= \sum_{j=1}^{n_1} \frac{\partial (y^j \circ F_1)}{\partial x^k} (p) \frac{\partial}{\partial \tilde{y}^j} \Big|_q + \sum_{j=1}^{n_2} \frac{\partial (z^j \circ F_2)}{\partial x^k} (p) \frac{\partial}{\partial \tilde{z}^j} \Big|_q. \end{aligned}$$

Exercise 6.2.0.2 then implies that

$$\begin{aligned}
\alpha_N \circ DG(p) \left(\frac{\partial}{\partial x^k} \right) &= \alpha_N \left(\sum_{j=1}^{n_1} \frac{\partial(y^j \circ F_1)}{\partial x^k}(p) \frac{\partial}{\partial \tilde{y}^j} \Big|_q + \sum_{j=1}^{n_2} \frac{\partial(z^j \circ F_2)}{\partial x^k}(p) \frac{\partial}{\partial \tilde{z}^j} \Big|_q \right) \\
&= \sum_{j=1}^{n_1} \frac{\partial(y^j \circ F_1)}{\partial x^k}(p) \alpha_N \left(\frac{\partial}{\partial \tilde{y}^j} \Big|_q \right) + \sum_{j=1}^{n_2} \frac{\partial(z^j \circ F_2)}{\partial x^k}(p) \alpha_N \left(\frac{\partial}{\partial \tilde{z}^j} \Big|_q \right) \\
&= \sum_{j=1}^{n_1} \frac{\partial(y^j \circ F_1)}{\partial x^k}(p) \left(\frac{\partial}{\partial y^j} \Big|_{F_1(p)}, 0 \right) + \sum_{j=1}^{n_2} \frac{\partial(z^j \circ F_2)}{\partial x^k}(p) \left(0, \frac{\partial}{\partial z^j} \Big|_{F_2(p)} \right) \\
&= \left(\sum_{j=1}^{n_1} \frac{\partial(y^j \circ F_1)}{\partial x^k}(p) \frac{\partial}{\partial y^j} \Big|_{F_1(p)}, \sum_{j=1}^{n_2} \frac{\partial(z^j \circ F_2)}{\partial x^k}(p) \frac{\partial}{\partial z^j} \Big|_{F_2(p)} \right) \\
&= \left(DF_1(p) \left(\frac{\partial}{\partial x^k} \Big|_p \right), DF_2(p) \left(\frac{\partial}{\partial x^k} \Big|_p \right) \right) \\
&= (DF_1(p), DF_2(p)) \left(\frac{\partial}{\partial x^k} \Big|_p \right).
\end{aligned}$$

Since $k \in [m]$ is arbitrary, we have that $\alpha_N \circ DG(p) = (DF_1(p), DF_2(p))$. □

6.3 The Cotangent Space

Definition 6.3.0.1. Let $p \in M$. We define the **cotangent space of M at p** , denoted T_p^*M , by

$$T_p^*M := (T_pM)^*$$

Definition 6.3.0.2. Let $f \in C^\infty(M)$. We define the **differential of f at p** , denoted $df_p : T_pM \rightarrow \mathbb{R}$, by

$$df_p(v) = v(f)$$

Exercise 6.3.0.3. Let $f \in C^\infty(M)$ and $p \in M$. Then $df_p \in T_p^*M$.

Proof. Let $v_1, v_2 \in T_pM$ and $\lambda \in \mathbb{R}$. Then

$$\begin{aligned} df_p(v_1 + \lambda v_2) &= (v_1 + \lambda v_2)f \\ &= v_1f + \lambda v_2f \\ &= df_p(v_1) + \lambda df_p(v_2) \end{aligned}$$

So that df_p is linear and hence $df_p \in T_p^*M$. □

Exercise 6.3.0.4. Let $(U, \phi) \in \mathcal{A}$ with $\phi = (x^1, \dots, x^n)$ and $p \in U$. Then for each $i, j \in \{1, \dots, n\}$,

$$dx_p^i \left(\frac{\partial}{\partial x^j} \Big|_p \right) = \delta_{i,j}$$

In particular, $\{dx_p^1, \dots, dx_p^n\}$ is the dual basis to $\left\{ \frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p \right\}$ and $T_p^*M = \text{span}\{dx_p^1, \dots, dx_p^n\}$.

Proof. Let $i, j \in \{1, \dots, n\}$. Then by definition,

$$\begin{aligned} \left[dx_p^i \left(\frac{\partial}{\partial x^i} \Big|_p \right) \right]_p &= \frac{\partial}{\partial x^i} \Big|_p x^i \\ &= \delta_{i,i} \end{aligned}$$

Exercise 6.3.0.5. Let $f \in C^\infty(M)$, (U, ϕ) a chart on M with $\phi = (x^1, \dots, x^n)$ and $p \in U$. Then

$$df_p = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p) dx_p^i$$

Proof. Since $\{dx_p^1, \dots, dx_p^n\}$ is a basis for T_p^*M , for each there exist $a_1(p), \dots, a_n(p) \in \mathbb{R}$ such that $df_p = \sum_{i=1}^n a_i(p) dx_p^i$. Therefore, we have that

$$\begin{aligned} df_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) &= \sum_{i=1}^n a_i(p) dx_p^i \left(\frac{\partial}{\partial x^i} \Big|_p \right) \\ &= a_j(p) \end{aligned}$$

By definition, we have that

$$\begin{aligned} df_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) &= \frac{\partial}{\partial x^i} \Big|_p f \\ &= \frac{\partial f}{\partial x^i}(p) \end{aligned}$$

So $a_j(p) = \frac{\partial f}{\partial x^j}(p)$ and

$$df_p = \sum_{i=1}^n \frac{\partial f}{\partial x^i}(p) dx_p^i$$

□

Chapter 7

Categorical Description of Manifolds

7.1 The Categories Man^0 , ManBnd^∞ and Man^∞

7.2 The Derivative as a Functor

Chapter 8

Immersions and Submersions

8.1 Maps of Constant Rank

Do this section assuming $\partial M, \partial N = \emptyset$

Definition 8.1.0.1. Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds, $F : M \rightarrow N$ a smooth map. We define the **rank map of F** , denoted $\text{rank } F : M \rightarrow \mathbb{N}_0$ by

$$\text{rank}_p F = \dim \text{Im } DF(p)$$

and F is said to have **constant rank** if for each $p, q \in M$, $\text{rank}_p F = \text{rank}_q F$. If F has constant rank, we define the **rank of F** , denoted $\text{rank } F$, by $\text{rank } F = \text{rank}_p F$ for $p \in M$.

Exercise 8.1.0.2. Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds of dimensions m and n respectively, $F \in C^\infty(M, N)$ and $p \in M$. Suppose that $\partial N = \emptyset$ and $\text{rank}_p F = k$. Then there exist $(U, \phi) \in \mathcal{A}$, $(V, \psi) \in \mathcal{B}$ and $A \in GL(k, \mathbb{R})$ such that for each $i, j \in \{1, \dots, k\}$,

$$([DF(p)]_{\phi, \psi})_{i,j} = A_{i,j}$$

Does the boundary need to be empty?

Proof. Define $q \in V$ by $q = F(p)$. Choose $(U, \phi') \in \mathcal{A}$ and $(V, \psi') \in \mathcal{B}$ such that $p \in U$, $q \in V$. Since $\partial N = \emptyset$, $\phi'(U) \subset \text{Int } \mathbb{H}_j^m$ and $\psi'(V) \subset \text{Int } \mathbb{H}_k^n$. Set $Z = [DF(p)]_{\phi', \psi'}$. By assumption, $\text{rank } Z = k$. Exercise 1.2.0.9 implies that there exist $\sigma \in S_m$, $\tau \in S_n$ and $A \in GL(k, \mathbb{R})$ such that for each $i, j \in \{1, \dots, k\}$,

$$(P_\tau Z P_\sigma^*)_{i,j} = A_{i,j}$$

Define $\phi : U \rightarrow (\sigma \cdot \phi')(U)$ and $\psi : V \rightarrow (\tau \cdot \psi')(V)$ by

$$\phi = \sigma \cdot \phi', \quad \psi = \tau \cdot \psi'$$

Exercise 4.1.0.7 implies that $(U, \phi) \in \mathcal{A}$, $(V, \psi) \in \mathcal{B}$ and Exercise 1.3.3.3 implies that

$$[DF(p)]_{\phi, \psi} = P_\tau Z P_\sigma^*$$

□

Exercise 8.1.0.3. Local Rank Theorem:

rework for $\mathbb{H}^m$ instead of $\mathbb{R}^m$ Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds of dimensions m and n respectively, $F \in C^\infty(M, N)$. Suppose that $\partial M, \partial N = \emptyset$, F has constant rank and $\text{rank } F = k$. Then for each $p \in M$, there exist $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U$, $F(U) \subset V$ and

$$\psi \circ F \circ \phi^{-1}(x^1, \dots, x^k, x^{k+1}, \dots, x^m) = (x^1, \dots, x^k, 0, \dots, 0)$$

Hint: Needs a hint

Proof. Let $p \in M$. The previous exercise implies that there exist $(U_0, \phi_0) \in \mathcal{A}$, $(V_0, \psi_0) \in \mathcal{B}$ and $L \in GL(k, \mathbb{R})$ such that $p \in U$, $F(p) \in V_0$ and for each $i, j \in \{1, \dots, k\}$,

$$([DF(p)]_{\phi_0, \psi_0})_{i,j} = L_{i,j}$$

Define $\hat{M} \subset \mathbb{R}^m$, $\hat{N} \subset \mathbb{R}^n$ and $\hat{F} : \hat{M} \rightarrow \hat{N}$ by $\hat{M} := \phi_0(U_0)$, $\hat{N} := \psi_0(V_0)$ and $\hat{F} := \psi_0 \circ F \circ \phi_0^{-1}$. Set $\hat{p} := \phi_0(p)$. Let (x, y) be the standard coordinates on $\mathbb{R}^m$, with $\pi_x : \mathbb{R}^m \rightarrow \mathbb{R}^k$ and $\pi_y : \mathbb{R}^m \rightarrow \mathbb{R}^{m-k}$ the standard projection maps. Write $\hat{p} = (x_0, y_0)$. There exist $Q : \hat{M} \rightarrow \mathbb{R}^k$ and $R : \hat{M} \rightarrow \mathbb{R}^{n-k}$ such that $\hat{F} = (Q, R)$. By construction, $[D_x Q(x_0, y_0)] = L$. Define $G : \hat{M} \rightarrow \mathbb{R}^m$ by $G(x, y) := (Q(x, y), y)$. Then

$$\begin{aligned} [DG(x_0, y_0)] &= \begin{pmatrix} [D_x Q(x_0, y_0)] & [D_x Q(x_0, y_0)] \\ [D_x \pi_y(x_0, y_0)] & [D_y \pi_y(x_0, y_0)] \end{pmatrix} \\ &= \begin{pmatrix} [D_x Q(x_0, y_0)] & [D_y Q(x_0, y_0)] \\ 0 & I \end{pmatrix} \\ &= \begin{pmatrix} L & [D_y Q(x_0, y_0)] \\ 0 & I \end{pmatrix} \end{aligned}$$

Hence

$$\begin{aligned} \det([DG(x_0, y_0)]) &= \det(L) \det(I) \\ &= \det(L) \\ &\neq 0 \end{aligned}$$

The inverse function theorem implies that there exist $\hat{U} \subset \hat{M}$ such that $\hat{U}$ is open, $\hat{p} \in \hat{U}$ and $G|_{\hat{U}} : \hat{U} \rightarrow G(\hat{U})$ is a diffeomorphism. Since

$$\{U_1 \times U_2 : U_1 \subset \mathbb{R}^k, U_2 \subset \mathbb{R}^{m-k} \text{ and } U_1, U_2 \text{ are open}\}$$

is a basis for the topology on $\mathbb{R}^m$, there exist $\hat{U}_1 \subset \mathbb{R}^k$ and $\hat{U}_2 \subset \mathbb{R}^{m-k}$ such that $\hat{U}_1, \hat{U}_2$ are open, $\hat{p} \in \hat{U}_1 \times \hat{U}_2$ and $\hat{U}_1 \times \hat{U}_2 \subset \hat{U}$. Set $\hat{U}_{12} := \hat{U}_1 \times \hat{U}_2$ and define $G_{12} : \hat{U}_{12} \rightarrow Q(\hat{U}_{12}) \times \hat{U}_2$ by $G_{12} := G|_{\hat{U}_{12}}$. Since $G|_{\hat{U}} : \hat{U} \rightarrow G(\hat{U})$ is a diffeomorphism, $\hat{U}_{12} \subset \hat{U}$ and

$$\begin{aligned} G(\hat{U}_{12}) &= G(\hat{U}_1 \times \hat{U}_2) \\ &= Q(\hat{U}_{12}) \times \hat{U}_2 \end{aligned}$$

we have that $G_{12} : \hat{U}_{12} \rightarrow Q(\hat{U}_{12}) \times \hat{U}_2$ is a diffeomorphism. Since G_{12} is a homeomorphism and π_x is open, $Q(\hat{U}_{12})$ is open. Since $G_{12}^{-1} : Q(\hat{U}_{12}) \times \hat{U}_2 \rightarrow \hat{U}_{12}$, there exist $A : Q(\hat{U}_{12}) \times \hat{U}_2 \rightarrow \hat{U}_1$ and $B : Q(\hat{U}_{12}) \times \hat{U}_2 \rightarrow \hat{U}_2$ such that A, B are smooth and $G_{12}^{-1} = (A, B)$. Define $\tilde{R} : Q(\hat{U}_{12}) \times \hat{U}_2 \rightarrow \mathbb{R}^{n-k}$ by $\tilde{R}(x, y) := R(A(x, y), y)$. Then $\tilde{R}$ is smooth. Let $(x, y) \in Q(\hat{U}_{12}) \times \hat{U}_2$. Then

$$\begin{aligned} (x, y) &= G_{12} \circ G_{12}^{-1}(x, y) \\ &= G(A(x, y), B(x, y)) \\ &= (Q(A(x, y), B(x, y)), B(x, y)) \end{aligned}$$

This implies that $B(x, y) = y$,

$$\begin{aligned} x &= Q(A(x, y), B(x, y)) \\ &= Q(A(x, y), y) \end{aligned}$$

and

$$\begin{aligned} G_{12}^{-1}(x, y) &= (A(x, y), B(x, y)) \\ &= (A(x, y), y) \end{aligned}$$

Therefore,

$$\begin{aligned} \hat{F} \circ G_{12}^{-1}(x, y) &= \hat{F}(A(x, y), y) \\ &= (Q(A(x, y), y), R(A(x, y), y)) \\ &= (x, R(A(x, y), y)) \\ &= (x, \tilde{R}(x, y)) \end{aligned}$$

We note that

$$\begin{aligned} [D(\hat{F} \circ G_{12}^{-1})(x, y)] &= \begin{pmatrix} [D_x \pi_x(x, y)] & [D_y \pi_x(x, y)] \\ [D_x \tilde{R}(x, y)] & [D_y \tilde{R}(x, y)] \end{pmatrix} \\ &= \begin{pmatrix} I & 0 \\ [D_x \tilde{R}(x, y)] & [D_y \tilde{R}(x, y)] \end{pmatrix} \end{aligned}$$

Since $G_{12}^{-1} : Q(\hat{U}_{12}) \times \hat{U}_2 \rightarrow \hat{U}_{12}$ is a diffeomorphism, we have that $[DG^{-1}(x, y)] \in GL(m, \mathbb{R})$. Since $\hat{F}$ has constant rank and $\text{rank } \hat{F} = k$, we have that

$$\begin{aligned} \text{rank}[D(\hat{F} \circ G_{12}^{-1})(x, y)] &= \text{rank}([D\hat{F}(G_{12}^{-1}(x, y))][DG_{12}^{-1}(x, y)]) \\ &= \text{rank}[D\hat{F}(G_{12}^{-1}(x, y))] \\ &= k \end{aligned}$$

Since $\text{rank} \begin{pmatrix} I \\ [D_x \tilde{R}(x, y)] \end{pmatrix} = k$, we have that $\text{rank} \begin{pmatrix} 0 \\ [D_y \tilde{R}(x, y)] \end{pmatrix} = 0$. Thus $[D_y \tilde{R}(x, y)] = 0$. Since $(x, y) \in Q(\hat{U}_{12}) \times \hat{U}_2$ is arbitrary, for each $(x, y) \in Q(\hat{U}_{12}) \times \hat{U}_2$,

$$\tilde{R}(x, y) = \tilde{R}(x, y_0)$$

Define $\tilde{S} : Q(\hat{U}_{12}) \rightarrow \mathbb{R}^{n-k}$ by $\tilde{S}(x) := \tilde{R}(x, y_0)$. Then $\tilde{S}$ is smooth and for each $(x, y) \in Q(\hat{U}_{12}) \times \hat{U}_2$,

$$\hat{F} \circ G_{12}^{-1}(x, y) = (x, \tilde{S}(x))$$

Let (a, b) be the standard coordinates on $\mathbb{R}^n$, with $\pi_a : \mathbb{R}^n \rightarrow \mathbb{R}^k$ and $\pi_b : \mathbb{R}^n \rightarrow \mathbb{R}^{n-k}$ the standard projection maps. Write $\hat{F}(\hat{p}) = (a_0, b_0)$. Set

$$\begin{aligned} \hat{V}_{12} &:= \pi_a|_{\hat{N}}^{-1}(Q(\hat{U}_{12})) \\ &= \pi_a^{-1}(Q(\hat{U}_{12})) \cap \hat{N} \end{aligned}$$

Since $Q(\hat{U}_{12})$ is open, $\hat{N}$ is open and π_a is continuous, we have that $\hat{V}_{12}$ is open. Since

$$\begin{aligned} Q(\hat{U}_{12}) &= \pi_a|_{\hat{N}} \circ \hat{F} \circ G^{-1}(Q(\hat{U}_{12}) \times \hat{U}_2) \\ &= \pi_a|_{\hat{N}} \circ \hat{F}(\hat{U}_{12}) \end{aligned}$$

we have that

$$\begin{aligned} \hat{F}(\hat{U}_{12}) &\subset \pi_a|_{\hat{N}}^{-1}(Q(\hat{U}_{12})) \\ &\subset \hat{V}_{12} \end{aligned}$$

In particular, $\hat{F}(\hat{p}) \in \hat{V}_{12}$. Define $H : Q(\hat{U}_{12}) \times \mathbb{R}^{n-k} \rightarrow Q(\hat{U}_{12}) \times \mathbb{R}^{n-k}$ by $H := (\pi_a, \pi_b - \tilde{S} \circ \pi_a)$, i.e. for each $(a, b) \in Q(\hat{U}_{12}) \times \mathbb{R}^{n-k}$, $H(a, b) = (a, b - \tilde{S}(a))$. Then H is a bijection and $H^{-1}(a, b) = (\pi_a, \pi_b + \tilde{S} \circ \pi_a)$. Thus H and H^{-1} are smooth and therefore H is a diffeomorphism. Define $H_{12} : \hat{V}_{12} \rightarrow H(\hat{V}_{12})$ by $H_{12} = H|_{\hat{V}_{12}}$. Then H_{12} is a diffeomorphism and for each $x, y \in Q(\hat{U}_{12} \times \hat{U}_2)$, $H_{12} \circ \hat{F} \circ G_{12}^{-1}(x, y) = (x, 0)$. Define $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$ by $U := \phi_0^{-1}(\hat{U}_{12})$, $V := \psi_0^{-1}(\hat{V}_{12})$, $\phi := G_{12} \circ \phi_0|_U$ and $\psi := H_{12} \circ \psi_0|_V$. **Show that $F(U) \subset V$.** Then for each $(x, y) \in \phi(U)$,

$$\begin{aligned} \psi \circ F \circ \phi^{-1}(x, y) &= H_{12} \circ \psi_0|_V \circ F \circ \phi_0|_U^{-1} \circ G_{12}^{-1}(x, y) \\ &= H_{12} \circ \hat{F} \circ G_{12}^{-1}(x, y) \\ &= (x, 0) \end{aligned}$$

need to start with compact chart domain and add constant so we stay in $\mathbb{H}^n$, i.e. need U to be compact, so set U_1 and U_2 to be compact, then U_{12} will be and thus U . $\square$

Exercise 8.1.0.4. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. Suppose that $\dim M = m$ and $\dim N = n$, F has constant rank and $\text{rank } F = r$. Then for each $p \in M$, there exist $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U$, $F(\text{cl } U) \subset V$, $\text{cl } U$ is compact and

$$\psi \circ F \circ \phi^{-1}(x^1, \dots, x^k, x^{k+1}, \dots, x^m) = (x^1, \dots, x^k, 0, \dots, 0)$$

Proof. content... □

Exercise 8.1.0.5. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. Suppose that F has constant rank.

- 1.
- 2.
- 3.

Proof. Set $m := \dim M$, $n := \dim N$ and $r := \text{rank } F$.

1. Let $p \in M$. The local rank theorem (Exercise 8.1.0.3) implies that there exists $(U_0, \phi_0) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$ such that $p \in U$, $F(U) \subset V$ and $\psi \circ F \circ \phi_0^{-1} = (\text{proj}_{[r]}^n, 0)$. Choose $\epsilon > 0$ such that $\bar{B}(\phi_0(p), \epsilon) \subset \phi(U)$. Set $U := \phi_0^{-1}(B(\phi_0(p), \epsilon))$. Since $\bar{B}(\phi_0(p), \epsilon)$ is compact, ϕ_0 is a homeomorphism and $\text{cl } U = \phi_0^{-1}(\bar{B}(\phi_0(p), \epsilon))$, we have that $\text{cl } U$ is compact and $\text{cl } U \subset U_0$.
 - 2.
 - 3.
-

Exercise 8.1.0.6. Global Rank Theorem:

Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. Suppose that F has constant rank.

- 1.
- 2.
- 3.

If F is surjective, then F is a $\mathbf{Man}^\infty$ -submersion,

Proof. Set $m := \dim M$, $n := \dim N$ and $r := \text{rank } F$. Suppose that F is surjective. For the sake of contradiction, suppose that F is not a $\mathbf{Man}^\infty$ submersion. Then $r < n$.

Let $p \in M$. The local rank theorem (Exercise 8.1.0.3) implies that there exists $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$ such that $p \in U$, $F(U) \subset V$ and $\psi \circ F \circ \phi = (\text{proj}_{[r]}^n, 0)$. □

Proof. Set $m := \dim M$, $n := \dim N$ and $r := \text{rank } F$.

1. Suppose that F is surjective. For the sake of contradiction, suppose that F is not a $\mathbf{Man}^\infty$ -submersion. Then $r < n$.
 - 2.
 - 3.
-

Definition 8.1.0.7. Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds, $F : M \rightarrow N$ a smooth map. Then F is said to be

- a **smooth immersion** if for each $p \in M$, $DF(p) : T_p M \rightarrow T_{F(p)} N$ is injective
- a **smooth submersion** if for each $p \in M$, $DF(p) : T_p M \rightarrow T_{F(p)} N$ is surjective

Exercise 8.1.0.8. Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds, $F : M \rightarrow N$ a smooth map. Let $p \in M$.

1. If that $DF(p)$ is injective, then there exists $U \subset M$ such that U is open and $F|_U$ is a smooth immersion.
2. If $DF(p)$ is surjective, then there exists $U \subset M$ such that U is open and $F|_U$ is a smooth submersion.

Proof.

1. Suppose that $DF(p)$ is injective. Exercise 8.1.0.3 implies that there exist $(U, \phi) \in \mathcal{A}$ and $(V, \psi) \in \mathcal{B}$ such that $p \in U$, $F(p) \in V$ and $([DF(p)]_{\phi, \psi})_{i,j}$
 2. Similar to (1).
-

8.2 Immersions

Definition 8.2.0.1. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$. Then F is said to be a $\mathbf{ManBnd}^\infty(U, N)$ -immersion if for each $p \in M$, $DF(p) : T_p M \rightarrow T_{F(p)} N$ is injective.

Note 8.2.0.2. When the context is clear, we say “immersion” instead of “ $\mathbf{ManBnd}^\infty(U, N)$ -immersion”.

Exercise 8.2.0.3. Local Immersion Theorem:

Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$ and $p \in M$. If $DF(p)$ is injective, then there exists $U \in \mathcal{T}_M$ such that $p \in U$ and $F|_U$ is an immersion.

Proof. Suppose that $DF(p)$ is injective. Set $m := \dim M$ and $n := \dim N$. Since $DF(p)$ is injective, $m \leq n$. Since $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$, there exists $(U_0, \phi_0) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$ such that $p \in U_0$, $F(U_0) \subset V$ and $\psi \circ F \circ \phi_0^{-1}$ is smooth. Define $F_0 : \phi_0(U_0) \rightarrow \psi(V)$ by $F_0 := \psi \circ F \circ \phi_0^{-1}$. **An ex in anlaysis notes** implies that $DF_0 \in \text{Hom}_{\mathbf{Man}^\infty}(U_0, L(\mathbb{R}^m, \mathbb{R}^n))$. Define $A : L(\mathbb{R}^m, \mathbb{R}^n) \rightarrow M(m, n)$ (notation?) by $A(e_i \otimes e_j^*) := E^{(ij)}$ (where $E^{(ij)}$ is $n \times m$ matrix of real numbers with (i, j) -th entry is 1 and the rest are 0) (Define A in prelim material) **A prev ex (make ex after smooth manifold section ex about $L(V, M)$ being a smooth manifold)** implies that $A \in \text{Hom}_{\mathbf{Man}^\infty}(L(\mathbb{R}^m, \mathbb{R}^n), M(m, n))$. Define $G : \phi_0(U_0) \rightarrow M(m, n)$ by $G := A \circ DF_0$. We note that $G = [DF(\cdot)]_{\phi_0, \psi}$ (show this). Define $a_0 \in \phi_0(U_0)$ by $a_0 := \phi_0(p)$. Since $\text{rank } G(a_0) = m$, there exists $P \in \text{Perm}(n), Q \in \text{Perm}(m)$ such that $\text{proj}_{m,m} P^* G(a_0) Q \in GL(m)$, where $\text{proj}_{m,m}$ means projecting $M \in M(m, n)$ onto $B \in M(m, m)$ given by $B(j, k) := M_{j,k}$ for $j, k \in [m]$. (define this in prelim material) Define $H : \phi_0(U_0) \rightarrow M(m, m)$ by $H(a) := \text{proj}_{m,m} P^* G(a) Q \in GL(m)$. Then $H \in \text{Hom}_{\mathbf{Man}^\infty}(\phi(U_0), M(m, m))$. Since $GL(m) \in \mathcal{T}_{M(m, m)}$ (ref ex), $H(a_0) \in GL(m)$, and H is continuous, we have that $H^{-1}(GL(m)) \in \mathcal{T}_{\phi_0(U_0)}$ and $a_0 \in H^{-1}(GL(m))$. Set $U := \phi_0^{-1}(H^{-1}(GL(m)))$ and $\phi := \phi_0|_U$. Then $U \in \mathcal{T}_M$, $p \in U$ and $F|_U$ is an immersion **work out explicitly. FINISH!!!!**

FINISH!!!!

FINISH!!!!

□

Exercise 8.2.0.4. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. Define $U \subset M$ by $U := \{p \in M : \text{rank } DF(p) = \dim M\}$. Then

1. $U \in \mathcal{T}_M$,
2. $F|_U$ is an immersion.

Proof.

1. Let $p \in U$. Then $\text{rank } DF(p) = M$. Hence Exercise 8.2.0.3 implies that there exists $V \in \mathcal{T}_M$ such that $p \in V$ and $F|_V$ is an immersion. Since $F|_V$ is a immersion, for each $x \in V$, $\text{rank } DF(x) = \dim M$. Hence $V \subset U$. Since $p \in U$ is arbitrary, we have that for each $p \in U$, there exists $V \in \mathcal{T}_M$ such that $p \in V$ and $V \subset U$. Hence $U \in \mathcal{T}_M$.
2. Let $p \in U$. By construction

$$\begin{aligned} \text{rank } DF|_U(p) &= \text{rank } DF(p) \\ &= \dim M. \end{aligned}$$

Hence $DF|_U(p)$ is injective. Since $p \in U$ is arbitrary, we have that for each $p \in U$, $DF(p)$ is injective. Hence $F|_U$ is an immersion.

□

Definition 8.2.0.5. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$. Then F is said to be a $\mathbf{ManBnd}^\infty$ -embedding if

1. F is a $\mathbf{ManBnd}^\infty$ -immersion,
2. $F|^{F(M)} \in \text{Iso}_{\mathbf{Top}}[(M, \mathcal{T}_M), (F(M), \mathcal{T}_N \cap F(M))]$.

Note 8.2.0.6. Here the topology on $F(M)$ is the subspace topology.

Exercise 8.2.0.7. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$. Suppose that F is an immersion. Then for each $U \in \mathcal{T}_M$, $F|_U$ is an immersion.

Proof. Let $p \in U$. Since $p \in M$ and F is an immersion, $\text{rank } DF(p) = \dim M$. Let $(U_0, \phi_0) \in \mathcal{A}_M$ and $(V', \psi') \in \mathcal{A}_N$. Define $(U', \phi') \in \mathcal{A}_M|_U$ by $U' := U \cap U_0$ and $(\phi' := \phi_0|_{U'})$. Since $\mathcal{A}_M|_U \subset \mathcal{A}_M$, we have that

$$\begin{aligned} \text{rank } D(F|_U)(p) &= \text{rank}[D(F|_U)(p)]_{\phi', \psi} \\ &= \text{rank}[DF(p)]_{\phi', \psi} \\ &= \text{rank } DF(p) \\ &= m \end{aligned}$$

Since $p \in U$ is arbitrary, we have that for each $p \in U$, $D(F|_U)(p)$ is injective. Hence $F|_U$ is an immersion. $\square$

Exercise 8.2.0.8. Local Embedding Theorem:

Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. Then F is an immersion iff for each $p \in M$, there exists $U \in \mathcal{T}_M$ such that $p \in U$ and $F|_U : U \rightarrow N$ is a $\mathbf{Man}^\infty$ -embedding. **generalize to $\mathbf{ManBnd}^\infty$ with local embedding theorem for manifolds with boundary with Lee pg 87**

Proof. Set $\dim M = m$ and $\dim N = n$.

• $(\implies)$:

Suppose that F is an immersion. Let $p \in M$.

- Let $p \in M$. Exercise 8.1.0.3 implies that there exists $(U_0, \phi_0) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$ such that $p \in U$, $F(U_0) \subset V$, and $\psi \circ F \circ \phi^{-1} = (\text{id}_{\phi(U_0)}, 0)$. Thus $\psi \circ F \circ \phi^{-1}$ is injective. Since ϕ, ψ are bijections and $F|_{U_0} = \psi^{-1} \circ (\psi \circ F \circ \phi^{-1}) \circ \phi$, we have that $F|_{U_0}$ is injective. Choose $K \subset U_0$ such that K is compact and $p \in \text{Int } K$. Since $F|_{U_0}$ is injective and continuous, $F|_K$ is injective and continuous. Since K is compact and N is Hausdorff, the **closed map lemma in the analysis notes section on compact spaces and continuity** implies that $F|_K : K \rightarrow F(K)$ is a homeomorphism. Set $U := \text{Int } K$. Then $F|_U : U \rightarrow F(U)$ is a homeomorphism. Since F is an immersion, $F|_U$ is an immersion. Hence $F|_U$ is a $\mathbf{Man}^\infty$ -embedding. **generalize to boundary using Lee pg 87**

• $(\impliedby)$:

Suppose that for each $p \in M$, there exists $U \in \mathcal{T}_M$ such that $p \in U$ and $F|_U : U \rightarrow N$ is a $\mathbf{Man}^\infty$ -embedding. Let $p \in M$. Then there exists $U \in \mathcal{T}_M$ such that $p \in U$ and $F|_U : U \rightarrow N$ is a $\mathbf{Man}^\infty$ -embedding. Since $F|_U$ is a $\mathbf{Man}^\infty$ -embedding, $F|_U$ is a $\mathbf{Man}^\infty$ -immersion. Thus $DF|_U(p) : T_p U \rightarrow T_p N$ is injective. Since $DF(p) = DF|_U(p)$, $DF(p) : T_p M \rightarrow T_p N$ is injective. Since $p \in M$ is arbitrary, we have that for each $p \in M$, $DF(p)$ is injective. Hence F is a $\mathbf{Man}^\infty$ -immersion. $\square$

Exercise 8.2.0.9. Let $(M, \mathcal{A})$ be a smooth manifold and $U \subset M$ open. Then the inclusion map $\iota_U : U \rightarrow M$ is a smooth embedding.

Proof. content... $\square$

Exercise 8.2.0.10. Let $(M, \mathcal{A})$ and $(N, \mathcal{B})$ be smooth manifolds, $p \in M$ and $q \in N$. Suppose that $\partial N = \emptyset$. Then

1. $\iota_q^M : M \rightarrow M \times N$ is a smooth embedding,
2. $\iota_p^N : N \rightarrow M \times N$ is a smooth embedding.

Proof.

1. Exercise 5.3.0.11 implies that ι_q^M is smooth. Let $p \in M$. Then
2. Similar to (1).

$\square$

Exercise 8.2.0.11. Let $M_1, M_2, N_1, N_2 \in \text{Obj}(\mathbf{Man}^\infty)$, $F_1 \in \text{Hom}_{\mathbf{Man}^\infty}(M_1, N_1)$ and $F_2 \in \text{Hom}_{\mathbf{Man}^\infty}(M_2, N_2)$.

1. If F_1 and F_2 are immersions, then $F_1 \times F_2$ is an immersion.
2. If F_1 and F_2 are embeddings, then $F_1 \times F_2$ is an embedding.

Proof.

1. Suppose that F_1 and F_2 are immersions. Set $n_1 := \dim N_1$ and $n_2 := \dim N_2$. Since F_1, F_2 are immersions, $\dim \operatorname{Im} DF_1(p_1) = n_1$ and $\dim \operatorname{Im} DF_2(p_2) = n_2$. Let $(p_1, p_2) \in M_1 \times M_2$. Then [cite exercise in section on products of tangent spaces](#)

$$\begin{aligned} \dim \operatorname{Im} D(F_1 \times F_2)(p_1, p_2) &= \dim \operatorname{Im} DF_1(p_1) \times DF_2(p_2) \\ &= n_1 + n_2 \\ &= \dim T_{F_1 \times F_2(p_1, p_2)} N_1 \times N_2. \end{aligned}$$

Hence $\operatorname{Im} D(F_1 \times F_2)(p_1, p_2) = T_{F_1 \times F_2(p_1, p_2)} N_1 \times N_2$. Since $(p_1, p_2) \in M_1 \times M_2$ is arbitrary, we have that for each $(p_1, p_2) \in M_1 \times M_2$, $D(F_1 \times F_2)(p_1, p_2)$ is injective. Thus $F_1 \times F_2$ is an immersion.

2. Suppose that F_1 and F_2 are embeddings. The previous part implies that $F_1 \times F_2$ is an immersion. Since F_1 and F_2 are embeddings, $F_1|^{F_1(M_1)} \in \operatorname{Iso}_{\mathbf{Top}}(M_1, F_1(M_1))$ and $F_2|^{F_2(M_2)} \in \operatorname{Iso}_{\mathbf{Top}}(M_2, F_2(M_2))$. Exercise ?? [\(cite exercise here\)](#) implies that

$$\begin{aligned} (F_1 \times F_2)|^{F_1 \times F_2(M_1 \times M_2)} &= F_1|^{F_1(M_1)} \times F_1|^{F_1(M_1)} \\ &\in \operatorname{Iso}_{\mathbf{Top}}(M_1 \times M_2, F_1(M_1) \times F_2(M_2)). \end{aligned}$$

Hence $F_1 \times F_2$ is a **Top**-embedding. Thus $F_1 \times F_2$ is an embedding.

FIX!!!

□

Exercise 8.2.0.12. Let $M, N_1, N_2 \in \operatorname{Obj}(\mathbf{Man}^\infty)$, $F_1 \in \operatorname{Hom}_{\mathbf{Man}^\infty}(M, N_1)$ and $F_2 \in \operatorname{Hom}_{\mathbf{Man}^\infty}(M, N_2)$.

1. If F_1 is an immersion or F_2 is an immersion, then (F_1, F_2) is an immersion.
2. If F_1 is an embedding or F_2 is an embedding, then (F_1, F_2) is an immersion.

Proof. Define $G \in \operatorname{Hom}_{\mathbf{ManBnd}^\infty}(M, N_1 \times N_2)$ by $G := (F_1, F_2)$.

1. Suppose that F_1 and F_2 are immersions. Set $m := \dim M$, $n_1 := \dim N_1$ and $n_2 := \dim N_2$. Let $p \in M$. Then there exists $(U, \phi) \in \mathcal{A}_M$, $(V_1, \psi_1) \in \mathcal{A}_{N_1}$ and $(V_2, \psi_2) \in \mathcal{A}_{N_2}$ such that $p \in U \cap F_1^{-1}(V_1) \cap F_2^{-1}(V_2)$. Define $(V, \psi) \in \mathcal{A}_{N_1} \otimes \mathcal{A}_{N_2}$ by $V := V_1 \times V_2$ and $\psi := \psi_1 \times \psi_2$. Write $\phi = (x^1, \dots, x^m)$, $\psi_1 = (y^1, \dots, y^{n_1})$, $\psi_2 = (z^1, \dots, z^{n_2})$ and $\psi = (\tilde{y}^1, \dots, \tilde{y}^{n_1}, \tilde{z}^1, \dots, \tilde{z}^{n_2})$.

Exercise 6.2.0.5 then implies that $[DG(p)]_{\phi, \psi} = \begin{pmatrix} [DF_1(p)]_{\phi, \psi_1} \\ [DF_2(p)]_{\phi, \psi_2} \end{pmatrix}$. Hence $\ker DG(p) = \ker DF_1(p) \cap DF_2(p)$.

- If F_1 is an immersion, then $\ker DF_1(p) = \{0\}$ and therefore

$$\begin{aligned} \ker DG(p) &= \ker DF_1(p) \cap DF_2(p) \\ &= \{0\}. \end{aligned}$$

Hence $DG(p)$ is injective.

- Similarly, if F_2 is an immersion, then $DG(p)$ is injective.

Hence $DG(p)$ is injective. Since $p \in M$ is arbitrary, we have that for each $p \in M$, $DG(p)$ is injective. Hence G is an immersion.

2. Suppose that F_1 and F_2 are embeddings. The previous part implies that G is an immersion. Since F_1 and F_2 are embeddings, $F_1|^{F_1(M)} \in \operatorname{Iso}_{\mathbf{Top}}(M, F_1(M))$ and $F_2|^{F_2(M)} \in \operatorname{Iso}_{\mathbf{Top}}(M, F_2(M))$. Exercise ?? [cite exercise here](#) implies that

$$\begin{aligned} G|^{G(M)} &= (F_1|^{F_1(M)}, F_1|^{F_1(M)}) \\ &\in \operatorname{Iso}_{\mathbf{Top}}(M, G(M)). \end{aligned}$$

Hence G is a **Top**-embedding. Thus G is an embedding. **FIX!!!**

□

Exercise 8.2.0.13. Local Representation of Immersions:

Let $M, N \in \operatorname{Obj}(\mathbf{Man}^\infty)$ and $F \in \operatorname{Hom}_{\mathbf{Man}^\infty}(M, N)$. Then F is an immersion iff for each $p \in M$, there exist $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$ such that $p \in U$, $\phi(U) = V$, and $\psi \circ F \circ \phi^{-1} = (\operatorname{id}_{\phi(U)}, 0)$.

Proof. **FINISH!!!**

□

Exercise 8.2.0.14. [Discuss Lemniscate \(pg 86 Lee\)](#)

8.3 Submersions

give boundary assumptions being empty

Definition 8.3.0.1. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$. Then F is said to be a **submersion** if for each $p \in M$, $DF(p) : T_p M \rightarrow T_{F(p)} N$ is surjective.

Exercise 8.3.0.2. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$ and $p \in M$. If $DF(p)$ is surjective, then there exists $U \in \mathcal{T}_M$ such that $p \in U$ and $F|_U$ is a smooth submersion.

Proof. content... □

Exercise 8.3.0.3. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. Define $U \subset M$ by $U := \{p \in M : \text{rank } DF(p) = \dim N\}$. Then

1. $U \in \mathcal{T}_M$,
2. $F|_U$ is a submersion.

Proof.

1. Let $p \in U$. Then $\text{rank } DF(p) = N$. Hence Exercise 8.3.0.2 implies that there exists $V \in \mathcal{T}_M$ such that $p \in V$ and $F|_V$ is a submersion. Since $F|_V$ is a submersion, for each $x \in V$, $\text{rank } DF(x) = \dim N$. Hence $V \subset U$. Since $p \in U$ is arbitrary, we have that for each $p \in U$, there exists $V \in \mathcal{T}_M$ such that $p \in V$ and $V \subset U$. Hence $U \in \mathcal{T}_M$.
2. Let $p \in U$. By construction

$$\begin{aligned} \text{rank } DF|_U(p) &= \text{rank } DF(p) \\ &= \dim N. \end{aligned}$$

Hence $DF|_U(p)$ is surjective. Since $p \in U$ is arbitrary, we have that for each $p \in U$, $DF(p)$ is surjective. Hence $F|_U$ is a submersion. □

Exercise 8.3.0.4. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$. Then $\pi_M : M \times N \rightarrow M$ and $\pi_N : M \times N \rightarrow N$ are submersions.

Proof. Exercise 6.2.0.1 implies that $[D\pi_M(p, q)]_{\phi, \phi_M} = [I_m, 0]$. Hence $\text{rank}[D\pi_M(p, q)]_{\phi, \phi_M} = m$. Since $\dim T_p M = m$, $D\pi_M(p, q) : M \times N \rightarrow T_p M$ is surjective. Since $(p, q) \in M \times N$ is arbitrary, we have that for each $(p, q) \in M \times N$, $D\pi_M(p, q)$ is surjective. Hence π_M is a submersion. □

Exercise 8.3.0.5. Let $E, M, N \in \text{Obj}(\mathbf{Man}^\infty)$, $F \in \text{Hom}_{\mathbf{Man}^\infty}(E, M)$, $G \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. If F, G are submersions, then $G \circ F$ is a submersion.

Proof. Suppose that F, G are submersions. Let $a \in E$. Then $DF(a)$ and $DG(F(a))$ are surjective. Since $D(G \circ F)(a) = DG(F(a)) \circ DF(a)$, we have that $D(G \circ F)(a)$ is surjective. Since $a \in E$ is arbitrary, we have that for each $a \in E$, $D(G \circ F)(a)$ is surjective. Hence $G \circ F$ is a submersion. □

Exercise 8.3.0.6. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. Then F is a submersion iff for each $p \in M$, there exists $U \in \mathcal{T}_M$ such that $p \in U$ and $F|_U$ is a submersion.

Proof. **FINISH!!!** □

Exercise 8.3.0.7. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ be smooth manifolds, $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$ a smooth map and $p \in M$.

1. If that $DF(p)$ is injective, then there exists $U \subset M$ such that U is open and $F|_U$ is a smooth immersion.
2. If $DF(p)$ is surjective, then there exists $U \subset M$ such that U is open and $F|_U$ is a smooth submersion.

Proof. **FINISH!!!** □

Note 8.3.0.8. We define $\text{proj}_{[n]}^{n+k} : \mathbb{R}^{n+k} \rightarrow \mathbb{R}^n$ by $\text{proj}_{[n]}^{n+k}(a^1, \dots, a^{n+k}) = (a^1, \dots, a^n)$.

Exercise 8.3.0.9. Local Representation of Submersions:

Let $E, M \in \text{Obj}(\mathbf{Man}^\infty)$ and $\pi \in \text{Hom}_{\mathbf{Man}^\infty}(E, M)$. Then π is a submersion iff for each $a \in E$, there exist $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_E$ such that $a \in V$, $U = \pi(V)$, and $\phi \circ \pi \circ \psi^{-1} = \text{proj}_{[n]}^{n+k}|_{\psi(V)}$.

Proof.

• $(\implies)$:

Suppose that π is a submersion. Set $n := \dim M$, $k := \dim E - n$. Let $a \in E$. Set $p := \pi(a)$. Since $\pi : E \rightarrow M$ is a submersion, π has constant rank and $\text{rank } \pi = n$. Exercise 8.1.0.3 implies that there exist $(V, \psi) \in \mathcal{A}_E$, $(U_0, \phi_0) \in \mathcal{A}_M$ such that $a \in V$, $\pi(V) \subset U_0$ and $\phi_0 \circ \pi \circ \psi^{-1} = \text{proj}_{[n]}^{n+k}|_{\psi(V)}$. Define $U := \phi_0^{-1}(\text{proj}_{[n]}^{n+k}(\psi(V)))$. Since $\text{proj}_{[n]}^{n+k}$ is open and $\psi(V)$ is open in $\mathbb{R}^{n+k}$, we have that $\text{proj}_{[n]}^{n+k}(\psi(V))$ is open in $\mathbb{R}^n$. Since ϕ_0 is a homeomorphism, U is open in M . Set $\phi := \phi_0|_U$. **a previous exercise in the section on smooth atlases** implies that $(U, \phi) \in \mathcal{A}_M$. By construction,

$$\begin{aligned} \pi(V) &= [\phi_0^{-1} \circ (\phi_0 \circ \pi \circ \psi^{-1}) \circ \psi](V) \\ &= \phi_0^{-1} \circ \text{proj}_{[n]}^{n+k} \circ \psi(V) \\ &= U. \end{aligned}$$

$$\begin{aligned} \phi \circ \pi \circ \psi^{-1} &= \phi_0|_U \circ \pi \circ \psi^{-1} \\ &= \phi_0 \circ \pi \circ \psi^{-1} \\ &= \text{proj}_{[n]}^{n+k}. \end{aligned}$$

Since $a \in E$ is arbitrary, we have that for each $a \in E$, there exist $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_E$ such that $a \in V$, $U = \pi(V)$, and $\phi \circ \pi \circ \psi^{-1} = \text{proj}_{[n]}^{n+k}|_{\psi(V)}$.

• $(\impliedby)$:

Conversely, suppose that for each $a \in E$, there exist $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_E$ such that $a \in V$, $U = \pi(V)$, and $\phi \circ \pi \circ \psi^{-1} = \text{proj}_{[n]}^{n+k}|_{\psi(V)}$. Let $a \in E$. By assumption, there exists $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_E$ such that $a \in V$, $U = \pi(V)$, and $\phi \circ \pi \circ \psi^{-1} = \text{proj}_{[n]}^{n+k}|_{\psi(V)}$. Since ϕ and ψ are diffeomorphisms, we have that

$$\begin{aligned} \text{rank } D\pi(a) &= \text{rank}[D\phi(\pi(a)) \circ D\pi(a) \circ D\psi^{-1}(\psi(a))] \\ &= \text{rank } D(\phi \circ \pi \circ \psi^{-1})(\psi(a)) \\ &= \text{rank } D \text{proj}_{[n]}^{n+k}|_{\psi(V)}(\psi(a)) \\ &= n \\ &= \dim T_{\pi(a)}M. \end{aligned}$$

Thus $D\pi(a) : T_a E \rightarrow T_{\pi(a)} M$ is surjective. Since $a \in E$ is arbitrary, we have that for each $a \in E$, $D\pi(a)$ is surjective. Hence π is a submersion. □

Exercise 8.3.0.10. Let $E, M \in \text{Obj}(\mathbf{Man}^\infty)$ and $\pi \in \text{Hom}_{\mathbf{Man}^\infty}(E, M)$.

1. If π is a submersion, then π is open.
2. If π is a surjective submersion, then π is a quotient map.

Proof.

1. Suppose that π is a submersion. Let $a \in E$. Exercise 8.3.0.9 implies that there exist $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_E$ such that
 - $a \in V$ and $U = \pi(V)$,

$$\bullet \phi \circ \pi \circ \psi^{-1} = \text{proj}_{[n]}^{n+k} |_{\psi(V)}.$$

Since $\text{proj}_{[n]}^{n+k}$ is open and $\psi(V)$ is open in $\mathbb{R}^{n+k}$, we have that $\text{proj}_{[n]}^{n+k} |_{\psi(V)}$ is open. Since ϕ, ψ are homeomorphisms and $\pi|_V = \phi^{-1} \circ \text{proj}_{[n]}^{n+k} |_{\psi(V)} \circ \psi$, we have that $\pi|_V$ is open. Since $a \in E$ is arbitrary, we have that for each $a \in E$, there exists $V \subset E$ such that V is open in E and $\pi|_V$ is open. [An exercise in the analysis notes section on subspace topology](#) implies that π is open.

2. Suppose that π is a surjective submersion. Part (1) implies that π is open. Since π is surjective, open and continuous, [an exercise in the analysis notes section on quotient maps](#) implies that π is a quotient map. □

Definition 8.3.0.11. Let $E, M \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $\pi \in \text{Hom}_{\mathbf{ManBnd}^\infty}(E, M)$ a surjection and $s : M \rightarrow E$. Then s is said to be a **ManBnd $^\infty$ -section** of π if

1. $s \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, E)$
2. s is a section of π

We define

$$\Gamma(\pi) := \{s \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, E) : s \text{ is a } \mathbf{ManBnd}^\infty\text{-section of } \pi.\}$$

Definition 8.3.0.12. Let $E, M \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $\pi \in \text{Hom}_{\mathbf{ManBnd}^\infty}(E, M)$, $U \in \mathcal{T}_M$ and $s : U \rightarrow E$. Then

- (U, s) is said to be a **smooth local section** of π if $s \in \Gamma(\pi|_{\pi^{-1}(U)})$,
- for each $p \in M$, we define

$$\Gamma_p(\pi) := \{(U, s) : (U, s) \text{ is a } \mathbf{ManBnd}^\infty\text{-local section of } \pi \text{ and } p \in U\}$$

Exercise 8.3.0.13. Local Section Theorem:

Let $E, M \in \text{Obj}(\mathbf{Man}^\infty)$ and $\pi \in \text{Hom}_{\mathbf{Man}^\infty}(E, M)$. Then π is a submersion iff for each $p \in M$ and $a \in \pi^{-1}(\{p\})$, there exists $(U, s) \in \Gamma_p(\pi)$ such that $s(p) = a$.

Proof. Set $n := \dim M$ and $k := \dim E - n$.

- $(\implies) :$

Suppose that π is a submersion. Let $p \in M$ and $a \in \pi^{-1}(\{p\})$. Then $\pi(a) = p$. Exercise 8.3.0.9 implies that there exist $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_E$ such that

- $a \in V$ and $U = \pi(V)$,
- $\phi \circ \pi \circ \psi^{-1} = \text{proj}_{[n]}^{n+k} |_{\psi(V)}.$

Set $\hat{x} := \text{proj}_{[n]}^{n+k}(\psi(a))$ and $\hat{y} := \text{proj}_{[-k]}^{n+k}(\psi(a))$ so that $\psi(a) = (\hat{x}, \hat{y})$. [An exercise in the analysis notes from the section on the product topology](#) implies that there exist $A \in \mathcal{T}_{\mathbb{R}^n}$ and $B \in \mathcal{T}_{\mathbb{R}^k}$ such that $(\hat{x}, \hat{y}) \in A \times B$ and $A \times B \subset \psi(V)$. We note that

$$\begin{aligned} \hat{x} &= \text{proj}_{[n]}^{n+k}(\psi(a)) \\ &= \phi \circ \pi \circ \psi^{-1}(\psi(a)) \\ &= \phi(\pi(a)) \\ &= \phi(p). \end{aligned}$$

Since $A \times B \subset \psi(V)$, we have that

$$\begin{aligned} A &= \text{proj}_{[n]}^{n+k}(A \times B) \\ &= \phi \circ \pi \circ \psi^{-1}(A \times B) \\ &\subset \phi \circ \pi \circ \psi^{-1}(\psi(V)) \\ &= \phi(\pi(V)) \\ &= \phi(U). \end{aligned}$$

We note that for each $(x^1, \dots, x^n) \in A$,

$$\begin{aligned} (x^1, \dots, x^n, \hat{y}) &\in A \times B \\ &\subset \psi(V). \end{aligned}$$

Define $\hat{s} : A \rightarrow \psi(V)$ by $\hat{s}(x^1, \dots, x^n) := (x^1, \dots, x^n, \hat{y})$. Then $\hat{s}$ is smooth. Define $s : \phi^{-1}(A) \rightarrow V$ by $s := \psi^{-1} \circ \hat{s} \circ \phi|_{\phi^{-1}(A)}$. Then s is smooth. Let $q \in \phi^{-1}(A)$. Set $x := \phi(q)$. Then

$$\begin{aligned} \pi \circ s(q) &= [\pi \circ (\psi^{-1} \circ \hat{s} \circ \phi|_{\phi^{-1}(A)})](q) \\ &= [\pi \circ (\psi^{-1} \circ \hat{s} \circ \phi|_{\phi^{-1}(A)})](\phi^{-1}(x)) \\ &= [\pi \circ (\psi^{-1} \circ \hat{s})](x) \\ &= [(\pi \circ \psi^{-1}) \circ \hat{s}](x) \\ &= (\phi^{-1} \circ \text{proj}_{[n]}^{n+k})(x, \hat{y}) \\ &= \phi^{-1}(x) \\ &= q \end{aligned}$$

Since $q \in \phi^{-1}(A)$ is arbitrary, we have that $\pi \circ s = \text{id}_{\phi^{-1}(A)}$ and therefore $(\phi^{-1}(A), s) \in \Gamma_p(\pi)$. By construction,

$$\begin{aligned} s(p) &= \psi^{-1} \circ \hat{s} \circ \phi(p) \\ &= \psi^{-1} \circ \hat{s}(\hat{x}) \\ &= \psi^{-1}(\hat{x}, \hat{y}) \\ &= \psi^{-1}(\psi(a)) \\ &= a. \end{aligned}$$

• ($\Leftarrow$) :

Suppose that for each $p \in M$ and $a \in \pi^{-1}(\{p\})$, there exists $(U, s) \in \Gamma_p(\pi)$ such that $s(p) = a$. Let $a \in E$. Set $p := \pi(a)$. Since $\pi \circ s = \text{id}_U$, we have that

$$\begin{aligned} \text{id}_{T_p U} &= D \text{id}_U(p) \\ &= D[\pi \circ s](p) \\ &= D\pi(s(p)) \circ Ds(p) \\ &= D\pi(a) \circ Ds(p). \end{aligned}$$

Thus

$$\begin{aligned} n &= \text{rank id}_{T_p U} \\ &= \text{rank}(D\pi(a) \circ Ds(p)) \\ &\leq \text{rank } D\pi(a) \\ &\leq n. \end{aligned}$$

Hence $\text{rank } D\pi(a) = n$. Since $a \in E$ is arbitrary, we have that for each $a \in E$, $\text{rank } D\pi(a) = \dim M$. Hence π is a submersion. □

Exercise 8.3.0.14. Let $E, M, N \in \text{Obj}(\mathbf{Man}^\infty)$, $\pi \in \text{Hom}_{\mathbf{Man}^\infty}(E, M)$ and $F : M \rightarrow N$. Suppose that π is a surjective submersion. Then $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$ iff $F \circ \pi \in \text{Hom}_{\mathbf{Man}^\infty}(E, N)$, in which case the following diagram commutes in $\mathbf{Man}^\infty$:

$$\begin{array}{ccc} E & & \\ \pi \downarrow & \searrow F \circ \pi & \\ M & \xrightarrow{F} & N \end{array}$$

Proof.

- $(\implies)$:

Suppose that F is smooth. Then clearly $F \circ \pi$ is smooth.

- $(\impliedby)$:

Suppose that $F \circ \pi$ is smooth. Let $p \in M$. Then there exists a local section $(U, s) \in \Gamma_p(\pi)$ such that $p \in U$. Since $F \circ \pi$ are smooth and s is smooth, we have that

$$\begin{aligned} (F \circ \pi) \circ s &= F \circ (\pi \circ s) \\ &= F \circ \text{id}_U \\ &= F|_U \end{aligned}$$

is smooth. Since $p \in M$ is arbitrary, we have that for each $p \in M$, there exists $U \subset M$ such that U is open in M , $p \in U$ and $F|_U$ is smooth. Thus F is smooth. □

Exercise 8.3.0.15. Let $(E, \mathcal{C})$ be a smooth manifold, M a topological manifold, $\mathcal{A}_1$ and $\mathcal{A}_2$ smooth structures on M and $\pi : E \rightarrow M$. Suppose that π is a surjective. If π is a $(\mathcal{C}, \mathcal{A}_1)$ -smooth submersion and π is a $(\mathcal{C}, \mathcal{A}_2)$ -smooth submersion, then $\mathcal{A}_1 = \mathcal{A}_2$. **clean up notation with $\mathcal{A}_E$ instead of $\mathcal{C}$**

Proof. Suppose that π is a $(\mathcal{C}, \mathcal{A}_1)$ -smooth submersion and π is a $(\mathcal{C}, \mathcal{A}_2)$ -smooth submersion. Since $\text{id}_M \circ \pi = \pi$ and π is $(\mathcal{C}, \mathcal{A}_2)$ -smooth, Exercise 8.3.0.14 implies that id_M is $(\mathcal{A}_1, \mathcal{A}_2)$ -smooth. Similarly, Since π is $(\mathcal{C}, \mathcal{A}_1)$ -smooth Exercise 8.3.0.14 implies that id_M is $(\mathcal{A}_2, \mathcal{A}_1)$ -smooth. Thus id_M is a $(\mathcal{A}_1, \mathcal{A}_2)$ diffeomorphism. Exercise 5.2.0.5 implies that $\mathcal{A}_1 = \mathcal{A}_2$. □

Exercise 8.3.0.16. Let $E, M, N \in \text{Obj}(\mathbf{Man}^\infty)$, $\pi \in \text{Hom}_{\mathbf{Man}^\infty}(E, M)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(E, N)$. Suppose that π is a surjective submersion. If for each $a, b \in E$, $\pi(a) = \pi(b)$ implies that $F(a) = F(b)$, then there exists a unique $\tilde{F} \in \text{Hom}(\mathbf{Man}^\infty)(M, N)$ such that $\tilde{F} \circ \pi = F$, i.e. the following diagram commutes:

$$\begin{array}{ccc} E & & \\ \pi \downarrow & \searrow F & \\ M & \xrightarrow{\tilde{F}} & N \end{array}$$

Proof. Exercise 8.3.0.10 implies that π is a quotient space. We define the relation $\sim_\pi$ on E by $a \sim_\pi b$ iff $\pi(a) = \pi(b)$. Let $p_\pi : E \rightarrow E/\sim_\pi$ be the projection map. **An exercise in the analysis notes section on quotient spaces** implies that there exists $h : E/\sim_\pi \rightarrow M$ such that h is a homeomorphism and $h \circ p_\pi = \pi$. Thus $p_\pi = h^{-1} \circ \pi$. By assumption, F is $\sim_\pi$ -invariant. **Another exercise in the analysis notes section on quotient spaces** implies that there exists a unique $\bar{F} : E/\sim_\pi \rightarrow N$ such that $\bar{F}$ is continuous and $\bar{F} \circ p_\pi = F$. Set $\tilde{F} := \bar{F} \circ h^{-1}$. Therefore,

$$\begin{aligned} \tilde{F} \circ \pi &= (\bar{F} \circ h^{-1}) \circ \pi \\ &= \bar{F} \circ (h^{-1} \circ \pi) \\ &= \bar{F} \circ p_\pi \\ &= F, \end{aligned}$$

i.e. the following diagram commutes:

$$\begin{array}{ccccc} & & E & & \\ & \swarrow F & \downarrow p_\pi & \searrow \pi & \\ N & \xleftarrow{\bar{F}} & E/\sim_\pi & \xleftarrow{h^{-1}} & M \end{array}$$

Since F is smooth and $\tilde{F} \circ \pi = F$, we have that $\tilde{F} \circ \pi$ is smooth, i.e. the following diagram commutes:

$$\begin{array}{ccc} E & & \\ \pi \downarrow & \searrow \tilde{F} \circ \pi & \\ M & \xrightarrow{\tilde{F}} & N \end{array}$$

Exercise 8.3.0.14 then implies that $\tilde{F}$ is smooth. □

Chapter 9

Submanifolds

need to figure out a more systematic way to handle restriction of codomains. Maybe for $F : A \rightarrow B$ with $F(A) \subset B' \subset B$, introduce notation $F|^{B'} : A \rightarrow B'$ by $F|^{B'}(x) = F(x)$. Try to phrase this in terms of composition, for example $F|_A' = F \circ \iota_A$, does it make sense to have $F|^{B'} = q \circ F$ with $q : B \rightarrow B'$ in a way that if $x \in B'$, then $q(x) = x$? According to <https://en.wikipedia.org/wiki/Corestriction>, $F|^{B'}$ is the unique map such that $\iota_{B'} \circ F|^{B'} = F$, which we can show exists., maybe dont call this corestriction, but constriction, can also dilate $F : A \rightarrow B'$ to $F : A \rightarrow B$

9.1 Introduction

Let $F : A \rightarrow B$, $B' \subset B$ and $F(A) \subset B'$. Define the constriction of F to B' by $F|^{B'} : A \rightarrow B'$ by $F|^{B'}(x) = F(x)$

Definition 9.1.0.1. Let $M, S \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $S \subset M$.

- Then S is said to be an **immersed submanifold** of M if the inclusion map $\iota_S : S \rightarrow M$ is an immersion.
- If S is an immersed submanifold of M , then M is said to be the **ambient manifold of S** .
- If S is an immersed submanifold of M , we define the **codimension of S with respect to M** , denoted $\text{codim}_M(S)$, by $\text{codim}_M(S) = \dim M - \dim S$.

Exercise 9.1.0.2. Let $M, N, S \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. Suppose that S is an immersed submanifold of M . Then $F|_S \in \text{Hom}_{\mathbf{Man}^\infty}(S, N)$.

Proof. Since S is an immersed submanifold of M , the inclusion $\iota_S \in \text{Hom}_{\mathbf{Man}^\infty}(S, M)$. Therefore

$$\begin{aligned} F|_S &= F \circ \iota \\ &\in \text{Hom}_{\mathbf{Man}^\infty}(S, N). \end{aligned}$$

□

Definition 9.1.0.3. Let $(M, \mathcal{T}_M, \mathcal{A}_M), (S, \mathcal{T}_S, \mathcal{A}_S) \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $S \subset M$. Then S is said to be an **embedded submanifold** of M if the inclusion map $\iota_S : (S, \mathcal{T}_S, \mathcal{A}_S) \rightarrow (M, \mathcal{T}_M, \mathcal{A}_M)$ is a $\mathbf{Man}^\infty$ -embedding.

Exercise 9.1.0.4. Let $M, S \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $S \subset M$. If S is an embedded submanifold of M , then S is an immersed submanifold of M .

Proof. Clear.

□

Exercise 9.1.0.5. Immersed Implies Locally Embedded:

Let $M, S \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $S \subset M$. Then S is an immersed submanifold of M iff for each $p \in S$, there exists $U \in \mathcal{T}_S$ such that $p \in U$ and U is an embedded submanifold of M .

Proof.

- $(\implies)$:

Suppose that S is an immersed submanifold of M . Then $\iota_S : S \rightarrow M$ is an immersion. Let $p \in S$. Since ι_S is an immersion, Exercise 8.2.0.8 implies that there exists $U \in \mathcal{T}_S$ such that $p \in U$ and $\iota_S|_U$ is a **Man** $^\infty$ -embedding. Since $\iota_S|_U = \iota_U$, we have that ι_U is a **Man** $^\infty$ -embedding and U is an embedded submanifold of M .

- $(\impliedby)$:

Suppose that for each $p \in S$, there exists $U \in \mathcal{T}_S$ such that $p \in U$ and U is an embedded submanifold of M . Let $p \in S$. By assumption, there exists $U \in \mathcal{T}_S$ such that $p \in U$ and U is an embedded submanifold of M . Thus ι_U is a **Man** $^\infty$ -embedding. Since $\iota_U = \iota_S|_U$, we have that $\iota_S|_U$ is a **Man** $^\infty$ -embedding. Since $p \in S$ is arbitrary, we have that for each $p \in S$, there exists $U \in \mathcal{T}_S$ such that $p \in U$ and $\iota_S|_U$ is a **Man** $^\infty$ -embedding. Exercise 8.2.0.8 implies that ι_S is an immersion. Thus S is an immersed submanifold of M . □

Exercise 9.1.0.6. Uniqueness of Topology for Embedded Submanifolds Let $(M, \mathcal{T}_M, \mathcal{A}_M), (S, \mathcal{T}_S, \mathcal{A}_S) \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $S \subset M$ and $(S, \mathcal{T}_S, \mathcal{A}_S)$ is an embedded submanifold of $(M, \mathcal{T}_M, \mathcal{A}_M)$. Then $\mathcal{T}_S = \mathcal{T}_M \cap S$.

Proof. Since $(S, \mathcal{T}_S, \mathcal{A}_S)$ is an embedded submanifold of $(M, \mathcal{T}_M, \mathcal{A}_M)$, $\iota_S \in \text{Iso}_{\text{Top}}[(S, \mathcal{T}_S), (S, \mathcal{T}_M \cap S)]$. [An exercise in the analysis notes section on subspaces](#) implies that $\mathcal{T}_S = \mathcal{T}_M \cap S$. □

Exercise 9.1.0.7. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$, $p \in M$ and $q \in N$. Then $M \times \{q\}$ and $N \times \{p\}$ are embedded submanifold of $M \times N$. [move to later in products section as corollary](#)

Proof. **FINISH!!!** □

Exercise 9.1.0.8. Let M, U be a smooth manifolds. Suppose that $U \subset M$. Then U is an embedded submanifold of M and $\text{codim}_M(U) = 0$ iff U is an open submanifold of M .

Proof.

- $(\implies)$:

Suppose that U is an embedded submanifold of M and $\text{codim}_M(U) = 0$. **FINISH!!!**

- $(\impliedby)$:

Suppose that U is an open submanifold of M . [need to say why \$U\$ is embedded](#) Exercise 3.2.1.6 and Definition 4.2.1.3 implies that $\dim U = n$, so that $\text{codim}_M(U) = 0$. □

Definition 9.1.0.9. Let $(M, \mathcal{A}), (S, \mathcal{B}) \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $S \subset M$ and $(S, \mathcal{B})$ is an embedded submanifold of $(M, \mathcal{A})$. Then $(S, \mathcal{B})$ is said to be **properly embedded** if $\iota_S : S \rightarrow M$ is proper.

Exercise 9.1.0.10. Let $(M, \mathcal{A}), (S, \mathcal{B}) \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $S \subset M$ and $(S, \mathcal{B})$ is an embedded submanifold of $(M, \mathcal{A})$. Then $(S, \mathcal{B})$ is properly embedded iff S is closed in M .

Proof.

- $(\implies)$:

Suppose that $(S, \mathcal{B})$ is properly embedded. Then $\iota_S : S \rightarrow M$ is proper. [An exercise in the analysis notes section on locally compact Hausdorff spaces](#) implies that ι_S is closed. Since S is closed in S and ι_S is closed, we have that $\iota_S(S)$ is closed in M . Since $\iota_S(S) = S$, we have that S is closed in M .

- $(\impliedby)$:

Conversely, suppose that S is closed in M . Let $K \subset M$. Suppose that K is compact in M . Since M is Hausdorff and S is closed in M , [an exercise in the analysis notes section on compactness](#) implies that $K \cap S$ is compact in M . [An exercise in the analysis notes section on compactness](#) implies that $K \cap S$ is compact in S . Since $\iota_S^{-1}(K) = K \cap S$, $\iota_S^{-1}(K)$ is compact in S . Since $K \subset M$ with K compact in M is arbitrary, we have that for each $K \subset M$, K is compact implies that $\iota_S^{-1}(K)$ is compact in S . Thus ι_S is proper. □

Definition 9.1.0.11. Let $n \in \mathbb{N}$ and $k \in [n]$. We define the k -**slice** of $\mathbb{R}^n$, denoted $\mathbb{S}^{n,k}$, by $\mathbb{S}^{n,k} := \{a \in \mathbb{R}^n : a^{k+1}, \dots, a^n = 0\}$.

Definition 9.1.0.12. Let $U \subset \mathbb{R}^n$ and $S \subset U$. Then S is said to be a k -**slice** of U if $S = U \cap \mathbb{S}^{n,k}$.

Exercise 9.1.0.13. show $\mathbb{S}^{n,k}$ is a k -slice of $\mathbb{R}^n$.

Proof. Clear. □

Definition 9.1.0.14. Let M be a smooth manifold, $S \subset M$ and $(U, \phi) \in \mathcal{A}_M$. Then (U, ϕ) is said to be a k -**slice chart** on S if $\phi(U \cap S)$ is a k -slice of $\phi(U)$. We define

$$\mathbb{S}^k(M; S) := \{(U, \phi) \in \mathcal{A}_M : (U, \phi) \text{ is a } k\text{-slice chart on } S\}$$

Exercise 9.1.0.15. Let M be a smooth manifold, $S \subset M$ and $(U, \phi) \in \mathcal{A}_M$ with $\phi = (x^1, \dots, x^n)$. If (U, ϕ) is a k -slice chart on S , then $\phi|_S = (x^1|_S, \dots, x^k|_S, 0, \dots, 0)$.

Proof. Clear. □

Definition 9.1.0.16. Let M be a smooth manifold and $S \subset M$. Then S is said to **satisfy the local k -slice condition with respect to M** if for each $p \in S$, there exists $(U, \phi) \in \mathbb{S}^k(M; S)$ such that $p \in U$.

Exercise 9.1.0.17. Let M, N be smooth manifolds and $S \subset M$. Suppose that $\dim M = m$, $\dim N = n$ and $M \subset N$. Then

1. $\mathbb{S}^k(M; S) \subset \mathbb{S}^k(N; S)$
- 2.

Proof. FINISH!!! □

Exercise 9.1.0.18. Let $U \subset \mathbb{R}^n$ and $S \subset U$. Suppose that S is a k -slice of U . Define $\text{proj}_{[k]}^n : \mathbb{R}^n \rightarrow \mathbb{R}^k$ by

$$\text{proj}(u^1, \dots, u^k, \dots, u^n) = (u^1, \dots, u^k)$$

Then $\pi_{[k]}^n|_S \rightarrow \text{proj}_k^n(S)$ is a diffeomorphism.

Proof. Clear. FINISH!!! □

Exercise 9.1.0.19. Let $M, S \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $S \subset M$. If S is a k -dimensional embedded submanifold of M , then S satisfies the local k -slice condition with respect to M .

Hint: Draw a picture

Proof. Set $n := \dim M$. Suppose that S is a k -dimensional embedded submanifold of M . Let $p \in S$. Since S is an embedded submanifold of M , the inclusion map $\iota : S \rightarrow M$ is an immersion. The local rank theorem (Exercise 8.1.0.3) implies that Then there exists $(U_0, \phi_0) \in \mathcal{A}_S$, $(V_0, \psi_0) \in \mathcal{A}_M$ such that $p \in U_0$, $\iota(p) \in V_0$, $\iota(U_0) \subset V_0$ and $\psi_0 \circ \iota \circ \phi_0^{-1} = (\text{id}_{\phi_0(U_0)}, 0)$. Since for each $q \in U_0$, $\iota(q) = q$, we have that $U_0 \subset V_0$ and $\psi_0 \circ \iota \circ \phi_0^{-1} = \psi_0 \circ \phi_0^{-1}$. Therefore for each $q \in U_0$,

$$\begin{aligned} \psi_0(q) &= \psi_0 \circ \phi_0^{-1}(\phi_0(q)) \\ &= \psi_0 \circ \iota \circ \phi_0^{-1}(\phi_0(q)) \\ &= (\text{id}_{\mathbb{R}^k}(\phi_0(q)), 0) \\ &= (\phi_0(q), 0) \end{aligned}$$

and in particular, $\psi_0(p) = (\phi_0(p), 0)$. Since $U_0 \in \mathcal{T}_S$ and $\mathcal{T}_S = \mathcal{T}_M \cap S$, there exists $U' \in \mathcal{T}_M$ such that $U_0 = U' \cap S$. [An exercise in the analysis notes in the section on product topology](#) implies that there exist $A_0 \in \mathcal{T}_{\mathbb{R}^k}$ and $B_0 \in \mathcal{T}_{\mathbb{R}^{n-k}}$ such that $(\phi(p), 0) \in A_0 \times B_0$ and $A_0 \times B_0 \subset \psi_0(V_0 \cap U') \cap [\phi_0(U_0) \times \mathbb{R}^{n-k}]$. Define $(V, \psi) \in \mathcal{A}_M$ by $V := \psi_0^{-1}(A_0 \times B_0)$ and $\psi := \psi_0|_V$. [A previous exercise in the subsection about smooth maps on subspaces](#) implies that $(V, \psi) \in \mathcal{A}_M$. Then $p \in V$.

- Let $y \in A_0 \times \{0\}$. Then there exists $a \in A_0$ such that $y = (a, 0)$. Since $A_0 \times B_0 \subset \phi_0(U_0) \times \mathbb{R}^{n-k}$, we have that $A_0 \subset \phi_0(U_0)$. In particular, $a \in \phi_0(U_0)$ and $\phi_0^{-1}(a) \in U_0$. Hence

$$\begin{aligned} y &= (a, 0) \\ &= \psi_0 \circ \phi_0^{-1}(a) \\ &\in \psi_0(U_0). \end{aligned}$$

By construction,

$$\begin{aligned} y &= (a, 0) \\ &= \psi_0(\psi_0^{-1}(a, 0)) \\ &\in \psi_0[\psi_0^{-1}(A_0 \times \{0\})] \\ &\subset \psi_0[\psi_0^{-1}(A_0 \times B_0)] \\ &= \psi_0(V). \end{aligned}$$

Therefore

$$\begin{aligned} y &\in \psi_0(U_0) \cap \psi_0(V) \\ &= \psi_0[(U_0) \cap V] \\ &= \psi_0[(U' \cap S) \cap V_0] \cap V \\ &= \psi_0(V \cap S). \end{aligned}$$

Since $y \in A_0 \times \{0\}$ is arbitrary, we have that $A_0 \times \{0\} \subset \psi_0(V \cap S)$.

- Conversely, we note that for each $q \in V \cap S$,

$$\begin{aligned} (\phi_0(q), 0) &= \psi_0(q) \\ &\in \psi_0(V \cap S) \\ &\subset \psi_0(V) \\ &= A_0 \times B_0, \end{aligned}$$

and therefore $\phi_0(V \cap S) \subset A_0$. Hence

$$\begin{aligned} \psi_0(V \cap S) &= \phi_0(V \cap S) \times \{0\} \\ &\subset A_0 \times \{0\}. \end{aligned}$$

Thus $A_0 \times \{0\} = \psi_0(V \cap S)$ and

$$\begin{aligned} \psi(V \cap S) &= \psi_0(V \cap S) \\ &= A_0 \times \{0\} \\ &= (A_0 \times B_0) \cap \mathbb{S}^{n,k} \\ &= \psi(V) \cap \mathbb{S}^{n,k}. \end{aligned}$$

Hence $\psi(V \cap S)$ is a k -slice of $\psi(V)$ and therefore $(V, \psi) \in \mathbb{S}^k(M; S)$. Since $p \in S$ is arbitrary, we have that for each $p \in S$, there exists $(V, \psi) \in \mathbb{S}^k(M; S)$ such that $p \in V$. Therefore S satisfies the local k -slice condition with respect to M . $\square$

Exercise 9.1.0.20. Let $(M, \mathcal{A}) \in \text{Obj}(\mathbf{Man}^\infty)$ and $S \subset M$. Suppose that $\dim M = n$ and S satisfies the local k -slice condition with respect to M . Then

1. for each $(U, \phi) \in \mathbb{S}^k(M; S)$, if $U \cap S \neq \emptyset$, then $(U \cap S, \pi_{n,k} \circ \phi|_{U \cap S}) \in X^k(S)$,
2. $(S, \mathcal{T}_M \cap S) \in \text{Obj}(\mathbf{Man}^0)$ and $\dim S = k$.

Proof.

1. Let $(U_0, \phi_0) \in \mathbb{S}^k(M; S)$. Suppose that $U_0 \cap S \neq \emptyset$. Set $U := U_0 \cap S$ and $\phi := \phi_0|_U$. Since $(U_0, \phi_0) \in \mathbb{S}^k(M; S)$, we have that

$$\begin{aligned}\phi_0(U) &= \phi_0(U_0 \cap S) \\ &= \phi_0(U_0) \cap \mathbb{S}^{n,k} \\ &\in \mathcal{T}_{\mathbb{R}^n} \cap \mathbb{S}^{n,k}\end{aligned}$$

(a) By assumption, $U_0 \in \mathcal{T}_M$. Therefore $U \in \mathcal{T}_M \cap S$.

(b) Since $(U_0, \phi_0) \in X^n(M, \mathcal{T}_M)$, $\phi_0(U_0) \in \mathcal{T}_{\mathbb{R}^n}$. Since $(U_0, \phi_0) \in \mathbb{S}^k(M; S)$, we have that

$$\begin{aligned}\phi_0(U_0 \cap S) &= \phi_0(U_0) \cap \mathbb{S}^{n,k} \\ &\in \mathcal{T}_{\mathbb{R}^n} \cap \mathbb{S}^{n,k} \\ &= \mathcal{T}_{\mathbb{S}^{n,k}}\end{aligned}$$

By a previous exercise, $\pi_{[k]}^n|_{\mathbb{S}^k}$ is a $(\mathcal{T}_{\mathbb{S}^{n,k}}, \mathcal{T}_{\mathbb{R}^k})$ -homeomorphism. Hence

$$\begin{aligned}\phi(U) &= \pi_{[k]}^n \circ \phi_0(U_0 \cap S) \\ &\in \mathcal{T}_{\mathbb{R}^k}\end{aligned}$$

(c) Since $\phi_0|_U$ is a $(\mathcal{T}_M \cap U, \mathcal{T}_{\mathbb{S}^{n,k}} \cap \phi_0(U_0))$ -homeomorphism and $\pi_{[k]}^n|_{\phi(U)}$ is a $(\mathcal{T}_{\mathbb{S}^{n,k}} \cap \phi_0(U_0), \mathcal{T}_{\mathbb{R}^k} \cap \phi(U))$ -homeomorphism, we have that ϕ is a $(\mathcal{T}_M \cap U, \mathcal{T}_{\mathbb{R}^k} \cap \phi(U))$ -homeomorphism.

Hence $(U, \phi) \in X^k(S)$.

2. (a) Since $(M, \mathcal{T}_M)$ is Hausdorff, $(S, \mathcal{T}_M \cap S)$ is Hausdorff.
 (b) Since $(M, \mathcal{T}_M)$ is second-countable, $(S, \mathcal{T}_M \cap S)$ is second-countable.
 (c) Let $p \in S$. Since S satisfies the local k -slice condition with respect to M , there exists $(U_0, \phi_0) \in \mathcal{A}$ such that $p \in U_0$ and $(U_0, \phi_0) \in \mathbb{S}^k(M; S)$. Set $U := U_0 \cap S$ and $\phi := \pi_{[k]}^n \circ \phi_0|_U$. Then $p \in U$ and the previous part implies that $(U, \phi) \in X^k(S, \mathcal{T}_M \cap S)$. Since $p \in S$ is arbitrary, we have that for each $p \in S$, there exists $(U, \phi) \in X^k(S, \mathcal{T}_M \cap S)$ such that $p \in U$. Hence S is locally Euclidean of dimension k .

Thus $(S, \mathcal{T}_M \cap S) \in \text{Obj}(\mathbf{Man}^0)$ and $\dim S = k$.

□

Definition 9.1.0.21. Let $(M, \mathcal{A}) \in \text{Obj}(\mathbf{Man}^\infty)$ and $S \subset M$. Suppose that $\dim M = n$ and S satisfies the local k -slice condition with respect to M . We define

$$\mathcal{A}|_S^0 := \{(U \cap S, \pi_{[k]}^n \circ \phi_{U \cap S}) : (U, \phi) \in \mathbb{S}^k(M; S)\}.$$

Exercise 9.1.0.22. Let $(M, \mathcal{A}) \in \text{Obj}(\mathbf{Man}^\infty)$ and $S \subset M$. Suppose that S satisfies the local k -slice condition with respect to M . Then

1. $\mathcal{A}|_S^0$ is an atlas on S ,
2. $\mathcal{A}|_S^0$ is smooth.

Proof.

1. The previous exercise implies that $\mathcal{A}|_S^0 \subset X^k(M, \mathcal{T}_M \cap S)$. Let $p \in S$. Since S satisfies the local k -slice condition with respect to M , there exists $(U_0, \phi_0) \in \mathbb{S}^k(M; S)$ such that $p \in U_0$. Set $U := U_0 \cap S$ and $\phi := \phi_0|_U$. By definition, $(U, \phi) \in \mathcal{A}|_S^0$. By construction, $p \in U$. Since $p \in S$ is arbitrary, we have that for each $p \in S$, there exists $(U, \phi) \in \mathcal{A}|_S^0$ such that $p \in U$. Hence $\mathcal{A}|_S^0$ is an atlas on S .

2. Let $(U, \phi), (V, \psi) \in \mathcal{A}_S^0$. Then there exist $(U_0, \phi_0), (V_0, \psi_0) \in \mathbb{S}^k(M; S)$ such that $U = U_0 \cap S$, $V = V_0 \cap S$, $\phi = \pi_{[k]}^n \circ \phi_0|_U$ and $\psi = \pi_{[k]}^n \circ \psi_0|_V$.

$$\begin{aligned}
\psi|_{U \cap V} \circ \phi|_{U \cap V}^{-1} &= (\pi_{[k]}^n|_{\psi_0(S \cap U_0 \cap V_0)} \circ \psi_0|_{S \cap (U_0 \cap V_0)}) \circ (\pi_{[k]}^n|_{\phi_0(S \cap U_0 \cap V_0)} \circ \phi_0|_{S \cap (U_0 \cap V_0)})^{-1} \\
&= (\pi_{[k]}^n|_{\psi_0(S \cap U_0 \cap V_0)} \circ \psi_0|_{S \cap (U_0 \cap V_0)}) \circ (\phi_0|_{S \cap (U_0 \cap V_0)}^{-1} \circ \pi_{[k]}^n|_{\phi_0(S \cap U_0 \cap V_0)}) \\
&= \pi_{[k]}^n|_{\psi_0(S \cap U_0 \cap V_0)} \circ [\psi_0|_{S \cap (U_0 \cap V_0)} \circ \phi_0|_{S \cap (U_0 \cap V_0)}^{-1}] \circ \pi_{[k]}^n|_{\phi_0(S \cap U_0 \cap V_0)}^{-1} \\
&= \pi_{[k]}^n|_{\psi_0(S \cap U_0 \cap V_0)} \circ [\psi_0|_{U_0 \cap V_0} \circ \phi_0|_{U_0 \cap V_0}^{-1}]|_{\phi_0(S \cap (U_0 \cap V_0))} \circ \pi_{[k]}^n|_{\phi_0(S \cap U_0 \cap V_0)}^{-1} \\
&= \pi_{[k]}^n|_{\psi_0(U \cap V)} \circ [\psi_0|_{U_0 \cap V_0} \circ \phi_0|_{U_0 \cap V_0}^{-1}]|_{\phi_0(U \cap V)} \circ \pi_{[k]}^n|_{\phi_0(U \cap V)}^{-1}
\end{aligned}$$

Since $\mathcal{A}$ is smooth, we have that $\psi_0|_{U_0 \cap V_0} \circ \phi_0|_{U_0 \cap V_0}^{-1}$ is smooth. Thus $(\psi_0|_{U_0 \cap V_0} \circ \phi_0|_{U_0 \cap V_0}^{-1})|_{\phi_0(U \cap V)}$ is smooth. A previous exercise implies that $\pi_{[k]}^n|_{\phi_0(U \cap V)}$ and $\pi_{[k]}^n|_{\psi_0(U \cap V)}$ are smooth. Thus $\psi|_{U \cap V} \circ \phi|_{U \cap V}^{-1}$ is smooth. Similarly, $\phi|_{U \cap V} \circ \psi|_{U \cap V}^{-1}$ is smooth. Hence $\psi|_{U \cap V} \circ \phi|_{U \cap V}^{-1}$ is a diffeomorphism and $(U, \phi), (V, \psi)$ are smoothly compatible. Since $(U, \phi), (V, \psi) \in \mathcal{A}_S^0$ are arbitrary, we have that for each $(U, \phi), (V, \psi) \in \mathcal{A}_S^0$, (U, ϕ) and (V, ψ) are smoothly compatible. Therefore $\mathcal{A}_S^0$ is smooth. $\square$

Definition 9.1.0.23. Let $(M, \mathcal{A}) \in \text{Obj}(\mathbf{Man}^\infty)$ and $S \subset M$. Suppose that S satisfies the local k -slice condition with respect to M . We define the **embedded smooth structure on S induced by $\mathcal{A}$** , denoted $\mathcal{A}|_S$, by

$$\mathcal{A}|_S := \alpha(\mathcal{A}_S^0).$$

Exercise 9.1.0.24. Let $(M, \mathcal{A}) \in \text{Obj}(\mathbf{Man}^\infty)$ and $S \subset M$. Suppose that S satisfies the local k -slice condition with respect to M . Then $(S, \mathcal{T}_M \cap S, \mathcal{A}|_S)$ is an embedded submanifold of $(M, \mathcal{T}_M, \mathcal{A})$,

Proof. By definition, ι_S is a topological embedding (check this). Let $p \in S$. Since S satisfies the local k -slice condition with respect to M , there exists $(V_0, \psi_0) \in \mathbb{S}^k(M; S)$ such that $p \in V_0$. Set $V := V_0 \cap S$ and $\psi := \pi_{[k]}^n \circ \psi_0|_V$. By definition,

$$\begin{aligned}
(V, \psi) &\in \mathcal{A}_S^0 \\
&\subset \mathcal{A}|_S.
\end{aligned}$$

Hence

$$\begin{aligned}
\psi_0 \circ \iota \circ \psi^{-1} &= \psi_0 \circ \psi^{-1} \\
&= \psi_0 \circ (\pi_{[k]}^n|_{\psi_0(V)} \circ \psi_0|_V)^{-1} \\
&= \psi_0 \circ \psi_0|_V^{-1} \circ \pi_{[k]}^n|_{\psi_0(V)}^{-1} \\
&= \pi_{[k]}^n|_{\psi_0(V)}^{-1}
\end{aligned}$$

A previous exercise in the section on immersions implies that $\pi_{[k]}^n|_{\psi_0(V)}^{-1}$ is an immersion and $\text{rank } \pi_{[k]}^n|_{\psi_0(V)}^{-1} = k$. Since $(V, \psi) \in \mathcal{A}$ and $(V_0, \psi_0) \in \mathcal{A}|_S$, an exercise in the section on smooth maps on submanifolds implies that ψ and ψ_0 are diffeomorphisms. Therefore

$$\begin{aligned}
\text{rank } D\iota(p) &= \text{rank } D(\psi_0 \circ \iota \circ \psi^{-1})(\psi(p)) \\
&= \text{rank } D(\psi_0 \circ \psi^{-1})(\psi(p)) \\
&= \text{rank } D(\pi_{[k]}^n|_{\psi_0(V)}^{-1})(\psi(p)) \\
&= k
\end{aligned}$$

Since $p \in S$ is arbitrary, we have that for each $p \in S$, $\text{rank } D\iota(p) = k$. Thus ι has constant rank and $\text{rank } \iota = k$. Since $\dim S = k$, an exercise in the section on maps of constant rank implies that ι is an immersion. Thus $(S, \mathcal{A}|_S)$ is an embedded submanifold of $(M, \mathcal{A})$. $\square$

Note 9.1.0.25. Let $(M, \mathcal{A}) \in \text{Obj}(\mathbf{Man}^\infty)$ and $S \subset M$. Suppose that S satisfies the local k -slice condition with respect to M . Unless otherwise specified, we equip S with $\mathcal{A}|_S$.

Exercise 9.1.0.26. Let $M, N, S \in \text{Obj}(\mathbf{Man}^\infty)$, $F \in \text{Hom}_{\mathbf{Man}^\infty}(N, M)$. Suppose that $S \subset M$ and S is an immersed submanifold of M , $F(N) \subset S$ and $F \in \text{Hom}_{\mathbf{Top}}(N, S)$. Then $F|_S \in \text{Hom}_{\mathbf{Man}^\infty}(N, S)$.

Hint: Note that $F = \iota_S \circ F|_S$.

Proof. Set $m := \dim M$, $k := \dim S$ and $n := \dim N$. Set $F_0 := F|_S$. We note that $\iota_S \circ F_0 = F$. Since S is an immersed submanifold of M , ι_S is an immersion. Let $p \in N$. Define $q \in S$ by $q := F(p)$. Exercise 8.2.0.8 implies that there exist $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_S$ such that $q \in V$, $\iota_S(V) \subset U$ and $\phi \circ \iota_S \circ \psi^{-1} = (\text{id}_{\psi(V)}, 0)$. Since F_0 is $(\mathcal{T}_N, \mathcal{T}_S)$ -continuous, $F_0^{-1}(V) \in \mathcal{T}_N$. Then there exists $(W, \eta) \in \mathcal{A}_N$ such that $p \in W$ and $W \subset F_0^{-1}(V)$. Define $\hat{F} : \eta(W) \rightarrow \phi(U)$ and $\hat{F}_0 : \eta(W) \rightarrow \psi(V)$ by $\hat{F} := \phi \circ F \circ \eta^{-1}$ and $\hat{F}_0 := \psi \circ F_0 \circ \eta^{-1}$. Since F is smooth, $\hat{F}$ is smooth. Then

$$\begin{aligned} (\hat{F}_0, 0) &= (\text{id}_{\psi(V)} \circ \hat{F}_0, 0) \\ &= (\text{id}_{\psi(V)}, 0) \circ \hat{F}_0 \\ &= (\phi \circ \iota_S \circ \psi^{-1}) \circ (\psi \circ F_0 \circ \eta^{-1}) \\ &= \phi \circ \iota_S \circ F_0 \circ \eta^{-1} \\ &= \phi \circ F \circ \eta^{-1} \\ &= \hat{F} \end{aligned}$$

Since $\hat{F}$ is smooth, we have that $\hat{F}_0$ is smooth. Since $p \in N$ is arbitrary, we have that for each $p \in N$, there exists $(W, \eta) \in \mathcal{A}_N$ and $(V, \psi) \in \mathcal{A}_S$ such that $p \in W$, $F_0(p) \in V$, $W \cap F_0^{-1}(V) \in \mathcal{T}_N$ and $\psi \circ F_0 \circ \eta^{-1}|_{W \cap F_0^{-1}(V)}$ is smooth. Exercise 5.1.0.10 implies that F_0 is smooth. $\square$

Exercise 9.1.0.27. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$, $F \in \text{Hom}_{\mathbf{Man}^\infty}(N, M)$ and $S \subset M$. Suppose that S is an embedded submanifold of M and $F(N) \subset S$. Then $F \in \text{Hom}_{\mathbf{Man}^\infty}(N, S)$.

Proof. Since S is an embedded submanifold of M , ι_S is a $\mathbf{Man}^\infty$ -embedding. Let $V \in \mathcal{T}_S$. Then

$$\begin{aligned} V &= \iota_S(V) \\ &\in \mathcal{T}_M \cap S. \end{aligned}$$

Therefore there exists $U \in \mathcal{T}_M$ such that $V = U \cap S$. Since F is $(\mathcal{T}_N, \mathcal{T}_M)$ -continuous, $F^{-1}(U) \in \mathcal{T}_N$. Hence

$$\begin{aligned} F^{-1}(V) &= F^{-1}(U \cap S) \\ &= F^{-1}(U) \cap F^{-1}(S) \\ &= F^{-1}(U) \cap N \\ &= F^{-1}(U) \\ &\in \mathcal{T}_N. \end{aligned}$$

Since $V \in \mathcal{T}_S$ is arbitrary, we have that for each $V \in \mathcal{T}_S$, $F^{-1}(V) \in \mathcal{T}_N$. Hence $F \in \text{Hom}_{\mathbf{Top}}(N, S)$. Since S is an embedded submanifold of M , S is an immersed submanifold of M . Exercise 9.1.0.26 then implies that $F \in \text{Hom}_{\mathbf{Man}^\infty}(N, S)$. $\square$

Exercise 9.1.0.28. Uniqueness of Topological and Smooth Structure for Embedded Submanifolds

Let $(M, \mathcal{T}_M, \mathcal{A}_M), (S, \mathcal{T}_S, \mathcal{A}_S) \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $S \subset M$. If $(S, \mathcal{T}_S, \mathcal{A}_S)$ is an embedded submanifold of $(M, \mathcal{T}_M, \mathcal{A}_M)$, then

1. $\mathcal{T}_S = \mathcal{T}_M \cap S$,
2. $\mathcal{A}_S = \mathcal{A}_M|_S$.

Proof. Suppose that $(S, \mathcal{T}_S, \mathcal{A}_S)$ is an embedded submanifold of $(M, \mathcal{T}_M, \mathcal{A}_M)$.

1. Since $(S, \mathcal{T}_S, \mathcal{A}_S)$ is an embedded submanifold of $(M, \mathcal{T}_M, \mathcal{A}_M)$, $\iota_S \in \text{Iso}_{\mathbf{Top}}[(S, \mathcal{T}_S), (S, \mathcal{T}_M \cap S)]$. [An exercise in the analysis notes section on subspaces](#) implies that $\mathcal{T}_S = \mathcal{T}_M \cap S$.

2. Define $\iota : S \rightarrow S$ by $\iota(p) := p$. Clearly, ι is a bijection. Since $(S, \mathcal{T}_S, \mathcal{A}_S)$ is an embedded submanifold of $(M, \mathcal{T}_M, \mathcal{A}_M)$, Exercise 9.1.0.19 implies that S satisfies the local k -slice condition with respect to M . Exercise 9.1.0.24 then implies that $(S, \mathcal{T}_M \cap S, \mathcal{A}_M|_S)$ is an embedded submanifold of $(M, \mathcal{T}_M, \mathcal{A}_M)$.
- Since $(S, \mathcal{T}_S, \mathcal{A}_S)$ is an embedded submanifold of $(M, \mathcal{T}_M, \mathcal{A}_M)$, $\iota \in \text{Hom}_{\mathbf{Man}^\infty}[(S, \mathcal{T}_S, \mathcal{A}_S), (M, \mathcal{T}_M, \mathcal{A}_M)]$. Since $\iota(S) = S$, Exercise 9.1.0.27 implies that $\iota \in \text{Hom}_{\mathbf{Man}^\infty}[(S, \mathcal{T}_S, \mathcal{A}_S), (S, \mathcal{T}_M \cap S, \mathcal{A}_M|_S)]$.
 - Since $(S, \mathcal{T}_M \cap S, \mathcal{A}_M|_S)$ is an embedded submanifold of $(M, \mathcal{T}_M, \mathcal{A}_M)$, $\iota^{-1} \in \text{Hom}_{\mathbf{Man}^\infty}[(S, \mathcal{T}_M \cap S, \mathcal{A}_M|_S), (M, \mathcal{T}_M, \mathcal{A}_M)]$. Since $\iota^{-1}(S) = S$, Exercise 9.1.0.27 again implies that $\iota^{-1} \in \text{Hom}_{\mathbf{Man}^\infty}[(S, \mathcal{T}_M \cap S, \mathcal{A}_M|_S), (S, \mathcal{T}_S, \mathcal{A}_S)]$.

Exercise 5.2.0.5 then implies that ι is a diffeomorphism and $\mathcal{A}_S = \mathcal{A}_M|_S$.

□

Exercise 9.1.0.29. Uniqueness of Smooth Structure for Immersed Submanifolds Let $(M, \mathcal{T}_M, \mathcal{A}_M) \in \text{Obj}(\mathbf{Man}^\infty)$, $(S, \mathcal{T}_S) \in \text{Obj}(\mathbf{Man}^0)$ and $\mathcal{A}_1, \mathcal{A}_2$ smooth structures on $(S, \mathcal{T}_S)$. Suppose that $S \subset M$. If $(S, \mathcal{T}_S, \mathcal{A}_1)$ and $(S, \mathcal{T}_S, \mathcal{A}_2)$ are immersed submanifolds of $(M, \mathcal{T}_M, \mathcal{A}_M)$, then $\mathcal{A}_1 = \mathcal{A}_2$.

Proof. Let $p \in S$. Since $(S, \mathcal{T}_S, \mathcal{A}_1)$, $(S, \mathcal{T}_S, \mathcal{A}_2)$ are immersed submanifolds of $(M, \mathcal{T}_M, \mathcal{A}_M)$, there exists $W_1, W_2 \in \mathcal{T}_S$ such that $p \in W_1 \cap W_2$ and $(W_1, \mathcal{T}_S \cap W_1, \mathcal{A}_1|_{W_1})$, $(W_2, \mathcal{T}_S \cap W_2, \mathcal{A}_2|_{W_2})$ are embedded submanifolds of $(M, \mathcal{T}_M, \mathcal{A}_M)$. Define $W \in \mathcal{T}_S$ by $W := W_1 \cap W_2$. Exercise ?? (reference previous exercise about open submanifolds here) implies that $(W, \mathcal{T}_S \cap W, \mathcal{A}_1|_W)$, $(W, \mathcal{T}_S \cap W, \mathcal{A}_2|_W)$ are embedded submanifolds of $(M, \mathcal{T}_M, \mathcal{A}_M)$. Exercise 9.1.0.28 then implies that $\mathcal{T}_S \cap W = \mathcal{T}_M \cap W$ and

$$\begin{aligned}\mathcal{A}_1|_W &= \mathcal{A}_M|_W \\ &= \mathcal{A}_2|_W.\end{aligned}$$

Since $\mathcal{A}_1|_W \subset \mathcal{A}_1$ and $\mathcal{A}_2|_W \subset \mathcal{A}_2$, we have that $\mathcal{A}_1|_W, \mathcal{A}_2|_W \subset \mathcal{A}_1 \cap \mathcal{A}_2$. Since $\mathcal{A}_1$ is an atlas on $(S, \mathcal{T}_S)$, there exists $(V', \psi') \in \mathcal{A}_1$ such that $p \in V'$. Define $(V, \psi) \in \mathcal{A}_1|_W$ by $V := V' \cap W$ and $\psi := \psi'|_{V' \cap W}$. Then $p \in V$ and

$$\begin{aligned}(V, \psi) &\in \mathcal{A}_1|_W \\ &\subset \mathcal{A}_1 \cap \mathcal{A}_2.\end{aligned}$$

Since $p \in S$ is arbitrary, we have that for each $p \in S$, there exists $(V, \psi) \in \mathcal{A}_1 \cap \mathcal{A}_2$ such that $p \in V$. The axiom of choice implies that there exists $\mathcal{A} \subset \mathcal{A}_1 \cap \mathcal{A}_2$ such that for each $p \in S$, there exists $(V, \psi) \in \mathcal{A}$ such that $p \in V$. Then $\mathcal{A}$ is a smooth atlas on $(S, \mathcal{T}_S)$. Since $\mathcal{A} \subset \mathcal{A}_1 \cap \mathcal{A}_2$, we have that

$$\begin{aligned}\mathcal{A}_1 &= \alpha(\mathcal{A}) \\ &= \mathcal{A}_2.\end{aligned}$$

□

Exercise 9.1.0.30. Let $M, S \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $S \subset M$ and S is an immersed submanifold of M . If for each $p \in S$, there exists $U \in \mathcal{T}_M$ such that $p \in U$ and $S \cap U$ is an embedded submanifold of U , then S is an embedded submanifold of M .

Proof. Suppose that for each $p \in S$, there exists $U \in \mathcal{T}_M$ such that $p \in U$ and $S \cap U$ is an embedded submanifold of U . Let $p \in S$. By assumption, there exists $U \in \mathcal{T}_M$ such that $p \in U$ and $S \cap U$ is an embedded submanifold of U . Since U is an embedded submanifold of M , we have that $S \cap U$ is an embedded submanifold of M (need exercise showing composition of embeddings is embedding?). Then $S \cap U$ satisfies the local k -slice condition with respect to M . Thus there exists $(V, \psi) \in \mathbb{S}^k(M; S \cap U)$ such that $p \in V$ and $V \subset U$. By definition of $\mathbb{S}^k(M; S \cap U)$, we have that

$$\begin{aligned}\psi(S \cap V) &= \psi(V \cap (S \cap U)) \\ &= \psi(V) \cap \mathbb{S}^{n,k}.\end{aligned}$$

Hence $(V, \psi) \in \mathbb{S}^k(M; S)$. Since $p \in S$ is arbitrary, we have that for each $p \in S$, there exists $(V, \psi) \in \mathbb{S}^k(M; S)$ such that $p \in V$. Hence S satisfies the local k -slice condition with respect to M . Thus S is an embedded submanifold of M . □

Exercise 9.1.0.31. talk about the boundary as an embedded submanifold. In particular if $\dim M = n$, then ∂M satisfies the local $n - 1$ -slice condition Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$. Then ∂M is an embedded submanifold of M .

Proof. content... **FINISH!!!**

□

9.2 Embedded Submanifolds

9.2.1 Images of Embeddings as Embedded Submanifolds

Note 9.2.1.1. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(N, M)$. Suppose that F is a $\mathbf{ManBnd}^\infty$ -embedding. We recall that Exercise 3.2.3.2 implies that $(F(N), \mathcal{T}_M \cap F(N)) \in \text{Obj}(\mathbf{Man}^0)$.

Exercise 9.2.1.2. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(N, M)$. Suppose that F is a $\mathbf{ManBnd}^\infty$ -embedding.

$$\mathcal{A}_0 := \{(F(V), \psi \circ F|_V^{-1}) : (V, \psi) \in \mathcal{A}_N\}$$

Then $\mathcal{A}_0$ is a smooth atlas on $(F(N), \mathcal{T}_M \cap F(N))$.

Proof. Set $n := \dim N$. We note that since F is a $\mathbf{ManBnd}^\infty$ -embedding, $F \in \text{Iso}_{\mathbf{Top}}((N, \mathcal{T}_N), (F(N), \mathcal{T}_M \cap F(N)))$. Since $\mathcal{A}_N \subset X^n(N, \mathcal{T}_N)$, Exercise 3.2.3.2 implies that $\mathcal{A}_0 \subset X^n(F(N), \mathcal{T}_M \cap F(N))$.

1. Let $p \in F(N)$. Then there exists $q \in N$ such that $F(q) = p$. Since $\mathcal{A}_N$ is an atlas on $(N, \mathcal{T}_N)$, there exists $(V, \psi) \in \mathcal{A}_N$ such that $q \in V$. Define $(U, \phi) \in \mathcal{A}_0$ by $U := F(V)$ and $\phi := \psi \circ F|_V^{-1}$. Then

$$\begin{aligned} p &= F(q) \\ &\in F(V) \\ &= U. \end{aligned}$$

Since $p \in F(N)$ is arbitrary, we have that for each $p \in F(N)$, there exists $(U, \phi) \in \mathcal{A}_0$ such that $p \in U$. Hence $\mathcal{A}_0$ is an atlas on $(F(N), \mathcal{T}_M \cap F(N))$.

2. Let $(U, \phi), (V, \psi) \in \mathcal{A}_0$. Then there exist $(U_0, \phi_0), (V_0, \psi_0) \in \mathcal{A}_N$ such that $U = F(U_0)$, $V = F(V_0)$, $\phi = \phi_0 \circ F|_{U_0}^{-1}$ and $\psi = \psi_0 \circ F|_{V_0}^{-1}$. Since $\mathcal{A}_N$ is a smooth atlas, $\psi_0|_{U_0 \cap V_0} \circ (\phi_0|_{U_0 \cap V_0})^{-1} \in \text{Iso}_{\mathbf{ManBnd}^\infty}(\phi_0(U_0 \cap V_0), \psi_0(U_0 \cap V_0))$. We note that

$$\begin{aligned} \phi(U \cap V) &= \phi_0 \circ F|_{U_0}^{-1}(F(U_0) \cap F(V_0)) \\ &= \phi_0 \circ F|_{U_0}^{-1}(F(U_0 \cap V_0)) \\ &= \phi_0(U_0 \cap V_0) \end{aligned}$$

and similarly, $\psi(U \cap V) = \psi_0(U_0 \cap V_0)$. Thus

$$\begin{aligned} \psi|_{U \cap V} \circ (\phi|_{U \cap V})^{-1} &= (\psi_0|_{U_0 \cap V_0} \circ F|_{V_0}^{-1})|_{F(U_0) \cap F(V_0)} \circ ((\phi_0 \circ F|_{U_0}^{-1})|_{F(U_0) \cap F(V_0)})^{-1} \\ &= (\psi_0|_{U_0 \cap V_0} \circ F|_{U_0 \cap V_0}^{-1}) \circ (F|_{U_0 \cap V_0}^{-1} \circ \phi_0|_{U_0 \cap V_0}^{-1}) \\ &= \psi_0|_{U_0 \cap V_0} \circ \phi_0|_{U_0 \cap V_0}^{-1} \\ &\in \text{Iso}_{\mathbf{ManBnd}^\infty}(\phi_0(U_0 \cap V_0), \psi_0(U_0 \cap V_0)) \\ &= \text{Iso}_{\mathbf{ManBnd}^\infty}(\phi(U \cap V), \psi(U \cap V)). \end{aligned}$$

Thus (U, ϕ) and (V, ψ) are smoothly compatible. Since $(U, \phi), (V, \psi) \in \mathcal{A}_0$ are arbitrary, we have that for each $(U, \phi), (V, \psi) \in \mathcal{A}_0$, (U, ϕ) and (V, ψ) are smoothly compatible. Hence $\mathcal{A}_0$ is a smooth atlas on $(F(N), \mathcal{T}_M \cap F(N))$. □

Definition 9.2.1.3. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(N, M)$. Suppose that F is a $\mathbf{ManBnd}^\infty$ -embedding. We define the **pushforward** of $\mathcal{A}_N$ to $F(N)$, denoted $F_*\mathcal{A}_N$, by

$$F_*\mathcal{A}_N := \alpha(\{(F(V), \psi \circ F|_V^{-1}) : (V, \psi) \in \mathcal{A}_N\})$$

Exercise 9.2.1.4. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(N, M)$. Suppose that F is a $\mathbf{Man}^\infty$ -embedding. Then

1. $F|^{F(N)} \in \text{Iso}_{\mathbf{ManBnd}^\infty}((N, \mathcal{A}_N), (F(N), F_*\mathcal{A}_N))$
2. $(F(N), F_*\mathcal{A}_N)$ is an embedded submanifold of M .

$$3. F_*\mathcal{A}_N = \mathcal{A}_M|_{F(N)}.$$

FINISH!!!

Proof. Define $G : N \rightarrow F(N)$ by $G := F|^{F(N)}$. Since F is a $\mathbf{ManBnd}^\infty$ -embedding, $G \in \text{Iso}_{\mathbf{Top}}((N, \mathcal{T}_N), (F(N), \mathcal{T}_M \cap F(N)))$.

1. • Since G is a homeomorphism, G is continuous. Let $(V, \psi) \in \mathcal{A}_N$ and $(U, \phi) \in F_*\mathcal{A}_N$. Then there exists $(U_0, \phi_0) \in \mathcal{A}_N$ such that $U = F(U_0)$ and $\phi = \phi_0 \circ G^{-1}$. We note that $G^{-1}(U) = U_0$ and

$$\begin{aligned} \phi \circ G \circ \psi|_{V \cap G^{-1}(U)}^{-1} &= \phi_0 \circ G^{-1} \circ G \circ \psi|_{V \cap U_0}^{-1} \\ &= \phi_0 \circ \psi|_{V \cap U_0}^{-1}. \end{aligned}$$

Thus, since $\phi_0 \circ \psi|_{V \cap U_0}^{-1}$ is smooth, $\phi \circ G \circ \psi|_{V \cap G^{-1}(U)}^{-1}$ is smooth. Since $(V, \psi) \in \mathcal{A}_N$ and $(U, \phi) \in F_*\mathcal{A}_N$ are arbitrary, we have that for each $(V, \psi) \in \mathcal{A}_N$ and $(U, \phi) \in F_*\mathcal{A}_N$, $\phi \circ G \circ \psi|_{V \cap G^{-1}(U)}^{-1}$ is smooth. Exercise 5.1.0.10 then implies that G is smooth.

- Similarly, G^{-1} is smooth.

Hence $G \in \text{Iso}_{\mathbf{ManBnd}^\infty}((N, \mathcal{A}_N), (F(N), F_*\mathcal{A}_N))$.

2. We note that $\iota_{F(N)} = F \circ G^{-1}$. Since G is a diffeomorphism, G is a $\mathbf{ManBnd}^\infty$ -embedding. Since F is a $\mathbf{ManBnd}^\infty$ -embedding, Exercise ?? **make exercise about compositions of embeddings being embeddings** implies that $\iota_{F(N)}$ is a $\mathbf{ManBnd}^\infty$ -embedding. Hence $(F(N), F_*\mathcal{A}_N)$ is an embedded submanifold of M .
3. Exercise 9.1.0.28 implies that $F_*\mathcal{A}_N = \mathcal{A}_M|_{F(N)}$.

□

Exercise 9.2.1.5. Let $M, S \in \text{Obj}(\mathbf{ManBnd}^\infty)$. Suppose that $S \subset M$. Then S is an embedded submanifold of M iff there exists $N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(N, M)$ such that F is a $\mathbf{ManBnd}^\infty$ -embedding and $F(N) = S$.

Proof. content... **FINISH!!! by two previous exercises.**

□

Exercise 9.2.1.6. Let $M \in \text{Obj}(\mathbf{Man}^\infty)$. Then $\Delta_M \in \text{Obj}(\mathbf{Man}^\infty)$ and Δ_M is an embedded submanifold of $M \times M$.

Proof. Define $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, M \times M)$ by $F := (\text{id}_M, \text{id}_M)$. Exercise 8.2.0.12 implies that F is an embedding. Since $F(M) = \Delta_M$, Exercise 9.2.1.4 implies that Δ_M is an embedded submanifold of $M \times M$.

□

9.2.2 Level Sets as Embedded Submanifolds

Exercise 9.2.2.1. Constant Rank Level Set Theorem:

Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$, $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$ and $q_0 \in F(M)$. Suppose F has constant rank and $\text{rank } F = r$. Then

1. $F^{-1}(\{q_0\})$ satisfies the local $(m - r)$ -slice condition with respect to M .
2. $(F^{-1}(\{q_0\}), \mathcal{T}_M \cap F^{-1}(\{q_0\}), \mathcal{A}_M|_{F^{-1}(\{q_0\})})$ is a properly embedded submanifold of M .

Proof.

1. Set $S := F^{-1}(\{q_0\})$. Let $p \in S$. Define $\text{proj}_{-r} : \mathbb{R}^m \rightarrow \mathbb{R}^r$ by $\text{proj}_{-r}(x^1, \dots, x^m) = (x^{m-r+1}, \dots, x^m)$. Since F has constant rank and $\text{rank } F = r$, Exercise 8.1.0.3 (the local rank theorem) **(add exercise about permutations on charts to get the 0's at the beginning)** implies that there exist $(U_0, \phi_0) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$ such that $p \in U$, $F(U) \subset V$, $\psi(q_0) = 0$ and $\psi \circ F \circ \phi_0^{-1} = (0, \text{proj}_{-r}|_{\phi_0(U_0)})$. Since $\phi(U_0) \in \mathcal{T}_{\mathbb{R}^m}$, **an exercise about bases of the product topology in the analysis notes** implies that there exists $A_0 \in \mathcal{T}_{\mathbb{R}^{m-r}}$ and $B_0 \in \mathcal{T}_{\mathbb{R}^r}$ such that $\phi_0(p) \in A_0 \times B_0$ and $A_0 \times B_0 \subset \phi(U_0)$. Set $U := \phi_0^{-1}(A_0 \times B_0)$ and $\phi := \phi_0|_U$. Then $(U, \phi) \in \mathcal{A}_M$, $p \in U$.

- By definition, $\phi(U) = A_0 \times B_0$. Hence $\text{proj}_{m-r}(\phi(U)) = A_0$. Since $U \subset U_0$, for each $p' \in U \cap S$,

$$\begin{aligned} 0 &= \psi(q_0) \\ &= \psi(F(p')) \\ &= \psi \circ F \circ \phi_0^{-1}(\phi_0(p')) \\ &= (0, \text{proj}^{-r}(\phi(p'))) \end{aligned}$$

Thus for each $p' \in U \cap S$, $\text{proj}^{-r}(\phi(p')) = 0$ and therefore

$$\begin{aligned} \phi(U \cap S) &\subset A_0 \times \{0\} \\ &= (A_0 \times B_0) \cap \mathbb{S}^{m,m-r} \\ &= \phi(U) \cap \mathbb{S}^{m,m-r}. \end{aligned}$$

- Let $y \in \phi(U) \cap \mathbb{S}^{m,m-r}$. Then there exists $p' \in U$ such that $\phi(p') = y$. Since $\phi(U) \cap \mathbb{S}^{m,m-r} = A_0 \times \{0\}$, there exists $a \in A_0$ such that $y = (a, 0)$. Let $p' \in (U \cap S)^c$. Since $p' \in U$, we have that $p' \in S^c$. Thus $F^{-1}(p') \neq q_0$. Since ϕ is injective,

$$\begin{aligned} 0 &= \psi(q_0) \\ &\neq \psi \circ F \circ \phi_0^{-1}(\phi_0(p')) \\ &= (0, \text{proj}_{-r}(\phi(p'))). \end{aligned}$$

Therefore $\text{proj}_{-r}(\phi(p')) \neq 0$. Hence $\phi(p') \in (\mathbb{S}^{m,m-r})^c$. Since $p' \in (U \cap S)^c$ is arbitrary, we have that

$$\begin{aligned} \phi(U \cap S)^c &= \phi((U \cap S)^c) \\ &\subset (\mathbb{S}^{m,m-r})^c \\ &\subset (\phi(U) \cap \mathbb{S}^{m,m-r})^c \end{aligned}$$

Thus $\phi(U) \cap \mathbb{S}^{m,m-r} \subset \phi(U \cap S)$.

Therefore $\phi(U \cap S) = \phi(U) \cap \mathbb{S}^{m,m-r}$ and $\phi(U \cap S)$ is a $(m-r)$ -slice of $\phi(U)$. Hence (U, ϕ) is an $(m-r)$ -slice chart on S . Since $p \in S$ is arbitrary, we have that for each $p \in S$, there exists $(U, \phi) \in \mathcal{A}_M$ such that $p \in U$ and (U, ϕ) is an $(m-r)$ -slice chart on S . So S satisfies the local $(m-r)$ -slice condition with respect to M .

2. Since F is $(\mathcal{T}_M, \mathcal{T}_N)$ -continuous and $\{q_0\}$ is closed in $(N, \mathcal{T}_N)$, we have that S is closed in $(M, \mathcal{T}_M)$. Exercise ?? (a previous exercise) implies that S is properly embedded.

□

Exercise 9.2.2.2. Submersion Level Set Theorem:

Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$, $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. Set $m := \dim M$ and $n := \dim N$. Suppose F is a submersion. Then for each $q \in N$,

1. $F^{-1}(\{q\})$ satisfies the local $(m-n)$ -slice condition with respect to M ,
2. $(F^{-1}(\{q\}), \mathcal{T}_M \cap F^{-1}(\{q\}), \mathcal{A}_M|_{F^{-1}(\{q\})})$ is a properly embedded submanifold of M .

Proof. Since F is a submersion, F has constant rank and $\text{rank } F = n$. Let $q \in N$. Exercise ?? (the previous exercise) implies that

1. $F^{-1}(\{q\})$ satisfies the local $(m-n)$ -slice condition with respect to M ,
2. $(F^{-1}(\{q\}), \mathcal{T}_M \cap F^{-1}(\{q\}), \mathcal{A}_M|_{F^{-1}(\{q\})})$ is a properly embedded submanifold of M .

□

Definition 9.2.2.3. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$, $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$ and $p \in M$ and $q \in N$. Then p is said to be a

- **regular point of F** if $DF(p) : T_p M \rightarrow T_{F(p)} N$ is surjective,

- **critical point of F** if p is not a regular point of F

and q is said to be a

- **regular value of F** if for each $x \in F^{-1}(\{q\})$, x is a regular point of F ,
- **critical value of F** if q is not a regular value of F .

Note 9.2.2.4. In particular, if $\dim M < \dim N$, then for each $p \in M$, p is a critical point of F and for each $q \in N$, if $F^{-1}(\{q\}) = \emptyset$, then q is a regular value of F .

Exercise 9.2.2.5. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. If F is a submersion, then for each $q \in N$, q is a regular value of F .

Proof. Suppose that F is a submersion. Let $q \in N$ and $p \in F^{-1}(\{q\})$. Since F is a submersion, $DF(p)$ is surjective. Hence p is a regular point of F . Since $p \in F^{-1}(\{q\})$ is arbitrary, we have that for each $p \in F^{-1}(\{q\})$, p is a regular point of F . Hence q is a regular value of F . Since $q \in N$ is arbitrary, we have that for each $q \in N$, q is a regular value of F . $\square$

Definition 9.2.2.6. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$, $S \subset M$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. Then S is said to be a **regular level set of F** if there exists $q \in N$ such that q is a regular value of F and $S = F^{-1}(\{q\})$.

Exercise 9.2.2.7. Regular Level Set Theorem:

Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$, $S \subset M$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. Set $m := \dim M$ and $n := \dim N$. Suppose that S is a regular level set of F . Then

1. S satisfies the local $(m - n)$ -slice condition with respect to M ,
2. $(S, \mathcal{T}_M \cap S, \mathcal{A}_M|_S)$ is a properly embedded submanifold of M .

Hint:

Define $U \subset M$ by $U := \{p \in M : \text{rank } DF(p) = \dim N\}$ and consider Exercise 8.3.0.3.

Proof. Define $U \subset M$ by $U := \{p \in M : \text{rank } DF(p) = \dim N\}$. Exercise 8.3.0.3 implies that $U \in \mathcal{T}_M$ and $F|_U$ is a submersion. Since S is a regular level set of F , there exists $q \in N$ such that q is a regular value of F and $S = F^{-1}(\{q\})$. Since q is a regular value of F , for each $x \in S$, x is a regular point of F . Thus for each $x \in S$, $DF(x)$ is surjective. Hence $S \subset U$. Since $F|_U$ is a submersion and

$$\begin{aligned} S &= S \cap U \\ &= F^{-1}(\{q\}) \cap U \\ &= F|_U^{-1}(\{q\}), \end{aligned}$$

Exercise 9.2.2.2 implies that S satisfies the local $(m - n)$ -slice condition with respect to U and S is a properly embedded submanifold of U . Since $U \in \mathcal{T}_M$, U is a properly embedded submanifold of M . Hence $F^{-1}(\{q\})$ is a properly embedded submanifold of M . (flesh out some of the last details here, like composition of proper maps is proper, composition of $\mathbf{Man}^\infty$ -embeddings is a $\mathbf{Man}^\infty$ -embedding, etc) FINISH!!! $\square$

Definition 9.2.2.8. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$, $S \subset M$, $U \in \mathcal{T}_M$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(U, N)$. Suppose that $S \subset M$. Then F is said to be a

- **local defining map for S** if $S \cap U$ is a regular level set of F ,
- **defining map for S** if F is a local defining map for S and $U = M$.

Exercise 9.2.2.9. Let $M \in \text{Obj}(\mathbf{Man}^\infty)$ and $S \subset M$. Set $m := \dim M$. Then S satisfies the local k -slice condition with respect to M iff for each $p \in S$, there exists $U \in \mathcal{T}_M$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(U, \mathbb{R}^{m-k})$ such that $p \in U$ and F is a local defining map for S .

Proof.

- $(\implies)$:

- Suppose that S satisfies the local k -slice condition with respect to M . Let $p \in S$. Then there exists $(U, \phi) \in \mathbb{S}^k(M; S)$ such that $p \in U$. Define $(U_0, \phi_0) \in \mathcal{A}_M|_S^0$ by $U_0 := U \cap S$ and $\phi_0 := \text{proj}_k^m \circ \phi|_{U_0}$. Set $r := m - k$ and define $F \in \text{Hom}_{\mathbf{Man}^\infty}(U, \mathbb{R}^r)$ by $F := \text{proj}_{[-r]}^m \circ \phi$. Then $F \in \text{Hom}_{\mathbf{Man}^\infty}(U, \mathbb{R}^r)$ and $p \in U$. By definition of $\mathbb{S}^k(M; S)$, $\phi(S \cap U) = \phi(U) \cap \mathbb{S}^{m,k}$. Hence

$$\begin{aligned} F(S \cap U) &= \pi_{[-r]}^m \circ \phi(S \cap U) \\ &= \pi_{[-r]}^m(\phi(U) \cap \mathbb{S}^{m,k}) \\ &= \{0\} \end{aligned}$$

Hence $S \cap U \subset F^{-1}(\{0\})$.

- Let $q \in F^{-1}(\{0\})$. Then $q \in U$ and $F(q) = 0$. Since

$$\begin{aligned} \phi(q) &= (\pi_{[k]}^m \circ \phi(q), F(q)) \\ &= (\pi_{[k]}^m \circ \phi(q), 0) \\ &\in \mathbb{S}^{m,k}, \end{aligned}$$

we have that

$$\begin{aligned} \phi(q) &\in \phi(U) \cap \mathbb{S}^{m,k} \\ &= \phi(S \cap U). \end{aligned}$$

Since ϕ is a bijection, $q \in S \cap U$. Since $q \in F^{-1}(\{0\})$ is arbitrary, we have that for each $q \in F^{-1}(\{0\})$, $q \in S \cap U$. Thus $F^{-1}(\{0\}) \subset S \cap U$.

Hence $F^{-1}(\{0\}) = S \cap U$. Let $q \in S \cap U$. Exercise 6.2.0.5 implies that $[D\phi(q)]_{\phi, \text{id}_{\mathbb{R}^m}} = \begin{pmatrix} [D\pi_{[k]}^m \circ \phi(q)]_{\phi, \text{id}_{\mathbb{R}^k}} \\ [DF(q)]_{\phi, \text{id}_{\mathbb{R}^r}} \end{pmatrix}$. Since $[D\phi(q)]_{\phi, \text{id}_{\mathbb{R}^m}}$ is a bijection, we have that $\text{rank}[DF(q)]_{\phi, \text{id}_{\mathbb{R}^r}} = r$. Thus $DF(q)$ is surjective. Since $q \in S \cap U$ is arbitrary, we have that for each $q \in U$, $DF(q)$ is surjective. Thus $0 \in \mathbb{R}^r$ is a regular value of F and $S \cap U$ is a regular level set of F . Hence F is a local defining map for $S \cap U$.

• ($\Leftarrow$):

Suppose that for each $p \in S$, there exists $U \in \mathcal{T}_M$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(U, \mathbb{R}^{m-k})$ such that $p \in U$ and F is a local defining map for S . Let $p \in S$. By assumption, there exists $U \in \mathcal{T}_M$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(U, \mathbb{R}^{m-k})$ such that $p \in U$ and F is a local defining map for S . Thus $S \cap U$ is a regular level set of F . Exercise 9.2.2.7 then implies that $S \cap U$ satisfies the local k -slice condition with respect to M . Since $p \in S$ is arbitrary, we have that for each $p \in S$, there exists $U \in \mathcal{T}_M$ such that $p \in U$ and $S \cap U$ satisfies the local k -slice condition with respect to M . (need exercise showing that if the local k -slice condition is satisfied locally, around each point in S , it is satisfied globally.) Exercise ?? implies that S satisfies the local k -slice condition with respect to M .

□

9.2.3 Submanifolds of Embedded Submanifolds

Exercise 9.2.3.1. Let $M, N, S \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $N \subset S$, $S \subset M$ and S is an embedded submanifold of M . Then N is an embedded submanifold of S iff N is an embedded submanifold of M .

Proof.

• ($\Rightarrow$):

Suppose that N is an embedded submanifold of S . Define $F \in \text{Hom}_{\mathbf{Man}^\infty}(N, S)$ by $F(p) := p$. Since N is an embedded submanifold of S , F is a $\mathbf{Man}^\infty$ -embedding. We note that $\iota_N = \iota_S \circ F$. Since S is an embedded submanifold of M , ι_S is a $\mathbf{Man}^\infty$ -embedding. Exercise ?? cite exercise about composition of embeddings is embedding then implies that ι_N is a $\mathbf{Man}^\infty$ -embedding. Since $\iota_N(N) = N$, Exercise 9.2.1.4 implies that N is an embedded submanifold of M .

• ($\Leftarrow$):

Suppose that N is an embedded submanifold of M . Define $F : N \rightarrow S$ by $F := \iota_N|_S$. Since $\iota_N \in \text{Hom}_{\mathbf{Man}^\infty}(N, M)$,

Exercise 9.1.0.27 implies that $F \in \text{Hom}_{\mathbf{Man}^\infty}(N, S)$. Since N is an embedded submanifold of M , $\iota_N \in \text{Hom}_{\mathbf{Man}^\infty}(N, M)$ is a $\mathbf{Man}^\infty$ -embedding. Let $p \in N$. Since $\iota_N = \iota_S \circ F$, we have that

$$\begin{aligned} D\iota_N(p) &= D\iota_S(F(p)) \circ DF(p) \\ &= D\iota_S(p) \circ DF(p). \end{aligned}$$

Since $DF(p) : T_p N \rightarrow T_p S$, $\dim[\text{Im } F(p)] \leq \dim T_p N$. Since ι_N is an immersion, $\dim[\text{Im } D\iota_N] = \dim T_p N$. Therefore

$$\begin{aligned} T_p N &= \dim[\text{Im } D\iota_N] \\ &\leq \min \left(\dim[\text{Im } D\iota_S(p)], \dim[\text{Im } DF(p)] \right) \\ &\leq \dim[\text{Im } DF(p)] \\ &\leq \dim T_p N. \end{aligned}$$

Hence $\dim[\text{Im } DF(p)] = \dim T_p N$ and $DF(p)$ is injective. Since $p \in N$ is arbitrary, we have that for each $p \in N$, $DF(p)$ is injective. Thus F is an immersion. Exercise 9.1.0.28 implies that $\mathcal{T}_N = \mathcal{T}_M \cap N$ and $\mathcal{T}_S = \mathcal{T}_M \cap S$. [An exercise in the analysis notes section on subspace topology](#) implies that $\mathcal{T}_M \cap N = (\mathcal{T}_M \cap S) \cap N$. Therefore

$$\begin{aligned} \mathcal{T}_N &= \mathcal{T}_M \cap N \\ &= (\mathcal{T}_M \cap S) \cap N \\ &= \mathcal{T}_S \cap N. \end{aligned}$$

Hence $F|_N \in \text{Iso}_{\mathbf{Top}}((N, \mathcal{T}_N), (N, \mathcal{T}_S \cap N))$. Thus F is a $\mathbf{Man}^\infty$ -embedding and N is an embedded submanifold of S . □

9.2.4 Products of Embedded Submanifolds

Exercise 9.2.4.1. Let $M, N, E, F \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $E \subset M$, $F \subset N$, E is an embedded submanifold of M and F is an embedded submanifold of N . Then $E \times F$ is an embedded submanifold of $M \times N$.

Proof. Since ι_E and ι_F are $\mathbf{Man}^\infty$ -embeddings, Exercise 8.2.0.11 implies that $\iota_E \times \iota_F$ is a $\mathbf{Man}^\infty$ -embedding. Since $\iota_E \times \iota_F = \iota_{E \times F}$, $E \times F$ is an embedded submanifold of $M \times N$. □

9.3 Immersed Submanifolds

9.4 The Tangent Space of Submanifolds

9.4.1 Introduction

Exercise 9.4.1.1. Let $M, S \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $S \subset M$ and S is an embedded submanifold of M . Set $n := \dim M$ and $k := \dim S$. Let $(U_0, \phi_0) \in \mathbb{S}^k(M; S)$ with $\phi_0 = (x^1, \dots, x^n)$. Set $U := U_0 \cap S$ and $\phi := \text{proj}_k^n \circ \phi_0|_U$ so that $(U, \phi) \in \mathcal{A}_M|_S^0$. Write $\phi = (\tilde{x}^1, \dots, \tilde{x}^k)$. Let $p \in U$. Then for each $j \in [k]$,

$$D(\iota_S)(p) \left(\frac{\partial}{\partial \tilde{x}^j} \Big|_p \right) = \frac{\partial}{\partial x^j} \Big|_p$$

Proof. Let $j \in [k]$ and $f \in C_p^\infty(M)$. By construction, $f \circ \phi_0^{-1} = f \circ \phi^{-1} \circ \pi_k^n$. Thus

$$\begin{aligned} D(\iota_S)(p) \left(\frac{\partial}{\partial \tilde{x}^j} \Big|_p \right) (f) &= \frac{\partial}{\partial \tilde{x}^j} \Big|_p (f \circ \iota_S) \\ &= \frac{\partial}{\partial u^j} \Big|_{\phi(p)} (f \circ \iota_S \circ \phi^{-1}) \\ &= \frac{\partial}{\partial u^j} \Big|_{\phi(p)} (f \circ \phi^{-1}) \\ &= \lim_{\epsilon \rightarrow 0} \frac{f \circ \phi^{-1}(\phi(p) + \epsilon e^j) - f \circ \phi^{-1}(\phi(p))}{\epsilon}, \quad (\text{in } \mathbb{R}^k) \\ &= \lim_{\epsilon \rightarrow 0} \frac{f \circ \phi_0^{-1}(\phi_0(p) + \epsilon e^j) - f \circ \phi_0^{-1}(\phi_0(p))}{\epsilon}, \quad (\text{in } \mathbb{R}^n) \\ &= \frac{\partial}{\partial x^j} \Big|_p f \end{aligned}$$

Since $f \in C_p^\infty(M)$ is arbitrary, we have that

$$D(\iota_S)(p) \left(\frac{\partial}{\partial \tilde{x}^j} \Big|_p \right) = \frac{\partial}{\partial x^j} \Big|_p.$$

□

Definition 9.4.1.2. Let $M, S \in \text{Obj}(\mathbf{Man}^\infty)$ and $p \in S$. Suppose that $S \subset M$ and S is an embedded submanifold of M . We define the **embedded tangent space of S with respect to M at p** , denoted $\tilde{T}_p(S, M)$, by

$$\tilde{T}_p(S, M) := \text{Im } D\iota_S^M(p).$$

Need to make official notation the inclusion $\iota_S^M : S \rightarrow M$, then make note that when context is clear, we write ι_S .

Note 9.4.1.3. When the context is clear, we write $\tilde{T}_p S$ in place of $\tilde{T}_p(S, M)$.

Note 9.4.1.4. Since ι_S is an immersion, $D\iota_S(p)|^{\tilde{T}_p S} \in \text{IsoVect}_{\mathbb{R}}(T_p S, \tilde{T}_p S)$ and $\dim \tilde{T}_p S = \dim T_p S$.

Exercise 9.4.1.5. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(N, M)$. Suppose that F is a $\mathbf{Man}^\infty$ -embedding. Then for each $p \in N$,

1. $DF|^{F(N)}(p) \in \text{IsoVect}_{\mathbb{R}}(T_p N, T_{F(p)} F(N))$
2. $\tilde{T}_{F(p)} F(N) = \text{Im } DF(p)$
3. $\dim \tilde{T}_{F(p)} F(N) = \dim T_p N$

Proof. Set $S := F(N)$ and $G := F|_S$. Let $p \in N$.

1. Exercise 9.2.1.4 implies that $G \in \text{Iso}_{\mathbf{Man}^\infty}(N, S)$. Thus $DG(p) \in \text{IsoVect}_{\mathbb{R}}(T_p N, T_{F(p)} S)$.

2. Since

$$\begin{aligned} DF(p) &= D[\iota_S \circ G](p) \\ &= D\iota_S(G(p)) \circ DG(p) \\ &= D\iota_S(F(p)) \circ DG(p), \end{aligned}$$

we have that

$$\begin{aligned} \text{Im } DF(p) &= D\iota_S(F(p)) \circ DG(p)(T_p N) \\ &= D\iota_S(F(p))(T_{F(p)} S) \\ &= \text{Im } D\iota_S(F(p)) \\ &= \tilde{T}_{F(p)} S. \end{aligned}$$

3. Since F is a $\mathbf{Man}^\infty$ -embedding, $DF(p)$ is injective. Thus

$$\begin{aligned} \dim \tilde{T}_{F(p)} S &= \dim \text{Im } DF(p) \\ &= \dim DF(p)(T_p N) \\ &= \dim T_p N. \end{aligned}$$

□

discuss how to identify $T_p M$ and $T_p U$ where $U \in \mathcal{T}_M$. Can use germs since derivations at a point are determined locally around that point. So in some sense even though $T_p M$ and $T_p U$ are isomorphic, they are isomorphic in a strong sense where we can define derivations on the germ at a point and discarding any nonlocal information about the functions at the point.

Need to define $T_p M$ in terms of germs, then explain how

Exercise 9.4.1.6. Let $M, S \in \text{Obj}(\mathbf{Man}^\infty)$ and $p \in S$. Suppose that $S \subset M$ and S is an embedded submanifold of M . Then $\dim \tilde{T}_p S = \dim T_p S$.

Proof. Define $F \in \text{Hom}_{\mathbf{Man}^\infty}(S, M)$ by $F := \iota_S$. Exercise 9.4.1.5 implies that

$$\begin{aligned} \dim \tilde{T}_p S &= \dim \tilde{T}_{F(p)} F(S) \\ &= \dim T_p S. \end{aligned}$$

□

Exercise 9.4.1.7. Let $M, N, S \in \text{Obj}(\mathbf{Man}^\infty)$, $U \in \mathcal{T}_M$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(U, N)$. Suppose that $S \subset M$, S is an embedded submanifold of M and F is a local defining map for S . Then for each $p \in S \cap U$, $\tilde{T}_p S = \ker DF(p)$.

Proof. Let $p \in S \cap U$.

- Since F is a local defining map for S , $S \cap U$ is a regular level set of F . Hence there exists $q \in N$ such that q is a regular value of F and $S \cap U = F^{-1}(\{q\})$. Thus $F|_{S \cap U}$ is constant. Hence

$$\begin{aligned} 0 &= D(F|_{S \cap U})(p) \\ &= D(F \circ \iota_{S \cap U})(p) \\ &= DF(p) \circ D\iota_{S \cap U}(p). \end{aligned}$$

and therefore $\text{Im } D\iota_{S \cap U}(p) \subset \ker DF(p)$. Since S is an embedded submanifold of M , $\mathcal{T}_S = \mathcal{T}_M \cap S$ and $S \cap U \in \mathcal{T}_S$. Then

$$\begin{aligned} \tilde{T}_p S &= \text{Im } D\iota_S(p) \\ &= D\iota_S(p)(T_p S) \\ &= D\iota_{S \cap U}(p)(T_p(S \cap U)) \text{ (need to rework definition of } T_p M \text{ using germs so that this step works out)} \\ &= \text{Im } D\iota_{S \cap U}(p) \\ &\subset \ker DF(p). \end{aligned}$$

- Set $m := \dim M$, $n := \dim N$ and $k := \dim S$. Since q is a regular value of F , $DF(p)$ is surjective. Exercise 9.2.2.7 implies that $k = m - n$ and therefore

$$\begin{aligned}
 \dim \ker DF(p) &= \dim T_p M - \dim \operatorname{Im} DF(p) \\
 &= \dim T_p M - \dim T_{F(p)} N \\
 &= m - n \\
 &= \dim T_p S \\
 &= \dim \operatorname{Im} D\iota_S(p) \\
 &= \dim \tilde{T}_p S.
 \end{aligned}$$

Since $\tilde{T}_p S \subset \ker DF(p)$ and $\dim \tilde{T}_p S = \dim \ker DF(p)$, we have that $\operatorname{Im} \tilde{T}_p S = \ker DF(p)$. □

Exercise 9.4.1.8. Let $M, S \in \operatorname{Obj}(\mathbf{Man}^\infty)$. Suppose that $S \subset M$ and S is an embedded submanifold of M . Let $(U_0, \phi_0) \in \mathbb{S}^k(M; S)$. Write $\phi_0 = (x^1, \dots, x^m)$. Define $\sigma \in S_m$ and $G_0 \in \operatorname{Hom}_{\mathbf{Man}^\infty}(U_0, \mathbb{R}^{m-k})$ by $\sigma := \begin{pmatrix} 1 & \dots & k & k+1 & \dots & n \\ k+1 & \dots & n & 1 & \dots & k \end{pmatrix}$ and $G_0 := \operatorname{proj}_{n-k} \circ (\sigma \cdot \phi_0)$. Then G_0 is a submersion and for each $q \in U_0 \cap S$, $\ker DG_0(q) = T_q S$.

Proof. Define $F_0 \in \operatorname{Hom}_{\mathbf{Man}^\infty}(U_0, \mathbb{R}^k)$ by $F_0 := \operatorname{proj}_k^m \circ \phi_0$.

- Since ϕ_0 is a diffeomorphism and $\phi_0 = (F_0, G_0)$, we have that for each $q \in U_0$,

$$\begin{aligned}
 [D\phi_0(q)]_{\phi_0, \operatorname{id}_{\mathbb{R}^m}} &= \begin{pmatrix} [DF_0(q)]_{\phi_0, \operatorname{id}_{\mathbb{R}^k}} \\ [DG_0(q)]_{\phi_0, \operatorname{id}_{\mathbb{R}^{m-k}}} \end{pmatrix} \\
 &= \begin{pmatrix} I_k & 0_{k, m-k} \\ 0_{m-k} & I_{m-k, m-k} \end{pmatrix}.
 \end{aligned}$$

Therefore, for each $q \in U_0$, $\operatorname{rank}[DG_0(q)]_{\phi_0, \operatorname{id}_{\mathbb{R}^{m-k}}} = m - k$ and $DG_0(q)$ is surjective. Hence G_0 is a submersion.

- Let $q \in U_0 \cap S$ and $j \in [k]$. Since $\phi_0(U_0)$ is open, there exists $\epsilon > 0$ such that for each $t \in (-\epsilon, \epsilon)$, $\phi_0(q) + te^j \in \phi_0(U_0)$. Since $\phi_0(U_0 \cap S) = \phi_0(U_0) \cap \mathbb{S}^{m,k}$, we have that

- $G_0(U_0 \cap S) = \{0\}$ and $G|_{U_0 \cap S} = 0$,
- for each $t \in (-\epsilon, \epsilon)$, $\phi_0(q) + te^j \in \phi_0(U_0) \cap \mathbb{S}^{m,k}$ and $\phi_0^{-1}(\phi_0(q) + te^j) \in U_0 \cap S$.

Since $G|_{U_0 \cap S} = 0$, we have that for each $f \in C^\infty(\mathbb{R}^{m-k})$,

$$\begin{aligned}
 DG\left(\frac{\partial}{\partial x^j} \Big|_q\right)(f) &= \frac{\partial}{\partial x^{k+j}} \Big|_q (f \circ G) \\
 &= \frac{d}{dt} \Big|_{t=0} \left[f \circ G \circ \phi_0^{-1}(\phi_0(q) + te^j) \right] \\
 &= 0.
 \end{aligned}$$

Since

$$T_q S = \left(\frac{\partial}{\partial x^j} \Big|_q : j \in [k] \right),$$

we have that $\ker G_0 = T_q S$. Since $q \in U_0 \cap S$ is arbitrary, we have that for each $q \in U_0 \cap S$, $\ker G_0 = T_q S$. □

9.4.2 Tangent Space of Products of Submanifolds

Exercise 9.4.2.1. Let $M \in \text{Obj}(\mathbf{Man}^\infty)$ and $p \in M$. Define $\alpha \in \text{Iso}_{\mathbf{Vect}_{\mathbb{R}}}(T_{(p,p)}(M \times M), T_p M \times T_p M)$ as in Exercise 6.2.0.2. Then $\alpha(\tilde{T}_{(p,p)}\Delta_M) = \Delta_{T_p M}$ and $\dim T_{(p,p)}\Delta_M = \dim T_p M$.

Proof. Set $n := \dim M$. Define $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, M \times M)$ by $F := (\text{id}_M, \text{id}_M)$. Exercise 8.2.0.12 implies that F is an embedding. Exercise 9.2.1.4 implies that $F|_{\Delta_M} \in \text{Iso}_{\mathbf{Man}^\infty}(M, \Delta_M)$. Set $G := F|_{\Delta_M}$. Let $(U, \phi) \in \mathcal{A}_M$. Write $\phi = (x^1, \dots, x^n)$. Define $(V, \psi) \in \mathcal{A}_M \otimes \mathcal{A}_M$ by $V := U \times U$ and $\psi := \phi \times \phi$. Write $\psi = (\tilde{y}^1, \dots, \tilde{y}^n, \tilde{z}^1, \dots, \tilde{z}^n)$. Exercise 6.2.0.6 implies that

$$\begin{aligned} \alpha \circ DF(p) \left(\frac{\partial}{\partial x^k} \Big|_p \right) &= (D \text{id}_M(p), D \text{id}_M(p)) \left(\frac{\partial}{\partial x^k} \Big|_p \right) \\ &= \left(\text{id}_{T_p M} \left[\frac{\partial}{\partial x^k} \Big|_p \right], \text{id}_{T_p M} \left[\frac{\partial}{\partial x^k} \Big|_p \right] \right) \\ &= \left(\frac{\partial}{\partial x^k} \Big|_p, \frac{\partial}{\partial x^k} \Big|_p \right). \end{aligned}$$

Hence $\alpha \circ DF(p)(T_p M) = \Delta_{T_p M}$. Exercise 9.4.1.5 then implies that

$$\begin{aligned} \alpha(\tilde{T}_{(p,p)}\Delta_M) &= \alpha \circ DF(p)(T_p M) \\ &= \Delta_{T_p M}. \end{aligned}$$

Since α and $DF(p)$ are injective, Exercise 9.4.1.5 and Exercise 9.4.1.6 imply that

$$\begin{aligned} \dim T_p M &= \dim \Delta_{T_p M} \\ &= \dim \alpha \circ DF(p)(T_p M) \\ &= \dim \alpha(\tilde{T}_{(p,p)}\Delta_M) \\ &= \dim \tilde{T}_{(p,p)}\Delta_M \\ &= \dim T_{(p,p)}\Delta_M. \end{aligned}$$

□

Exercise 9.4.2.2. Let $M, S \in \text{Obj}(\mathbf{Man}^\infty)$ and $p \in S$. Suppose that $S \subset M$ and S is an immersed submanifold of M . Define $\alpha_M \in \text{Iso}_{\mathbf{Vect}_{\mathbb{R}}}(T_{(p,p)}(M \times M), T_p M \times T_p M)$ as in Exercise 6.2.0.2. Then for each $p \in M$,

$$\alpha_M(\tilde{T}_{(p,p)}(\Delta_S, M \times M)) = \Delta_{\tilde{T}_p(S, M)}.$$

Proof. Set $n := \dim M$. Define $\alpha_S \in \text{Iso}_{\mathbf{Vect}_{\mathbb{R}}}(T_{(p,p)}(S \times S), T_p S \times T_p S)$ as in Exercise 6.2.0.2. Exercise 9.4.2.1 implies that $\alpha_S(\tilde{T}_{(p,p)}(\Delta_S, S \times S)) = \Delta_{T_p S}$. Exercise 6.2.0.4 then implies that $\alpha_M \circ D\iota_{S \times S}(p, p) = D\iota_S(p) \times D\iota_S(p) \circ \alpha_S$, i.e. following diagram commutes:

$$\begin{array}{ccc} T_{(p,p)}(S \times S) & \xrightarrow{D(\iota_{S \times S})(p, p)} & T_{(p,p)}(M \times M) \\ \alpha_S \downarrow & & \downarrow \alpha_M \\ T_p S \times T_p S & \xrightarrow{D\iota_S(p) \times D\iota_S(p)} & T_p M \times T_p M \end{array}$$

Since $[D\iota_S^M(p) \times D\iota_S^M(p)](\Delta_{T_p S}) = \Delta_{\tilde{T}_p(S, M)}$, we have that

$$\begin{aligned}
 \alpha_M(\tilde{T}_{(p,p)}(\Delta_S, M \times M)) &= \alpha_M \circ D\iota_{\Delta_S}^{M \times M}(p, p)(T_{(p,p)}\Delta_S) \\
 &= \alpha_M \circ D\iota_{S \times S}^{M \times M}(p, p) \circ D\iota_{\Delta_S}^{S \times S}(p, p)(T_{(p,p)}\Delta_S) \\
 &= \alpha_M \circ D\iota_{S \times S}^{M \times M}(p, p)(\tilde{T}_{(p,p)}(\Delta_S, S \times S)) \\
 &= [D\iota_S^M(p) \times D\iota_S^M(p)] \circ \alpha_S(\tilde{T}_{(p,p)}(\Delta_S, S \times S)) \\
 &= [D\iota_S^M(p) \times D\iota_S^M(p)](\Delta_{T_p S}) \\
 &= \Delta_{\tilde{T}_p(S, M)}
 \end{aligned}$$

□

Exercise 9.4.2.3. Let $M, S \in \text{Obj}(\mathbf{Man}^\infty)$ and $p \in S$. Define $\text{proj}_1 \in \text{Hom}_{\mathbf{Man}^\infty}(M \times M, M)$ as in (cite def). Suppose that $S \subset M$ and S is an immersed submanifold of M . Then

$$D\text{proj}_1(p, p)(\tilde{T}_{(p,p)}\Delta_S) = \tilde{T}_p S.$$

Proof. Exercise 6.2.0.2 Exercise 9.4.1.1 Exercise 6.2.0.6 **FINISH!!!**

□

Exercise 9.4.2.4. Let $M, N, S, E \in \text{Obj}(\mathbf{Man}^\infty)$, $p \in S$ and $q \in E$. Suppose that $S \subset M$, S is an immersed submanifold of M , $E \subset N$, E is an immersed submanifold of N . Define $\alpha \in \text{Iso}_{\mathbf{Vect}_\mathbb{R}}(T_{(p,p)}(M \times M), T_p M \times T_p M)$. Then

$$\alpha(\tilde{T}_{(p,q)}(S \times E)) = \tilde{T}_p S \times \tilde{T}_q E.$$

Proof. Define $\alpha_0 \in \text{Iso}_{\mathbf{Vect}_\mathbb{R}}(T_{(p,q)}(S \times E), T_p S \times T_q E)$ and $\alpha \in \text{Iso}_{\mathbf{Vect}_\mathbb{R}}(T_{(p,q)}(M \times N), T_p M \times T_q N)$ as in Exercise 6.2.0.2. Exercise 6.2.0.4 then implies that $\alpha \circ D\iota_{S \times E}(p, q) = D\iota_S(p) \times D\iota_E(q) \circ \alpha_0$, i.e. following diagram commutes:

$$\begin{array}{ccc}
 T_{(p,q)}(S \times E) & \xrightarrow{D(\iota_{S \times E})(p, q)} & T_{(p,q)}(M \times N) \\
 \downarrow \alpha_0 & & \downarrow \alpha \\
 T_p S \times T_q E & \xrightarrow{D\iota_S(p) \times D\iota_E(q)} & T_p M \times T_q N
 \end{array}$$

Hence

$$\begin{aligned}
 \tilde{T}_p S \times \tilde{T}_q E &= D\iota_S(p)(T_{p_M} S) \times D\iota_E(q)(T_{p_N} E) \\
 &= [D\iota_S(p) \times D\iota_E(q)](T_p S \times T_q E) \\
 &= [D\iota_S(p) \times D\iota_E(q)] \circ \alpha_0(T_{(p,q)}(S \times E)) \\
 &= \alpha \circ D\iota_{S \times E}(p, q)(T_{(p,q)}(S \times E)) \\
 &= \alpha(\tilde{T}_{(p,q)}(S \times E)).
 \end{aligned}$$

□

9.5 Transverse Submanifolds

Definition 9.5.0.1. Let $M, S, S_2 \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $S_1, S_2 \subset M$, S_1, S_2 are immersed submanifolds of M . Then S_1 and S_2 are said to be

- **transverse** if for each $p \in S_1 \cap S_2$, $T_p M = \tilde{T}_p S_1 + \tilde{T}_p S_2$
- **strictly transverse** if for each $p \in S_1 \cap S_2$, $T_p M = \tilde{T}_p S_1 \oplus \tilde{T}_p S_2$ (maybe change vocab here from strict to strong or something)

Exercise 9.5.0.2. Define $S_1, S_2 \subset \mathbb{R}^n$ by $S_1 := \{(a, 0) \in \mathbb{R}^n : a \in \mathbb{R}^k\}$ and $S_2 := \{(0, b) \in \mathbb{R}^n : b \in \mathbb{R}^{n-k}\}$. Then S_1 and S_2 are strictly transverse.

Proof. Define $\phi_0, \psi_0 : \mathbb{R}^n \rightarrow \mathbb{R}^n$ by $\phi_0(a^1, \dots, a^n) := (a^1, \dots, a^n)$ and $\psi_0(a^1, \dots, a^k, a^{k+1}, \dots, a^n) := (a^{k+1}, \dots, a^n, a^1, \dots, a^k)$. Write $\phi_0 = (x^1, \dots, x^n)$ and $\psi_0 = (y^1, \dots, y^n)$. Then $(\mathbb{R}^n, \phi_0) \in \mathbb{S}^k(\mathbb{R}^n, S_1)$ and $(\mathbb{R}^n, \psi_0) \in \mathbb{S}^k(\mathbb{R}^n, S_2)$. Set $\phi := \pi_{[k]}^n \circ \phi_0|_{S_1}$ and $\psi := \pi_{[n-k]}^n \circ \psi_0|_{S_2}$. Write $\phi = (\tilde{x}^1, \dots, \tilde{x}^k)$ and $\psi = (\tilde{y}^1, \dots, \tilde{y}^{n-k})$. Exercise 9.4.1.1 implies that for each $j \in [k]$,

$$\begin{aligned} D\iota_{S_1}(0) \left(\frac{\partial}{\partial \tilde{x}^j} \Big|_0 \right) &= \frac{\partial}{\partial x^j} \Big|_0 \\ &= \frac{\partial}{\partial u^j} \Big|_0 \end{aligned}$$

and for each $j \in [n-k]$

$$\begin{aligned} D\iota_{S_2}(0) \left(\frac{\partial}{\partial \tilde{y}^j} \Big|_0 \right) &= \frac{\partial}{\partial y^j} \Big|_0 \\ &= \frac{\partial}{\partial u^{k+j}} \Big|_0. \end{aligned}$$

Hence

$$\begin{aligned} T_0(\mathbb{R}^n) &= \text{span} \left\{ \frac{\partial}{\partial u^j} \Big|_0 : j \in [k] \right\} \oplus \text{span} \left\{ \frac{\partial}{\partial u^{k+j}} \Big|_0 : j \in [n-k] \right\} \\ &= \text{Im } D\iota_{S_1}(0) \oplus \text{Im } D\iota_{S_2}(0) \\ &= \tilde{T}_0 S_1 \oplus \tilde{T}_0 S_2. \end{aligned}$$

Since $S_1 \cap S_2 = \{0\}$, we have that for each $p \in S_1 \cap S_2$, $T_p(\mathbb{R}^n) = \tilde{T}_0 S_1 \oplus \tilde{T}_0 S_2$. Hence S_1 and S_2 are strictly transverse. $\square$

Exercise 9.5.0.3. Let $M, S \in \text{Obj}(\mathbf{Man}^\infty)$ and $p \in S$. Suppose that $S \subset M$, S is an embedded submanifold of M and $\dim S < \dim M$. Then there exists $S' \in \text{Obj}(\mathbf{Man}^\infty)$ such that $S' \subset M$, S' is an immersed submanifold of M , $p \in S'$ and S, S' are strictly transverse.

Proof. Set $n := \dim M$ and $k := \dim S$. Then there exists $(U, \phi) \in \mathcal{A}_M|_S^0$ such that $p \in U$ and $\phi(p) = 0$. Then there exists $(U_0, \phi_0) \in \mathbb{S}^k(M; S)$ such that $U = U_0 \cap S$ and $\phi = \text{proj}_k^n \circ \phi_0|_U$. Thus $\phi_0(p) = 0$. Write $\phi_0 = (x^1, \dots, x^n)$ and $\phi = (\tilde{x}^1, \dots, \tilde{x}^k)$. Define $B, B' \subset \mathbb{R}^n$ by $B := \{(a, 0) \in \mathbb{R}^n : a \in \mathbb{R}^k\} \cap \phi_0(U_0)$ and $B' := \{(0, b) \in \mathbb{R}^n : b \in \mathbb{R}^{n-k}\} \cap \phi_0(U_0)$. Then

$$\begin{aligned} B &= \phi_0(U_0) \cap \mathbb{S}^{n,k} \\ &= \phi_0(U_0 \cap V) \\ &= \phi_0(U) \end{aligned}$$

Define $U' \subset M$, $\sigma \in S_n$ and $\psi_0 : U_0 \rightarrow \sigma \cdot \phi_0(U_0)$ by $U' := \phi_0^{-1}(B')$, $\sigma := \begin{pmatrix} 1 & \dots & k & k+1 & \dots & n \\ k+1 & \dots & n & 1 & \dots & k \end{pmatrix}$ and $\psi_0 := \sigma \cdot \phi_0$. Then

need exercise saying U' is embedded submanifold of M , $(U_0, \psi_0) \in \mathcal{A}_M$ and

$$\begin{aligned}
 \psi_0(U_0 \cap U') &= \psi_0(U') \\
 &= \sigma \cdot \phi_0(U') \\
 &= \sigma \cdot B' \\
 &= \sigma \cdot [\phi_0(U_0) \cap \{(0, b) \in \mathbb{R}^n : b \in \mathbb{R}^{n-k}\}] \\
 &= \sigma \cdot \phi_0(U_0) \cap \sigma \cdot \{(0, b) \in \mathbb{R}^n : b \in \mathbb{R}^{n-k}\} \\
 &= \psi_0(U_0) \cap \mathbb{S}^{n, n-k}.
 \end{aligned}$$

Thus $(U_0, \psi_0) \in \mathbb{S}^{n-k}(M; U')$. Write $\psi_0 = (y^1, \dots, y^n)$. Define $(U', \psi') \in \mathcal{A}_M|_{U'}$ by $\psi' := \pi_{n-k}^n \circ \psi_0|_{U'}$. Write $\psi' = (\tilde{y}^1, \dots, \tilde{y}^{n-k})$. Since $B \cap B' = \{0\}$,

$$\begin{aligned}
 U \cap U' &= \phi_0^{-1}(B) \cap \phi_0^{-1}(B') \\
 &= \phi_0^{-1}(B \cap B') \\
 &= \phi_0^{-1}(\{0\}) \\
 &= p.
 \end{aligned}$$

Exercise 9.4.1.1 implies that for each $j \in [k]$

$$D\iota_U(p) \left(\frac{\partial}{\partial \tilde{x}^j} \Big|_p \right) = \frac{\partial}{\partial x^j} \Big|_p$$

and for each $j \in [n-k]$

$$\begin{aligned}
 D\iota_{U'}(p) \left(\frac{\partial}{\partial \tilde{y}^j} \Big|_p \right) &= \frac{\partial}{\partial y^j} \Big|_p \\
 &= \frac{\partial}{\partial x^{k+j}} \Big|_p.
 \end{aligned}$$

Therefore

$$\begin{aligned}
 T_p M &= \text{span} \left\{ \frac{\partial}{\partial x^j} \Big|_p : j \in [k] \right\} \oplus \text{span} \left\{ \frac{\partial}{\partial x^{k+j}} \Big|_p : j \in [n-k] \right\} \\
 &= \text{Im } D\iota_U(p) \oplus \text{Im } D\iota_{U'}(p).
 \end{aligned}$$

Set $S' := U'$. Since $U \in \mathcal{T}_V$ and $V \in \mathcal{T}_S$, we have that $U \in \mathcal{T}_S$. Thus

$$\begin{aligned}
 T_p M &= \text{Im } D\iota_U(p) \oplus \text{Im } D\iota_{U'}(p) \\
 &= \text{Im } D\iota_S(p) \oplus \text{Im } D\iota_{S'}(p) \\
 &= \tilde{T}_p S \oplus \tilde{T}_p S'.
 \end{aligned}$$

Let $q \in S \cap S'$. Then

$$\begin{aligned}
 q &\in S' \\
 &= U' \\
 &\subset U_0.
 \end{aligned}$$

and therefore

$$\begin{aligned}
 q &\in U_0 \cap S \\
 &= U.
 \end{aligned}$$

Hence

$$\begin{aligned} q &\in U \cap U' \\ &= \{p\}. \end{aligned}$$

Since $q \in S \cap S'$ is arbitrary, we have that $S \cap S' \subset \{p\}$. Since $\{p\} \subset S \cap S'$, we have that $S \cap S' = \{p\}$. Thus for each $q \in S \cap S'$, $T_q M = \tilde{T}_q S \oplus \tilde{T}_q S'$ and S, S' are strictly transverse. $\square$

Note 9.5.0.4. If S is not embedded, we would only know that there is $\tilde{S} \subset S$ such that $p \in \tilde{S}$ and $\tilde{S}$ is an embedded submanifold of M . However, in this case, following the same proof as above with $(U, \phi) \in \mathcal{A}_M|_{\tilde{S}}^0$ and $q \in S \cap S'$, we may not be able to show that $U_0 \cap S = U$ since we only know that $U_0 \cap \tilde{S} = U$. Therefore we cannot show that $S \cap S' = \{p\}$ and that for each $q \in S \cap S'$, $T_q S + T_q S' = T_p M$.

Exercise 9.5.0.5. Let $M, S \in \text{Obj}(\mathbf{Man}^\infty)$ and $p \in S$. Suppose that $S \subset M$, S is an embedded submanifold of M and $\dim S < \dim M$. Then there exists $S' \in \text{Obj}(\mathbf{Man}^\infty)$ such that $S' \subset M$, S' is an embedded submanifold of M , $p \in S'$ and S, S' are strictly transverse.

Proof. Similar proof to previous ex, but use the fact that immersed submanifolds are locally embedded. **FINISH!!!!** $\square$

Exercise 9.5.0.6. Let $M, N, S_1, S_2, E_1, E_2 \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $S_1, S_2 \subset M$, S_1, S_2 are immersed submanifolds of M , $E_1, E_2 \subset N$ and E_1, E_2 are immersed submanifolds of N . If S_1, S_2 are strictly transverse and E_1, E_2 are strictly transverse, then $S_1 \times E_1$ and $S_2 \times E_2$ are strictly transverse. (need exercise saying product of immersed submanifolds is immersed submanifold of product)

Proof. Suppose that S_1, S_2 are strictly transverse and E_1, E_2 are strictly transverse. Set $m := \dim M$, $n := \dim N$, and for each $j \in [2]$, $m_j := \dim S_j$ and $n_j := \dim E_j$. Let

$$\begin{aligned} (p_M, p_N) &\in (S_1 \times E_1) \cap (S_2 \times E_2) \\ &= (S_1 \cap S_2) \times (E_1 \cap E_2). \end{aligned}$$

Since S_1 and S_2 are strictly transverse, $T_{p_M} M = \tilde{T}_{p_M} S_1 \oplus \tilde{T}_{p_M} S_2$. Hence $m = m_1 + m_2$. Since E_1 and E_2 are strictly transverse, $T_{p_N} N = \tilde{T}_{p_N} E_1 \oplus \tilde{T}_{p_N} E_2$. Hence $n = n_1 + n_2$. Since $\tilde{T}_{p_M} S_1 \times \tilde{T}_{p_N} E_1, \tilde{T}_{p_M} S_2 \times \tilde{T}_{p_N} E_2 \subset T_{p_M} M \times T_{p_N} N$, we have that $\dim[(\tilde{T}_{p_M} S_1 \times \tilde{T}_{p_N} E_1) + (\tilde{T}_{p_M} S_2 \times \tilde{T}_{p_N} E_2)] \leq m + n$. Since

$$\begin{aligned} (\tilde{T}_{p_M} S_1 \times \tilde{T}_{p_N} E_1) \cap (\tilde{T}_{p_M} S_2 \times \tilde{T}_{p_N} E_2) &= (\tilde{T}_{p_M} S_1 \cap \tilde{T}_{p_M} S_2) \times (\tilde{T}_{p_N} E_1 \cap \tilde{T}_{p_N} E_2) \\ &= \{0\} \times \{0\} \\ &= \{(0, 0)\}, \end{aligned}$$

we have that

$$\begin{aligned} \dim[(\tilde{T}_{p_M} S_1 \times \tilde{T}_{p_N} E_1) + (\tilde{T}_{p_M} S_2 \times \tilde{T}_{p_N} E_2)] &= \dim(\tilde{T}_{p_M} S_1 \times \tilde{T}_{p_N} E_1) + \dim(\tilde{T}_{p_M} S_2 \times \tilde{T}_{p_N} E_2) \\ &= m_1 + n_1 + m_2 + n_2 \\ &= m + n. \end{aligned}$$

Hence

$$T_{p_M} M \times T_{p_N} N = (\tilde{T}_{p_M} S_1 \times \tilde{T}_{p_N} E_1) \oplus (\tilde{T}_{p_M} S_2 \times \tilde{T}_{p_N} E_2).$$

Exercise 9.4.2.4 implies that $\tilde{T}_{p_M} S_1 \times \tilde{T}_{p_N} E_1 = \alpha(\tilde{T}_{(p_M, p_N)}(S_1 \times E_1))$ and $\tilde{T}_{p_M} S_2 \times \tilde{T}_{p_N} E_2 = \alpha(\tilde{T}_{(p_M, p_N)}(S_2 \times E_2))$. Therefore

$$\begin{aligned} T_{(p_M, p_N)}(M \times N) &= \alpha^{-1}(T_{p_M} M \times T_{p_N} N) \\ &= \alpha^{-1}[(\tilde{T}_{p_M} S_1 \times \tilde{T}_{p_N} E_1) \oplus (\tilde{T}_{p_M} S_2 \times \tilde{T}_{p_N} E_2)] \\ &= \alpha^{-1}[\alpha(\tilde{T}_{(p_M, p_N)}(S_1 \times E_1)) \oplus \alpha(\tilde{T}_{(p_M, p_N)}(S_2 \times E_2))] \\ &= \alpha^{-1} \circ \alpha[\tilde{T}_{(p_M, p_N)}(S_1 \times E_1) \oplus \tilde{T}_{(p_M, p_N)}(S_2 \times E_2)] \\ &= \tilde{T}_{(p_M, p_N)}(S_1 \times E_1) \oplus \tilde{T}_{(p_M, p_N)}(S_2 \times E_2). \end{aligned}$$

Since $(p_M, p_N) \in (S_1 \times E_1) \cap (S_2 \times E_2)$ is arbitrary, we have that for each $(p_M, p_N) \in (S_1 \times E_1) \cap (S_2 \times E_2)$, $T_{(p_M, p_N)}(M \times N) = \tilde{T}_{(p_M, p_N)}(S_1 \times E_1) \oplus \tilde{T}_{(p_M, p_N)}(S_2 \times E_2)$. Thus $S_1 \times E_1$ and $S_2 \times E_2$ are strictly transverse. $\square$

Definition 9.5.0.7. Let $M, N, S \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(N, M)$. Suppose that $S \subset M$, S is an immersed submanifold of M . Then F is said to be **transverse to S** if for each $p \in F^{-1}(S)$,

$$DF(p)(T_p N) + \tilde{T}_{F(p)} S = T_{F(p)} M$$

Exercise 9.5.0.8. Let $M, N, S \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(N, M)$. Suppose that $S \subset M$, S is an immersed submanifold of M . If F is a submersion, then F is transverse to S .

Proof. Suppose that F is a submersion. Let $p \in F^{-1}(S)$. Since F is a submersion,

$$\begin{aligned} DF(p)(T_p N) + \tilde{T}_{F(p)} S &= T_{F(p)} M + \tilde{T}_{F(p)} S \\ &= T_{F(p)} M. \end{aligned}$$

Since $p \in F^{-1}(S)$ is arbitrary, we have that for each $p \in F^{-1}(S)$, $DF(p)(T_p N) + \tilde{T}_{F(p)} S = T_{F(p)} M$. Hence F is transverse to S . $\square$

Exercise 9.5.0.9. Let $M, N, S \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(N, M)$. Suppose that $S \subset M$, S is an embedded submanifold of M . If F is transverse to S , then $F^{-1}(S)$ is an embedded submanifold of N and $\text{codim } F^{-1}(S) = \text{codim } S$.

Proof. Suppose that F is transverse to S . Set $m := \dim M$, $n := \dim N$ and $k := \dim S$. Let $p \in F^{-1}(S)$. Since S is an embedded submanifold of M , there exists $(U_0, \phi_0) \in \mathbb{S}^k(M; S)$ such that $F(p) \in U_0$. Define $\sigma \in S_m$ and $G_0 \in \text{Hom}_{\mathbf{Man}^\infty}(U_0, \mathbb{R}^{m-k})$ by $\sigma := \begin{pmatrix} 1 & \dots & k & k+1 & \dots & n \\ k+1 & \dots & n & 1 & \dots & k \end{pmatrix}$ and $G_0 := \text{proj}_{n-k} \circ (\sigma \cdot \phi_0)$. Since $\phi_0 \in \mathbb{S}^k(M; S)$, $\phi_0(S \cap U_0) = \phi_0(U_0) \cap \mathbb{S}^{m,k}$. Therefore $G_0(S \cap U_0) = \{0\}$ and $G_0^{-1}(\{0\}) = S \cap U_0$. Exercise 9.4.1.8 implies that G_0 is a submersion and for each $q \in U_0 \cap S$, $\ker DG_0(q) = \tilde{T}_q S$. Define $V \in \mathcal{T}_N$ and $G \in \text{Hom}_{\mathbf{Man}^\infty}(V, \mathbb{R}^{m-k})$ by $V := F^{-1}(U_0)$ and $G := G_0 \circ F|_V$. Then

$$\begin{aligned} G^{-1}(\{0\}) &= (G_0 \circ F|_V)^{-1}(\{0\}) \\ &= F|_V^{-1}(G_0^{-1}(\{0\})) \\ &= F|_V^{-1}(S \cap U_0) \\ &= V \cap F^{-1}(S \cap U_0) \\ &= V \cap F^{-1}(S) \cap F^{-1}(U_0) \\ &= V \cap F^{-1}(S). \end{aligned}$$

Let $p' \in V \cap F^{-1}(S)$. Since F is transverse to S , we have $DF(p')(T_{p'} N) + \tilde{T}_{F(p')} S = T_{F(p')} M$. Since G_0 is a submersion and $\ker DG_0(F(p')) = \tilde{T}_{F(p')} S$, we have that

$$\begin{aligned} DG(p')(T_{p'} N) &= DG_0(F(p'))[DF(p')(T_{p'} N)] \\ &= DG_0(F(p'))[DF(p')(T_{p'} N) + \ker DG_0(F(p'))] \\ &= DG_0(F(p'))[DF(p')(T_{p'} N) + \tilde{T}_{F(p')} S] \\ &= DG_0(F(p'))[T_{F(p')} M] \\ &= \text{Im } DG_0(F(p')) \\ &= \tilde{T}_{G_0(F(p'))} \mathbb{R}^{m-k} \\ &= \tilde{T}_{G(p')} \mathbb{R}^{m-k} \end{aligned}$$

Hence $DG(p')$ is surjective. Since $p' \in V \cap F^{-1}(S)$ is arbitrary, we have that for each $p' \in V \cap F^{-1}(S)$, $DG(p')$ is surjective. Hence 0 is a regular value of G and $V \cap F^{-1}(S)$ is a regular level set of G . Hence G is a local defining map for $F^{-1}(V)$. Since $p \in F^{-1}(S)$ is arbitrary, we have that for each $p \in F^{-1}(S)$, there exists $V \in \mathcal{T}_N$ and $G \in \text{Hom}_{\mathbf{Man}^\infty}(V, \mathbb{R}^{m-k})$ such that $p \in V$ and G is a local defining map for $F^{-1}(S)$. Exercise 9.2.2.9 then implies that $F^{-1}(S)$ is an embedded submanifold of N and

$$\begin{aligned} \dim F^{-1}(S) &= \dim G^{-1}(\{0\}) \\ &= \dim N - \dim \mathbb{R}^{m-k}. \end{aligned}$$

Hence

$$\begin{aligned}
 \operatorname{cod} F^{-1}(S) &= \dim N - \dim F^{-1}(S) \\
 &= \dim N - (\dim N - \dim \mathbb{R}^{m-k}) \\
 &= m - k \\
 &= \operatorname{cod} S.
 \end{aligned}$$

□

Note 9.5.0.10. (check) I dont think the previous fact remains true in the case that S is immersed. Since immersed submanifolds are locally embedded, you can use the previous result to show that $F^{-1}(S)$ is locally embedded and therefore immersed, maybe make this an exercise?

Exercise 9.5.0.11. Let $M, N, S \in \operatorname{Obj}(\mathbf{Man}^\infty)$ and $F \in \operatorname{Hom}_{\mathbf{Man}^\infty}(N, M)$. Suppose that $S \subset M$, S is an embedded submanifold of M . If F is a submersion, then $F^{-1}(S)$ is an embedded submanifold of N .

Proof. Suppose that F is a submersion. Exercise 9.5.0.8 implies that F is transverse to S . Exercise 9.5.0.9 then implies that $F^{-1}(S)$ is an embedded submanifold of N . □

Exercise 9.5.0.12. Let $M, S_1, S_2 \in \operatorname{Obj}(\mathbf{Man}^\infty)$. Suppose that $S_1, S_2 \subset M$ and S_1, S_2 are embedded submanifolds of M . If S_1 and S_2 are transverse, then $S_1 \cap S_2$ is an embedded submanifold of M and $\dim S_1 \cap S_2 = \dim S_1 + \dim S_2 - \dim M$.

Proof. Suppose that S_1 and S_2 are transverse. We note that $S_1 \cap S_2 = \iota_{S_1}^{-1}(S_2)$. Let $p \in \iota_{S_1}^{-1}(S_2)$. Since S_1 and S_2 are transverse, we have that

$$\begin{aligned}
 D\iota_{S_1}(p)(T_p S_1) + \tilde{T}_p S_2 &= \tilde{T}_p S_1 + \tilde{T}_p S_2 \\
 &= T_p M.
 \end{aligned}$$

Since $p \in \iota_{S_1}^{-1}(S_2)$ is arbitrary, we have that for each $p \in \iota_{S_1}^{-1}(S_2)$, $D\iota_{S_1}(p)(T_p S_1) + \tilde{T}_p S_2 = T_p M$ and ι_{S_1} is transverse to S_2 . Since $S_1 \cap S_2 = \iota_{S_1}^{-1}(S_2)$, Exercise 9.5.0.9 implies that $S_1 \cap S_2$ is an embedded submanifold of S_1 and

$$\dim S_1 - \dim S_1 \cap S_2 = \dim M - \dim S_2.$$

Since S_1 is an embedded submanifold of M , Exercise ?? **An exercise in the subsection of submanifolds of embedded submanifolds** then implies that $S_1 \cap S_2$ is an embedded submanifold of M and

$$\dim S_1 \cap S_2 = \dim S_1 + \dim S_2 - \dim M.$$

□

Note 9.5.0.13. The previous result about $\dim S_1 \cap S_2$ is analogous to the dimension of the intersection of subspaces or measure of the intersection of measurable subsets or the log of the lcm and log of the gcd.

Chapter 10

Quotient Manifolds

10.1 Introduction

Note 10.1.0.1. Let $M, R \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that $R \subset M \times M$. We denote the projection maps from $M \times M \rightarrow M$ by $\text{proj}_1, \text{proj}_2$. If R is an equivalence relation on M , then

1. $\text{proj}_1|_R, \text{proj}_2|_R$ are surjective
2. $\text{proj}_1|_R$ is a submersion iff $\text{proj}_2|_R$ is a submersion

the submersion assumption for $\text{proj}_1|_R$ may not be necessary, but NEED TO PROVE IT. Though thinking on it, for general embedded submanifold $R \subset M \times M$, it may be the case that $T_{(p,p)}R$ has enough dimension to map surjectively onto T_pM , however if R is an equivalence relation on M , then maybe this is not an issue.

Exercise 10.1.0.2. Let $M, R \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that R is a properly embedded submanifold of $M \times M$, R is an equivalence relation on M , and $\text{proj}_2|_R$ is a submersion. Then

1. for each $U \in \mathcal{T}_M$, $\pi^{-1}(\pi(U)) = \text{proj}_1((M \times U) \cap R)$,
2. $\pi : M \rightarrow M/R$ is open,
3. M/R is Hausdorff.

Proof.

1. Let $U \in \mathcal{T}_M$ and $x \in M$. Then

$$\begin{aligned}
 x \in \pi^{-1}(\pi(U)) &\iff \pi(x) \in \pi(U) \\
 &\iff \text{there exists } u \in U \text{ such that } \pi(x) = \pi(u) \\
 &\iff \text{there exists } u \in U \text{ such that } (x, u) \in R \\
 &\iff \text{there exists } u \in U \text{ such that } (x, u) \in (M \times U) \cap R \\
 &\iff x \in \text{proj}_1((M \times U) \cap R)
 \end{aligned}$$

Hence $\pi^{-1}(\pi(U)) = \text{proj}_1((M \times U) \cap R)$. Since $U \in \mathcal{T}_M$ is arbitrary, we have that for each $U \in \mathcal{T}_M$, $\pi^{-1}(\pi(U)) = \text{proj}_1((M \times U) \cap R)$.

2. Let $U \in \mathcal{T}_M$. Then $(M \times U) \cap R \in \mathcal{T}_R$. Since $\text{proj}_1|_R$ is a surjective submersion, Exercise 8.3.0.10 implies that $\text{proj}_1|_R$ is open. Part (1) implies that for each $U \in \mathcal{T}_M$,

$$\begin{aligned}
 \pi^{-1}(\pi(U)) &= \text{proj}_1((M \times U) \cap R) \\
 &= \text{proj}_1|_R((M \times U) \cap R) \\
 &\in \mathcal{T}_M
 \end{aligned}$$

Since π is a quotient map, [an exercise in the analysis notes section on the quotient topology](#) implies that π is open.

3. Since R is properly embedded [an exercise in the section on embedded submanifolds](#) implies that R is closed in $M \times M$. [An exercise in the analysis notes section on separation axioms on quotient spaces](#) implies that M/R is Hausdorff. □

Exercise 10.1.0.3. Let $M, R \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that R is a properly embedded submanifold of $M \times M$, R is an equivalence relation on M , and $\text{proj}_1|_R$ is a submersion. Then for each $p \in M$, $\pi(p)$ is a properly embedded submanifold of M and $\dim \pi(p) = \dim R - \dim M$.

Hint: For each $p \in M$, $\pi(p) = \text{proj}_1|_R(\text{proj}_2|_R^{-1}(\{p\}))$ and $\text{proj}_1|_{M \times \{p\}} \in \text{Iso}_{\mathbf{Man}^\infty}(M \times \{p\}, M)$.

Proof. Let $p \in M$. Exercise ?? implies that $\text{proj}_1 : M \times M \rightarrow M$ is a submersion. Exercise ?? implies that $M \times \{p\}$ is an embedded submanifold of $M \times M$. Exercise ?? implies that $\text{proj}_2|_R$ is a submersion. Since $\text{proj}_2|_R$ is a surjective submersion, Exercise ?? implies that $\text{proj}_2|_R^{-1}(\{p\})$ is a properly embedded submanifold of R and $\dim \text{proj}_2|_R^{-1}(\{p\}) = \dim R - \dim M$. Since R is an embedded submanifold of $M \times M$, Exercise ?? [\(need to make\) exercise in section on embedded submanifolds subsection on subspaces](#) implies that $\text{proj}_2|_R^{-1}(\{p\})$ is an embedded submanifold of $M \times M$. Since $\text{proj}_2|_R^{-1}(\{p\}) \subset M \times \{p\}$ Exercise 9.2.3.1 implies that $\text{proj}_2|_R^{-1}(\{p\})$ is an embedded submanifold of $M \times \{p\}$. Since $\text{proj}_1|_{M \times \{p\}} \in \text{Iso}_{\mathbf{Man}^\infty}(M \times \{p\}, M)$ is a diffeomorphism and $\pi(p) = \text{proj}_1|_{M \times \{p\}}(\text{proj}_2|_R^{-1}(\{p\}))$, Exercise ?? [make exercise in the section on embedded submanifolds](#) implies that $\pi(p)$ is an embedded submanifold of M and $\dim \pi(p) = \dim R - \dim M$. □

Exercise 10.1.0.4. Let $M, R, S' \in \text{Obj}(\mathbf{Man}^\infty)$ and $p \in S'$. Suppose that $R \subset M \times M$, R is a properly embedded submanifold of $M \times M$, R is an equivalence relation on M , $\text{proj}_1|_R$ is a submersion, $S' \subset M$, S' is an embedded submanifold of M , $\dim S' = \dim M - \dim \pi(p)$ and S' is transverse to $\pi(p)$. Define $Z \subset R$ by $Z := \text{proj}_2|_R^{-1}(S')$. Then

1. $(p, p) \in Z$
2. Z is an embedded submanifold of R and $\dim Z = \dim M$
3. $D\text{proj}_1|_Z(p, p) \in \text{Iso}_{\mathbf{Vect}_{\mathbb{R}}}(T_{(p,p)}Z, T_p M)$

Hint:

- (a) $\Delta_{S'}$ and $\pi(p) \times \{p\}$ are embedded submanifolds of Z
- (b) $T_{(p,p)}Z = \tilde{T}_{(p,p)}(\Delta_{S'}, Z) \oplus \tilde{T}_{(p,p)}((\pi(p) \times \{p\}), Z)$

Proof.

1. Since R is an equivalence relation on M and $p \in S'$, we have that $p \in M$ and therefore $(p, p) \in R$. Since $p \in S'$ and $\text{proj}_2|_R(p, p) = p$, we have that

$$\begin{aligned} (p, p) &\in \text{proj}_2|_R^{-1}(S') \\ &= Z. \end{aligned}$$

2. [since \$\text{proj}_1|_R\$ is a submersion and \$R\$ is an equivalence relation, \$\text{proj}_2|_R\$ is a submersion.](#) Since S' is an embedded submanifold of M , Exercise 9.5.0.11 implies that Z is an embedded submanifold of R . Exercise ?? [Another exercise in the section on level sets as embedded submanifolds](#) implies that

$$\begin{aligned} \dim Z &= \dim M \times M - \dim M \\ &= 2 \dim M - \dim M \\ &= \dim M \\ &= \dim M. \end{aligned}$$

3. (a) • We note that Exercise 9.2.1.6 implies that $\Delta_{S'}$ is an embedded submanifold of $S' \times S'$. Since S' is an embedded submanifold of M , Exercise 9.2.4.1 implies that $S' \times S'$ is an embedded submanifold of $M \times M$. Exercise 9.2.3.1 then implies that $\Delta_{S'}$ is an embedded submanifold of $M \times M$. Also by definition, $\Delta_{S'} \subset R$ and for each $(x, x) \in \Delta_{S'}$,

$$\begin{aligned} \text{proj}_2(x, x) &= x \\ &\in S'. \end{aligned}$$

Hence

$$\begin{aligned}\Delta_{S'} &\subset \text{proj}_2^{-1}(S') \cap R \\ &= \text{proj}_2|_R^{-1}(S') \\ &= Z.\end{aligned}$$

- Define $N \in \text{Obj}(\mathbf{Man}^\infty)$ by $N := N$. Exercise 10.1.0.3 implies that $\pi(p)$ is an embedded submanifold of M . Exercise ?? (need exercise) implies that $\{p\}$ is an embedded submanifold of M . Exercise 9.2.4.1 then implies that N is an embedded submanifold of $M \times M$. We note that by definition, $N \subset R$. Since for each $(a, p) \in N$,

$$\begin{aligned}\text{proj}_2(a, p) &= p \\ &\in S',\end{aligned}$$

we have that

$$\begin{aligned}N &\subset \text{proj}_2^{-1}(S') \cap R \\ &= \text{proj}_2|_R^{-1}(S') \\ &= Z.\end{aligned}$$

Since Z is an embedded submanifold of R and R is an embedded submanifold of $M \times M$, Exercise 9.2.3.1 implies that Z is an embedded submanifold of $M \times M$. Since $\Delta_{S'}, N \subset Z$ and $\Delta_{S'}, N$ are embedded submanifolds of $M \times M$, Exercise 9.2.3.1 then implies that $\Delta_{S'}, N$ are embedded submanifolds of Z .

- (b) Define $\alpha \in \text{Iso}_{\mathbf{Vect}_{\mathbb{R}}}(T_{(p,p)}M \times M, T_pM \times T_pM)$ as in Exercise 6.2.0.2. Exercise 9.4.2.2 and Exercise 9.4.2.4 then imply that

$$\begin{aligned}\alpha(\tilde{T}_{(p,p)}(\Delta_{S'}, M \times M) \cap \tilde{T}_{(p,p)}(N, M \times M)) &= \alpha(\tilde{T}_{(p,p)}(\Delta_{S'}, M \times M)) \cap \alpha(\tilde{T}_{(p,p)}(N, M \times M)) \\ &= \Delta_{\tilde{T}_p(S', M)} \cap [\tilde{T}_p(\pi(p), M) \times \tilde{T}_p(\{p\}, M)] \\ &= \Delta_{\tilde{T}_p(S', M)} \cap [\tilde{T}_p(\pi(p), M) \times \{0\}] \\ &= \{(0, 0)\}.\end{aligned}$$

Since α is an isomorphism,

$$\begin{aligned}\{0\} &= \tilde{T}_{(p,p)}(\Delta_{S'}, M \times M) \cap \tilde{T}_{(p,p)}(N, M \times M) \\ &= D\iota_Z^{M \times M}(p, p)[\tilde{T}_{(p,p)}(\Delta_{S'}, Z)] \cap D\iota_Z^{M \times M}(p, p)[\tilde{T}_{(p,p)}(N, Z)] \\ &= D\iota_Z^{M \times M}(p, p)[\tilde{T}_{(p,p)}(\Delta_{S'}, Z) \cap \tilde{T}_{(p,p)}(N, Z)].\end{aligned}$$

Injectivity of $D\iota_Z^{M \times M}(p, p)$ implies that $\tilde{T}_{(p,p)}(\Delta_{S'}, Z) \cap \tilde{T}_{(p,p)}(N, Z) = \{0\}$ and therefore

$$\begin{aligned}\dim[\tilde{T}_{(p,p)}(\Delta_{S'}, Z) + \tilde{T}_{(p,p)}(N, Z)] &= \dim D\iota_Z^{M \times M}(p, p)[\tilde{T}_{(p,p)}(\Delta_{S'}, Z) + \tilde{T}_{(p,p)}(N, Z)] \\ &= \dim[\tilde{T}_{(p,p)}(\Delta_{S'}, M \times M) + \tilde{T}_{(p,p)}(N, M \times M)] \\ &= \dim \tilde{T}_{(p,p)}(\Delta_{S'}, M \times M) + \dim \tilde{T}_{(p,p)}(N, M \times M).\end{aligned}$$

We note that Exercise 9.4.1.6 and Exercise 9.4.2.1 imply that

$$\begin{aligned}\dim \tilde{T}_{(p,p)}(\Delta_{S'}, M \times M) &= \dim T_{(p,p)}\Delta_{S'} \\ &= \dim \Delta_{T_p S'} \\ &= \dim T_p S' \\ &= \dim \tilde{T}_p(S', M).\end{aligned}$$

Exercise 6.2.0.4 Exercise 9.4.2.4 and the injectivity of α imply that

$$\begin{aligned}\dim \tilde{T}_{(p,p)}(N, M \times M) &= \dim \alpha(\tilde{T}_{(p,p)}(N, M \times M)) \\ &= \dim[\tilde{T}_p(\pi(p), M) \times \tilde{T}_p(\{p\}, M)] \\ &= \dim \tilde{T}_p(\pi(p), M) + \dim \{0\} \\ &= \dim \tilde{T}_p(\pi(p), M).\end{aligned}$$

Since S' is transverse to $\pi(p)$,

$$\begin{aligned} \dim[\tilde{T}_{(p,p)}(\Delta_{S'}, Z) + \tilde{T}_{(p,p)}(N, Z)] &= \dim \tilde{T}_{(p,p)}(\Delta_{S'}, M \times M) + \dim \tilde{T}_{(p,p)}(N, M \times M) \\ &= \dim \tilde{T}_p(S', M) + \dim \tilde{T}_p(\pi(p), M) \\ &= \dim T_p M \\ &= \dim T_p Z. \end{aligned}$$

Thus $T_p Z = \tilde{T}_{(p,p)}(\Delta_{S'}, Z) \oplus \tilde{T}_{(p,p)}(N, Z)$.

Exercise ?? implies that

$$\begin{aligned} D \text{proj}_1|_Z(p, p)[T_{(p,p)}Z] &= D(\text{proj}_1 \circ \iota_Z^{M \times M})(p, p)[T_{(p,p)}Z] \\ &= D \text{proj}_1(p, p) \circ D\iota_Z^{M \times M}(p, p)[T_{(p,p)}Z] \\ &= D \text{proj}_1(p, p) \circ D\iota_Z^{M \times M}(p, p)[\tilde{T}_{(p,p)}(\Delta_{S'}, Z) \oplus \tilde{T}_{(p,p)}(N, Z)] \\ &= D \text{proj}_1(p, p)(D\iota_Z^{M \times M}(p, p)[\tilde{T}_{(p,p)}(\Delta_{S'}, Z)] \oplus D\iota_Z^{M \times M}(p, p)[\tilde{T}_{(p,p)}(N, Z)]) \\ &= D \text{proj}_1(p, p)(\tilde{T}_{(p,p)}(\Delta_{S'}, M \times M) \oplus \tilde{T}_{(p,p)}(N, M \times M)) \\ &= D \text{proj}_1(p, p)[\tilde{T}_{(p,p)}(\Delta_{S'}, M \times M)] + D \text{proj}_1(p, p)[\tilde{T}_{(p,p)}(N, M \times M)] \\ &= \tilde{T}_p(S', M) + \tilde{T}_p(\pi(p), M) \text{ (need to show)} \\ &= T_p M \end{aligned}$$

Thus $D \text{proj}_1|_Z(p, p)$ is surjective. Since $\dim Z = \dim M$, we have that $D \text{proj}_1|_Z(p, p)$ is a bijection. Hence $D \text{proj}_1|_Z(p, p) \in \text{Iso}_{\text{Vect}_{\mathbb{R}}}(T_{(p,p)}Z, T_p M)$. **FIX!!!**

□

10.2 Godement's Theorem

Definition 10.2.0.1. Let $M, R, S \in \text{Obj}(\mathbf{Man}^\infty)$, $U \in \mathcal{T}_M$ and $q \in \text{Hom}_{\mathbf{Man}^\infty}(U, S)$. Suppose that R is a properly embedded submanifold of $M \times M$, R is an equivalence relation on M , $S \subset U$ and S is a properly embedded submanifold of U . Then (S, q) is said to be a **R -slice of U** if for each $p \in U$, $\pi(p) \cap S = \{q(p)\}$

Exercise 10.2.0.2. $O(n)$ acting on $\mathbb{R}^n$, $U = \mathbb{R}^n$, $S = \{te_1 : t \geq 0\}$ and $q(x) = \|x\|e_1$. **FINISH!!! clean up**

Proof.

□

Exercise 10.2.0.3. Slice Theorem: Let $M, R \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that R is a properly embedded submanifold of $M \times M$, R is an equivalence relation on M . Then for each $p \in M$, there exists $U \in \mathcal{T}_M$, $S \in \text{Obj}(\mathbf{Man}^\infty)$ and $q \in \text{Hom}_{\mathbf{Man}^\infty}(U, S)$ such that $p \in U$, $S \subset U$ and S is a properly embedded submanifold of U and (S, q) is a R -slice of U .

Proof. Let $p \in M$. Define $S' \in \text{EmbSubman}^\infty(M)$ by $S' := \pi(p)$. Then $\dim S' = \text{???}$. Exercise 9.5.0.5 implies that there exists $S \in \text{EmbSubman}^\infty(M)$, such that $p \in S$ and S, S' are strictly transverse. Define $Z \in \text{EmbSubman}^\infty(R)$ by $Z := \text{proj}_2|_R^{-1}(S)$. Exercise 10.1.0.4 implies that (verify assumption in this ex) $D\text{proj}_1|_Z(p, p) \in \text{Iso}_{\mathbf{Vect}_\mathbb{R}}(T_{(p,p)}Z, T_pM)$. ref ex on local diffeomorphisms implies that there exists **FINISH!!!**

□

Chapter 11

The Tangent and Cotangent Bundles

11.1 Introduction

Definition 11.1.0.1. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$. Set $n := \dim M$. We define the **tangent bundle of M** , denoted TM , by

$$TM = \coprod_{p \in M} T_p M$$

and we define the **tangent bundle projection**, denoted $\pi_{TM} : TM \rightarrow M$, by

$$\pi_{TM}(p, v) := p$$

Let $(U, \phi) \in \mathcal{A}_M$ with $\phi = (x^1, \dots, x^n)$. We define $\tilde{\phi} : \pi^{-1}(U) \rightarrow \phi(U) \times \mathbb{R}^n$ by

$$\tilde{\phi}\left(p, \sum_{j=1}^n \xi^j \frac{\partial}{\partial x^j} \Big|_p\right) := (\phi(p), \xi^1, \dots, \xi^n)$$

Note 11.1.0.2. When the context is clear, we write π in place of π_{TM} .

Exercise 11.1.0.3. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $(U, \phi) \in \mathcal{A}_M$. Set $n := \dim M$. Then

- π is surjective,
- for each $A \subset U$, $\tilde{\phi}(\pi^{-1}(A)) = \phi(A) \times \mathbb{R}^n$.

Proof. **FINISH!!!**

□

Exercise 11.1.0.4. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$. Then there exists a unique topology $\mathcal{T}_{TM}$ on TM and smooth structure $\mathcal{A}_{TM}$ on $(TM, \mathcal{T}_{TM})$ such that $(TM, \mathcal{T}_{TM}, \mathcal{A}_{TM}) \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $(\pi^{-1}(U_\alpha), \tilde{\phi}_\alpha)_{\alpha \in \Gamma} \subset \mathcal{A}_{TM}$ and $\pi \in \text{Hom}_{\mathbf{ManBnd}^\infty}(TM, M)$.

Proof. Write $\mathcal{A}_M = (U_\alpha, \phi_\alpha)_{\alpha \in \Gamma}$.

(a) Let $\alpha \in \Gamma$. Since $U_\alpha \in \mathcal{T}_M$ and ϕ_α is a homeomorphism, $\phi_\alpha(U_\alpha) \in \mathcal{T}_{\mathbb{H}_n^n}$. Hence

$$\begin{aligned} \tilde{\phi}_\alpha(\pi^{-1}(U_\alpha)) &= \phi_\alpha(U_\alpha) \times \mathbb{R}^n \\ &\in \mathbb{H}_n^{2n}. \end{aligned}$$

(b) Let $\alpha, \beta \in \Gamma$. Since $U_\alpha, U_\beta \in \mathcal{T}_M$, we have that $U_\alpha \cap U_\beta \in \mathcal{T}_M$. Since ϕ_α is a homeomorphism, and $\phi_\alpha(U_\alpha \cap U_\beta) \in \mathcal{T}_{\mathbb{H}_n^n}$. Therefore

$$\begin{aligned} \tilde{\phi}_\alpha(\pi^{-1}(U_\alpha) \cap \pi^{-1}(U_\beta)) &= \tilde{\phi}_\alpha(\pi^{-1}(U_\alpha \cap U_\beta)) \\ &= \phi_\alpha(U_\alpha \cap U_\beta) \times \mathbb{R}^n \\ &\in \mathcal{T}_{\mathbb{H}_n^{2n}}. \end{aligned}$$

(c) Let $\alpha, \beta \in \Gamma$. Write $\phi_\alpha = (x^1, \dots, x^n)$. Then $\tilde{\phi}_\alpha : \pi^{-1}(U_\alpha) \rightarrow \phi_\alpha(U_\alpha) \times \mathbb{R}^n$ is a bijection with

$$\tilde{\phi}_\alpha^{-1}(a, \xi^1, \dots, \xi^n) = \left(\phi_\alpha^{-1}(a), \sum_{j=1}^n \xi^j \frac{\partial}{\partial x^j} \Big|_{\phi_\alpha^{-1}(a)} \right).$$

(d) Let $\alpha, \beta \in \Gamma$. Write $\phi_\alpha = (x^1, \dots, x^n)$ and $\phi_\beta = (y^1, \dots, y^n)$. Set $f_\alpha := \tilde{\phi}_\alpha|_{\pi^{-1}(U_\alpha) \cap \pi^{-1}(U_\beta)}$ and $f_\beta := \tilde{\phi}_\beta|_{\pi^{-1}(U_\alpha) \cap \pi^{-1}(U_\beta)}$. Let $(a, \xi^1, \dots, \xi^n) \in \phi_\alpha(U_\alpha \cap U_\beta) \times \mathbb{R}^n$. Then

$$\begin{aligned} f_\beta \circ f_\alpha^{-1}(a, \xi^1, \dots, \xi^n) &= \tilde{\phi}_\beta \left(\phi_\alpha^{-1}(a), \sum_{j=1}^n \xi^j \frac{\partial}{\partial x^j} \Big|_{\phi_\alpha^{-1}(a)} \right) \\ &= \tilde{\phi}_\beta \left(\phi_\alpha^{-1}(a), \sum_{k=1}^n \left[\sum_{j=1}^n \xi^j \frac{\partial y^k}{\partial x^j}(\phi_\alpha^{-1}(a)) \right] \frac{\partial}{\partial y^k} \Big|_{\phi_\alpha^{-1}(a)} \right) \\ &= \left(\phi_\beta(\phi_\alpha^{-1}(a)), \sum_{j=1}^n \xi^j \frac{\partial y^1}{\partial x^j}(\phi_\alpha^{-1}(a)), \dots, \sum_{j=1}^n \xi^j \frac{\partial y^n}{\partial x^j}(\phi_\alpha^{-1}(a)) \right). \end{aligned}$$

Since $(U_\alpha, \phi_\alpha), (U_\beta, \phi_\beta) \in \mathcal{A}_M$, we have that $(U_\alpha, \phi_\alpha), (U_\beta, \phi_\beta)$ are smoothly compatible. Hence $\phi_\beta \circ \phi_\alpha|_{U_\alpha \cap U_\beta}^{-1}$ is smooth. In particular, for each $k \in [n]$, $y^k \circ \phi|_{U_\alpha \cap U_\beta}^{-1}$ is smooth. By definition, for each $a \in \phi_\alpha(U_\alpha \cap U_\beta)$ and $j, k \in [n]$, we have that $\frac{\partial y^k}{\partial x^j}(\phi_\alpha^{-1}(a)) = \frac{\partial}{\partial w^j}[y^k \circ \phi_\alpha|_{U_\alpha \cap U_\beta}^{-1}](a)$. Hence for each $j, k \in [n]$, $\frac{\partial y^k}{\partial x^j} \circ \phi_\alpha|_{U_\alpha \cap U_\beta}^{-1}$ is smooth. Thus $\tilde{\phi}_\beta|_{\pi^{-1}(U_\alpha) \cap \pi^{-1}(U_\beta)} \circ \tilde{\phi}_\alpha|_{\pi^{-1}(U_\alpha) \cap \pi^{-1}(U_\beta)}^{-1}$ is smooth.

(e) Since $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$, M is second-countable. Thus M is Lindelof. Since $(U_\alpha, \phi_\alpha)_{\alpha \in A}$ is an atlas on M , $(U_\alpha)_{\alpha \in \Gamma}$ is an open cover of M . Hence there exists $\Gamma' \subset \Gamma$ such that Γ' is countable and $M \subset \bigcup_{\alpha \in \Gamma'} U_\alpha$. Hence

$$\begin{aligned} TM &= \pi^{-1}(M) \\ &\subset \pi^{-1}\left(\bigcup_{\alpha \in \Gamma'} U_\alpha\right) \\ &= \bigcup_{\alpha \in \Gamma'} \pi^{-1}(U_\alpha). \end{aligned}$$

(f) Let $(p_1, v_1), (p_2, v_2) \in TM$.

- Suppose that $p_1 \neq p_2$. Since $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$, M is Hausdorff. Thus there exist $U'_1, U'_2 \in \mathcal{T}_M$ such that $p_1 \in U'_1$, $p_2 \in U'_2$ and $U'_1 \cap U'_2 = \emptyset$. Since $(U_\alpha)_{\alpha \in \Gamma}$ is an open cover of M , there exist $\alpha'_1, \alpha'_2 \in \Gamma$ such that $p_1 \in U_{\alpha'_1}$ and $p_2 \in U_{\alpha'_2}$. Set $U_1 := U'_1 \cap U_{\alpha'_1}$, $U_2 := U'_2 \cap U_{\alpha'_2}$, $\phi_1 := \phi_{\alpha'_1}|_{U_1}$ and $\phi_2 := \phi_{\alpha'_2}|_{U_2}$. Exercise ?? (reference ex here) implies that $(U_1, \phi_1), (U_2, \phi_2) \in \mathcal{A}_M$. Hence there exists $\alpha_1, \alpha_2 \in \Gamma$ such that $(U_1, \phi_1) = (U_{\alpha_1}, \phi_{\alpha_1})$ and $(U_2, \phi_2) = (U_{\alpha_2}, \phi_{\alpha_2})$. By construction, $p_1 \in U_{\alpha_1}$, $p_2 \in U_{\alpha_2}$ and $U_{\alpha_1} \cap U_{\alpha_2} = \emptyset$. Therefore $(p_1, v_1) \in \pi^{-1}(U_{\alpha_1})$, $(p_2, v_2) \in \pi^{-1}(U_{\alpha_2})$ and

$$\begin{aligned} \pi^{-1}(U_{\alpha_1}) \cap \pi^{-1}(U_{\alpha_2}) &= \pi^{-1}(U_{\alpha_1} \cap U_{\alpha_2}) \\ &= \pi^{-1}(\emptyset) \\ &= \emptyset. \end{aligned}$$

- Suppose that $p_1 = p_2$. Since $\mathcal{A}_M$ is an atlas on M , there exists $\alpha \in \Gamma$ such that $p_1 \in U_\alpha$. Since $p_1 = p_2$, we have that $(p_1, v_1), (p_2, v_2) \in \pi^{-1}(U_\alpha)$.

Exercise 4.1.0.14 implies that there exists a unique topology $\mathcal{T}_{TM}$ on TM and smooth structure $\mathcal{A}_{TM}$ on $(TM, \mathcal{T}_{TM})$ such that $(TM, \mathcal{T}_{TM}, \mathcal{A}_{TM}) \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $(\pi^{-1}(U_\alpha), \tilde{\phi}_\alpha)_{\alpha \in \Gamma} \subset \mathcal{A}_{TM}$.

Let $(p, v) \in TM$. Since $(\pi^{-1}(U_\alpha), \tilde{\phi}_\alpha)_{\alpha \in \Gamma} \subset \mathcal{A}_{TM}$ is an atlas on TM , there exists $\alpha \in \Gamma$ such that $(p, v) \in \pi^{-1}(U_\alpha)$. Set $U := \pi^{-1}(U_\alpha)$, $V := U_\alpha$, $\phi := \tilde{\phi}_\alpha$ and $\psi := \phi_\alpha$. $(U, \phi) \in \mathcal{A}_{TM}$, $(V, \psi) \in \mathcal{A}_M$, $(p, v) \in U$, $\pi(p, v) \in V$ and

$$\begin{aligned} U \cap \pi^{-1}(V) &= \pi^{-1}(U_\alpha) \cap \pi^{-1}(U_\alpha) \\ &= \pi^{-1}(U_\alpha) \\ &\in \mathcal{T}_{TM}. \end{aligned}$$

Write $\phi_\alpha = (x^1, \dots, x^n)$. Then for each $(a, \xi^1, \dots, \xi^n) \in \tilde{\phi}_\alpha(\pi^{-1}(U_\alpha))$,

$$\begin{aligned} \psi \circ \pi \circ \phi|_{U \cap \pi^{-1}(V)}^{-1}(a, \xi^1, \dots, \xi^n) &= \phi_\alpha \circ \pi \circ \tilde{\phi}_\alpha|_{\pi^{-1}(U_\alpha)}^{-1}(a, \xi^1, \dots, \xi^n) \\ &= \phi_\alpha \circ \pi \left(\phi_\alpha^{-1}(a), \sum_{j=1}^n \xi^j \frac{\partial}{\partial x^j} \Big|_{\phi_\alpha^{-1}(a)} \right) \\ &= \phi_\alpha(\phi_\alpha^{-1}(a)) \\ &= \text{id}_{\phi_\alpha(U_\alpha)}(a) \end{aligned}$$

Hence $\psi \circ \pi \circ \phi|_{U \cap \pi^{-1}(V)}^{-1} = \text{id}_{\phi_\alpha(U_\alpha)}$ which is smooth. Exercise 5.1.0.10 implies that π is smooth. $\square$

Exercise 11.1.0.5. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$. Then $\pi : TM \rightarrow M$ is a submersion.

Proof. Let $(p, v) \in TM$. Choose $(U, \phi) \in \mathcal{A}_M$ such that $p \in U$. Set $V := \pi^{-1}(U)$ and $\psi := \tilde{\phi}$. Then $(V, \psi) \in \mathcal{A}_{TM}$, $(p, v) \in V$, $U = \pi(V)$,

$$\begin{aligned} \psi(V) &= \tilde{\phi}(\pi^{-1}(U)) \\ &= \phi(U) \times \mathbb{R}^n, \end{aligned}$$

and since π is surjective,

$$\begin{aligned} \pi(V) &= \pi(\pi^{-1}(U)) \\ &= U. \end{aligned}$$

Since for each $(a, \xi^1, \dots, \xi^n) \in \psi(V)$,

$$\begin{aligned} \phi \circ \pi \circ \psi^{-1}(a, \xi^1, \dots, \xi^n) &= \phi \circ \pi \left(\phi^{-1}(a), \sum_{j=1}^n \xi^j \frac{\partial}{\partial x^j} \Big|_{\phi^{-1}(a)} \right) \\ &= \phi(\phi^{-1}(a)) \\ &= a \\ &= \text{proj}_{[n]}^{2n}(a), \end{aligned}$$

we have that $\phi \circ \pi \circ \psi^{-1} = \text{proj}_{[n]}^{2n}(a)|_{\psi(V)}$. Since $(p, v) \in TM$ is arbitrary, we have that for each $(p, v) \in TM$, there exists $(U, \phi) \in \mathcal{T}_M$, $(V, \psi) \in \mathcal{T}_{TM}$ such that $(p, v) \in V$, $U = \pi(V)$ and $\phi \circ \pi \circ \psi^{-1} = \text{proj}_{[n]}^{2n}|_{\psi(V)}$. Exercise 8.3.0.9 implies that π is a submersion. $\square$

Exercise 11.1.0.6. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $(U, \phi) \in \mathcal{A}_M$. Set $n := \dim M$ and write $\phi = (x^1, \dots, x^n)$ and $\tilde{\phi} = (\tilde{x}^1, \dots, \tilde{x}^n, \tilde{y}^1, \dots, \tilde{y}^n)$. Then for each $(p, v) \in \pi^{-1}(U)$,

1. $[D\pi(p, v)]_{\tilde{\phi}, \phi} = \begin{pmatrix} I_n & 0_n \end{pmatrix}$
2. $\ker D\pi(p, v) = \text{span} \left\{ \frac{\partial}{\partial \tilde{y}^j} \Big|_{(p, v)} : j \in [n] \right\}$

Proof. 1. **The previous exercise** Exercise ?? implies that for each $(p, v) \in \pi^{-1}(U)$, $\phi \circ \pi \circ \tilde{\phi}^{-1} = \text{proj}_{[n]}^{2n}|_{\phi(U) \times \mathbb{R}^n}$. Hence

$$\begin{aligned} [D\pi(p, v)]_{\tilde{\phi}, \phi} &= [D \text{proj}_{[n]}^{2n}(p, v)] \\ &= \begin{pmatrix} I_n & 0_n \end{pmatrix}. \end{aligned}$$

2. Clear from previous part. $\square$

Definition 11.1.0.7. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$. We define the **pushforward of F** , denoted by $DF : TM \rightarrow TN$ by

$$DF(p, v) := (F(p), DF(p)(v))$$

Note 11.1.0.8. Another common notation for DF is TF .

Exercise 11.1.0.9. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$. Then

1. $\pi_{TN} \circ DF = F \circ \pi_{TM}$, i.e. the following diagram commutes:

$$\begin{array}{ccc} TM & \xrightarrow{DF} & TN \\ \pi_{TM} \downarrow & & \downarrow \pi_{TN} \\ M & \xrightarrow{F} & N \end{array}$$

2. for each $V \in \mathcal{T}_N$, $DF^{-1}(\pi_{TN}^{-1}(V)) = \pi_{TM}^{-1}(F^{-1}(V))$

Proof.

1. We note that for each $(p, v) \in TM$,

$$\begin{aligned} \pi_{TN} \circ DF(p, v) &= \pi_{TN}(F(p), DF(p)(v)) \\ &= F(p) \\ &= F \circ \pi_{TM}(p, v). \end{aligned}$$

Thus $\pi_{TN} \circ DF = F \circ \pi_{TM}$.

2. Let $V \in \mathcal{T}_N$. Then

$$\begin{aligned} DF^{-1}(\pi_{TN}^{-1}(V)) &= (\pi_{TN} \circ DF)^{-1}(V) \\ &= (F \circ \pi_{TM})^{-1}(V) \\ &= \pi_{TM}^{-1}(F^{-1}(V)). \end{aligned}$$

□

Exercise 11.1.0.10. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$. Then $DF \in \text{Hom}_{\mathbf{ManBnd}^\infty}(TM, TN)$.

Proof. Let $(p, v) \in TM$. Since $\mathcal{A}_M$ is an atlas on M and $\mathcal{A}_N$ is an atlas on N , there exist $(U, \phi) \in \mathcal{A}_M$ and $(V, \psi) \in \mathcal{A}_N$ such that $p \in U$ and $F(p) \in V$. Since $p \in U$, $(p, v) \in \pi_{TM}^{-1}(U)$. The previous exercise implies that $DF^{-1}(\pi_{TN}^{-1}(V)) = \pi_{TM}^{-1}(F^{-1}(V))$. Since F is smooth, $U \cap F^{-1}(V) \in \mathcal{T}_M$. Since π_{TM} is smooth, we have that

$$\begin{aligned} \pi_{TM}^{-1}(U) \cap DF^{-1}(\pi_{TN}^{-1}(V)) &= \pi_{TM}^{-1}(U) \cap \pi_{TM}^{-1}(F^{-1}(V)) \\ &= \pi_{TM}^{-1}(U \cap F^{-1}(V)) \\ &\in \mathcal{T}_{TM}. \end{aligned}$$

Set $m := \dim M$, $n := \dim N$ and write $\phi = (x^1, \dots, x^m)$ and $\psi = (y^1, \dots, y^n)$. Then for each $(a, \xi^1, \dots, \xi^m) \in \tilde{\phi}[\pi_{TM}^{-1}(U) \cap DF^{-1}(\pi_{TN}^{-1}(V))]$, we have that

$$\begin{aligned} \tilde{\psi} \circ DF \circ \tilde{\phi}^{-1}(a, \xi^1, \dots, \xi^m) &= \tilde{\psi} \circ DF \left(\phi^{-1}(a), \sum_{j=1}^n \xi^j \frac{\partial}{\partial x^j} \Big|_{\phi^{-1}(a)} \right) \\ &= \tilde{\psi} \left(F \circ \phi^{-1}(a), \sum_{j=1}^n \xi^j DF(\phi^{-1}(a)) \left(\frac{\partial}{\partial x^j} \Big|_{\phi^{-1}(a)} \right) \right) \\ &= \tilde{\psi} \left(F \circ \phi^{-1}(a), \sum_{j=1}^n \xi^j \left[\sum_{k=1}^n \frac{\partial(y^k \circ F)}{\partial x^j}(\phi^{-1}(a)) \frac{\partial}{\partial y^k} \Big|_{F \circ \phi^{-1}(a)} \right] \right) \\ &= \tilde{\psi} \left(F \circ \phi^{-1}(a), \sum_{k=1}^n \left[\sum_{j=1}^n \xi^j \frac{\partial(y^k \circ F)}{\partial x^j}(\phi^{-1}(a)) \right] \frac{\partial}{\partial y^k} \Big|_{F \circ \phi^{-1}(a)} \right) \\ &= \left(\psi \circ F \circ \phi^{-1}(a), \sum_{j=1}^n \xi^j \frac{\partial(y^1 \circ F)}{\partial x^j}(\phi^{-1}(a)), \dots, \sum_{j=1}^n \xi^j \frac{\partial(y^n \circ F)}{\partial x^j}(\phi^{-1}(a)) \right). \end{aligned}$$

Thus $\tilde{\psi} \circ DF \circ \tilde{\phi}^{-1}|_{\pi_{TM}^{-1}(U) \cap DF^{-1}(\pi_{TN}^{-1}(V))}$ is smooth. Exercise 5.1.0.10 implies that DF is smooth. (maybe add more details here). □

Exercise 11.1.0.11. Let $M, N, E \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$ and $G \in \text{Hom}_{\mathbf{ManBnd}^\infty}(N, E)$. Then

1. for each $p \in M$, $DF|_{\{p\} \times T_p M} = \text{id}_{\{p\}} \times DF(p)$.
2. $D(G \circ F) = DG \circ DF$
3. $D(\text{id}_M) = \text{id}_{TM}$
4. $F \in \text{Iso}_{\mathbf{ManBnd}^\infty}(M, N)$ implies that $DF \in \text{Iso}_{\mathbf{ManBnd}^\infty}(TM, TN)$ and $D(F^{-1}) = DF^{-1}$.

Proof.

- 1.
- 2.
- 3.
- 4.

FINISH!!!

□

11.2 Cotangent Bundle

Chapter 12

Vector and Covector Fields

12.1 Vector Fields

Definition 12.1.0.1. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$. We define the **vector fields on** M , denoted $\mathfrak{X}(M)$, by $\mathfrak{X}(M) := \Gamma(\pi_{TM})$.

Exercise 12.1.0.2. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $X : M \rightarrow TM$. If X is a section of π_{TM} , then for each $p \in M$, $X(p) \in \{p\} \times T_p M$.

Proof. Suppose that X is a section of π_{TM} . Let $p \in M$. Since $X(p) \in TM$, there exists $q \in M$ and $v \in T_q M$ such that $X(p) = (q, v)$. Since X is a section of π_{TM} ,

$$\begin{aligned} p &= \text{id}_M(p) \\ &= \pi_{TM} \circ X(p) \\ &= \pi_{TM}(q, v) \\ &= q. \end{aligned}$$

Hence

$$\begin{aligned} X(p) &= (p, v) \\ &\in \{p\} \times T_p M. \end{aligned}$$

actually just reference exercise in set theory section

□

Note 12.1.0.3.

- When the context is clear, we may write X_p in place of $X(p)$.
- If

$$\begin{aligned} X_p &= (p, v) \\ &\in \{p\} \times T_p M, \end{aligned}$$

we identify X_p with v . So we may write $X_p \in T_p M$ or $X_p \in TM$.

Exercise 12.1.0.4. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $U \in \mathcal{T}_M$. Then

1. $X(U) \subset TU$
2. for each $X \in \mathfrak{X}(M)$, $X|_U^{TU} \in \mathfrak{X}(U)$.

Proof. (need to make exercise and develop tangent space in terms of germs so tangent space is independent of neighborhood of point and $T_p U = T_p M$, then make exercise in section on tangent bundle that $TU \subset TM$)

1. An ex in tangent bundle intro section implies that for each $p \in U$,

$$\begin{aligned} X_p &\in \{p\} \times T_p M \\ &= \{p\} \times T_p U \\ &\subset TU. \end{aligned}$$

Thus $X(U) \subset TU$. Thus there exists $v \in T_p M$ such that

2. • Since $X \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, TM)$ and $\iota_U \in \text{Hom}_{\mathbf{ManBnd}^\infty}(U, M)$, we have that

$$\begin{aligned} X|_U &= X \circ \iota_U \\ &\in \text{Hom}_{\mathbf{ManBnd}^\infty}(U, TM). \end{aligned}$$

Since

$$\begin{aligned} X|_U(U) &= X(U) \\ &\subset TU. \end{aligned}$$

and TU is an open submanifold of TM (make/ref ex about this), a prev ex in embedded submanifolds implies that $X|_U^{TU} \in \text{Hom}_{\mathbf{ManBnd}^\infty}(U, TU)$.

- Since for each $p \in U$,

$$\begin{aligned} \pi_{TU} \circ X|_U(p) &= \pi_{TM} \circ X|_U(p) \\ &= \pi_{TM} \circ [X \circ \iota_U](p) \\ &= [\pi_{TM} \circ X] \circ \iota_U(p) \\ &= \text{id}_M \circ \iota_U(p) \\ &= \iota_U(p) \\ &= p \\ &= \text{id}_U(p). \end{aligned}$$

Thus $\pi_{TU} \circ X|_U^{TU} = \text{id}_U$.

Hence $X|_U^{TU} \in \mathfrak{X}(U)$.

□

Definition 12.1.0.5. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $(U, \phi) \in \mathcal{A}_M$. Set $n := \dim M$ and write $\phi = (x^1, \dots, x^n)$. Let $X : M \rightarrow TM$. Suppose that X is a section of π_{TM} . For each $j \in [n]$, we define the j -th component function of X with respect to (U, ϕ) , denoted $X^j : U \rightarrow \mathbb{R}$, by $X^j(p) := dx_p^j(X_p)$.

Note 12.1.0.6. In particular, for each $p \in U$, $[\tilde{\phi} \circ X](p) = (\phi(p), X_p^1, \dots, X_p^n)$ and

$$X_p = \sum_{j=1}^n X^j(p) \frac{\partial}{\partial x^j} \Big|_p.$$

Exercise 12.1.0.7. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $(U, \phi) \in \mathcal{A}_M$. Set $n := \dim M$ and write $\phi = (x^1, \dots, x^n)$. Let $X : M \rightarrow TM$. Suppose that X is a section of π_{TM} . Then $X|_U \in \mathfrak{X}(U)$ iff for each $j \in [n]$, $X^j \in C^\infty(U)$.

Proof.

- ($\implies$):

Suppose that X is smooth. Then $\tilde{\phi} \circ X \circ \phi^{-1}$ is smooth. Since $\tilde{\phi} \circ X \circ \phi^{-1} = (\text{id}_{\phi(U)}, X^1 \circ \phi^{-1}, \dots, X^n \circ \phi^{-1})$, we have that for each $j \in [n]$, $X^j \circ \phi^{-1}$ is smooth. Hence for each $j \in [n]$, X^j is smooth.

- ($\impliedby$):

Suppose that for each $j \in [n]$, X^j is smooth. Then for each $j \in [n]$, $X^j \circ \phi^{-1}$ is smooth. Since $\tilde{\phi} \circ X \circ \phi^{-1} = (\text{id}_{\phi(U)}, X^1 \circ \phi^{-1}, \dots, X^n \circ \phi^{-1})$, we have that $\tilde{\phi} \circ X \circ \phi^{-1}$ is smooth. Since $X|_U = \tilde{\phi}^{-1} \circ [\tilde{\phi} \circ X \circ \phi^{-1}] \circ \phi$, we have that $X|_U$ is smooth.

□

Exercise 12.1.0.8. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $X : M \rightarrow TM$. Set $n := \dim M$. Suppose that X is a section of π_{TM} . Then $X \in \mathfrak{X}(M)$ iff for each $(U, \phi) \in \mathcal{A}_M$, $X^1, \dots, X^n \in C^\infty(U)$.

Proof. Since X is smooth iff for each $(U, \phi) \in \mathcal{A}_M$, $X|_U$ is smooth, [the previous exercise](#) implies that $X \in \mathfrak{X}(M)$ iff for each $(U, \phi) \in \mathcal{A}_M$, $X^1, \dots, X^n \in C^\infty(U)$. **reword** □

Exercise 12.1.0.9. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $(U, \phi) \in \mathcal{A}_M$. Set $n := \dim M$ and write $\phi = (x^1, \dots, x^n)$. Then for each $j \in [n]$, $\frac{\partial}{\partial x^j} \in \mathfrak{X}(U)$.

Proof. Let $j \in [n]$. Define $X : U \rightarrow TM$ by $X_p := \frac{\partial}{\partial x^j} \Big|_p$. Clearly, X is a section of π_{TU} . Since for each $k \in [n]$, $X^k = \delta_{j,k}$, [the previous exercise](#) implies that $X \in \mathfrak{X}(U)$. □

Definition 12.1.0.10. Let $X, Y \in \mathfrak{X}(M)$ and $f \in C^\infty(M)$. We define

- $fX : M \rightarrow TM$ by

$$(fX)_p = f(p)X_p$$

- $X + Y : M \rightarrow TM$ by

$$(X + Y)_p = X_p + Y_p$$

Exercise 12.1.0.11. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$. Then

1. for each $f \in C^\infty(M)$ and $X, Y \in \mathfrak{X}(M)$,

- (a) $fX \in \mathfrak{X}(M)$
- (b) $X + Y \in \mathfrak{X}(M)$

2. $\mathfrak{X}(M) \in \text{Obj}(\mathbf{Mod}_{C^\infty(M)})$.

Proof.

1. Let $f \in C^\infty(M)$, $X, Y \in \mathfrak{X}(M)$ and $(U, \phi) \in \mathcal{A}_M$. Set $n := \dim M$ and write $\phi = (x^1, \dots, x^n)$.

- (a) Clearly fX is a section of π_{TM} . Since

$$\begin{aligned} (fX)|_U &= f|_U \sum_{j=1}^n X^j \frac{\partial}{\partial x^j} \\ &= \sum_{j=1}^n f|_U X^j \frac{\partial}{\partial x^j}, \end{aligned}$$

we have that for each $j \in [n]$, $(fX)^j = f|_U X^j$. Since $f|_U, X^j \in C^\infty(U)$, $f|_U X^j \in C^\infty(U)$. [a previous exercise](#) implies that $(fX)|_U$ is smooth. Since $(U, \phi) \in \mathcal{A}_M$ is arbitrary, we have that for each $(U, \phi) \in \mathcal{A}_M$, $(fX)|_U$ is smooth. Hence fX is smooth and $fX \in \mathfrak{X}(M)$.

- (b) Clearly $X + Y$ is a section of π_{TM} . Since

$$\begin{aligned} (X + Y)|_U &= \sum_{j=1}^n X^j \frac{\partial}{\partial x^j} + \sum_{j=1}^n Y^j \frac{\partial}{\partial x^j} \\ &= \sum_{j=1}^n (X^j + Y^j) \frac{\partial}{\partial x^j} \end{aligned}$$

we have that for each $j \in [n]$, $(X + Y)^j = X^j + Y^j$. Since $X^j, Y^j \in C^\infty(U)$, $X^j + Y^j \in C^\infty(U)$. [a previous exercise](#) implies that $(X + Y)|_U$ is smooth. Since $(U, \phi) \in \mathcal{A}_M$ is arbitrary, we have that for each $(U, \phi) \in \mathcal{A}_M$, $(X + Y)|_U$ is smooth. Hence $X + Y$ is smooth and $X + Y \in \mathfrak{X}(M)$.

2. Clearly by previous part.

□

12.2 Vector Fields as Derivations on $C^\infty(M)$

Definition 12.2.0.1. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $D : C^\infty(M) \rightarrow C^\infty(M)$. Then D is said to be a **derivation on $C^\infty(M)$** if

- (linearity):
for each $f, g \in C^\infty(M)$ and $\lambda \in \mathbb{R}$, $D(f + \lambda g) = D(f) + \lambda D(g)$,
- (Leibnizianity):
for each $f, g \in C^\infty(M)$, $D(fg) = fD(g) + D(f)g$.

We define

$$\text{Deriv}^\infty(M) := \{D : C^\infty(M) \rightarrow C^\infty(M) : D \text{ is a derivation on } C^\infty(M)\}.$$

Exercise 12.2.0.2. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $D \in \text{Deriv}^\infty(M)$.

Definition 12.2.0.3. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$.

- For each $D_1, D_2 \in \text{Deriv}^\infty(M)$, we define $D_1 + D_2 : C^\infty(M) \rightarrow C^\infty(M)$ by

$$[D_1 + D_2](g) := D_1(g) + D_2(g).$$

- For each $D \in \text{Deriv}^\infty(M)$ and $f \in C^\infty(M)$, we define $fD : C^\infty(M) \rightarrow C^\infty(M)$ by

$$[fD](g) := fD(g).$$

Exercise 12.2.0.4. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$. Then

1. for each $D_1, D_2 \in \text{Deriv}^\infty(M)$ and $f \in C^\infty(M)$,
 - (a) $D_1 + D_2 \in \text{Deriv}^\infty(M)$,
 - (b) $fD_1 \in \text{Deriv}^\infty(M)$.
2. $\text{Deriv}^\infty(M) \in \text{Obj}(\mathbf{Mod}_{C^\infty(M)})$.

Proof.

1. (a)
- (b)
- 2.

FINISH!!!

□

Definition 12.2.0.5. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $X : M \rightarrow TM$. Suppose that X is a section of π_{TM} . For each $f \in C^\infty(M)$, we define $Xf : M \rightarrow \mathbb{R}$ by

$$(Xf)_p := X_p(f).$$

Exercise 12.2.0.6. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $X : M \rightarrow TM$ and $(U, \phi) \in \mathcal{A}_M$. Suppose that X is a section of π_{TM} . Set $n := \dim M$ and write $\phi = (x^1, \dots, x^n)$. Then

$$X|_U = \sum_{j=1}^n (X|_U(x^j)) \frac{\partial}{\partial x^j}$$

Proof. We have that for each $k \in [n]$,

$$\begin{aligned} X|_U(x^k) &= \sum_{j=1}^n X^j \frac{\partial}{\partial x^j}(x^k) \\ &= \sum_{j=1}^n X^j \delta_{j,k} \\ &= X^k. \end{aligned}$$

Hence

$$X|_U = \sum_{j=1}^n (X|_U(x^j)) \frac{\partial}{\partial x^j}.$$

□

Exercise 12.2.0.7. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $X \in \mathfrak{X}(M)$. Then for each $f \in C^\infty(M)$, $Xf \in C^\infty(M)$.

Proof. Let $(U, \phi) \in \mathcal{A}_M$. Set $n := \dim M$ and write $\phi = (x^1, \dots, x^n)$. Then **need exercise about how Xf only depends on neighborhood of p , maybe already exists in tangent space section, need reference** implies that for each $p \in U$,

$$\begin{aligned} [X|_U f|_U](p) &= X_p(f) \\ &= \left[\sum_{j=1}^n X^j(p) \frac{\partial}{\partial x^j} \right]_p f \\ &= \sum_{j=1}^n X^j(p) \frac{\partial f}{\partial x^j}(p) \\ &= \left[\sum_{j=1}^n X^j \frac{\partial f}{\partial x^j} \right](p). \end{aligned}$$

Since $X|_U \in \mathfrak{X}(U)$, and $f|_U \in C^\infty(U)$, we have that for each $j \in [n]$, $X^j \frac{\partial f}{\partial x^j} \in C^\infty(U)$. Thus $\sum_{j=1}^n X^j \frac{\partial f}{\partial x^j} \in C^\infty(U)$. Hence $X|_U f|_U \in C^\infty(U)$. Since $(Xf)|_U = X|_U f|_U$, we have that $(Xf)|_U \in C^\infty(U)$. Since $(U, \phi) \in \mathcal{A}_M$ is arbitrary, we have that for each $U \in \mathcal{T}_M$, $(Xf)|_U \in C^\infty(U)$. Thus $Xf \in C^\infty(M)$. □

Definition 12.2.0.8. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $X \in \mathfrak{X}(M)$. We define the **derivation on $C^\infty(M)$ induced by X** , denoted $D^X : C^\infty(M) \rightarrow C^\infty(M)$, by $D^X(f) := Xf$.

Exercise 12.2.0.9. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $X \in \mathfrak{X}(M)$. Then $D^X \in \text{Deriv}^\infty(M)$.

Proof.

- Let $f, g \in C^\infty(M)$ and $\lambda \in \mathbb{R}$. Then for each $p \in M$,

$$\begin{aligned} D^X(f + \lambda g) &= X(f + \lambda g)(p) \\ &= X_p(f + \lambda g) \\ &= X_p f + \lambda X_p g \\ &= (Xf)(p) + \lambda (Xg)(p) \\ &= [Xf + \lambda Xg](p) \\ &= [D^X(f) + \lambda D^X(g)](p) \end{aligned}$$

Hence $D^X(f + \lambda g) = D^X(f) + \lambda D^X(g)$ and $D^X : C^\infty(M) \rightarrow C^\infty(M)$ is linear.

- Let $f, g \in C^\infty(M)$. Then for each $p \in M$,

$$\begin{aligned} [D^X(fg)](p) &= [X(fg)](p) \\ &= X_p(fg) \\ &= (X_p f)g(p) + f(p)X_p(g) \\ &= (Xf)(p)g(p) + f(p)(Xg)(p) \\ &= [(Xf)g + f(Xg)](p) \\ &= D^X(f)g + fD^X(g). \end{aligned}$$

Hence $D^X(fg) = D^X(f)g + fD^X(g)$ and $D^X : C^\infty(M) \rightarrow C^\infty(M)$ is Leibnizian.

Thus $D^X \in \text{Deriv}^\infty(M)$. □

Definition 12.2.0.10. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$. We define the **Derivation map**, denoted $\text{Der} : \mathfrak{X}(M) \rightarrow \text{Deriv}^\infty(M)$, by $\text{Der}(X) := D^X$.

Exercise 12.2.0.11. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$. Then $\text{Der} \in \text{Hom}_{\mathbf{Mod}_{C^\infty(M)}}(\mathfrak{X}(M), \text{Deriv}^\infty(M))$.

Proof. Let $X, Y \in \mathfrak{X}(M)$ and $f, g \in C^\infty(M)$. Then for each $p \in M$,

$$\begin{aligned} [D^{X+fY}(g)](p) &= ([X + fY]g)(p) \\ &= [X + fY]_p(g) \\ &= [X_p + f(p)Y_p](g) \\ &= X_p(g) + f(p)Y_p(g) \\ &= (Xg)(p) + [f(Yg)](p) \\ &= [Xg + f(Yg)](p) \\ &= [D^X(g) + fD^Y(g)](p). \end{aligned}$$

Hence $D^{X+fY}(g) = D^X(g) + fD^Y(g)$. Since $g \in C^\infty(M)$ is arbitrary, we have that

$$\begin{aligned} \text{Der}(X + fY) &= D^{X+fY} \\ &= D^X + fD^Y \\ &= \text{Der}(X) + f\text{Der}(Y). \end{aligned}$$

Thus Der is $C^\infty(M)$ -linear. □

Exercise 12.2.0.12. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $X : M \rightarrow TM$. Suppose that X is a section of π_{TM} . Then the following are equivalent:

1. X is smooth
2. for each $f \in C^\infty(M)$, $Xf \in C^\infty(M)$
3. for each $U \in \mathcal{T}_M$ $f \in C^\infty(U)$, $X|_U(f) \in C^\infty(U)$

Proof.

- (1) $\implies$ (2):
Suppose that X is smooth. Let $f \in C^\infty$ and $(U, \phi) \in \mathcal{A}_M$. Then

$$\begin{aligned} X|_U f|_U &= \left[\sum_{j=1}^n X^j \frac{\partial}{\partial x^j} \right] f|_U \\ &= \sum_{j=1}^n X^j \frac{\partial}{\partial x^j} (f|_U) \\ &= \sum_{j=1}^n X^j \frac{\partial f|_U}{\partial x^j}. \end{aligned}$$

Since X and f are smooth, for each $j \in [n]$, $X^j, \frac{\partial f|_U}{\partial x^j} \in C^\infty(U)$. Hence $X|_U f|_U$ is smooth. Since $X|_U f|_U = (Xf)|_U$, we have that $(Xf)|_U$ is smooth. Since $U \in \mathcal{T}_M$ is arbitrary, we have that for each $U \in \mathcal{T}_M$, $(Xf)|_U$ is smooth. Exercise ?? **A previous exercise** implies that Xf is smooth.

- (2) $\implies$ (3):
Clear. **maybe add details, maybe bump function.**
- (3) $\implies$ (1):
FINISH!!!

□

Definition 12.2.0.13. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $D \in \text{Deriv}^\infty(M)$. For each $p \in M$, we define $X_p^D : C^\infty(M) \rightarrow \mathbb{R}$ by $X_p^D(f) := D(f)(p)$.

Exercise 12.2.0.14. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $D \in \text{Deriv}^\infty(M)$. Then for each $p \in M$, $X_p^D \in T_p M$.

Proof. Let $p \in M$.

- (linearity):

Let $f, g \in C^\infty$ and $\lambda \in \mathbb{R}$. Then

$$\begin{aligned} X_p^D(f + \lambda g) &= D(f + \lambda g)(p) \\ &= [D(f) + \lambda D(g)](p) \\ &= D(f)(p) + \lambda D(g)(p) \\ &= X_p^D(f) + \lambda X_p^D(g). \end{aligned}$$

- (Leibnizianity):

Let $f, g \in C^\infty(M)$. Then

$$\begin{aligned} X_p^D(fg) &= D(fg)(p) \\ &= [(Df)g + f(Dg)](p) \\ &= Df(p)g(p) + f(p)Dg(p) \\ &= X_p^D(f)g(p) + f(p)X_p^D(g). \end{aligned}$$

Thus $X_p^D \in T_p M$. □

Definition 12.2.0.15. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $D \in \text{Deriv}^\infty(M)$. We define $X^D : M \rightarrow TM$ by $X^D(p) := (p, X_p^D)$.

Exercise 12.2.0.16. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $D \in \text{Deriv}^\infty(M)$. Then $X^D \in \mathfrak{X}(M)$.

Proof. By construction X^D is a section of π_{TM} . Let $(U, \phi) \in \mathcal{A}_M$. Set $n := \dim M$ and write $\phi = (x^1, \dots, x^n)$. Then for each $j \in [n]$,

$$\begin{aligned} (X^D)^j &= X^D|_U(x^j) \\ &= D|_U(x^j) \quad (\text{need to define and show something like } D|_U \in \text{Deriv}^\infty(U)) \\ &\in C^\infty(U) \end{aligned}$$

(maybe need to make more rigorous with a bump function or maybe talk about restrictions of derivations, doesnt feel clean here). Then ref prev ex on smoothness of derivations and vector fields □

Exercise 12.2.0.17. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$. Then $\text{Der} \in \text{Iso}_{\text{Mod}_{C^\infty(M)}}(\mathfrak{X}(M), \text{Deriv}^\infty(M))$.

Proof.

- (injectivity):

Let $X, Y \in \mathfrak{X}(M)$. Suppose that $\text{Der}(X) = \text{Der}(Y)$. Let $(U, \phi) \in \mathcal{A}_M$. Set $n := \dim M$ and write $\phi = (x^1, \dots, x^n)$. Then for each $j \in [n]$,

$$\begin{aligned} X^j &= X|_U(x^j) \\ &= D^{X|_U}(x^j) \\ &= D^{Y|_U}(x^j) \\ &= Y|_U(x^j) \\ &= Y^j. \end{aligned}$$

Hence $X|_U = Y|_U$. Since $(U, \phi) \in \mathcal{A}_M$ is arbitrary, for each $U \in \mathcal{T}_M$, $X|_U = Y|_U$. Thus $X = Y$. Since $X, Y \in \mathfrak{X}(M)$ are arbitrary, we have that Der is injective

- **(subjectivity):**

Let $D \in \text{Deriv}^\infty(M)$. Define $X \in \mathfrak{X}(M)$ by $X := X^D$. Then for each $f \in C^\infty(M)$,

$$\begin{aligned} \text{Der}(X)(f) &= D^X(f) \\ &= Xf \\ &= X^D(f) \\ &= D(f). \end{aligned}$$

Hence $\text{Der}(X) = D$. Thus for each $D \in \text{Deriv}^\infty(M)$, there exists $X \in \mathfrak{X}(M)$ such that $\text{Der}(X) = D$. Thus Der is surjective.

Thus $\text{Der} \in \text{Iso}_{\mathbf{Mod}_{C^\infty(M)}}(\mathfrak{X}(M), \text{Deriv}^\infty(M))$. □

Note 12.2.0.18. For each $X \in \mathfrak{X}(M)$ and $D \in \text{Deriv}^\infty(M)$, we identify

- X with $D^X \in \text{Deriv}^\infty(M)$,
- D with $X^D \in \mathfrak{X}(M)$

via the isomorphism **above in prev ex.**

12.3 The Commutator of Vector Fields

Definition 12.3.0.1. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $X, Y \in \mathfrak{X}(M)$. We define $XY : C^\infty(M) \rightarrow C^\infty(M)$ by $XY(f) := X(Yf)$.

Exercise 12.3.0.2. There exist $X, Y \in \mathfrak{X}(\mathbb{R}^2)$ such that $XY \notin \text{Deriv}^\infty(\mathbb{R}^2)$.

Proof. Set $X := \frac{\partial}{\partial x^1}$ and $Y := \frac{\partial}{\partial x^2}$. Then $XY = \frac{\partial^2}{\partial x^1 \partial x^2}$. Define $f, g \in C^\infty(\mathbb{R}^2)$ by $f(x^1, x^2) := x^1$ and $g(x^1, x^2) := x^2$. Then

$$\begin{aligned} XY(fg) &= \frac{\partial^2}{\partial x^1 \partial x^2}(fg) \\ &= \frac{\partial}{\partial x^1} \left[\frac{\partial(fg)}{\partial x^2} \right] \\ &= \frac{\partial}{\partial x^1} \left[\frac{\partial f}{\partial x^2} g + f \frac{\partial g}{\partial x^2} \right] \\ &= \frac{\partial^2 f}{\partial x^1 \partial x^2} g + \frac{\partial f}{\partial x^2} \frac{\partial g}{\partial x^1} + \frac{\partial f}{\partial x^1} \frac{\partial g}{\partial x^2} + f \frac{\partial^2 g}{\partial x^1 \partial x^2} \\ &= \frac{\partial^2 f}{\partial x^1 \partial x^2} g + f \frac{\partial^2 g}{\partial x^1 \partial x^2} + 1 \\ &= [XY(f)]g + fXY(g) + 1 \\ &\neq [XY(f)]g + fXY(g). \end{aligned}$$

Thus XY is not Leibnizian and therefore $XY \notin \text{Deriv}^\infty(M)$. □

Definition 12.3.0.3. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $X, Y \in \mathfrak{X}(M)$. We define the **commutator on** $\mathfrak{X}(M)$, denoted $[\cdot, \cdot] : C^\infty(M) \times C^\infty(M) \rightarrow C^\infty(M)$, by

$$[X, Y] := XY - YX.$$

Exercise 12.3.0.4. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $X, Y \in \mathfrak{X}(M)$. Then $[X, Y] \in \mathfrak{X}(M)$.

Proof. Let $f, g \in C^\infty(M)$. Then

- (linearity):
Let $f, g \in C^\infty(M)$ and $\lambda \in \mathbb{R}$. Then

$$\begin{aligned} [X, Y](f + \lambda g) &= (XY - YX)(f + \lambda g) \\ &= XY(f + \lambda g) - YX(f + \lambda g) \\ &= X(Yf + \lambda Yg) - Y(Xf + \lambda Xg) \\ &= XY(f) + \lambda XY(g) - (YX(f) + \lambda YX(g)) \\ &= XY(f) - YX(f) + \lambda(XY(g) - YX(g)) \\ &= (XY - YX)(f) + \lambda(XY - YX)(g) \\ &= [X, Y](f) + \lambda[X, Y](g). \end{aligned}$$

Thus $[X, Y]$ is $\mathbb{R}$ -linear.

- (Leibnizianity):

$$\begin{aligned} (XY)(fg) &= X(Y(fg)) \\ &= X((Yf)g + f(Yg)) \\ &= X((Yf)g) + X(f(Yg)) \\ &= [X(Yf)]g + (Yf)(Xg) + (Xf)(Yg) + f[X(Yg)] \\ &= [(XY)(f)]g + (Yf)(Xg) + (Xf)(Yg) + f[(XY)(g)]. \end{aligned}$$

Similarly, $(YX)(fg) = [(YX)(f)]g + (Xf)(Yg) + (Yf)(Xg) + f[(YX)(g)]$. Hence

$$\begin{aligned}
 [X, Y](fg) &= (XY - YX)(fg) \\
 &= XY(fg) - YX(fg) \\
 &= [(XY)(f)]g + (Yf)(Xg) + (Xf)(Yg) + f[(XY)(g)] - ([YX)(f)]g + (Xf)(Yg) + (Yf)(Xg) + f[(YX)(g)] \\
 &= [(XY)(f)]g - [(YX)(f)]g + f[(XY)(g)] - f[(YX)(g)] \\
 &= [(XY)(f) - (YX)(f)](g) + f[(XY)(g) - (YX)(g)] \\
 &= [(XY - YX)(f)]g + f[(XY - YX)(g)] \\
 &= ([X, Y](f))g + f([X, Y](g)).
 \end{aligned}$$

Thus $[X, Y]$ is Leibnizian.

Hence $[X, Y] \in \mathfrak{X}(M)$. □

Exercise 12.3.0.5. Jacobi Identity:

Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $X, Y, Z \in \mathfrak{X}(M)$. Then

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

Proof. Let **FINISH!!!** □

12.4 Vector Fields and Smooth Maps

Given two smooth manifolds M, N and a smooth map $F : M \rightarrow N$, [a previous section explains](#) how at a specific point $p \in M$, the derivative $DF(p) : T_p M \rightarrow T_{F(p)} N$ allows us to push a vector v tangent to M at p forward to a vector $DF(p)(v)$ tangent to N at $F(p)$. In light of the machinery developed in [tangent bundle and vector field sections](#) thus far, it is natural to ask if the tangent bundle map $DF : TM \rightarrow TN$ would us to generalize this idea and push a vector field X on M forward to a vector field $F_* X$ on N . In general, this is not the case and the obstruction is captured in the following exercise.

Exercise 12.4.0.1. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$. Then for each $X \in \mathfrak{X}(M)$, $\pi_{TN} \circ DF \circ X = F$.

Proof. Let $X \in \mathfrak{X}(M)$. Since

$$\begin{aligned} X &\in \mathfrak{X}(M) \\ &= \Gamma(\pi_{TM}), \end{aligned}$$

we have that $\pi_{TM} \circ X = \text{id}_M$. [A prev ex in tangent bundle intro section implies that](#) $\pi_{TN} \circ DF = F \circ \pi_{TM}$. Therefore

$$\begin{aligned} \pi_{TN} \circ (DF \circ X) &= (\pi_{TN} \circ DF) \circ X \\ &= (F \circ \pi_{TM}) \circ X \\ &= F \circ (\pi_{TM} \circ X) \\ &= F \circ \text{id}_M \\ &= F. \end{aligned}$$

□

The previous exercise exposes a sufficient assumption on F that would allow us to push forward vector fields on M to vector fields on N , namely that F be a diffeomorphism.

Exercise 12.4.0.2. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $F \in \text{Iso}_{\mathbf{ManBnd}^\infty}(M, N)$. Then for each $X \in \mathfrak{X}(M)$, $DF \circ X \circ F^{-1} \in \mathfrak{X}(N)$.

Proof. Let $X \in \mathfrak{X}(M)$. Define $Y : N \rightarrow TN$ by $Y := DF \circ X \circ F^{-1}$. Since DF , F and F^{-1} are smooth, we have that Y is smooth. [The previous exercise](#) implies that

$$\begin{aligned} \pi_{TN} \circ Y &= \pi_{TN} \circ (DF \circ X \circ F^{-1}) \\ &= (\pi_{TN} \circ DF \circ X) \circ F^{-1} \\ &= F \circ F^{-1} \\ &= \text{id}_N. \end{aligned}$$

Hence

$$\begin{aligned} DF \circ X \circ F^{-1} &= Y \\ &\in \mathfrak{X}(Y). \end{aligned}$$

Since $X \in \mathfrak{X}(M)$ is arbitrary, we have that for each $X \in \mathfrak{X}(M)$, $DF \circ X \circ F^{-1} \in \mathfrak{X}(N)$. □

Definition 12.4.0.3. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $F \in \text{Iso}_{\mathbf{ManBnd}^\infty}(M, N)$, $f \in C^\infty(M)$ and $X \in \mathfrak{X}(M)$. We define the

- **pushforward of f by F** , denoted $F_* f \in C^\infty(N)$, by $F_* f := f \circ F^{-1}$,
- **pushforward of X by F** , denoted $F_* X \in \mathfrak{X}(N)$, by $F_* X := DF \circ X \circ F^{-1}$.

Exercise 12.4.0.4. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $F \in \text{Iso}_{\mathbf{ManBnd}^\infty}(M, N)$. Then for each $X, Y \in \mathfrak{X}(M)$ and $f \in C^\infty(M)$, $F_*(X + fY) = F_* X + (F_* f)F_* Y$.

Proof. Let $X, Y \in \mathfrak{X}(M)$, $\lambda \in \mathbb{R}$ and $q \in N$. Define $p \in M$ by $p := F^{-1}(q)$. Since $DF|_{\{p\} \times T_p M} = \text{id}_{\{p\}} \times DF(p)$, and $DF(p) : T_p M \rightarrow T_q N$ is $\mathbb{R}$ -linear, we have that $DF|_{\{p\} \times T_p M}$ is $\mathbb{R}$ -linear and

$$\begin{aligned}
[F_*(X + fY)]_q &= [DF \circ (X + fY) \circ F^{-1}](q) \\
&= [DF \circ (X + fY)](p) \\
&= DF([X + fY]_p) \\
&= DF(p, v^{[X+fY]_p}) \\
&= DF(p)(v^{X_p} + f(p)Y_p) \\
&= DF(p)(v^{X_p} + f(p)v^{Y_p}) \\
&= DF(p)(v^{X_p}) + f(p)DF(p)(v^{Y_p}) \\
&= DF(p, v^{X_p}) + f(p)DF(p, v^{Y_p}) \\
&= DF(X_p) + f(p)DF(Y_p) \\
&= [DF \circ X](p) + f(p)[DF \circ Y](p) \\
&= [DF \circ X \circ F^{-1}](q) + f \circ F^{-1}(q)[DF \circ Y \circ F^{-1}](q) \\
&= [F_*X + (F_*f)F_*Y]_q.
\end{aligned}$$

Since $q \in N$ is arbitrary, we have that $F_*(X + fY) = F_*X + (F_*f)F_*Y$. □

Let X be a vector field on M and Y a vector field on N . If $F : M \rightarrow N$ is a diffeomorphism, then by the definition of F_*X ([ref prev def](#)), asking whether $Y = F_*X$ is equivalent to asking whether $Y \circ F = DF \circ X$.

If however F is not a diffeomorphism, this second characterization provides a relation between X and Y that still captures the idea of Y being a pushforward of X by F “in spirit”. We formalize this idea in the following definition.

Definition 12.4.0.5. [move to vector fields section](#)

Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$, $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$. Then Y is said to be **F -related to X** if $Y \circ F = DF \circ X$.

Exercise 12.4.0.6. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $F \in \text{Iso}_{\mathbf{ManBnd}^\infty}(M, N)$. Then for each $X \in \mathfrak{X}(M)$, there exists a unique $Y \in \mathfrak{X}(N)$ such that X is F -related to Y .

Proof.

• **Existence:**

Define $Y \in \mathfrak{X}(N)$ by $Y := F_*X$. Then

$$\begin{aligned}
Y \circ F &= (DF \circ X \circ F^{-1}) \circ F \\
&= (DF \circ X) \circ (F^{-1} \circ F) \\
&= (DF \circ X) \circ \text{id}_M \\
&= DF \circ X.
\end{aligned}$$

Hence Y is F -related to X .

• **Uniqueness:**

Let $Y' \in \mathfrak{X}(N)$. Suppose that Y' is F -related to X . Then

$$\begin{aligned}
Y' &= Y' \circ \text{id}_N \\
&= Y' \circ (F \circ F^{-1}) \\
&= (Y' \circ F) \circ F^{-1} \\
&= (DF \circ X) \circ F^{-1} \\
&= F_*X \\
&= Y.
\end{aligned}$$

Since $Y' \in \mathfrak{X}(N)$ such that Y is F -related to X is arbitrary, we have that for each $Y' \in \mathfrak{X}(N)$, Y is F -related to X implies that $Y' = Y$.

□

If F is a diffeomorphism, then it is straightforward to show that for each $f \in C^\infty(N)$, $[F_*X]f = X(f \circ F) \circ F^{-1}$. Thus when identifying $\mathfrak{X}(N)$ with $\text{Deriv}^\infty(N)$, asking whether $Y = F_*X$ is equivalent to asking whether for each $f \in C^\infty(N)$, $Yf \circ F = X(f \circ F)$.

If however F is not a diffeomorphism, this second characterization provides a derivation-centered formulation of F -relatedness. We formalize this idea in the following exercise.

Exercise 12.4.0.7. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$, $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$. Then Y is F -related to X iff for each $f \in C^\infty(N)$, $(Yf) \circ F = X(f \circ F)$.

Proof. We note that for each $f \in C^\infty(N)$ and $p \in M$,

$$\begin{aligned} [(Yf) \circ F](p) &= (Yf)(F(p)) \\ &= Y_{F(p)}f \\ &= [Y \circ F](p)(f) \end{aligned}$$

and

$$\begin{aligned} [X(f \circ F)](p) &= X_p(f \circ F) \\ &= [DF(X_p)](f) \\ &= [DF \circ X](p)(f). \end{aligned}$$

• ($\implies$):

Suppose that Y is F -related to X . Let $f \in C^\infty(N)$. Then for each $p \in M$,

$$\begin{aligned} [(Yf) \circ F](p) &= [Y \circ F](p)(f) \\ &= [DF \circ X](p)(f) \\ &= [X(f \circ F)](p). \end{aligned}$$

Thus $(Yf) \circ F = X(f \circ F)$. Since $f \in C^\infty(N)$ is arbitrary, we have that for each $f \in C^\infty(N)$, $(Yf) \circ F = X(f \circ F)$.

• ($\impliedby$):

Suppose that for each $f \in C^\infty(N)$, $(Yf) \circ F = X(f \circ F)$. Let $p \in M$. Then for each $f \in C^\infty(N)$,

$$\begin{aligned} [Y \circ F](p)(f) &= [(Yf) \circ F](p) \\ &= [X(f \circ F)](p) \\ &= [DF \circ X](p)(f). \end{aligned}$$

Thus $[Y \circ F](p) = [DF \circ X](p)$. Since $p \in M$ is arbitrary, we have for each $p \in M$, $[Y \circ F](p) = [DF \circ X](p)$. Therefore $Y \circ F = DF \circ X$. Hence Y is F -related to X .

□

Exercise 12.4.0.8. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$, $X_1, X_2 \in \mathfrak{X}(M)$ and $Y_1, Y_2 \in \mathfrak{X}(N)$. If Y_1 is F -related to X_1 and Y_2 is F -related to X_2 , then $[Y_1, Y_2]$ is F -related to $[X_1, X_2]$.

Proof. Suppose that Y_1 is F -related to X_1 and Y_2 is F -related to X_2 . Let $f \in C^\infty(N)$. Define $g_1, g_2 \in C^\infty(N)$ by $g_1 := Y_1f$ and $g_2 := Y_2f$. Then

$$\begin{aligned} ([Y_1, Y_2](f)) \circ F &= [(Y_1Y_2 - Y_2Y_1)(f)] \circ F \\ &= [Y_1Y_2(f) - Y_2Y_1(f)] \circ F \\ &= [Y_1Y_2(f)] \circ F - [Y_2Y_1(f)] \circ F \\ &= (Y_1g_2) \circ F - (Y_2g_1) \circ F \\ &= X_1(g_2 \circ F) - X_2(g_1 \circ F) \\ &= X_1([Y_2f] \circ F) - X_2([Y_1f] \circ F) \\ &= X_1(X_2(f \circ F)) - X_2(X_1(f \circ F)) \\ &= X_1X_2(f \circ F) - X_2X_1(f \circ F) \\ &= (X_1X_2 - X_2X_1)(f \circ F) \\ &= [X_1, X_2](f \circ F). \end{aligned}$$

Since $f \in C^\infty(N)$ is arbitrary, we have that for each $f \in C^\infty(N)$, $([Y_1, Y_2](f)) \circ F = [X_1, X_2](f \circ F)$. The **prev ex** implies that $[Y_1, Y_2]$ is F -related to $[X_1, X_2]$. $\square$

Definition 12.4.0.9. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $\Gamma \subset \text{Aut}_{\mathbf{ManBnd}^\infty}(M)$ and $X \in \mathfrak{X}(M)$.

1. Then X is said to be Γ -**invariant** if for each $\phi \in \Gamma$, X is ϕ -related to X .

2. We define $\mathfrak{X}^\Gamma(M) \subset \mathfrak{X}(M)$ by $\mathfrak{X}^\Gamma(M) := \{X \in \mathfrak{X}(M) : X \text{ is } \Gamma\text{-invariant}\}$.

Note 12.4.0.10. When $\Gamma = \{\phi\}$, we may say “ ϕ -invariant” instead of “ $\{\phi\}$ -invariant”.

Exercise 12.4.0.11. Let $M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $\Gamma \subset \text{Aut}_{\mathbf{ManBnd}^\infty}(M)$. Then $\mathfrak{X}^\Gamma(M)$ is a vector subspace of $\mathfrak{X}(M)$

Proof. Let $X, Y \in \mathfrak{X}^\Gamma(M)$, $\lambda \in \mathbb{R}$. Then for each $\phi \in \Gamma$,

$$\begin{aligned} \phi_*(X + \lambda Y) &= \phi_*X + \lambda \phi_*Y \quad (\text{by prev ex}) \\ &= X + \lambda Y \quad (\text{since } X, Y \in \mathfrak{X}^\Gamma(M)). \end{aligned}$$

Hence $X + \lambda Y \in \mathfrak{X}^\Gamma(M)$. Since $X, Y \in \mathfrak{X}^\Gamma(M)$ and $\lambda \in \mathbb{R}$ are arbitrary, we have that for each $X, Y \in \mathfrak{X}^\Gamma(M)$ and $\lambda \in \mathbb{R}$, $X + \lambda Y \in \mathfrak{X}^\Gamma(M)$. Thus $\mathfrak{X}^\Gamma(M)$ is a vector subspace of $\mathfrak{X}(M)$. $\square$

12.5 1-Forms

Definition 12.5.0.1. Let $\omega : M \rightarrow T^*M$. Then ω is said to be a **1-form on M** if for each $p \in M$, $\omega_p \in T_p^*M$. For each $X \in \mathfrak{X}(M)(M)$, we define $\omega(X) : M \rightarrow \mathbb{R}$ by

$$\omega(X)_p = \omega_p(X_p)$$

and ω is said to be **smooth** if for each $X \in \mathfrak{X}(M)(M)$, $\omega(X)$ is smooth. The set of smooth 1-forms on M is denoted $\Gamma_1(M)$.

Definition 12.5.0.2. Let $f \in C^\infty(M)$ and $\alpha, \beta \in \mathfrak{X}(M)(M)$. We define

- $f\alpha \in \Gamma_1(M)$ by

$$(f\omega)_p = f(p)\omega_p$$

- $\alpha + \beta \in \mathfrak{X}(M)(M)$ by

$$(\alpha + \beta)_p = \alpha_p + \beta_p$$

Exercise 12.5.0.3. The set $\Gamma_1(M)$ is a $C^\infty(M)$ -module.

Proof. Clear. □

Part III

Lie Theory

Chapter 13

Lie Groups

13.1 Introduction

Definition 13.1.0.1. change/move this somewhere?, this def works for any set

Let $G \in \mathbf{Man}^\infty \cap \mathbf{Grp}$. For each $g \in G$, we define $\iota_g^l : G \rightarrow G \times G$ and $\iota_g^r : G \rightarrow G \times G$ by $\iota_g^l(x) = (g, x)$ and $\iota_g^r(x) = (x, g)$ respectively.

Note 13.1.0.2. If $G \in \mathbf{Man}^\infty$, Exercise 5.3.0.11 implies that for each $g \in G$, $\iota_g^l, \iota_h^r \in C^\infty(G, G \times G)$.

Definition 13.1.0.3. Let $(G, \text{mult}) \in \mathbf{Grp}$. We define the **inversion map**, denoted $\text{inv} : G \rightarrow G$, by $\text{inv}(g) := g^{-1}$.

Note 13.1.0.4. When the context is clear, we write gh in place of $\text{mult}(g, h)$.

Definition 13.1.0.5. Let $G \in \mathbf{Man}^\infty$ and $\text{mult} : G \times G \rightarrow G$. Then (G, mult) is said to be a **Lie group** if

1. $(G, \text{mult}) \in \mathbf{Grp}$,
2. $\text{mult} \in C^\infty(G \times G, G)$,
3. $\text{inv} \in C^\infty(G, G)$.

Note 13.1.0.6. When the context is clear, we write G in place of (G, mult) .

Definition 13.1.0.7. Let G be a Lie group and $g \in G$. We define the **left and right translation maps**, denoted $l_g : G \rightarrow G$ and $r_g : G \rightarrow G$ respectively, by $l_g(x) = gx$ and $r_g(x) = xg^{-1}$.

Exercise 13.1.0.8. Let G be a Lie group. Then for each $g \in G$,

1. $l_g^{-1} = l_{g^{-1}}$ and $r_g^{-1} = r_{g^{-1}}$,
2. $l_g, r_g \in C^\infty(G, G)$,
3. $l_g, r_g \in \text{Aut}_{\mathbf{Man}^\infty}(G)$.

Proof. Let $g \in G$.

1. Clear
2. Since G is a Lie group, mult is smooth. Since $l_g = \text{mult} \circ \iota_g^l$ and $r_g = \text{mult} \circ \iota_{g^{-1}}^r$, we have that l_g and r_g are smooth.
3. Since $l_g \in C^\infty(G, G)$ and

$$\begin{aligned} l_g^{-1} &= l_{g^{-1}} \\ &\in C^\infty(G, G), \end{aligned}$$

we have that $l_g \in \text{Aut}_{\mathbf{Man}^\infty}(G)$. Similarly, $r_g \in \text{Aut}_{\mathbf{Man}^\infty}(G)$.

□

Exercise 13.1.0.9. Let $G \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that G is a Lie Group. Then $\partial G = \emptyset$.

Proof. Let $g \in G$. Since $\mathcal{A}_G$ is a smooth atlas, there exists $(U_0, \phi_0) \in \mathcal{A}_G$ such that $e \in U_0$. There exists $x \in U_0$ such that $x \in \text{Int } G$ (add details). Set $U := U_0 \cap \text{Int } G$. Since $U_0, \text{Int } G \in \mathcal{T}_G$, $x \in U_0$ and $x \in \text{Int } G$, we have that $U \in \mathcal{T}_G$ and $x \in U$. Set $\phi := \phi_0|_U$. Exercise ?? (exercise in section on open submanifolds) implies that $(U, \phi) \in \mathcal{A}_G$. Since $l_{gx^{-1}}$ is a diffeomorphism, $l_{gx^{-1}}$ is a homeomorphism. Hence

$$\begin{aligned} g &= l_{gx^{-1}}(x) \\ &\in l_{gx^{-1}}(U) \\ &\subset \text{Int } G \end{aligned}$$

Since $g \in G$ is arbitrary, we have that for each $g \in G$, $g \in \text{Int } G$. Thus $\text{Int } G = G$ and Exercise ?? (ref ex from intro to topological manifolds) implies that

$$\begin{aligned} \partial G &= (\text{Int } G)^c \\ &= \emptyset. \end{aligned}$$

□

Exercise 13.1.0.10. Let $G \in \text{Obj}(\mathbf{Man}^\infty)$. Suppose that G is a group. Define $f : G \times G \rightarrow G$ by $f(g, h) = gh^{-1}$. Then G is a Lie group iff $f \in C^\infty(G \times G, G)$.

Proof.

- ($\implies$):
Suppose that G is a Lie group. Then mult is smooth and inv is smooth. Thus $\text{id}_G \times \text{inv}$ is smooth. Since $f = \text{mult} \circ (\text{id}_G \times \text{inv})$, we have that f is smooth.
- ($\impliedby$):
Suppose that f is smooth. Since $\text{inv} = f \circ \iota_e^l$, inv is smooth. Therefore $\text{id}_G \times \text{inv}$ is smooth and since $\text{mult} = f \circ (\text{id}_G \times \text{inv})$, mult is smooth. Since mult and inv are smooth, G is a Lie group.

□

Exercise 13.1.0.11. Let $G, H \in \mathbf{Man}^\infty$ and $\phi : G \rightarrow H$. Suppose that G, H are Lie groups. Then ϕ is said to be a **Lie group homomorphism** if $\phi \in \text{Hom}_{\mathbf{Man}^\infty}(G, H) \cap \text{Hom}_{\mathbf{Grp}}(G, H)$.

Definition 13.1.0.12. We define the category of Lie groups, denoted **LieGrp**, by

- $\text{Obj}(\mathbf{LieGrp}) = \{G : G \text{ is a Lie group}\}$
- For $G_1, G_2 \in \text{Obj}(\mathbf{LieGrp})$,

$$\text{Hom}_{\mathbf{LieGrp}}(G_1, G_2) = \text{Hom}_{\mathbf{Man}^\infty}(G_1, G_2) \cap \text{Hom}_{\mathbf{Grp}}(G_1, G_2)$$

- For
 - $G_1, G_2, G_3 \in \text{Obj}(\mathbf{LieGrp})$
 - $\phi_{12} \in \text{Hom}_{\mathbf{LieGrp}}(G_1, G_2)$
 - $\phi_{23} \in \text{Hom}_{\mathbf{LieGrp}}(G_2, G_3)$

we define $\phi_{23} \circ_{\mathbf{LieGrp}} \phi_{12} \in \text{Hom}_{\mathbf{LieGrp}}(G_1, G_3)$ by

$$\phi_{23} \circ_{\mathbf{LieGrp}} \phi_{12} = \phi_{23} \circ_{\mathbf{Set}} \phi_{12}$$

Exercise 13.1.0.13. We have that **LieGrp** is a subcategory of **Grp** and $\mathbf{Man}^\infty$.

Proof. FINISH!!!

□

Exercise 13.1.0.14. Let $G, H \in \text{Obj}(\mathbf{LieGrp})$ and $\phi \in \text{Hom}_{\mathbf{LieGrp}}(G, H)$. Then ϕ has constant rank.

Proof. Let $g \in G$. Since ϕ is a homomorphism, we have that for each $x \in G$, $\phi(gx) = \phi(g)\phi(x)$. Thus $\phi \circ l_g = l_{\phi(g)} \circ \phi$, i.e. the following diagram commutes:

$$\begin{array}{ccc} G & \xrightarrow{\phi} & H \\ l_g \downarrow & & \downarrow l_{\phi(g)} \\ G & \xrightarrow{\phi} & H \end{array}$$

Let $x \in G$. Then

$$\begin{aligned} D\phi(gx) \circ Dl_g(x) &= D(\phi \circ l_g)(x) \\ &= D(l_{\phi(g)} \circ \phi) \\ &= Dl_{\phi(g)}(\phi(x)) \circ D\phi(x) \end{aligned}$$

Since $l_g \in \text{Aut}_{\mathbf{Man}^\infty}(G)$, $l_{\phi(g)} \in \text{Aut}_{\mathbf{Man}^\infty}(H)$, we have that $Dl_g(x) \in \text{Iso}_{\mathbf{Vect}_{\mathbb{R}}}(T_x G, T_{gx} G)$ and $Dl_{\phi(g)}(\phi(x)) \in \text{Iso}_{\mathbf{Vect}_{\mathbb{R}}}(T_{\phi(x)} H, T_{\phi(g)\phi(x)} H)$. Hence

$$\begin{aligned} \text{rank } D\phi(gx) &= \text{rank } D\phi(gx) \circ Dl_g(x) \\ &= \text{rank } Dl_{\phi(g)}(\phi(x)) \circ D\phi(x) \\ &= \text{rank } D\phi(x) \end{aligned}$$

Since $x \in G$ is arbitrary, for each $x \in G$, $\text{rank } D\phi(gx) = \text{rank } D\phi(x)$. In particular, $\text{rank } D\phi(g) = \text{rank } D\phi(e)$. Since $g \in G$ is arbitrary, for each $g \in G$, $\text{rank } D\phi(g) = \text{rank } D\phi(e)$ and ϕ has constant rank. $\square$

Exercise 13.1.0.15. Let $G, H \in \text{Obj}(\mathbf{LieGrp})$ and $\phi \in \text{Hom}_{\mathbf{LieGrp}}(G, H)$. Then $\phi \in \text{Iso}_{\mathbf{LieGrp}}(G, H)$ iff ϕ is a bijection.

Proof. global rank theorem **FINISH!!!** $\square$

Definition 13.1.0.16. Let $G, H \in \text{Obj}(\mathbf{LieGrp})$ and $\phi \in \text{Hom}_{\mathbf{LieGrp}}(G, H)$. Then ϕ is said to be a

- **LieGrp-immersion** if ϕ is a $\mathbf{Man}^\infty$ -immersion
- **LieGrp-embedding** if ϕ is a $\mathbf{Man}^\infty$ -embedding

Exercise 13.1.0.17. Let $G, H \in \text{Obj}(\mathbf{LieGrp})$ and $\phi \in \text{Hom}_{\mathbf{LieGrp}}(G, H)$. Suppose that ϕ is a **LieGrp-immersion**. If G is compact, then ϕ is a **LieGrp-embedding**.

13.2 Lie Subgroups

Definition 13.2.0.1. Let $G, H \in \text{Obj}(\mathbf{LieGrp})$. Suppose that $G \leq H$. Then H is said to be an

- **immersed Lie subgroup** of G if G is an immersed submanifold of H ,
- **embedded Lie subgroup** of G if G is an embedded submanifold of H .

Definition 13.2.0.2. content...

Exercise 13.2.0.3. Let $G, H \in \text{Obj}(\mathbf{LieGrp})$. Suppose that $G \leq H$.

13.3 Product Lie Groups

Definition 13.3.0.1. Let $G, H \in \text{Obj}(\mathbf{LieGrp})$. Suppose that $G \subset H$. Then G is said to be a **Lie subgroup** of H if

1. $G \leq H$
2. G is an immersed submanifold of H . **FIX!!!**

13.4 Representations of Lie Groups

Chapter 14

Lie Algebras

14.1 Introduction

Definition 14.1.0.1. Let $V \in \text{Obj}(\mathbf{Vect}_{\mathbb{K}})$ and $[\cdot, \cdot] : V \times V \rightarrow V$. Then

- $[\cdot, \cdot]$ is said to be a **$\mathbb{K}$ -Lie bracket on V** if
 1. **(bilinearity):** for each $x, y, z \in V$ and $\lambda \in \mathbb{K}$, $[x + \lambda y, z] = [x, z] + \lambda[y, z]$
 2. **(antisymmetry):** for each $x, y \in V$, $[x, y] = -[y, x]$
 3. **(Jacobi identity):** for each $x, y, z \in V$, $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$
- $(V, [\cdot, \cdot])$ is said to be a **$\mathbb{K}$ -Lie Algebra** if $[\cdot, \cdot]$ is a $\mathbb{K}$ -Lie bracket on V .

Note 14.1.0.2. When the context is clear, we may say

- “Lie bracket” instead of “ $\mathbb{K}$ -Lie bracket”,
- “Lie algebra” instead of “ $\mathbb{K}$ -Lie algebra”.

Exercise 14.1.0.3. Let $M \in \text{Obj}(\mathbf{Man}^{\infty})$. Then $(\mathfrak{X}(M), [\cdot, \cdot])$ is an $\mathbb{R}$ -Lie Algebra.

Proof. Clear by ?? (make exercise in section on vector fields about $[\cdot, \cdot]$). □

14.2 Lie Subalgebras

Definition 14.2.0.1. Let $(V, [\cdot, \cdot]_V), (W, [\cdot, \cdot]_W)$ be $\mathbb{K}$ -Lie algebras. Then $(W, [\cdot, \cdot]_W)$ is said to be a **Lie subalgebra of $(V, [\cdot, \cdot]_V)$** if

1. $W \subset V$
2. $[\cdot, \cdot]_W = [\cdot, \cdot]_V|_{W \times W}$.

Exercise 14.2.0.2. Let $(V, [\cdot, \cdot])$ be a Lie algebra and $W \subset V$. If

- (a) W is a vector subspace of V ,
- (b) for each $x, y \in W$, $[x, y] \in W$,

then $(W, [\cdot, \cdot]|_{W \times W})$ is a Lie subalgebra of $(V, [\cdot, \cdot])$.

Proof. Suppose that W is a vector subspace of V and for each $x, y \in W$, $[x, y] \in W$. Define $[\cdot, \cdot]_W : W \times W \rightarrow W$ by $[\cdot, \cdot]_W := [\cdot, \cdot]|_{W \times W}$.

- 1. **(bilinearity):** Let $x, y, z \in W$ and $\lambda \in \mathbb{K}$. (a) implies that $\lambda y \in W$. Therefore

$$\begin{aligned} [x + \lambda y, z]_W &= [x + \lambda y, z] \\ &= [x, z] + \lambda[y, z] \quad (\text{since } [\cdot, \cdot] \text{ is a Lie bracket on } V) \\ &= [x, z]_W + \lambda[y, z]_W \quad (\text{by definition.}) \end{aligned}$$

Since $x, y, z \in W$ and $\lambda \in \mathbb{K}$ are arbitrary, we have that for each $x, y, z \in W$ and $\lambda \in \mathbb{K}$, $[x + \lambda y, z]_W = [x, z]_W + \lambda[y, z]_W$.

2. **(antisymmetry):** For each $x, y \in W$,

$$\begin{aligned} [x, y]_W &= [x, y] \\ &= -[y, x] \quad (\text{since } [\cdot, \cdot] \text{ is a Lie bracket on } V) \\ &= -[y, x]_W \end{aligned}$$

3. **(Jacobi identity):** For each $x, y, z \in W$,

$$\begin{aligned} [x, [y, z]_W]_W + [y, [z, x]_W]_W + [z, [x, y]_W]_W \\ &= [x, [y, z]] + [y, [z, x]] + [z, [x, y]] \\ &= 0 \quad (\text{since } [\cdot, \cdot] \text{ is a Lie bracket on } V). \end{aligned}$$

Thus $[\cdot, \cdot]_W$ is a Lie bracket on W .

- Since W is a vector space and $[\cdot, \cdot]_W$ is a Lie bracket on W , $(W, [\cdot, \cdot]_W)$ is a Lie algebra.

1. By assumption, $W \subset V$.

2. By definition, $[\cdot, \cdot]_W = [\cdot, \cdot]|_{W \times W}^W$.

Thus $(W, [\cdot, \cdot]_W)$ is a Lie subalgebra of $(V, [\cdot, \cdot])$. □

Exercise 14.2.0.3. exercise about intersection of two lie subalgebras is a lie subalgebra

Proof. FINISH!!! □

Exercise 14.2.0.4. Let $M \in \text{Man}^\infty$ and $\Gamma \subset \text{Aut}_{\text{Man}^\infty}(M)$. Then $\mathfrak{X}^\Gamma(M)$ is a Lie subalgebra of $\mathfrak{X}(M)$.

Proof.

1. A prev ex implies that $\mathfrak{X}^\Gamma(M)$ is a vector subspace of $\mathfrak{X}(M)$.
2. Let $X, Y \in \mathfrak{X}^\Gamma(M)$. Then

$$\begin{aligned} \phi_*[X, Y] &= [\phi_*X, \phi_*Y] \quad (\text{by ex in section on vectorfields and smooth maps}) \\ &= [X, Y] \quad (\text{since } X, Y \in \mathfrak{X}^\Gamma(M)). \end{aligned}$$

Hence $[X, Y] \in \mathfrak{X}^\Gamma(M)$. Since $X, Y \in \mathfrak{X}^\Gamma(M)$ are arbitrary, we have that for each $X, Y \in \mathfrak{X}^\Gamma(M)$, $[X, Y] \in \mathfrak{X}^\Gamma(M)$. Thus $\mathfrak{X}^\Gamma(M)$ is a Lie subalgebra of $\mathfrak{X}(M)$. □

14.3 Lie Algebra of a Lie Group

Definition 14.3.0.1. Let G be a Lie group. We define the **Lie algebra of G** , denoted $\text{Lie}(G) \subset \mathfrak{X}(G)$, by $\text{Lie}(G) := \mathfrak{X}^{(L_g)_{g \in G}}(G)$.

Exercise 14.3.0.2. Let G be a Lie group. Then $\text{eval}_e \in \text{Iso}_{\text{Vect}_{\mathbb{R}}}(\text{Lie}(G), T_e G)$.

Proof. • **injectivity:**

Let $X \in \ker \text{eval}_e$. Then

$$\begin{aligned} X_e &= \text{eval}_e(X) \\ &= 0. \end{aligned}$$

Since $X \in \text{Lie}(G)$, we have that for each $g \in G$,

$$\begin{aligned}
 X_g &= X_{L_g(e)} \\
 &= [X \circ L_g](e) &= [DL_g \circ X](e) \quad (\text{since } (L_g)_*X = X) \\
 &= DL_g(X_e) \\
 &= D(L_g)(e)(0) \\
 &= 0.
 \end{aligned}$$

Hence $X = 0$. Since $X \in \ker \text{eval}_e$ is arbitrary, we have that for each $X \in \ker \text{eval}_e$, $X = 0$. Thus $\ker \text{eval}_e = \{0\}$ and eval_e is injective.

• **surjectivity:**

Let

□

Part IV

Bundles

Chapter 15

Fiber Bundles

15.1 Introduction

15.1.1 Local Trivializations

Note 15.1.1.1. Let M, F be sets, we write $\text{proj}_1 : M \times F \rightarrow M$ to denote the projection onto M .

Definition 15.1.1.2. Let $E, M \in \text{Obj}(\mathbf{Set})$, $\pi \in \text{Epi}_{\mathbf{Set}}(E, M)$ and $A \subset M$. We define the **fibers of π over A** , denoted $E_A^\pi \subset E$ by $E_A := \pi^{-1}(A)$.

Note 15.1.1.3. When the context is clear, we may write E_A in place of E_A^π .

Definition 15.1.1.4. Let $E, M, F \in \text{Obj}(\mathbf{Set})$, $\pi \in \text{Hom}_{\mathbf{Set}}(E, M)$ a surjection, $U \subset M$ and $\Phi : E_U \rightarrow U \times F$.

- Then (U, Φ) is said to be a **local trivialization of π over U with fiber F** if
 1. Φ is a bijection
 2. $\text{proj}_1 \circ \Phi = \pi|_{E_U}$, i.e. the following diagram commutes:

$$\begin{array}{ccc} E_U & \xrightarrow{\Phi} & U \times F \\ & \searrow \pi & \downarrow \text{proj}_1 \\ & & U \end{array}$$

- We define $\text{LocTriv}(\pi, F) := \{(U, \Phi) : U \subset M, \Phi \in (U \times F)^{E_U} \text{ and } (U, \Phi) \text{ is a local trivialization of } \pi \text{ over } U \text{ with fiber } F\}$.

Exercise 15.1.1.5. Let $E, M, F \in \text{Obj}(\mathbf{Set})$ and $\pi \in \text{Hom}_{\mathbf{Set}}(E, M)$ a surjection, $U \subset M$ and $\Phi : \pi^{-1}(U) \rightarrow U \times F$ a local trivialization with respect to π of E over U with fiber F . Then for each $A \subset U$,

$$\Phi(E_A) = A \times F.$$

Hint: consider $\Phi^{-1}(A \times F)$

Proof. Let $A \subset U$. Since $\text{proj}_1^{-1}(A) = A \times F$, we have that

$$\begin{aligned} \Phi^{-1}(A \times F) &= \Phi^{-1}(\text{proj}_1^{-1}(A)) \\ &= (\text{proj}_1 \circ \Phi)^{-1}(A) \\ &= (\pi|_{\pi^{-1}(U)})^{-1}(A) \\ &= \pi^{-1}(A) \cap \pi^{-1}(U) \\ &= \pi^{-1}(A \cap U) \\ &= \pi^{-1}(A) \end{aligned}$$

Since Φ is a bijection, we have that

$$\begin{aligned}\Phi(\pi^{-1}(A)) &= \Phi \circ \Phi^{-1}(A \times F) \\ &= A \times F\end{aligned}$$

□

15.1.2 $\mathbf{Man}^0$ Fiber Bundles

Definition 15.1.2.1. Let $E, M, F \in \text{Obj}(\mathbf{Man}^0)$ and $\pi \in \text{Epi}_{\mathbf{Man}^0}(E, M)$, $U \subset M$ and $\Phi : E_U \rightarrow U \times F$.

- Then (U, Φ) is said to be a **$\mathbf{Man}^0$ -local trivialization of π over U with fiber F** if
 1. $U \in \mathcal{T}_M$,
 2. $(U, \Phi) \in \text{LocTriv}(\pi, F)$,
 3. $\Phi \in \text{Iso}_{\mathbf{Top}}(E_U, U \times F)$ is a homeomorphism
- We define $\text{LocTriv}^0(\pi, F) := \{(U, \Phi) \in \text{LocTriv}(\pi, F) : (U, \Phi) \text{ is a } \mathbf{Man}^0\text{-local trivialization of } \pi \text{ over } U \text{ with fiber } F\}$.

Definition 15.1.2.2. Let $E, M, F \in \text{Obj}(\mathbf{Man}^0)$ and $\pi \in \text{Epi}_{\mathbf{Man}^0}(E, M)$. Then (E, M, π, F) is said to be a **$\mathbf{Man}^0$ fiber bundle with total space E , base space M , fiber F and projection π** if for each $p \in M$, there exist $(U, \Phi) \in \text{LocTriv}^0(\pi, F)$ such that $U \in \mathcal{T}_M$ and $\Phi : \pi^{-1}(U) \rightarrow U \times F$ such that $p \in U$.

Definition 15.1.2.3. Define category **$\mathbf{Bun}^0$ FINISH!!!**

Exercise 15.1.2.4. $\mathbf{Man}^0$ Fiber Bundle Chart Lemma:

Let $E \in \text{Obj}(\mathbf{Set})$, $M, F \in \text{Obj}(\mathbf{Man}^0)$, $\pi \in \text{Epi}_{\mathbf{Set}}(E, M)$, Γ an index set and for each $\alpha \in \Gamma$, $(U_\alpha, \Phi_\alpha) \in \text{LocTriv}(\pi, F)$. Set $n := \dim M$ and $k := \dim F$. Suppose that

- for each $\alpha \in \Gamma$, $U_\alpha \in \mathcal{T}_M$
- $M \subset \bigcup_{\alpha \in \Gamma} U_\alpha$
- for each $\alpha, \beta \in \Gamma$, $\Phi_\beta|_{\pi^{-1}(U_\alpha \cap U_\beta)} \circ (\Phi_\alpha|_{\pi^{-1}(U_\alpha \cap U_\beta)})^{-1} : (U_\alpha \cap U_\beta) \times F \rightarrow (U_\alpha \cap U_\beta) \times F$ is continuous.

Then there exist a unique $\mathcal{T}_E \in \text{Top}(E)$ such that

1. $(E, \mathcal{T}_E) \in \text{Obj}(\mathbf{Man}^0)$ and $\dim E = n + k$,
2. $\pi \in \text{Epi}(E, M)$ and
3. for each $\alpha \in \Gamma$, $(U_\alpha, \Phi_\alpha) \in \text{LocTriv}^0(\pi, F)$
4. $(E, M, \pi, F) \in \text{Obj}(\mathbf{Bun}^0)$.

Proof.

1. For $\alpha \in \Gamma$, we define $X_\alpha^n(M, \mathcal{T}_M) \subset X^n(M, \mathcal{T}_M)$ by

$$X_\alpha^n(M, \mathcal{T}_M) = \{(V^M, \psi^M) \in X^n(M, \mathcal{T}_M) : V^M \subset U_\alpha\}$$

Choose index sets $(\Pi_\alpha^M)_{\alpha \in \Gamma}$ and Π^F such that for each $\alpha \in \Gamma$, $X_\alpha^n(M, \mathcal{T}_M) = (V_{\alpha, \mu}^M, \psi_{\alpha, \mu}^M)_{\mu \in \Pi_\alpha^M}$ and $X^k(F, \mathcal{T}_F) = (V_\nu^F, \psi_\nu^F)_{\nu \in \Pi^F}$. Set $\Pi^M = \coprod_{\alpha \in \Gamma} \Pi_\alpha^M$ and $\Pi^E = \Pi^M \times \Pi^F$. For $(\alpha, \mu, \nu) \in \Pi^E$, we define $V_{\alpha, \mu, \nu}^E \subset E$ and $\psi_{\alpha, \mu, \nu}^E : V_{\alpha, \mu, \nu}^E \rightarrow \psi_{\alpha, \mu}^M(V_{\alpha, \mu}^M) \times \psi_\nu^F(V_\nu^F)$ by

- $V_{\alpha, \mu, \nu}^E = \Phi_\alpha^{-1}(V_{\alpha, \mu}^M \times V_\nu^F)$
- $\psi_{\alpha, \mu, \nu}^E = (\psi_{\alpha, \mu}^M \times \psi_\nu^F) \circ \Phi_\alpha|_{V_{\alpha, \mu, \nu}^E}$

We have the following:

- For each $(\alpha, \mu, \nu) \in \Pi^E$, $\psi_{\alpha, \mu, \nu}^E(V_{\alpha, \mu, \nu}^E) = \psi_{\alpha, \mu}^M(V_{\alpha, \mu}^M) \times \psi_{\nu}^F(V_{\nu}^F)$ and thus $\psi_{\alpha, \mu, \nu}^E(V_{\alpha, \mu, \nu}^E) \in \mathcal{T}_{\mathbb{H}^{n+k}}$
- For each $(\alpha_1, \mu_1, \nu_1), (\alpha_2, \mu_2, \nu_2) \in \Pi^E$,

$$\begin{aligned}
\psi_{\alpha_1, \mu_1, \nu_1}^E(V_{\alpha_1, \mu_1, \nu_1}^E \cap V_{\alpha_2, \mu_2, \nu_2}^E) &= (\psi_{\alpha_1, \mu_1}^M \times \psi_{\nu_1}^F) \circ \Phi_{\alpha_1}|_{V_{\alpha_1, \mu_1, \nu_1}^E} (\Phi_{\alpha_1}^{-1}([V_{\alpha_1, \mu_1}^M \times V_{\nu_1}^F] \cap [V_{\alpha_2, \mu_2}^M \times V_{\nu_2}^F])) \\
&= (\psi_{\alpha_1, \mu_1}^M \times \psi_{\nu_1}^F)([V_{\alpha_1, \mu_1}^M \times V_{\nu_1}^F] \cap [V_{\alpha_2, \mu_2}^M \times V_{\nu_2}^F]) \\
&= (\psi_{\alpha_1, \mu_1}^M \times \psi_{\nu_1}^F)([V_{\alpha_1, \mu_1}^M \cap V_{\alpha_2, \mu_2}^M] \times [V_{\nu_1}^F \cap V_{\nu_2}^F]) \\
&= \psi_{\alpha_1, \mu_1}^M(V_{\alpha_1, \mu_1}^M \cap V_{\alpha_2, \mu_2}^M) \times \psi_{\nu_1}^F(V_{\nu_1}^F \cap V_{\nu_2}^F) \\
&\in \mathcal{T}_{\mathbb{H}^{n+k}}
\end{aligned}$$

- For each $(\alpha, \mu, \nu) \in \Pi^E$, $\psi_{\alpha, \mu, \nu}^E : V_{\alpha, \mu, \nu}^E \rightarrow \psi_{\alpha, \mu}^M(V_{\alpha, \mu}^M) \times \psi_{\nu}^F(V_{\nu}^F)$ is a bijection
- Let $(\alpha_1, \mu_1, \nu_1), (\alpha_2, \mu_2, \nu_2) \in \Pi^E$. For notational convenience, set $\psi_1^E = \psi_{\alpha_1, \mu_1, \nu_1}^E$, $\psi_2^E = \psi_{\alpha_2, \mu_2, \nu_2}^E$, $V^E = V_{\alpha_1, \mu_1, \nu_1}^E \cap V_{\alpha_2, \mu_2, \nu_2}^E$, $V^M = V_{\alpha_1, \mu_1}^M \cap V_{\alpha_2, \mu_2}^M$ and $V^F = V_{\nu_1}^F \cap V_{\nu_2}^F$. Then $\psi_2|_{V^E} \circ (\psi_1|_{V^E})^{-1} : \psi_1(V^E) \rightarrow \psi_2(V^E)$ is given by

$$\begin{aligned}
\psi_2^E|_{V^E} \circ (\psi_1^E|_{V^E})^{-1} &= [(\psi_{\alpha_2, \mu_2}^M|_{V^M} \times \psi_{\nu_2}^F|_{V^F}) \circ \Phi_{\alpha_2}|_{V^E}] \circ [(\psi_{\alpha_1, \mu_1}^M|_{V^M} \times \psi_{\nu_1}^F|_{V^F}) \circ \Phi_{\alpha_1}|_{V^E}]^{-1} \\
&= [(\psi_{\alpha_2, \mu_2}^M|_{V^M} \times \psi_{\nu_2}^F|_{V^F}) \circ \Phi_{\alpha_2}|_{V^E}] \circ [(\Phi_{\alpha_1}|_{V^E})^{-1} \circ (\psi_{\alpha_1, \mu_1}^M|_{V^M} \times \psi_{\nu_1}^F|_{V^F})^{-1}] \\
&= (\psi_{\alpha_2, \mu_2}^M|_{V^M} \times \psi_{\nu_2}^F|_{V^F}) \circ [\Phi_{\alpha_2}|_{V^E} \circ (\Phi_{\alpha_1}|_{V^E})^{-1}] \circ (\psi_{\alpha_1, \mu_1}^M|_{V^M} \times \psi_{\nu_1}^F|_{V^F})^{-1}
\end{aligned}$$

Since $\Phi_{\alpha_2}|_{V^E} \circ (\Phi_{\alpha_1}|_{V^E})^{-1}$ is continuous, we have that $\psi_{\alpha_2, \mu_2, \nu_2}^E|_{V_{\alpha_1, \mu_1, \nu_1}^E \cap V_{\alpha_2, \mu_2, \nu_2}^E} \circ (\psi_{\alpha_1, \mu_1, \nu_1}^E|_{V_{\alpha_1, \mu_1, \nu_1}^E \cap V_{\alpha_2, \mu_2, \nu_2}^E})^{-1} : \psi_{\alpha_1, \mu_1, \nu_1}^E(V_{\alpha_1, \mu_1, \nu_1}^E \cap V_{\alpha_2, \mu_2, \nu_2}^E) \rightarrow \psi_{\alpha_2, \mu_2, \nu_2}^E(V_{\alpha_1, \mu_1, \nu_1}^E \cap V_{\alpha_2, \mu_2, \nu_2}^E)$ is continuous.

- A previous exercise in the section on topological manifolds implies that $(V_{\alpha, \mu}^M)_{(\alpha, \mu) \in \Pi^M}$ is an open cover of M and $(V_{\nu}^F)_{\nu \in \Pi^F}$ is an open cover of F . Since M, F are second-countable M, F are Lindelöf and there exists $S^M \subset \Pi^M$, $S^F \subset \Pi^F$ such that S^M, S^F are countable, $(V_{\alpha, \mu}^M)_{(\alpha, \mu) \in S^M}$ is an open cover of M and $(V_{\nu}^F)_{\nu \in S^F}$ is an open cover of F . Then $S^M \times S^F$ is countable and $(V_{\alpha, \mu}^M \times V_{\nu}^F)_{(\alpha, \mu, \nu) \in S^M \times S^F}$ is an open cover of $M \times F$. Let $a \in E$. Set $p = \pi(a)$. Choose $(\alpha, \mu) \in S^M$ such that $p \in V_{\alpha, \mu}^M$. Since $V_{\alpha, \mu}^M \subset U_{\alpha}$, $a \in E_{U_{\alpha}}$ which implies that

$$\begin{aligned}
p &= \pi(a) \\
&= \text{proj}_1 \circ \Phi_{\alpha}(a)
\end{aligned}$$

Set $q = \text{proj}_2 \circ \Phi_{\alpha}(a)$. Choose $\nu \in S^F$ such that $q \in V_{\nu}^F$. Then

$$\begin{aligned}
\Phi_{\alpha}(a) &= (\text{proj}_1 \circ \Phi_{\alpha}(a), \text{proj}_2 \circ \Phi_{\alpha}(a)) \\
&= (p, q) \\
&\in V_{\alpha, \mu}^M \times V_{\nu}^F
\end{aligned}$$

Thus

$$\begin{aligned}
a &\in \Phi_{\alpha}^{-1}(V_{\alpha, \mu}^M \times V_{\nu}^F) \\
&= V_{\alpha, \mu, \nu}^E
\end{aligned}$$

Since $a \in E$ is arbitrary, we have that for each $a \in E$, there exists $(\alpha, \mu, \nu) \in S^M \times S^F \subset \Pi^E$ such that $a \in V_{\alpha, \mu, \nu}^E$. Thus

$$E \subset \bigcup_{(\alpha, \mu, \nu) \in S^M \times S^F} V_{\alpha, \mu, \nu}^E$$

- Let $a_1, a_2 \in E$. For now, suppose that $\pi(a_1) \neq \pi(a_2)$. Set $p_1 = \pi(a_1)$ and $p_2 = \pi(a_2)$. Since M is Hausdorff, there exist $(\alpha_1, \mu_1), (\alpha_2, \mu_2) \in \Pi^M$ such that $p_1 \in V_{\alpha_1, \mu_1}^M$, $p_2 \in V_{\alpha_2, \mu_2}^M$ and $V_{\alpha_1, \mu_1}^M \cap V_{\alpha_2, \mu_2}^M = \emptyset$. Set $q_1 = \text{proj}_2 \circ \Phi_{\alpha_1}(a_1)$ and $q_2 = \text{proj}_2 \circ \Phi_{\alpha_2}(a_2)$. Choose $\nu_1, \nu_2 \in \Pi^F$ such that $q_1 \in V_{\nu_1}^F$ and $q_2 \in V_{\nu_2}^F$. Then similarly to the previous part, $a_1 \in V_{\alpha_1, \mu_1, \nu_1}^E$ and $a_2 \in V_{\alpha_2, \mu_2, \nu_2}^E$ and

therefore

$$\begin{aligned}
 V_{\alpha_1, \mu_1, \nu_1}^E \cap V_{\alpha_2, \mu_2, \nu_2}^E &= \Phi_{\alpha_1}^{-1}(V_{\alpha_1, \mu_1}^M \times V_{\nu_1}^F) \cap \Phi_{\alpha_2}^{-1}(V_{\alpha_2, \mu_2}^M \times V_{\nu_2}^F) \\
 &\subset \pi^{-1}(V_{\alpha_1, \mu_1}^M) \cap \pi^{-1}(V_{\alpha_2, \mu_2}^M) \\
 &= \pi^{-1}(V_{\alpha_1, \mu_1}^M \cap V_{\alpha_2, \mu_2}^M) \\
 &= \pi^{-1}(\emptyset) \\
 &= \emptyset
 \end{aligned}$$

Now suppose that $\pi(a_1) = \pi(a_2)$. Set $p = \pi(a_1)$. Then there exists $(\alpha, \mu) \in \Pi^M$ such that $p \in V_{\alpha, \mu}^M \subset U_\alpha$.

For now, suppose that $\text{proj}_2 \circ \Phi_\alpha(a_1) \neq \text{proj}_2 \circ \Phi_\alpha(a_2)$. Set $q_1 = \text{proj}_2 \circ \Phi_\alpha(a_1)$ and $q_2 = \text{proj}_2 \circ \Phi_\alpha(a_2)$.

Since F is Hausdorff, there exist $\nu_1, \nu_2 \in \Pi^F$ such that $q_1 \in V_{\nu_1}^F$ and $q_2 \in V_{\nu_2}^F$ and $V_{\nu_1}^F \cap V_{\nu_2}^F = \emptyset$. Then $a_1 \in V_{\alpha, \mu, \nu_1}^E$, $a_2 \in V_{\alpha, \mu, \nu_2}^E$ and

$$\begin{aligned}
 V_{\alpha, \mu, \nu_1}^E \cap V_{\alpha, \mu, \nu_2}^E &= \Phi_\alpha^{-1}(V_{\alpha, \mu}^M \times V_{\nu_1}^F) \cap \Phi_\alpha^{-1}(V_{\alpha, \mu}^M \times V_{\nu_2}^F) \\
 &= \Phi_\alpha^{-1}([V_{\alpha, \mu}^M \times V_{\nu_1}^F] \cap [V_{\alpha, \mu}^M \times V_{\nu_2}^F]) \\
 &= \Phi_\alpha^{-1}([V_{\alpha, \mu}^M \cap V_{\alpha, \mu}^M] \times [V_{\nu_1}^F \cap V_{\nu_2}^F]) \\
 &= \Phi_\alpha^{-1}(V_{\alpha, \mu}^M \times [V_{\nu_1}^F \cap V_{\nu_2}^F]) \\
 &= \Phi_\alpha^{-1}(V_{\alpha, \mu}^M \times \emptyset) \\
 &= \Phi_\alpha^{-1}(\emptyset) \\
 &= \emptyset
 \end{aligned}$$

Now, suppose that $\text{proj}_2 \circ \Phi_\alpha(a_1) = \text{proj}_2 \circ \Phi_\alpha(a_2)$. Set $q = \text{proj}_2 \circ \Phi_\alpha(a_1)$. Choose $\nu \in \Pi^F$ such that $q \in V_\nu^F$. Since

$$\begin{aligned}
 \Phi_\alpha(a_1) &= (\text{proj}_1 \circ \Phi_\alpha(a_1), \text{proj}_2 \circ \Phi_\alpha(a_1)) \\
 &= (p, q) \\
 &= (\text{proj}_1 \circ \Phi_\alpha(a_2), \text{proj}_2 \circ \Phi_\alpha(a_2)) \\
 &= \Phi_\alpha(a_2)
 \end{aligned}$$

we have that $a_1 = a_2$ and $a_1, a_2 \in V_{\alpha, \mu, \nu}^E$. Therefore, for each $a_1, a_2 \in E$, there exists $(\alpha, \mu, \nu) \in \Pi^E$ such that $p, q \in V_{\alpha, \mu, \nu}^E$ or there exist $(\alpha_1, \mu_1, \nu_1), (\alpha_2, \mu_2, \nu_2) \in \Pi^E$ such that $a_1 \in V_{\alpha_1, \mu_1, \nu_1}^E$, $a_2 \in V_{\alpha_2, \mu_2, \nu_2}^E$ and $V_{\alpha_1, \mu_1, \nu_1}^E \cap V_{\alpha_2, \mu_2, \nu_2}^E = \emptyset$.

The topological manifold chart lemma (Exercise 3.1.0.37) implies that there exists a unique topology $\mathcal{T}_E$ on E such that $(E, \mathcal{T}_E) \in \text{Obj}(\mathbf{Man}^0)$, $\dim E = n + k$ and $(V_{\alpha, \mu, \nu}^E, \psi_{\alpha, \mu, \nu}^E)_{(\alpha, \mu, \nu) \in \Pi^E} \subset X^{n+k}(E, \mathcal{T}_E)$.

2. Let $\alpha \in \Gamma$. By assumption $U_\alpha \in \mathcal{T}_M$. Let $\mu \in \Pi_\alpha^M$ and $\nu \in \Pi^F$. Then $(\alpha, \mu, \nu) \in \Pi^E$. Since

- $\psi_{\alpha, \mu, \nu}^E : V_{\alpha, \mu, \nu}^E \rightarrow \psi_{\alpha, \mu}^M(V_{\alpha, \mu}^M) \times \psi_\nu^F(V_\nu^F)$ is a homeomorphism
- $\psi_{\alpha, \mu}^M \times \psi_\nu^F : V_{\alpha, \mu}^M \times V_\nu^F \rightarrow \psi_{\alpha, \mu}^M(V_{\alpha, \mu}^M) \times \psi_\nu^F(V_\nu^F)$ is a homeomorphism
- $\Phi_\alpha|_{V_{\alpha, \mu, \nu}^E} : V_{\alpha, \mu, \nu}^E \rightarrow V_{\alpha, \mu}^M \times V_\nu^F$ is given by $\Phi_\alpha|_{V_{\alpha, \mu, \nu}^E} = (\psi_{\alpha, \mu}^M \times \psi_\nu^F)^{-1} \circ \psi_{\alpha, \mu, \nu}^E$,

we have that $\Phi_\alpha|_{V_{\alpha, \mu, \nu}^E} : V_{\alpha, \mu, \nu}^E \rightarrow V_{\alpha, \mu}^M \times V_\nu^F$ is a homeomorphism. Since $\mu \in \Pi_\alpha^M$ and $\nu \in \Pi^F$ are arbitrary we have that for each $\mu \in \Pi_\alpha^M$ and $\nu \in \Pi^F$, $\Phi_\alpha|_{V_{\alpha, \mu, \nu}^E} : V_{\alpha, \mu, \nu}^E \rightarrow V_{\alpha, \mu}^M \times V_\nu^F$ is a homeomorphism. Since $(V_{\alpha, \mu}^M)_{\mu \in \Pi_\alpha^M}$ is an open cover of U_α and

$(V_{\alpha,\mu}^M \times V_\nu^F)_{(\mu,\nu) \in \Pi_\alpha^M \times \Pi^F}$ is an open cover of $U_\alpha \times F$, we have that

$$\begin{aligned}
 E_{U_\alpha} &= \pi^{-1} \left(\bigcup_{\mu \in \Pi_\alpha^M} V_{\alpha,\mu}^M \right) \\
 &= \bigcup_{\mu \in \Pi_\alpha^M} \pi^{-1}(V_{\alpha,\mu}^M) \\
 &= \bigcup_{\mu \in \Pi_\alpha^M} \Phi_\alpha^{-1}(V_{\alpha,\mu}^M \times F) \\
 &= \bigcup_{\mu \in \Pi_\alpha^M} \Phi_\alpha^{-1} \left(V_{\alpha,\mu}^M \times \left[\bigcup_{\nu \in \Pi^F} V_\nu^F \right] \right) \\
 &= \bigcup_{\mu \in \Pi_\alpha^M} \Phi_\alpha^{-1} \left(\bigcup_{\nu \in \Pi^F} [V_{\alpha,\mu}^M \times V_\nu^F] \right) \\
 &= \bigcup_{\mu \in \Pi_\alpha^M} \left[\bigcup_{\nu \in \Pi^F} \Phi_\alpha^{-1}(V_{\alpha,\mu}^M \times V_\nu^F) \right] \\
 &= \bigcup_{(\mu,\nu) \in \Pi_\alpha^M \times \Pi^F} V_{\alpha,\mu,\nu}^E
 \end{aligned}$$

Hence $E_{U_\alpha} \in \mathcal{T}_E$, $(V_{\alpha,\mu,\nu}^E)_{(\mu,\nu) \in \Pi_\alpha^M \times \Pi^F}$ is an open cover of E_{U_α} and Φ_α is a local homeomorphism. Since Φ_α is a bijection, [ref ex in topology notes](#) implies that $\Phi_\alpha \in \text{Iso}_{\text{Top}}(E_{U_\alpha}, U_\alpha \times F)$. Since $\alpha \in \Gamma$ is arbitrary, we have that for each $\alpha \in \Gamma$, $\Phi_\alpha \in \text{Iso}_{\text{Top}}(E_{U_\alpha}, U_\alpha \times F)$. Let $(\alpha, \mu, \nu) \in \Pi^E$. Since

- $V_{\alpha,\mu,\nu}^E \subset E_{U_\alpha}$
- $\text{proj}_1 : M \times F \rightarrow M$ is continuous
- $\Phi_\alpha : E_{U_\alpha} \rightarrow U_\alpha \times F$ is continuous
- $\pi|_{V_{\alpha,\mu,\nu}^E} = \text{proj}_1 \circ \Phi|_{V_{\alpha,\mu,\nu}^E}$

we have that $\pi|_{V_{\alpha,\mu,\nu}^E} : V_{\alpha,\mu,\nu}^E \rightarrow V_{\alpha,\mu}^M$ is continuous. Since $(\alpha, \mu, \nu) \in \Pi^E$ is arbitrary and $(V_{\alpha,\mu,\nu}^E)_{(\alpha,\mu,\nu) \in \Pi^E}$ is an open cover of E , [ref ex in topology notes](#) implies that $\pi \in \text{Hom}_{\mathbf{Man}^0}(E, M)$. Since π is surjective, $\pi \in \text{Epi}_{\mathbf{Man}^0}(E, M)$.

3. Let $\alpha \in \Gamma$. Since $E, M, F \in \text{Obj}(\mathbf{Man}^0)$, $\pi \in \text{Epi}_{\mathbf{Man}^0}(E, M)$ and

- $U_\alpha \in \mathcal{T}_M$,
- $(U_\alpha, \Phi_\alpha) \in \text{LocTriv}(\pi, F)$,
- $\Phi_\alpha \in \text{Iso}_{\text{Top}}(E_{U_\alpha}, U_\alpha \times F)$,

we have that $(U_\alpha, \Phi_\alpha) \in \text{LocTriv}^0(\pi, F)$. Since $\alpha \in \Gamma$ is arbitrary, we have that for each $\alpha \in \Gamma$, $(U_\alpha, \Phi_\alpha) \in \text{LocTriv}^0(\pi, F)$.

4. Let $p \in M$. By assumption, there exists $\alpha \in \Gamma$ such that $p \in U_\alpha$. The previous part implies that $(U_\alpha, \Phi_\alpha) \in \text{LocTriv}^0(\pi, F)$. Since $p \in M$ is arbitrary, we have that for each $p \in M$, there exists $\alpha \in \Gamma$ such that $p \in U_\alpha$ and $(U_\alpha, \Phi_\alpha) \in \text{LocTriv}^0(\pi, F)$. Hence $(E, M, \pi, F) \in \text{Obj}(\mathbf{Bun}^0)$.

□

15.1.3 $\mathbf{Man}^\infty$ Fiber Bundles

Definition 15.1.3.1. Let $E, M, F \in \text{Obj}(\mathbf{Man}^\infty)$ and $\pi \in \text{Hom}_{\mathbf{Man}^\infty}(E, M)$ a surjection, $U \subset M$ and $\Phi : \pi^{-1}(U) \rightarrow U \times F$. Then (U, Φ) is said to be a **smooth fiber bundle local trivialization with respect to π of E over U with fiber F** if

1. U is open in M
2. (U, Φ) is a local trivialization of E over U with fiber F with respect to π

3. Φ is a diffeomorphism

Definition 15.1.3.2. Let $E, M, F \in \text{Obj}(\mathbf{Man}^\infty)$ and $\pi \in \text{Hom}_{\mathbf{Man}^\infty}(E, M)$ a surjection. Then (E, M, π, F) is said to be a **$\mathbf{Man}^\infty$ fiber bundle with total space E , base space M , fiber F and projection π** if for each $p \in M$, there exist $U \in \mathcal{T}_M$ and $\Phi : \pi^{-1}(U) \rightarrow U \times F$ such that $p \in U$ and (U, Φ) is a smooth local trivialization of E over U with fiber F . For $p \in M$, we define the **fiber over p** , denoted E_p , by $E_p = \pi^{-1}(\{p\})$.

Exercise 15.1.3.3. $\mathbf{Man}^\infty$ Fiber Bundle Chart Lemma:

Let $E \in \text{Obj}(\mathbf{Set})$, $M, F \in \text{Obj}(\mathbf{Man}^\infty)$, $\pi \in \text{Epi}_{\mathbf{Set}}(E, M)$, Γ an index set and for each $\alpha \in \Gamma$, $(U_\alpha, \Phi_\alpha) \in \text{LocTriv}(\pi, F)$. Set $n := \dim M$ and $k := \dim F$. Suppose that

- for each $\alpha \in \Gamma$, $U_\alpha \in \mathcal{T}_M$
- $M \subset \bigcup_{\alpha \in \Gamma} U_\alpha$
- for each $\alpha, \beta \in \Gamma$, $\Phi_\beta|_{\pi^{-1}(U_\alpha \cap U_\beta)} \circ (\Phi_\alpha|_{\pi^{-1}(U_\alpha \cap U_\beta)})^{-1} : (U_\alpha \cap U_\beta) \times F \rightarrow (U_\alpha \cap U_\beta) \times F$ is smooth.

Then there exists a unique topology $\mathcal{T}_E$ on E and smooth structure $\mathcal{A}_E \subset X^{n+k}(M, \mathcal{T}_E)$ on E such that

1. $(E, \mathcal{T}_E)$ is an $n + k$ -dimensional topological manifold and $(E, \mathcal{T}_E, \mathcal{A}_E)$ is a smooth manifold,
2. for each $\alpha \in \Gamma$, $\pi^{-1}(U_\alpha) \in \mathcal{T}_E$ and $\Phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times F$ is a diffeomorphism
3. $\pi : E \rightarrow M$ is smooth
4. (E, M, π, F) is a **$\mathbf{Man}^\infty$ fiber bundle**

Proof. Exercise 15.1.2.4 implies that there exists a unique topology $\mathcal{T}_E$ on E such that

- $(E, \mathcal{T}_E)$ is a $n + k$ -dimensional topological manifold
- for each $\alpha \in \Gamma$, $\pi^{-1}(U_\alpha) \in \mathcal{T}_E$ and $\Phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times F$ is a homeomorphism
- $\pi : E \rightarrow M$ is continuous
- (E, M, π, F) is an **$\mathbf{Man}^0$ fiber bundle**

1. Define $(V_{\alpha, \mu, \nu}^E, \psi_{\alpha, \mu, \nu}^E)_{(\alpha, \mu, \nu) \in \Pi^E} \subset X^{n+k}(E, \mathcal{T}_E)$ as in the proof of the **$\mathbf{Man}^0$ fiber bundle chart lemma**. Let $(\alpha_1, \mu_1, \nu_1), (\alpha_2, \mu_2, \nu_2) \in \Pi^E$. For notational convenience, set $\psi_1^E = \psi_{\alpha_1, \mu_1, \nu_1}^E$, $\psi_2^E = \psi_{\alpha_2, \mu_2, \nu_2}^E$, $V^E = V_{\alpha_1, \mu_1, \nu_1}^E \cap V_{\alpha_2, \mu_2, \nu_2}^E$, $V^M = V_{\alpha_1, \mu_1}^M \cap V_{\alpha_2, \mu_2}^M$ and $V^F = V_{\nu_1}^F \cap V_{\nu_2}^F$. Then $\psi_2|_{V^E} \circ (\psi_1|_{V^E})^{-1} : \psi_1(V^E) \rightarrow \psi_2(V^E)$ is given by

$$\begin{aligned} \psi_2^E|_{V^E} \circ (\psi_1^E|_{V^E})^{-1} &= [(\psi_{\alpha_2, \mu_2}^M|_{V^M} \times \psi_{\nu_2}^F|_{V^F}) \circ \Phi_{\alpha_2}|_{V^E}] \circ [(\psi_{\alpha_1, \mu_1}^M|_{V^M} \times \psi_{\nu_1}^F|_{V^F}) \circ \Phi_{\alpha_1}|_{V^E}]^{-1} \\ &= [(\psi_{\alpha_2, \mu_2}^M|_{V^M} \times \psi_{\nu_2}^F|_{V^F}) \circ \Phi_{\alpha_2}|_{V^E}] \circ [(\Phi_{\alpha_1}|_{V^E})^{-1} \circ (\psi_{\alpha_1, \mu_1}^M|_{V^M} \times \psi_{\nu_1}^F|_{V^F})^{-1}] \\ &= (\psi_{\alpha_2, \mu_2}^M|_{V^M} \times \psi_{\nu_2}^F|_{V^F}) \circ [\Phi_{\alpha_2}|_{V^E} \circ (\Phi_{\alpha_1}|_{V^E})^{-1}] \circ (\psi_{\alpha_1, \mu_1}^M|_{V^M} \times \psi_{\nu_1}^F|_{V^F})^{-1} \end{aligned}$$

Since $\Phi_{\alpha_2}|_{V^E} \circ (\Phi_{\alpha_1}|_{V^E})^{-1}$ is smooth, we have that $\psi_{\alpha_2, \mu_2, \nu_2}^E|_{V_{\alpha_1, \mu_1, \nu_1}^E \cap V_{\alpha_2, \mu_2, \nu_2}^E} \circ (\psi_{\alpha_1, \mu_1, \nu_1}^E|_{V_{\alpha_1, \mu_1, \nu_1}^E \cap V_{\alpha_2, \mu_2, \nu_2}^E})^{-1} : \psi_{\alpha_1, \mu_1, \nu_1}^E(V_{\alpha_1, \mu_1, \nu_1}^E \cap V_{\alpha_2, \mu_2, \nu_2}^E) \rightarrow \psi_{\alpha_2, \mu_2, \nu_2}^E(V_{\alpha_1, \mu_1, \nu_1}^E \cap V_{\alpha_2, \mu_2, \nu_2}^E)$ is smooth. Since $(\alpha_1, \mu_1, \nu_1), (\alpha_2, \mu_2, \nu_2) \in \Pi^E$ are arbitrary, we have that $(V_{\alpha, \mu, \nu}^E, \psi_{\alpha, \mu, \nu}^E)_{(\alpha, \mu, \nu) \in \Pi^E}$ is a smooth atlas on E . **An exercise in the section on smooth manifolds** implies that there exists a unique smooth structure $\mathcal{A}_E$ on E such that $(E, \mathcal{A}_E)$ is an $n + k$ -dimensional smooth manifold.

2. Let $\alpha \in \Gamma$. By assumption $U_\alpha \in \mathcal{T}_M$. Let $\mu \in \Pi_\alpha^M$ and $\nu \in \Pi^F$. Then $(\alpha, \mu, \nu) \in \Pi^E$. Since

- $\psi_{\alpha, \mu, \nu}^E : V_{\alpha, \mu, \nu}^E \rightarrow \psi_{\alpha, \mu}^M(V_{\alpha, \mu}^M) \times \psi_\nu^F(V_\nu^F)$ is a diffeomorphism
- $\psi_{\alpha, \mu}^M \times \psi_\nu^F : V_{\alpha, \mu}^M \times V_\nu^F \rightarrow \psi_{\alpha, \mu}^M(V_{\alpha, \mu}^M) \times \psi_\nu^F(V_\nu^F)$ is a diffeomorphism
- $\Phi_\alpha|_{V_{\alpha, \mu, \nu}^E} : V_{\alpha, \mu, \nu}^E \rightarrow V_{\alpha, \mu}^M \times V_\nu^F$ is given by $\Phi_\alpha|_{V_{\alpha, \mu, \nu}^E} = (\psi_{\alpha, \mu}^M \times \psi_\nu^F)^{-1} \circ \psi_{\alpha, \mu, \nu}^E$,

we have that $\Phi_\alpha|_{V_{\alpha, \mu, \nu}^E} : V_{\alpha, \mu, \nu}^E \rightarrow V_{\alpha, \mu}^M \times V_\nu^F$ is a diffeomorphism. Since $\mu \in \Pi_\alpha^M$ and $\nu \in \Pi^F$ are arbitrary and $(V_{\alpha, \mu, \nu}^E)_{(\mu, \nu) \in \Pi_\alpha^M \times \Pi^F}$ is an open cover of $\pi^{-1}(U_\alpha)$, we have that $\Phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times F$ is a local diffeomorphism. Since Φ_α is a bijection, Φ_α is a diffeomorphism. Since $\alpha \in \Gamma$ is arbitrary, we have that for each $\alpha \in \Gamma$, $\Phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times F$ is a diffeomorphism.

3. Let $(\alpha, \mu, \nu) \in \Pi^E$. Since

- $V_{\alpha, \mu, \nu}^E \subset \pi^{-1}(U_\alpha)$
- $\text{proj}_1 : M \times F \rightarrow M$ is smooth
- $\Phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times F$ is smooth
- $\pi|_{V_{\alpha, \mu, \nu}^E} = \text{proj}_1 \circ \Phi|_{V_{\alpha, \mu, \nu}^E}$

we have that $\pi|_{V_{\alpha, \mu, \nu}^E} : V_{\alpha, \mu, \nu}^E \rightarrow V_{\alpha, \mu}^M$ is smooth. Since $(\alpha, \mu, \nu) \in \Pi^E$ is arbitrary and $(V_{\alpha, \mu, \nu}^E)_{(\alpha, \mu, \nu) \in \Pi^E}$ is an open cover of E , we have that $\pi : E \rightarrow M$ is smooth.

4. Let $p \in M$. By assumption, there exists $\alpha \in \Gamma$ such that $p \in U_\alpha$, $U_\alpha \in \mathcal{T}_M$. Since $E, M, F \in \text{Obj}(\mathbf{Man}^\infty)$, $\pi \in \text{Hom}_{\mathbf{Man}^\infty}(E, M)$ is a surjection, and

- U_α is open
- (U_α, Φ_α) is a local trivialization with respect to π of E over U_α with fiber F
- $\Phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times F$ is a diffeomorphism

we have that (U_α, Φ_α) is a smooth local trivialization with respect to π of E over U_α with fiber F . Since $p \in M$ is arbitrary, (E, M, π, F) is a $\mathbf{Man}^\infty$ fiber bundle. □

Definition 15.1.3.4. Let (E_1, M_1, π_1, F_1) and (E_2, M_2, π_2, F_2) be $\mathbf{Man}^\infty$ fiber bundles, $\Phi \in \text{Hom}_{\mathbf{Man}^\infty}(E_1, E_2)$ and $\phi \in \text{Hom}_{\mathbf{Man}^\infty}(M_1, M_2)$. Then (Φ, ϕ) is said to be a **smooth bundle morphism** from (E_1, M_1, π_1, F_1) to (E_2, M_2, π_2, F_2) if $\pi_2 \circ \Phi = \phi \circ \pi_1$, i.e. the following diagram commutes:

$$\begin{array}{ccc} E_1 & \xrightarrow{\Phi} & E_2 \\ \pi_1 \downarrow & & \downarrow \pi_2 \\ M_1 & \xrightarrow{\phi} & M_2 \end{array}$$

Exercise 15.1.3.5. Let (E_1, M_1, π_1, F_1) and (E_2, M_2, π_2, F_2) be $\mathbf{Man}^\infty$ fiber bundles, $\Phi \in \text{Hom}_{\mathbf{Man}^\infty}(E_1, E_2)$ and $\phi \in \text{Hom}_{\mathbf{Man}^\infty}(M_1, M_2)$. If (Φ, ϕ) is a smooth bundle morphism from (E_1, M_1, π_1, F_1) to (E_2, M_2, π_2, F_2) , then for each $p \in M_1$, $\Phi((E_1)_p) \subset (E_2)_{\phi(p)}$.

Proof. Suppose that (Φ, ϕ) is a smooth bundle morphism from (E_1, M_1, π_1, F_1) to (E_2, M_2, π_2, F_2) . Let $p \in M_1$ and $y \in \Phi((E_1)_p)$. Then there exists $x \in (E_1)_p$ such that $y = \Phi(x)$. Since $x \in (E_1)_p$, we have that $\pi_1(x) = p$. Since (Φ, ϕ) is a smooth bundle morphism from (E_1, M_1, π_1, F_1) to (E_2, M_2, π_2, F_2) , we have that $\pi_2 \circ \Phi = \phi \circ \pi_1$. Therefore

$$\begin{aligned} \pi_2(y) &= \pi_2(\Phi(x)) \\ &= \pi_2 \circ \Phi(x) \\ &= \phi \circ \pi_1(x) \\ &= \phi(p) \end{aligned}$$

Thus

$$\begin{aligned} y &\in \pi_2^{-1}(\phi(p)) \\ &= (E_2)_{\phi(p)} \end{aligned}$$

Since $y \in \Phi((E_1)_p)$ is arbitrary, we have that $\Phi((E_1)_p) \subset (E_2)_{\phi(p)}$. □

Definition 15.1.3.6. We define the category of $\mathbf{Man}^\infty$ fiber bundles, denoted $\mathbf{Bun}^\infty$, by

- $\text{Obj}(\mathbf{Bun}^\infty) := \{(E, M, \pi, F) : (E, M, \pi, F) \text{ is a } \mathbf{Man}^\infty \text{ fiber bundle}\}$
- For $(E_1, M_1, \pi_1, F_1), (E_2, M_2, \pi_2, F_2) \in \text{Obj}(\mathbf{Bun}^\infty)$,

$$\begin{aligned} \text{Hom}_{\mathbf{Bun}^\infty}((E_1, M_1, \pi_1, F_1), (E_2, M_2, \pi_2, F_2)) &:= \\ \{(\Phi, \phi) : (\Phi, \phi) \text{ is a smooth bundle morphism from } (E_1, M_1, \pi_1, F_1) &\text{ to } (E_2, M_2, \pi_2, F_2)\} \end{aligned}$$

- For
 - $(E_1, M_1, \pi_1, F_1), (E_2, M_2, \pi_2, F_2), (E_3, M_3, \pi_3) \in \text{Obj}(\mathbf{Bun}^\infty)$
 - $(\Phi_{12}, \phi_{12}) \in \text{Hom}_{\mathbf{Bun}^\infty}((E_1, M_1, \pi_1, F_1), (E_2, M_2, \pi_2, F_2))$
 - $(\Phi_{23}, \phi_{23}) \in \text{Hom}_{\mathbf{Bun}^\infty}((E_2, M_2, \pi_2, F_2), (E_3, M_3, \pi_3))$

we define $(\Phi_{23}, \phi_{23}) \circ (\Phi_{12}, \phi_{12}) \in \text{Hom}_{\mathbf{Bun}^\infty}((E_1, M_1, \pi_1, F_1), (E_3, M_3, \pi_3))$ by

$$(\Phi_{23}, \phi_{23}) \circ (\Phi_{12}, \phi_{12}) := (\Phi_{23} \circ \Phi_{12}, \phi_{23} \circ \phi_{12})$$

Exercise 15.1.3.7. We have that $\mathbf{Bun}^\infty$ is a full subcategory of $(\text{id}_{\mathbf{Man}^\infty} \downarrow \text{id}_{\mathbf{Man}^\infty})$.

Proof. Set $\mathcal{C} = (\text{id}_{\mathbf{Man}^\infty} \downarrow \text{id}_{\mathbf{Man}^\infty})$. We note that

- $\text{Obj}(\mathbf{Bun}^\infty) \subset \text{Obj}(\mathcal{C})$
- for each $(E_1, M_1, \pi_1, F_1), (E_2, M_2, \pi_2, F_2) \in \text{Obj}(\mathbf{Bun}^\infty)$,

$$\text{Hom}_{\mathbf{Bun}^\infty}((E_1, M_1, \pi_1, F_1), (E_2, M_2, \pi_2, F_2)) = \text{Hom}_{\mathcal{C}}((E_1, M_1, \pi_1, F_1), (E_2, M_2, \pi_2, F_2))$$

So $\mathbf{Bun}^\infty$ is a full subcategory of $\mathcal{C}$. □

Exercise 15.1.3.8. Let $(E, M, \pi, F) \in \mathbf{Bun}^\infty$. Then π is a submersion.

Proof. Let $a \in E$. Set $p := \pi(a)$. Since $(E, M, \pi, F) \in \mathbf{Bun}^\infty$, there exists $U \in \mathcal{T}_M$ and $\Phi \in \text{Hom}_{\mathbf{Man}^\infty}(\pi^{-1}(U), U \times F)$ such that $p \in U$ and (U, Φ) is a smooth fiber bundle local trivialization of E over U with fiber F with respect to π . Then Φ is a diffeomorphism and $\text{proj}_1 \circ \Phi = \pi|_{\pi^{-1}(U)}$. Exercise 8.3.0.4 implies that $\text{proj}_1 : U \times F \rightarrow U$ is a submersion. Since Φ is a diffeomorphism, Φ is a submersion. Exercise 8.3.0.5 then implies that $\pi|_{\pi^{-1}(U)}$ is a submersion. Since $a \in E$ is arbitrary, we have that for each $a \in E$, there exists $V \in \mathcal{T}_E$ such that $a \in V$ and $\pi|_V$ is a submersion. (cite exercise) Exercise ?? implies that π is a submersion. □

Exercise 15.1.3.9. Let $(E, M, \pi, F) \in \mathbf{Bun}^\infty$ and (U, Φ) a local trivialization of E over U . For each $p \in M$,

1. E_p is an embedded submanifold of E ,
2. $\Phi|_{E_p} : E_p \rightarrow \{p\} \times F$ is a diffeomorphism.

Proof. Let $p \in M$.

1. Since $E_p = \pi^{-1}(\{p\})$ and π is a surjective submersion Exercise ?? ref exercise in section on submersion implies that E_p is an embedded submanifold of E .
2. Exercise ?? ref exercise in section on immersed submanifolds implies that $\Phi|_{E_p}$ is a diffeomorphism. □

Exercise 15.1.3.10. Let $(E, M, \pi, F) \in \mathbf{Bun}^\infty$, (U, Φ) a local trivialization of E over U and (V, Ψ) a local trivialization of E over V . Then

1. $\text{proj}_1 \circ \Psi|_{\pi^{-1}(U \cap V)} \circ \Phi|_{\pi^{-1}(U \cap V)}^{-1} = \text{proj}_1$
2. there exists $\sigma \in \text{Hom}_{\mathbf{Man}^\infty}((U \cap V) \times F, F)$ such that $\Psi|_{\pi^{-1}(U \cap V)} \circ \Phi|_{\pi^{-1}(U \cap V)}^{-1} = (\text{proj}_1, \sigma)$ and for each $p \in U \cap V$, $\sigma(p, \cdot) \in \text{Aut}_{\mathbf{Man}^\infty}(F)$.

Proof.

1. By definition and Exercise 15.1.1.5, the following diagram commutes:

$$\begin{array}{ccccc} (U \cap V) \times F & \xleftarrow{\Phi} & \pi^{-1}(U \cap V) & \xrightarrow{\Psi} & (U \cap V) \times F \\ & \searrow \text{proj}_1 & \downarrow \pi & \swarrow \text{proj}_1 & \\ & & U \cap V & & \end{array}$$

Therefore $\text{proj}_1 \circ \Psi|_{\pi^{-1}(U \cap V)} \circ (\Phi|_{\pi^{-1}(U \cap V)})^{-1} = \text{proj}_1$.

2. Define $\sigma, \tau \in \text{Hom}_{\mathbf{Man}^\infty}((U \cap V) \times F, F)$ by $\sigma := \text{proj}_2 \circ \Psi|_{\pi^{-1}(U \cap V)} \circ (\Phi|_{\pi^{-1}(U \cap V)})^{-1}$ and $\tau := \text{proj}_2 \circ \Phi|_{\pi^{-1}(U \cap V)} \circ (\Psi|_{\pi^{-1}(U \cap V)})^{-1}$. Part (1) implies that for each $(p, x) \in (U \cap V) \times F$,

$$\begin{aligned} \Psi|_{\pi^{-1}(U \cap V)} \circ (\Phi|_{\pi^{-1}(U \cap V)})^{-1}(p, x) &= (\text{proj}_1(p, x), \sigma(p, x)) \\ &= (p, \sigma(p, x)). \end{aligned}$$

Similarly, for each $(p, x) \in (U \cap V) \times F$, $\Phi|_{\pi^{-1}(U \cap V)} \circ (\Psi|_{\pi^{-1}(U \cap V)})^{-1}(p, x) = (p, \tau(x))$. Let $p \in U \cap V$ and $x \in F$. Set $\sigma_p := \sigma \circ \iota_p^F$ and $\tau_p := \tau \circ \iota_p^F$. Exercise 8.2.0.10 implies that σ_p and τ_p are smooth (clean up a bit here). Then

$$\begin{aligned} (p, x) &= \text{id}_{(U \cap V) \times F}(p, x) \\ &= [\Psi|_{\pi^{-1}(U \cap V)} \circ (\Phi|_{\pi^{-1}(U \cap V)})^{-1}] \circ [\Phi|_{\pi^{-1}(U \cap V)} \circ (\Psi|_{\pi^{-1}(U \cap V)})^{-1}](p, x) \\ &= (p, \sigma(\Phi|_{\pi^{-1}(U \cap V)} \circ (\Psi|_{\pi^{-1}(U \cap V)})^{-1}(p, x))) \\ &= (p, \sigma(p, \tau(p, x))) \\ &= (p, \sigma_p \circ \tau_p(x)) \end{aligned}$$

Since $x \in F$ is arbitrary, we have that for each $x \in F$, $\text{id}_F(x) = \sigma_p \circ \tau_p(x)$. Thus $\sigma_p \circ \tau_p = \text{id}_F$. Similarly, $\tau_p \circ \sigma_p = \text{id}_F$. Thus σ_p is a bijection and $\sigma_p^{-1} = \tau_p$. Therefore $\sigma_p \in \text{Aut}_{\mathbf{Man}^\infty}(F)$. Since $p \in U \cap V$ is arbitrary, we have that for each $p \in U \cap V$, $\sigma(p, \cdot) \in \text{Aut}_{\mathbf{Man}^\infty}(F)$. □

15.1.4 cocycles

Definition 15.1.4.1. Let $(E, M, \pi, F) \in \mathbf{Bun}^\infty$, A an index set and for each $\alpha \in A$, (U_α, Φ_α) a smooth local trivializations of E . Then $(U_\alpha, \Phi_\alpha)_{\alpha \in A}$ is said to be a **smooth fiber bundle atlas on** (E, M, π, F) if for each $p \in M$, there exists $\alpha \in A$ such that $p \in U_\alpha$.

Definition 15.1.4.2. Let $(E, M, \pi, F) \in \text{Obj}(\mathbf{Bun}^\infty)$, A an index set and $(U_\alpha, \Phi_\alpha)_{\alpha \in A}$ a smooth fiber bundle atlas on (E, M, π, F) . For each $\alpha, \beta \in A$, we define $U_{\alpha, \beta} \subset M$ and $\Phi_{\alpha, \beta} : U_{\alpha, \beta} \times F \rightarrow U_{\alpha, \beta} \times F$ by

- $U_{\alpha, \beta} = U_\alpha \cap U_\beta$
- $\Phi_{\alpha, \beta} = \Phi_\alpha|_{U_{\alpha, \beta}} \circ \Phi_\beta|_{U_{\alpha, \beta}}^{-1}$

Exercise 15.1.4.3. Let $(E, M, \pi, F) \in \text{Obj}(\mathbf{Bun}^\infty)$, A an index set and $(U_\alpha, \Phi_\alpha)_{\alpha \in A}$ a smooth fiber bundle atlas on (E, M, π, F) . Then for each $\alpha, \beta \in A$ and $p \in U_{\alpha, \beta}$, $\Phi_{\alpha, \beta}(p, \cdot) \in \text{Aut}_{\mathbf{Man}^\infty}(F)$.

Proof. Let $\alpha, \beta \in \Gamma$ and $p \in U_{\alpha, \beta}$. Since **FINISH**, basically reference the previous exercise □

15.2 Subbundles

Definition 15.2.0.1. Let $(E, M, \pi_E, F), (E', M, \pi', F') \in \text{Obj}(\mathbf{Bun}^\infty)$. Then (E', M, π', F') is said to be a **subbundle of** (E, M, π_E, F) if

1. $E' \subset E$
2. $\pi' = \pi|_{E'}$
3. for each $(U, \Phi) \in \text{LocTriv}(E, \pi, F, U)$

15.3 Product Bundles

Definition 15.3.0.1.

15.4 Vertical and Horizontal Subbundles

Definition 15.4.0.1. Let $(E, M, \pi) \in \text{Obj}(\mathbf{Bun}^\infty)$. We define the **vertical bundle associated to** (E, M, π) , denoted $(VE, M, \pi_V) \in \mathbf{Bun}^\infty$, by

$$VE = \coprod_{q \in E} \ker D\pi(q)$$

relocate this to after tangent bundle is introduced

Exercise 15.4.0.2. Let $(M, \mathcal{A})$ be an n -dimensional smooth manifold and $(U, \phi) \in \mathcal{A}$ with $\phi = (x^1, \dots, x^n)$, $(\pi^{-1}(U), \Phi_\phi) \in \mathcal{A}_{TM}$ the induced chart on TM with $\Phi_\phi = (\tilde{x}^1, \dots, \tilde{x}^n, \tilde{y}^1, \dots, \tilde{y}^n)$. Then

$$V(TM)|_{\pi^{-1}(U)} = \coprod_{(p, \xi) \in \pi^{-1}(U)} \text{span} \left\{ \frac{\partial}{\partial \tilde{y}^j} \Big|_{(p, \xi)} : j \in \{1, \dots, n\} \right\}$$

Split into smaller exercises

Proof. Let $f \in C^\infty(M)$ and $(u^1, \dots, u^n, v^1, \dots, v^n)$ the standard coordinates on $\mathbb{R}^n \times \mathbb{R}^n$. We note that by definition, $\Phi_\phi(p, \xi) = (\phi(p), \psi(\xi))$ where $\psi : \bigcup_{p \in U} T_p M \rightarrow \mathbb{R}^n$ is given by

$$\psi \left(\sum_{j=1}^n \xi^j \frac{\partial}{\partial x^j} \Big|_p \right) = (\xi^1, \dots, \xi^n)$$

$$\begin{aligned} x^k \circ \pi \circ \Phi_\phi^{-1}(u, v) &= x^k \circ \pi(\phi^{-1}(u), \psi^{-1}(v)) \\ &= x^k \circ \phi^{-1}(u) \end{aligned}$$

Therefore

$$\begin{aligned} \frac{\partial}{\partial \tilde{x}^i} \Big|_{(p, \xi)} [x^k \circ \pi] &= \frac{\partial}{\partial u^i} \Big|_{\Phi_\phi(p, \xi)} [x^k \circ \pi \circ \Phi_\phi^{-1}] \\ &= \frac{\partial}{\partial u^i} \Big|_{(\phi(p), \psi(\xi))} [x^k \circ \pi \circ \Phi_\phi^{-1}] \\ &= \frac{\partial}{\partial u^i} \Big|_{\phi(p)} [x^k \circ \phi^{-1}] \\ &= \frac{\partial}{\partial x^i} \Big|_p x^k \\ &= \delta_{i, k} \end{aligned}$$

and

$$\begin{aligned} \frac{\partial}{\partial \tilde{y}^i} \Big|_{(p, \xi)} [x^k \circ \pi] &= \frac{\partial}{\partial v^i} \Big|_{\Phi_\phi(p, \xi)} [x^k \circ \pi \circ \Phi_\phi^{-1}] \\ &= \frac{\partial}{\partial v^i} \Big|_{(\phi(p), \psi(\xi))} [x^k \circ \pi \circ \Phi_\phi^{-1}] \\ &= \frac{\partial}{\partial v^i} \Big|_{\phi(p)} [x^k \circ \phi^{-1}] \\ &= 0 \end{aligned}$$

This implies that for each $i \in \{1, \dots, n\}$, we have that

$$\begin{aligned}
 D\pi(p, \xi) \left(\frac{\partial}{\partial \tilde{x}^i} \Big|_{(p, \xi)} \right) (f) &= \frac{\partial}{\partial \tilde{x}^i} \Big|_{(p, \xi)} f \circ \pi \\
 &= \sum_{k=1}^n \frac{\partial f}{\partial x^k} (\pi(p, \xi)) \frac{\partial x^k \circ \pi}{\partial \tilde{x}^i} (p, \xi) \\
 &= \sum_{k=1}^n \frac{\partial f}{\partial x^k} (p) \delta_{i,k} \\
 &= \frac{\partial f}{\partial x^i} (p)
 \end{aligned}$$

and

$$\begin{aligned}
 D\pi(p, \xi) \left(\frac{\partial}{\partial \tilde{y}^i} \Big|_{(p, \xi)} \right) (f) &= \frac{\partial}{\partial \tilde{y}^i} \Big|_{(p, \xi)} f \circ \pi \\
 &= \sum_{k=1}^n \frac{\partial f}{\partial x^k} (\pi(p, \xi)) \frac{\partial x^k \circ \pi}{\partial \tilde{y}^i} (p, \xi) \\
 &= \sum_{k=1}^n \frac{\partial f}{\partial x^k} (p) 0 \\
 &= 0
 \end{aligned}$$

Hence

$$\begin{aligned}
 V(TM)|_{\pi^{-1}(U)} &= \coprod_{(p, \xi) \in \pi^{-1}(U)} \ker D\pi(p, \xi) \\
 &= \coprod_{(p, \xi) \in \pi^{-1}(U)} \text{span} \left\{ \frac{\partial}{\partial \tilde{y}^j} \Big|_{(p, \xi)} : j \in \{1, \dots, n\} \right\}
 \end{aligned}$$

□

Chapter 16

Vector Bundles

16.1 Introduction

16.1.1 $\mathbf{Man}^\infty$ Vector Bundles

Note 16.1.1.1. Let M be a set and $p \in M$. We endow $\{p\} \times \mathbb{R}^n$ with the natural vector space structure such that $\{p\} \times \mathbb{R}^k \cong \mathbb{R}^k$.

Definition 16.1.1.2. Let $E, M \in \mathbf{Obj}(\mathbf{Man}^\infty)$ and $\pi \in \mathbf{Hom}_{\mathbf{Man}^\infty}(E, M)$ a surjection, $U \subset M$ and $\Phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^k$. Then (U, Φ) is said to be a **smooth vector bundle local trivialization of E over U** if

1. U is open in M
2. (U, Φ) is a smooth local trivialization of E over U with fiber $\mathbb{R}^k$ (Definition 15.1.3.1)
3. for each $q \in U$, $\Phi|_{E_q} \in \mathbf{Iso}_{\mathbf{Vect}_{\mathbb{R}}}(E_q, \mathbb{R}^k)$

Definition 16.1.1.3. Let $E, M \in \mathbf{Obj}(\mathbf{Man}^\infty)$ and $\pi \in \mathbf{Hom}_{\mathbf{Man}^\infty}(E, M)$ a surjection. Then (E, M, π) is said to be a **rank- k smooth vector bundle** if

1. $(E, M, \pi, \mathbb{R}^k) \in \mathbf{Obj}(\mathbf{Bun}^\infty)$
2. for each $p \in M$, E_p is a k -dimensional real vector space and there exists $U \in \mathcal{T}_M$ and $\Phi \in \mathbf{Hom}_{\mathbf{Man}^\infty}(\pi^{-1}(U), U \times \mathbb{R}^k)$ such that
 - (a) $p \in U$
 - (b) (U, Φ) is a smooth vector bundle local trivialization of E over U

In this case we define the **rank of (E, M, π)** , denoted $\mathbf{rank}(E, M, \pi)$, by $\mathbf{rank}(E, M, \pi) = k$.

Exercise 16.1.1.4. Let (E, M, π) be a rank- k smooth vector bundle, (U, Φ) a local trivialization of E over U and (V, Ψ) a smooth vector bundle local trivialization of E over V . Then

1. $\mathbf{proj}_1 \circ \Psi|_{\pi^{-1}(U \cap V)} \circ \Phi|_{\pi^{-1}(U \cap V)}^{-1} = \mathbf{proj}_1$
2. there exists $\tau \in \mathbf{Hom}_{\mathbf{Man}^\infty}(U \cap V, GL(k, \mathbb{R}))$ such that for each $(p, v) \in (U \cap V) \times \mathbb{R}^k$, $\Psi|_{\pi^{-1}(U \cap V)} \circ \Phi|_{\pi^{-1}(U \cap V)}^{-1}(p, v) = (p, \tau(p)(v))$.

Proof. Exercise 15.1.3.10 implies that there exists $\sigma \in \mathbf{Hom}_{\mathbf{Man}^\infty}((U \cap V) \times \mathbb{R}^k, \mathbb{R}^k)$ such that $\Psi|_{\pi^{-1}(U \cap V)} \circ \Phi|_{\pi^{-1}(U \cap V)}^{-1} = (\mathbf{proj}_1, \sigma)$ and for each $p \in U \cap V$, $\sigma(p, \cdot) \in \mathbf{Aut}_{\mathbf{Man}^\infty}(\mathbb{R}^k)$. Define $\tau : U \cap V \rightarrow \mathbf{Aut}_{\mathbf{Man}^\infty}(\mathbb{R}^k)$ by $\tau(p) = \sigma(p, \cdot)$. Since (U, Φ) , (V, Ψ) are smooth vector bundle local trivializations, for each $q \in U \cap V$, $\Phi|_{E_q} \rightarrow \{q\} \times \mathbb{R}^k$ and $\Psi|_{E_q} \rightarrow \{q\} \times \mathbb{R}^k$ are linear isomorphism. Let $q \in U \cap V$. Since $\Psi|_{E_q} \circ \Phi|_{E_q}^{-1} : \{q\} \times \mathbb{R}^k \rightarrow \{q\} \times \mathbb{R}^k$, is a vector space isomorphism and for each $v \in \mathbb{R}^k$,

$$\begin{aligned} \Psi|_{E_q} \circ \Phi|_{E_q}^{-1}(q, v) &= (q, \sigma(q, v)) \\ &= (q, \tau(q)(v)), \end{aligned}$$

we have that $\tau(q) \in GL(k, \mathbb{R})$. **need to show τ is smooth, use hint in book, make exercise in a previous section about actions** □

the fiber bundle construction theorems don't actually construct a fiber bundle, they just show that a given set is one and characterize the topology and smooth structure under some assumptions, maybe go back and rename them to "characterization theorem" and then actually have a construction theorem. then here, introduce a characterization theorem and then have a separate short construction theorem.

Exercise 16.1.1.5. Smooth Vector Bundle Chart Lemma:

Let $M \in \text{Obj}(\mathbf{Man}^\infty)$ and $(E_p)_{p \in M} \subset \text{Obj}(\mathbf{Vect}_{\mathbb{R}})$. Set $n := \dim M$. Suppose that for each $p \in M$, $\dim E_p = k$. We define $E \in \text{Obj}(\mathbf{Set})$ and $\pi \in \text{Hom}_{\mathbf{Set}}(E, M)$ by

$$E = \coprod_{p \in M} E_p$$

and $\pi(p, v) = p$. Let Γ an index set and for each $\alpha \in \Gamma$, $U_\alpha \subset M$ and $\Phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^k$. Set $n := \dim M$ and $k := \dim F$. Suppose that

1. for each $\alpha \in \Gamma$, $U_\alpha \in \mathcal{T}_M$
2. $M \subset \bigcup_{\alpha \in \Gamma} U_\alpha$
3. for each $\alpha \in \Gamma$, there exists $\Phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^k$ such that
 - $\Phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^k$ is a bijection
 - for each $q \in U_\alpha$, $\Phi_\alpha|_{E_q} : E_q \rightarrow \{q\} \times \mathbb{R}^k$ is a vector space isomorphism
4. for each $\alpha, \beta \in \Gamma$, there exists $\tau_{\alpha, \beta} : U_\alpha \cap U_\beta \rightarrow GL(k, \mathbb{R})$ such that
 - $\tau_{\alpha, \beta} : U_\alpha \cap U_\beta \rightarrow GL(k, \mathbb{R})$ is smooth
 - $\Phi_\alpha|_{\pi^{-1}(U_\alpha \cap U_\beta)} \circ (\Phi_\beta|_{\pi^{-1}(U_\alpha \cap U_\beta)})^{-1} : (U_\alpha \cap U_\beta) \times \mathbb{R}^k \rightarrow (U_\alpha \cap U_\beta) \times \mathbb{R}^k$ is given by $\Phi_\alpha|_{\pi^{-1}(U_\alpha \cap U_\beta)} \circ (\Phi_\beta|_{\pi^{-1}(U_\alpha \cap U_\beta)})^{-1}(p, v) = (p, \tau_{\alpha, \beta}(p)(v))$.

Then there exists a unique topology $\mathcal{T}_E$ on E and smooth structure $\mathcal{A}_E$ on $(E, \mathcal{T}_E)$ such that

1. $(E, \mathcal{T}_E)$ is an $(n + k)$ -dimensional topological manifold and $(E, \mathcal{T}_E, \mathcal{A}_E)$ is a smooth manifold
2. for each $\alpha \in \Gamma$, (U_α, Φ_α) is a diffeomorphism
3. $\pi : E \rightarrow M$ is smooth
4. (E, M, π) is a rank- k $\mathbf{Man}^\infty$ vector bundle.

Proof. Let $\alpha \in \Gamma$ and $a \in \pi^{-1}(U_\alpha)$. By definition, there exists $q \in U_\alpha$ and $v_0 \in E_q$ such that $a = (q, v_0)$. Since $\Phi_\alpha|_{E_q} : E_q \rightarrow \{q\} \times \mathbb{R}^k$ is a vector space isomorphism, there exists $v \in \mathbb{R}^k$ such that $\Phi_\alpha(q, v_0) = (q, v)$. Then

$$\begin{aligned} \text{proj}_1 \circ \Phi_\alpha(a) &= \text{proj}_1 \circ \Phi_\alpha(q, v_0) \\ &= \text{proj}_1(q, v) \\ &= q \\ &= \pi(q, v_0) \\ &= \pi(a). \end{aligned}$$

Since $a \in \pi^{-1}(U_\alpha)$ is arbitrary, we have that $\text{proj}_1 \circ \Phi_\alpha = \pi|_{\pi^{-1}(U_\alpha)}$. Therefore (U_α, Φ_α) is a local trivialization of E over U_α with fiber $\mathbb{R}^k$ with respect to π .

such that need to show that (U_α, Φ_α) smooth vector bundle local trivialization of E over U with fiber $\mathbb{R}^k$ with respect to π here using the cocycle condition. Let $\alpha \in A$.

1. By assumption, Φ_α is a bijection

2. $\text{proj}_1 \circ \Phi_\alpha = \pi|_{\pi^{-1}(U_\alpha)}$, i.e. the following diagram commutes:

$$\begin{array}{ccc} \pi^{-1}(U_\alpha) & \xrightarrow{\Phi_\alpha} & U_\alpha \times \mathbb{R}^k \\ & \searrow \pi & \downarrow \text{proj}_1 \\ & & U_\alpha \end{array}$$

then Exercise 15.1.3.3 implies that there exist a unique topology $\mathcal{T}_E$ on E and smooth structure $\mathcal{A}_E \subset X^{n+k}(M, \mathcal{T}_E)$ on E such that

1. $(E, \mathcal{T}_E)$ is an $n + k$ -dimensional topological manifold and $(E, \mathcal{T}_E, \mathcal{A}_E)$ is a smooth manifold,
2. for each $\alpha \in \Gamma$, $\pi^{-1}(U_\alpha) \in \mathcal{T}_E$ and $\Phi_\alpha : \pi^{-1}(U_\alpha) \rightarrow U_\alpha \times \mathbb{R}^k$ is a diffeomorphism,
3. $\pi : E \rightarrow M$ is smooth,
4. $(E, M, \pi, \mathbb{R}^k)$ is an $\mathbf{Man}^\infty$ fiber bundle.
 - As noted above, $(E, M, \pi, \mathbb{R}^k) \in \text{Obj}(\mathbf{Bun}^\infty)$.
 - Let $p \in M$, Clearly E_p is a k -dimensional real vector space. By assumption, there exists $\alpha \in \Gamma$ such that
 - (a) $p \in U_\alpha$.
 - (b) As noted above, (U_α, Φ_α) is a smooth local trivialization of E over U with fiber $\mathbb{R}^k$ with respect to π .
 - (c) Let $q \in U_\alpha$. By assumption, $\Phi|_{E_q} : E_q \rightarrow \{p\} \times \mathbb{R}^k$ is a vector space isomorphism.

FINISH!!!

□

Definition 16.1.1.6. Let (E_1, M_1, π_1) and (E_2, M_2, π_2) be rank- k_1 and rank- k_2 smooth vector bundles respectively, $(\Phi, \phi) \in \text{Hom}_{\mathbf{Bun}^\infty}((E_1, M_1, \pi_1, \mathbb{R}^{k_1}), (E_2, M_2, \pi_2, \mathbb{R}^{k_2}))$. Then (Φ, ϕ) is said to be a **smooth vector bundle morphism** from (E_1, M_1, π_1) to (E_2, M_2, π_2) if for each $p \in M_1$, $\Phi|_{(E_1)_p} : (E_1)_p \rightarrow (E_2)_{\phi(p)}$ is linear.

Definition 16.1.1.7. We define the category of smooth vector bundles, denoted $\mathbf{VecBun}^\infty$, by

- $\text{Obj}(\mathbf{VecBun}^\infty) := \{(E, M, \pi) : (E, M, \pi) \text{ is a smooth vector bundle}\}$
- For $(E_1, M_1, \pi_1), (E_2, M_2, \pi_2) \in \text{Obj}(\mathbf{VecBun}^\infty)$ with $\text{rank}(E_1, M_1, \pi_1) = k_1$ and $\text{rank}(E_2, M_2, \pi_2) = k_2$,

$$\begin{aligned} \text{Hom}_{\mathbf{VecBun}^\infty}((E_1, M_1, \pi_1), (E_2, M_2, \pi_2)) &:= \{(\Phi, \phi) \in \text{Hom}_{\mathbf{Bun}^\infty}((E_1, M_1, \pi_1, \mathbb{R}^{k_1}), (E_2, M_2, \pi_2, \mathbb{R}^{k_2})) : \\ &\quad (\Phi, \phi) \text{ is a smooth vector bundle morphism from} \\ &\quad (E_1, M_1, \pi_1) \text{ to } (E_2, M_2, \pi_2)\} \end{aligned}$$

Exercise 16.1.1.8. We have that $\mathbf{VecBun}^\infty$ is a subcategory of $\mathbf{Bun}^\infty$.

Proof. We note that

- $\text{Obj}(\mathbf{VecBun}^\infty) \subset \text{Obj}(\mathbf{Bun}^\infty)$
- for each $(E_1, M_1, \pi_1), (E_2, M_2, \pi_2) \in \text{Obj}(\mathbf{VecBun}^\infty)$ with $\text{rank}(E_1, M_1, \pi_1) = k_1$ and $\text{rank}(E_2, M_2, \pi_2) = k_2$,

$$\text{Hom}_{\mathbf{VecBun}^\infty}((E_1, M_1, \pi_1), (E_2, M_2, \pi_2)) = \text{Hom}_{\mathbf{Bun}^\infty}((E_1, M_1, \pi_1, \mathbb{R}^{k_1}), (E_2, M_2, \pi_2, \mathbb{R}^{k_2}))$$

FINISH!!!

So $\mathbf{Bun}^\infty$ is a subcategory of $\mathcal{C}$.

□

Exercise 16.1.1.9. Let $M \in \text{Obj}(\mathbf{Man}^\infty)$. Set $n := \dim M$, $E := M \times \mathbb{R}^k$ and define $\pi : E \rightarrow M$ by $\pi(p, x) := p$. Then (E, M, π) is a rank- k smooth vector bundle.

Proof.

1. For each $p \in M$, $E_p = \{p\} \times \mathbb{R}^k$ is an n -dimensional real vector space.
2. Let $p \in M$. Set $U = M$. Then $\pi^{-1}(U) = E$. Define $\Phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^k$ by $\Phi = \text{id}_E$. Then (U, Φ) is a smooth local trivialization of E over U .
3. Let $p \in M$. Then $\Phi|_{E_p} : E_p \rightarrow \{p\} \times \mathbb{R}^k$ is clearly an isomorphism.

□

16.1.2 Subbundles

Definition 16.1.2.1. Let $(E, M, \pi_E), (D, M, \pi_D) \in \text{Obj}(\mathbf{VecBun}^\infty)$. Then (D, M, π_D) is said to be a **subbundle of** (E, M, π_E) if

1. D is an embedded submanifold of E
2. $\pi_E|_D = \pi_D$
3. for each $p \in M$, D_p is a subspace of E_p .

Exercise 16.1.2.2. Local Frame Criterion:

FINISH!!!

16.1.3 Direct Sum Bundles

Definition 16.1.3.1. Let $(E_1, M, \pi_1), (E_2, M, \pi_2) \in \text{Obj}(\mathbf{VecBun}^\infty)$. We define the **tensor product of** (E_1, M, π_1) and (E_2, M, π_2) , denoted $(E_1 \otimes E_2, M, \pi)$, by

16.1.4 Tensor Product Bundles

Definition 16.1.4.1. Let $(E_1, M, \pi_1), (E_2, M, \pi_2) \in \text{Obj}(\mathbf{VecBun}^\infty)$. Set

$$\bullet \quad E_1 \otimes E_2 := \coprod_{p \in M} (E_1)_p \otimes (E_2)_p$$

$$\bullet \quad \pi : E_1 \otimes E_2 \rightarrow M \text{ by} \quad \pi(p, v) = p$$

We define the **tensor product bundle of** (E_1, M, π_1) and (E_2, M, π_2) , denoted $(E_1 \otimes E_2, M, \pi)$.

16.1.5 Hom Bundles

Definition 16.1.5.1. Let $(E_1, M, \pi_1), (E_2, M, \pi_2) \in \text{Obj}(\mathbf{VecBun}^\infty)$. Set

$$\bullet \quad \text{Hom}(E_1, E_2) := \coprod_{p \in M} L((E_1)_p, (E_2)_p)$$

$$\bullet \quad \pi : E_1 \otimes E_2 \rightarrow M \text{ by} \quad \pi(p, v) = p$$

We define the **Hom bundle of** (E_1, M, π_1) and (E_2, M, π_2) , denoted $(\text{Hom}(E_1, E_2), M, \pi)$, by $\text{Hom}(E_1, E_2)$.

need to show the hom and tensor bundles are bundle isomorphic, then use that to define a covariant derivative from a connection

Chapter 17

The Tangent and Cotangent Bundle

17.1 The Tangent Bundle

Definition 17.1.0.1. We define the **tangent bundle of M** , denoted TM , by

$$TM = \coprod_{p \in M} T_p M$$

We denote the natural projection map by $\pi : TM \rightarrow M$.

Definition 17.1.0.2. Let $(U, \phi) \in \mathcal{A}$ with $\phi = (x^1, \dots, x^n)$. Define $\tilde{U} \subset TM$ and $\tilde{\phi} : \tilde{U} \rightarrow \phi(U) \times \mathbb{R}^n$ by

- $\tilde{U} = \pi^{-1}(U)$
-

$$\begin{aligned} \tilde{\phi} \left(\sum_{i=1}^n v^i \frac{\partial}{\partial x^i} \Big|_p \right) &= (\phi(p), v) \\ &= (x^1(p), \dots, x^n(p), v^1, \dots, v^n) \end{aligned}$$

Exercise 17.1.0.3. Let $(U, \phi) \in \mathcal{A}$ with $\phi = (x^1, \dots, x^n)$. Then $\tilde{\phi} : \tilde{U} \rightarrow \phi(U) \times \mathbb{R}^n$ is a bijection.

17.2 The cotangent Bundle

Definition 17.2.0.1. We define the **cotangent bundle of M** , denoted T^*M , by

$$T^*M = \coprod_{p \in M} T_p^*M$$

17.3 The (r, s) -Tensor Bundle

Definition 17.3.0.1. 1. the **cotangent bundle of M** , denoted T^*M , by

$$T^*M = \coprod_{p \in M} T_p^*M$$

2. the **(r, s) -tensor bundle of M** , denoted $T_s^r M$, by

$$T_s^r M = \coprod_{p \in M} T_s^r(T_p M)$$

3. the **k -alternating tensor bundle of M** , denoted $\Lambda^k(M)$, by

$$\Lambda^k M = \coprod_{p \in M} \Lambda^k(T_p M)$$

17.4 Vector Fields

Definition 17.4.0.1. Let $X : M \rightarrow TM$. Then X is said to be a **vector field on M** if for each $p \in M$, $X_p \in T_p M$. For $f \in \mathbb{C}^\infty(M)$, we define $Xf : M \rightarrow \mathbb{R}$ by

$$(Xf)_p = X_p(f)$$

and X is said to be **smooth** if for each $f \in \mathbb{C}^\infty(M)$, Xf is smooth.

We denote the set of smooth vector fields on M by $\Gamma^1(M)$.

Exercise 17.4.0.2.

17.5 (r, s) -Tensor Fields

Definition 17.5.0.1. Let $\alpha : M \rightarrow T_s^r M$. Then α is said to be an (r, s) -**tensor field on M** if for each $p \in M$, $\alpha_p \in T_p^r(T_p M)$. For each $\omega \in \Gamma_1(M)^r$ and $X \in \Gamma^1(M)^s$, we define $\alpha(\omega, X) : M \rightarrow \mathbb{R}$ by

$$\alpha(\omega, X)_p = \alpha_p(\omega_p, X_p)$$

and α is said to be **smooth** if for each $\omega \in \Gamma_1(M)^r$ and $X \in \Gamma^1(M)^s$, $\alpha(\omega, X)$ is smooth. The set of smooth (r, s) -tensor fields on M is denoted $T_s^r(M)$.

Definition 17.5.0.2. Let $f \in C^\infty(M)$ and $\alpha, \beta \in T_s^r(M)$. We define

- $f\alpha : M \rightarrow T_s^r M$ by

$$(f\alpha)_p = f(p)\alpha_p$$

- $\alpha + \beta : M \rightarrow T_s^r M$ by

$$(\alpha + \beta)_p = \alpha_p + \beta_p$$

Exercise 17.5.0.3. Let $f \in C^\infty(M)$ and $\alpha, \beta \in T_s^r(M)$. Then

1. $f\alpha \in T_s^r(M)$ by

$$(f\alpha)_p = f(p)\alpha_p$$

2. $\alpha + \beta \in T_s^r(M)$ by

$$(\alpha + \beta)_p = \alpha_p + \beta_p$$

Proof. Clear. □

Exercise 17.5.0.4. The set $T_s^r(M)$ is a $C^\infty(M)$ -module. □

Proof. Clear. □

Definition 17.5.0.5. Let $\alpha_1 \in \Gamma_{s_1}^{r_1}(M)$ and $\alpha_2 \in \Gamma_{s_2}^{r_2}(M)$. We define the **tensor product of α with β** , denoted $\alpha \otimes \beta : M \rightarrow T_{s_1+s_2}^{r_1+r_2} M$, by

$$(\alpha \otimes \beta)_p = \alpha_p \otimes \beta_p$$

Exercise 17.5.0.6. Let $\alpha_1 \in \Gamma_{s_1}^{r_1}(M)$ and $\alpha_2 \in \Gamma_{s_2}^{r_2}(M)$. Then $\alpha_1 \otimes \alpha_2 \in \Gamma_{s_1+s_2}^{r_1+r_2}(M)$

Proof. Let $\omega_1 \in \Gamma_1(M)^{r_1}$, $\omega_2 \in \Gamma_1(M)^{r_2}$, $X_1 \in \Gamma^1(M)^{s_1}$ and $X_2 \in \Gamma^1(M)^{s_2}$. By definition,

$$\alpha_1 \otimes \alpha_2(\omega_1, \omega_2, X_1, X_2) = \alpha_1(\omega_1, X_1)\alpha_2(\omega_2, X_2)$$

This implies that $\alpha_1 \otimes \alpha_2$ is smooth since α_1 and α_2 are smooth by assumption. □

Definition 17.5.0.7. We define the **tensor product**, denoted $\otimes : \Gamma_{s_1}^{r_1}(M) \times \Gamma_{s_2}^{r_2}(M) \rightarrow \Gamma_{s_1+s_2}^{r_1+r_2}(M)$ by

$$(\alpha_1, \alpha_2) \mapsto \alpha_1 \otimes \alpha_2$$

Exercise 17.5.0.8. The tensor product $\otimes : \Gamma_{s_1}^{r_1}(M) \times \Gamma_{s_2}^{r_2}(M) \rightarrow \Gamma_{s_1+s_2}^{r_1+r_2}(M)$ is associative. □

Proof. Clear. □

Exercise 17.5.0.9. The tensor product $\otimes : \Gamma_{s_1}^{r_1}(M) \times \Gamma_{s_2}^{r_2}(M) \rightarrow \Gamma_{s_1+s_2}^{r_1+r_2}(M)$ is $C^\infty(M)$ -bilinear. □

Proof. Clear. □

Definition 17.5.0.10. Let $(N, \mathcal{B})$ be a smooth manifold, $F : M \rightarrow N$ a smooth map and $\alpha \in \Gamma_k^0(N)$. We define the **pullback of α by F** , denoted $F^*\alpha \in \Gamma_k^0(M)$, by

$$(F^*\alpha)_p(v_1, \dots, v_k) = \alpha_{F(p)}(DF_p(v_1), \dots, DF_p(v_k))$$

for $p \in M$ and $v_1, \dots, v_k \in T_p M$

Exercise 17.5.0.11. Let $(M, \mathcal{A})$, $(N, \mathcal{B})$ and $(L, \mathcal{C})$ be smooth manifolds, $F : M \rightarrow N$ and $G : N \rightarrow L$ smooth maps, $\alpha \in \Gamma_k^0(N)$, $\beta \in \Gamma_l^0(N)$, $\gamma \in \Gamma_k^0(L)$ and $f \in C^\infty(N)$. Then

1. $F^*(f\alpha) = (f \circ F)F^*\alpha$
2. $F^*(\alpha \otimes \beta) = F^*\alpha \otimes F^*\beta$
3. $F^*(\alpha + \beta) = F^*\alpha + F^*\beta$
4. $(G \circ F)^*\gamma = F^*(G^*\gamma)$
5. $id_N^*\alpha = \alpha$

Proof.

1.

$$\begin{aligned} [F^*(f\alpha)]_p(v_1, \dots, v_k) &= (f\alpha)_{F(p)}(DF_p(v_1), \dots, DF_p(v_k)) \\ &= f(F(p))\alpha_{F(p)}(DF_p(v_1), \dots, DF_p(v_k)) \\ &= (f \circ F)(p)(F^*\alpha)_p(v_1, \dots, v_k) \end{aligned}$$

So that $F^*(f\alpha) = (f \circ F)F^*\alpha$

2.

$$F^*$$

□

Definition 17.5.0.12.

Exercise 17.5.0.13.

Proof.

□

Exercise 17.5.0.14. Let $\alpha \in T_s^r(M)$ and $(U, \phi) \in \mathcal{A}$ with $\phi = (x^1, \dots, x^n)$. Then there exist $(f_J^I)_{I \in \mathcal{I}_r, J \in \mathcal{I}_s} \subset C^\infty(M)$ such that

$$\alpha|_U = \sum_{(I, J) \in \mathcal{I}_r \times \mathcal{I}_s} f_J^I \partial_{x^{\otimes I}} \otimes dx^{\otimes J}$$

Proof. Let $p \in M$. Then $\omega_p \in T_s^r(T_p M)$ and $\left\{ \partial_{x^{\otimes I}}|_p \otimes dx_p^{\otimes J} \right\}$ is a basis of $T_s^r(T_p M)$. So there exist $(f_J^I(p))_{I \in \mathcal{I}_r, J \in \mathcal{I}_s} \subset \mathbb{R}$ such that

$$\omega_p = \sum_{(I, J) \in \mathcal{I}_r \times \mathcal{I}_s} f_J^I(p) \partial_{x^{\otimes I}}|_p \otimes dx_p^{\otimes J}$$

Let $(K, L) \in \mathcal{I}_r \times \mathcal{I}_s$. Then

$$\begin{aligned} \alpha_p(dx_p^K, \partial_{x^L}|_p) &= \sum_{(I, J) \in \mathcal{I}_r \times \mathcal{I}_s} f_J^I(p) \partial_{x^{\otimes I}}|_p \otimes dx_p^{\otimes J}(dx_p^K, \partial_{x^L}|_p) \\ &= \sum_{(I, J) \in \mathcal{I}_r \times \mathcal{I}_s} f_J^I(p) \partial_{x^{\otimes I}}|_p(dx_p^K) dx_p^{\otimes J}(\partial_{x^L}|_p) \\ &= f_L^K(p) \end{aligned}$$

By assumption, the map $p \mapsto \alpha(dx^K, \partial_{x^L})_p$ is smooth, so that $f_L^K \in C^\infty(U)$.

□

Definition 17.5.0.15.

17.6 Differential Forms

Definition 17.6.0.1. We define

$$\Lambda^k(TM) = \coprod_{p \in M} \Lambda^k(T_p M)$$

Definition 17.6.0.2. Let $\omega : M \rightarrow \Lambda^k(TM)$. Then ω is said to be a **k -form on M** if for each $p \in M$, $\omega_p \in \Lambda^k(T_p M)$. For each $X \in \Gamma^1(M)^k$, we define $\omega(X) : M \rightarrow \mathbb{R}$ by

$$\omega(X)_p = \omega_p(X_p)$$

and ω is said to be **smooth** if for each $X \in \Gamma^1(M)^k$, $\omega(X)$ is smooth. The set of smooth k -forms on M is denoted $\Omega^k(M)$.

Note 17.6.0.3. Observe that

1. $\Omega^k(M) \subset \Gamma_k^0(M)$
2. $\Omega^0(M) = C^\infty(M)$

Exercise 17.6.0.4. The set $\Omega^k(M)$ is a $C^\infty(M)$ -submodule of $\Gamma_k^0(M)$.

Proof. Clear. □

Definition 17.6.0.5. Define the **exterior product**

$$\wedge : \Omega^k(M) \times \Omega^l(M) \rightarrow \Omega^{k+l}(M)$$

by

$$(\alpha \wedge \beta)_p = (\alpha)_p \wedge (\beta)_p$$

Note 17.6.0.6. For $f \in \Omega^0(M)$ and $\alpha \in \Omega^k(M)$, we have that $f \wedge \alpha = f\alpha$.

Exercise 17.6.0.7. The exterior product $\wedge : \Omega^k(M) \times \Omega^l(M) \rightarrow \Omega^{k+l}(M)$ is well defined.

Proof. Let $\alpha \in \Omega^k(M)$, $\beta \in \Omega^l(M)$, $(x^i)_{i=1}^k \subset \Gamma^1(M)$, $(y^j)_{j=1}^l \subset \Gamma^1(M)$ and $p \in M$. Then

$$\begin{aligned} \alpha \wedge \beta(X_1, \dots, X_{k+l})_p &= (\alpha \wedge \beta)_p(X_1(p), \dots, X_{k+l}(p)) \\ &= \frac{(k+l)!}{k!l!} \text{Alt}(\alpha_p \otimes \beta_p)(X_1(p), \dots, X_{k+l}(p)) \\ &= \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \sigma(\alpha_p \otimes \beta_p)(X_1(p), \dots, X_{k+l}(p)) \\ &= \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) (\alpha_p \otimes \beta_p)(X_{\sigma(1)}(p), \dots, X_{\sigma(k+l)}(p)) \\ &= \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \alpha_p(X_{\sigma(1)}(p), \dots, X_{\sigma(k)}(p)) \beta(X_{\sigma(k+1)}(p), \dots, X_{\sigma(k+l)}(p)) \\ &= \frac{1}{k!l!} \sum_{\sigma \in S_{k+l}} \text{sgn}(\sigma) \alpha_p(X_{\sigma(1)}(p), \dots, X_{\sigma(k)}(p)) \beta(X_{\sigma(k+1)}(p), \dots, X_{\sigma(k+l)}(p)) \end{aligned}$$

□

Exercise 17.6.0.8. The exterior product $\wedge : \Omega^k(M) \times \Omega^l(M) \rightarrow \Omega^{k+l}(M)$ is $C^\infty(M)$ -bilinear.

Proof.

1. $C^\infty(M)$ -linearity in the first argument:

Let $\alpha \in \Omega^k(M)$, $\beta, \gamma \in \Omega^l(M)$, $f \in C^\infty(M)$ and $p \in M$. Bilinearity of $\wedge : \Lambda^k(T_p M) \times \Lambda^l(T_p M) \rightarrow \Lambda^{k+l}(T_p M)$ implies that

$$\begin{aligned} [(\beta + f\gamma) \wedge \alpha]_p &= (\beta + f\gamma)_p \wedge \alpha_p \\ &= (\beta_p + f(p)\gamma_p) \wedge \alpha_p \\ &= \beta_p \wedge \alpha_p + f(p)(\gamma_p \wedge \alpha_p) \\ &= [\beta \wedge \alpha + f(\gamma \wedge \alpha)]_p \end{aligned}$$

So that

$$(\beta + f\gamma) \wedge \alpha = \beta \wedge \alpha + f(\gamma \wedge \alpha)$$

and $\wedge : \Omega^k(M) \times \Omega^l(M) \rightarrow \Omega^{k+l}(M)$ is $C^\infty(M)$ -linear in the first argument.

2. $C^\infty(M)$ -linearity in the second argument:

Similar to (1).

□

Note 17.6.0.9. All of the results from multilinear algebra apply here.

Definition 17.6.0.10. We define the **exterior derivative** $d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$ inductively by

1. $d(d\alpha) = 0$ for $\alpha \in \Omega^p(M)$
2. $df(X) = Xf$ for $f \in \Omega^0(M)$
3. $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta$ for $\alpha \in \Omega^p(M)$ and $\beta \in \Omega^q(M)$
4. extending linearly

Exercise 17.6.0.11. Let (U, ϕ) be a chart on M with $\phi = (x^1, \dots, x^n)$. Then on U , for each $i, j \in \{1, \dots, n\}$,

$$dx^i \left(\frac{\partial}{\partial x^j} \right) = \delta_{i,j}$$

In particular, for each $p \in U$, $\{dx_p^1, \dots, dx_p^n\}$ is the dual basis to $\left\{ \frac{\partial}{\partial x^1} \Big|_p, \dots, \frac{\partial}{\partial x^n} \Big|_p \right\}$ and $T_p^* M = \text{span}\{dx_p^1, \dots, dx_p^n\}$.

Proof. Let $p \in U$ and $i, j \in \{1, \dots, n\}$. Then by definition,

$$\begin{aligned} \left[dx^i \left(\frac{\partial}{\partial x^j} \right) \right]_p &= \left(\frac{\partial}{\partial x^j} x^i \right)_p \\ &= \frac{\partial}{\partial x^j} \Big|_p x^i \\ &= \delta_{i,j} \end{aligned}$$

□

Exercise 17.6.0.12. Let $f \in C^\infty(M)$ and (U, ϕ) be a chart on M with $\phi = (x^1, \dots, x^n)$. Then

$$df|_U = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i$$

Proof. Let $p \in U$. Since $\{dx^1, \dots, dx^n\}$ is a basis for $\Lambda(T_p M)$, for each there exist $a_1(p), \dots, a_n(p) \in \mathbb{R}$ such that $df_p = \sum_{i=1}^n a_i(p) dx_p^i$. Therefore, we have that

$$\begin{aligned} df_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) &= \sum_{i=1}^n a_i(p) dx_p^i \left(\frac{\partial}{\partial x^i} \Big|_p \right) \\ &= a_j(p) \end{aligned}$$

By definition, we have that

$$\begin{aligned} df_p \left(\frac{\partial}{\partial x^i} \Big|_p \right) &= \frac{\partial}{\partial x^i} \Big|_p f \\ &= \frac{\partial f}{\partial x^j}(p) \end{aligned}$$

So $a_j(p) = \frac{\partial f}{\partial x^j}(p)$ and

$$df_p = \sum_{i=1}^n \frac{\partial f}{\partial x^j}(p) dx_p^i$$

Therefore

$$df|_U = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i$$

□

Exercise 17.6.0.13. Let $f \in \Omega^0(M)$. If f is constant, then $df = 0$.

Proof. Suppose that f is constant. Let $p \in M$. Choose $(U, \phi) \in \mathcal{A}$ such that $p \in U$. Write $\phi = (x_1, \dots, x_n)$. Then for each $i \in \{1, \dots, n\}$,

$$\frac{\partial}{\partial x^i} \Big|_p f = 0$$

This implies that

$$\begin{aligned} df_p &= \sum_{i=1}^n \frac{\partial f}{\partial x^j}(p) dx_p^i \\ &= 0 \end{aligned}$$

□

Exercise 17.6.0.14.

Definition 17.6.0.15. Let $(U, \phi) \in \mathcal{A}$ with $\phi = (x^1, \dots, x^n)$ and $I = (i_1, \dots, i_k) \in \mathcal{I}_n^{\wedge k}$. We define

$$dx^i = dx^{i_1} \wedge \dots \wedge dx^{i_k} \in \Omega^k(M)$$

and we define

$$\frac{\partial}{\partial x^i} = \left(\frac{\partial}{\partial x^{i_1}}, \dots, \frac{\partial}{\partial x^{i_k}} \right)$$

Note 17.6.0.16. We have that

1.

$$dx^i \left(\frac{\partial}{\partial x^j} \right) = \delta_{I,J}$$

2. Since $\frac{\partial}{\partial x^i} \in \Gamma(U)^k$, by definition, for each $\omega \in \Omega^k(U)$,

$$\omega\left(\frac{\partial}{\partial x^i}\right) \in C^\infty(U)$$

Exercise 17.6.0.17. Let $\omega \in \Omega^k(M)$ and (U, ϕ) be a chart on M with $\phi = (x^1, \dots, x^n)$. Then

$$\omega = \sum_{I \in \mathcal{I}_n^{\wedge k}} \omega\left(\frac{\partial}{\partial x^i}\right) dx^i$$

Proof. Let $p \in U$. Since $\{dx_p^i : I \in \mathcal{I}_n^{\wedge k}\}$ is a basis for $\Lambda^k(T_p M)$, there exists $(f_I(p))_{I \in \mathcal{I}} \subset \mathbb{R}$ such that $\omega_p = \sum_{I \in \mathcal{I}_n^{\wedge k}} f_I(p) dx_p^i$. So for each $J \in \mathcal{I}_n^{\wedge k}$,

$$\begin{aligned} \omega\left(\frac{\partial}{\partial x^j}\right) &= \sum_{I \in \mathcal{I}_n^{\wedge k}} f_I dx^i \left(\frac{\partial}{\partial x^j}\right) \\ &= f_J \end{aligned}$$

□

Exercise 17.6.0.18. Let $\omega \in \Omega^k(M)$ and (U, ϕ) be a chart on M with $\phi = (x^1, \dots, x^n)$. If $\omega = \sum_{I \in \mathcal{I}_n^{\wedge k}} f_I dx^i$, then

$$d\omega = \sum_{I \in \mathcal{I}_n^{\wedge k}} \sum_{i=1}^n \frac{\partial f_I}{\partial x^i} dx^i \wedge dx^i$$

.

Proof. First we note that

$$\begin{aligned} d(f_I dx^i) &= df_I \wedge dx^i + (-1)^0 f_I d(dx^i) \\ &= df_I \wedge dx^i \\ &= \left(\sum_{i=1}^n \frac{\partial f_I}{\partial x^i} dx^i \right) \wedge dx^i \\ &= \sum_{i=1}^n \frac{\partial f_I}{\partial x^i} dx^i \wedge dx^i \end{aligned}$$

Then we extend linearly.

□

Definition 17.6.0.19. Let $(N, \mathcal{B})$ be a smooth manifold and $F : M \rightarrow N$ be a diffeomorphism. Define the **pullback of F** , denoted $F^* : \Omega^k(N) \rightarrow \Omega^k(M)$ by

$$(F^*\omega)_p(v_1, \dots, v_k) = \omega_{F(p)}(DF_p(v_1), \dots, DF_p(v_k))$$

for $\omega \in \Omega^k(N)$, $p \in M$ and $v_1, \dots, v_k \in T_p M$

17.7 Vector Bundle Valued Differential Forms

change notation in earlier sections so that $\Lambda^k(V^*)$ is k -forms instead of $\Lambda^k(V)$

Definition 17.7.0.1. Let $(E, M, \pi) \in \text{Obj}(\mathbf{VecBun}^\infty)$. For each $k \in \mathbb{N}_0$, we define the E -valued k -forms on M , denoted $\Omega^k(M; E)$ by $\Omega^k(M; E) := \Gamma(\Lambda^k T^* M \otimes E)$.

Note 17.7.0.2. Let $(E, M, \pi) \in \text{Obj}(\mathbf{VecBun}^\infty)$ and $V \in \text{Obj}(\mathbf{Vect}_\mathbb{R})$. Then we write $\Omega^k(M; V)$ in place of $\Omega^k(M; M \times V)$.

Chapter 18

The Tangent Bundle

18.1 The Tangent Bundle

Definition 18.1.0.1. Let $(M, \mathcal{A}_M)$ be an n -dimensional smooth manifold. We define the **tangent bundle of M** , denoted TM , by

$$TM = \coprod_{p \in M} T_p M$$

and we define the **tangent bundle projection**, denoted $\pi : TM \rightarrow M$, by

$$\pi(p, v) = p$$

Let $(U, \phi) \in \mathcal{A}_M$ with $\phi = (x^1, \dots, x^n)$. We define $\Phi_\phi : \pi^{-1}(U) \rightarrow \mathbb{R}^{2n}$ by

$$\Phi_\phi \left(p, \sum_{j=1}^n \xi^j \frac{\partial}{\partial x^j} \Big|_p \right) = (\phi(p), \xi^1, \dots, \xi^n)$$

We define $\mathcal{T}_{TM} = \tau_{TM}(\iota_p : p \in M)$.

Exercise 18.1.0.2. $\psi : \bigcup_{p \in U} T_p M \rightarrow \mathbb{R}^n$ is given by

$$\psi \left(\sum_{j=1}^n \xi^j \frac{\partial}{\partial x^j} \Big|_p \right) = (\xi^1, \dots, \xi^n)$$

$$\begin{aligned} x^k \circ \pi \circ \Phi_\phi^{-1}(u, v) &= x^k \circ \pi(\phi^{-1}(u), \psi^{-1}(v)) \\ &= x^k \circ \phi^{-1}(u) \end{aligned}$$

Therefore

$$\begin{aligned} \frac{\partial}{\partial \tilde{x}^i} \Big|_{(p, \xi)} [x^k \circ \pi] &= \frac{\partial}{\partial u^i} \Big|_{\Phi_\phi(p, \xi)} [x^k \circ \pi \circ \Phi_\phi^{-1}] \\ &= \frac{\partial}{\partial u^i} \Big|_{(\phi(p), \psi(\xi))} [x^k \circ \pi \circ \Phi_\phi^{-1}] \\ &= \frac{\partial}{\partial u^i} \Big|_{\phi(p)} [x^k \circ \phi^{-1}] \\ &= \frac{\partial}{\partial x^i} \Big|_p x^k \\ &= \delta_{i,k} \end{aligned}$$

and

$$\begin{aligned}
 \left. \frac{\partial}{\partial \tilde{y}^i} \right|_{(p, \xi)} [x^k \circ \pi] &= \left. \frac{\partial}{\partial v^i} \right|_{\Phi_\phi(p, \xi)} [x^k \circ \pi \circ \Phi_\phi^{-1}] \\
 &= \left. \frac{\partial}{\partial v^i} \right|_{(\phi(p), \psi(\xi))} [x^k \circ \pi \circ \Phi_\phi^{-1}] \\
 &= \left. \frac{\partial}{\partial v^i} \right|_{\phi(p)} [x^k \circ \phi^{-1}] \\
 &= 0
 \end{aligned}$$

This implies that for each $i \in \{1, \dots, n\}$, we have that

$$\begin{aligned}
 D\pi(p, \xi) \left(\left. \frac{\partial}{\partial \tilde{x}^i} \right|_{(p, \xi)} \right) (f) &= \left. \frac{\partial}{\partial \tilde{x}^i} \right|_{(p, \xi)} f \circ \pi \\
 &= \sum_{k=1}^n \frac{\partial f}{\partial x^k}(\pi(p, \xi)) \frac{\partial x^k \circ \pi}{\partial \tilde{x}^i}(p, \xi) \\
 &= \sum_{k=1}^n \frac{\partial f}{\partial x^k}(p) \delta_{i,k} \\
 &= \frac{\partial f}{\partial x^i}(p)
 \end{aligned}$$

and

$$\begin{aligned}
 D\pi(p, \xi) \left(\left. \frac{\partial}{\partial \tilde{y}^i} \right|_{(p, \xi)} \right) (f) &= \left. \frac{\partial}{\partial \tilde{y}^i} \right|_{(p, \xi)} f \circ \pi \\
 &= \sum_{k=1}^n \frac{\partial f}{\partial x^k}(\pi(p, \xi)) \frac{\partial x^k \circ \pi}{\partial \tilde{y}^i}(p, \xi) \\
 &= \sum_{k=1}^n \frac{\partial f}{\partial x^k}(p) 0 \\
 &= 0
 \end{aligned}$$

Hence

$$\begin{aligned}
 V(TM)|_{\pi^{-1}(U)} &= \coprod_{(p, \xi) \in \pi^{-1}(U)} \ker D\pi(p, \xi) \\
 &= \coprod_{(p, \xi) \in \pi^{-1}(U)} \text{span} \left\{ \left. \frac{\partial}{\partial \tilde{y}^j} \right|_{(p, \xi)} : j \in \{1, \dots, n\} \right\}
 \end{aligned}$$

Definition 18.1.0.3. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. We define the **push-forward of F** , denoted $F_* : TM \rightarrow TN$, by $F_*(p, v) = (F(p), DF(p)(v))$.

Exercise 18.1.0.4. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. Then $F_* \in \text{Hom}_{\mathbf{Man}^\infty}(TM, TN)$.

Proof.

□

Definition 18.1.0.5. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. We define the **tangent functor**, denoted $T : \mathbf{Man}^\infty \rightarrow \mathbf{Man}^\infty$, by

- $T(M) = TM$
- $TF = F_*$

Exercise 18.1.0.6. Let $M, N \in \text{Obj}(\mathbf{Man}^\infty)$ and $F \in \text{Hom}_{\mathbf{Man}^\infty}(M, N)$. Then $T : \mathbf{Man}^\infty \rightarrow \mathbf{Man}^\infty$ is a functor.

Proof. content...

□

18.2 Vector Fields

Exercise 18.2.0.1.

Part V

Differential Geometry

Chapter 19

Principle Bundles

19.1 Introduction

define $\triangleleft$ -invariance and $(\triangleleft_1, \triangleleft_2)$ -equivariance

Definition 19.1.0.1. Let X be a set and G a group. We define the **trivial right action on $X \times G$** , denoted $\triangleleft_{X \times G}^{\text{Triv}} : (X \times G) \times G \rightarrow X \times G$, by

$$(x, g) \triangleleft_{X \times G}^{\text{Triv}} h = (x, gh)$$

Exercise 19.1.0.2. Let $(P, X, \pi, G) \in \text{Obj}(\mathbf{Bun}^\infty)$ and $\triangleleft \in \text{Hom}_{\mathbf{Man}^\infty}(P \times G, P)$. Suppose that $\triangleleft$ is a right group action. Then π is $\triangleleft$ -invariant iff for each $x \in X$, $P_x \triangleleft G = P_x$.

Proof.

- $(\implies)$:
Suppose that π is $\triangleleft$ -invariant. Let $x \in X$, $p \in P_x$ and $g \in G$. Then

$$\begin{aligned} \pi(p \triangleleft g) &= \pi(p) \\ &= x. \end{aligned}$$

Hence $p \triangleleft g \in P_x$. Since $p \in P_x$ and $g \in G$ are arbitrary, we have that $P_x \triangleleft G \subset P_x$. Let $p \in P_x$. Then

$$\begin{aligned} p &= p \triangleleft e \\ &\in P_x \triangleleft G. \end{aligned}$$

Since $p \in P_x$ is arbitrary, we have that $P_x \subset P_x \triangleleft G$. Hence $P_x \triangleleft G = P_x$. Since $x \in X$ is arbitrary, we have that for each $x \in X$, $P_x \triangleleft G = P_x$.

- $(\impliedby)$:
Conversely, suppose that for each $x \in X$, $P_x \triangleleft G = P_x$. Let $p \in P$ and $g \in G$. Set $x := \pi(p)$. Since $p \in P_x$, by assumption, we have that

$$\begin{aligned} p \triangleleft g &\in P_x \triangleleft G \\ &= P_x. \end{aligned}$$

Therefore

$$\begin{aligned} \pi(p \triangleleft g) &= x \\ &= \pi(p). \end{aligned}$$

Since $p \in P$ and $g \in G$ are arbitrary, we have that for each $p \in P$ and $g \in G$, $\pi(p \triangleleft g) = \pi(p)$. Hence π is $\triangleleft$ -invariant. □

Definition 19.1.0.3. Let $(P, X, \pi, G) \in \text{Obj}(\mathbf{Bun}^\infty)$ and $\triangleleft \in \text{Hom}_{\mathbf{Man}^\infty}(P \times G, P)$. Suppose that

- G is a Lie group
- $\triangleleft$ a right group action
- π is $\triangleleft$ -invariant.

For each $x \in X$, we define the **right action of G on P_x induced by $\triangleleft$** , denoted $\triangleleft_x$, by $\triangleleft_x := \triangleleft|_{P_x \times G}$.

Exercise 19.1.0.4. Let $(P, X, \pi, G) \in \text{Obj}(\mathbf{Bun}^\infty)$ and $\triangleleft \in \text{Hom}_{\mathbf{Man}^\infty}(P \times G, P)$. Suppose that

- G is a Lie group
- $\triangleleft$ a right group action
- π is $\triangleleft$ -invariant.

Then for each $x \in X$, $\triangleleft_x : P_x \times G \rightarrow P_x$ is a smooth group action.

Proof. Let $x \in X$, $g, h \in G$ and $p \in P_x$.

- Then

$$\begin{aligned} p \triangleleft_x (gh) &= p \triangleleft (gh) \\ &= (p \triangleleft g) \triangleleft h \\ &= (p \triangleleft_x g) \triangleleft_x h \end{aligned}$$

and

$$\begin{aligned} p \triangleleft_x e &= p \triangleleft e \\ &= p. \end{aligned}$$

Since $g, h \in G$ and $p \in P_x$ is arbitrary, we have that $\triangleleft_x$ is a group action.

- Since π is a surjective submersion,

FINISH!!!, need previous exercise showing P_x is a smooth embedded submanifold of P in a fiber bundle and therefore the restriction of a smooth map to a smooth embedded submanifold is smooth.

□

Definition 19.1.0.5. Let $P, X, G \in \text{Obj}(\mathbf{Man}^\infty)$, $\pi \in \text{Hom}_{\mathbf{Man}^\infty}(P, X)$ a surjection, $\triangleleft \in \text{Hom}_{\mathbf{Man}^\infty}(P \times G, P)$, $U \in \mathcal{T}_X$ and $\Phi \in \text{Hom}_{\mathbf{Man}^\infty}(\pi^{-1}(U), U \times G)$. Suppose that

- G is a Lie Group,
- $\triangleleft$ is a right group action,
- π is $\triangleleft$ -invariant.

Then (U, Φ) is said to be a **smooth principle bundle local trivialization of P over U with respect to π and $\triangleleft$** if

1. (U, Φ) is a smooth fiber bundle local trivialization of P over U with fiber G with respect to π
2. Φ is $(\triangleleft, \triangleleft_{U \times G}^{\text{Triv}})$ -equivariant

Definition 19.1.0.6. Let $P, X, G \in \text{Obj}(\mathbf{Man}^\infty)$ and $\pi \in \text{Hom}_{\mathbf{Man}^\infty}(P, X)$ a surjection and $\triangleleft \in \text{Hom}_{\mathbf{Man}^\infty}(P \times G, P)$. Suppose that

- G is a Lie Group,
- $\triangleleft$ is a right group action.

Then $(P, X, \pi, G, \triangleleft)$ is said to be a **Man[∞]** principle bundle with total space P , base space X , structure group G , projection π and action $\triangleleft$ if

1. $(P, X, \pi, G) \in \text{Obj}(\mathbf{Bun}^\infty)$,
2. π is $\triangleleft$ -invariant,
3. for each $x \in X$,
 - (a) $\triangleleft_x : P_x \times G \rightarrow P_x$ is transitive and free,
 - (b) there exists $U \in \mathcal{T}_X$ and $\Phi \in \text{Hom}_{\mathbf{Man}^\infty}(\pi^{-1}(U), U \times G)$ such that (U, Φ) is a smooth principle bundle local trivialization of P over U with respect to π and $\triangleleft$.

Exercise 19.1.0.7. Exercise 15.1.3.10

FINISH!!!

Definition 19.1.0.8. We define the category of **Man[∞]**-principle bundles, denoted **PrinBun[∞]**, by

- $\text{Obj}(\mathbf{PrinBun}^\infty) := \{(P, X, \pi, G, \triangleleft) : (P, X, \pi, G) \text{ is a } \mathbf{Man}^\infty\text{-principal bundle}\}$
- For $(P_1, X_1, \pi_1, G_1, \triangleleft_1), (P_2, X_2, \pi_2, G_2, \triangleleft_2) \in \text{Obj}(\mathbf{PrinBun}^\infty)$,

$$\begin{aligned} \text{Hom}_{\mathbf{Bun}^\infty}((P_1, X_1, \pi_1, G_1, \triangleleft_1), (P_2, X_2, \pi_2, G_2, \triangleleft_2)) &:= \\ \{(\Phi, \phi) : (\Phi, \phi) \text{ is a smooth bundle morphism from } (E_1, M_1, \pi_1, F_1) \text{ to } (E_2, M_2, \pi_2, F_2)\} \end{aligned}$$

- For
 - $(E_1, M_1, \pi_1, F_1), (E_2, M_2, \pi_2, F_2), (E_3, M_3, \pi_3) \in \text{Obj}(\mathbf{Bun}^\infty)$
 - $(\Phi_{12}, \phi_{12}) \in \text{Hom}_{\mathbf{Bun}^\infty}((E_1, M_1, \pi_1, F_1), (E_2, M_2, \pi_2, F_2))$
 - $(\Phi_{23}, \phi_{23}) \in \text{Hom}_{\mathbf{Bun}^\infty}((E_2, M_2, \pi_2, F_2), (E_3, M_3, \pi_3))$

we define $(\Phi_{23}, \phi_{23}) \circ (\Phi_{12}, \phi_{12}) \in \text{Hom}_{\mathbf{Bun}^\infty}((E_1, M_1, \pi_1, F_1), (E_3, M_3, \pi_3))$ by

$$(\Phi_{23}, \phi_{23}) \circ (\Phi_{12}, \phi_{12}) := (\Phi_{23} \circ \Phi_{12}, \phi_{23} \circ \phi_{12})$$

FINISH!!!

Chapter 20

de Rham Cohomology

20.1 TO DO

1. de Rham cohomology
2. de Rham homology
3. in de Rham homology, measures on the manifold can be identified with the 0th Homology, group
4. think about how the other homology groups can be used in statistics

20.2 Introduction

Note 20.2.0.1. We recall that $d : \Omega^*(M) \rightarrow \Omega^*(M)$ satisfies the properties:

1. $d^2 = 0$
- 2.
- 3.

Definition 20.2.0.2. Let M be an n -dimensional smooth manifold. For $k \in \{1, \dots, n\}$, we define the

- **k -th coboundary operator**, denoted $d^k : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$, by $d^k = d|_{\Omega^k(M)}$
-
-

Part VI

Variational Geometry

Chapter 21

Jet Bundles

Note 21.0.0.1. The motivation is the following: Suppose you have a manifold M and $L : C^\infty(M) \rightarrow \mathbb{R}$, given by $L(\phi) := \int_M [\phi^2 + \sum_{i=1}^n (\partial_{x_i}^2 \phi)^2] dx^1 \wedge \cdots \wedge dx^n$. At first glance it might seem we would need infinite dimensional tools to analyze L since its domain is $C^\infty(M)$ which is infinite dimensional. Fortunately for us, we can reduce and analyze L with the usual finite dimensional toolset if we make one very powerful observation: L only depends on ϕ in through its derivatives of order ≤ 2 . Specifically this means that if ϕ_1 and ϕ_2 have the same derivatives of order ≤ 2 , then $L(\phi_1) = L(\phi_2)$. This provides us with an equivalence relation $\sim$ on $C^\infty(M)$ and when we reduce $C^\infty(M)$ modulo this equivalence relation, we obtain a finite dimensional space $M/\sim$ and factor $\bar{L} : M/\sim \rightarrow \mathbb{R}$ such that $L = \bar{L} \circ \pi_\sim$. We can then analyze L by analyzing $\bar{L}$. These constructions will allow us to talk about differential equations and variational optimization in a more unified geometric framework.

21.1 Fibered Manifolds

Definition 21.1.0.1. Let $E, M \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $\pi \in \text{Hom}_{\mathbf{ManBnd}^\infty}(E, M)$. Then (E, M, π) is said to be a **fibered manifold** if π is a surjective submersion.

Definition 21.1.0.2. Let $E, F, M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$ and $\pi \in \text{Hom}_{\mathbf{ManBnd}^\infty}(E, M)$, $\tau \in \text{Hom}_{\mathbf{ManBnd}^\infty}(F, N)$, $\Phi \in \text{Hom}_{\mathbf{ManBnd}^\infty}(E, F)$ and $\phi \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$. Suppose that (E, M, π) and (F, N, τ) are fibered manifolds. Then (Φ, ϕ) is said to be a **fibered manifold morphism** if $\tau \circ \Phi = \phi \circ \pi$, i.e. the following diagram commutes:

$$\begin{array}{ccc} E & \xrightarrow{\Phi} & F \\ \pi \downarrow & & \downarrow \tau \\ M & \xrightarrow{\phi} & N \end{array}$$

Note 21.1.0.3. We write $\text{proj}_1^n : \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^n$ to denote the projection onto M .

- Define fibered manifold morphism and category
- Define set of atlas charts which are fibered
- define jet bundles

Definition 21.1.0.4. Let (E, M, π) be a fibered manifold and $(V, \psi) \in \mathcal{A}_E$. Set $n := \dim M$ and $k := \dim E - n$. Then (V, ψ) is said to be a **π -fibered chart on E** if there exists $(U, \phi) \in \mathcal{A}_M$ such that

1. $U = \pi(V)$
2. $\phi \circ \pi|_V = \pi_{[n]}^{n+k} \circ \psi$,
i.e. if $\psi = (y^1, \dots, y^{n+k})$ and $\phi = (x^1, \dots, x^n)$, then $\psi = (x^1 \circ \pi|_V, \dots, x^n \circ \pi|_V, y^{n+1}, \dots, y^{n+k})$.

We define $\mathcal{A}_E^\pi := \{(U, \phi) \in \mathcal{A}_E : (U, \phi) \text{ is } \pi\text{-fibered}\}$.

Exercise 21.1.0.5. Let (E, M, π) be a smooth fibered manifold. Suppose that $\partial E, \partial M = \emptyset$. Then for each $a \in E$, there exists $(V, \psi) \in \mathcal{A}_E^\pi$ such that $a \in V$.

Hint: local rank theorem [reference ex from submersions section](#)

Proof. Set $n := \dim M$, $k := \dim E - n$. Let $a \in E$. Set $p := \pi(a)$. Since $\pi : E \rightarrow M$ is a submersion, π has constant rank and $\text{rank } \pi = n$. Exercise 8.1.0.3 implies that there exist $(V, \psi) \in \mathcal{A}_E$, $(U_0, \phi_0) \in \mathcal{A}_M$ such that $a \in V$, $p \in U_0$, $\pi(V) \subset U_0$ and $\phi_0 \circ \pi \circ \psi^{-1} = \text{proj}_n^{n+k}|_{\psi(V)}$. Hence $\phi_0 \circ \pi = \text{proj}_n^{n+k} \circ \psi$. Define $U := \pi(V)$ and $\phi := \phi_0|_U$. [An exercise in the section on submersions](#) implies that π is open. Hence $U \in \mathcal{T}_M$ and $(U, \phi) \in \mathcal{A}_M$. By construction,

1. $U = \pi(V)$

2. $\phi \circ \pi|_V = \text{proj}_n^{n+k} \circ \psi$

Hence (V, ψ) is a π -fibered chart on E . □

Exercise 21.1.0.6. Let (E, M, π) be a smooth fibered manifold and $a \in E$ and $(U_0, \phi_0) \in \mathcal{A}_E^\pi$. Set $n := \dim M$ and $k := \dim E - n$. Since $(U, \phi) \in \mathcal{A}_E^\pi$, there exists $(U, \phi) \in \mathcal{A}_M$ such that $\pi(U_0) = U$ and $\phi \circ \pi = \pi_{[n]}^{n+k} \circ \phi_0$. Suppose that $\partial E, \partial M = \emptyset$ and $a \in U_0$. Write $\phi_0 = (x^1, \dots, x^n, v^1, \dots, v^k)$ and $\phi = (\tilde{x}^1, \dots, \tilde{x}^1)$. Then for each $j \in [n]$ and $l \in [k]$,

$$D\pi(a) \left(\frac{\partial}{\partial x^j} \Big|_a \right) = \frac{\partial}{\partial \tilde{x}^j} \Big|_{\pi(a)}, \quad D\pi(a) \left(\frac{\partial}{\partial v^l} \Big|_a \right) = 0.$$

Proof. Let $j \in [n]$, $l \in [k]$ and $f \in C^\infty(M)$. Set $p := \pi(a)$. Then

$$\begin{aligned} D\pi(a) \left(\frac{\partial}{\partial x^j} \Big|_a \right) (f) &= \frac{\partial}{\partial x^j} \Big|_a (f \circ \pi) \\ &= \frac{\partial}{\partial x^j} \Big|_a (f \circ \phi^{-1} \circ \phi \circ \pi) \\ &= \frac{\partial}{\partial x^j} \Big|_a (f \circ \phi^{-1} \circ \pi_{[n]}^{n+k} \circ \phi_0) \\ &= \frac{\partial}{\partial u^j} \Big|_{\phi_0(a)} (f \circ \phi^{-1} \circ \pi_{[n]}^{n+k} \circ \phi_0 \circ \phi_0^{-1}) \\ &= \frac{\partial}{\partial u^j} \Big|_{\phi_0(a)} (f \circ \phi^{-1} \circ \pi_{[n]}^{n+k}) \\ &= \sum_{l=1}^n \frac{\partial(f \circ \phi^{-1})}{\partial u^l} (\pi_{[n]}^{n+k}(\phi_0(a))) \frac{\partial(\pi_l^n \circ \pi_{[n]}^{n+k})}{\partial u^j} (\phi_0(a)) \\ &= \sum_{l=1}^n \frac{\partial(f \circ \phi^{-1})}{\partial u^l} (\phi \circ \pi(a)) \frac{\partial(\pi_l^{n+k})}{\partial u^j} (\phi_0(a)) \\ &= \sum_{l=1}^n \frac{\partial(f \circ \phi^{-1})}{\partial u^l} (\phi(p)) \delta_{l,j} \\ &= \frac{\partial(f \circ \phi^{-1})}{\partial u^j} (\phi(p)) \\ &= \frac{\partial}{\partial \tilde{x}^j} \Big|_p f \end{aligned}$$

and similarly,

$$\begin{aligned}
 D\pi(a) \left(\frac{\partial}{\partial v^l} \Big|_a \right) (f) &= \frac{\partial}{\partial v^l} \Big|_a (f \circ \pi) \\
 &= \frac{\partial}{\partial u^{n+l}} \Big|_{\phi_0(a)} (f \circ \phi^{-1} \circ \pi_{[n]}^{n+k}) \\
 &= \sum_{j=1}^n \frac{\partial(f \circ \phi^{-1})}{\partial u^j} (\phi \circ \pi(a)) \frac{\partial(\pi_j^{n+k})}{\partial u^{n+l}} (\phi_0(a)) \\
 &= 0
 \end{aligned}$$

Since $f \in C^\infty(M)$ is arbitrary, we have that

$$D\pi(a) \left(\frac{\partial}{\partial x^j} \Big|_a \right) = \frac{\partial}{\partial \tilde{x}^j} \Big|_{\pi(a)}, \quad D\pi(a) \left(\frac{\partial}{\partial v^l} \Big|_a \right) = 0.$$

FINISH!!! (math scribbles) □

Exercise 21.1.0.7. Let $(E, M, \pi, F) \in \text{Obj}(\mathbf{Bun}^\infty)$. Then (E, M, π) is a smooth fibered manifold.

Proof. Since $(E, M, \pi, F) \in \text{Obj}(\mathbf{Bun}^\infty)$, π is surjective. **An exercise in the section on smooth fiber bundles** implies that π is a submersion. Since π is a surjective submersion, (E, M, π) is a smooth fibered manifold. □

Exercise 21.1.0.8. Let (E, M, π) be a smooth fibered manifold, $(V, \psi) \in \mathcal{A}_E^\pi$ and $r \geq 1$. Set $n := \dim M$ and $k := \dim E - n$ and $N := n + k \sum_{j=0}^r \binom{n-1+j}{j}$. Then there exists $(V_0, \psi_0) \in \mathcal{A}_M$ such that $V_0 = \pi(V)$ and $\text{proj}_n^{n+k} \circ \psi = \psi_0 \circ \pi$. Write $\psi = (\tilde{x}^1, \dots, \tilde{x}^n, \tilde{y}^1, \dots, \tilde{y}^k)$ and $\psi_0 = (\hat{x}^1, \dots, \hat{x}^n)$. Then for each $p \in V_0$, $a \in \pi^{-1}(\{p\}) \cap V$ and $(C_\alpha^\sigma)_{\sigma \in [k], \alpha \in \mathbb{N}_{\leq r}^n} \in \mathbb{R}^{N-k-n}$, there exists $(U, s) \in \Gamma_p(\pi)$ such that $U \subset V_0$, $s(p) = a$ and for each $\sigma \in [k]$, $\alpha \in \mathbb{N}_{\leq r}^n$, $\partial_\alpha [\tilde{y}^\sigma \circ s](p) = C_\alpha^\sigma$.

Hint: same idea as Exercise 8.3.0.13

Proof. Let $p \in V_0$, $a \in \pi^{-1}(\{p\}) \cap V$ and $(C_\alpha^\sigma)_{\sigma \in [k], \alpha \in \mathbb{N}_{\leq r}^n} \in \mathbb{R}^{N-k-n}$. Then $\pi(a) = p$, $a \in V$ and $\psi_0 \circ \pi \circ \psi^{-1} = \text{proj}_n^{n+k}|_{\psi(V)}$. Set $\hat{u}_0 := \text{proj}_{[n]}^{n+k}(\psi(a))$ and $\hat{v}_0 := \text{proj}_{[-k]}^{n+k}(\psi(a))$ so that $\psi(a) = (\hat{u}_0, \hat{v}_0)$. **An exercise in the analysis notes from the section on the product topology** implies that there exist $A \in \mathcal{T}_{\mathbb{R}^n}$ and $B \in \mathcal{T}_{\mathbb{R}^k}$ such that $(\hat{u}_0, \hat{v}_0) \in A \times B$ and $A \times B \subset \psi(V)$. We note that

$$\begin{aligned}
 \hat{u}_0 &= \text{proj}_{[n]}^{n+k}(\psi(a)) \\
 &= \psi_0 \circ \pi \circ \psi^{-1}(\psi(a)) \\
 &= \psi_0(\pi(a)) \\
 &= \psi_0(p).
 \end{aligned}$$

Since $A \times B \subset \psi(V)$, we have that

$$\begin{aligned}
 A &= \text{proj}_{[n]}^{n+k}(A \times B) \\
 &= \psi_0 \circ \pi \circ \psi^{-1}(A \times B) \\
 &\subset \psi_0 \circ \pi \circ \psi^{-1}(\psi(V)) \\
 &= \psi_0(\pi(V)) \\
 &= \psi_0(V_0)
 \end{aligned}$$

and therefore $\psi_0^{-1}(A) \subset V_0$. We note that for each $(\hat{u}^1, \dots, \hat{u}^n) \in A$,

$$\begin{aligned}
 (\hat{u}^1, \dots, \hat{u}^n, \hat{v}_0) &\in A \times B \\
 &\subset \psi(V).
 \end{aligned}$$

For each $\sigma \in [k]$, we define $C_{0 \oplus n}^\sigma \in \mathbb{N}_{0, \leq r}^n$ by $C_{0 \oplus n}^\sigma := \tilde{y}^\sigma(a)$. For each $\sigma \in [k]$, define $\hat{v}^\sigma : \psi_0(V_0) \rightarrow \mathbb{R}$ by

$$\hat{v}^\sigma(\hat{u}) := \sum_{k=0}^r \frac{1}{k!} \sum_{\substack{\alpha \in \mathbb{N}_{\leq r}^n \\ |\alpha|=k}} C_\alpha^\sigma (\hat{u} - \hat{u}_0)^\alpha.$$

We note that for each $\sigma \in [k]$, $\hat{v}^\sigma$ is smooth. For each $\sigma \in [k]$, define $v^\sigma : V_0 \rightarrow \mathbb{R}$ by $v^\sigma := \hat{v}^\sigma \circ \psi_0$. Since

$$\begin{aligned} \psi_0(p) &= \psi_0 \circ \pi(a) \\ &= \psi_0 \circ \pi \circ \psi^{-1}(\psi(a)) \\ &= \text{proj}_{[n]}^{n+k}(\psi(a)) \\ &= \hat{u}_0, \end{aligned}$$

we have that

$$\begin{aligned} p &= \psi_0^{-1}(\hat{u}_0) \\ &\in \psi_0^{-1}(A) \end{aligned}$$

and for each $q \in V_0$,

$$v^\sigma(q) = \sum_{k=0}^r \frac{1}{k!} \sum_{\substack{\alpha \in \mathbb{N}_{\leq r}^n \\ |\alpha|=k}} C_\alpha^\sigma (\psi_0(q) - \psi_0(p))^\alpha.$$

Define $v : V_0 \rightarrow \mathbb{R}^k$ by $v := (v^\sigma)_{\sigma \in [k]}$. Then $U \subset V_0$ and $p \in U$. Since for each $\sigma \in [k]$, $\hat{v}^\sigma$ is smooth, we have that v is smooth. We note that for each $\sigma \in [k]$,

$$\begin{aligned} v^\sigma(p) &= \sum_{k=0}^r \frac{1}{k!} \sum_{\substack{\alpha \in \mathbb{N}_{\leq r}^n \\ |\alpha|=k}} C_\alpha^\sigma (\psi_0(q) - \psi_0(p))^\alpha \\ &= C_{0 \oplus n}^\sigma \\ &= \tilde{y}^\sigma(a). \end{aligned}$$

Since v is continuous, $\psi_0^{-1}(A) \in \mathcal{T}_M$, $B \in \mathcal{T}_{\mathbb{R}^k}$, $p \in \psi_0^{-1}(A)$ and

$$\begin{aligned} v(p) &= (v^\sigma(p))_{\sigma \in [k]} \\ &= (\tilde{y}^\sigma(p))_{\sigma \in [k]} \\ &= \hat{v}_0 \\ &\in B, \end{aligned}$$

there exists $U \subset \psi_0^{-1}(A)$ such that $p \in U$ and $v(U) \subset B$. Define $\hat{s} : \psi_0(U) \rightarrow \psi(V)$ by $\hat{s}(u) := (u, \hat{v}(u))$. Then $\hat{s}$ is smooth. Define $s : U \rightarrow V$ by $s := \psi^{-1} \circ \hat{s} \circ \psi_0|_U$. Then s is smooth. Let $q \in U$. Set $x := \psi_0(q)$. Then

$$\begin{aligned} \pi \circ s(q) &= [\pi \circ (\psi^{-1} \circ \hat{s} \circ \psi_0|_U)](q) \\ &= [\pi \circ (\psi^{-1} \circ \hat{s} \circ \psi_0|_U)](\psi_0^{-1}(x)) \\ &= [\pi \circ (\psi^{-1} \circ \hat{s})](x) \\ &= [(\pi \circ \psi^{-1}) \circ \hat{s}](x) \\ &= (\psi_0^{-1} \circ \text{proj}_{[n]}^{n+k})(x, \hat{v}(x)) \\ &= \psi_0^{-1}(x) \\ &= q \end{aligned}$$

Since $q \in \psi_0^{-1}(A)$ is arbitrary, we have that $\pi \circ s = \text{id}_{\psi_0^{-1}(A)}$ and therefore $(U, s) \in \Gamma_p(\pi)$. By construction,

$$\begin{aligned} s(p) &= \psi^{-1} \circ \hat{s} \circ \psi_0(p) \\ &= \psi^{-1} \circ \hat{s}(\hat{u}_0) \\ &= \psi^{-1}(\hat{u}_0, \hat{v}_0) \\ &= \psi^{-1}(\psi(a)) \\ &= a \end{aligned}$$

and for each $\sigma \in [k]$, $\alpha \in \mathbb{N}_{\leq r}^n$,

$$\begin{aligned} \partial_\alpha[\tilde{y}^\sigma \circ s](p) &= \partial_\alpha[\tilde{y}^\sigma \circ \psi^{-1} \circ \hat{s} \circ \psi_0|_U](p) \\ &= \partial_\alpha[\text{proj}_{n+\sigma}^{n+k} \circ \psi \circ \psi^{-1} \circ \hat{s} \circ \psi_0|_U](p) \\ &= \partial_\alpha[\text{proj}_{n+\sigma}^{n+k} \circ \hat{s} \circ \psi_0|_U](p) \\ &= \partial_\alpha[\hat{v}^\sigma \circ \psi_0|_U](p) \\ &= \partial_\alpha[v^\sigma|_U](p) \\ &= C_\alpha^\sigma. \end{aligned}$$

□

21.2 Contact Order

Definition 21.2.0.1. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $A, B \in \mathcal{T}_M$, $F \in \text{Hom}_{\mathbf{Man}^\infty}(A, N)$, $G \in \text{Hom}_{\mathbf{Man}^\infty}(B, N)$, $p \in A \cap B$ and $r \in \mathbb{N}_0$. Set $m := \dim M$ and $n := \dim N$. Then F and G are said to have a **contact of order r at p** if there exists $(U, \phi) \in \mathcal{A}_M|_{A \cap B}$ with $\phi = (x^1, \dots, x^m)$ and $(V, \psi) \in \mathcal{A}_N$ with $\psi = (y^1, \dots, y^n)$ such that $p \in U$, $F(p), G(p) \in V$ and for each $j \in [n]$ and $\alpha \in \mathbb{N}_0$, $|\alpha| \leq r$ implies that

$$\frac{\partial^{|\alpha|}(y^j \circ F)}{\partial x^\alpha}(p) = \frac{\partial^{|\alpha|}(y^j \circ G)}{\partial x^\alpha}(p)$$

Exercise 21.2.0.2. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$. Set $m := \dim M$ and $n := \dim N$. For $a \in \mathbb{N}_0$, we define

$$A_a := \{(\beta, \gamma, \delta, t, v) : \beta, \delta \in \mathbb{N}_0^m, \gamma \in \mathbb{N}_0^n, |\beta|, |\gamma|, |\delta| \leq a, t \in [m], v \in [n]\}.$$

Then

1. For each $j \in [n]$ and $\alpha \in \mathbb{N}_0^n$, there exists a $P_{j,\alpha} \in \mathbb{R}[X_{\beta,v}, X_\gamma, X_{\delta,t} : (\beta, \gamma, \delta, t, v) \in A_{|\alpha|}]$ such that for each $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$, $(U, \phi), (\tilde{U}, \tilde{\phi}) \in \mathcal{A}_M$, $(V, \psi), (\tilde{V}, \tilde{\psi}) \in \mathcal{A}_N$ with $\phi = (x^1, \dots, x^m)$, $\tilde{\phi} = (\tilde{x}^1, \dots, \tilde{x}^m)$, $\psi = (y^1, \dots, y^n)$, $\tilde{\psi} = (\tilde{y}^1, \dots, \tilde{y}^n)$ and $p \in (U \cap \tilde{U}) \cap F^{-1}(V \cap \tilde{V})$,

$$\frac{\partial^{|\alpha|}(\tilde{y}^j \circ F)}{\partial \tilde{x}^\alpha}(p) = P_{j,\alpha} \left(\frac{\partial^{|\beta|}(y^v \circ F)}{\partial x^\beta}(p), \frac{\partial^{|\gamma|}\tilde{y}^j}{\partial y^\gamma}(F(p)), \frac{\partial^{|\delta|}x^t}{\partial \tilde{x}^\delta}(p) : (\beta, \gamma, \delta, t, v) \in A_{|\alpha|} \right)$$

2. Let $F, G \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$, $r \in \mathbb{N}_0$ and $p_0 \in M$. Suppose that F and G have a contact of order r at p_0 . Let $(U, \phi), (\tilde{U}, \tilde{\phi}) \in \mathcal{A}_M$, $(V, \psi), (\tilde{V}, \tilde{\psi}) \in \mathcal{A}_N$ with $\phi = (x^1, \dots, x^m)$, $\tilde{\phi} = (\tilde{x}^1, \dots, \tilde{x}^m)$, $\psi = (y^1, \dots, y^n)$, $\tilde{\psi} = (\tilde{y}^1, \dots, \tilde{y}^n)$. If $p_0 \in (U \cap \tilde{U}) \cap F^{-1}(V \cap \tilde{V})$, then for each $j \in [n]$ and $\alpha \in \mathbb{N}_0^m$,

$$\frac{\partial^{|\alpha|}(y^j \circ F)}{\partial x^\alpha}(p_0) = \frac{\partial^{|\alpha|}(y^j \circ G)}{\partial x^\alpha}(p_0)$$

iff for each $j \in [n]$ and $\alpha \in \mathbb{N}_0^m$,

$$\frac{\partial^{|\alpha|}(\tilde{y}^j \circ F)}{\partial \tilde{x}^\alpha}(p_0) = \frac{\partial^{|\alpha|}(\tilde{y}^j \circ G)}{\partial \tilde{x}^\alpha}(p_0)$$

Proof.

1. **Base Case:**

The claim is clear for $|\alpha| = 0$.

- Induction Step:**

Let $a \in \mathbb{N}$. Suppose that for each $j \in [n]$ and $\alpha \in \mathbb{N}_0^m$, $|\alpha| = a - 1$ implies that there exists $P_{j,\alpha} \in \mathbb{R}[X_{\xi\beta,\xi v}, X_{\xi\gamma}, X_{\xi\delta,\xi t} : \xi \in A_{|\alpha|}]$ such that for each $F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(M, N)$, $(U, \phi), (\tilde{U}, \tilde{\phi}) \in \mathcal{A}_M$, $(V, \psi), (\tilde{V}, \tilde{\psi}) \in \mathcal{A}_N$ with $\phi = (x^1, \dots, x^m)$, $\tilde{\phi} = (\tilde{x}^1, \dots, \tilde{x}^m)$, $\psi = (y^1, \dots, y^n)$, $\tilde{\psi} = (\tilde{y}^1, \dots, \tilde{y}^n)$ and $p \in (U \cap \tilde{U}) \cap F^{-1}(V \cap \tilde{V})$,

$$\frac{\partial^{|\alpha|}(\tilde{y}^j \circ F)}{\partial \tilde{x}^\alpha}(p) = P_{j,\alpha} \left(\frac{\partial^{|\beta|}(y^v \circ F)}{\partial x^\beta}(p), \frac{\partial^{|\gamma|}\tilde{y}^j}{\partial y^\gamma}(F(p)), \frac{\partial^{|\delta|}x^t}{\partial \tilde{x}^\delta}(p) : (\beta, \gamma, \delta, t, v) \in A_{a-1} \right).$$

Let $j \in [n]$, $\alpha \in \mathbb{N}_0^m$ and $(U, \phi), (\tilde{U}, \tilde{\phi}) \in \mathcal{A}_M$, $(V, \psi), (\tilde{V}, \tilde{\psi}) \in \mathcal{A}_N$ with $\phi = (x^1, \dots, x^m)$, $\tilde{\phi} = (\tilde{x}^1, \dots, \tilde{x}^m)$, $\psi = (y^1, \dots, y^n)$, $\tilde{\psi} = (\tilde{y}^1, \dots, \tilde{y}^n)$. Suppose that $|\alpha| = a$. Since $a > 0$, there exists $l_0 \in [m]$ and $\alpha_0 \in \mathbb{N}_0$ such that $\alpha = \alpha_0 + e_{l_0}$. Since $P_{j,\alpha_0} \in \mathbb{R}[X_{\xi\beta,\xi v}, X_{\xi\gamma}, X_{\xi\delta,\xi t} : \xi \in A_{|\alpha_0|}]$, there exist $(c_\xi)_{\xi \in A_{|\alpha_0|}} \subset \mathbb{R}$ and $(\mu_\xi, \sigma_\xi, \tau_\xi)_{\xi \in A_{|\alpha_0|}} \subset \mathbb{N}_0^3$ such that

$$P_{j,|\alpha_0|}(X_{\xi\beta,\xi v}, X_{\xi\gamma}, X_{\xi\delta,\xi t} : \xi \in A_{|\alpha_0|}) = \sum_{\xi \in A_{|\alpha_0|}} c_\xi X_{\xi\beta,\xi v}^{\mu_\xi} X_{\xi\gamma}^{\sigma_\xi} X_{\xi\delta,\xi t}^{\tau_\xi}.$$

Then

$$\begin{aligned}
\frac{\partial^{|\alpha|}(\tilde{y}^j \circ F)}{\partial \tilde{x}^\alpha} &= \frac{\partial}{\partial \tilde{x}^{l_0}} \left[\frac{\partial^{|\alpha_0|}(\tilde{y}^j \circ F)}{\partial \tilde{x}^{\alpha_0}} \right] \\
&= \frac{\partial}{\partial \tilde{x}^{l_0}} P_{j, \alpha_0} \left(\frac{\partial^{|\xi_\beta|}(y^v \circ F)}{\partial x^{\xi_\beta}}, \frac{\partial^{|\xi_\gamma|}\tilde{y}^j}{\partial y^{\xi_\gamma}} \circ F, \frac{\partial^{|\xi_\delta|}x^{\xi_t}}{\partial \tilde{x}^{\xi_\delta}} : \xi \in A_{|\alpha_0|} \right) \\
&= \frac{\partial}{\partial \tilde{x}^{l_0}} \left[\sum_{\xi \in A_{|\alpha_0|}} c_\xi \left(\frac{\partial^{|\xi_\beta|}(y^{\xi_v} \circ F)}{\partial x^{\xi_\beta}} \right)^{\mu_\xi} \left(\frac{\partial^{|\xi_\gamma|}\tilde{y}^j}{\partial y^{\xi_\gamma}} \circ F \right)^{\sigma_\xi} \left(\frac{\partial^{|\xi_\delta|}x^{\xi_t}}{\partial \tilde{x}^{\xi_\delta}} \right)^{\tau_\xi} \right] \\
&= \sum_{\xi \in A_{|\alpha_0|}} c_\xi \frac{\partial}{\partial \tilde{x}^{l_0}} \left[\left(\frac{\partial^{|\xi_\beta|}(y^{\xi_v} \circ F)}{\partial x^{\xi_\beta}} \right)^{\mu_\xi} \left(\frac{\partial^{|\xi_\gamma|}\tilde{y}^j}{\partial y^{\xi_\gamma}} \circ F \right)^{\sigma_\xi} \left(\frac{\partial^{|\xi_\delta|}x^{\xi_t}}{\partial \tilde{x}^{\xi_\delta}} \right)^{\tau_\xi} \right] \\
&= \sum_{\xi \in A_{|\alpha_0|}} c_\xi \left[\left(\frac{\partial}{\partial \tilde{x}^{l_0}} \left[\frac{\partial^{|\xi_\beta|}(y^{\xi_v} \circ F)}{\partial x^{\xi_\beta}} \right]^{\mu_\xi} \right) \left(\frac{\partial^{|\xi_\gamma|}\tilde{y}^j}{\partial y^{\xi_\gamma}} \circ F \right)^{\sigma_\xi} \left(\frac{\partial^{|\xi_\delta|}x^{\xi_t}}{\partial \tilde{x}^{\xi_\delta}} \right)^{\tau_\xi} \right. \\
&\quad + \left(\frac{\partial^{|\xi_\beta|}(y^{\xi_v} \circ F)}{\partial x^{\xi_\beta}} \right)^{\mu_\xi} \left(\frac{\partial}{\partial \tilde{x}^{l_0}} \left[\frac{\partial^{|\xi_\gamma|}\tilde{y}^j}{\partial y^{\xi_\gamma}} \circ F \right]^{\sigma_\xi} \right) \left(\frac{\partial^{|\xi_\delta|}x^{\xi_t}}{\partial \tilde{x}^{\xi_\delta}} \right)^{\tau_\xi} \\
&\quad + \left. \left(\frac{\partial^{|\xi_\beta|}(y^{\xi_v} \circ F)}{\partial x^{\xi_\beta}} \right)^{\mu_\xi} \left(\frac{\partial^{|\xi_\gamma|}\tilde{y}^j}{\partial y^{\xi_\gamma}} \circ F \right)^{\sigma_\xi} \left(\frac{\partial}{\partial \tilde{x}^{l_0}} \left[\frac{\partial^{|\xi_\delta|}x^{\xi_t}}{\partial \tilde{x}^{\xi_\delta}} \right]^{\tau_\xi} \right) \right] \\
&= \sum_{\xi \in A_{|\alpha_0|}} c_\xi \left[\mu_\xi \left(\frac{\partial^{|\xi_\beta|}(y^{\xi_v} \circ F)}{\partial x^{\xi_\beta}} \right)^{\mu_\xi - 1} \left(\sum_{k=1}^m \frac{\partial x^k}{\partial \tilde{x}^{l_0}} \frac{\partial^{|\xi_\beta|+1}(y^{\xi_v} \circ F)}{\partial x^{\xi_\beta + e_k}} \right) \left(\frac{\partial^{|\xi_\gamma|}\tilde{y}^j}{\partial y^{\xi_\gamma}} \circ F \right)^{\sigma_\xi} \left(\frac{\partial^{|\xi_\delta|}x^{\xi_t}}{\partial \tilde{x}^{\xi_\delta}} \right)^{\tau_\xi} \right. \\
&\quad + \left(\frac{\partial^{|\xi_\beta|}(y^{\xi_v} \circ F)}{\partial x^{\xi_\beta}} \right)^{\mu_\xi} \sigma_\xi \left(\frac{\partial^{|\xi_\gamma|}\tilde{y}^j}{\partial y^{\xi_\gamma}} \circ F \right)^{\sigma_\xi - 1} \left(\sum_{s=1}^n \sum_{k=1}^m \left[\frac{\partial^{|\xi_\gamma|+1}\tilde{y}^j}{\partial y^{\xi_\gamma + e_s}} \circ F \right] \frac{\partial(y^s \circ F)}{\partial x^k} \frac{\partial x^k}{\partial \tilde{x}^{l_0}} \right) \left(\frac{\partial^{|\xi_\delta|}x^{\xi_t}}{\partial \tilde{x}^{\xi_\delta}} \right)^{\xi_\delta} \\
&\quad + \left. \left(\frac{\partial^{|\xi_\beta|}(y^{\xi_v} \circ F)}{\partial x^{\xi_\beta}} \right)^{\mu_\xi} \left(\frac{\partial^{|\xi_\gamma|}\tilde{y}^j}{\partial y^{\xi_\gamma}} \circ F \right)^{\sigma_\xi} \tau_\xi \left(\frac{\partial^{|\xi_\delta|}x^{\xi_t}}{\partial \tilde{x}^{\xi_\delta}} \right)^{\tau_\xi - 1} \left(\frac{\partial^{|\xi_\delta|+1}x^{\xi_t}}{\partial \tilde{x}^{\xi_\delta + e_{l_0}}} \right) \right] \\
&= P_{j, \alpha} \left(\frac{\partial^{|\xi_\beta|}(y^{\xi_v} \circ F)}{\partial x^{\xi_\beta}}, \frac{\partial^{|\xi_\gamma|}\tilde{y}^j}{\partial y^{\xi_\gamma}} \circ F, \frac{\partial^{|\xi_\delta|}x^{\xi_t}}{\partial \tilde{x}^{\xi_\delta}} : \xi \in A_{|\alpha|} \right)
\end{aligned}$$

2. Suppose that $p_0 \in (U \cap \tilde{U}) \cap F^{-1}(V \cap \tilde{V})$.

• ($\implies$) :

Suppose that for each $j \in [n]$ and $\alpha \in \mathbb{N}_0^n$, $|\alpha| \leq r$ implies that

$$\frac{\partial^{|\alpha|}(y^j \circ F)}{\partial x^\alpha}(p) = \frac{\partial^{|\alpha|}(y^j \circ G)}{\partial x^\alpha}(p).$$

Let $j \in [n]$ and $\alpha \in \mathbb{N}_0^n$. Suppose that $|\alpha| \leq r$. Then

$$\begin{aligned}
\frac{\partial^{|\alpha|}(\tilde{y}^j \circ F)}{\partial \tilde{x}^\alpha}(p_0) &= P_{j, \alpha} \left(\frac{\partial^{|\beta|}(y^v \circ F)}{\partial x^\beta}(p_0), \frac{\partial^{|\gamma|}\tilde{y}^j}{\partial y^\gamma}(F(p_0)), \frac{\partial^{|\delta|}x^t}{\partial \tilde{x}^\delta}(p_0) : (\beta, \gamma, \delta, t, r) \in A_{|\alpha|} \right) \\
&= P_{j, \alpha} \left(\frac{\partial^{|\beta|}(y^v \circ G)}{\partial x^\beta}(p_0), \frac{\partial^{|\gamma|}\tilde{y}^j}{\partial y^\gamma}(G(p_0)), \frac{\partial^{|\delta|}x^t}{\partial \tilde{x}^\delta}(p_0) : (\beta, \gamma, \delta, t, r) \in A_{|\alpha|} \right) \\
&= \frac{\partial^{|\alpha|}(\tilde{y}^j \circ G)}{\partial \tilde{x}^\alpha}(p_0).
\end{aligned}$$

• ($\impliedby$) :

Similar to the previous part.

□

Definition 21.2.0.3. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $p \in M$, $q \in N$ and $r \in \mathbb{N}_0$. We define

- $C_p^\infty(M, N) := \{F \in \text{Hom}_{\mathbf{ManBnd}^\infty}(U, N) : U \in \mathcal{T}_M, p \in U\}$ **move this definition to before definition of contact order at p and rework that definition using this notation.**
- $\sim_r \subset C_p^\infty(M, N) \times C_p^\infty(M, N)$ by $F \sim_r G$ iff F and G have a contact of order r at p .

Exercise 21.2.0.4. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $p \in M$, $q \in N$ and $r \in \mathbb{N}_0$. Then $\sim_r$ is an equivalence relation on $C_p^\infty(M, N)$.

Proof. Set $m := \dim M$ and $n := \dim N$. **use previous exercise and definition**

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□

Definition 21.2.0.5. Let $M, N \in \text{Obj}(\mathbf{ManBnd}^\infty)$, $p \in M$, $q \in N$, $r \in \mathbb{N}_0$ and $F \in C_{(p,q)}^\infty(M, N)$. We define the

- **r -jet of F at p** , denoted $J_p^r F$, by $J_p^r F := [F]_{\sim_r}$
- **r -jets with source p** , denoted $J_p^r(M, N)$, by $J_p^r(M, N) := C_p^\infty(M, N)/\sim_r$

Definition 21.2.0.6. Let (E, M, π) be a smooth fibered manifold and $r \in \mathbb{N}_0$. We define the **r -th jet manifold of π** , denoted $J^r \pi$, by $J^r \pi := J_p^r$.

Exercise 21.2.0.7. Let (E, M, π) be a smooth fibered manifold and $r \in \mathbb{N}_0$. Set $n := \dim M$ and $k := \dim E - n$. Then for each $p \in M$ and $a \in \pi^{-1}(p)$, there exists $(U, \sigma) \in \Gamma_p(\pi)$ such that $s(p) = a$ and for each $\sigma \in [k]$, $\alpha \in \mathbb{N}_0$, $\partial_\alpha[y^\sigma \circ s](p) = C_\alpha^\sigma$. Exercise 5.4.0.14 Exercise 8.3.0.13. **FINISH!!**, maybe make iff submersion like in local section theorem?

21.3 Jet Bundles of Fiber Bundles

Definition 21.3.0.1. Let $(E, M, \pi, F) \in \text{Obj}(\mathbf{Bun}^\infty)$, $r \in \mathbb{N}_0$ and $(V, \psi) \in \mathcal{A}_E^\pi$. Set $n := \dim M$ and $k := \dim E - n$. Write $\psi = (x^1, \dots, x^n, y^1, \dots, y^k)$. We define the r -th jet manifold of π , denoted $J^r \pi$, by $J^r \pi := \coprod_{p \in M} J_p^r(M, E) \cap \Gamma_p(\pi)$. We define the r -th jet bundle projection of $J^r \pi$ onto M , denoted $\pi_r : J^r \pi \rightarrow M$, by $\pi_r(p, s) := p$.

Note 21.3.0.2. Let $(E, M, \pi, F) \in \text{Obj}(\mathbf{Bun}^\infty)$ and $(U, \phi) \in \mathcal{A}_E^\pi$. Set $n := \dim M$ and $k := \dim E - n$. Then there exists $(U_0, \phi_0) \in \mathcal{A}_M$ such that $U_0 = \pi(U)$ and $\text{proj}_n^{n+k} \circ \psi = \psi_0 \circ \pi$.

- Let $p \in U_0$ and $s \in \Gamma_p(\pi)$. Since

$$\begin{aligned} \pi \circ s(p) &= p \\ &\in U_0, \end{aligned}$$

we have that $s(p) \in \pi^{-1}(U_0)$.

- Let $b_1, b_2 \in \pi_r^{-1}(U_0)$. Suppose that $b_1 = b_2$. Then there exists $p_1, p_2 \in U_0$ and $s_1, s_2 \in \Gamma_p(\pi)$ such that $b_1 = J_{p_1}^r$ and $b_2 = J_{p_2}^r$. Then

$$\begin{aligned} p_1 &= \pi_r(J_{p_1} s_1) \\ &= \pi_r(J_{p_2} s_2) \\ &= p_2. \end{aligned}$$

Let $J_{p_1}^r s_1, J_{p_2}^r s_2 \in \pi_r^{-1}(U_0)$. Suppose that $J_{p_1}^r s_1 = J_{p_2}^r s_2$. Then

$$\begin{aligned} p_1 &= \pi_r(J_{p_1} s_1) \\ &= \pi_r(J_{p_2} s_2) \\ &= p_2. \end{aligned}$$

try to define $J_p^r s$ as a bundle, specifically a disjoint union over p $J^r \pi := \coprod_{p \in M} J_p^r(M, E) \cap \Gamma_p(\pi)$. We would identify $J_p^r s$ with $(p, J_p^r s)$ in the disjoint union. Define projective system of projections $(\pi_{r,k})_{r \leq k} \in \prod_{r \leq k} \text{Hom}_{\mathbf{Top}}(J^k \pi, J^r \pi)$ defined by $\pi_{r,k}(J_p^k s) := J_p^r s$.

Can we define the smooth structure on $J^r \pi$ inductively to be the unique smooth structure such that $(\pi_{r,k})_{r \leq k}$ are smooth?

(on that note, can we define the product atlas to be the unique atlas such that $\pi_M : M \times N \rightarrow M$ and $\pi_N : M \times N \rightarrow N$ are smooth?)

Definition 21.3.0.3. Let $(E, M, \pi, F) \in \text{Obj}(\mathbf{Bun}^\infty)$ and $(V, \psi) \in \mathcal{A}_E^\pi$. Set $n := \dim M$ and $k := \dim E - n$. Then there exists $(V_0, \psi_0) \in \mathcal{A}_M$ such that $V_0 = \pi(V)$ and $\text{proj}_n^{n+k} \circ \psi = \psi_0 \circ \pi$. Write $\psi = (\hat{x}^1, \dots, \hat{x}^n, \tilde{y}^1, \dots, \tilde{y}^k)$ and $\psi_0 = (\hat{x}^1, \dots, \hat{x}^n)$. For $j \in [n]$, $\sigma \in [k]$ and $\alpha \in \mathbb{N}_0^n$ with $|\alpha| \leq r$, define $x^j, y^\sigma, y_\alpha^\sigma : J^r \pi \rightarrow \mathbb{R}$ by $x^j(p, J_p^r s) := \hat{x}^j(p)$, $y^\sigma(p, J_p^r s) := \tilde{y}^\sigma(s(p))$ and $y_\alpha^\sigma(p, J_p^r s) := \partial_\alpha(\tilde{y}^\sigma \circ s)(p)$. Set $N := n + k \sum_{j=0}^r \binom{n-1+j}{j}$. We define the **jet manifold chart induced by** (V, ψ) , denoted $\Psi_\psi : J^r \rightarrow \mathbb{R}^N$ by $\Psi_\psi := (x^j, y^\sigma, y_\alpha^\sigma : j \in [n], \sigma \in [k], \alpha \in \mathbb{N}, |\alpha| \leq r)$.

Note 21.3.0.4. Since $\text{proj}_n^{n+k} \circ \psi = \psi_0 \circ \pi$ and $s \in \Gamma_p(\pi)$, we have that

$$\begin{aligned} x^j(p, J_p^r s) &= \hat{x}^j(p) \\ &= \hat{x}^j \circ \text{id}_M(p) \\ &= \hat{x}^j \circ \pi \circ s(p) \\ &= \tilde{x}^j \circ s(p) \end{aligned}$$

so that the definition of x^j is consistent with that of y^σ and y_α^σ .

Definition 21.3.0.5. Let $(E, M, \pi, F) \in \text{Obj}(\mathbf{Bun}^\infty)$, $(U_\phi, \phi) \in \mathcal{A}_E^\pi$ and $(U_{0,\Phi}, \Phi)$ a smooth fiber bundle local trivialization with respect to π of E over $U_{0,\Phi}$ with fiber F . Set $n := \dim M$ and $k := \dim E - n$. Then there exists $(U_{0,\phi}, \phi_0) \in \mathcal{A}_M$ such that $U_{0,\phi} = \pi(U_\phi)$ and $\text{proj}_n^{n+k} \circ \phi = \phi_0 \circ \pi$. Write $\phi = (\hat{x}^1, \dots, \hat{x}^n, \tilde{y}^1, \dots, \tilde{y}^k)$. Define $U_{\Phi,\phi} \in \mathcal{T}_M$ by $U_{\Phi,\phi} := U_{0,\phi} \cap U_{0,\Phi}$. Set $N := \dim J^r \pi$. We define the **r -th jet bundle local trivialization induced by** (U_ϕ, ϕ) and $(U_{0,\Phi}, \Phi)$, denoted $\Phi_{\Phi,\phi}^r : \pi_r^{-1}(U_{\Phi,\phi}) \rightarrow U_{\Phi,\phi} \times (F \times \mathbb{R}^{N-(n+k)})$, by

$$\Phi_{\Phi,\phi}^r(p, J_p^r s) := (\Phi \circ s(p), (\partial_\alpha[\tilde{y}^\sigma \circ s](p))_{\sigma,\alpha}).$$

Exercise 21.3.0.6. Let $(E, M, \pi, F) \in \text{Obj}(\mathbf{Bun}^\infty)$, $(U_\phi, \phi) \in \mathcal{A}_E^\pi$ and $(U_{0,\Phi}, \Phi)$ a smooth fiber bundle local trivialization with respect to π of E over $U_{0,\Phi}$ with fiber F . Set $n := \dim M$ and $k := \dim E - n$. Then there exists $(U_{0,\phi}, \phi_0) \in \mathcal{A}_M$ such that $U_{0,\phi} = \pi(U_\phi)$ and $\text{proj}_n^{n+k} \circ \phi = \phi_0 \circ \pi$. Write $\phi = (\hat{x}^1, \dots, \hat{x}^n, \tilde{y}^1, \dots, \tilde{y}^k)$. Define $U_{\Phi,\phi} \in \mathcal{T}_M$ by $U_{\Phi,\phi} := U_{0,\phi} \cap U_{0,\Phi}$. Then

1. $\Phi_{\Phi,\phi}^r$ is well-defined,
2. $\Phi_{\Phi,\phi}^r$ is bijective,
3. $\text{proj}_{U_{\Phi,\phi}} \circ \Phi_{\Phi,\phi}^r = \pi_r|_{\pi_r^{-1}(U_{\Phi,\phi})}$,
4. $(U_{\Phi,\phi}, \Phi_{\Phi,\phi}^r)$ is a local trivialization with respect to π_r of $J^r \pi$ over $U_{\Phi,\phi}$ with fiber $F \times \mathbb{R}^{N-k-n}$.

Proof. Write $\phi_0 = (\hat{x}^1, \dots, \hat{x}^n)$.

1. Let $b_1, b_2 \in \pi_r^{-1}(U_{\Phi,\phi})$. Suppose that $b_1 = b_2$. Since $b_1, b_2 \in \pi_r^{-1}(U_{\Phi,\phi})$, there exists $p_1, p_2 \in U_{\Phi,\phi}$, $s_1 \in \Gamma_{p_1}$ and $s_2 \in \Gamma_{p_2}$ such that $b_1 = (p_1, J_{p_1}^r s_1)$ and $b_2 = (p_2, J_{p_2}^r s_2)$. Therefore

$$\begin{aligned} p_1 &= \pi_r(p_1, J_{p_1}^r s_1) \\ &= \pi_r(b_1) \\ &= \pi_r(b_2) \\ &= \pi_r(p_2, J_{p_2}^r s_2) \\ &= p_2. \end{aligned}$$

Thus for each $j \in [n]$, $\hat{x}^j(p_1) = \hat{x}^j(p_2)$. Since $s_1 \sim_{r,p} s_2$, we have that for each $\sigma \in [k]$ and $\alpha \in \mathbb{N}^n$ with $|\alpha| \leq r$, $\tilde{y}^\sigma \circ s_1(p_1) = \tilde{y}^\sigma \circ s_2(p_2)$ and $\partial_\alpha[\tilde{y}^\sigma \circ s_1](p_1) = \partial_\alpha[\tilde{y}^\sigma \circ s_2](p_2)$. Therefore

$$\begin{aligned} s_1(p_1) &= \phi^{-1} \circ \phi \circ s_1(p_1) \\ &= \phi^{-1}(\hat{x}^1 \circ s_1(p_1), \dots, \hat{x}^n \circ s_1(p_1), \tilde{y}^1 \circ s_1(p_1), \dots, \tilde{y}^k \circ s_1(p_1)) \\ &= \phi^{-1}(\hat{x}^1(p_1), \dots, \hat{x}^n(p_1), \tilde{y}^1 \circ s_1(p_1), \dots, \tilde{y}^k \circ s_1(p_1)) \\ &= \phi^{-1}(\hat{x}^1(p_2), \dots, \hat{x}^n(p_2), \tilde{y}^1 \circ s_2(p_2), \dots, \tilde{y}^k \circ s_2(p_2)) \\ &= \phi^{-1}(\hat{x}^1 \circ s_2(p_2), \dots, \hat{x}^n \circ s_2(p_2), \tilde{y}^1 \circ s_2(p_2), \dots, \tilde{y}^k \circ s_2(p_2)) \\ &= \phi^{-1} \circ \phi \circ s_2(p_2) \\ &= s_2(p_2). \end{aligned}$$

Hence $\Phi \circ s_1(p_1) = \Phi \circ s_2(p_2)$. Therefore

$$\begin{aligned}\Phi_{\Phi,\phi}^r(b_1) &= \Phi_{\Phi,\phi}^r(p_1, J_{p_1}^r s_1) \\ &= (\Phi \circ s_1(p_1), (\partial_\alpha[\tilde{y}^\sigma \circ s_1](p_1))_{\sigma,\alpha}) \\ &= (\Phi \circ s_2(p_2), (\partial_\alpha[\tilde{y}^\sigma \circ s_2](p_2))_{\sigma,\alpha}) \\ &= \Phi_{\Phi,\phi}^r(p_2, J_{p_2}^r s_2) \\ &= \Phi_{\Phi,\phi}^r(b_2).\end{aligned}$$

Since $b_1, b_2 \in \pi_r^{-1}(U_{\Phi,\phi})$ are arbitrary, we have that for each $b_1, b_2 \in \pi_r^{-1}(U_{\Phi,\phi})$, $b_1 = b_2$ implies that $\Phi_{\Phi,\phi}^r(b_1) = \Phi_{\Phi,\phi}^r(b_2)$. Hence $\Phi_{\Phi,\phi}^r$ is well-defined.

2. • **(injectivity):**

Let $b_1, b_2 \in \pi_r^{-1}(U_{\Phi,\phi})$. Suppose that $\Phi_{\Phi,\phi}^r(b_1) = \Phi_{\Phi,\phi}^r(b_2)$. Since $b_1, b_2 \in \pi_r^{-1}(U_{\Phi,\phi})$, there exists $p_1, p_2 \in U_{\Phi,\phi}$, $s_1 \in \Gamma_{p_1}$ and $s_2 \in \Gamma_{p_2}$ such that $b_1 = (p_1, J_{p_1}^r s_1)$ and $b_2 = (p_2, J_{p_2}^r s_2)$. Then

$$\begin{aligned}(\Phi \circ s_1(p_1), (\partial_\alpha[\tilde{y}^\sigma \circ s_1](p_1))_{\sigma,\alpha}) &= \Phi_{\Phi,\phi}^r(p_1, J_{p_1}^r s_1) \\ &= \Phi_{\Phi,\phi}^r(b_1) \\ &= \Phi_{\Phi,\phi}^r(b_2) \\ &= \Phi_{\Phi,\phi}^r(p_2, J_{p_2}^r s_2) \\ &= (\Phi \circ s_2(p_2), (\partial_\alpha[\tilde{y}^\sigma \circ s_2](p_2))_{\sigma,\alpha}).\end{aligned}$$

Thus $\Phi \circ s_1(p_1) = \Phi \circ s_2(p_2)$ and for each $\sigma \in [k]$ and $\alpha \in \mathbb{N}^n$ with $|\alpha| \leq r$, $\partial_\alpha[\tilde{y}^\sigma \circ s_1](p_1) = \partial_\alpha[\tilde{y}^\sigma \circ s_2](p_2)$. Since Φ is a bijection, we have that

$$\begin{aligned}s_1(p_1) &= \Phi^{-1} \circ \Phi \circ s_1(p_1) \\ &= \Phi^{-1} \circ \Phi \circ s_2(p_2) \\ &= s_2(p_2).\end{aligned}$$

Since $s_1, s_2 \in \Gamma_p(\pi)$, we then have that

$$\begin{aligned}p_1 &= \pi(s_1(p_1)) \\ &= \pi(s_2(p_2)) \\ &= p_2.\end{aligned}$$

Since $s_1(p_1) = s_2(p_2)$, we have that for each $\sigma \in [k]$,

$$\begin{aligned}\tilde{y}^\sigma \circ s_1(p_1) &= \text{proj}_\sigma^k \circ \phi(s_1(p_1)) \\ &= \text{proj}_\sigma^k \circ \phi(s_2(p_2)) \\ &= \tilde{y}^\sigma \circ s_2(p_2).\end{aligned}$$

Therefore $s_1 \sim_{r,p} s_2$ and $J_{p_1}^r s_1 = J_{p_2}^r s_2$. Hence

$$\begin{aligned}b_1 &= (p_1, J_{p_1}^r s_1) \\ &= (p_2, J_{p_2}^r s_2) \\ &= b_2.\end{aligned}$$

Since $b_1, b_2 \in \pi_r^{-1}(U_{\Phi,\phi})$ are arbitrary, we have that for each $b_1, b_2 \in \pi_r^{-1}(U_{\Phi,\phi})$, $\Phi_{\Phi,\phi}^r(b_1) = \Phi_{\Phi,\phi}^r(b_2)$ implies that $b_1 = b_2$. Thus $\Phi_{\Phi,\phi}^r$ is injective.

• **(surjectivity):**

Let $(p, f, g) \in U_{\Phi,\phi} \times F \times \mathbb{R}^{N-k-n}$. Define $a \in \pi^{-1}(U_{\Phi,\phi})$ by $a := \Phi^{-1}(p, f)$. We note that

$$\begin{aligned}\pi(a) &= \text{proj}_{U_{\Phi,\phi}} \circ \Phi(a) \\ &= \text{proj}_{U_{\Phi,\phi}} \circ \Phi(\Phi^{-1}(p, f)) \\ &= \text{proj}_{U_{\Phi,\phi}}(p, f) \\ &= p.\end{aligned}$$

Exercise 21.1.0.8 implies that there exists $s \in \Gamma_p(\pi)$ such that $s(p) = a$ and $\partial_\alpha[\tilde{y}^\sigma \circ s](p) = g_\alpha^\sigma$. Then $(p, J_p s) \in \pi_r^{-1}(U_{\Phi, \phi})$ and

$$\begin{aligned}\Phi_{\Phi, \phi}(p, J_p s) &= (\Phi \circ s(p), (\partial_\alpha[\tilde{y}^\sigma \circ s](p))_{\sigma, \alpha}) \\ &= (\Phi(a), g) \\ &= (p, f, g)\end{aligned}$$

Since $(p, f, g) \in U_{\Phi, \phi} \times F \times \mathbb{R}^{N-k-n}$ is arbitrary, we have that for each $(p, f, g) \in U_{\Phi, \phi} \times F \times \mathbb{R}^{N-k-n}$, there exists $(p, J_p s) \in \pi_r^{-1}(U_{\Phi, \phi})$ such that $\Phi_{\Phi, \phi}(p, J_p s) = (p, f, g)$. Hence $\Phi_{\Phi, \phi}$ is surjective.

Since $\Phi_{\Phi, \phi}$ is injective and surjective, $\Phi_{\Phi, \phi}$ is a bijection.

3. Let $(p, J_p^r s) \in \pi_r^{-1}(U_{\Phi, \phi})$. Since $\text{proj}_{U_{\Phi, \phi}}^{U_{\Phi, \phi} \times F} \circ \Phi = \pi|_{U_{\Phi, \phi}}$, we have that

$$\begin{aligned}\text{proj}_{U_{\Phi, \phi}}^{U_{\Phi, \phi} \times F} \Phi(s(p)) &= \pi \circ s(p) \\ &= p.\end{aligned}$$

Therefore there exists $f \in F$ such that $\Phi(s(p)) = (p, f)$. Hence

$$\begin{aligned}\text{proj}_{U_{\Phi, \phi}}^{U_{\Phi, \phi} \times F \times \mathbb{R}^{N-k-n}} \circ \Phi_{\Phi, \phi}^r(p, J_p^r s) &= \text{proj}_{U_{\Phi, \phi}}^{U_{\Phi, \phi} \times F \times \mathbb{R}^{N-k-n}} ((\Phi \circ s(p), (\partial_\alpha[\tilde{y}^\sigma \circ s](p))_{\sigma, \alpha})) \\ &= \text{proj}_{U_{\Phi, \phi}}^{U_{\Phi, \phi} \times F} \circ \Phi \circ s(p) \\ &= \text{proj}_{U_{\Phi, \phi}}^{U_{\Phi, \phi} \times F}(p, f). \\ &= \pi_r(p, J_p^r s).\end{aligned}$$

Since $(p, J_p^r s) \in \pi_r^{-1}(U_{\Phi, \phi})$ is arbitrary, we have that for each $(p, J_p^r s) \in \pi_r^{-1}(U_{\Phi, \phi})$, $\text{proj}_{U_{\Phi, \phi}}^{U_{\Phi, \phi} \times F \times \mathbb{R}^{N-k-n}} \circ \Phi_{\Phi, \phi}^r(p, J_p^r s) = \pi_r(p, J_p^r s)$. Thus

$$\text{proj}_{U_{\Phi, \phi}}^{U_{\Phi, \phi} \times F \times \mathbb{R}^{N-k-n}} \circ \Phi_{\Phi, \phi}^r = \pi_r|_{\pi_r^{-1}(U_{\Phi, \phi})}.$$

4. Since $\Phi_{\Phi, \phi}$ is a bijection and $\text{proj}_{U_{\Phi, \phi}}^{U_{\Phi, \phi} \times F \times \mathbb{R}^{N-k-n}} \circ \Phi_{\Phi, \phi} = \pi_r|_{\pi_r^{-1}(U_{\Phi, \phi})}$, we have that $(U_{\Phi, \phi}, \Phi_{\Phi, \phi})$ is a local trivialization with respect to π_r of $J^r \pi$ over $U_{\Phi, \phi}$ with fiber F .

□

Exercise 21.3.0.7. Let $(E, M, \pi, F) \in \text{Obj}(\mathbf{Bun}^\infty)$. Then there exists a unique topology $\mathcal{T}_{J^r \pi}$ on $J^r \pi$ and smooth structure $\mathcal{A}_{J^r \pi} \subset X^{n+k}(J^r \pi, \mathcal{T}_{J^r \pi})$ on $J^r \pi$ such that

1. $(J^r \pi, \mathcal{T}_{J^r \pi})$ is an $n + k$ -dimensional topological manifold and $(J^r \pi, \mathcal{T}_{J^r \pi}, \mathcal{A}_{J^r \pi})$ is a smooth manifold,
2. for each $(U_\phi, \phi) \in \mathcal{A}_E^\pi$ and smooth fiber bundle local trivialization (U_Φ, Φ) of E over U_Φ with fiber F with respect to π fiber F **create notation for local trivializations and bundle atlases to avoid writing this over and over**. $\pi_r^{-1}(U_\alpha) \in \mathcal{T}_{J^r \pi}$ and $\Phi_{\Phi, \phi} : \pi_r^{-1}(U_{\Phi, \phi}) \rightarrow U_{\Phi, \phi} \times F \times \mathbb{R}^{N-k-n}$ is a diffeomorphism
3. $\pi_r : J^r \pi \rightarrow M$ is smooth
4. $(J^r \pi, M, \pi_r, F \times \mathbb{R}^{N-k-n}) \in \text{Obj}(\mathbf{Bun}^\infty)$

IDEA: make exercise about $(J^r \pi, E, \pi_{0,r})$ being a fiber bundle, make another exercise in the section on fiber bundles that if (E, M, π_M^E, F) is a fiber bundle and (G, E, π_E^G, F) is a fiber bundle, then $(G, M, \pi_{M,G}, F)$ where $\pi_{M,E} \circ \pi_{E,G} := \pi_{M,G}$ is a fiber bundle, then apply that exercise here?, might be cleaner/more general

Proof.

- for each $\alpha \in \Gamma$, $U_\alpha \in \mathcal{T}_M$
- $M \subset \bigcup_{\alpha \in \Gamma} U_\alpha$

- for each $\alpha \in \Gamma$, (U_α, Φ_α) is a local trivialization with respect to π of E over U_α with fiber F
- for each $\alpha, \beta \in \Gamma$, $\Phi_\beta|_{\pi^{-1}(U_\alpha \cap U_\beta)} \circ (\Phi_\alpha|_{\pi^{-1}(U_\alpha \cap U_\beta)})^{-1} : (U_\alpha \cap U_\beta) \times F \rightarrow (U_\alpha \cap U_\beta) \times F$ is smooth.

Let $p \in M$. Since π is surjective, there exists $a \in E$ such that $\pi(a) = p$. Exercise 21.1.0.5 implies that there exists $(U, \phi) \in \mathcal{A}_E^\pi$ such that $a \in U$. Then there exists $(U_0, \phi_0) \in \mathcal{A}_M$ such that $U_0 = \pi(U)$ and $\phi_0 \circ \pi = \text{proj}_n^{n+k} \circ \phi$. Since $(E, M, \pi, F) \in \text{Obj}(\mathbf{Bun}^\infty)$, there exists $U'_0 \in \mathcal{T}_M$ and $\Phi : \pi^{-1}(U'_0) \rightarrow U'_0 \times F$ such that $p \in U'_0$ and (U'_0, Φ) is a smooth local trivialization of E over U with fiber F . We define $V \in \mathcal{T}_M$ by $V := U_0 \cap U'_0$

- By construction
-
-
-

Exercise 15.1.3.3 **FINISH!!!**

□

Exercise 21.3.0.8. Let $s_1, s_2 \in \Gamma_p(\pi)$. Write $\phi_0 = (x^1, \dots, x^n, v^1, \dots, v^k)$ and $\psi_0 = (y^1, \dots, y^n, \omega^1, \dots, \omega^k)$, $\phi = (\tilde{x}^1, \dots, \tilde{x}^n)$ and $\psi = (\tilde{y}^1, \dots, \tilde{y}^n)$. Then for each $j \in [n]$ and $l \in [k]$,

$$\frac{\partial}{\partial \tilde{x}^j} \Big|_{\pi(a)} (v^l \circ s_1) = \frac{\partial}{\partial \tilde{x}^j} \Big|_{\pi(a)} (v^l \circ s_2)$$

iff for each $j' \in [n]$ and $l' \in [k]$,

$$\frac{\partial}{\partial \tilde{y}^{j'}} \Big|_{\pi(a)} (\omega^{l'} \circ s_1) = \frac{\partial}{\partial \tilde{y}^{j'}} \Big|_{\pi(a)} (\omega^{l'} \circ s_2).$$

Proof. Set $p := \pi(a)$.

• ($\implies$) :

Suppose that for each $j \in [n]$ and $l \in [k]$,

$$\frac{\partial}{\partial \tilde{x}^j} \Big|_p (v^l \circ s_1) = \frac{\partial}{\partial \tilde{x}^j} \Big|_p (v^l \circ s_2).$$

Let $j' \in [j]$ and $l' \in [k]$. Then

$$\begin{aligned} \frac{\partial}{\partial \tilde{y}^{j'}} \Big|_p (\omega^{l'} \circ s_1) &= \sum_{m=1}^n \frac{\partial \tilde{x}^m}{\partial \tilde{y}^{j'}}(a) \frac{\partial}{\partial \tilde{x}^m} \Big|_p (\omega^{l'} \circ s_1) \\ &= \sum_{m=1}^n \frac{\partial \tilde{x}^m}{\partial \tilde{y}^{j'}}(a) \left[\sum_{j=1}^n \frac{\partial \omega^{l'}}{\partial x^j}(s_1(p)) \frac{\partial}{\partial \tilde{x}^m} \Big|_p (x^j \circ s_1) + \sum_{l=1}^k \frac{\partial \omega^{l'}}{\partial v^l}(s_1(p)) \frac{\partial}{\partial \tilde{x}^m} \Big|_p (v^l \circ s_1) \right] \\ &= \sum_{m=1}^n \frac{\partial \tilde{x}^m}{\partial \tilde{y}^{j'}}(a) \left[\sum_{j=1}^n \frac{\partial \omega^{l'}}{\partial x^j}(s_1(p)) \frac{\partial}{\partial \tilde{x}^m} \Big|_p (x^j \circ s_1) + \sum_{l=1}^k \frac{\partial \omega^{l'}}{\partial v^l}(s_1(p)) \frac{\partial}{\partial \tilde{x}^m} \Big|_p (v^l \circ s_2) \right] \end{aligned}$$

FINISH!!!, need to get rid of fibered charts, contact order is defined more generally, should move this exercise to the smooth maps section

• ($\impliedby$:)

□

need to go over multi index notation for partial derivatives

Definition 21.3.0.9. Let (E, M, π) be a smooth fibered manifold.

Exercise 21.3.0.10.

Chapter 22

Connections

22.1 Ehresmann Connections

Definition 22.1.0.1. Let $(P, X, \pi, G, \triangleleft) \in \text{Obj}(\mathbf{PrinBun}^\infty)$ and $p \in P$. Set $x := \pi(p)$. We define the **verticle tangent space of P at p** , denoted V_p , by $V_p := T_p(P_x)$.

Exercise 22.1.0.2. Let $(P, X, \pi, G, \triangleleft) \in \text{Obj}(\mathbf{PrinBun}^\infty)$. For each $p \in P$, $V_p = \ker D\pi(p)$.

Proof. Let $p \in P$. Set $x := \pi(p)$. ref ex about tangent space of subamnifold being the kernel of derivative

□

22.2 Koszul Connections

Definition 22.2.0.1.

- Let $(E, M, \pi) \in \text{Obj}(\mathbf{VecBun}^\infty)$ and $\nabla : \Gamma(E) \rightarrow \Gamma(T^*M \otimes E)$. Then ∇ is said to be a **Koszul connection on E** if for each $f \in C^\infty(M)$ and $s \in \Gamma(E)$, $\nabla(fs) = df \otimes s + f \nabla s$.
- We define $\text{KosConn}(E) := \{\nabla : \Gamma(E) \rightarrow \Gamma(T^*M \otimes E) : \nabla \text{ is a Koszul connection}\}$.

Exercise 22.2.0.2. content...

Definition 22.2.0.3. Let $(E, M, \pi) \in \text{Obj}(\mathbf{VecBun}^\infty)$ and $\nabla \in \text{KosConn}$. We define the **covariant derivative induced by ∇** , denoted $\nabla : \mathfrak{X}(M) \times \Gamma(E) \rightarrow \Gamma(E)$, by $\nabla(X, s) := \nabla(s)$

Definition 22.2.0.4. Let $(E, M, \pi) \in \text{Obj}(\mathbf{VecBun}^\infty)$, $\nabla_1 : \Gamma(E) \rightarrow \Gamma(T^*M \otimes E)$ and $\nabla_2 : \mathfrak{X}(M) \times \Gamma(E) \rightarrow \Gamma(E)$. Then

- ∇_1 is said to be a **type-1 Koszul connection on E** if for each $f \in C^\infty(M)$ and $s \in \Gamma(E)$, $\nabla_1(fs) = df \otimes s + f \nabla_1 s$.
- ∇_2 is said to be a **type-2 Koszul connection on E** if
 1. for each $s \in \Gamma(E)$, $\nabla(\cdot, s)$ is $C^\infty(M)$ -linear
 2. for each $X \in \mathfrak{X}(M)$, $\nabla(X, \cdot)$ is $\mathbb{R}$ -linear
 3. for each $X \in \mathfrak{X}(M)$, $s \in \Gamma(E)$ and $f \in C^\infty(M)$,

$$\nabla(X, fs) = f \nabla(X, s) + X(f)s$$

- We define T

Exercise 22.2.0.5. Let $(E, M, \pi) \in \text{Obj}(\mathbf{VecBun}^\infty)$. There exists $\phi : \text{Conn}_1 \rightarrow \text{Conn}_2$ such that ϕ is a bijection.

Proof. • Let $\nabla_1 \in \text{Conn}_1$, $X \in \mathfrak{X}(M)$ and $s \in \Gamma(E)$. Set $\nabla_2(X, s) := \nabla_1(s)(X)$. □

Exercise 22.2.0.6. We define $\text{Conn}_1(E) := \{\nabla_1 : \Gamma(E) \rightarrow \Gamma(T^*M \otimes E) : \nabla \text{ is a Koszul connection}\}$.

Proof. content... □

Note 22.2.0.7. We identify type-1 and type-2 Koszul connections.

Definition 22.2.0.8. Let $(E, M, \pi) \in \text{Obj}(\mathbf{VecBun}^\infty)$ be a smooth vector bundle and $\nabla : \Gamma(E) \rightarrow \Gamma(T^*M \otimes E)$. Then ∇ is said to be a **Koszul connection on E in the second representation** if

1. ∇ is $\mathbb{R}$ -linear
2. for each $s \in \Gamma(E)$ and $f \in C^\infty(M)$,

$$\nabla(fs) = f \nabla s + df \otimes s$$

Exercise 22.2.0.9. There exists a bijection $\phi : \text{Conn}_1 \rightarrow \text{Conn}_2$.

Proof. Let $\nabla \in \text{Conn}_1$. We define $\phi(\nabla) : \mathfrak{X}(M) \times \Gamma(E) \rightarrow \Gamma(E)$ by

$$\phi(\nabla)(X, s) = (\nabla s)(X)$$

FINISH!!! □

Note 22.2.0.10. When the context is clear, we will write $\nabla_X Y$ in place of $\nabla(X, Y)$ and we will refer to ∇ as a connection.

Exercise 22.2.0.11. Let $(E, M, \pi) \in \text{Obj}(\mathbf{VecBun}^\infty)$, ∇ a connection on E , $X \in \mathfrak{X}(M)$ and $Y \in \Gamma(E)$. If $X = 0$ or $Y = 0$, then $\nabla_X Y = 0$.

Proof.

- If $X = 0$, then

$$\begin{aligned}\nabla_X Y &= \nabla_{0X} Y \\ &= 0 \nabla_X Y \\ &= 0\end{aligned}$$

- Similarly, if $Y = 0$, then $\nabla_X Y = 0$.

□

Exercise 22.2.0.12. Let (E, M, π) be a smooth vector bundle, ∇ a connection on E , $X \in \mathfrak{X}(M)$, $Y \in \Gamma(E)$ and $p \in M$. If $X \sim_p 0$ or $Y \sim_p 0$, then $[\nabla_X Y]_p = 0$.

Proof.

- Suppose that $X \sim_p 0$. Then there exists $U \subset M$ such that U is open and $X|_U = 0$. Choose $\phi \in C^\infty(M)$ such that $\text{supp } \phi \subset U$ and $\phi \sim_p 1$. Then $\phi X = 0$. The previous exercise implies that $\nabla_{\phi X} Y = 0$. Therefore

$$\begin{aligned}\nabla_X Y &= \nabla_{\phi X + (1-\phi)X} Y \\ &= \nabla_{\phi X} Y + \nabla_{(1-\phi)X} Y \\ &= 0 + (1-\phi) \nabla_X Y \\ &= (1-\phi) \nabla_X Y\end{aligned}$$

Hence

$$\begin{aligned}[\nabla_X Y]_p &= [(1-\phi) \nabla_X Y]_p \\ &= (1-\phi(p))[\nabla_X Y]_p \\ &= 0\end{aligned}$$

- Suppose that $Y \sim_p 0$. Then there exists $U \subset M$ such that U is open and $Y|_U = 0$. Choose $\phi \in C^\infty(M)$ such that $\text{supp } \phi \subset U$ and $\phi \sim_p 1$. Then $\phi Y = 0$. The previous exercise implies that $\nabla_X \phi Y = 0$. Since $\phi \sim_p 1$, we have that $1-\phi \sim_p 0$. Thus $X(1-\phi) \sim_p 0$ and

$$\begin{aligned}\nabla_X Y &= \nabla_X [\phi Y + (1-\phi)Y] \\ &= \nabla_X [\phi Y] + \nabla_X [(1-\phi)Y] \\ &= \nabla_X [(1-\phi)Y] \\ &= (1-\phi) \nabla_X Y + [X(1-\phi)] \nabla_X Y\end{aligned}$$

Hence

$$\begin{aligned}[\nabla_X Y]_p &= (1-\phi(p))[\nabla_X Y]_p + [X(1-\phi)](p)[\nabla_X Y]_p \\ &= 0\end{aligned}$$

□

Exercise 22.2.0.13. Let (E, M, π) be a smooth vector bundle and ∇ a connection on E . Then for each $X_1, X_2 \in \mathfrak{X}(M)$ and $Y_1, Y_2 \in \Gamma(E)$, $X_1 \sim_p X_2$ and $Y_1 \sim_p Y_2$ implies that $[\nabla_{X_1} Y_1]_p = [\nabla_{X_2} Y_2]_p$.

Proof. Let $X_1, X_2 \in \mathfrak{X}(M)$ and $Y_1, Y_2 \in \Gamma(E)$. Suppose that $X_1 \sim_p X_2$ and $Y_1 \sim_p Y_2$. Define $X \in \mathfrak{X}(M)$ and $Y \in \Gamma(E)$ by $X = X_2 - X_1$ and $Y = Y_2 - Y_1$. Then $X \sim_p 0$ and $Y \sim_p 0$. The previous exercise implies that $[\nabla_X Y_1]_p = 0$ and $[\nabla_{X_2} Y]_p = 0$.

Therefore

$$\begin{aligned}
 [\nabla_{X_1} Y_1]_p &= [\nabla_{X_1} Y_1]_p + [\nabla_X Y_1]_p \\
 &= [\nabla_{X_1} Y_1 + \nabla_X Y_1]_p \\
 &= [\nabla_{X_1+X} Y_1]_p \\
 &= [\nabla_{X_2} Y_1]_p \\
 &= [\nabla_{X_2} Y_1]_p + [\nabla_{X_2} Y]_p \\
 &= [\nabla_{X_2} Y_1 + \nabla_{X_2} Y]_p \\
 &= [\nabla_{X_2} (Y_1 + Y)]_p \\
 &= [\nabla_{X_2} Y_2]_p
 \end{aligned}$$

□

Exercise 22.2.0.14. Let (E, M, π) be a smooth vector bundle, ∇ a connection on E and $U \subset M$. If U is open, then there exists a unique connection $\nabla^U : \mathfrak{X}(U) \times \Gamma(E|_U) \rightarrow \Gamma(E|_U)$ such that for each $X \in \mathfrak{X}(M)$ and $Y \in \Gamma(E)$,

$$\nabla_{X|_U}^U Y|_U = (\nabla_X Y)|_U$$

Chapter 23

Semi-Riemannian Geometry

23.1 Metric Tensors

Definition 23.1.0.1. Let M be a manifold and $g \in \Gamma(\Sigma^2 M)$. Then g is said to be nondegenerate if for each $p \in M$, g_p is nondegenerate.

Definition 23.1.0.2. Let M be a manifold and $g \in \Gamma(\Sigma^2 M)$.

- Then g is said to be a **metric tensor field** on M if
 1. g is nondegenerate,
 2. g has constant index.
- If g is a metric tensor field on M , then (M, g) is said to be a **semi-Riemannian manifold**.

Definition 23.1.0.3.

23.2 Curvature

Definition 23.2.0.1. Define Interval
FINISH!!!

Definition 23.2.0.2. Let $(E, M, \pi) \in \text{Obj}(\mathbf{Bun}^\infty)$, $I \subset \mathbb{R}$ an interval, $\alpha \in \text{Hom}_{\mathbf{Man}^\infty}(I, M)$ and $\gamma \in \text{Hom}_{\mathbf{Man}^\infty}(I, E)$. Then γ is said to be a **section of E over α** if $\pi \circ \gamma = \alpha$. We denote the set of sections of E over α by $\Gamma(E, \alpha)$.

Definition 23.2.0.3. Let $(E, M, \pi) \in \text{Obj}(\mathbf{Bun}^\infty)$, $I \subset \mathbb{R}$ an interval, $\alpha \in \text{Hom}_{\mathbf{Man}^\infty}(I, M)$ and $\gamma \in \Gamma(E, \alpha)$. Then γ is said to be **extendible** if there exists $U \in \mathcal{N}_{\alpha(I)}$ and $\tilde{\gamma} \in \Gamma(E|_U)$ such that U is open and $\tilde{\gamma} \circ \alpha = \gamma$.

Exercise 23.2.0.4. figure 8 not extendible FINISH!!!

Exercise 23.2.0.5. Let $(E, M, \pi) \in \text{Obj}(\mathbf{VecBun}^\infty)$, ∇ a connection on E , $I \subset \mathbb{R}$ an interval and $\alpha \in \text{Hom}_{\mathbf{Man}^\infty}(I, M)$. There exists a unique $D_\alpha : \Gamma(E, \alpha) \rightarrow \Gamma(E, \alpha)$ such that

1. for each $\lambda \in \mathbb{R}$ and $\gamma, \sigma \in \Gamma(E, \alpha)$,

$$D_\alpha(\gamma + \lambda\sigma) = D_\alpha\gamma + \lambda D_\alpha\sigma$$

2. for each $f \in C^\infty(I)$ and $\gamma \in \Gamma(E, \alpha)$,

$$D_\alpha(f\gamma) = f'\gamma + fD_\alpha\gamma$$

3. for each $\gamma \in \Gamma(E)$, if $\tilde{\gamma}$ extends γ , then

$$D_\alpha\gamma = \nabla_{\alpha'}\gamma$$

Proof.

□

Chapter 24

Riemannian Geometry

Definition 24.0.0.1. Let M be a smooth manifold and $g \in T_2^0(M)$ a metric tensor on M . We define $\hat{g} \in T_0^2(M)$ by $\hat{g}(\omega, \eta) = g(\phi_g^{-1}(\omega), \phi_g^{-1}(\eta))$.

Exercise 24.0.0.2. content...

Exercise 24.0.0.3. Let (M, g) be a semi-Riemannian manifold and $(U, \phi) \in \mathcal{A}$. Then the induced metric $\langle \rangle_{T^*M \otimes TM}$ on $T^*M \otimes TM$ is given by

$$\left\langle dx^i \otimes \frac{\partial}{\partial x^k}, dx^j \otimes \frac{\partial}{\partial x^l} \right\rangle_{T^*M \otimes TM} = g^{i,j} g_{kl}$$

Proof. We have that

$$\begin{aligned} \left\langle dx^i \otimes \frac{\partial}{\partial x^k}, dx^j \otimes \frac{\partial}{\partial x^l} \right\rangle_{T^*M \otimes TM} &= \langle dx^i, dx^j \rangle_{T^*M} \left\langle \frac{\partial}{\partial x^k}, \frac{\partial}{\partial x^l} \right\rangle_{TM} \\ &= g^{i,j} g_{k,l} \end{aligned}$$

□

Exercise 24.0.0.4. Let (M, g) be an n -dimensional Riemannian manifold.

1. There exists $\lambda \in \Omega^n(M)$ such that for each orthonormal frame $e_1, \dots, e_n$,

$$\lambda(e_1, \dots, e_n) = 1$$

Hint: Choose a frame $z_1, \dots, z_n$ on M with corresponding dual frame $\zeta^1, \dots, \zeta^n$. Define

$$\lambda = \det[g(z_i, z_j)]^{1/2} \zeta^1 \wedge \dots \wedge \zeta^n$$

2. Let $N \in \mathfrak{X}(M)$ be the outward pointing normal to ∂M and $X \in \mathfrak{X}(M)$. Then

$$\int_M \operatorname{div} X \lambda = \int_{\partial M} g(X, N) \tilde{\lambda}$$

3. For each $u \in \mathbb{C}^\infty(M)$ and $X \in \mathfrak{X}(M)$, we have that

$$\operatorname{div}(uX) = u \operatorname{div}(X) + du(X)$$

and therefore

$$\int_M du(X) \lambda = \int_{\partial M} u g(X, N) \tilde{\lambda} - \int_M u \operatorname{div}(X) \lambda$$

Proof.

1. Let $z_1, \dots, z_n$ be a frame on M and $\zeta^1, \dots, \zeta^n$ with corresponding dual frame $\zeta^1, \dots, \zeta^n$. Define

$$\lambda = \det[g(z_i, z_j)]^{1/2} \zeta^1 \wedge \dots \wedge \zeta^n$$

Let $e_1, \dots, e_n$, be an orthonormal frame on M with corresponding dual coframe $\epsilon^1, \dots, \epsilon^n$. Let $i, j \in \{1, \dots, n\}$. Then there exist $(a_{k,i}) \subset \mathbb{R}$ such that $\zeta^i = \sum_{k=1}^n a_{k,i} \epsilon^k$. Then

$$\begin{aligned} \hat{g}(\epsilon^j, \zeta^i) &= \sum_{k=1}^n a_{k,i} \hat{g}(\epsilon^j, \epsilon^k) \\ &= \sum_{k=1}^n a_{k,i} g(\phi_g^{-1}(\epsilon^j), \phi_g^{-1}(\epsilon^k)) \\ &= \sum_{k=1}^n a_{k,i} g(e_j, e_k) \\ &= \sum_{k=1}^n a_{k,i} \delta_{j,k} \\ &= a_{j,i} \end{aligned}$$

which implies that

$$\begin{aligned} \delta_{i,j} &= \zeta^i(z_j) \\ &= \sum_{k=1}^n a_{k,i} \epsilon^k(z_j) \\ &= \sum_{k=1}^n a_{k,i} g(e_k, z_j) \\ &= \sum_{k=1}^n \hat{g}(\epsilon^k, \zeta^i) g(e_k, z_j) \end{aligned}$$

Define $U, V \in \mathbb{R}^{n \times n}$ by $U_{i,k} = \hat{g}(\zeta^i, \epsilon^k)$ and $V_{k,j} = g(e_k, z_j)$. Then from above, we have that $UV = I$. Since $U, V \in \mathbb{R}^{n \times n}$, $VU = I$. Hence $U = V^{-1}$. Since

$$\begin{aligned} \zeta^i(e_j) &= \sum_{k=1}^n a_{k,i} \epsilon^k(e_j) \\ &= \sum_{k=1}^n a_{k,i} \delta_{k,j} \\ &= a_{j,i} \\ &= \hat{g}(\epsilon^j, \zeta^i) \\ &= U_{i,j} \end{aligned}$$

and

$$\begin{aligned}
g(z_i, z_j) &= \left(\sum_{k=1}^n g(e_k, z_i) e_k, \sum_{l=1}^n g(e_l, z_j) e_l \right) \\
&= \sum_{k=1}^n \sum_{l=1}^n g(e_k, z_i) g(e_l, z_j) g(e_k, e_l) \\
&= \sum_{k=1}^n \sum_{l=1}^n g(e_k, z_i) g(e_l, z_j) \delta_{k,l} \\
&= \sum_{k=1}^n g(e_k, z_i) g(e_k, z_j) \\
&= (V^* V)_{i,j}
\end{aligned}$$

we have that

$$\begin{aligned}
\lambda(e_1, \dots, e_n) &= \det[g(z_i, z_j)]^{1/2} \zeta^1 \wedge \dots \wedge \zeta^n(e_1, \dots, e_n) \\
&= \det[g(z_i, z_j)]^{1/2} \det[\zeta^i(e_j)] \\
&= \det(V^* V)^{1/2} \det U \\
&= \det V (\det V)^{-1} \\
&= 1
\end{aligned}$$

2. Choose an orthonormal frame $e_1, \dots, e_{n-1} \in \mathfrak{X}(\partial M)$ with dual coframe $\epsilon^1, \dots, \epsilon^{n-1}$. Define $\nu \in \Omega^1(M)$ to be the dual covector to N . We note that $N, e_1, \dots, e_{n-1}$ is an orthonormal frame on $\mathfrak{X}(M)$. Let $X_1, \dots, X_{n-1} \in \mathfrak{X}(\partial M)$. Since for each $j \in \{1, \dots, n-1\}$, $X_j \in \mathfrak{X}(\partial M)$ and for each $p \in \partial M$, $N_p \in (T_p \partial M)^\perp$, we have that for each $j \in \{1, \dots, n-1\}$, $g(X_j, N) = 0$. This implies that

$$\begin{aligned}
\iota^* \iota_X \lambda(X_1, \dots, X_{n-1}) &= \lambda(X, X_1, \dots, X_{n-1}) \\
&= \nu \wedge \epsilon^1 \wedge \dots \wedge \epsilon^{n-1}(X, X_1, \dots, X_{n-1}) \\
&= \det \begin{pmatrix} \nu(X) & \nu(X_1) & \dots & \nu(X_{n-1}) \\ \epsilon^1(X) & \epsilon^1(X_1) & \dots & \epsilon^1(X_{n-1}) \\ \vdots & \vdots & \ddots & \vdots \\ \epsilon^{n-1}(X) & \epsilon^{n-1}(X_1) & \dots & \epsilon^{n-1}(X_{n-1}) \end{pmatrix} \\
&= \det \begin{pmatrix} g(X, N) & g(X_1, N) & \dots & g(X_{n-1}, N) \\ \epsilon^1(X) & \epsilon^1(X_1) & \dots & \epsilon^1(X_{n-1}) \\ \vdots & \vdots & \ddots & \vdots \\ \epsilon^{n-1}(X) & \epsilon^{n-1}(X_1) & \dots & \epsilon^{n-1}(X_{n-1}) \end{pmatrix} \\
&= g(X, N) \det(\epsilon^i(X_j)) \\
&= g(X, N) \epsilon^1 \wedge \dots \wedge \epsilon^{n-1}(X_1, \dots, X_{n-1}) \\
&= g(X, N) \tilde{\lambda}(X_1, \dots, X_{n-1})
\end{aligned}$$

Therefore $\iota^* \iota_X \lambda = g(X, N) \tilde{\lambda}$ and

$$\begin{aligned}
\int_M \operatorname{div} X \lambda &= \int_M d(\iota_X \lambda) \\
&= \int_{\partial M} \iota^*(\iota_X \lambda) \\
&= \int_{\partial M} g(X, N) \tilde{\lambda}
\end{aligned}$$

3. We note that

$$\begin{aligned} 0 &= \iota_X(du \wedge \lambda) \\ &= \iota_X(du) \wedge \lambda - du \wedge (\iota_X \lambda) \\ &= du(X)\lambda - du \wedge (\iota_X \lambda) \end{aligned}$$

which implies that

$$\begin{aligned} \operatorname{div}(uX)\lambda &= d(\iota_{uX}\lambda) \\ &= d(u\iota_X\lambda) \\ &= du \wedge (\iota_X\lambda) + u d(\iota_X\lambda) \\ &= du(X)\lambda + u \operatorname{div}(X)\lambda \\ &= [du(X) + u \operatorname{div}(X)]\lambda \end{aligned}$$

This implies that $\operatorname{div}(uX) = du(X) + u \operatorname{div}(X)$. From before, we have that

$$\begin{aligned} \int_M du(X)\lambda &= \int_M \operatorname{div}(uX)\lambda - \int_M u \operatorname{div}(X)\lambda \\ &= \int_{\partial M} g(uX, N)\tilde{\lambda} - \int_M u \operatorname{div}(X)\lambda \\ &= \int_{\partial M} u g(X, N)\tilde{\lambda} - \int_M u \operatorname{div}(X)\lambda \end{aligned}$$

□

Exercise 24.0.0.5.

$$\operatorname{div}(X) = \sum_{j=1}^n (\nabla_{\partial_j} X)^j$$

Proof. We have that

$$\begin{aligned} \nabla_{\partial_i}(X) &= \sum_{j=1}^n \nabla_{\partial_i}(X^j \partial_j) \\ &= \sum_{j=1}^n \left[X^j \nabla_{\partial_i} \partial_j + \partial_i(X^j) \partial_j \right] \\ &= \sum_{j=1}^n \left[X^j \left(\sum_{k=1}^n \Gamma_{i,j}^k \partial_k \right) + \partial_i(X^j) \partial_j \right] \\ &= \sum_{j=1}^n \left[X^j \left(\sum_{k=1}^n \Gamma_{i,j}^k \partial_k \right) + \partial_i(X^j) \partial_j \right] \\ &= \sum_{j=1}^n X^j \left(\sum_{k=1}^n \Gamma_{i,j}^k \partial_k \right) + \sum_{j=1}^n \partial_i(X^j) \partial_j \\ &= \sum_{k=1}^n \left(\sum_{j=1}^n X^j \Gamma_{i,j}^k \right) \partial_k + \sum_{k=1}^n \partial_i(X^k) \partial_k \\ &= \sum_{k=1}^n \left[\left(\sum_{j=1}^n X^j \Gamma_{i,j}^k \right) + \partial_i(X^k) \right] \partial_k \end{aligned}$$

so that $(\nabla_{\partial_i}(X))^i = \left(\sum_{j=1}^n X^j \Gamma_{i,j}^i \right) + \partial_i(X^i)$. We note that

$$\begin{aligned} \operatorname{div}(X) &= \sum_{i=1}^n \operatorname{div}(X^i \partial_i) \\ &= \sum_{i=1}^n [X^i \operatorname{div}(\partial_i) + dx^i(\partial_i)] \\ &= \sum_{i=1}^n [X^i \operatorname{div}(\partial_i) + 1] \end{aligned}$$

Since $\lambda = [\det g(\partial_i, \partial_j)]^{1/2} dx^1 \wedge \cdots \wedge dx^n = (\det g)^{1/2} dx$, we have that

$$\begin{aligned} d(\iota_{\partial_i} \lambda) &= d((\det g)^{1/2} \iota_{\partial_i} dx) \\ &= d[(\det g)^{1/2}] \iota_{\partial_i} dx + (\det g)^{1/2} d(\iota_{\partial_i} dx) \\ &= d[(\det g)^{1/2}] \sum_{k=1}^n (-1)^{k-1} dx^1 \wedge \cdots \wedge \widehat{dx^k} \wedge \cdots \wedge dx^n + (\det g)^{1/2} \sum_{k=1}^n (-1)^{k-1} dx^1 \wedge \cdots \wedge \widehat{dx^k} \wedge \cdots \wedge dx^n \end{aligned}$$

FINISH!!!

□

Exercise 24.0.0.6. Let (M, g) be a Riemannian manifold.

1. For each $u, v \in C^\infty(M)$. Then

(a)

$$\int_M u \Delta v \lambda + \int_M g(\nabla u, \nabla v) \lambda = \int_{\partial M} u N(v) \tilde{\lambda}$$

(b)

$$\int_M [u \Delta v - v \Delta u] \lambda = \int_{\partial M} [u N(v) - v N(u)] \tilde{\lambda}$$

2. (a) If $\partial M \neq \emptyset$, then for each $u, v \in C^\infty(M)$, u and v are harmonic and $u|_{\partial M} = v|_{\partial M}$ implies that $u = v$.

(b) If $\partial M = \emptyset$, then for each $u \in C^\infty(M)$, u is harmonic implies that u is constant.

Proof.

1. Let $u, v \in C^\infty(M)$. Then

(a)

$$\begin{aligned} \int_M u \Delta v \lambda &= \int_M u \operatorname{div}(\nabla v) \lambda \\ &= \int_{\partial M} u g(\nabla v, N) \tilde{\lambda} - \int_M du(\nabla v) \lambda \\ &= \int_{\partial M} u dv(N) \tilde{\lambda} - \int_M g(\nabla u, \nabla v) \lambda \\ &= \int_{\partial M} u N(v) \tilde{\lambda} - \int_M g(\nabla u, \nabla v) \lambda \end{aligned}$$

(b) From above, we have that

$$\begin{aligned}
 \int_M [u\Delta v - v\Delta u]\lambda &= \int_M u\Delta v\lambda - \int_M v\Delta u\lambda \\
 &= \int_{\partial M} uN(v)\tilde{\lambda} - \int_M g(\nabla u, \nabla v)\lambda - \left(\int_{\partial M} vN(u)\tilde{\lambda} - \int_M g(\nabla v, \nabla u)\lambda \right) \\
 &= \int_{\partial M} uN(v)\tilde{\lambda} - \int_{\partial M} vN(u)\tilde{\lambda} \\
 &= \int_{\partial M} [uN(v) - vN(u)]\tilde{\lambda}
 \end{aligned}$$

2. (a) Suppose that $\partial M \neq \emptyset$. Let $u, v \in C^\infty(M)$. Suppose that u and v are harmonic and $u|_{\partial M} = v|_{\partial M}$. Then $u - v$ is harmonic and

$$\begin{aligned}
 \int_M \|\nabla(u - v)\|_g^2 \lambda &= \int_M g(\nabla(u - v), \nabla(u - v))\lambda \\
 &= 0 + \int_M g(\nabla(u - v), \nabla(u - v))\lambda \\
 &= \int_M (u - v)\Delta(u - v)\lambda + \int_M g(\nabla(u - v), \nabla(u - v))\lambda \\
 &= \int_{\partial M} (u - v)N(u - v)\tilde{\lambda} \\
 &= 0
 \end{aligned}$$

Thus $\nabla(u - v) = 0$ and $u - v$ is constant. Since $u|_{\partial M} = v|_{\partial M}$, we have that $u - v = 0$ and thus $u = v$.

- (b) Suppose that $\partial M = \emptyset$. Let $u \in C^\infty(M)$. Suppose that u is harmonic. Then

$$\begin{aligned}
 \int_M \|\nabla u\|_g^2 \lambda &= \int_M g(\nabla u, \nabla u)\lambda \\
 &= 0 + \int_M g(\nabla u, \nabla u)\lambda \\
 &= \int_M u\Delta u\lambda + \int_M g(\nabla u, \nabla u)\lambda \\
 &= \int_{\partial M} (u - v)g(\nabla(u - v), N)\tilde{\lambda} \\
 &= 0
 \end{aligned}$$

Therefore $\nabla u = 0$ and u is constant.

□

Chapter 25

Symplectic Geometry

25.1 TODO

- Discuss symplectic and poisson geometry on semi-riemannian manifold, the goal is to get hamiltonian mechanics on spacetime with equations of motion on extended phase-space and try to weyl quantize or generally deform quantize to get an operator formulation of the equations of motion. Then get corresponding differtial schrodinger-type equation

25.2 Symplectic Manifolds

Definition 25.2.0.1. Let $M \in \text{Obj}(\mathbf{Man}^\infty)$ and $\omega \in \Omega^2(M)$. Then ω is said to be **symplectic** if

1. ω is nondegenerate
2. ω is closed

Chapter 26

Extra

Definition 26.0.0.1. When working in $\mathbb{R}^n$, we introduce the formal objects $dx^1, dx_2, \dots, dx^n$. Let $I = (i_1, i_2, \dots, i_k) \in \mathcal{I}_{k,n}$ and $\phi : \mathbb{R}^k \rightarrow \mathbb{R}^n$. Write $\phi = (\phi_1, \phi_2, \dots, \phi_n)$. We formally define $dx^i = dx_{i_1} \wedge dx_{i_2} \wedge \dots \wedge dx_{i_k}$ and $\phi_I = (\phi_{i_1}, \phi_{i_2}, \dots, \phi_{i_k})$.

Definition 26.0.0.2. Let $k \in \{0, 1, \dots, n\}$. We define a $C^\infty(\mathbb{R}^n)$ -module of dimension $\binom{n}{k}$, denoted $\Gamma^k(\mathbb{R}^n)$ to be

$$\Phi_k(\mathbb{R}^n) = \begin{cases} C^\infty(\mathbb{R}^n) & k = 0 \\ \text{span}\{dx^i : I \in \mathcal{I}_{k,n}\} & k \geq 1 \end{cases}$$

For each $\omega \in \Phi_k(\mathbb{R}^n)$ and $\chi \in \Gamma^l(\mathbb{R}^n)$, we may form their **exterior product**, denoted by $\omega \wedge \chi \in \Gamma^{k+l}(\mathbb{R}^n)$. Thus the exterior product is a map $\wedge : \Phi_k(\mathbb{R}^n) \times \Gamma^l(\mathbb{R}^n) \rightarrow \Gamma^{k+l}(\mathbb{R}^n)$. The exterior product is characterized by the following properties:

1. the exterior product is bilinear
2. for each $\omega \in \Phi_k(\mathbb{R}^n)$ and $\chi \in \Gamma^l(\mathbb{R}^n)$, $\omega \wedge \chi = -\chi \wedge \omega$
3. for each $\omega \in \Phi_k(\mathbb{R}^n)$, $\omega \wedge \omega = 0$
4. for each $f \in C^\infty(\mathbb{R}^n)$ and $\omega \in \Phi_k(\mathbb{R}^n)$, $f \wedge \omega = f\omega$

We call $\Phi_k(\mathbb{R}^n)$ the differential k -forms on $\mathbb{R}^n$. Let ω be a k -form on $\mathbb{R}^n$. If $k \geq 1$, then for each $I \in \mathcal{I}_{k,n}$, there exists $f_I \in C^\infty(\mathbb{R}^n)$ such that $\omega = \sum_{I \in \mathcal{I}_{k,n}} f_I dx^i$

Note 26.0.0.3. The terms $dx^1, dx_2, \dots, dx^n$ are are a sort of place holder for the coordinates of a point $x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$. When we work with functions $\phi : \mathbb{R}^k \rightarrow \mathbb{R}^n$, we will have different coordinates and to avoid confusion, we will write $\{du^1, du_2, \dots, du_k\}$ when referencing the coordinates on $\mathbb{R}^k$ and $\{dx^1, dx_2, \dots, dx^n\}$ when referencing the coordinates on $\mathbb{R}^n$.

Exercise 26.0.0.4. Let $B_{n \times n} = (b_{i,j}) \in [C^\infty(M)]^{n \times n}$ be an $n \times n$ matrix. Then

$$\bigwedge_{i=1}^n \left(\sum_{j=1}^n b_{i,j} dx^j \right) = (\det B) dx^1 \wedge dx_2 \wedge \dots \wedge dx^n$$

Proof. Bilinearity of the exterior product implies that

$$\begin{aligned}
 \bigwedge_{i=1}^n \left(\sum_{j=1}^n b_{i,j} dx^j \right) &= \left(\sum_{j=1}^n b_{1,j} dx^j \right) \wedge \left(\sum_{j=1}^n b_{2,j} dx^j \right) \wedge \cdots \wedge \left(\sum_{j=1}^n b_{n,j} dx^j \right) \\
 &= \sum_{j_1, \dots, j_n=1}^n \left(\prod_{i=1}^n b_{i,j_i} \right) dx_{j_1} \wedge dx_{j_2} \wedge \cdots \wedge dx_{j_n} \\
 &= \sum_{j_1 \neq \dots \neq j_n} \left(\prod_{i=1}^n b_{i,j_i} \right) dx_{j_1} \wedge dx_{j_2} \wedge \cdots \wedge dx_{j_n} \\
 &= \left[\sum_{\sigma \in S_n} \text{sgn}(\sigma) \left(\prod_{i=1}^n b_{i,\sigma(i)} \right) \right] dx_1 \wedge dx_2 \wedge \cdots \wedge dx_n \\
 &= (\det B) dx_1 \wedge dx_2 \wedge \cdots \wedge dx_n
 \end{aligned}$$

□

Definition 26.0.0.5. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a 0-form on $\mathbb{R}^n$. We define a 1-form, denoted df , on $\mathbb{R}^n$ by

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x^i} dx^i$$

Let $\omega = \sum_{I \in \mathcal{I}_{k,n}} f_I dx^i$ be a k -form on $\mathbb{R}^n$. We can define a differential $k+1$ -form, denoted $d\omega$, on $\mathbb{R}^n$ by

$$d\omega = \sum_{I \in \mathcal{I}_{k,n}} df_I \wedge dx^i$$

Exercise 26.0.0.6. On $\mathbb{R}^3$, put

1. $\omega_0 = f_0$,
2. $\omega_1 = f_1 dx^1 + f_2 dx_2 + f_3 dx_3$,
3. $\omega_2 = f_1 dx_2 \wedge dx_3 - f_2 dx^1 \wedge dx_3 + f_3 dx^1 \wedge dx_2$

Show that

1. $d\omega_0 = \frac{\partial f_0}{\partial x^1} dx^1 + \frac{\partial f_0}{\partial x^2} dx_2 + \frac{\partial f_0}{\partial x^3} dx_3$
2. $d\omega_1 = \left(\frac{\partial f_3}{\partial x^2} - \frac{\partial f_2}{\partial x^3} \right) dx_2 \wedge dx_3 + \left(\frac{\partial f_3}{\partial x^1} - \frac{\partial f_1}{\partial x^3} \right) dx^1 \wedge dx_3 + \left(\frac{\partial f_2}{\partial x^1} - \frac{\partial f_1}{\partial x^2} \right) dx^1 \wedge dx_2$
3. $d\omega_2 = \left(\frac{\partial f_1}{\partial x^1} + \frac{\partial f_2}{\partial x^2} + \frac{\partial f_3}{\partial x^3} \right) dx^1 \wedge dx_2 \wedge dx_3$

Proof. Straightforward. □

Exercise 26.0.0.7. Let $I \in \mathcal{I}_{k,n}$. Then there is a unique $I_* \in \mathcal{I}_{n-k,n}$ such that $dx^i \wedge dx_{I_*} = dx^1 \wedge dx_2 \wedge \cdots \wedge dx^n$.

Definition 26.0.0.8. We define a linear map $*$: $\Phi_k(\mathbb{R}^n) \rightarrow \Gamma^{n-k}(\mathbb{R}^n)$ called the **Hodge *-operator** by

$$* \sum_{I \in \mathcal{I}_{k,n}} f_I dx^i = \sum_{I \in \mathcal{I}_{k,n}} f_I dx_{I_*}$$

Definition 26.0.0.9. Let $\phi : \mathbb{R}^k \rightarrow \mathbb{R}^n$ be smooth. Write $\phi = (\phi_1, \phi_2, \dots, \phi_n)$. We define $\phi^* : \Phi_k(\mathbb{R}^n) \rightarrow \Phi_k(\mathbb{R}^k)$ via the following properties:

1. for each 0-form f on $\mathbb{R}^n$, $\phi^* f = f \circ \phi$

2. for $i = 1, \dots, n$, $\phi^* dx^i = d\phi_i$
3. for an s -form ω , and a t -form χ on $\mathbb{R}^n$, $\phi^*(\omega \wedge \chi) = (\phi^*\omega) \wedge (\phi^*\chi)$
4. for l -forms ω, χ on $\mathbb{R}^n$, $\phi^*(\omega + \chi) = \phi^*\omega + \phi^*\chi$

Exercise 26.0.0.10. Let $M \subset \mathbb{R}^n$ be a k -dimensional smooth submanifold of $\mathbb{R}^n$, $\phi : U \rightarrow V$ a smooth parametrization of M , $\omega = \sum_{I \in \mathcal{I}_{k,n}} f_I dx^i$ an k -form on $\mathbb{R}^n$. Then

$$\phi^*\omega = \left(\sum_{I \in \mathcal{I}_{k,n}} (f_I \circ \phi)(\det v\phi_I) \right) du^1 \wedge du_2 \wedge \dots \wedge du_k$$

Proof. By definition,

$$\begin{aligned} \phi^*\omega &= \phi^* \sum_{I \in \mathcal{I}_{k,n}} f_I dx^i \\ &= \sum_{I \in \mathcal{I}_{k,n}} (\phi^* f_I) \phi^* dx^i \\ &= \sum_{I \in \mathcal{I}_{k,n}} (f_I \circ \phi) d\phi_I \end{aligned}$$

A previous exercise tells us that for each $I \in \mathcal{I}_{k,n}$,

$$\begin{aligned} d\phi_I &= d\phi_{i_1} \wedge d\phi_{i_2} \wedge \dots \wedge d\phi_{i_n} \\ &= \left(\sum_{j=1}^n \frac{\partial \phi_{i_1}}{\partial u^j} du^j \right) \wedge \left(\sum_{j=1}^n \frac{\partial \phi_{i_2}}{\partial u^j} du^j \right) \wedge \dots \wedge \left(\sum_{j=1}^n \frac{\partial \phi_{i_k}}{\partial u^j} du^j \right) \\ &= (\det v\phi_I) du^1 \wedge du_2 \wedge \dots \wedge du_k \end{aligned}$$

Therefore

$$\begin{aligned} \phi^*\omega &= \sum_{I \in \mathcal{I}_{k,n}} (f_I \circ \phi) d\phi_I \\ &= \sum_{I \in \mathcal{I}_{k,n}} (f_I \circ \phi)(\det v\phi_I) du^1 \wedge du_2 \wedge \dots \wedge du_k \\ &= \left(\sum_{I \in \mathcal{I}_{k,n}} (f_I \circ \phi)(\det v\phi_I) \right) du^1 \wedge du_2 \wedge \dots \wedge du_k \end{aligned}$$

□

26.1 Integration of Differential Forms

Definition 26.1.0.1. Let $U \subset \mathbb{R}^k$ be open and $\omega = f dx^1 \wedge dx_2 \wedge \dots \wedge dx_k$ a k -form on $\mathbb{R}^k$. Define

$$\int_U \omega = \int_U f dx$$

Definition 26.1.0.2. Let $M \subset \mathbb{R}^n$ be a k -dimensional oriented smooth submanifold of $\mathbb{R}^n$, ω a k -form on $\mathbb{R}^n$ and $\phi : U \rightarrow V$ a local smooth, orientation-preserving parametrization of M . Define

$$\int_V \omega = \int_U \phi^*\omega$$

Exercise 26.1.0.3.

Theorem 26.1.0.4. Stokes Theorem:

Let $M \subset \mathbb{R}^n$ be a k -dimensional oriented smooth submanifold of $\mathbb{R}^n$ and ω a $k-1$ -form on $\mathbb{R}^n$. Then

$$\int_{\partial M} \omega = \int_M d\omega$$

Summation

Asymptotic Notation

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